GMDC/ENV/BW/121/2025-26
September 24, 2025

To,
The Member Secretary (Coal Mining),
Ministry of Environment, Forests & Climate Change
Indira Paryavaran Bhawan, Aliganj, Jor bagh Road
New Delhi-110003

Subject: **Submission of Application for Environment Clearance w.r.t Baitarni West Opencast Coal Mine with production capacity of 15 Million TPA in an area of 1196 ha at Tehsil- Chhendipada, District-Angul (Odisha) of M/s. Gujarat Mineral Development Corporation Ltd. (GMDC)**

Ref:

1. **TOR granted by MoEF&CC vide letter reference No. IA-J-11015/81/2017-IA-II(M) dated 22nd November 2021 to M/s Odisha Mining Corporation Ltd. (OMC) and Transfer of ToR to Ms. GMDC vide Letter. No. IA-J-11015/81/2017-IA-II(M) dated 11th November 2023.**
2. **Minutes of the 23rd Meeting of Expert Appraisal Committee (EAC–Coal Mining Sector) held on 03.04.2025 (Agenda No. 23.1).**

Dear Sir,

With reference to the above and the Minutes of the 23rd Meeting of Expert Appraisal Committee (EAC–Coal Mining Sector) held on 03.04.2025 (Agenda No. 23.1), we are submitting a fresh application for the appraisal of Environmental Clearance of Baitarni West Opencast Coal Mine with production capacity of 15 Million TPA. We are submitting herewith Final EIA/EMP Report duly incorporating the points as mentioned in the MOM of the 23rd EAC (Coal) meeting as given below:

1. Final EIA/EMP Report along with all requisite annexures
2. Public Hearing Proceedings
3. CAF, Form I (Part A, Part B & Part C)

We humbly request you to kindly consider our proposal in the upcoming EAC (Coal) meeting for grant of Environmental Clearance.

Thanking you,

Yours faithfully,
For M/s Gujarat Mineral Development Corporation Ltd.

General Manager (Environment)
(Authorized Signatory)

Encl.: As above

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PROJECT SHIKHAR
A STRATEGIC TRANSFORMATION INITIATIVE AT GMDC LTD.

Final Environmental Impact Assessment Report

BAITARNI WEST OPENCAST COAL MINE WITH PRODUCTION CAPACITY OF 15 MTPA IN AN AREA OF 1196 HA, CHHENDIPADA, DISTRICT-ANGUL (ODISHA)

Schedule: 1 (a), Mining of Minerals, Category: A

(Study Period: 1st October to 31st December-2023)

(Post Monsoon Season)

ToR Letter Granted vide: No. IA-J-11015/81/2017-IA-II(M) dated 22nd November 2021

Transfer of ToR vide: No. IA-J-11015/81/2017-IA-II(M) dated 11th November 2023

Project ID-GRC/EIA/Sep-23/11



M/s Gujarat Mineral Development Corporation Ltd (GMDC)

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Consultant

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GRC INDIA TRAINING & ANALYTICAL LABORATORY

(NABL ACCREDITED & MOEFCC NO. S.O. 388 (E))

A unit of GRC India

Validity: 25.04.2025

(Submission –September 2025)

UNDERTAKING

(As per OM No. J.11013/41/2006/IA-II (I) Dated 5/10/2011 by MoEFCC, New Delhi, Govt. of India)

I, Chirag Arvindkumar Shah, General Manager (Environment) give this undertaking to the effect that conditions laid down in ToR prescribed by MoEF&CC, New Delhi vide letter No. IA-J-11015/81/2017-IA-II(M) Dated 22.11.2021 & Transfer to Ms. GMDC vide Letter. No. IA-J-11015/81/2017-IA-II(M) dated 11th November 2023 for preparation of EIA for Baitarni West Opencast Coal Mine with production capacity of 15 Million TPA in an area of 1196 ha at Tehsil-Chhendipada, District-Angul (Odisha) have been complied in the EIA report.

In order to assess the likely impacts of the proposed project, M/s. Gujarat Mineral Development Corporation Ltd (GMDC), entrusted M/s GRC India Pvt Ltd, Noida to facilitate in the process of Environmental Impact Assessment (EIA) for various environmental attributes which are likely to be affected and to prepare a detailed Environmental Management Plan (EMP).

We hereby submit the undertaking to MoEF&CC, New Delhi as per OM No. J.11013/41/2006/IA-II (I), Dated 05/10/2011 that the contents and information submitted in this EIA report is owned by M/s. Gujarat Mineral Development Corporation Ltd. (GMDC). The EIA Report fully complies with all the points mentioned in the TOR prescribed by MoEF&CC. The data submitted in this EIA Report is true and factually correct.

For M/s. Gujarat Mineral Development Corporation Ltd. (GMDC)

22/01/25

Name: Chirag Arvindkumar Shah
Designation: General Manager (Environment)

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PROJECT SHIKHAR
A STRATEGIC TRANSFORMATION INITIATIVE AT GMDC LTD.

Declaration by Experts contributing to the EIA/EMP report of **Baitarni West Opencast Coal Mine with production capacity of 15 MTPA in an area of 1196 ha, Chhendipada, District-Angul (Odisha) by M/s Gujrat Mineral Development Corporation Ltd (GMDC) (Category-1 (a) project as per EIA Notification 2006)**

We, hereby, certify that we were part of the EIA team in the following capacity that developed the above EIA.

Sector	:	**Mining of Minerals**	**Signature**
Name of EIA Coordinator	:	Dr. K. L. Satapathy	
Associate to EIA Coordinator	:	Ms. Mudita Tomar Singh	
Period of Involvement	:	**October 2023 to till date**	

Functional area experts:

S. No.	Functional Area	Name of the expert/s	Involvement (period and task**)	Signature and date
1	FAE-AP	Dr. Dhiraj Kr Singh Ms. Monika Gupta (Cat- B)	Oct-2023 to till date	
2	FAE-WP	Mr. Sonu Mr. Saurabh Gola (Cat- B)	Oct-2023 to till date	
3	FAE-SHW	Dr. Dhiraj Kr Singh	Oct-2023 to till date	
4	FAE-SE	Mr. Rajiv Rastogi	Oct-2023 to till date	
5	FAE-EB	Dr. Ashok Kr Rathoure	Oct-2023 to till date	
6	FAE-HG	Prof. Tapan Majumder	Oct-2023 to till date	
7	FAE-GEO	Prof. Tapan Majumder	Oct-2023 to till date	
8	FAE-SC	Mr. N. P. S. Varde	Oct-2023 to till date	
9	FAE-AQ	Dr. K. L. Satapathy Ms. Monika Gupta (Cat- B)	Oct-2023 to till date	
10	FAE- NV	Dr. K. L. Satapathy	Oct-2023 to till date	
11	FAE – LU	Mr. N. P. S. Varde	Oct-2023 to till date	
12	FAE – RH	Mr. Kalidas Choudhury	Oct-2023 to till date	

*One TM against each FAE may be shown

**Please attach additional sheet if required

Declaration by the Head of the accredited consultant organization/authorized person

I, Dr. Dhiraj Kr. Singh hereby, confirm that the above-mentioned experts prepared the EIA/EMP report of **Baitarni West Opencast Coal Mine with production capacity of 15 MTPA in an area of 1196 ha, Chhendipada, District-Angul (Odisha) by M/s Gujrat Mineral Development Corporation Ltd (GMDC)**

I also confirm that EIA Coordinator (EC) has gone through the report, and the consultant organization shall be fully accountable for any misleading information.

It is certified that no unethical practices, plagiarism involved in carrying out the work and external data/text has not been used without proper acknowledgement while preparing this EIA report.

Signature:

Name: **Dr. Dhiraj Kr. Singh**

Designation: **Managing Director**

Name of the EIA consultant organization: **Grass Roots Research & Creation India (P) Ltd**

NABET Certificate No. & Issue Date: **NABET/EIA/21-24/SA 0211, dated: -15.09.2021 and Valid till: - 15.08.2024.**

Certificate of Plagiarism check

Title of EIA Report:	Baitarni West Opencast Coal Mine with production capacity of 15 MTPA in an area of 1196 ha at Tehsil- Chhendipada, District-Angul (Odisha)
Name of Accredited Organization:	M/s GRC India Pvt. Ltd, Noida
Unique Identification Number:	GRC/EIA/Sep-23/11
Name of EIA Coordinator:	Dr. K. L. Satapathy
Name of the Software:	Plagiarism Checker X
Date of Check:	23.01.2025
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EIA Coordinator:

Name : Dr. K. L. Satapathy

Signature :

Head of ACO / Authorized person:

Name : Dr. Dhiraj Kr. Singh (MD)

Signature :

Name of the EIA Consultant Organization: M/s GRC India Pvt. Ltd, Noida, 201301

NABET Certificate no. & Issue date: NABET/EIA/24-27/RA 0354, dated: 11.11.2024 valid till 16.08.2027.

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Contents

CHAPTER-1 : INTRODUCTION 1

1.1 PURPOSE OF REPORT 1
1.2 IDENTIFICATION OF PROJECT & PROJECT PROPONENT 2
1.2.1 IDENTIFICATION OF PROJECT 2
1.3 PROJECT PROPONENT 4
1.4 BRIEF DESCRIPTION OF NATURE, SIZE, LOCATION OF THE PROJECT AND ITS IMPORTANCE TO THE COUNTRY, REGION 5
1.4.1 NATURE & SIZE OF PROJECT 5
1.4.2 LOCATION OF PROJECT 5
1.5 IMPORTANCE OF PROJECT TO THE COUNTRY & REGION 6
1.6 PRODUCTION AND IMPORTS 7
1.7 OVERVIEW OF INDIAN COAL SECTOR 7
1.8 DEMAND AND SUPPLY GAP 8
1.9 IMPORT VS INDIGENOUS PRODUCTION 9
1.10 EXPORT POSSIBILITY 9
1.11 DOMESTIC/EXPORT MARKETS 9
1.12 BRIEF HISTORY OF PROJECT 9
1.12.1 DETAILS OF MINING LEASE 10
1.12.2 MINING PLAN 10
1.12.3 AIR AND WATER CONSENTS 10
1.12.4 FOREST CLEARANCE 10
1.13 SCOPE OF THE STUDY 10
1.14 STRUCTURE OF EIA REPORT 11
1.15 LAWS APPLICABLE TO THIS PROJECT 13

CHAPTER-2 : PROJECT DESCRIPTION 20

2.1 TYPE OF THE PROJECT 20
2.2 NEED OF THE PROJECT 20
2.3 LOCATION OF THE PROJECT (MAPS SHOWING GENERAL LOCATION, SPECIFIC LOCATION, PROJECT BOUNDARY & PROJECT SITE LAYOUT) 21
2.3.1 MINE BOUNDARY 21
2.3.2 DISTANCE FROM WATER BODIES 21
2.3.3 LIST OF INDUSTRIES WITHIN 10 KM RADIUS OF THE PLANT SITE 28
2.4 SIZE OR MAGNITUDE OF OPERATION (INCL. ASSOCIATED ACTIVITIES REQUIRED BY OR FOR THE PROJECT) 29
2.4.1 SIZE OR MAGNITUDE OF OPERATION 29
2.4.2 PROJECT COST 29
2.5 LAND USE 30
2.5.1 PRE-MINING LAND TYPE 35
2.5.2 ENERGY 36
2.5.3 EMPLOYMENT GENERATION 36
2.5.4 PROPOSED SCHEDULE FOR APPROVAL AND IMPLEMENTATION 37
2.6 PROJECT DESCRIPTION INCLUDING DRAWINGS SHOWING PROJECT LAYOUT, COMPONENTS OF PROJECT ETC. 39
2.7 GEOLOGY OF AREA 43
2.7.1 REGIONAL GEOLOGY 43

2.7.2 LOCAL GEOLOGY, STRUCTURE, STRATIGRAPHIC SEQUENCE, CHARACTERISTICS OF THE LITHO-LOGICAL UNITS (COAL SEAMS /PARTINGS/OVERBURDEN) 45
2.8 STATUS OF EXPLORATION OF THE BLOCK 46
2.8.1 FURTHER EXPLORATION 48
2.9 COAL RESOURCE / RESERVE ESTIMATION 49
2.9.1 LIFE OF THE MINE 49
2.9.2 GRADE OF COAL 49
2.10 TECHNOLOGY AND PROCESS DESCRIPTION 50
2.10.1 PROPOSED METHOD OF MINING 50
2.10.2 SELECTION OF TECHNOLOGY 50
2.10.3 BENCH PARAMETERS 50
2.10.4 MINE LAYOUT 51
2.10.5 ACCESS TRENCH 51
2.10.6 DRILLING AND BLASTING 52
2.10.7 MINE DEVELOPMENT 52
2.10.8 TRANSPORT LAYOUT 53
2.10.9 PRODUCTION PLAN 54
2.10.10 EQUIPMENT/MACHINERIES 55
2.10.11 WASTE GENERATION AND MANAGEMENT 56
2.10.12 TOPSOIL GENERATION AND MANAGEMENT 59
2.10.13 COAL HANDLING ARRANGEMENT 60
2.10.14 SITE SERVICES 62
2.11 OCCUPATIONAL HEALTH ISSUES 63
2.12 DIVERSION/ RECHANNELING OF NALLAH/CANAL/RIVER/ROAD 63
2.13 REHABILITATION & RESETTLEMENT 63
2.14 DESCRIPTION OF MITIGATION MEASURES INCORPORATED INTO THE PROJECT TO MEET ENVIRONMENTAL STANDARDS, ENVIRONMENTAL OPERATING CONDITIONS, OR OTHER EIA REQUIREMENTS. 63
2.15 USE OF NATURAL RESOURCES 69
2.16 SAFETY MANAGEMENT AND CONSERVATION 69
2.17 ECONOMIC PARAMETERS 69
2.18 ASSESSMENT OF NEW & UNTESTED TECHNOLOGY FOR THE RISK OF TECHNOLOGICAL FAILURE. 70

CHAPTER-3 : DESCRIPTION OF THE ENVIRONMENT 71

3.1 TOPOGRAPHY 72
3.2 DRAINAGE 74
3.3 GEOLOGY & GEOMORPHOLOGY 75
3.4 LOCAL GEOLOGY 75
3.5 GEOMORPHOLOGY 75
3.6 SOIL CHARACTERSTICS 75
3.7 SEISMIC STATUS 76
3.8 BASELINE STUDY AREA AND PERIOD 76
3.9 ESTABLISHMENT OF BASELINE STUDY FOR VALUED ENVIRONMENTAL COMPONENTS, AS IDENTIFIED IN THE SCOPE 78
3.9.1 MICROMETEOROLOGY 78
3.9.2 SOURCE OF METEOROLOGICAL DATA 79
3.9.3 TEMPERATURE 79

3.9.4 HUMIDITY 79
3.9.5 RAINFALL 79
3.9.6 WIND SPEED AND WIND DIRECTION 79
3.9.7 WINDROSE 80
3.9.8 INTERPRETATION OF MICROMETEOROLOGICAL DATA 81
3.9.9 AIR ENVIRONMENT 81
3.9.10 SELECTION OF SAMPLING LOCATIONS 81
3.9.11 FREQUENCY AND PARAMETERS FOR SAMPLING 82
3.9.12 METHODOLOGY FOR SAMPLING AND ANALYSIS 82
3.9.13 AIR QUALITY OF STUDY AREA 82
3.9.14 INTERPRETATION OF AMBIENT AIR QUALITY DATA 87
3.10 NOISE ENVIRONMENT 88
3.10.1 OBSERVATIONS 92
3.10.2 INTERPRETATION OF NOISE DATA 92
3.11 WATER ENVIRONMENT 92
3.11.1 RECONNAISSANCE SURVEY 92
3.11.2 SAMPLING AND ANALYSIS 93
3.11.3 METHOD OF WATER SAMPLING 93
3.11.4 GROUND WATER AND SURFACE WATER RESOURCES 93
3.11.5 SUMMARY OF GROUND WATER QUALITY 105
3.11.6 INTERPRETATION OF GROUND WATER QUALITY DATA 105
3.11.7 SUMMARY OF SURFACE WATER QUALITY 106
3.11.8 INTERPRETATION OF SURFACE WATER QUALITY DATA 106
3.12 SOIL ENVIRONMENT 107
3.12.4 INTERPRETATION OF SOIL ANALYSIS DATA 111
3.13 LAND USE PATTERN OF STUDY AREA 112
3.14 SOCIO-ECONOMICS 121
3.14.1 DESCRIPTION OF THE STUDY AREA (10 KM RADIUS AREA) 121
3.14.2 BASE LINE DATA 121
3.14.3 ANGUL DISTRICT AS A WHOLE 130
3.15 ECOLOGY AND BIODIVERSITY 159

CHAPTER-4 : ANTICIPATED ENVIRONMENTAL IMPACTS AND ITS MITIGATION MEASURES 177

4.1 INTRODUCTION 177
4.2 IDENTIFICATION & CHARACTERIZATION OF IMPACTS 178
4.2.1 IMPACTS DUE TO PROJECT LOCATION 179
4.2.2 IMPACTS DUE TO PROJECT DESIGN 179
4.2.3 IMPACT DURING CONSTRUCTION 179
4.2.4 RESIDENTIAL COLONY 179
4.2.5 SERVICE BUILDING 179
4.2.6 ROADS 179
4.2.7 EMBANKMENT 179
4.2.8 RAILWAY SIDING 180
4.2.9 IN-PIT CONVEYOR BELT 183
4.2.10 IMPACT DURING OPERATION PHASE 186
4.3 ANTICIPATED ENVIRONMENTAL IMPACTS & MITIGATION MEASURES DURING OPERATIONAL PHASE 186

4.3.1 IMPACT ON LAND USE 187
4.3.2 IMPACT ON AIR QUALITY 189
4.4 IMPACT DUE TO TRANSPORT 207
4.4.1 TRANSPORTATION OF COAL 207
4.4.2 TRAFFIC STUDY 207
4.5 AIR MODELLING 213
4.6 IMPACTS OF WATER POLLUTION AND ITS MITIGATION MEASURES 219
4.6.1 TREATMENT OF ACIDIC MINE WATER 222
4.6.2 IMPACT OF MINING ON HYDROLOGY 222
4.6.3 IMPACT OF MINING ON SURFACE WATER 224
4.6.4 DIVERSION PLAN OF SUPAI NALA 228
4.6.5 ESTIMATION OF MINE SEEPAGE AND ADVANCED DEWATERING PLAN 233
4.6.6 IMPACT OF MINE DEWATERING/ ABSTRACTION AND MITIGATION MEASURES 235
4.6.7 TREATMENT & DISPOSAL OF WATER FROM MINE 235
4.6.8 WATER CONSERVATION MEASURES 237
4.6.9 ASSESSMENT OF IMPACT AND CONTROL MEASURES ON HYDROLOGY & HYDROGEOLOGY 237
4.6.10 GROUNDWATER RESOURCES POTENTIAL 238
4.6.11 STAGE OF GROUNDWATER EXTRACTION 238
4.6.12 IMPACT ON AQUIFER SYSTEM 239
4.6.13 GROUND WATER FLOW AND AQUIFER INTERACTION WITH SURFACE WATER BODIES 240
4.6.14 WATER HARVESTING & ARTIFICIAL RECHARGE 241
4.6.15 CONSERVATION MEASURES 242
4.6.16 FUTURE STRATEGIES 243
4.7 IMPACTS OF NOISE/VIBRATIONS AND MITIGATION MEASURES 243
4.7.1 PREDICTION OF IMPACT DUE TO THE PROPOSED ACTIVITY 243
4.7.2 MITIGATION MEASURES 245
4.8 IMPACT DUE TO GROUND VIBRATIONS, AIR BLAST & FLY ROCK 246
4.8.1 GROUND VIBRATION 246
4.9 IMPACTS DUE TO SOLID WASTE GENERATION & ITS MANAGEMENT 247
4.9.1 IMPACT ON TOPSOIL 247
4.9.2 TOPSOIL GENERATION AND MANAGEMENT 248
4.10 IMPACTS ON ECOLOGY & BIODIVERSITY AND ITS MITIGATION MEASURES 248
4.11 GREENBELT DEVELOPMENT 258
4.11.1 PLANTATION FOR GREEN COVER 258
4.12 MINE CLOSURE PLAN 259
4.13 IMPACT ON LAND USE AND LANDSCAPE 259
4.13.1 IMPACT ON LAND USE PATTERN 259
4.13.2 CONTROL MEASURES TO RESTORE LAND USE & LANDSCAPE 260
4.13.3 POST MINING LAND USE 260
4.13.4 LAND DEGRADATION AND TECHNICAL RECLAMATION 263
4.13.5 RECLAMATION & RESTORATION OF THE LEASE AREA 266
4.14 CARBON FOOTPRINT STUDY 269
4.15 ASSESSMENT OF ANNUAL MINE CLOSURE COST 275
4.16 PUBLIC HEALTH IMPLICATIONS OF THE PROJECT 280
4.17 SPECIFIC PROGRAMME FOR UPLIFTMENT OF SC/ST POPULATION OF THE STUDY AREA 280
4.18 PUBLIC CONSULTATION 281
4.19 MERCURY CONTENT 281

4.20 HAZARDOUS SUBSTANCE MANAGEMENT 281
4.21 CORPORATE SOCIAL RESPONSIBILITY 282
4.22 WILDLIFE CONSERVATION PLAN 282
4.23 SITE SPECIFIC WILDLIFE MANAGEMENT PLAN 284
4.24 SITE SPECIFIC MANAGEMENT PLAN 284
4.25 IMPACT ON SOCIO-ECONOMIC PROFILE 284
4.26 IMPACT ON BIODIVERSITY 287
4.27 ENVIRONMENTAL IMPACT MATRIX 295
4.27.1 IMPACT MATRIX WITH QUANTITATIVE RATING 295

CHAPTER-5 : ANALYSIS OF ALTERNATIVES TECHNOLOGY & SITE 272

5.1 INTRODUCTION 272
5.2 PROCESS/TECHNOLOGY ALTERNATIVE 272
5.2.1 MINING METHODOLOGY 272
5.2.1.2 EQUIPMENT SELECTION 273
5.3 ASSESSMENT OF NEW AND UNTESTED TECHNOLOGY 274

CHAPTER-6 : ENVIRONMENTAL MONITORING PROGRAM 275

6.1 INTRODUCTION 275
6.2 SCOPE OF THE ENVIRONMENT MONITORING PROGRAMME 276
6.3 KEY PARAMETERS TO BE MONITORED 276
6.4 ENVIRONMENTAL MONITORING PROGRAMME FOR KEY PARAMETERS 277
6.4.1MONITORING OF PLANTATION 278
6.4.2 MONITORING OF OCCUPATIONAL HEALTH 278
6.5 MONITORING SCHEDULE OF ENVIRONMENTAL PARAMETERS 279
6.6 ENVIRONMENTAL CELL 279
6.7 BUDGETARY PROVISIONS FOR MONITORING 280
6.8 LABORATORY FACILITIES AND EQUIPMENT 284
6.9 ENVIRONMENTAL AUDITING 284

CHAPTER-7 : ADDITIONAL STUDIES 287

7.1 PUBLIC HEARING 287
7.2 RISK ASSESSMENT AND DISASTER MANAGEMENT PLAN 336
7.3 OPERATION OF MINING AND OTHER EQUIPMENT 337
7.3.4 RISK ASSESSMENT FOR MINING ACTIVITY & OVERBURDEN DUMPING OPERATION 339
7.4 DISASTER MANAGEMENT PLAN 342
7.4.1 PRE –DISASTER ACTIONS 343
7.4.1.1 EMERGENCY ORGANIZATION 343
7.4.1.2 EMERGENCY COMMUNICATION 344
7.4.1.3 RESPONSIBILITIES OF KEY PERSONNEL DURING EMERGENCY 344
7.4.1.4 EMERGENCY FACILITIES 346
7.4.1.5 OFF-SITE EMERGENCY PREPAREDNESS PLAN 347
7.5 POST DISASTER ANALYSIS AND EVALUATION 348
7.6 OCCUPATIONAL HEALTH AND SAFETY 348
7.6.1 IDENTIFICATION OF WORK-RELATED OCCUPATIONAL HEALTH HAZARDS 348
7.6.2 MEASURES FOR MITIGATION OF OCCUPATIONAL HEALTH HAZARDS-PROJECT PERSONNEL 350

7.6.3 MEASURES FOR MITIGATION OF HEALTH HAZARDS-GENERAL PUBLIC 350
7.7 RESETTLEMENT AND REHABILITATION 350
7.8 SOCIAL IMPACT ASSESSMENT 351

CHAPTER-8 : PROJECT BENEFITS 353

8.1 INTRODUCTION 353
8.2 SOCIAL BENEFITS 353
8.2.1 EMPLOYMENT 353
8.3 ENVIRONMENTAL BENEFITS 354
8.4 IMPROVEMENT IN PHYSICAL INFRASTRUCTURE & COMMUNITY DEVELOPMENT 355
8.5 IMPROVEMENT IN SOCIAL INFRASTRUCTURE 355
8.6 EMPLOYMENT POTENTIAL 356
8.7 OTHER TANGIBLE BENEFITS 356
8.8 CORPORATE ENVIRONMENT RESPONSIBILITY (CER) 356
8.9 CORPORATE SOCIAL RESPONSIBILITY (CSR) 357
8.10 CONCLUSION 359

CHAPTER-9 : ENVIRONMENTAL COST BENEFIT ANALYSIS 361

9.1 INTRODUCTION 361
9.2 ENVIRONMENTAL COST BENEFIT ANALYSIS 361
9.3 INTRODUCTION 361
9.4 EXPENDITURE TO BE INCURRED TOWARDS CSR 364
9.5 COMPENSATORY AFFORESTATION AND WILDLIFE MANAGEMENT COST 364
9.6 RAINWATER HARVESTING 364
9.7 COMMUNITY DEVELOPMENT FOR PERIPHERAL VILLAGES 364
9.8 BIOLOGICAL RECLAMATION COST 364
9.9 MINE CLOSURE COST 364
9.10 ENVIRONMENTAL COST BENEFIT ANALYSIS 365

CHAPTER-10 : ENVIRONMENTAL MANAGEMENT PLAN 371

10.1 INTRODUCTION 371
10.2 ELEMENTS OF ENVIRONMENTAL MANAGEMENT PLAN 371
10.2.1 QUALITY ENVIRONMENT HEALTH AND SAFETY POLICY (QEHS) 372
10.2.1.1 OBJECTIVES AND TARGETS 373
10.2.2 OPERATIONAL AND EMERGENCY PROCEDURES 374
10.2.3 ANTICIAPTED POTENTIAL IMPACTS & SUGGESTED MITIGATION MEASURES 375
10.3 GREENBELT DEVELOPMENT 379
10.4 PLAN FOR IMPLEMENTATION & MONITORING OF THE SUGGESTED MITIGATION MEASURES 379
10.4.1 ENVIRONMENT MANAGEMENT CELL 380
10.4.2 ENVIRONMENT MANAGEMENT PLAN BUDGET 380
10.5 CONCLUSION 381

CHAPTER-11 : SUMMARY & CONCLUSION 382

11.1 DETAILED CHRONOLOGY ON ALLOCATION OF COAL BLOCK 382
11.2 PROJECT DETAILS 383
11.3 ANTICIPATED ENVIRONMENTAL IMPACTS AND MITIGATION MEASURES 389

11.4 ENVIRONMENT MANAGEMENT PLAN 390
11.5 OVERALL JUSTIFICATION FOR IMPLEMENTATION OF THE PROJECT 390
11.6 EXPLANATION ON HOW ADVERSE EFFECTS ARE MITIGATED 391
11.7 CONCLUSION 391

CHAPTER-12 : DISCLOSURE OF CONSULTANT ENGAGED 392

12.0 GENERAL 392
12.1 INTRODUCTION OF ORGANIZATION 392
12.3 EIA CORDINATOR AND TEAM MEMBER 393
12.4 LABORATORY INVOLVED FOR BASELINE MONITORING AND OTHER ANALYSIS 394

LIST OF FIGURES

FIGURE 2-1: THE LOCATION OF THE LEASE AREA/GENERAL LOCATION 23
FIGURE 2-2: LOCATION OF PROJECT 24
FIGURE 2-3: PILLAR-COORDINATE MAP 25
FIGURE 2-4:-LOCATION PLAN OF BAITARNI WEST COAL BLOCK IN TALCHER COAL FIELD 26
FIGURE 2-5: VICINITY MAP (500M) SHOWING THE SITE AND SURROUNDINGS 27
FIGURE 2-6: STUDY AREA MAP (10 KM RADIUS) 27
FIGURE 2-7 ECO SENSITIVE MAP OF 15 KM STUDY AREA 28
FIGURE 2-8 MAP SHOWING NEARBY INDUSTRIES 29
FIGURE 2-9 SURFACE PLAN 30
FIGURE 2-10 PIE CHART SHOWING THE BREAKUP OF BROAD LAND TYPE 31
FIGURE 2-11 PIE CHART SHOWING THE BREAKUP OF TENANCY LAND 32
FIGURE 2-12 PIE CHART SHOWING THE BREAKUP OF GOVT. NON-FOREST LAND 32
FIGURE 2-13 PIE CHART SHOWING THE BREAKUP OF FOREST LAND 33
FIGURE 2-14 TYPE OF LAND 34
FIGURE 2-15 PIE CHART & GRAPH SHOWING THE BREAKUP OF PRE-MINING LAND TYPE (IN HA) 36
FIGURE 2-16 GEOLOGICAL MAP OF THE STUDY AREA 45
FIGURE 2-17 COAL FLOW DIAGRAM 61
FIGURE 3-1 VARIATION IN LAND ELEVATIONS ABOVE MSL OF DISTRICT ANGUL 72
FIGURE 3-2 CONTOUR MAP OF THE STUDY AREA 73
FIGURE 3-3 CONTOUR MAP OF THE PROJECT SITE 73
FIGURE 3-4 DRAINAGE PATTERN OF THE STUDY AREA 74
FIGURE 3-5 GEOMORPHOLOGY OF THE STUDY AREA 75
FIGURE 3-6 GEOMORPHOLOGY OF THE STUDY AREA 76
FIGURE 3-7 WIND ROSE DIAGRAM 80
FIGURE 3-8 MAP SHOWING THE STUDY REGION WITH LOCATIONS FOR AMBIENT AIR SAMPLING STATIONS 81
FIGURE 3-9 PM_{10} CONCENTRATION IN $\mu G/M^3$ 85
FIGURE 3-10 $PM_{2.5}$ CONCENTRATION IN $\mu G/M^3$ 85
FIGURE 3-11 NO_2 CONCENTRATION IN $\mu G/M^3$ 86
FIGURE 3-12 SO_2 CONCENTRATION IN $\mu G/M^3$ 86
FIGURE 3-13 CO CONCENTRATION IN $\mu G/M^3$ 87
FIGURE 3-14 NOISE MONITORING LOCATIONS MAP 90
FIGURE 3-15 GRAPHICAL REPRESENTATION OF NOISE LEVEL IN THE STUDY AREA 92
FIGURE 3-16 GROUND WATER QUALITY SAMPLING LOCATIONS 96
FIGURE 3-17 SURFACE WATER QUALITY SAMPLING LOCATIONS 97
FIGURE 3-18 MAP SHOWING THE STUDY REGION WITH SOIL SAMPLING LOCATIONS 108
FIGURE 3-19 MAJOR LAND USE/LAND COVER CATEGORIES OF THE STUDY AREA 116
FIGURE 3-20 FCC MAP OF THE STUDY AREA 116
FIGURE 3-21 LULC MAP OF THE STUDY AREA 117
FIGURE 3-22 FCC MAP OF THE SITE 118
FIGURE 3-23 LULC MAP OF THE PROJECT SITE 119
FIGURE 3-24 DEM MAP OF THE STUDY AREA 120
FIGURE 3-25 DEM MAP OF THE LEASE AREA 120
FIGURE 3-26 PIE CHART DEPICTING THE CASTES 127
FIGURE 3-27 PIE CHART DEPICTING THE SEX RATIO 128
FIGURE 3-28 PIE CHART DEPICTING THE LITERACY RATE 128
FIGURE 3-29 PIE CHART DEPICTING THE %OF WORKERS 129
FIGURE 3-30 ECOLOGICAL MAP OF ORISSA 163
FIGURE 3-31 FOREST MAP OF ODISHA 163
FIGURE 3-32 FOREST MAP OF ANGUL 164

FIGURE 3-33 DISTRICT MAP 164
FIGURE 3-34 CATCHMENT MAP OF ANGUL 165
FIGURE 3-35 ANIMAL HUSBANDRY IN ANGUL 165
FIGURE 4-1 EMBANKMENT CROSS SECTION 180
FIGURE 4-2 KEY PLAN OF BAITARNI WEST RAIL TERMINAL 183
FIGURE 4-3 LOCATION OF IN-PIT BELT CONVEYOR 185
FIGURE 4-4 PROPOSED LAND USE 188
FIGURE 4-5 ISOPLETHS FOR 24HR INCREMENTAL CONC. (MG/M^3) FOR PM_{10} 201
FIGURE 4-6 ISOPLETHS FOR 24HR INCREMENTAL CONC. (MG/M^3) FOR $PM_{2.5}$ 202
FIGURE 4-7 ISOPLETHS FOR 24HR INCREMENTAL CONC. (MG/M^3) FOR SO_2 203
FIGURE 4-8 ISOPLETHS FOR 24HR INCREMENTAL CONC. (MG/M^3) FOR NO_2 204
FIGURE 4-9 PROPOSED DEDICATED ROAD TRANSPORTATION ROUTE 211
FIGURE 4-10 TRANSPORTATION ROUTE MAP 212
FIGURE 4-11 GEOTAGGED PHOTOGRAPHS OF TRANSPORTATION ROUTE 213
FIGURE 4-12 SPATIAL DISTRIBUTION OF PREDICTED GLCS OF PM_{10} DUE TO TRANSPORTATION 215
FIGURE 4-13 SPATIAL DISTRIBUTION OF PREDICTED GLCS OF $PM_{2.5}$ DUE TO TRANSPORTATION 216
FIGURE 4-14 SPATIAL DISTRIBUTION OF PREDICTED GLCS OF NO_2 DUE TO TRANSPORTATION 217
FIGURE 4-15 SPATIAL DISTRIBUTION OF PREDICTED GLCS OF CO DUE TO TRANSPORTATION 218
FIGURE 4-16 WATER BALANCE FOR TOTAL WATER REQUIREMENT 220
FIGURE 4-17 WATER BALANCE FOR INITIAL WATER REQUIREMENT 221
FIGURE 4-18 WATERSHED AFFECTED BY THE MINE PIT (QUARRY) 224
FIGURE 4-19: PROPOSED ALIGNMENT OF DIVERSION CHANNELS (SWALES) AND BUNDS 231
FIGURE 4-20 HYDROGEOLOGICAL MAP OF STUDY AREA 239
FIGURE 4-21 DEPTH TO WATER LEVEL MAP PRE-MONSOON APRIL 2023 (SOURCE: CGWB) 240
FIGURE 4-22 DEPTH TO WATER LEVEL MAP POST-MONSOON NOV 2023 (SOURCE: CGWB) 241
FIGURE 4-23 CONTOUR OF RESULTANT NOISE IN DBA 245
FIGURE 4-24 POST MINING LAND USE MAP & FINAL STAGE DUMP PLAN 261
FIGURE 4-25 LAND DEGRADED VS YEAR GRAPH 265
FIGURE 4-26 TECHNICALLY RECLAIMED AREA VS YEAR GRAPH 265
FIGURE 4-27 YEAR WISE TOTAL LAND DEGRADED VS TECHNICALLY RECLAIMED GRAPH 266
FIGURE 4-28 RECLAMATION PLAN (FINAL MINE CLOSURE PLAN) 268
FIGURE 6-1 ORGANOGRAM OF ENVIRONMENTAL CELL 280
FIGURE 7-1 NOTICE FOR PUBLIC HEARING PUBLISHED ON 23.08.24 IN PRAGATIVADI & IN THE NEW INDIAN EXPRESS 288
FIGURE 7-2 PHOTOGRAPHS OF PUBLIC HEARING 289
FIGURE 7-3 THE ON-SITE EMERGENCY ORGANIZATION CHART FOR VARIOUS EMERGENCIES 347
FIGURE 8-1 PHOTOGRAPHS OF CSR ACTIVITIES DONE 359

LIST OF TABLES

TABLE 1:1 STRUCTURE OF EIA AS PER NOTIFICATION 2006 11
TABLE 1:2 APPLICABLE LAWS 14
TABLE 2:1 DISTANCE FROM WATER BODIES 21
TABLE 2:2 INDUSTRY/COAL MINES LIST WITHIN 10 KM 28
TABLE 2:3 AREA AND LAND TYPE 30
TABLE 2:4 VILLAGE WISE GRAZING LAND 33
TABLE 2:5 BREAKUP OF PRE-MINING LAND TYPE (INDICATIVE) 35
TABLE 2:6 PROPOSED SCHEDULE FOR APPROVAL AND IMPLEMENTATION 37
TABLE 2:7 SALIENT FEATURES OF THE PROJECT 39
TABLE 2:8 STRATIGRAPHIC SUCCESSION OF TALCHER COALFIELD 44
TABLE 2:9 AGENCY-WISE BOREHOLES DRILLED IN BAITARNI WEST BLOCK, TALCHER COALFIELD 47
TABLE 2:10 POTENTIAL SEAM WISE NUMBER OF BOREHOLES INTERSECTION AREA AND BOREHOLE DENSITY 47
TABLE 2:11 HEAVY METAL ANALYSIS IN COAL SAMPLES OF HINGULA OCP 49
TABLE 2:12 TENTATIVE COAL PRODUCTION PLAN 54
TABLE 2:13 TYPE OF EQUIPMENT/ HEMM PROPOSED 55
TABLE 2:14 WASTE MANAGEMENT (FIGURES IN MILLION M^3) (TENTATIVE) 57
TABLE 2:15 TOPSOIL MANAGEMENT (FIGURES IN MILLION M^3) – (TENTATIVE) 59
TABLE 3:1 FREQUENCY OF ENVIRONMENTAL MONITORING 78
TABLE 3:2 AMBIENT AIR SAMPLING STATIONS 80
TABLE 3:3 METHODOLOGY FOR SAMPLING AND ANALYSIS 82
TABLE 3:4 SUMMARY OF AMBIENT AIR QUALITY IN THE STUDY AREA (DATE OF SAMPLING & ANALYSIS – 01.10.2023 TO 31.12.2023) 83
TABLE 3:5 NOISE MONITORING LOCATIONS WITHIN THE STUDY AREA 89
TABLE 3:6 NOISE LEVELS IN STUDY AREA (DATE OF MONITORING – 03.10.2023) 91
TABLE 3:7 WATER SAMPLING LOCATIONS (GROUND & SURFACE WATER) 93
TABLE 3:8 GROUND WATER ANALYSIS RESULT (DATE OF SAMPLING – 03.10.2023, DATE OF ANALYSIS – 06.10.2023 TO 20.10.2023) 98
TABLE 3:9 SURFACE WATER ANALYSIS (DATE OF SAMPLING – 03.10.2023, DATE OF ANALYSIS – 06.10.2023 TO 20.10.2023) 102
TABLE 3:10 SOIL SAMPLING LOCATION 107
TABLE 3:11 SOIL QUALITY RESULT (DATE OF SAMPLING – 03.10.2023, DATE OF ANALYSIS – 06.10.2023 TO 20.10.2023) 108
TABLE 3:12 SOIL CHARACTERISTICS 111
TABLE 3:13 MAJOR LAND USE/LAND COVER CATEGORIES OF THE STUDY AREA 114
TABLE 3:14 LULC OF THE LEASE AREA 119
TABLE 3:15 SUMMARY OF DEMOGRAPHIC FEATURES OF RURAL AREA UNDER STUDY 122
TABLE 3:16 DEMOGRAPHIC FEATURES AND OTHER STATISTICS FOR THE STUDY AREA (RURAL AREA) AS PER CENSUS 2011 129
TABLE 3:17 AREA IRRIGATED BY DIFFERENT SOURCES (HA) 137
TABLE 3:18 SOURCE WISE IRRIGATION POTENTIAL IN ANGUL DISTRICT 138
TABLE 3:19 BLOCK WISE MIPS IN ANGUL DISTRICT 138
TABLE 3:20 LIST OF VILLAGES FALLING WITHIN 10 KM FROM THE BOUNDARY OF THE LEASE 149
TABLE 3:21 CHARACTERISTICS OF TREE LAYER VEGETATION AT CORE ZONE FOREST 166
TABLE 3:22 CHARACTERISTICS OF TREE LAYER VEGETATION AT KOSALA RF 167
TABLE 3:23 CHARACTERISTICS OF TREE LAYER VEGETATION AT DEBINAGAR PF 167
TABLE 3:24 GENERAL LIST OF FLORA IN 10 KM STUDY AREA 168
TABLE 3:25 AVIFAUNA IN STUDY AREA 170
TABLE 3:26 BUTTERFLIES IN THE STUDY AREA 172

TABLE 3:27 REPTILES AND AMPHIBIAN IN STUDY AREA 173
TABLE 3:28 MAMMALS IN STUDY AREA 174
TABLE 3:29 FISHES WITHIN STUDY AREA 175
TABLE 3:30 PLANKTONIC ABUNDANCE IN AQUATIC ENVIRONMENT 176
TABLE 4:1 SOURCE OF ORIGIN/DESTINATION OF TRAFFIC 180
TABLE 4:2 YEAR WISE PRODUCTION (PROPOSED) 180
TABLE 4:3 TIME-BOUND ACTION PLAN FOR IMPLEMENTATION OF IN-PIT CONVEYOR BELT 183
TABLE 4:4 PROPOSED MINING LAND-USE (IN HA) 187
TABLE 4:5 EMISSION OF PARTICULATE MATTER (PM_{10}) FOR PEAK PRODUCTION CAPACITIES FROM DIFFERENT MINING ACTIVITIES 195
TABLE 4:6 EMISSION OF PARTICULATE MATTER ($PM_{2.5}$) FOR PEAK PRODUCTION CAPACITIES FROM DIFFERENT MINING ACTIVITIES 196
TABLE 4:7 CALCULATED EMISSION RATE FOR VARIOUS SOURCE GROUP FOR SO_2 AND NO_2 197
TABLE 4:8 INCREMENTAL GROUND LEVEL CONCENTRATIONS OF PM_{10} 197
TABLE 4:9 INCREMENTAL GROUND LEVEL CONCENTRATIONS OF $PM_{2.5}$ 198
TABLE 4:10 INCREMENTAL GROUND LEVEL CONCENTRATIONS OF SO_2 199
TABLE 4:11 INCREMENTAL GROUND LEVEL CONCENTRATIONS OF NO_2 200
TABLE 4:12 TRAFFIC SCENARIO 208
TABLE 4:13 TRAFFIC SCENARIO 209
TABLE 4:14 LEVEL OF SERVICE CRITERIA 209
TABLE 4:15 RECOMMENDATIONS ON TRAFFIC CAPACITY 209
TABLE 4:16 MODIFIED TRAFFIC SCENARIO 210
TABLE 4:17 SOURCE OF ORIGIN/DESTINATION OF TRAFFIC 211
TABLE 4:18 EMISSION RATE FOR MODELING DAILY EMISSIONS FROM TRANSPORTATION EMISSION FACTORS (GM/KM/HR) 214
TABLE 4:19 PREDICTED INCREMENTAL GROUND LEVEL CONCENTRATIONS (GLC) DUE TO ADDITIONAL TRAFFIC LOAD 214
TABLE 4:20 TOTAL WATER REQUIREMENT 220
TABLE 4:21 INITIAL WATER REQUIREMENT 220
TABLE 4:22: DIVERSION CHANNEL (SWALE) DESIGN SPECIFICATION 229
TABLE 4:23: BUND/EMBANKMENT DESIGN SPECIFICATION 229
TABLE 4:24 TOTAL MINE SEEPAGE 234
TABLE 4:25 TOTAL MINE WATER DUE TO MONSOON RAINFALL 234
TABLE 4:26 TOTAL MINE SEEPAGE INCLUDING MINE PIT RAINFALL 235
TABLE 4:27 LAND USE DETAIL BAITARANI WEST OCP BUFFER ZONE 238
TABLE 4:28 RECHARGE COMPUTATION BY ARTIFICIAL POND 242
TABLE 4:29 RESULTANT NOISE LEVELS 244
TABLE 4:30 ECOLOGICAL IMPACT ASSESSMENT 249
TABLE 4:31 BIOLOGICALLY RECLAIMED AREA (IN HA) 258
TABLE 4:32 POST MINING LAND-USE (IN HA) 263
TABLE 4:33 TENTATIVE LAND DEGRADATION AND TECHNICAL RECLAMATION (CUMULATIVE AREA HA) 264
TABLE 4:34 ABANDONMENT COST: COST OF ACTIVITIES TO BE TAKEN UP FOR CLOSURE OF THE MINE 276
TABLE 4:35 AMOUNT TO BE DEPOSITED IN ESCROW ACCOUNT 278
TABLE 4:36 AMOUNT TO BE DEPOSITED IN ESCROW ACCOUNT (YEAR WISE BREAKUP) 278
TABLE 4:37 IMPACT ON PRESENT STATE ENVIRONMENT DUE TO MINING WITH MITIGATION MEASURES 270
TABLE 6:1 ENVIRONMENT MONITORING PROGRAM 277
TABLE 6:2 IMPORTANT RECORDS TO BE MAINTAINED BY EMC 283
TABLE 6:3 LIST OF EQUIPMENT REQUIRED FOR ENVIRONMENTAL LABORATORY 284
TABLE 6:4 STEPS INVOLVED IN ENVIRONMENTAL AUDIT 286

TABLE 7:1 ISSUES RAISED IN THE PUBLIC HEARING AND COMMITMENTS BY M/S GUJARAT MINERAL DEVELOPMENT CORPORATION LTD (GMDC) 290
TABLE 7:2 ACTION PLAN TO ADDRESS ISSUES RAISED IN PUBLIC HEARING AS PER MOEF&CC O.M. DATED 30/09/2020 309
TABLE 7:3 SUMMARY OF AMOUNT EXPENDITURE 336
TABLE 7:4 ACCIDENT DUE TO THE OPERATION OF MINING AND OTHER EQUIPMENT 337
TABLE 7:5 DETAILS OF THE ACTIVITIES AND ASSOCIATED PRINCIPAL ENVIRONMENTAL AND OCCUPATIONAL RISKS 348
TABLE 7:6 CASTE –WISE DISTRIBUTION OF PAFS/ PDFS (AS PER SOCIO ECONOMIC SURVEY) 351
TABLE 8:1 PROPOSED CSR BUDGET 357
TABLE 8:2 CSR FUND UTILIZED IN VARIOUS HEAD 358
TABLE 9:1COST ESTIMATE OF EMP 362
TABLE 10:1 SUMMARY OF ANTICIPATED POTENTIAL IMPACTS & SUGGESTED MITIGATION MEASURES 375
TABLE 10:2 CAPITAL PROJECT COST PROPOSED 380
TABLE 11:1 SALIENT FEATURES OF THE PROJECT 383
TABLE 11:2 DESCRIPTION OF BASELINE ENVIRONMENT 388

List of Annexures

S. No	Annexure	Name of Document
1.	Annexure I	PH Proceedings
2.	Annexure II	Forest Diversion Proposal forwarded to MoEF&CC
3.	Annexure III	FC Schedule & Status
4.	Annexure IV	Hydrological Report & Diversion of Supai Nala
5.	Annexure V	Clarification with respect to Urban Area
6.	Annexure VI	Photographs from Drone Video
7.	Annexure VII	Schedule & Plan of R&R
8.	Annexure VIII	CGWA NOC
9.	Annexure IX	Part II Filled by DFO showing no corridor of National Park and WLS
10.	Annexure X	Certified Wildlife Map
11.	Annexure XI	Letter from PCCF
12.	Annexure XII	Lab Report
13.	Annexure XIII	NABL Certificate
14.	Annexure XIV	ToR Letter
15.	Annexure XV	Stage of Land Acquisition
16.	Annexure XVI	Approved SSWLCP
17.	Annexure XVII	External Dump Cross section plan
18.	Annexure XVIII	Coal Analysis report
19.	Annexure XIX(a)	Approval of Mining Plan
20.	Annexure XIX(b)	Vesting Order
21.	Annexure XX	Land Rights Section 11
22.	Annexure XXI	Land Rights Section 19
23.	Annexure XXII	Wind Rose Oct-Dec 2022
24.	Annexure XXIII	Quality Environment Health & Safety Policy
25.	Annexure XXIV	NABET Certificate

Abbreviations

OMC-M/s Odisha Mining Corporation Ltd
GMDC- Gujarat Mineral Development Corporation
TOR- Terms Of Reference
EIA – Environmental Impact Assessment
EMP- Environmental Management Plan
MoEF&CC – Ministry of Environment, Forest and Climate Change
GPCL - Gujarat Power Corporation Ltd.
KSEB - Kerala State Electricity Board
OHPC - Orissa Hydro Power Corporation
MOC- Ministry of Coal
PSU – Public Sector Undertaking
PFR – Prefeasibility Report
BSE&NSE – Bombay Stock Exchange
NSE – National Stock Exchange
MTPA – Million Tonnes Per Anum
BT – Billion Tonnes
MT - Million Tonnes
GW – Ground Water
OGL – Open General Licence
CPCB – Central Pollution Control Board
DGFT- Directorate General of Foreign Trade
PCB – Pollution Control Board
E-Waste – Electronic Waste
CCO - Coal Controller Organisation
DMG – Directorate of Mining and Geology
CPA - Critically Polluted Area
CBA – Coal Bearing Area
SPCB – State Pollution Control Board
OB – Over Burden
CSR – Corporate Social Responsibility
CER – Corporate Environmental Responsibility
MLD – Million Liter per Day
GSI – Geological Survey of India
CHP – Coal Handling Plant
ML – Mining Lease
OC – Open Cast
UG – Under Ground
TPH – Tonnes Per Hour
Mty – Million Tonnes per Year
PP – Project Proponent
HEMM – Heavy Earth Moving Machine
DGMS – Director General of Mine and Safety
HFL – Highest Flood Level
NABL – National Accreditation Board for Testing and Calibration Laboratories
NAAQM – National Ambient Air Quality Monitoring
IMD – India Meteorological Department
APHA – American Public Health Association
QOL – Quality of Life
QLI – Quality of Life Index
QCI – Quality Council of India
NABET – National Accreditation Board for Education and Training
COPD – Chronic Obstructive Pulmonary Disease
RF - Relative frequency
RD - Relative density
RDO- Relative Dominance
IUCN – International Union for Conservation of Nature

USGS – United States Geological Survey
SMS - Site Mixed Slurry
EMC - Environment Management Cell
ECC - Emergency Control Center
IC – Incident Controler
PPE - Personal Protection Equipment
PDF - Project Displaced Families
CBA - Cost-Benefit Analysis
CEP - Corporate Environment Policy

Point wise Reply to ToR Points

S. No.	Specific Conditions (TOR)	Compliance
1	Public Consultation, including public hearing, shall be conducted through concerned SPCB as per the provisions /procedure contained in the EIA Notification, 2006 for information of the stakeholders about the present coal mining operations inviting comments and their redressal. All the issues raised, and PP reply should be incorporated in EIA.	Noted. Public Hearing was conducted by Odisha State Pollution Control Board on 24.09.2024. All the issues raised by the public along with the reply of PP is discussed in Section 7.1 of Chapter 7 of Final EIA. Proceedings of Public Hearing are enclosed as **Annexure I.**
2	Stage-I Forest Clearance for diversion for non-forestry activity shall be submitted at the time of submission of EIA.	The ML area of 1196 ha includes 64.3723 ha of forest land (4.469 ha of RF and 59.9033 ha of revenue forest land). Stage-I Forest Clearance for diversion for non- forestry activity has been applied online vide single window no-SW/171065/2024. Govt. of Odisha forwarded Forest diversion proposal to MoEF&CC vide letter no. FE-DIV-FLD-0005-2025-1477/FE&CC dated 15.01.25. IRO visit for Site & CA land is completed. EDS raised by MoEF&CC dated 26.06.25 and it is closed on 26.08.25. As on date EDS closed by DFO & forwarded to RCCF, Angul dated 17.09.25. Copy of letter is attached as **Annexure II**. The detailed status & proposed schedule of the proposal is enclosed as **Annexure III.**
3	Natural forest located adjacent to the Kumbhira stream towards south at Handigora shall not be disturbed.	The said area located adjacent to Kumbhira stream towards South at Handigora is non-forest & a part of Gochar land which will be used for external dumping area as per the approved mine plan. However, the forest area in the Northwest, North & East side shall be kept untouched as a green belt.
4	PP shall not divert the Kumbhira stream and will not disturb the forest area or green patch located towards	As per the approved Mine Plan there is no proposal for diversion of Kumbhira stream. Embankment of 3 km length &

	south bank of the Kumbhira stream. Adequate protection measures with 30 mts embankment and greenbelt for Kumbhira stream and permanent fencing for green patch shall be proposed in EIA Report.	height of 3m above HFL will be provided along the Kumbhira Nala for its protection. Toe-walls, retaining walls with weepholes at strategic stretches will be constructed to arrest the sediments and prevent silting of the water courses by the run-off during rains. Greenbelt will be developed along the embankment. Details are discussed in Section 4.2.7 of Chapter 4.
5	A detailed hydrological survey of the Supai stream, regarding its catchment area, flow volume and length of the stretch to be diverted with water augmentation plan (develop artificial pond) to be provided with proper diversion plan in EIA/EMP report.	Supai stream is a seasonal stream, and no water is available December onwards. A detailed hydrological survey of the Supai Nala, regarding its catchment area, flow volume and length of the stretch to be diverted with water augmentation plan along with development of artificial pond has been provided in Section 4.6.4 of Chapter 4. Detailed hydrological survey of the Supai stream is enclosed as **Annexure IV.**
6	PP shall propose alternate land of suitable area for grazing land in replacement of 118 ha of grazing land in project area with water bodies.	As per the existing norm, the said area is declared under urban area (relevant page of the document dated 21.01.25 is attached as **Annexure V).** The Grazing land of 118.8397 ha is in patches & used by the villagers in the mine lease area for grazing of their cattles. They will be displaced in phases as per the approved R&R plan including grazing of their cattles. Grazing land will also be catered in the R&R colony for those villagers. Adequate grazing land shall be provided as per the state OGLS Act & Policy. The breakup of 118.8397 ha of grazing land, village-wise, is provided in Table 2.4 of Chapter 2 of the Final EIA.
7	PP shall prepare the Mining Plan in such a manner that condition prescribed by EAC shall intact from environment point of view. EIA-EMP	Mine Plan & Mine closure Plan of the proposed project has been prepared to minimize its adverse impact over environment. Mining Plan & Mine Closure

	shall accordingly be prepared on the suggested stipulation with point-wise compliance & in accordance with recommendations of Mining Plan	Plan of Baitarani (W) OCP has been approved by MoC vide letter dated 10-11-2021. EIA-EMP has been prepared as per the approved mining plan. Land use, land cover, reclamation & rehabilitation, waste management, etc are as per the approved mine plan.
8	PP should submit the real time aerial footage and video of the Mining lease area made through drone with a special focus on the area adjacent to the rivers.	The real time aerial footage and video of the Mining lease area through drone with a special focus on the area adjacent to the rivers is prepared and will be shown during EC presentation. Few Photographs from Drone Video are attached as **Annexure VI.**
9	In addition to existing data already collected (if any), the Cumulative Impact Assessment Study, carrying capacity, ecosystem services study and biodiversity study of the area shall be carried over by project proponent. PP shall collect one season baseline data of all environmental parameters and shall compare with the data of earlier data collected for cumulative assessment of area. Air pollution impact predication shall be conducted by considering the maximum values.	Cumulative Impact Assessment Study is done considering the base line ambient air quality data for 10 km radius area where the contribution from all other projects are monitored. The biodiversity study of the area is done and presented in Chapter 3. Baseline study is done for the period October to December, 2023. The results of baseline ambient air quality analysis data are compared with CPCB monitoring station & are discussed in Section 3.9.14 of Chapter 3. Air pollution impact prediction is done for the proposed operation and along the transport route. The GLC was estimated through AERMOD cloud. The details are discussed in Chapter- 4.
10	PP shall explore the possibilities of utilization of OB material for different purposes (in construction of roads, manufacture of artificial sand, aggregates, use for farmers etc.) and accordingly Plan shall be included in EIA/EMP Report.	By the end of life of the mine, it is estimated that a total of 713.53 million m^3 of OB shall be generated, out of which 140.25 million m^3 OB will remain in external dump, while the remaining 573.3 million m^3 OB will be accommodated in internal dump. No internal backfilling is possible up to year 10 as mine just reaches the ultimate quarry floor with a

		nominal area at bottom. As the quarry expands towards south and north- west, some backfilling can be started but major part of waste removed will have to be taken to external dump. From Year-14 onwards, possibly no external dumping is required as total area will be available for backfilling. The utilization will be done for construction of embankments, road, filling etc. as discussed in Chapter 2.
11	PP shall submit design details of all Air Pollution control equipment (APCEs) to be Implemented as part of Environment Management Plan vis-a-vis reduction in concentration of emission for each APCEs.	Closed conveyor system for Coal transportation. Mobile and fixed water sprinkling system, water cannons, wheel washing system will be used to control fugitive dust emissions. Further, the design details of APCEs if any used will be provided.
12	Inpit conveyor belt with silo loading should be proposed and installed for transportation of coal till railway siding. No transportation of coal by trucks/dumpers shall be proposed in EIA/EMP.	Out of 15.0 MTPA, 12.0 MTPA of coal will be transported through conveyor belt. Remaining 3.0 MTPA will be dispatched by trucks/dumpers through tarpaulin covered trucks for road sale to nearby local consumers. Conveyor system shall be used to transport the coal up to railway siding. A dedicated bitumen road/Tar Road shall be made up to the nearest MDR/NH for transportation. Installation of in pit conveyor belt is possible in 10/11th year of mine operation as suitable area will be available for construction & operation of the in-pit conveyor. Details are discussed in Section 4.2.8 & 4.2.9. of Chapter 4.

13	PP shall submit detailed project report for implementation of railway siding for evacuation of coal with its target date of completion. Target date should be such that railway siding should be operational within 2 years of commissioning of mine operations. Forest Clearance shall be submitted if railway siding land comes under forest land	PP will make own private railway siding for evacuation of coal within the ML area. This private siding will be connected, through the Common Rail Corridor, to the East- Coast Railway Line at Jarpara Railway Station. Construction of the siding shall be completed by the time of despatch of the coal (3rd year of mine operation when it will achieve a production of 6.0 MTPA). The above quantity of coal will move to different stations in the state of Gujarat as well as to destinations in state of Odisha, West Bengal, Jharkhand and Andhra Pradesh. Production of coal from Baitarni West mine has been targeted in FY- 2026 and will increase gradually which will touch to 15 MTPA in FY-2030. Implementation, schedule & status of railway siding is discussed in Section 4.2.8 of Chapter 4.
14	Wind rose pattern in the area should be reviewed and accordingly location of AAQMS shall be planned by the collection of air quality data. Monitoring location for collecting baseline data should cover overall the 10 KM buffer zone i.e. dispersed in 10 km buffer area.	AAQ sampling locations are planned based on the micrometeorological data of October-December 2022. Subsequently the locations are finalized based on the current season (October-December 2023) data, which covers a 10 km buffer zone. Wind rose diagram is shown in Figure 3.7 & Monitoring location for collecting baseline data is given in Table 3.2 of Chapter 3.
15	Project proponent to prepare Environmental Cost Benefit Analysis for the project in EIA/EMP Report.	Total cost of the project is Rs. 3855.00 Cr. Cost of production is Rs.337.11 per tonne. CSR cost is approx. Rs. 1 Cr/Annum (Minimum 2% of the net profit over the preceding three financial years of the organisation) which shall be used in sectors like healthcare, skill training, development of rural infrastructure etc. R&R cost is Rs. 727.94 crores. NPV cost is 6.16 Cr & Compensatory Afforestation cost is 5.19 Cr.

		Details of Environmental Cost Benefit Analysis are discussed in Chapter 9.
16	PP shall provide the details of mining technology/methodology proposed to be adopted for coal mining operations and its associated environmental benefits of using from Climate Change perspective by i.e. the likely emissions of greenhouse gases from the mining operations to be estimated with the modelling for future prediction related to the climate of that study area.	Opencast Mining Method shall be adopted. Coal winning by Surface Miner, pay loader & tipper and OB removal by Shovel-Dumper combination. Details are discussed in Section 2.10 of Chapter 2. Emissions of greenhouse gases from the mining operations are discussed in Section 4.14 of Chapter 4.
17	Detailed Social Impact Assessment shall be prepared in villages for Rehabilitation and Resettlement. R &R Activity shall be proposed with timeline and allotted fund with the approval of District Commissioner/collector.	Detailed Socio economic as a part of R&R has been completed & the summary is mentioned in Section 7.8 of Chapter 7. As on date site selection for the development of the rehabilitation colony has been completed and is awaiting RPDAC approval. Payment of R&R is going on. The detailed status of R&R with timeline & allotted funds is enclosed as **Annexure VII.**
18	Permission for ground water withdrawal shall be obtained from Central Ground Water Authority (CGWA) only for mining activity.	Permission of ground water has been obtained vide NOC no. NOC/MIN/OD/2025/534/MIN dated 29.07.25 & copy of same is attached as **Annexure VIII.**
19	Heavy metals including other parameters in surface water quality shall be analyzed and provided in EIA Report. Further, detailed mineralogical and chemical composition of the mineral and percentage of free silica from a NABL/MoEF&CC accredited laboratory	The details for this is provided in Table 3.9 of Chapter 3.
20	PP shall be submitting R &R in respect of SCs /STs and other weaker sections of the society in the study area, a need based sample survey, family- wise, should be undertaken to assess their requirements, and action programs prepared and submitted accordingly,	Refer Item No. 17 of Specific ToR condition.

	integrating the sectoral programmes of line departments of the State Government.	
21	PP should clearly bring out that what is the specific diesel consumption ~ (Liters/Tonne of total excavation & mineral) and steps to be taken for reduction of the same. Year-wise target for reduction in the specific diesel consumption needs to be submitted.	The PP will use HSD with low sulfur, with the latest efficient machinery for low consumption of diesel. However, efforts will be made to replace diesel-based machinery with CNG/LNG/ Electric Vehicle based machinery in future subject to their availability and techno economic feasibility for further reduction in diesel consumption. The actual consumption along with the target for reduction will be submitted after Mine is in operation.
22	PP shall provide provision of integrated mine plan and mine reclamation cum landform/landscape plan for both Underground and open cast coal mining projects. The plan must show the predicted post mining reclaimed and reformed surface by regarding and reshaping to reduce its height as close to the original surface level and proper sloping benching and terracing of external dump should be clearly brought out in the post mine closure plan. This would also include water management strategies such as surface water catchment and drainage paths etc. of post mining land surface. The final mine void shall be reduced and brought as near as ground so that land can be restored and reclaimed	Integrated Mine Plan for open cast mining and Mine closure Plan has been prepared and approved by MoC vide letter dated 10-11-2021. Details of Mine closure activities and post closure land use have been given in Section 4.12 of Chapter 4. Out of total Mining lease area of 1196 ha, 1029.58 ha will be reclaimed as plantation area, 38.49 ha will be covered under water body, 31.85 ha will be under public/company use, 64.37 ha Forest land will be returned back to forest Dept. and 31.71 ha will be undisturbed area. No underground mining is proposed in this proposal.
23	PP shall propose to use LNG/CNG based mining machineries and trucks for mining operation and transportation of coal.	As mentioned in point number 21, efforts will be made to replace diesel-based machinery with CNG/LNG/Electric Vehicle based machinery in future subject to their availability and techno economic feasibility.
24	PP shall submit letter from PCCF that mine does not fall under corridors of	The ML area does not fall under any Eco-Sensitive Zone, National Park, Sanctuary,

	any National Park and Wildlife Sanctuary and does not involve any violation of forest area and wild sanctuary with certified map showing distance of nearest sanctuary	Biosphere Reserve, Tiger Reserve and Elephant Corridor. Letter from PCCF is attached. (Please refer to **Annexure XI, X & IX** for Letter, Map, etc.)
25	Details of toe wall and garland drain to be constructed along the OB dump	Garland drains and toe wall will be constructed all along the external OB dump. Details discussed in Section 7.3.4.1 of Chapter 7.
26	Reclamation to be done using geo-texturing technique of the dumps close to habitation and a cause of visual intrusion.	Year wise reclamation details have been given in Table 4.31 of Chapter 4. Geo-texturing technique will be adopted for stability of the dump.
27	Impact of proposed project/activity on hydrological regime of the area shall be assessed and report be submitted. Hydrological studies as per GEC 2015 guidelines to be prepared and submitted.	Impact of proposed project/activity on hydrological regime of the area as per GEC 2015 guidelines has been prepared and incorporated in Chapter 4.
28	PP should bring out the awareness campaign to be carried out on various Environmental issues, practical training facility to be provided to the environmental engineer/diploma holders, mining engineer/diploma holders, geologists, and other trades related to mining operations. Target for the same needs to be submitted.	Awareness campaign will be carried out on various Environmental issues, practical training facility to be provided to the environmental engineer/diploma holders, geologists, and other trades related to mining operations shall be provided. **iCEM:** iCEM, "International Center of Excellence in Mining Safety and Automation (iCEM), Ahmedabad is an initiative of Government of Gujarat (GoG) being promoted by GoG through GMDC Ltd as Principal Promoter. The centre is envisioned to focus on Skill Development, Corporate Development, Demonstration of Technology and Best Practices, Education and Training, Research& Development, Technology Incubation & Entrepreneurship Development in Mining sector through continued innovation and excellence to facilitate industrial development in a joint

		venture with industrial partners. GMDC, through iCEM is in continually working on capacity building through various training programs. The details of the same is given in Section 10.2.2.1 of Chapter 10.
29	Details of Fog mist sprayer (static water sprinklers) at coal stock yard and along the permanent haul road.	Adequate nos. of fixed type Mist spray will be provided at Coal stock yard and Truck Mounted /mobile Mist Spray Sprinkling Arrangement will be provided for Haul Roads to control airborne dust and particulate matter generated by mining activities such as coal handling, transportation, and storage. Details are given in Table 9.1 of Chapter 9.
30	Details of black topping of permanent haul roads.	Wherever possible Black topping of permanent haul roads shall be done as provided in mine closure plan. Details discussed in Section 4.2.6 of Chapter 4.
31	The PP should submit the photograph of monitoring stations & sampling locations. The photograph should bear the date, time, latitude & longitude of the monitoring station/sampling location. In addition to this PP should submit the original test reports and certificates of the labs which will analyse the samples.	Photograph of monitoring stations / sampling locations bearing the date, time, latitude & longitude of the monitoring station/sampling location are provided in Chapter 3. Original test reports and certificates of the labs are attached as **Annexure XII & XIII** respectively.
32	No trucks or vehicles used for transportation of Coal to be passed by village roads or roads located near to the villages	Dedicated road of 2.10 km length will be made as shown in the route map in Figure 4.9 of Chapter 4.
33	The socio-economic study to be conducted with actual survey report and a comparative assessment to be provided from the census data of 2011- part B to be provided in EIA/ EMP report also economic status of the study area and what economically project will contribute should be clearly mention. The study should also include	The details of socioeconomic survey along with census data, infrastructural facilities and amenities present in the study area and a comparative assessment with census data of 2011 part A with photographs are provided in Section 3.14 of Chapter 3. CSR Activities to be carried out are

	the status of infrastructural facilities and amenities present in the study area and a comparative assessment with census data of 2011 part A to be provided and to link it with the initialization and quantification of need based survey for CSR activities to be followed.	discussed in Section 8.9 of Chapter 8.
34	A detailed traffic study along with presence of habitation in 100 mts distance from both side / of road, the impact on the air quality with its proper measures and plan of action with timeline for widening of road. The project will increase the no. of vehicle along the road which will indirectly contribute to carbon emission so what will be the compensatory action plan should be clearly spell out in EIA/ EMP report	Out of 15 MTPA of Coal, 12 MTPA shall be transported through rail and PP will have its own private siding within lease. Balance 3 MTPA of coal shall be transported by road. The proponent will make dedicated coal transportation road (which will not pass through any village having habitation) to connect the MDR connecting to SH-63. The proposed dedicated road transportation route is approximately 2.1 km long, as shown in Figure 4.9 of Chapter 4. Details of traffic study along with dispersion modelling are discussed in Section 4.4.2 of Chapter 4.
35	PP had proposed a separate colony of 1222 houses in Form -1, it should have proper rainwater harvesting system, green belt area with roads, solar lights and STP and the detail to be provided with EIA/ EMP report.	There will be no township for staff in this proposal. However, if required then the project proponent will approach to SEIAA at that time. The wastewater from office shall be treated in STP.
General Conditions		
S.No.	**General Conditions**	**Remarks**
1	All documents should be properly indexed, page numbered.	Complied
2	Period/date of data collection should be clearly indicated.	Complied
3	Authenticated English translation of all material provided in regional languages	Complied
4	After the preparation of the draft EIA-	Complied

	EMP Report as per the aforesaid TOR, the proponent shall get the Public Hearing conducted as prescribed in the EIA Notification 2006 and take necessary action for obtaining environmental clearance under the provisions of the EIA Notification 2006.	
5	The letter/application for EC should quote the Ministry's file No. and attach a copy of the letter prescribing the ToR.	Complied
6	The copy of the letter received from the Ministry on the ToR prescribed for the project should be attached as an annexure to the final EIA-EMP Report.	Enclosed as **Annexure XIV.**
7	General Instructions for the preparation and presentation before the EAC of ToR/EC projects of Coal Sector should be incorporated/followed.	Complied
8	The aforesaid ToR has a validity of Four years only.	Noted
9	Grant of ToR does not necessarily mean grant of EC.	Noted
10	Grant of ToR to the present project does not necessarily mean grant of TOR/EC to the captive/linked project.	Noted
11	Grant of ToR to the present project does not necessarily mean grant of approvals under the Forest (Conservation) Act, 1980 or the Wildlife (Protection) Act, 1972.	Noted
12	Grant of EC is also subject to circulars issued under the EIA Notification 2006, which are available on the Ministry's website: www.envfor.nic.in	Noted
GENERIC TOR FOR AN OPENCAST COALMINE PROJECT for EC		
1	An EIA-EMP Report shall be prepared for 15 MTPA rated capacity in an ML/project area of 1196 ha based on the generic structure specified in Appendix III of the EIA Notification, 2006.	EIA-EMP report for a rated production of 15.0 MTPA has been prepared.

2	An EIA-EMP Report would be prepared for 15 MTPA rated capacity to cover the impacts and environment management plan for the project specific activities on the environment of the region, and the environmental quality encompassing air, water, land, biotic community, etc. through collection of data and information, generation of data on impacts including prediction modeling for 15 MTPA of coal production based on approved project/Mining Plan for 15 MTPA. Baseline data collection can be for any season (three months) except monsoon.	EIA-EMP Report is prepared for 15 MTPA rated capacity to assess the impacts. Environment management plan for the project specific activities on the environment of the region, and the environmental quality encompassing air, water, land, biotic community, etc. are prepared. Data and information, generation of data on impacts including prediction modelling for 15 MTPA of coal production based on approved project/Mining Plan for 15 MTPA is done. Baseline data collection was done for post monsoon season (three months) October-December, 2023.
3	A toposheet specifying locations of the State, District and Project site should be provided.	Toposheet specifying locations is provided in Figure 2.6 of Chapter 2.
4	A Study area map of the core zone (project area) and 10 km area of the buffer zone (1: 50,000 scale) clearly delineating the major topographical features such as the land use, surface drainage pattern including rivers/streams/nullahs/canals, locations of human habitations, major constructions including railways, roads, pipelines, major industries/mines and other polluting sources. In case of ecologically sensitive areas such as Biosphere Reserves/National Parks/WL Sanctuaries/ Elephant Reserves, forests (Reserved/Protected), migratory corridors of fauna, and areas where endangered fauna and plants of medicinal and economic importance found in the 15 km study area should be given.	The study area map of the core zone & 10 km area of the Buffer Zone is provided in Figure 2.5 & Figure 2.8 of Chapter 2. Eco sensitive Map of 15 km study area is provided in Figure 2.7 of Chapter 2.
5	Land use map (1: 50,000 scale) based on a recent satellite imagery of the study area may also be provided with explanatory note on the land use.	Land use map based on latest cloud free satellite imagery with explanatory note on the land use has been provided in Figure 3.21 of Chapter 3.

6	Map showing the core zone delineating the agricultural land (irrigated and un-irrigated, uncultivable land as defined in the revenue records, forest areas (as per records), along with other physical features such as water bodies, etc should be furnished.	Map showing the core zone delineating the agricultural land (irrigated and un-irrigated, uncultivable land as defined in the revenue records, forest areas (as per records), along with other physical features such as water bodies, etc. is provided in Figure 3.23 of Chapter 3.
7	A contour map showing the area drainage of the core zone and 25 km of the study area (where the water courses of the core zone ultimately join the major rivers/streams outside the lease/project area) should also be clearly indicated in the separate map.	Contour map and drainage map for the study area and ML area are shown in Figure 3.2 & 3.4 respectively of Chapter 3.
8	A detailed Site plan of the mine showing the proposed break-up of the land for mining. operations such as the quarry area, OB dumps, green belt, safety zone, buildings, infrastructure, CHP, ETP, Stockyard, township/colony (within and adjacent to the ML), undisturbed area -if any, and landscape features such as existing roads, drains/natural water bodies to be left undisturbed along with any natural drainage adjoining the lease /project areas, and modification of thereof in terms of construction of embankments/bunds, proposed diversion/re-channelling of the water courses, etc., approach roads, major haul roads, etc should be indicated.	The ML area of 1196 ha. will have breakup (in ha. till life of the mine) as per the land use given below: Excavation Area - 623.71 Embankment - 16.85 Green Belt - 86.5 Undisturbed - 31.707 Water Reservoir Near Pit - 7.89 External Dump - 215.84 Road And Infrastructure Area - 186.783 Garland Drains - 8.38 Safety Zone - 16 Topsoil Dump - 2.34 Surface Plan is provided in Figure 2.9 of Chapter 2. Proposed land use details are provided in Chapter 4 section 4.3.1.2.
9	In case of any proposed diversion of nallah/canal/river, the proposed route of diversion /modification of drainage and their realignment, construction of embankment etc. should also be shown on the map as per the approval of Irrigation and flood control Department of the concerned state.	Nala Diversion Plan is shown in Figure 4.19 of Chapter 4. A small seasonal Supai nala flowing in the NE direction of the ML area is proposed to be diverted for mining operation as per the approved mine plan. A detailed hydrological report including catchment area, diversion plan etc. is prepared and

		attached with the EIA. Since the mine operation will reach the said nala in 10th year of operation, the diversion of the nala will be made only after approval from the concerned department well in advance. All relevant control measures shall be taken till that time and after diversion.
10	Similarly if the project involves diversion of any road/railway line passing through the ML/project area, the proposed route of diversion and its realignment should be shown in the map along with the status of the approval of the competent authority.	No Diversion of road/railway line is envisaged.
11	Break up of lease/project area as per different land uses and their stage of acquisition should Be provided. Landuse details for opencast project should be given as per the following table: Sl. No. Landuse Within ML area (ha) Outside ML area (ha) Total Agricultural land Forest land Wasteland Grazing land Surface water bodies Settlements Others (specify) TOTAL	Existing land use breakup of ML area (in ha) is given below: Road - 31.4086 Barren - 44.3739 Community - 5.3926 Grazing - 118.8397 Others - 78.8479 Reserve Forest - 4.469 Revenue Forest - 59.9033 Homestead - 32.7729 Agricultural - 784.8499 Water Bodies - 35.1422 Break up of lease/project area as per different land uses is given in Table No-2.3 of Chapter 2. Stage of land acquisition for first 5 years will be 347.09 ha. The details of acquisition for private land & govt. land is attached as **Annexure XV.**
12	Break-up of lease/project area as per mining plan should be provided.	Refer Point No 11 of the General ToR Conditions.
13	Impact of changes in the land use due to the project if the land is predominantly agricultural land/forestland/grazing land, should be provided.	The changes in land use of ML area shall be reclaimed/rehabilitated in the form of Plantation, Water Body, Public Use, and Forest Land (Returned). Plantation will be 1029.583 ha, Water Body will be 38.488

		ha, Public Use will be 31.85 ha and Forest Land (Returned) will be 64.372 ha. 31.707 ha of land shall be Undisturbed. Impact of changes in the land use is given in in Section 4.3.1 of Chapter 4.
14	One-season (other than monsoon) primary baseline data on environmental quality - air (PM10, PM2.5, SO2, NO2 and heavy metals such as Hg, Pb, Cr, As, etc), noise, water (surface and groundwater), soil - along with one-season met data coinciding with the same season for AAQ collection period should be provided.	Baseline data on environmental quality is provided for the period of October 2023 to December 2023 in Chapter 3.
15	Map (1: 50, 000 scale) of the study area (core and buffer zone) showing the location of various sampling stations superimposed with location of habitats, other industries/mines, polluting sources, should be provided. The number and location of the sampling stations in both core and buffer zones should be selected on the basis of size of lease/project area, the proposed impacts in the downwind (air)/downstream (surface water)/groundwater regime (based on flow). One station should be in the upwind/upstream/non- impact/non-polluting area as a control station. The monitoring should be as per CPCB guidelines and parameters for water testing for both ground water and surface water as per ISI standards and CPCB classification wherever applicable. Observed values should be provided along with the specified standards.	All the maps are provided in Chapter 3. Sampling locations for air, noise, water, and soil are shown in the Figure 3.8, 3.14, 3.16 and 3.17 and respectively as per CPCB guidelines. The parameters for air, water, noise, soil etc. are monitored/analysed as per the CPCB/MoEF&CC Guidelines.
16	Study on the existing flora and fauna in the study area (10km) should be carried out by an institution of relevant	Study for the existing flora and fauna in the study area has been carried out for the period of October 2023 to December

<table>
<tr><td></td><td>discipline. The list of flora and fauna duly authenticated separately for the core and study area and a statement clearly specifying whether the study area forms a part of the migratory corridor of any endangered fauna should be given. If the study area has endangered flora and fauna, or if the area is occasionally visited or used as a habitat by Schedule-I species, or if the project falls within 15 km of an ecologically sensitive area, or used as a migratory corridor then a Comprehensive Conservation Plan along with the appropriate budgetary provision should be prepared and submitted with EIA-EMP Report; and comments/observation from the CWLW of the State Govt. should also be obtained and furnished.</td><td>2023, the details for which are provided in Section 3.15 of Chapter 3. List of flora and fauna in the surrounding mine lease area is provided in the SSWLCP approved by PCCF & Chief Wildlife Warden. Approved SSWLCP is attached as Annexure XVI.

As per the forest record there are 29 numbers Schedule I Species namely Indian Elephant, Bengal Monitor, Indian Chameleon, Common Rat Snake, Common Indian Krait, Indian Cobra, Russell’s Viper, etc. listed in the study area. ln this context, A comprehensive Regional Wildlife conservation plan (WLCP) is under preparation at DFO level. PP will deposit the proportionate amount as per the direction of Forest Department. Further Site-Specific Conservation Plan has been prepared & approved by PCCF & Chief Wildlife Warden, Odisha. Copy of SSWLCP is attached as Annexure XVI. A budgetary provision of 37.26 Cr (Including 5% of the total financial outlay of the SSWCP to be deposited in Wildlife Society of Odisha) is earmarked for 10 years.</td></tr>
<tr><td>17</td><td>Details of mineral reserves, geological status of the study area and the seams to be worked, ultimate working depth and progressive stage-wise working scheme until the end of mine life should be provided on the basis of the approved rated capacity and calendar plans of production from the approved Mining Plan. Geological maps and sections should be included. The Progressive mine development and Conceptual Final Mine Closure Plan should also be shown in figures. Details of mine plan and mine closure plan approval of Competent Authority</td><td><table>
<tr><td>Gross Geological Reserve of the block “MT”</td><td>662.94</td></tr>
<tr><td>Net Geological Reserve of the block “MT”</td><td>602.1106</td></tr>
<tr><td>Minable Reserve of the block “MT”</td><td>485.06</td></tr>
<tr><td>Blocked Reserve “MT”</td><td>117.0413</td></tr>
<tr><td>Corresponding extractable reserve of the block “MT”</td><td>468.27</td></tr>
<tr><td>Percentage of Extraction</td><td>77.771</td></tr>
<tr><td>Balance Reserve (as on Base Date)</td><td>468.2700</td></tr>
</table>These details are provided in Section 2.6 of Chapter 2.</td></tr>
</table>

	should be furnished for green field and expansion projects.	The Final Mine Closure Plan is provided in Section 4.12 of Chapter 4, also refer point no 13 of general ToR conditions.
18	Details of mining methods, technology, equipment to be used, etc., rationale for selection of specified technology and equipment proposed to be used vis-à-vis the potential impacts. should be provided.	Proposed method of mining is Opencast Mining Method of Coal winning by Surface Miner, pay loader & tipper and OB removed by Shovel-Dumper combination. The details for the same have been provided in Section 2.10 of Chapter 2.
19	Impact of mining on hydrology, modification of natural drainage, diversion and channeling of the existing rivers/water courses flowing though the ML and adjoining the lease/project and the impact on the existing users and impacts of mining operations thereon.	Assessment of impact on hydrology and control measures have been provided in Section 4.6.2 of Chapter 4.
20	Detailed water balance should be provided. The break-up of water requirement for the various mine operations should be given separately.	Total water demand for the project is 3.2 MLD. Initially the project will have a requirement of 0.26 MLD, the water balance is provided in Figure 4.16 of Chapter 4.
21	Source of water for use in mine, sanction of the Competent Authority in the State Govt. and impacts vis-à-vis the competing users in the upstream and downstream of the project site should be given.	Details have been provided in Section 4.6 of Chapter 4. Initially the project will meet the requirement of water of 0.26 MLD from ground water. After that the project will use mine seepage water after proper treatment. Application for permission of ground water is already made vide application no. MIN/OD/2025/534 dated 09.01.25 & copy of same is attached as **Annexure VIII.**
22	Impact of mining and water abstraction from the mine on the hydrogeology and groundwater regime within the core zone and 10 km buffer zone including long-term monitoring measures should be provided. Details of rainwater harvesting and measures	The area is under safe zone. Details have been provided in Section 4.6.2 of Chapter 4. Kumbhira Nala will remain as it is & Supai Nala will also remain within the ML area after diversion. Apart from the above an

	for recharge of groundwater should be reflected in case there is a declining trend of groundwater availability and/or if the area falls within dark/grey zone.	area of 22.21 ha of mine out area will be converted to water body in the post mining land use. All the above will recharge the ground water.
23	Impact of blasting, noise and vibrations should be given.	The major adverse impact and its control measures have been provided in Section 4.7 of Chapter 4. 93% of the total excavation will be using surface miner and only 7% through drilling and blasting
24	Impacts of mining on the AAQ and predictions based on modelling using the ISCST-3 (Revised) or latest model should be provided.	The AERMOD cloud model has been setup and run for study period (3 months) to find out the cumulative maximum ground level concentration of $PM_{10,}$ $M_{2.5,}$ SO_2 & NO_2 to be generated from the different activities in the mining areas. Assessment of impact and control measures on air quality have been provided in Section 4.3.2 of Chapter 4. Incremental GLC for PM_{10} being 1.6771 $\mu g/m^3$, 0.5590 $\mu g/m^3$ for $PM_{2.5}$, 0.534 $\mu g/m^3$ for SO_2 and 0.169 $\mu g/m^3$ for NO_2.
25	Impacts of mineral transportation within the mining area and outside the lease/project along with flow-chart indicating the specific areas generating fugitive emissions should be provided. Impacts of transportation, handling, transfer of mineral and waste on air quality, generation of effluents from workshop etc, management plan for maintenance of HEMM and other machinery/equipment should be given. Details of various facilities such as rest areas and canteen for workers and effluents/pollution load emanating from these activities should also be provided.	Coal Flow diagram is shown in Figure 2.17 of Chapter 2. Coal Handling and dispatch arrangement have been provided in Section 2.10.13 of Chapter 2. 12 MTPA out of 15 MTPA shall be transported through railway and only 3 MTPA shall be transported through road to meet the local demand. A dedicated coal evacuation route of length 2.10 km is proposed for the coal transport. These details are provided in Section 4.4 of Chapter 4. Details of site facilities such as rest areas and canteen for workers are discussed in Section 2.10.14 of Chapter 2.

26	Effort be made to reduce/eliminate road transport of coal inside and outside mine and for mechanized loading of coal through CHP/ Silo into wagons and trucks/tippers.	Coal Handling and dispatch arrangement have been provided in Section 2.10.13 of Chapter 2. 12 MTPA out of 15 MTPA shall be transported through railway and only 3 MTPA shall be transported through road to meet the local demand. A dedicated coal evacuation road of length 2.10 km is proposed for the coal transport.
27	Details of waste OB and topsoil generated as per the approved calendar programme, and their management shown in figures as well explanatory notes tables giving progressive development and mine closure plan, green belt development, backfilling programme and conceptual post mining land use should be given. OB dump heights and terracing based on slope stability studies with a max of 28o angle as the ultimate slope should be given. Sections of final dumps (both longitudinal and cross section) with relation to the adjacent area should be shown.	During the entire life of mine of 37 years (including construction period), around 1.866 million m3 of topsoil and 713.534 million m3 of Overburden (OB) will be generated. The entire quantities of waste generation will be utilized for backfilling the excavated areas. The land after backfilling will be biologically reclaimed. The generated topsoil will be utilized for plantation activity on the backfilled area, external dump, and embankment. The details of stage wise generation of overburden/topsoil waste are provided in Section 2.10.11 of Chapter 2. Sections of final dumps (both longitudinal and cross section) with relation to the adjacent area is attached as **Annexure XVII.**
28	Efforts be made for maximising progressive internal dumping of O.B., sequential mining, External dump on coal bearing area and later rehandling into the mine void. To reduce land degradation.	Mine closure details have been provided in Section 4.12 of Chapter 4. Backfilling shall be started from 14th year. 1 External OB Dump of 215.84 ha with maximum total height of 90 m and quantity of OB 140.25 million m^3 and Internal OB Dump of 623.71 ha with quantity of OB 573.3 million m^3 is envisaged for backfilling. By the end of mine life, 140.25 Mcum OB will remain in external dump, while the remaining 573.3 Mcum OB will be accommodated in internal dump. The dump top and slope areas will be covered

		with soil and plantation will be done for afforestation and stability. The top RL of internal dump will be 190m towards north and central part of the quarry and 220m towards southern area.
29	Impact of change in land use due to mining operations and plan for restoration of the mined area to its original land use should be provided.	Impact of land use discussed in Section 4.13 of Chapter 4.
30	Progressive Green belt and ecological restoration /afforestation plan (both in text, figures and in the tabular form as per the format of MOEFCC given below) and selection of species (native) based on original survey/land-use should be given. Table 1: Stage-wise Landuse and Reclamation Area (ha) S.N. Land use Category Present (1st Year) 5th Year 10th Year 20th Year 24th Year (end of mine life)* Backfilled Area(Reclaimed with plantation) Excavated Area (not reclaimed)/void External OB dump Reclaimed with plantation) Reclaimed Top soil dump Green Built Area Undisturbed area (brought under plantation) Roads (avenue plantation) Area around buildings and Infrastructure TOTAL	Progressive Green Belt Plan will be as per the schedule given below: Year Cumulative Area (in Ha.) Y-1 0 Y-3 8.09 Y-5 27.84 Y-10 64.46 Y-15 159.55 Y-20 192.52 Y-25 305.75 Y-30 328.65 Y-35 436.38 Y-37 574.35 Y-40 1029.583 Details for the same are provided in Table 4.31 of Chapter 4.

	* As a representative example Table 2: Stage Wise Cumulative Plantation No. YEAR* Green Belt External Dump Backfilled Area Others (Undisturbed Area/etc) TOTAL 1st year 3rd year 5th year 10th year 15th year 20th year 25th year 30th year 34th year(end of mine life) 10. 34- 37th Year (Postmining) * As a representative example	
31	Conceptual Final Mine Closure Plan and post mining land use and restoration of land/habitat to the pre- mining status should be provided. A Plan for the ecological restoration of the mined out area and post mining land use should be prepared with detailed cost provisions. Impact and management of wastes and issues of re-handling (wherever applicable) and backfilling and progressive mine closure and reclamation should be furnished. Table 3: Post-Mining Land Use Pattern of ML/Project Area (ha) S.N. Land use during Mining Land Use (ha) Plantation Water Body Public Use Undisturbed TOTAL External OB Dump Topsoil Dump Excavation Roads	Mine closure details have been provided in Section 4.12 of Chapter 4. Post-Mining Land Use is provided in Table 4.32 of Chapter 4.

	Built up area Green Belt Undisturbed Area TOTAL	
32	Flow chart of water balance should be provided. Treatment of effluents from workshop, township, domestic wastewater, mine water discharge, etc. should be provided. Details of STP in colony and ETP in mine should be given. Recycling of water to the max. possible extent should be done.	The workshop effluent will be routed through oil & grease trap and treated to the discharge standards through ETP and reused in the workshop. There will be no township for staff in this proposal. However, if required then the project proponent will approach to SEIAA at that time. The wastewater from office will be treated in STP. Water balance is shown in Figure 4.16 of Chapter 4.
33	Occupational health issues. Baseline data on the health of the population in the impact zone and measures for occupational health and safety of the personnel and manpower in the mine should be given.	Occupational health issues have been provided in Section 7.6 of Chapter 7.
34	Risk Assessment and Disaster Preparedness and Management Plan should be provided.	Details of Risk Assessment and Disaster preparedness and management plan have been provided in Section 7.2 & Section 7.4 respectively of Chapter 7.
35	Integration of the Env. Management Plan with measures for minimizing use of natural resources - water, land, energy, etc. should be carried out.	Environment Management Plan has been provided in Chapter 10.
36	Cost of EMP (capital and recurring) should be included in the project cost and for progressive and final mine closure plan.	Capital Cost: 10072.9 Lakhs Recurring Cost: 315.7 Lakhs Cost of EMP is provided in Table 9.1 of Chapter 9.
37	Details of R&R. Detailed project specific R&R Plan with data on the existing socioeconomic status of the population (including tribals, SC/ST, BPL families) found in the study area and broad plan for resettlement of the displaced	Socioeconomic survey as a part of base line study was done and the details are discussed in Section 3.14 of Chapter 3. The detailed project specific R&R Plan and status of R&R is enclosed as **Annexure VII.**

	population, site for the resettlement colony, alternate livelihood concerns/employment for the displaced people, civic and housing amenities being offered, etc and costs along with the schedule of the implementation of the R&R Plan should be given.	
38	CSR Plan along with details of villages and specific budgetary provisions (capital and recurring) for specific activities over the life of the project should be given.	Estimated CSR budget is Rs. 1 Cr/Annum. (Minimum 2% of the net profit over the preceding three financial years of the organisation) Details have been provided in Section 8.9 of Chapter 8.
39	Corporate Environment Responsibility: a) The Company must have a well laid down Environment Policy approved by the Board of Directors. b) The Environment Policy must prescribe for standard operating process/procedures to bring into focus any infringements/deviation/violation of the environmental or forest norms/conditions. c) The hierarchical system or Administrative Order of the company to deal with environmental issues and for ensuring compliance with the environmental clearance conditions must be furnished. d) To have proper checks and balances, the company should have a well laid down system of reporting of non-compliances/violations of environmental norms to the Board of Directors of the company and/or shareholders or stakeholders at large.	The details have been provided in Section 8.8 of Chapter 8.
40	Details on Public Hearing should cover	Details of Public Hearing are discussed in

	the information relating to notices issued in the newspaper, proceedings/minutes of Public Hearing, the points raised by the general public and commitments made by the proponent and the action proposed with budgets in suitable time frame. These details should be presented in a tabular form. If the Public Hearing is in the regional language, an authenticated English Translation of the same should be provided.	Section 7.1 of Chapter 7. Public Hearing Proceedings are enclosed as **Annexure I.**
41	In built mechanism of self- monitoring of compliance of environmental regulations should be indicated.	For self-monitoring of environmental regulations, compliance monitoring shall be done. A separate Environmental Cell will be there to ensure effective monitoring of the Key Environmental Aspects of the Project on a day-to-day basis. Key components of such mechanisms include Environmental Management Systems (EMS), Internal Audits and Inspections, Self-Reporting Mechanisms, Environmental Performance Reviews, etc. The details have been provided in section 10.4 of Chapter 10 & in Chapter 6.
42	Status of any litigations/ court cases filed/pending on the project should be provided.	There is no litigation pending.
43	Submission of sample test analysis of Characteristics of coal: This should include details on grade of coal and other characteristics such as ash content, S and heavy metals including levels of Hg, As, Pb, Cr etc.	Analysis of Coal & Heavy Metal is provided in Table 2.11 of Chapter 2. Coal Analysis report of Baitarni West block, Talcher coalfield is attached as **Annexure XVIII.**
44	Copy of clearances/approvals such as Forestry clearances, Mining Plan Approval, mine closer plan approval. NOC from Flood and Irrigation Dept. (if req.), etc. wherever applicable.	Stage-I Forest Clearance for diversion for non-forestry activity has been applied online vide single window no-SW/171065/2024. Govt. of Odisha forwarded Forest diversion proposal to MoEF&CC vide letter no. FE-DIV-FLD-

	FOREST CLEARANCE: Details on the Forest Clearance should be given as per the format given: TOTAL ML/PROJECT AREA (ha) TOTAL FORESTLAND (ha) Date of FC Extent of forestland Balance area for which FC is yet to be obtained Status of application for. diversion of forestland If more than, provide details of each FC	0005-2025-1477/FE&CC dated 15.01.25. IRO visit for Site & CA land is completed. EDS raised by MoEF&CC dated 26.06.25 and it is closed on 26.08.25. As on date EDS closed by DFO & forwarded to RCCF, Angul dated 17.09.25. Copy of letter is attached as **Annexure II**. The detailed status of the proposal is enclosed as **Annexure III.** Approval of Mining Plan is attached as **Annexure XIX(a).** Vesting Order is enclosed as **Annexure XIX(b).**
45	Study on subsidence, measures for mitigation/prevention of subsidence, modelling subsidence prediction and its use during mine operation, safety issues.	This is an open cast mine project, so it is not applicable.
46	Impacts of CHP, if any on air and water quality. A flow chart of water uses and whether the unit can be made a zero-discharge unit.	Coal Handling and dispatch arrangement have been provided in Section 2.10.13 of Chapter 2. Flow chart of water uses is shown in Figure 4.16 of Chapter 4. No discharge will be done outside the mine lease area.

CHAPTER-1 : INTRODUCTION

1.1 PURPOSE OF REPORT

Baitarni West Coal block of Talcher Coalfield with production capacity of 15 MTPA in an area of 1196 ha by M/s Gujarat Mineral Development Corporation Ltd. located at Village Chhendipada Jangal, Chhendipada RF, Handigora, Machhakuta, Machhakuta Jangal, Porapara, Porapara Jangal, Tentuloi-Korasahi Tehsil Chhendipada District Angul (Odisha).

The Coal mine project was allotted to GMDC, the successful bidder through vesting order issued on 08.06.23.

The ML area is 1196 ha which includes 64.3723 ha of forest land (4.469 ha of RF and 59.903 ha of revenue land)

Mining Plan for 15 MTPA(Normative) and 22.50 MTPA(Peak) for Open Cast Mechanized operation was approved and transferred in the name of GMDC.

TOR issued to OMC is transferred to GMDC on 6.11.23 on the same terms and conditions as stipulated in the TOR granted to OMC.

Land Rights have been obtained by Gujarat Mineral Development Corporation under Sec 11(1) of CBA & henceforth Sec 19 of CBA has been published.

The mine was earlier allotted to M/s Odisha Mining Corporation Ltd. The Ministry of Environment, Forest and Climate Change has granted Terms of Reference to Baitarni West Opencast Coal Mine of OMC with production capacity of 15 MTPA in an area of 1196 ha vide its letter with File No. IA-J-11015/81/2017- IA-II (M) dated 22nd November, 2021 with certain terms and conditions mentioned therein. Vesting order was issued in favor of Gujarat Mineral Development Corporation, being successful bidder, vide order NA-104/4/2023-NA, dated 08/06/2023, after auction of the mine , for the purpose of sale of coal including sale to affiliates and related parties, utilization of coal for any purpose including but not limited to captive consumption, coal gasification, coal liquefaction and export of coal. M/s Gujarat Mineral Development Corporation Ltd accordingly applied for transfer of Terms of Reference and has submitted the affidavit dated 22nd September, 2023 stating to comply with all the Environmental Safeguards as per the terms and conditions as stipulated in Terms of Reference (ToR) granted on 22.11.2021 to OMC Ltd. MoEF&CC transferred the TOR from M/s Odisha Mining Corporation Ltd to M/s Gujarat Mineral Development Corporation Ltd., on the same terms and conditions as mentioned in the TOR issued vide file no : IA-J-11015/81/2017- IA-II(M) dated 06.11.23.

Coal is one of the most principal mineral fuels known to be a fossil fuel. It is an organic rock which content mostly carbon, but also hydrogen, sulfur, oxygen as well some inorganic minerals and water.

Coal mining has adverse impacts on land, air, water, biodiversity, loss of livelihoods etc. due to displacement and encroachment of land. Hence mining and related activities have to be done in a scientific and systematic manner to ensure minimum or negligible damage to the environment. EIA is an important procedure for ensuring that the likely effects due to new development on the environment are fully understood and takes into account before the development starts. It will analyze the probable changes in the physical, biophysical and socio-economic environment of the proposed project. The prediction and evaluation of the environmental consequences enable the planner to plan better to avoid irreparable damage to the environment and ensure sustainable development.

This EIA report is being prepared for environmental clearance of above mine as per the EIA notification 14 September 2006 and its subsequent amendment to operate the mine for coal production and its utilization for any purpose including but not limited to captive consumption, coal gasification, coal liquefaction and export of coal.

GMDC Ltd. has engaged EIA consultant M/s. Grass Roots Research & Creation India Pvt Ltd., Noida (Accredited by QCI/NABET) for preparation of EIA/EMP report.

Baseline monitoring studies were conducted during the post monsoon season October to December 2023 in core and buffer zone (10 Km radius) of Baitarni West Opencast Coal Mine. The draft EIA/EMP report is prepared based on the Specific as well as Standard Terms of Reference issued by MoEF&CC vide File No. IA-J-11015/81/2017- IA-II (M) dated 22/11/2021 and vide file no: IA-J-11015/81/2017- IA-II(M) dated 06.11.23.

1.2 IDENTIFICATION OF PROJECT & PROJECT PROPONENT

1.2.1 IDENTIFICATION OF PROJECT

Baitarni West Coal Mine of GMDC comes under Talcher Coal Field was earlier allotted to Gujarat Power Corporation Ltd. (GPCL), Kerala State Electricity Board (KSEB) and Orissa Hydro Power Corporation (OHPC) by Ministry of Coal, Govt. of India, vides letter no. 13016/8/2007-CA-I dated 25.07.2007. Later on Supreme Court of India through its judgment dated 25th August, 2014 read with its order dated 24th Sept 2014 cancelled allotment of 204 coal blocks including Baitarni West Coal Mine.

Accordingly, its allotment to the prior allottee was cancelled in pursuant to the Supreme Court Judgment. MoC vide order no. 13016/2014-CA-III. dt. 16.03.2016 notified Baitarni West Coal Mine for allotment to the State PSUs for sale of coal.

Based on the above order of the MoC, GoI, Govt. of Odisha vide letter no. 2831/SM, dtd.05.04.2016 advised OMC Ltd a Gold category state PSU to apply for allocation of Baitarni West Coal Mine.

Accordingly, OMC Ltd entered into the allotment agreement with the nominated authority on dt. 24.08.2016 after fulfilling all the conditions as required by allotment authority. The nominated authority MoC, GoI issued the allotment order to OMC Ltd. vide order no. 103/3/2016/N.A dtd.29.09.2016. Again, a Corrigendum No. 01 for the said allotment is issued by MOC vide 103/16/2015/NA dated 14.08.2020.

As per the earlier approved Mining Plan vide letter dt. 07-10-2009, Form-I and Prefeasibility report for Baitarni West Coal Mine (15.0 MTPA) capacity was prepared and considered in 19th Expert Appraisal Committee (EAC) meeting held on 26th September 2017.

EAC had recommended Terms of Reference (TOR) for Baitarni West Coal Mine (15MTPA) vide letter no. J-11015/81/2017-IA.II (M) dated 29th September, 2017 issued by MoEF&CC, New Delhi. However, due to some changes in the Mining Plan, final EIA/EMP could not be submitted to MoEF&CC by OMC.

Accordingly, as per the changes a revised Mining Plan of Baitarni (W) has been prepared and approved by MoC vide letter no. SWCS/Baitarni West Coal Mine/OR/AN/2021/CCO-MCBA dt. 10-11-2021.

Further an application for fresh TOR was submitted (by OMC) along with Form 1 and PFR vide Online Proposal No. IA/OR/CMIN/234993/202 dated 21st October 2021.

Accordingly approved mining plan and mine closure plan (first modification) is transferred to Gujarat Mineral Development Corporation vide vesting order NA-104/4/2023-NA, dated 08/06/2023.

The ToR has been granted for 1196 ha & the mine plan is also approved for 1196 ha. The mining plan letter no. SWCS/BaitarniWestCoalMine/OR/AN/2021/ CCOMCBA dated 10.11.2021 issued to OMC by Ministry of Coal, Government of India mentioning "Lease area 1196 Ha and Project area 1196 Ha has been considered and approval of the Central Government there on is here by conveyed under Section 5(2)(b) of the Mines & Minerals (Development and Regulation); Act 1957". The same mining plan for 1196 Ha area has been vested to GMDC. Thus, the area stated in the ToR, approved mining plan, EC application and in the EIA/EMP report is same i.e. 1196 Ha.

The ToR issued vide File No. IA-J-11015/81/2017- IA-II(M) dated 22.11.2021 for ML area of 1196 ha. However, the project area mentioned in the TOR was for 1267.293 ha out of which 71.293 ha is outside ML area proposed for

township/community/other use. However, GMDC has no proposal of Township in the present EC proposal. EIA has been prepared for 1196 ha area only.

There will be direct sale of coal to meet the demand of the present and future thermal power stations, as well as MSME units and other industries.

Mode of transport of major quantity of coal will be through belt conveyor up to dispatch point or RLS Rail transport, thereafter through the proposed Common Rail Corridor Project in Talcher Coalfield.

A small quantity of coal will be despatched through road for sale to local customers.

1.3 PROJECT PROPONENT

Gujarat Mineral Development Corporation Limited (GMDC) is a major Indian state-owned minerals and lignite mining company based in Gujarat. GMDC was founded in 1963. The company has operational experience of over 60 years. GMDC's product portfolio spans across mining, value added products and power. Its power portfolio includes clean energy sources such as solar and wind besides thermal power. Its product range includes essential energy minerals like lignite, base metals and industrial minerals like bauxite and fluorspar.

In 1999, the government of Gujarat being the sole owner, divested 26% of its stake and the GMDC became a listed entity (BSE&NSE) and is occupying a position within the top fortune 500 companies in our country with an annual turnover surpassing 10 billion with considerable annual growth rate.

GMDC's mining activities are spread across Gujarat in Kutch, Devbhoomi Dwarka, Panchmahal, Vadodara, Bhavnagar, Bharuch, Surat and Chhotaudepur districts of the State. It is currently mining minerals like Lignite with five operational lignite mines, Bauxite (11 operating mines), Fluorspar, Manganese, Ball Clay, Silica Sand, Bentonitic Clay and Limestone. GMDC also value adds to minerals through works such as Pyrite removal from Lignite, Beneficiation of Bauxite, Beneficiation of Low-Grade Manganese and Beneficiation of Fluorspar. The Company has set up a 250 MW lignite based Thermal Power Station at Nani Chher in Kutch as a forward integration, Wind power plant of 200.9 MW at Maliya, Jodiya, Godsar, Bhanvad, Bada, Verbala, Rojmal and Solar Power plant of 5 MW at Panandhro Project. GMDC's existing Metallic mineral portfolio includes Fluorspar, Manganese, Bauxite, and Multimetals in addition to other associated minerals like silica sand, Bentonite, Ball Clay and Limestone.

GMDC has undertaken a strategic transformation exercise over the last year with a view to achieve growth milestones, diversity its product portfolio, leverage existing assets, provide inputs to the industry and contribute to the growth of the state's economy.

As part of its strategic transformation exercise, GMDC is already in the process of expanding its mining operations by setting up six new lignite- based projects in Kutch and South Gujarat. Further, GMDC through its metal division also intends to expand and develop metal mining portfolio by developing existing metal mining projects and exploring new opportunities in metal mining sector in India. A number of other projects are in the pipeline based on identified thrust areas.

Leveraging its capabilities in lignite thus, one of the key strategic diversifications for GMDC is in the area of domestic coal mining, where it would like to access domestic coal mostly occurring in the eastern part of the country. As part of this strategy, it has bid for a number of coal blocks in the recently concluded commercial mine auctions by the Ministry of Coal, Govt. of India. The PSU won the bids for Burapahar block and the Baitarani West coal block in Odisha.

1.4 BRIEF DESCRIPTION OF NATURE, SIZE, LOCATION OF THE PROJECT AND ITS IMPORTANCE TO THE COUNTRY, REGION

1.4.1 NATURE & SIZE OF PROJECT

Baitarni West Opencast Coal Mine is an opencast coal mining project with production capacity of 15 MTPA in an area of 1196 ha of M/s GMDC located at villages Chhendipada Jangal, Chhendipada RF, Handigora, Machhakuta, Machhakuta Jangal, Porapara, Porapara Jangal, Tentuloi Korasahi, Tehsil Chhendipada, District Angul (Odisha).

The proposed project activity falls under the category of "Mining of Minerals" and categorized as "Category-A" under project activity 1(a) of EIA Notification 2006 & its amendment till date.

1.4.2 LOCATION OF PROJECT

The Baitarni West Coal Block is located in the North central part of Talcher coalfield. Co-ordinates:
NW: 21°05'26.87" N, 84°50'12.08" E
NE: 21°05'17.37" N, 84°51'26.13" E
SW: 21°01'44.84" N, 84°50'39.59" E
SE: 21°01'44.74" N, 84°51'40.70" E

The block is under Survey of India topo-sheet No. F45M16 on RF 1:50,000 and falls within Angul district.

Angul, the district head quarter, is the nearest town at a distance about 43 kms from northeastern boundary of the block.

Angul railway station is the nearest railway station. The project is well connected by an all-weather road.

1.5 IMPORTANCE OF PROJECT TO THE COUNTRY & REGION

Coal mining is extraction of coal deposits from the surface of Earth or from underground. Coal has been used since the Bronze Age, 3,000 to 4,000 years ago, and was the basic energy source that fuelled the Industrial Revolution of the 18th and 19th centuries.

As much of the world lacks access to modern, clean energy, coal is still essential to alleviating worldwide energy poverty. The challenge the industry faces lies in developing technologies and pathways to zero emissions, especially carbon dioxide, which scientists identify as a factor in climate change. Coal also serves many industrial uses.

Coal is one of the main sources of energy in India. Accordingly, affordable and sustainable supply of coal is inextricably linked to the goal of ensuring energy security in India. It currently accounts for 55% to 70 % of India's total energy consumption (CSO 2013).

Coal Companies have been taking assistance from International Agencies/ companies for guiding and monitoring the adoption of best mining practices and technology for coal mining.

India is the world's second largest producer, consumer and importer of coal after China. In 2020, India had the fifth-largest coal reserves of over 111 billion tonnes (BT), just below China which had the fourth-largest reserves of 143 BT. Though coal reserves are comparable in quantity, China produced 3.9 BT of coal in 2020, five times more than India's production of 759 million tons (MT). In November 2021 and again in April-May 2022, a crisis of power supply attributed to low coal stocks at thermal power generation plants led the Indian government to push generators to import coal. This advice came at a time when seaborne thermal coal prices were at their highest levels. An increase in the use of imported coal not only contradicts India's strategy of self-reliance for energy security but also compromises the quest for affordability, an idea that underpins most of India's energy policy choices.

With India's power demand expected to double in the coming decade, coal power generation may not yet have reached its peak. Historically, coal plants have accounted for 70 to 80% of the country's total power generation. This share is estimated to remain above 50 per cent even in 2030. Further, India's current installed coal capacity of 209 GW is set to expand, with around 57 GW of new capacity in the pipeline. Coal is a reliable energy source, especially when compared with the seasonal and diurnal variability of renewables. Abundant in India, it is considered important for the nation's energy security and is a key source of revenue for the government.

1.6 PRODUCTION AND IMPORTS

Raw coal (coking and non-coking) production in India increased from 341.272 MT in 2002-03 to 777.31 MT in 2021-22 implying an annual growth rate of just over 8 percent. Most of the growth came from an increase in production of thermal coal. Historically this has meant the import of only coking coal as reserves were not adequate. However, as demand for power accelerated, coal was put under open general license (OGL) in 1993 that initiated thermal coal imports. Until the mid-2000s, volume of coking coal imports exceeded volume of thermal coal imports. This changed in 2005-06 when India imported 21.695 MT of thermal coal compared to 16.891 MT of coking coal. It was attributed to consumer (thermal power generators) preference for coal quality. Thermal coal imports accelerated with construction of imported coal-based coastal power plants. Between 2002-03 and 2019-20 (pre-pandemic year), coking coal imports increased from about 12.947 MT to 51.833 MT, whilst thermal coal imports grew from just 10.313 MT to over 196.704 MT in the same period.

1.7 OVERVIEW OF INDIAN COAL SECTOR

The coal mining and coal-fired thermal power generation sectors are two of the core industries and together contribute ~10% to India's Index of Industrial Production (IIP), affirming their importance to the economy. Further, India's logistics industry, sponge iron industry, aluminum industry among several others, as on date, depend on India's domestic coal industry. Economic activities in three eastern states (Chhattisgarh, Jharkhand & Odisha) are significantly dependent on coal. Hence, the importance of the coal sector in India, not just in terms of energy source for the country but also for the socio-economic role it plays, cannot be denied.

COAL PRODUCTION

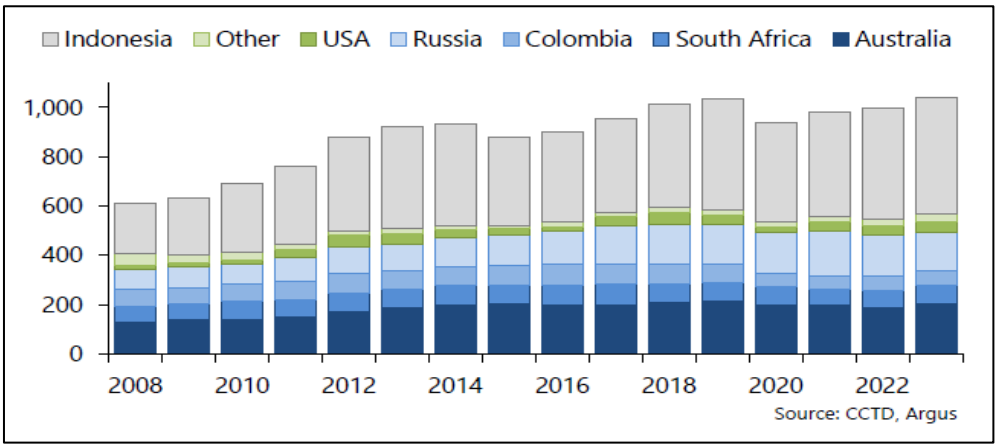
Through sustained programme of investment and greater thrust on application of modern technologies, it has been possible to raise the All-India production of coal at 778.21 million tonnes in 2021-22. The all-India Production of coal during 2022-23 was 893.19 MT with a positive growth of 14.77%.

Coal India Limited (CIL) and its subsidiaries accounted for 622.63 MT during 2021-22 as compared to a production of 596.22 MT in 2020-21 showing a positive growth of 4.4%. Coal production of CIL during 2022-23 was 703.20 MT with a positive growth of 12.94%.

INDIA COAL PRODUCTION GROWTH



SEABORNE COAL SUPPLY





Seaborne Coal supply

Source: https://coal.gov.in

1.8 DEMAND AND SUPPLY GAP

The overall long-term demand of coal is closely linked to the performance of the end-use sectors. In India, the end-use sectors of coal mainly include electricity, iron-steel and cement. Demand from the unorganized small-scale sector comprising primarily of the brick and ceramic industry is relatively less.

As per the Report of the Working Group for coal & lignite in the terminal year of XII Plan, the total demand of non-coking coal in different projections is assessed at 913.30 million tonnes. The production of non-coking coal is projected at 683.30 million tonnes. The total supply projection of coal through various sources i.e. Coal India Ltd in 540.66 million tonnes; Singareni Collieries Company Ltd in 57.0 million tonnes and others are in 85.64 million tonnes. This leaves a gap of 230.0 million tonnes of non-coking coal to be met through imports.
(Source: Indian Minerals Yearbook 2017, IBM, March 2018).

Some of this shortfall will be met by supplies from captive coal blocks and rest through imports. Captive coal mining in India was, gradually, being permitted by amending the Coal Blocks Nationalization Act, primarily in iron and steel making, power generation and cement production. Hence, it became important for India to secure coal through imports from international market to meet their significantly rising coal demand.

1.9 IMPORT VS INDIGENOUS PRODUCTION

The Coal Block is allotted to M/s Gujarat Mineral Development Corporation by the Nominated Authority for the sale of coal. There is no restriction to carry on mining operations for own consumption, sale or for any other purpose. The coal produced from the Coal Block can be sold by the project proponent in any manner as may be decided by the project proponent including sale to Affiliates and related parties, utilization of coal for any purpose including but not limited to captive consumption, Coal Gasification, Coal Liquefaction and export of coal.

1.10 EXPORT POSSIBILITY

Based on the requirement, export aspects may be explored. Accordingly, necessary clearance/certification will be obtained from the competent authority, if required.

1.11 DOMESTIC/EXPORT MARKETS

The coal will be supplied to captive end use power plant and may be supplied to cater domestic/export markets through sale.

1.12 BRIEF HISTORY OF PROJECT

It has been covered under **Section1.1** above in detail. The Mining Plan is approved, and TOR has been transferred to GMDC and accordingly EIA/EMP is prepared. As per the Vesting Order Issued to GMDC, Mining Plan is deemed to be transferred to GMDC.

The status of statutory clearances (also refer item 1.1 and 1.2.1) is given below:

1.12.1 DETAILS OF MINING LEASE

As per Gazette Notification U/s 11(1) of CBA Act, 1957-land measuring 1192.2930 ha and all rights in or over the said land shall vest in the Government Company (Odisha Mining Corporation Limited) vide Ministry of Coal S.O. No. 195 dt 17/02/2022. The nominated authority, GoI, Ministry of coal, has issued vesting order in favour of Gujarat Mineral Development Corporation, being successful bidder; vide order NA-104/4/2023-NA, dated 08/06/2023(after auction of the mine). Land Rights have been obtained by Gujarat Mineral Development Corporation under Sec 11(1) of CBA & henceforth Sec 19 of CBA has been published. Copy of Sec 11(1) of CBA & Sec 19 of CBA are enclosed as **Annexure XX & XXI** respectively.

1.12.2 MINING PLAN

Revised Mining Plan of Baitarni West has been prepared and approved by MoC for 1196.0 ha vide letter no. SWCS/Baitarni West CoalMine/OR/AN/2021/CCO-MCBA dt. 10-11-2021.

1.12.3 AIR AND WATER CONSENTS

The consent to operate for discharge of effluents under Water (Prevention and Control of Pollution) Act 1974 and emission under Air (Prevention and Control of Pollution) Act 1981 will be obtained separately after Environmental clearance granted. The CTO will be obtained accordingly.

1.12.4 FOREST CLEARANCE

64.3723 ha of area is under forest land. Stage-I Forest Clearance for diversion for non-forestry activity has been applied online vide single window no-SW/171065/2024. Govt. of Odisha forwarded Forest diversion proposal to MoEF&CC vide letter no. FE-DIV-FLD-0005-2025-1477/FE&CC dated 15.01.25. IRO visit for Site & CA land is completed. EDS raised by MoEF&CC dated 26.06.25 and it is closed on 26.08.25. As on date EDS closed by DFO & forwarded to RCCF, Angul dated 17.09.25. Copy of letter is attached as **Annexure II**. The detailed status of the proposal is enclosed as **Annexure III.**

1.13 SCOPE OF THE STUDY

The core zone in the study area comprises of the proposed mine lease area (1196 ha) and buffer zone covering the area within 10 km radial distance from the proposed mine lease boundary. The scope of study broadly includes:

- Collection of data relevant to the study area and literature review
- Undertaking environmental monitoring, so as to establish the current environmental status of the study area viz. air, noise, water, land, biological and socio-economic components, and other parameters of human interest.
- Identification of various existing pollution loads due to various activities.

- Evaluation of the impacts on various environmental attributes in the study area by using scientifically developed and widely accepted environmental impact assessment methodologies.
- Documentation of the environment management plan (EMP), outlining the measures required to improve the environmental quality; and
- Identification of the critical environmental attributes required to be monitored
- To prepare a post-project monitoring programme for checking and regulating the environmental quality within the limits stipulated by the State Pollution Control Board and Ministry of Environment & Forests and other statutory agencies.

This Final EIA/EMP has been prepared based on field monitoring, collection of secondary data, as per the TOR issued by the MoEF&CC and based on public hearing proceedings.

1.14 STRUCTURE OF EIA REPORT

Table 1:1 Structure of EIA as Per Notification 2006

S. No	EIA Structure	Contents
1.	Introduction	• Purpose of the Report • Identification of Project and Project Proponent • Brief description of nature, size, location of the project and its importance to the country, region • Production & Imports • Overview of Indian Coal Sector • Scope of study-details of regulatory scoping carried out (As per terms of reference)

S. No	EIA Structure	Contents
2.	Project Description	• Condensed description of those aspects of the project (based on project feasibility study), likely to cause environmental effects. • Description contains the details of the following: • Type of Project • Need for the Project • Location details showing general location, specific location, project boundary and project site layout. • Size or Magnitude of Operation • Project description including drawings showing project layout, components of project etc. • Proposed schedule for approval and implementation • Technology and Process Description • Schematic representations of the feasibility drawings, which give information important for EIA purpose Description of mitigation measures incorporated into the project to meet environmental standards, environmental operating conditions, or other EIA requirements
3.	Description of the Environment	• Study Area, Period, Components and Methodology • Establishment of baseline for valued Environmental Components, as identified in the scope • Study Period • Base maps of all Environmental Components
4.	Anticipated Environmental Impacts and Mitigation Measures	• Details of Investigated Environmental impacts due to project location, possible accidents, project design, project construction, regular operations • Measures for minimizing and/or offsetting adverse impacts identified • Irreversible and Irretrievable commitments of environmental components • Assessment of significance of impacts (Criteria for determining significance, Assigning significance) • Impact scores and Mitigation Measures • Air Modeling
5.	Analysis of Alternatives (Technology and Site)	• In case, the scoping exercise results in need for alternatives: • Description of each alternative • Selection of alternative

S. No	EIA Structure	Contents
6.	Environmental Monitoring Program	• Technical aspects of environmental monitoring for the effectiveness of mitigation measures (including measurement methodologies, frequency, location, data analysis, reporting schedules, emergency procedures, detailed budget and procurement schedules)
7.	Additional Studies	• Risk Assessment • DMP
8.	Project Benefits	• Improvements in physical infrastructure • Improvements in social infrastructure • Employment potential–Skilled; semi-skilled and unskilled • Other tangible benefits of the project
9.	Environmental Cost Benefit Analysis	• An analytical way to make an educated decision regarding the commencement of mining activity or any other similar activity • involves a comparison of the costs of an action with considerations of the benefits associated with that action • assists the regulators to evaluate the benefits and challenges imposed by the upcoming activity in commercial terms with respect to the impact on the environmental scenario such as human wellbeing, quality of life and environmental wellbeing • aims to maximize economic efficiency at a point where marginal benefits and marginal costs are equal
10.	Environment Management Plan	• Description of the administrative aspects of ensuring that mitigation measures are implemented, and their effectiveness monitored, after approval of the EIA • Explanation of how, adverse effects have been mitigated
11.	Summary and Conclusion	Overall justification for implementation of the project
12.	Disclosure of Consultant Engaged	The names of the Consultants engaged with their brief resume and nature of consultancy rendered

1.15 LAWS APPLICABLE TO THIS PROJECT

Table 1:2 Applicable Laws

Legal Channel	Responsible Ministries/ Bodies	Objective of Legislation	Action Plan
The Water (Prevention & Control of Pollution) Acts 1974/ Rules 1975	CPCB, Odisha Pollution Control Board	The prevention and control of water pollution and also maintaining or restoring the wholesomeness of water	• Not to discharge any effluent, which are not confirming to standards, prescribed by Odisha Pollution Control Board into any stream, well, sewers or land • Not to discharge air pollutant(s) in excess of standards, prescribed by the Odisha Pollution Control Board • Obtain 'Consent to Establish' prior to establish any process, operation or treatment system • Obtain 'Consent to Operate' prior to operation of system, which is likely to discharge effluent • Apply & Obtain 'Consent to Operate' • Comply with conditions as prescribed under consents
The Air (Prevention & Control of Pollution) Acts 1981/ Rules 1982	CPCB, Odisha Pollution Control Board	The prevention, control and abatement of air pollution	
The Environment (Protection) Acts 1986/Rules 1986 The Environmental Impact Assessment (EIA) Notification, 2006	MoEF&CC, CPCB, Odisha Pollution Control Board	Protection and Improvement of the Environment	• Prevent discharge or emission of environmental pollutants in excess of the prescribed standards • Submit 'Environmental Statement' every year • Obtain prior "Environmental Clearance' from MoEF&CC in case of

Legal Channel	Responsible Ministries/ Bodies	Objective of Legislation	Action Plan
			new project or for Modernization / Expansion
Hazardous and Other Wastes (Management and Trans boundary Movement) Rules, 2016	MoEF, CPCB, Odisha Pollution Control Board	Management & Handling of hazardous wastes in line with the Basel convention	• It is the responsibility of the occupier to identify the hazardous wastes in their units and ensure proper handling and disposal PP to take all steps to contain contamination, prevent accident and limit consequences on human being and environment • Obtain authorization from Odisha Pollution Control Board and comply with the conditions. • Maintain records of Hazardous Waste generated in Form-3 and submit yearly return for generation, treatment, recycling, disposal etc., to Odisha Pollution Control Board in Form-4 • Used Oil to be send / sold to the registered recycler, re-processor, registered authorized facility • Shall be transported in accordance with the rule • Site storage is allowed for 90 days only
Batteries (Management	Odisha Pollution Control	To control the hazardous waste generation (lead	• It is the responsibility of the generator to ensure, used batteries

Legal Channel	Responsible Ministries/ Bodies	Objective of Legislation	Action Plan
and Handling) Rules, 2001.	Board, CPCB and MoEF&CC	waste) from used lead acid batteries	are not disposed of in any manner other than depositing with dealer, manufacturer, importer, re-conditioner registered recycler or at designated collection center • Submit half yearly return for disposal of used batteries to State PCB by 30th June & 31st December, every year • In case of auction, ensure batteries are auctioned to the registered recycler only • File half yearly return for the auction • Maintain record for such auction
E-Waste (Management) Rules, 2016	Odisha Pollution Control Board, CPCB and MoEF&CC	To recycle /manage the electronic waste from the industry	• Consumers or bulk consumers of electrical and electronic equipment listed in Schedule I shall ensure that e-waste generated by them is channelized through collection centre or dealer of authorised producer or dismantler or recycler or through the designated take back service provider of the producer to authorised dismantler or recycler • Bulk consumers of electrical and electronic

Legal Channel	Responsible Ministries/ Bodies	Objective of Legislation	Action Plan
			equipment listed in Schedule I shall maintain records of e-waste generated by them in Form-2 and make such records available for scrutiny by the concerned State Pollution Control Board • Consumers or bulk consumers of electrical and electronic equipment listed in Schedule I shall ensure that such end-of-life electrical and electronic equipment are not admixed with e-waste containing radioactive material as covered under the provisions of the Atomic Energy Act, 1962 (33 of 1962) and rules made there under • Bulk consumers of electrical and electronic equipment listed in Schedule I shall file annual returns in Form-3, to the concerned State Pollution Control Board on or before the 30th day of June following the financial year to which that return relates. In case of the bulk consumer with multiple offices in a State, one annual return combining information from all the offices shall be filed to

Legal Channel	Responsible Ministries/ Bodies	Objective of Legislation	Action Plan
			the concerned State Pollution Control Board on or before the 30th day of June following the financial year to which that return relates
The Solid Waste Management Rules, 2016	Odisha Pollution Control Board , CPCB	To manage/ utilize the generated solid waste without damaging the environment and surroundings	• Segregate waste in to three streams, Wet (Biodegradable), Dry (Plastic, Paper, metal, wood, etc.) and domestic hazardous wastes (diapers, napkins, empty containers of cleaning agents, mosquito repellents, etc.) and handover segregated wastes to authorized rag-pickers or waste collectors or local bodies
Noise Pollution (Regulation and Control) Rules, 2000 and its amendments	CPCB, Odisha Pollution Control Board, MoEF&CC	To maintain the noise levels with respect to the place/equipment / industry	• Noise Quality Monitoring & submission of reports on weekly/monthly basis • Providing Ear plugs and Muffs to the workers working in noise prone areas • Dampening the source noise level or making the noise characteristics less annoying by providing suitable enclosures and barriers
The Central Motor Vehicle Rules, 1989	Ministry of Shipping, Road	To consolidate and amend the law relating to	• Ensure compliance to safety provisions in the transport vehicle

Legal Channel	Responsible Ministries/ Bodies	Objective of Legislation	Action Plan
	Transport and Highways	motor vehicles including to regulate the transportation of dangerous goods with a view to prevent loss of life or damage to the environment	carrying dangerous and hazardous substances inside works • Display of emergency information panels at front, back and both side of vehicle • Every transporter to ensure safe transportation of dangerous/ hazardous goods • Earthing chain for grounding, any prevalent static charge • All motor vehicle entering the works shall have properly maintained brakes, lights, signal system for brakes, blinkers and registration number displayed, and valid Pollution under Control Certificate
Mineral Concession Rules, 1960.	Coal Controller Organisation (CCO), State Mining Department and District Administration	Guidelines for mining of minerals in the lease area	• Mineral will be extracted as per the Approved mining plan
Mines & Minerals Development Restoration and Rehabilitation Act 1957	DMG, CCO, State Mining Department and District Admn	Regulation of mines and the development of minerals	• Development, Restoration and Rehabilitation fund will be deposited in Escrow Account with be spend as per Mines & minerals Development

Legal Channel	Responsible Ministries/ Bodies	Objective of Legislation	Action Plan
			Restoration and Rehabilitation Act 1957.
Forest (Conservation) Act, 1980	Forest Department, MoEF&CC	Diversion of entire forest land located within the mining lease/project is to be obtained before execution/renewal of mining lease	• Stage-I Forest Clearance for diversion for non-forestry activity has been applied online vide single window no-SW/171065/2024. Govt. of Odisha forwarded Forest diversion proposal to MoEF&CC vide letter no. FE-DIV-FLD-0005-2025-1477/FE&CC dated 15.01.25. IRO visit for Site & CA land is completed. EDS raised by MoEF&CC dated 26.06.25 and it is closed on 26.08.25. As on date EDS closed by DFO & forwarded to RCCF, Angul dated 17.09.25.
Wildlife protection Act 1972	Forest Department, MoEF&CC	Protection of plants and animal species	• Schedule – I species will be looked into.
Land Acquisition, Rehabilitation and Resettlement Act, 2013 and Coal Bearing Areas (Acquisition and Development) Act, 1957	Administration	Right to Fair Compensation and Transparency in Land Acquisition	• All rules will be followed

----------x----------

The project area is a part of Talcher coal field in Chhendipada Tehsil, Angul District, Odisha and fall in eight number of villages.

Two seasonal streams, Kumbhira and Supai are flowing across the ML area. A detailed hydrological survey of the Supai Nala, regarding its catchment area, flow volume and length of the stretch to be diverted with water augmentation plan along with development of artificial pond has been done.

Major part of the lease area is agriculture land.

Average grade of the coal is G-12

Number of seams -18

Stripping Ratio- 1.5278 m3 per tonne

Extractable reserve is 468.27 MT

Life of the mine is 37 years

Open cast mining method with Surface miner, Pay loader and Tipper, OB removal by Shovel Dumper combination.

Only 7% drilling and blasting and 93% shall be through Surface miner.

12 MTPA coal transportation through rail and 3 MTPA through road.

Project cost-3855 Cr

CHAPTER-2 : PROJECT DESCRIPTION

2.1 TYPE OF THE PROJECT

Baitarni West Coal block of Talcher Coalfield is an open cast mine with a target to produce 15 MTPA of coal. The purpose of coal production is for sale of coal including sale to affiliates and related parties, utilization of coal for any purpose including but not limited to captive consumption, coal gasification, coal liquefaction and export of coal.

As per EIA notification dated 14th September 2006 and its amendments, the project falls under Activity 1 (a) Mining of Minerals. As per MoEF&CC Notification No. S.O. 1886(E) dated 20th April 2022, all the Coal Mining Projects with lease area more than 500 ha has been classified as Category "A" projects. Therefore, the project will be considered at central level.

2.2 NEED OF THE PROJECT

It is discussed in Chapter 1.3.2. India is the world's second largest producer, consumer, and importer of coal after China.

Historically, coal plants have accounted for 70 to 80 per cent of the country's total power generation. This share is estimated to remain above 50 per cent even in 2030. Further, India's current installed coal capacity of 209 GW is set to expand, with around 57 GW of new capacity in the pipeline. Coal is a reliable energy source, especially when compared with the seasonal and diurnal variability of renewables. Abundant in India, it is considered important for the nation's energy security and is a key source of revenue for the government.

2.3 LOCATION OF THE PROJECT (MAPS SHOWING GENERAL LOCATION, SPECIFIC LOCATION, PROJECT BOUNDARY & PROJECT SITE LAYOUT).

The Baitarni West Coal Block is located in the North central part of Talcher coalfield.

The map showing the location of Baitarni West Coal block in the Talcher coalfield is given below. The project area is covered under Survey of India Toposheet No: F45M16 on RF 1:50,000 and is bounded by the geographical coordinates:
NW: 21°05'26.87" N, 84°50'12.08" E
NE: 21°05'17.37" N, 84°51'26.13" E
SW: 21°01'44.84" N, 84°50'39.59" E
SE: 21°01'44.74" N, 84°51'40.70" E

Project does not fall in the Critically Polluted Area (CPA) where the MoEF&CC vide its OM dated 13th January 2010 has imposed moratorium on grant of environment clearance.

The project area is in Chhendipada Tehsil, Angul District, Odisha covering Villages
(i) Chhendipada Jangal-(Part)
(ii) Handigora- (Part)
(iii) Machhakuta - (Part)
(iv) Machhakuta Jangal- (Part)
(v) Porapara- (Part)
(vi) Porapara Jangal -(Full)
(vii)Tentuloi-Korasahi- (Part)
(viii) Chhendipada RF - (Part)

2.3.1 MINE BOUNDARY

North-After leaving a surface barrier of 45 m from major district road.
West-15m from common block boundary between Baitarni (West) and Machhakuta Block.
South-Keeping mine floor boundary at faults F1-F1, the surface boundary has been arrived with a quarry slope angle of around 40° to south.
East-25m from western bank of Kumbhira nullah. Surface boundary has been smoothened at sharp bends turns of nullah. The barrier is provided for access road and embankment along Kumbhira nullah to prevent inundation.

2.3.2 DISTANCE FROM WATER BODIES

Table 2:1 Distance from Water Bodies

Name of the Water Bodies	Distance & Direction
Kumbhira Nala (seasonal)	Within mine lease area
Supai Nala (seasonal)	Within mine lease area
Gauduni Nala	1.2 km/E

Kusal Nala	1 km/W
Bainsajaria Nala	1.5 km/NW
Shagarhia Nala	3.0 km/NW
Singhda Jhor	5.0/S
Aunli Nadi	5.5 km/N
Mandalia Nala	9.5 km/W
Nuhurang nadi	9 km, NW
Tikra river	9 km, NNE

ROAD/RAIL/AIR LINK

District headquarter Angul, located on National Highway 42 (Bhubaneswar to Sambalpur), is the nearest town at a distance of about 43 kms from the northeastern boundary of the block.

The Angul-Chhendipada-Baghadia-Kanloi metalled road runs along the northern boundary of the block. (Also, there is Block Development as well as Tehsil office at Chhendipada village, which is located about 2 kms from the eastern boundary of the block).

Existing major District Road from Jarapada (in Angul-Sambalpur Highway) up to Tentuloi (about 18 km long) may be used for communication and road transportation to Baitarni-West coal mine.

Another prominent village, Baghadia, is located around 4 kms from northwestern boundary of the block along Angul-Chhendipada-Baghadia-Kanloi metalled road. The village Kosala, where the **Exploration Camp of CMPDI, RI-7** is located, lies about 15 kms southeast of the block.

The **Kaniah Super Thermal Power Station of National Thermal Power Corporation** is connected at a distance of 25 kms through fair weather / all-weather roads from Kosala village.

A **Captive Power Plant and a Smelter Plant of National Aluminium Corporation** are located on either side of National Highway 42 in Angul township.

The Talcher town is located about 45 kms away from the block and is connected by fair/all weathered road through Kosala village.

NTPC has another thermal power plant near the outskirts of Angul. The block is connected to the state capital Bhubaneswar through metalled road up to Angul, then by National Highway 42 up to Cuttack and thereafter through National Highway 5, with total distance of about 185 kms.

The nearest **railhead** is Angul, which is about 42 kms. from the northeastern boundary of block and is connected to Sambalpur-Talcher-Bhubaneswar railway line.

The nearest **port** Paradip at Bay of Bengal is situated at a distance of 250 kms from the block.

The **Biju Pattanaik International Airport**, located at the southern end of the state capital Bhubaneswar, is the nearest airport from the Baitarni West block.

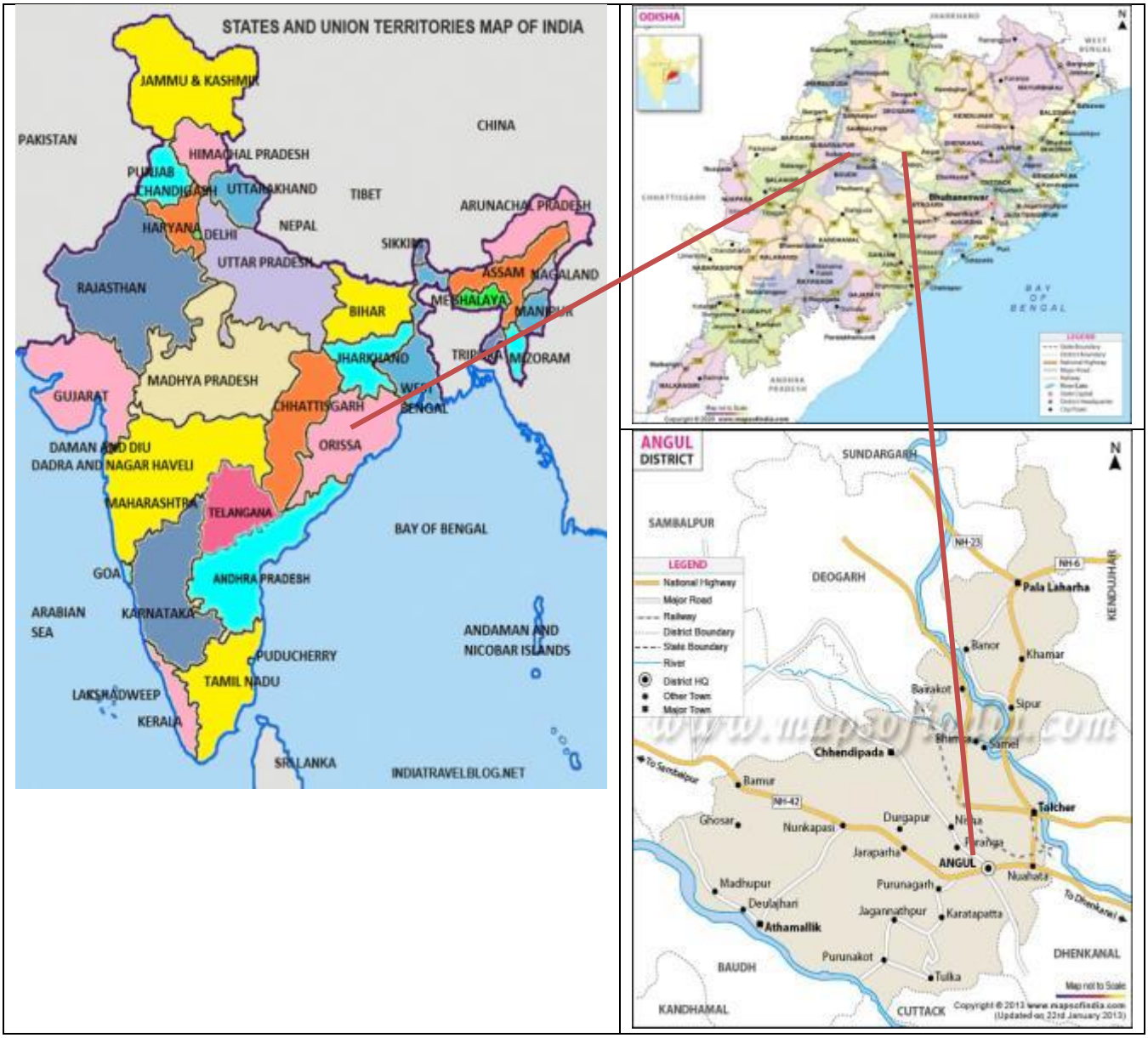


Figure 2-1: The location of the lease area/General location

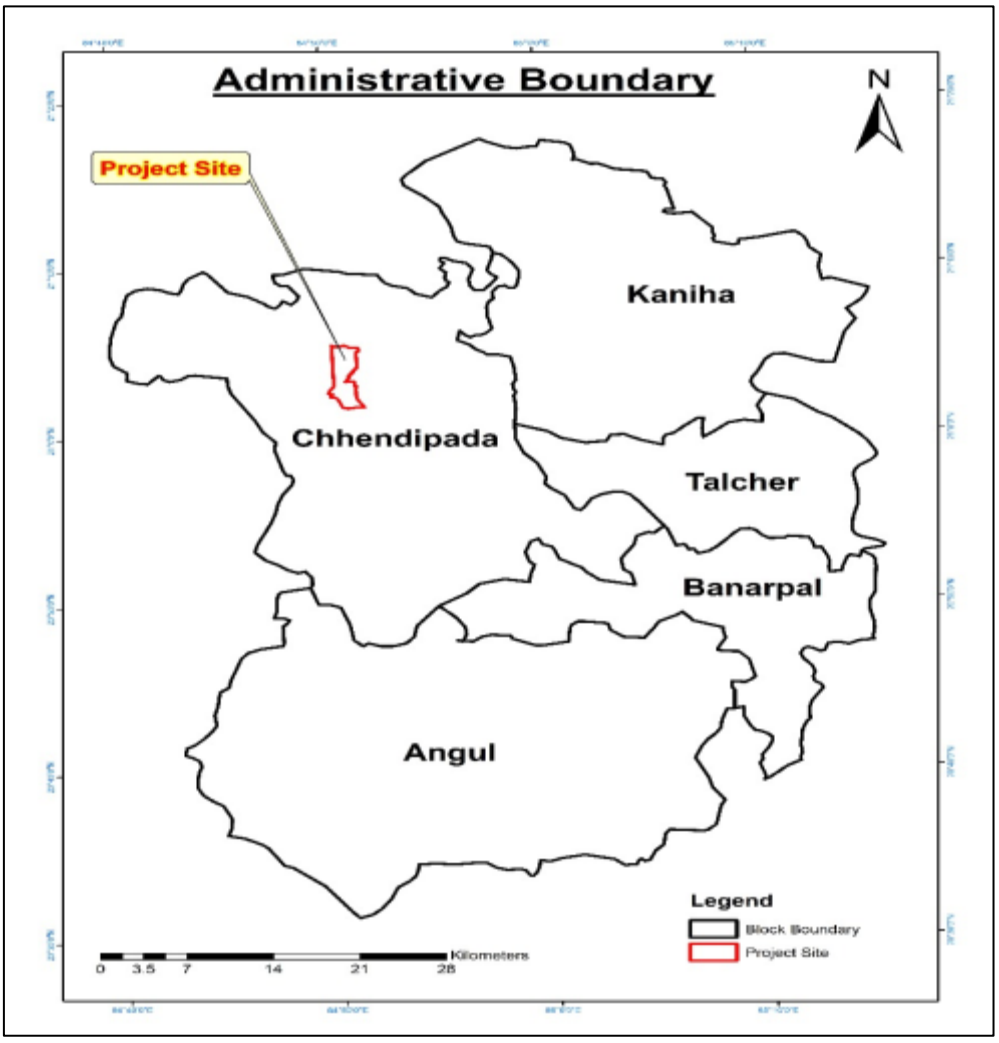
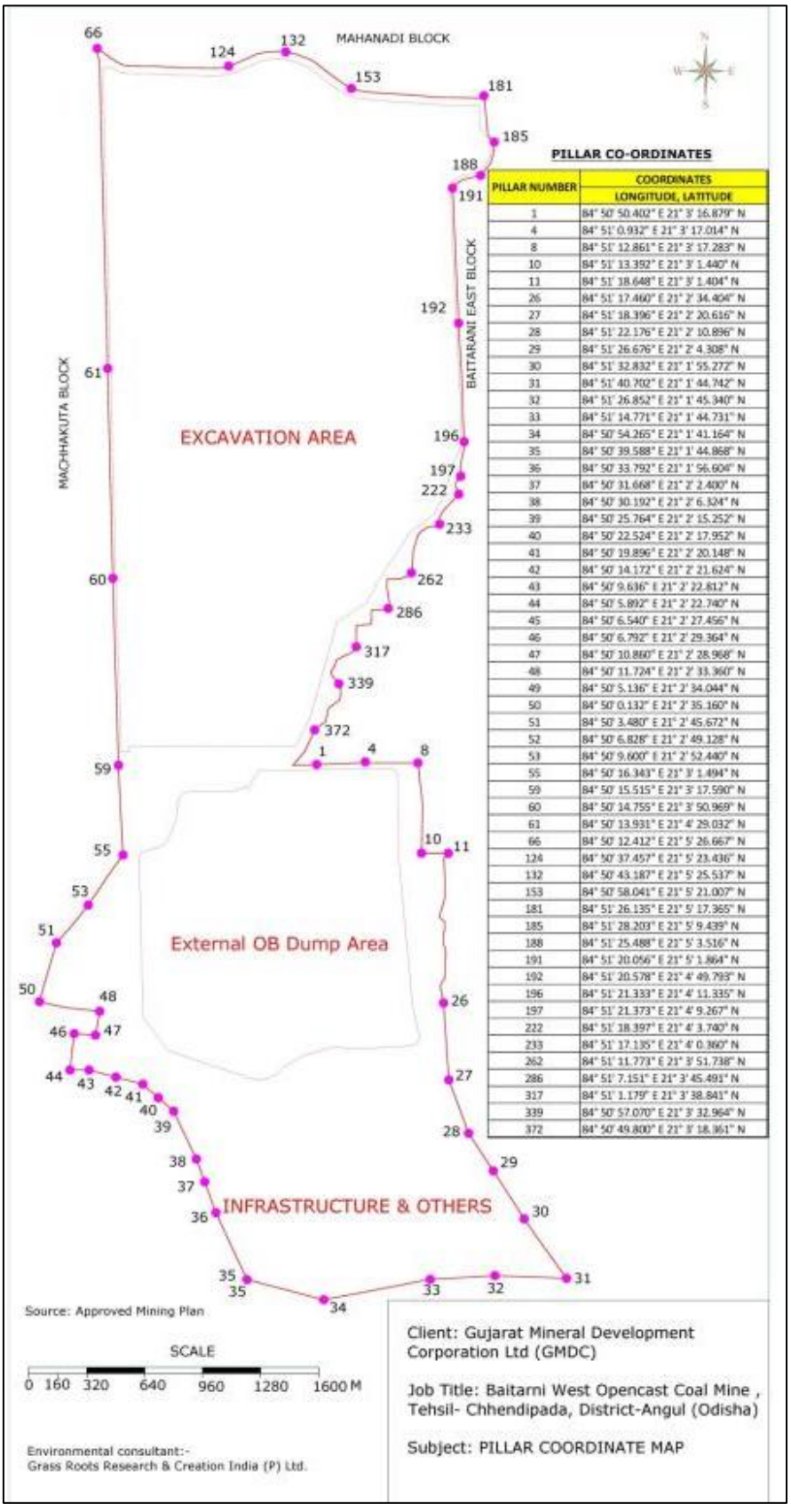




Figure 2-2: Location of Project



PILLAR CO-ORDINATES

PILLAR NUMBER	COORDINATES LONGITUDE, LATITUDE
1	84° 50' 50.402" E 21° 3' 16.879" N
4	84° 51' 0.932" E 21° 3' 17.014" N
8	84° 51' 12.861" E 21° 3' 17.283" N
10	84° 51' 13.392" E 21° 3' 1.440" N
11	84° 51' 18.648" E 21° 3' 1.404" N
26	84° 51' 17.460" E 21° 2' 34.404" N
27	84° 51' 18.396" E 21° 2' 20.616" N
28	84° 51' 22.176" E 21° 2' 10.896" N
29	84° 51' 26.676" E 21° 2' 4.308" N
30	84° 51' 32.832" E 21° 1' 55.272" N
31	84° 51' 40.702" E 21° 1' 44.742" N
32	84° 51' 26.852" E 21° 1' 45.340" N
33	84° 51' 14.771" E 21° 1' 44.731" N
34	84° 50' 54.265" E 21° 1' 41.164" N
35	84° 50' 39.588" E 21° 1' 44.868" N
36	84° 50' 33.792" E 21° 1' 56.604" N
37	84° 50' 31.668" E 21° 2' 2.400" N
38	84° 50' 30.192" E 21° 2' 6.324" N
39	84° 50' 25.764" E 21° 2' 15.252" N
40	84° 50' 22.524" E 21° 2' 17.952" N
41	84° 50' 19.896" E 21° 2' 20.148" N
42	84° 50' 14.172" E 21° 2' 21.624" N
43	84° 50' 9.636" E 21° 2' 22.812" N
44	84° 50' 5.892" E 21° 2' 22.740" N
45	84° 50' 6.540" E 21° 2' 27.456" N
46	84° 50' 6.792" E 21° 2' 29.364" N
47	84° 50' 10.860" E 21° 2' 28.968" N
48	84° 50' 11.724" E 21° 2' 33.360" N
49	84° 50' 5.136" E 21° 2' 34.044" N
50	84° 50' 0.132" E 21° 2' 35.160" N
51	84° 50' 3.480" E 21° 2' 45.672" N
52	84° 50' 6.828" E 21° 2' 49.128" N
53	84° 50' 9.600" E 21° 2' 52.440" N
55	84° 50' 16.343" E 21° 3' 1.494" N
59	84° 50' 15.515" E 21° 3' 17.590" N
60	84° 50' 14.755" E 21° 3' 50.969" N
61	84° 50' 13.931" E 21° 4' 29.032" N
66	84° 50' 12.412" E 21° 5' 26.667" N
124	84° 50' 37.457" E 21° 5' 23.436" N
132	84° 50' 43.187" E 21° 5' 25.537" N
153	84° 50' 58.041" E 21° 5' 21.007" N
181	84° 51' 26.135" E 21° 5' 17.365" N
185	84° 51' 28.203" E 21° 5' 9.439" N
188	84° 51' 25.488" E 21° 5' 3.516" N
191	84° 51' 20.056" E 21° 5' 1.864" N
192	84° 51' 20.578" E 21° 4' 49.793" N
196	84° 51' 21.333" E 21° 4' 11.335" N
197	84° 51' 21.373" E 21° 4' 9.267" N
222	84° 51' 18.397" E 21° 4' 3.740" N
233	84° 51' 17.135" E 21° 4' 0.360" N
262	84° 51' 11.773" E 21° 3' 51.738" N
286	84° 51' 7.151" E 21° 3' 45.491" N
317	84° 51' 1.179" E 21° 3' 38.841" N
339	84° 50' 57.070" E 21° 3' 32.964" N
372	84° 50' 49.800" E 21° 3' 18.961" N

Figure 2-3: Pillar-coordinate Map

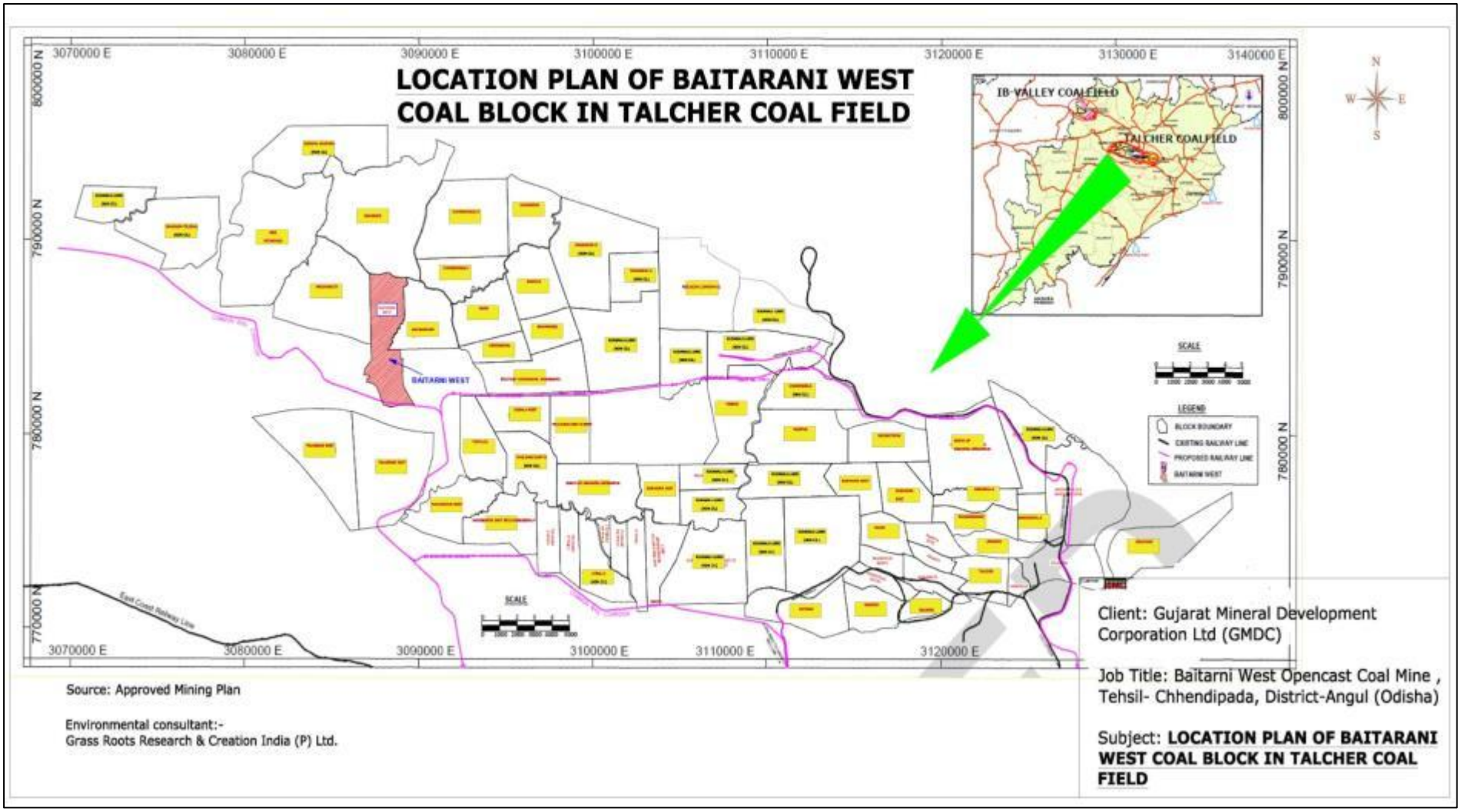


Figure 2-4:-Location Plan of Baitarni West Coal Block in Talcher Coal Field

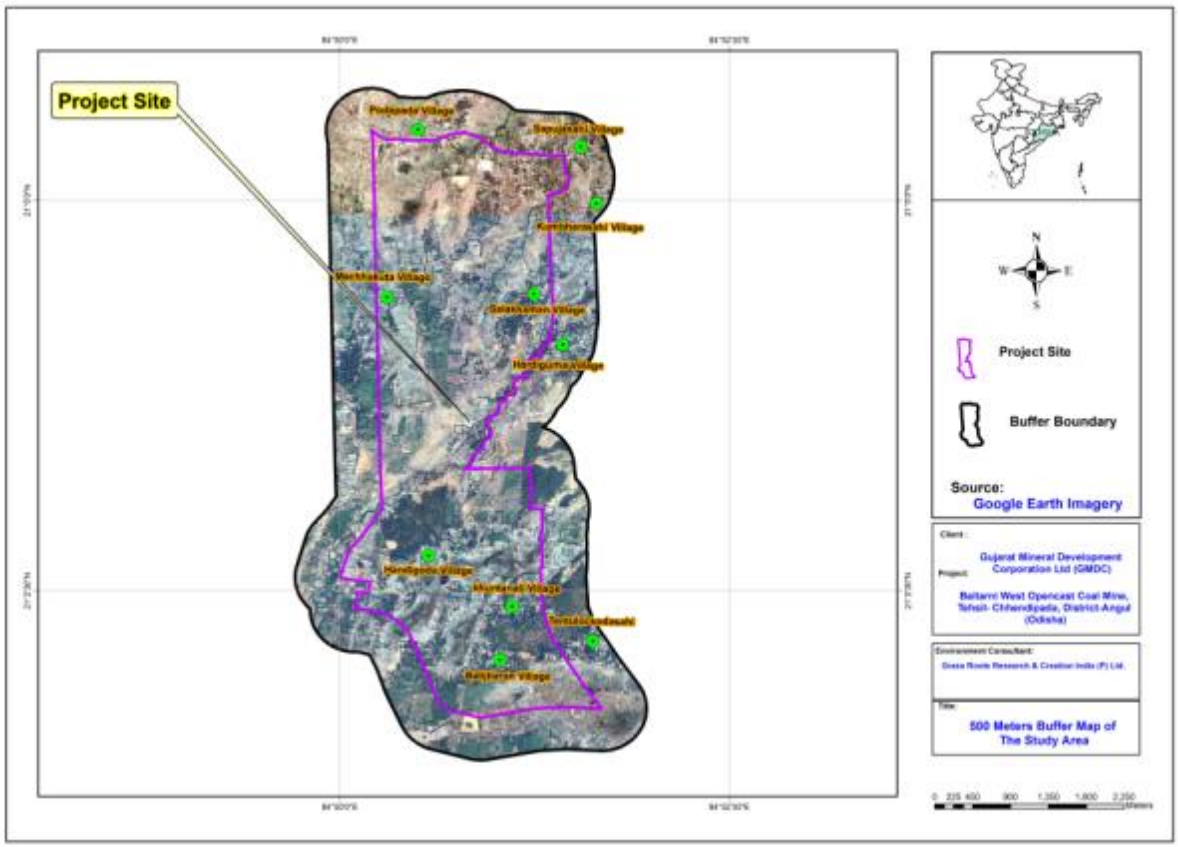


Figure 2-5: Vicinity Map (500m) Showing the Site and Surroundings

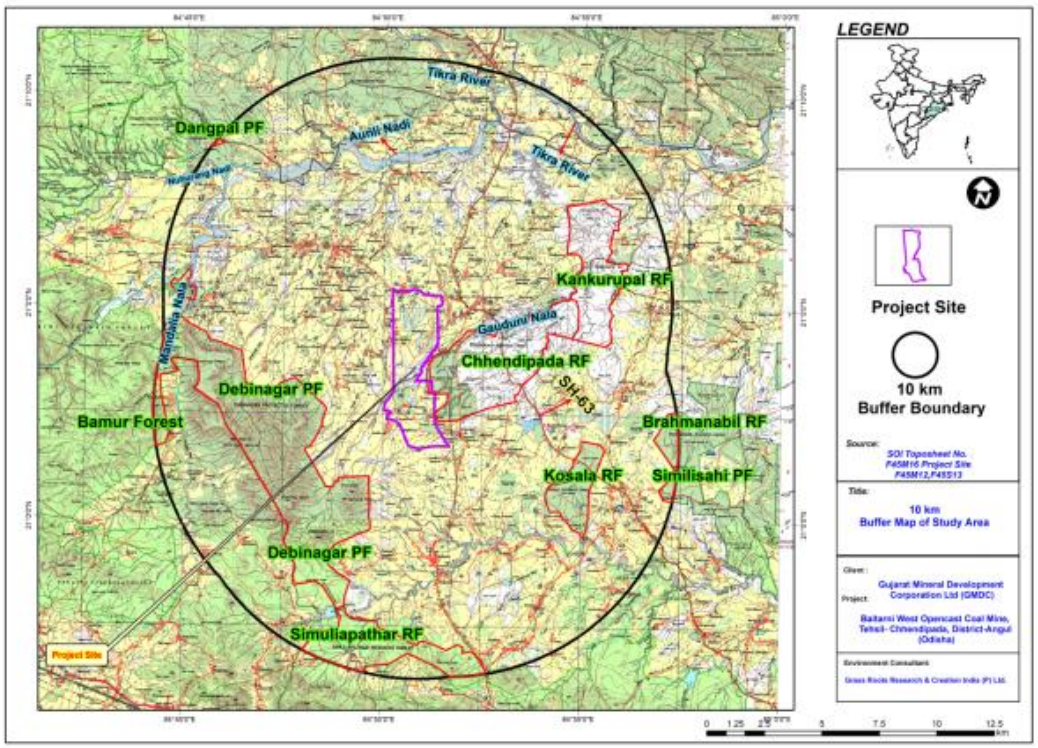


Figure 2-6: Study Area Map (10 km radius)

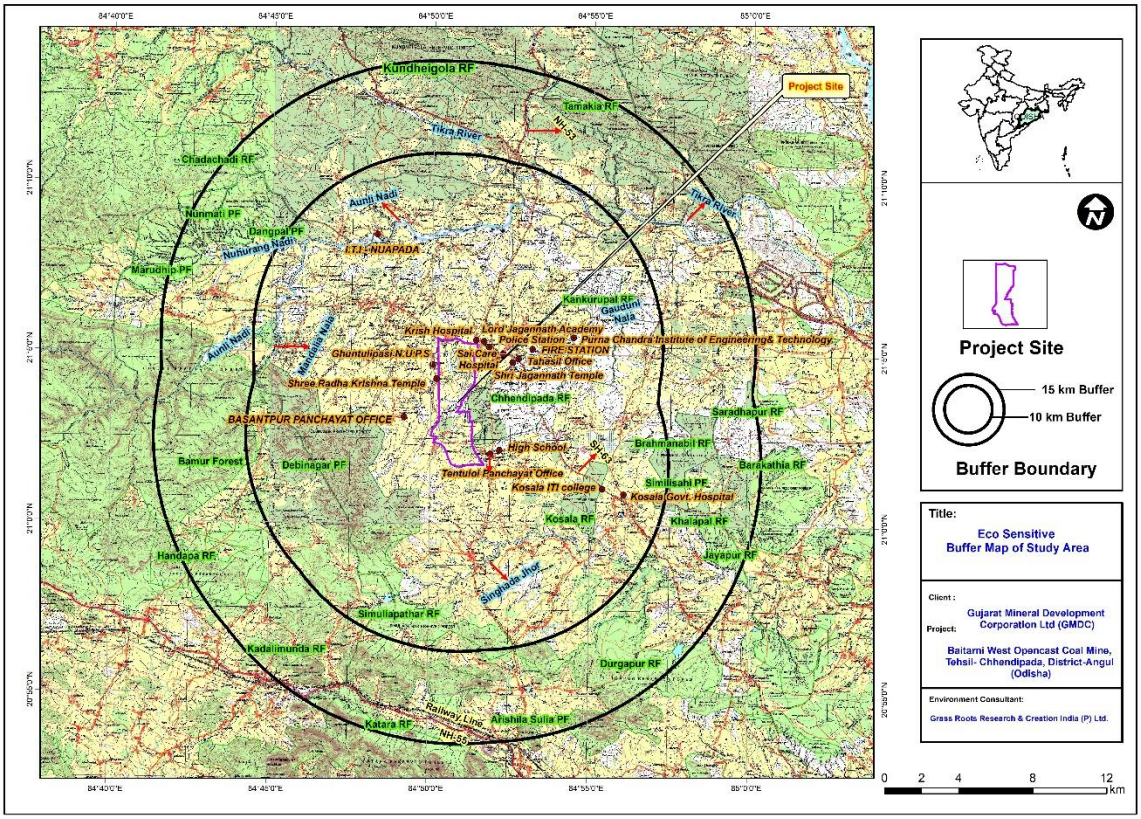


Figure 2-7 Eco sensitive Map of 15 km study area

2.3.3 LIST OF INDUSTRIES WITHIN 10 KM RADIUS OF THE PLANT SITE

The list of Industries within 10 km radius of the plant site is given in Table 2.2.

Table 2:2 Industry/Coal Mines list within 10 km

S. No.	Name of Industry	Direction	Distance (km)
1	Pratima Group of Industry	E	8
2	Behera Bio Briquette	E	7
3	Seven Hill Minerals	E	7.1
4	Saralia LPG Plant	E	3.65
5	M/s Pradhan Fly Ash Bricks	E	4.1
6	Maa Bayani Rice Mill	E	8
7	Dilli Rice Mill	WNW	2.9
8	Shriya Mudhi Mill	WNW	3
9	Tutu Mill	W	2.6

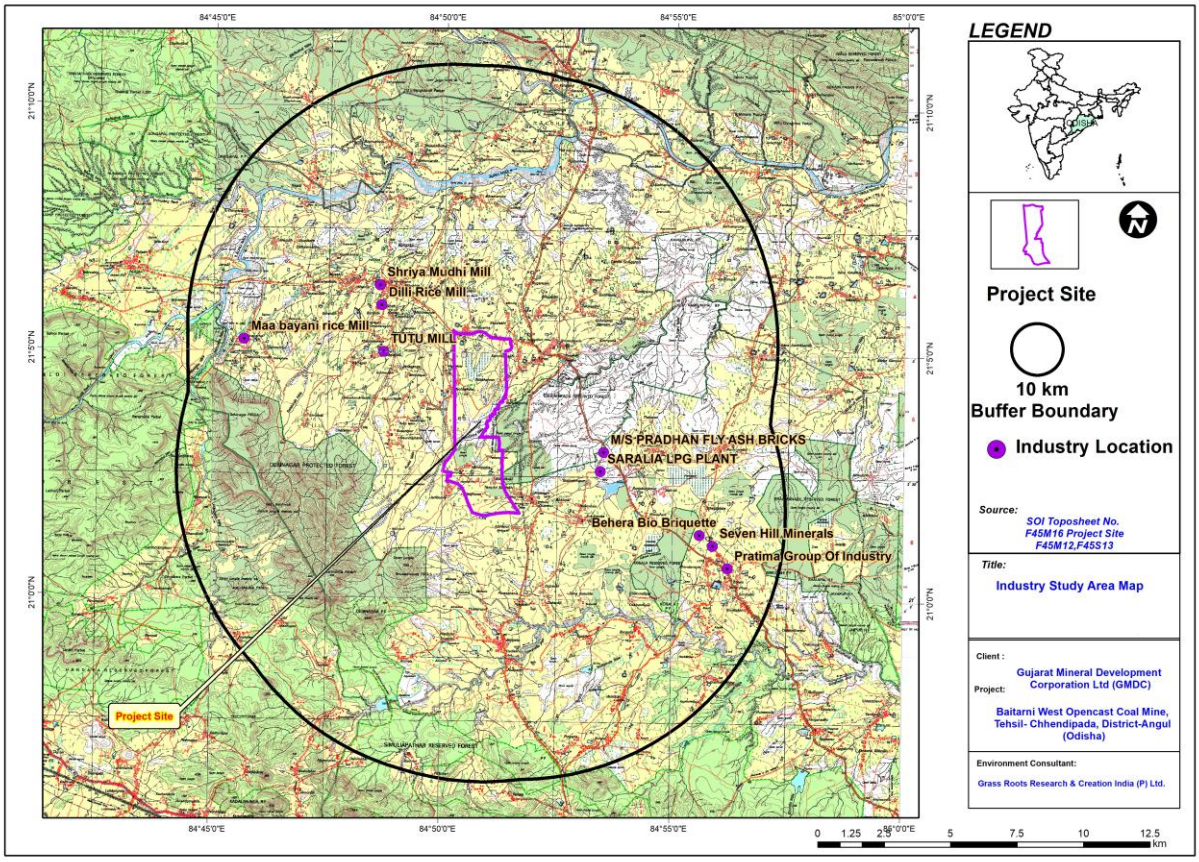


Figure 2-8 Map showing nearby Industries

2.4 SIZE OR MAGNITUDE OF OPERATION (INCL. ASSOCIATED ACTIVITIES REQUIRED BY OR FOR THE PROJECT)

2.4.1 SIZE OR MAGNITUDE OF OPERATION

The proposed lease area of 1196 ha shall be used for mining & allied activities. The proposed annual coal production from this mine shall be 15 MTPA. It is envisaged to achieve the peak rated capacity of mine from 7th Year of mine operations. The coal production from mine is through an opencast method.

2.4.2 PROJECT COST

Total cost of the project is Rs. 3855.00 Cr. Cost of production is Rs.337.11 per tonne; CSR Cost is approx. Rs. 1 Cr/Annum (Minimum of 2% of the net profit over the preceding three financial years of the organisation) which shall be used in sectors like healthcare, skill training, development of rural infrastructure etc. R&R cost is Rs. 727.94 crores.

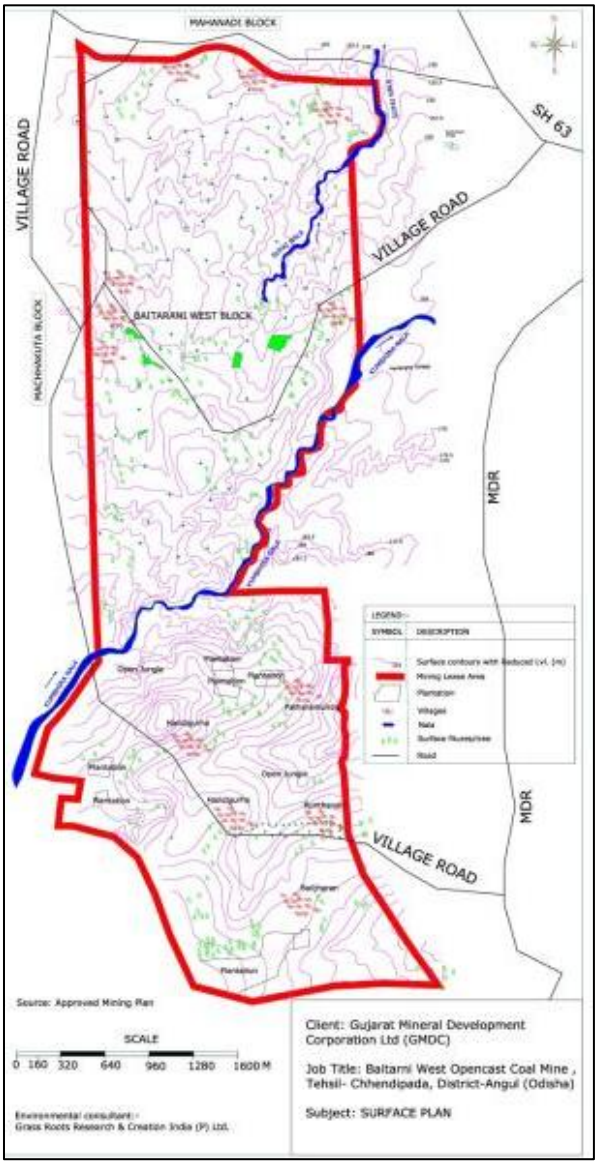


Figure 2-9 Surface Plan

2.5 LAND USE

Mine lease Area: 1196 ha.

AREA AND LAND TYPE

The breakup of Land Type with its further breakup is given in the table. The total area of 1196 ha is mainly divided into three Land Type-Tenancies, Govt. Non-Forest and Forest.

Table 2:3 Area and land type

Land Type		**Area (ha)**
Tenancy	Homestead	25.1027
	Agricultural	784.8499
	Water Bodies	6.6207
	Road	0.182
	Barren	20.2674

	Others	71.63
Sub-Total		**908.6527**
Govt. non-Forest	Home stead	7.6702
	Water Bodies	28.5215
	Road	31.2266
	Barren	24.1065
	Community	5.3926
	Grazing	118.8397
	Others	7.2179
Sub-Total		**222.975**
Forest	Reserve	4.469
	Revenue	59.9033
Sub-Total		**64.3723**
Total		**1196**

Source of Data: *Record of Rights (RoR) from Tehsil Office as Incorporated in the approved mine plan*

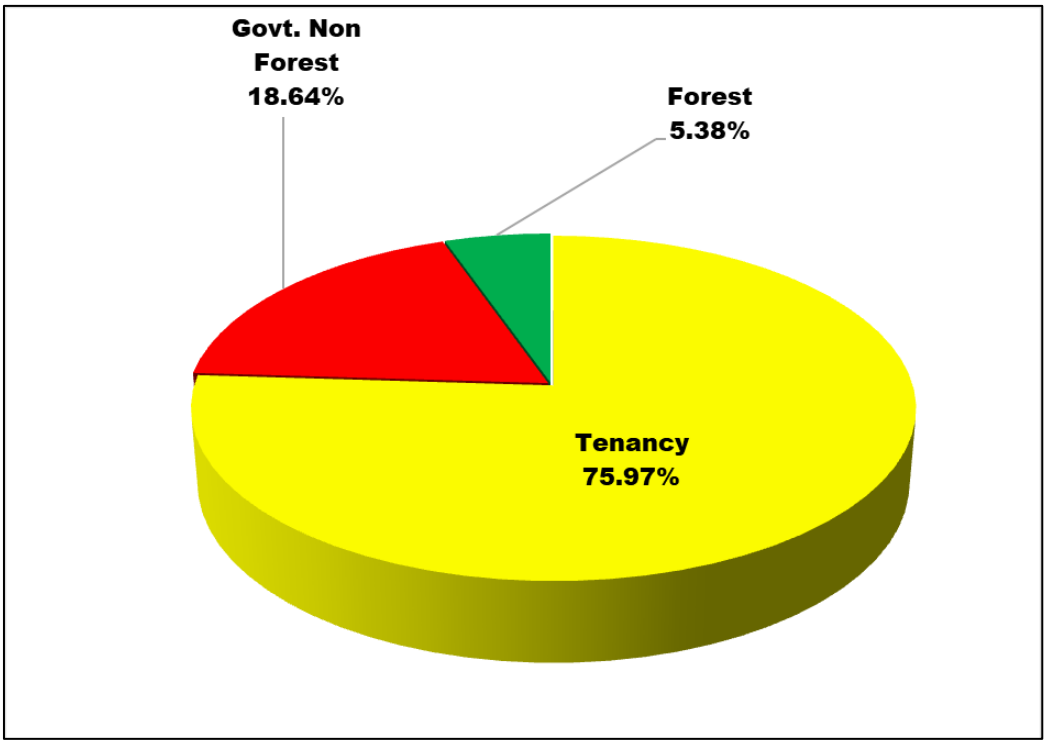


Figure 2-10 Pie Chart Showing the breakup of Broad Land Type

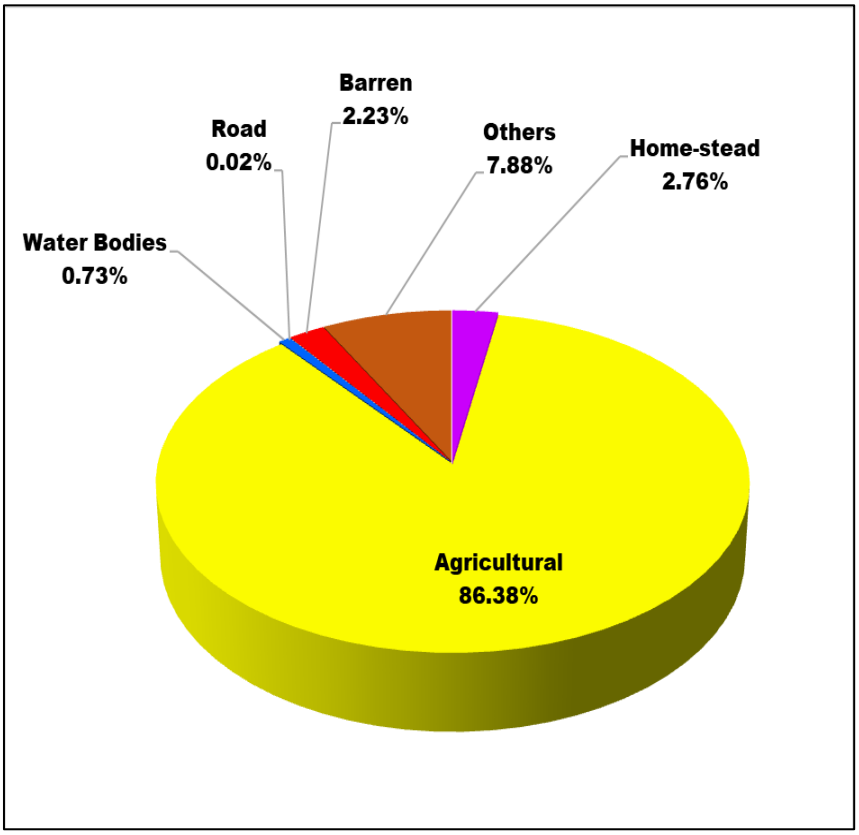


Figure 2-11 Pie Chart Showing the breakup of Tenancy Land

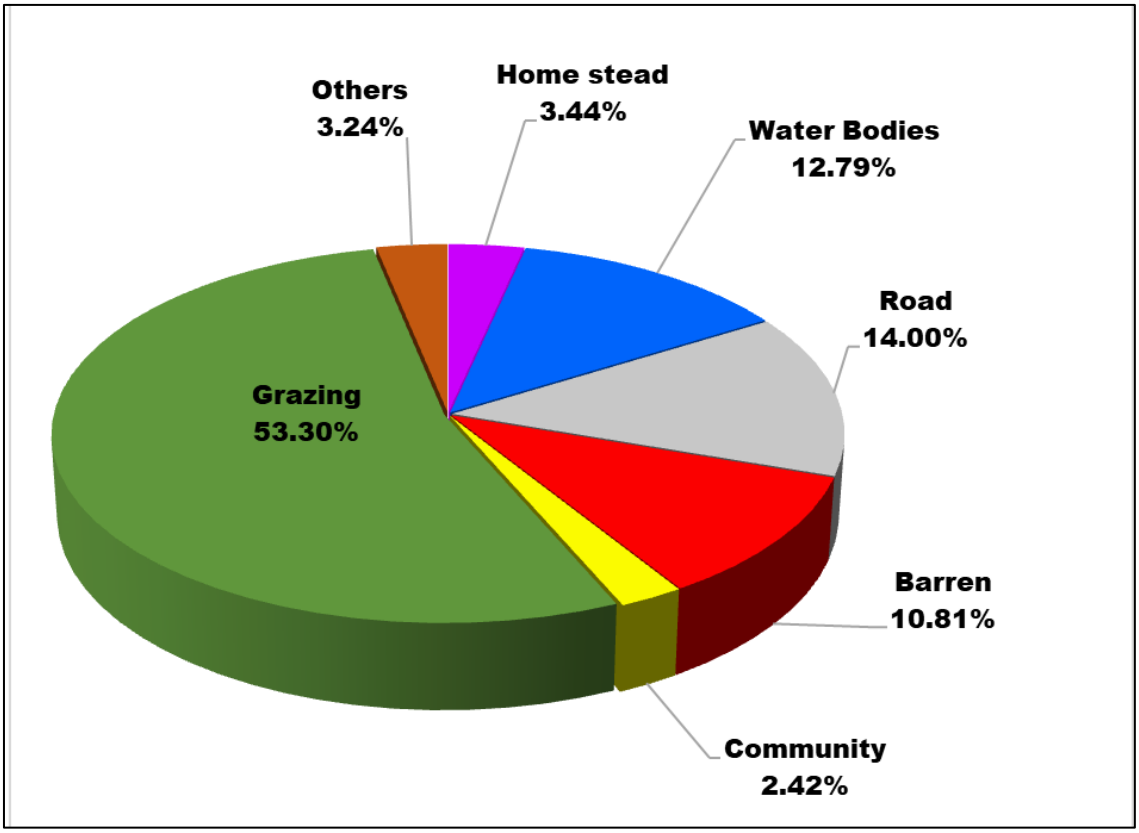


Figure 2-12 Pie Chart Showing the breakup of Govt. Non-Forest Land

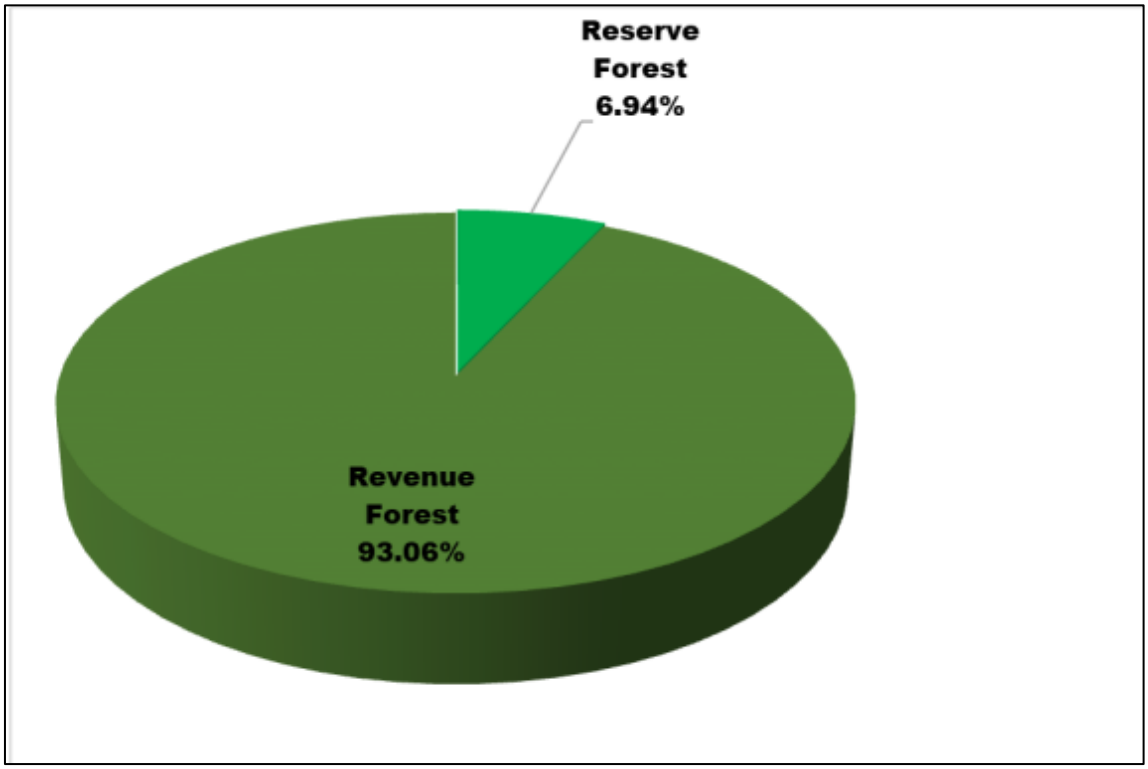


Figure 2-13 Pie Chart Showing the breakup of Forest Land

Village wise grazing land is given below:

Table 2:4 Village wise grazing land

S. No.	**Villages**	**Area (Ha)**
1	Chhendipada Jangal	24.4379
2	Handigora	47.0128
3	Machhakuta	2.6225
4	Machhakuta Jangal	8.347
5	Porapara	10.838
6	Porapara Jangal	5.9517
7	Tentuloi-Korasahi	19.6298
	Total	**118.8397**

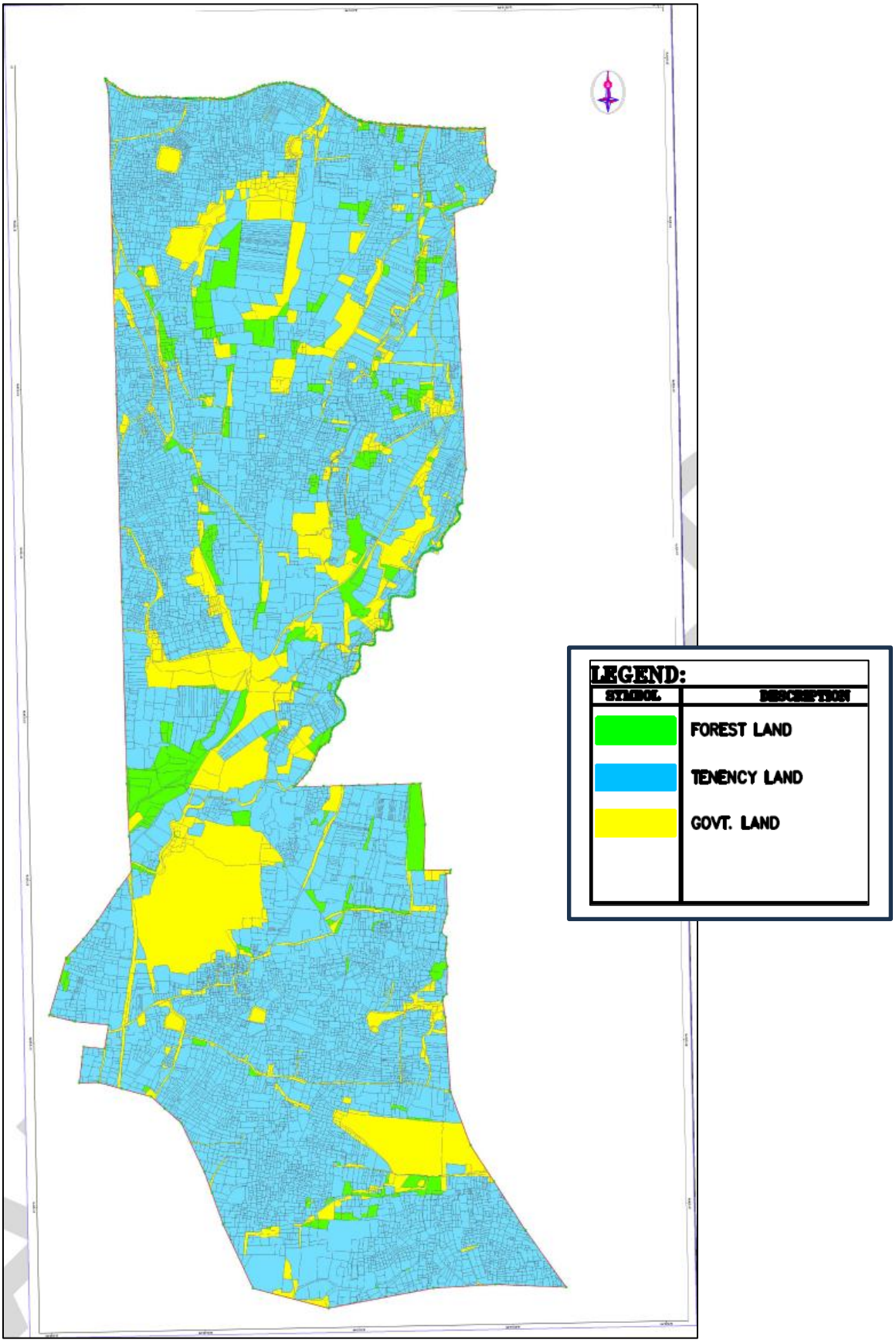


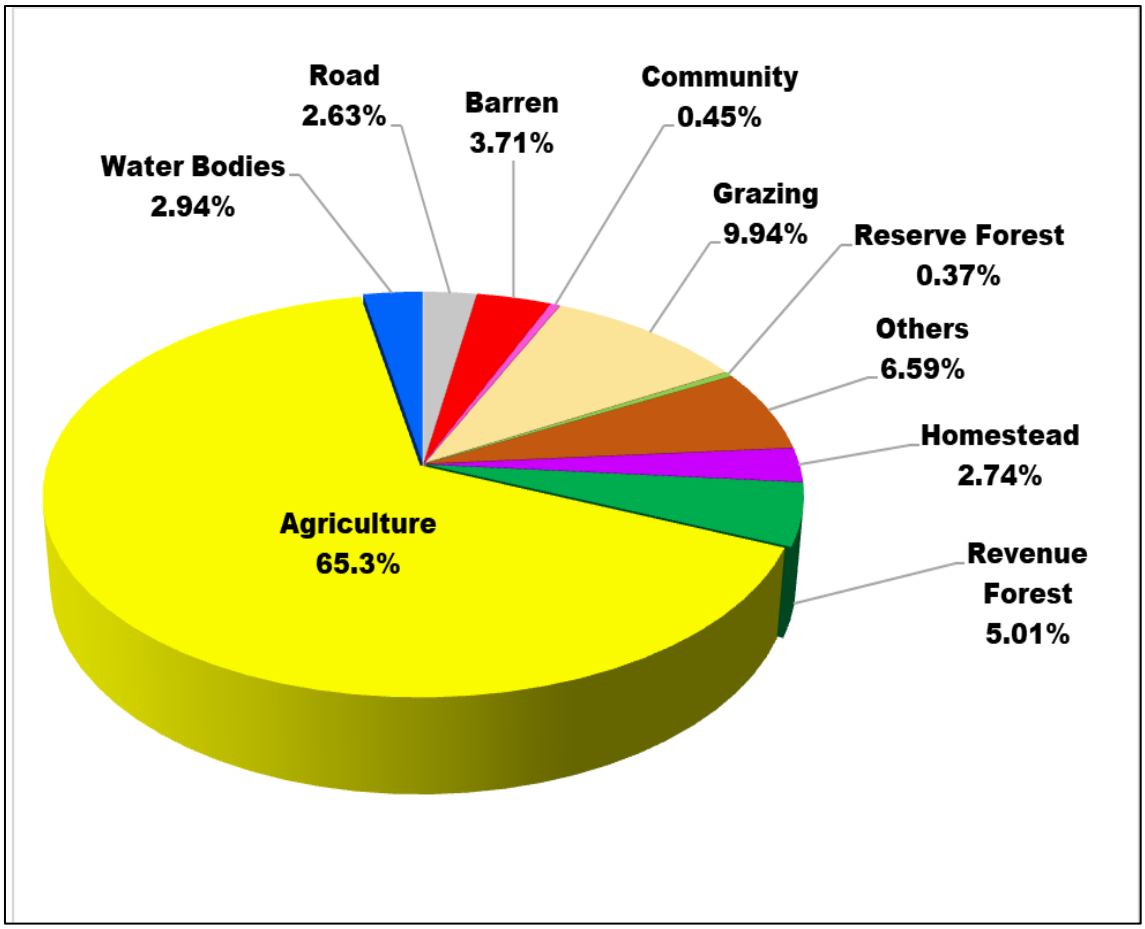
Figure 2-14 TYPE OF LAND

2.5.1 PRE-MINING LAND TYPE

The pre-mining Land type of above-mentioned area is given in the following table. The major percent of land occupied is agriculture land as depicted in the Table-2.5.

Table 2:5 Breakup of pre-mining land type (indicative)

S.No.	Land Type	Existing/pre-Mining Use	Area (ha)
1	Road	Existing/pre-Mining Use	31.4086
2	Barren	Existing/pre-Mining Use	44.3739
3	Community	Existing/pre-Mining Use	5.3926
4	Grazing	Existing/pre-Mining Use	118.8397
5	Others	Existing/pre-Mining Use	78.8479
6	Reserve Forest	Existing/pre-Mining Use	4.469
7	Revenue Forest	Existing/pre-Mining Use	59.9033
8	Homestead	Existing/pre-Mining Use	32.7729
9	Agricultural	Existing/pre-Mining Use	784.8499
10	Water Bodies	Existing/pre-Mining Use	35.1422
		TOTAL	**1196**



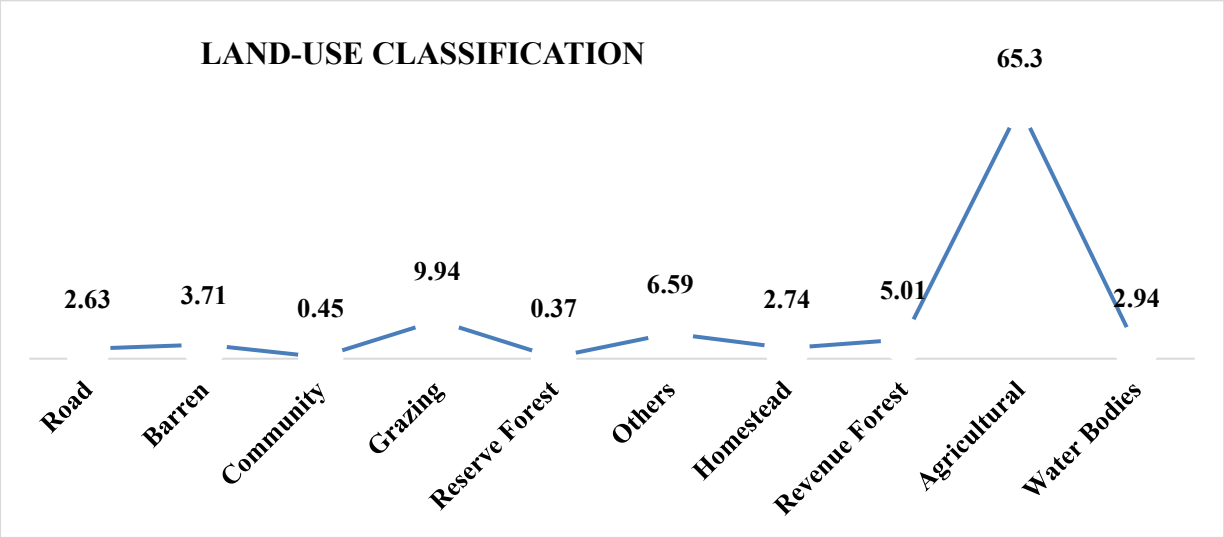


Figure 2-15 Pie Chart & Graph Showing the breakup of Pre-mining Land Type (in ha)

2.5.2 ENERGY

Power Supply

For the initial mine operation, 11 KV power supply shall be obtained through a link from the Handigoda rural line, within the mine lease boundary of the coal block. The proposed 33 KV transmission power line is still at the conceptual stage, as the alignment of the 132 KV line originating from the Boinda Substation under OPTCL is yet to be finalized.

Solar Power: Efforts shall be made to encourage the use of solar power. Around 20% of the total consumption is planned to be met from solar power.

Fuel consumption

The requirement of HSD will be 2.53 Litre/tonne of coal production and will be sourced from HPCL & IOCL.

Action for reduction of Diesel

The PP will use HSD with low sulfur, with the latest efficient machinery for low consumption of diesel.

However, efforts will be made to replace diesel-based machinery with CNG/LNG based machinery in future subject to their availability and techno economic feasibility for further reduction in diesel consumption.

The actual consumption along with the target for reduction will be submitted after Mine is in operation.

2.5.3 EMPLOYMENT GENERATION

1752 manpower will be deployed as direct employment. There will be around 5000 more indirect manpower shall be deployed for the related ancillary activities.

Breakup of manpower:

S. No.	Type of Employees	Manpower Required
1.	Contractual	1483
2.	Departmental	269
Total		**1752**

2.5.4 PROPOSED SCHEDULE FOR APPROVAL AND IMPLEMENTATION

Table 2:6 Proposed schedule for approval and implementation

Description	Status	Submission Of Application	Expected Approval	Expected Implementation By PP
Transfer Of TOR to GMDC	Application for transfer as Form -8 submitted on Parivesh	23.09.2023	06.11.2023 (Approved)	Final EIA/EMP prepared
Mining Lease	Land Rights have been obtained by Gujarat Mineral Development Corporation under Sec 11(1) of CBA & henceforth Sec 19 of CBA has been published.	After Receiving EC		Immediate
Mining Plan	Revised Mining Plan of Baitarni (W) Has Been Prepared and Approved by MoC Vide Letter No. SWCS/Baitarni West Coal Mine/OR/AN/2021/ CCO-MCBA Dt. 10-11-2021.	Mining Plan of Baitarni (W) Has Been Prepared and Approved by MoC	As Per the Vesting Order Approved Mining Plan Is Deemed to Be Transferred To M/S GMDC	Immediate
Forest Clearance	64.3723 ha of Forest land is	**Under process**	**Under process**	Immediate

	involved in the Project. Stage-I Forest Clearance for diversion for non-forestry activity has been applied online vide single window no-SW/171065/2024. Govt. of Odisha forwarded Forest diversion proposal to MoEF&CC vide letter no. FE-DIV-FLD-0005-2025-1477/FE&CC dated 15.01.25. IRO visit for Site & CA land is completed. EDS raised by MoEF&CC dated 26.06.25 and it is closed on 26.08.25. As on date EDS closed by DFO & forwarded to RCCF, Angul dated 17.09.25.			
Consent To Establish/Operate	This will be obtained from Odisha SPCB after getting EC.	After receiving EC		Immediate
Air And Water Consents	The Consent to Operate for discharge of effluents under water (Prevention and Control of Pollution) Act 1974 and emission under Air (Prevention and Control of Pollution) Act 1981 will be obtained from Odisha SPCB after getting EC.	After receiving EC		Immediately

2.6 PROJECT DESCRIPTION INCLUDING DRAWINGS SHOWING PROJECT LAYOUT, COMPONENTS OF PROJECT ETC.

The Salient feature of the project is given in Table-2.7.

Table 2:7 Salient Features of the project

A	**General Details**	
1	Name of the Company	**M/s Gujarat Mineral Development Corporation Ltd (**GMDC)
2	Project Location	Village-Chhendipada Jangal, Chhendipada RF Handigora, Machhakuta, Machhakuta Jangal, Porapara, Porapara Jangal, Tentulo - Korasahi, Tehsil- Chhendipada, District-Angul (Odisha)
3	Coordinates	Co-ordinates: NW: 21°05'26.87" N, 84°50'12.08" E NE: 21°05'17.37"N, 84°51'26.13"E SW: 21°01'44.84"N, 84°50'39.59"E SE: 21°01'44.74" N, 84°51'40.70" E
4	Category of project as per EIA Notification 2006	1 (a), Mining of Minerals "A".
5	Proponent Details	Registered Address: Gujarat Mineral Development Corporation Ltd (GMDC), Khanij Bhawan, 132 ft Ring Road, Vastrapur, Ahmedabad, Gujarat, 380052 Correspondence Address: Gujarat Mineral Development Corporation Ltd (GMDC), Khanij Bhawan, 132 ft Ring Road, Vastrapur, Ahmedabad, Gujarat, 380052 E-mail: gm.env.co@gmdcltd.com
6	Type of Project	1 (a), Mining of Minerals "A"
7	Production capacity	15 MTPA
B	**Mine details**	
1	Forest land	64.3723 ha
2	Non-forestland	222.975 ha
3	Tenancies	908.6527 ha
4	Area:	Block Area: 1196 ha Mining Lease Area: 1196 ha
5	Technology	Opencast mining is proposed by Surface miners (SM) -FEL-Truck for coal and drill blast for OB.
6	Area of the block	1196 ha
7	Gross geological Reserve	662.94 MT
8	Net Geological Reserve	602.1106 MT
9	Blocked Reserve	117.0413 MT

10	Mineable Reserve	485.06 MT
11	Extractable Reserve	468.27 MT
12	% of Extraction/ recovery	77.771%
13	Reserve depleted (Till the base date)	0 MT
14	Balance Extractable Reserve	468.27 MT
15	Average grade	G12 (3946CV)
16	Ultimate depth of Mining (m)	Maximum-245 Minimum-160
17	Bench width	24-36 m for overburden and 24-34 m for coal
18	Bench height	10-12 m
19	Mining technology	Opencast Mining Method of Coal winning by Surface Miner, pay loader& tipper and OB removed by Shovel-Dumper combination.
20	Total Life of mine	37 years including construction period
21	Period of mining lease	Land Rights have been obtained by Gujarat Mineral Development Corporation under Sec 11(1) of CBA & henceforth Sec 19 of CBA has been published.
22	R & R Involved	Total 7 villages namely Chhendipada Jangal (Part), Handigora (Part), Machhakuta (Part), Machhakuta Jangal (Part), Porapara (Part), Porapara Jangal (Full), Tentuloi-Korasahi (Part) are going to get affected by proposed project. Total Project Affected Families (PAF) is 2664 Nos and Project Displaced Families (PDF) is 1629 Nos. These displaced families will be resettled and rehabilitated socially, culturally and economically as per the latest R&R policy (2006 & subsequent amendments) of the Govt. of Odisha.
23	No. of affected families	2664 & 1629 no of PAFs & PDFs respectively due to proposed project.
C	**Details of Seams**	
1	No. of seams	18
2	Seams available as per GR	Seam-IX, Seam- I A TOP, Seam- I A BOTT, Seam- I B, Seam- I C, Seam- I D, Seam- IG, Seam- IE, Seam-IF, Seam-X, Seam-VI/VII/VII, Seam-IV/V, Seam-III

<table>
<tr><td></td><td></td><td>TOP(TOP), Seam-III TOP (BOTT), Seam-III TOP (T&B Comb.), Seam-III Middle, Seam- III BOTT (TOP), Seam-III BOT (BOT)</td></tr>
<tr><td>3</td><td>Seams not considered for Mining with Reasons</td><td>
<table>
<tr><th>S. No.</th><th>Seams</th><th>Reason</th></tr>
<tr><td>1</td><td>Seam- IG</td><td>Thin, unworkable in major area, occurs around 50m below seam ID</td></tr>
<tr><td>2</td><td>Seam- IE</td><td>Thin, unworkable, not considered in GR</td></tr>
<tr><td>3</td><td>Seam-IF</td><td>Thin, unworkable, not considered in GR</td></tr>
<tr><td>4</td><td>Seam-X</td><td>Incrops, Local development, Deteriorated. Not considered for reserve estimation.</td></tr>
</table>
</td></tr>
<tr><td>4</td><td>Thickness of seam</td><td>0.08 m to 33.94m</td></tr>
<tr><td>5</td><td>Grade of Coal</td><td>G-12</td></tr>
<tr><td>6</td><td>Stripping Ratio</td><td>1.5278 m^3/tonne</td></tr>
<tr><td>7</td><td>Average Gradient</td><td>2^O to 7°</td></tr>
<tr><td>D</td><td colspan="2">Details of Dumps</td></tr>
<tr><td>1</td><td>No of Dumps</td><td>1 External OB Dump of 215.84 ha with maximum total height of 90 m and quantity of OB 140.25 million m^3 and Internal OB Dump of 623.71 ha with quantity of OB 573.3 million m^3 is envisaged for backfilling.
By the end of mine life, 140.25 Mcum OB will remain in external dump, while the remaining 573.3 Mcum OB will be accommodated in internal dump. The top RL of internal dump will be 190m towards north and central part of the quarry and 220m towards southern area.
1 Topsoil dump of area 2.34 ha with quantity of topsoil 1.866 Mcum. However at the Final Stage, Topsoil shall be utilized for plantation in external dump 0.647 Mcum, internal dump 1.199 Mcum & embankment 0.021 Mcum.</td></tr>
<tr><td>2</td><td>Area of Dump</td><td>Internal dump = 623.71 ha
External Dump = 215.84 ha</td></tr>
<tr><td>3</td><td>Height of Dump</td><td>There will be mainly three tiers of OB dump, each with a maximum height of 30 m, with a maximum total height of 90 m.</td></tr>
</table>

		RL of top of the external dump at final shape will be 290 m.
E	**Environmental Sensitivity**	
1	Nearest District Headquarter	Angul 43 km from southeastern boundary in SE Direction
2	Nearest Airport	Biju Pattanaik International Airport, 131.7 kms in SE Direction
3	Nearest Railway station	Angul 42 kms from southeastern boundary in SE Direction
4	Nearest National/ State Highway	SH 63 which is 0.8 km in E Direction
5	Nearby Water Bodies	Rivers/Nalla: Kumbhira Nala flows from south-east (seasonal). Supai Nala flows (seasonal) in the core zone. Gauduni Nala – 1.2 km, E Kusal Nala – 1 km, W Bainsajaria Nala – 1.5 km, NW Shagarhia Nala – 3.0 km, NW Singhda Jhor – 5.0 km, S Aunli Nadi – 5.5 km, N Mandalia Nala – 9.5 km, W Nuhurang nadi -9 km, NW Tikra river- 9 km, NNE A small seasonal nala Supai, is flowing across the northern part of the block from south to north near north-eastern boundary of the block.
6	Nala Diversion	Supai stream is a seasonal stream, and no water is available December onwards. A detailed hydrological survey of the Supai Nala, regarding its catchment area, flow volume and length of the stretch to be diverted with water augmentation plan along with development of artificial pond has been done. The Nala shall be diverted in the 10th year of mine operation.
7	National Park/ Wildlife sanctuary/ Biosphere reserve/ Tiger Reserve/ migratory routes for Birds/Corridor	Following RF/PF are present within 10km study area as per SOI toposheet. i) Chhendipada RF – Core zone ii) Debinagar PF – 2.5 km, SW iii) Kamkurupal RF – 5.0 km, E iv) Kosala RF – 4.6 km, ES v) Simulipathar RF – 7.8 km, S vi) Brahmanabil RF- 8.8 km, E

		vii) Dangpal PF- 9.7 km, NW viii) Khalapal RF- 9.3 km, E ix) Bamur Forest- 9.3 km, W x) Katara RF – 14.3 km, SSW Letter from PCCF is attached as **Annexure XI.**
F	**Project Cost**	Total cost of the project is Rs. 3855.00 Cr. Cost of production is Rs.337.11 per tonne, Rs. CSR cost is approx. 1 Cr/Annum (2% of the net profit over the preceding three financial years of the organisation) which shall be used in sectors like healthcare, skill training, development of rural infrastructure etc. R&R cost is Rs. 727.94 crores.
G	**Manpower requirement**	1752 direct employment
H	**Water requirement**	3.62 MLD (potable 1.02 MLD, industrial water 2.6MLD). Initial requirement of potable water and industrial water shall be 0.26 MLD which will be met out from ground water. After wards the demand of water will be met from the mine inflow water.
I	**Water source**	Through borewell and mine seepage water.
J	**Power requirement**	For the initial mine operation, 11 KV power supply shall be obtained through a link from the Handigoda rural line, within the mine lease boundary of the coal block. The proposed 33 KV transmission power line is still at the conceptual stage, as the alignment of the 132 KV line originating from the Boinda Substation under OPTCL is yet to be finalized.
K	**EMP Cost**	Capital Cost: 10072.9 Lakhs Recurring Cost: 315.7 Lakhs
L	**CSR Cost**	Approx. Rs. 1 Cr/Annum (Minimum of 2% of the net profit over the preceding three financial years of the organisation) which shall be used in sectors like healthcare, skill training, development of rural infrastructure etc.

Source: Approved Mine Plan

2.7 GEOLOGY OF AREA

2.7.1 REGIONAL GEOLOGY

Regional geological setup of the area, local geology, structure, stratigraphic sequence, characteristics of the Lithological units (coal seams/ partings/ overburden).

Recent explorations in this region have extensively enhanced the geological knowledge of the coalfield, such as:

i. Discovery of coal bearing Barakar Formation in the extreme northwestern part of the basin.
ii. Re-demarcation of Barakar - Kamthi Formation boundary.
iii. Extension of Barakar Formation towards further west of Radhikapur block.
iv. Confirmation of northerly dip right up to the northern limit of the basin and
v. Confirmation of huge potential of power grade coal in the northern part of the basin

GSI indicated, vide Bulletin Series A, no.45, Coalfields of India, Volume II (1982), that the coalfield occupies an area of about 1860 sq km as per following break-ups:

i. Total area of coalfield: 1860 sq km
ii. Area covered by Kamthi Formation: 605 sq km
iii. Area covered by Lower Karharbari Formation (beyond Seam I incrop): 75 sq km.
iv. Area unknown or unlikely to be Potential: 180 sq km.
v. Potential coal bearing area (excluding Kamthis, Lower Karharbari Formation and area Unlikely to be potential): 1000 sq km.

Stratigraphical sequence in Talcher coalfield, as per GSI (1997), is given in the following **Table 2-8.**

Table 2:8 Stratigraphic succession of Talcher coalfield

Group	**Age**	**Formation**	**Thickness (m)**		**Lithology**
			rom	**To**	
	Quaternary	Quaternary deposits	-	40+	Colluvial fills, sand, silt and clay of older alluvium, older and younger floodplain deposits, channel fills etc.
	Cenozoic	Laterite	-	-	Laterites, lateritised detrital pebble bed.
GONDWANA SUPER GROUP	Upper Permian to Lower Triassic	Undifferentiated Kamthi Formation	-	575+	Fine to medium grained light grey to reddish sandstone and shale at the base and pale greenish sandstone with rare shale and pink clay bands, ferruginous coarse grained to pebbly sandstone at top.
---------- Unconformity ----------					
	Upper Permian	Barren Measures	-	50+	Greenish grey to buff coloured pebbly, coarse to medium grained highly

					ferruginous sandstone with variable proportions of fresh K-feldspar.

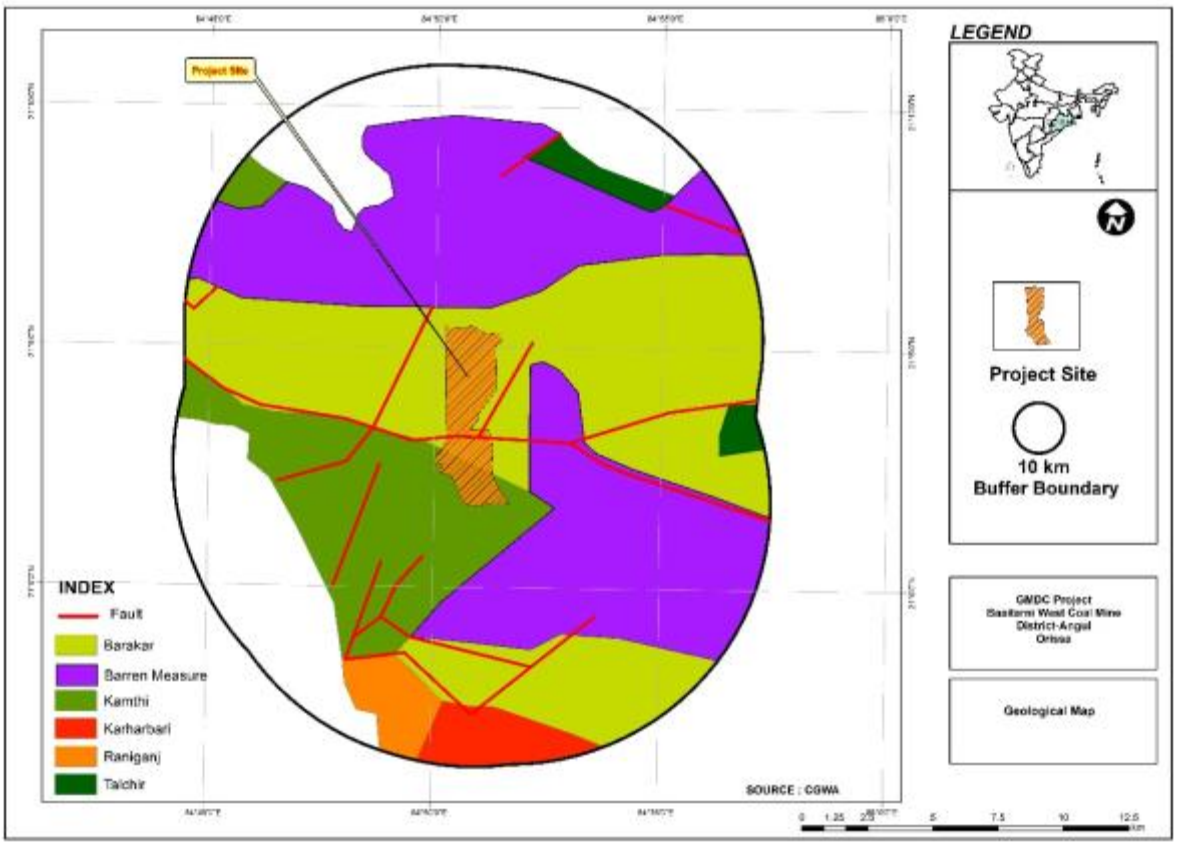


Figure 2-16 Geological Map of the study Area

2.7.2 LOCAL GEOLOGY, STRUCTURE, STRATIGRAPHIC SEQUENCE, CHARACTERISTICS OF THE LITHO-LOGICAL UNITS (COAL SEAMS /PARTINGS/OVERBURDEN)

Out of total 1196 ha, 672.147 ha is fully explored & details are available only for this area as described below:

The structure of Baitarani West block is highly disturbed in about 2 sq km area towards the north central part and extreme southern part of the block. A total of 17 faults have been delineated in Baitarni West block.

The stretch of formation in the block generally exhibits E-W strike which swings to NE-SW in the eastern part. The dip is generally towards north but is towards NW in the eastern part. The minimum and maximum dip amount ranges between >1degree to around 8 degree (in the NE and North-central part).

However, the general dip amount varies between >1 degree to 4 degree. Barring the north-eastern, central and towards the southern fringe where the dip amount

is relatively high (>4-degree -7 degree), rest of the major area has moderately low dip.

In this block out of 10 coal seams 20 nos. of correlatable coal horizons/splits i.e., from Seam-I to seam-X are identified. Seams VI/VII/VIII occurs as single seam throughout the block. This is the most important seam in this block which incrops repeatedly due to presence of number of strike faults and attains a combined thickness of about 26-30m. Seam-X is not considered potential as it mostly consists of carbonaceous shale and is impersistent in nature with localized developments. Seams IX and IV+V are comparatively thin seams. Lower seams do not have incrops. Seam-III is also a major seam of economic importance which occurs in 4-5 splits with considerable thickness. Seam-II group of coal horizons are impersistent in this block with localized developments and are not considered for reserve estimation in geological report. Whenever Seam X and sections of seam-II are found workable the same will be mined separately as coal. Seam-I group occurs in 8 correlatable splits. In the proposed mining plan, all the horizons of seam-I which are persistent throughout the block are considered for opencast mining due to better recovery on one hand and high cost and risks of underground mining on the other. Lower three splits of seam-I are mostly impersistent and develop locally with high incremental stripping ratio, so they are not considered for opencast mining.

Geological Block Area - 1196 ha -Resources and detailed geological information available in area of 672.147 ha. Remaining area is partly regionally explored. Block boundary as communicated by CMPDI HQ due to modification in bounding coordinate through Corrigendum issued by MoC vide 103/16/2015/NA dated 14.08.2020.

2.8 STATUS OF EXPLORATION OF THE BLOCK

Out of total 1196 ha, 672.147 ha is fully explored. A total of 131 boreholes have been drilled in the Baitarni West block which includes 4 boreholes drilled by GSI. It may be mentioned that out of total 127 boreholes drilled in the block, 15 CMBT boreholes involving 2901.70 m and 112 CMBI borehole involving 23136.95 m had been drilled. In addition, four GSI boreholes are also available in the block. In view of the nature of occurrence of coal seams, Baitarni West block has been assessed for both quarriable and underground potentiality for Barakar and Karharbari seams respectively after due consultation with the project planners. The Barakar seams from VIII to III in descending order are found favourable for quarriable prospect and Karharbari seam I splits are considered for underground exploitation.

The present Geological Report on Baitarni West Block incorporates the finding of 131 inside boreholes in conjunction with the information of the 14 adjacent boreholes drilled around the northern, eastern and western proximities of the block.

Table 2:9 Agency-wise boreholes drilled in Baitarni West block, Talcher coalfield

Sl No	Agency	Period	Drilling		Type of Exploration	Status of documentation
			No. of Borehole (series)	Meterage		
1	GSI	March'91 toAug'93	4 (TCP)	1403.40	Regional exploration	Report on Chhendipada (West)-Patrapara block Talcher CF, Orissa-1998
2	CMPDI	July'93 Dec'93 (Ist phase)	15 (CMBT)	2901.70	Scout drilling in north-central partof block	
3	CMPDI	July'98 to Jan'02 (II phase)	112	23136.95	Detailed exploration	
Sub Total (CMPDI)			**127**	**26038.65**		**Present Report**
Total (GSI & CMPDI)			**131**	**27442.05**		

Source: Approved Mine Plan and Mine Closure Plan

A total of 131 boreholes have been drilled in an area of 6.44 sq. km [**6.72Sq km** (as per Geo-referencing & conversion to WGS-84)] which works out thedensity to be 20 boreholes per sq. km.

Table 2:10 Potential seam wise number of boreholes intersection area and borehole density

Sl. No.	Name of coal seam	Details of intersection				Area of occurrence (sq km)	Borehole density per sq km
		Full seam	Incrop	Parting	Total		
1	IX	70	4	5	79	3.5	23
2	VI/VII/VII	86	30	8	124	6.04	21
3	IV/V	107	1	6	114	6.15	19
4	III TOP (TOP)	27	-	-	27	1.21	22
5	III TOP (BOTT)	27	-	-	27	0.79	34
6	III TOP (T&B Comb.)	92	-	7	99	4.92	20

7	III Middle	119	-	7	126	6.15	20
8	III BOTT (TOP)	118	-	8	124	6.15	20
9	III BOT (BOT)	118	-	8	126	6.15	20
10	II A TOP	92	-	4	96	6.15	16
11	I A BOTT	92	-	4	96	6.15	16
12	I B	90	-	4	94	6.15	15
13	I C	92	-	1	93	6.15	15
14	I D	88	-	-	88	6.15	14
15	I G	49	-	-	49	6.15	8

Source: Approved Mine Plan and Mine Closure Plan

2.8.1 FURTHER EXPLORATION

Coal in the southern part of the block (523.853 ha) beyond detailed explored area cannot be extracted through OC Method so soon as per Office Memorandum of Ministry of Coal dated 14.03.2018 (F.NO 110/08/2017/NA, O/o Nominated authority, Ministry of Coal, sub-Summary of 2nd meeting of monitoring committee held on 27.02.2018).

The southern part of the block where external dump and infrastructure are located is regionally explored. Boreholes present in this area indicate that coal is much deep-seated (more than 300 m). So, in the near future, this area may not be amenable to profitable opencast mining considering high initial depth, stripping ratio as per report submitted to MoC by CMPDI.

During the regional exploration in the southern part, it is seen that the top-most coal seam in this area occurs at a depth of 450-460 mtr from the surface. Opencast mining of such high- ash, thermal grade coal from such higher depths is not practiced now as this is not amenable to profitable opencast mining. In the event of any change in the economic scenario to mine the coal becomes viable, then, the external dump can be re-handled as per condition of the approved mining plan. Although extraction of such high-ash, thermal grade coal from such higher depths is not techno-economically feasible, still if this area contains mineable coal reserves and if any party is allotted this area for mining eventually, then PP would need to re-handle/remove the external O.B. dump located in this south part. This was one of the conditions in the approval of the earlier Mining Plan. At present, there is no available technology (opencast or underground) to extract such high-ash, non-coking coal from such higher depths which can be adopted by PP for winning coal in this area economically.

In compliance with the additional clarification sought by the Internal Committee of MoC, GoI in its meeting dt. 09.08.2021 for approval of Mining Plan (including

Mine Closure Plan) of Baitarni-West Coal Mine, a scheme has been prepared for exploration of the south-side of Kumbhira Nala to rule out the presence of economically mineable coal seams before starting dumping of overburden/ construction of infrastructure.

In future 7 boreholes will be explored with 3575 meterage. Overall borehole density within the block (no. / Sq. km) approx. is 20 (within fully explored area).

2.9 COAL RESOURCE / RESERVE ESTIMATION

As per Coal Resource / Reserve Estimation, procedure & practice, May 1989, the reserve of coal is classified into 'Proved', 'Indicated & 'Inferred' categories.

The reserves within an area falling within a radius of 200 meters around boreholes area categorized as 'Proved'.The 'Indicated' reserves are calculated within a radius of 1000 meters around boreholes. The 'Inferred' reserves are calculated beyond 1000 meters and within 2000 meters around boreholes.

Gross Geological Reserve of theblock "MT"	662.94
Net Geological Reserve of theblock "MT"	602.1106
Minable Reserve of the block "MT"	485.06
Blocked Reserve "MT"	117.0413
Corresponding extractable reserveof the block "MT"	468.27
Percentage of Extraction	77.771
Reserve already depleted (Basedate of Mining Plan)	0
Balance Reserve (as on Base Date)	468.2700

2.9.1 LIFE OF THE MINE

As per the proposed production of 15 MTPA the overall life of the mine shall be 37 years.

2.9.2 GRADE OF COAL

The average grade of coal has been considered as G-12.

Table 2:11 Heavy Metal Analysis in Coal Samples of Hingula OCP

SI. No	Test Parameters	Unit of measurement	Test Method	Specifications	Results
1	Cu	mg/kg	USEPA 6010 D	NA	54
2	Cr	mg/kg	USEPA 6010 D	NA	59
3	Co	mg/kg	USEPA 6010 D	NA	10
4	Zn	mg/kg	USEPA 6010 D	NA	83
5	V	mg/kg	USEPA 6010 D	NA	121
6	Ni	mg/kg	USEPA 6010 D	NA	19

7	Pb	mg/kg	USEPA 6010 D	NA	9
8	Fe	%	USEPA 6010 D	NA	0.95
9	As	mg/kg	USEPA 6010 D	NA	BLQ (LOQ = 2.5)
10	Cd	mg/kg	USEPA 6010 D	NA	BLQ (LOQ = 2.5)
11	Hg	mg/kg	USEPA 200.8	NA	BLQ (LOQ = 2.5)

2.10 TECHNOLOGY AND PROCESS DESCRIPTION

2.10.1 PROPOSED METHOD OF MINING

Thick seams occur at shallow depth in wide areas having power grade coal reserve. It can be best extracted by opencast mining method deploying surface miners, mechanical shovels of high capacity and dumpers of matching sizes.

2.10.2 SELECTION OF TECHNOLOGY

The shovel-dumper system is very flexible and also offers convenient mining operations to deal with sudden occurrences of unworkable or poor-quality patches and change of floor position due to repeated faulting. It also offers flexibility for easy transition to any other technology or equipment configuration. The technology is well known and advantageous to get skilled manpower.

Surface-miner excavation is still limited to winning coal only. There are many advantages in this technology, of avoiding drilling, blasting, crushing of coal and related environmental hazards, improvement in grade of ROM coal by removing thin bands, clean surface for transport etc.

But shovel-dumper system has to continue along with for excavation of overburden. Ideally, surface miners require working space of about 400 m length and about 50 m width for its optimum use. The high-wall angle required is comparatively flatter to shovel-dumper system. This bench geometry makes the overall working angle flatter which will generate more overburden compared to shovel-dumper system if the seams are thin with thick intermediate parting. But as seam VI/VII/VIII is combined thick seam with number of bands and occurs at a comparatively shallow depth, surface miner will be introduced in this seam and some other sections of moderately thick sections of seam-III. Other thin seams and lower seams can also be worked by surface miner system depending on geological conditions & machine parameters.

As the seam gradient is relatively flat (around 2 degrees to 7 degree), coal and the parting will be worked by parallel slicing method (working along seam floor) whereas top overburden will be removed by level slicing method.

2.10.3 BENCH PARAMETERS

Benches will be aligned along general strike. Bench floor should follow own seam floor/roof or that of adjacent seam. Main bench parameters for above mentioned equipment are:

Maximum bench height : 10-12 m for 10-12 cum hydraulic face shovel
Bench width: 24-36 m for overburden and 24-34 m for coal
Working angle :70 degree with horizontal for individual working bench
Again, bench dimensions may vary with different equipment deployed.

2.10.4 MINE LAYOUT

Total available area within quarry limits as described in earlier section will be mined. Due to closely spaced repeated faulting, seams VI/VII/VIII occur at shallow depth at south-eastern part towards north of F5-F5 and again at north-eastern part towards north of F15-F15. In year 3 the initial access trench will be driven from south-eastern corner of the block to touch the floor of thick upper seam VI/VII/VIII towards north of fault F5-F5. In year5 the main access trench will be started near extreme southwestern corner of the block which will be prepared to reach seam III BOT(Bot) and later seam ID at the same place where the mining is started in year 3. Mining activity initially will be advanced towards north-west and then in year 5 mining will also be advanced towards sectors south of F5-F5. Mine will be advanced towards west, completing the southern half of the block. In later years, the northern half will be excavated till the end of mine life.

An external dump is proposed towards south of the quarry. CHP, Workshop and RLS are proposed towards south of quarry mouth but beyond 300 m blasting safety zone.

For approach to the property, one road will be constructed outside the ML area, from south-eastern side corner of the ML area to connect to the existing Jarpara-Chhendipada MDR (Major District Road).

The alignment of the Western Wing of the proposed Common Rail Corridor for Talcher Coalfield passes on the south side of the ML area of Baitarni-West coal mine. For rail evacuation of coal from Baitarni-West coal mine, the project proponent will have its own private siding on the south side of the ML area. This private siding will be connected, through the Common Rail Corridor, to the East-Coast Railway Line at Jarpara Railway Station.

2.10.5 ACCESS TRENCH

A long access trench of around 2.6km is required to reach the floor of the bottommost coal horizon which lies almost at a depth of 160m in this block. So, it is not possible to reach the bottommost seam in the initial years and hence initial production will be met from thick upper seams with simultaneous deepening of access trench up to lower-most seam.

2.10.6 DRILLING AND BLASTING

About 93 % of coal will be produced through surface-miner and balance 7% of coal will be produced through drilling-blasting.

Surface-miner excavation is still limited to winning coal only. There are many advantages in this technology, of avoiding drilling, blasting, crushing of coal and related environmental hazards, improvement in grade of ROM coal by removing thin bands, clean surface for transport etc.

Moderately thick to thick horizons of overburden are proposed to be excavated by 10-12 cum hydraulic shovels/ backhoe in combination with 100 T dumpers after proper drilling and blasting in overburden. In later years higher size of hydraulic shovel-dumpers (20 cum hydraulic shovel with 190t dumper) may be introduced in thick parting between seam-I & III. The proposed sizes and combination will reduce congestion for such high-capacity mining activity and increase the overall working angle for lesser external dump.

Two types of drills have been proposed. 250 mm drills will be deployed for benches in thicker parting and top overburden, 160 mm drills will be deployed for thinner parting horizons.

2.10.7 MINE DEVELOPMENT

Stage of land acquisition for first 5 years will be 347.09 ha. The details of acquisition for private land & govt. land is attached as **Annexure XV**.

The first two years **(Year1 & Year2),** after receipt of the statutory clearances, are kept for land acquisition and establishment of infrastructure. Removal of overburden and coal extraction will commence from **Year 3**. In this year, a temporary access trench will be driven to touch the in-crop of thick upper seam VI/VII/VIII. Overburden generated will be transported to area identified for external dump located south of Kumbhira nala at a distance of 550 m from quarry mouth. Width of mine, after Year 3, is about 600 m, with a dip-rise length of about 250 m.

In **Year 4,** the same access trench will be used to transport coal from seam VI/VII/VIII. Working will be advanced towards north-west with increasing width of working up to 900m. All overburden is taken to external dump.

During **Year 5**, the main access trench will be started from south-west corner of the block; the length of this access trench will be 900 m at the end of this year with bottom RL at 150m. But coal extraction this year will still be limited to the working of seam VI/VII/VIII and the initial access trench will be used to transport the coal to the surface. Some quantity of overburden towards south of fault F5-F5 up to main access trench will be removed in this year. All overburden will be taken to external dump.

During **Year 6**, the main access trench will be connected to the initial access trench and will be deepened to touch the floor of seam III Top (Top &Bot.) in the initial working area. Some of the coal from seam VI/VII/VIII in the down-thrown area towards south of F5-F5 will also be extracted. Ultimate mine floor is not reached as yet. So, no internal dumping is possible. Overburden removal towards both north-west and towards south will be continued.

During **Year 7**, the main access trench reaches mine floor at seam III Bot (Bot) on north of fault of F5-F5. This is the bottommost floor of seam-III group. Mine-depth in this year, will be around 105m. In this year, mine will be advanced towards both north-west and south side. From this year, coal production will be possible from all the seams up to bottom section of seam-III. Seam VI/VII/VIII will be worked in the full strike-length of the block of around 1800m towards north of F5-F5. Coal extraction from seam VI/VII/VIII will be started towards western portion, between fault F4-F4 and F5-F5. Ultimate mine floor is not reached as yet, as there is a thick parting between seam-I and seam-III. So, no internal dumping is possible. At the end of this year, the mine will reach its target production level of 15 Mt per annum.

Mining up to bottom section of seam-III will be continued for another year. In this year the working of upper benches is advanced further towards north-east adjacent to Kumbhira nala. This advance in the north-east provides enough space in the eastern batter to construct a road in the batter to touch the lowermost workable seam-ID floor. This road will be started from +60m level of main access trench in eastern slope of the mine to +5m level where floor of seam I-D is found at a minimum depth of 185m. After reaching the floor of the bottommost seam-I D at the same place where mining of seam VI/VII/VIII is scheduled to start at higher level), benches will be advanced towards north-west.

Internal dumping will be started in 11th year when enough space is generated towards north of F5-F5. During this period, coal extraction from sections of seam-III will be advanced further south of F5-F5 and F4-F4.

When coal benches of seam ID are sufficiently advanced towards north of F5-F5, all the coal up to seam-ID, will be extracted towards south of F5-F5 up to southernmost boundary fault. After exhaustion of coal in this area, all the area beyond south of F5-F5 will be used for internal dumping.
In general, the surface level of eastern part is lower; so, sump can be formed towards eastern part when large area of quarry is opened-up. The lowest bench will be aligned at 1 in 200 gradients, dipping towards the sump to avoid water logging. Change in strike direction is observed between lower and upper seams. Such change is also observed between either side of a fault. As such, all benches cannot be laid parallel.

2.10.8 TRANSPORT LAYOUT

The haul-road laid on floor of quarry will be used for transporting coal from seam-I (all splits). Other major uses are transport, installation and maintenance of main pumps at sump. To facilitate transport of coal from all other seams and overburden to dumps, flank roads are to be developed on all sides of quarry slope. Horizontal flank roads are to be developed at 30 m vertical intervals matching with levels of internal dumps. Main access trench and haul road from surface to floor of lower most seam will be maintained in the eastern slope of the quarry. This haul road will touch the floor of seam I-D at a depth of 160m near borehole CMBI-035 towards north of F15-F15. Another branch from this road is constructed from +60m level to floor of seam ID at +5m level towards north of F5-F5. Coal from all other seams will be transported along benches to nearest level of flank roads and by travelling through horizontal or inclined roads, will join in main access trench to come out to surface.

2.10.9 PRODUCTION PLAN

Calendar Plan/Production Plan for the entire life of the mine is given in the following table 2.12.

Table 2:12 Tentative Coal production Plan

Year of Operation	Coal Production Schedule (MT)			OB (Mcum)	SR
	UG	OC	Total		
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0.9	0.9	3.55	3.9444
4	0	3	3	4.54	1.5133
5	0	6	6	11.22	1.87
6	0	10.5	10.5	15.99	1.5229
7	0	15	15	15.84	1.056
8	0	15	15	16.07	1.0713
9	0	15	15	18.2	1.2133
10	0	15	15	18.2	1.2133
11	0	15	15	20.88	1.392
12	0	15	15	24.75	1.65
13	0	15	15	24.75	1.65
14	0	15	15	24.75	1.65
15	0	15	15	24.75	1.65
16	0	15	15	24.75	1.65
17	0	15	15	24.75	1.65
18	0	15	15	24.75	1.65
19	0	15	15	24.75	1.65
20	0	15	15	24.75	1.65
21	0	15	15	24.75	1.65
22	0	15	15	24.75	1.65

23	0	15	15	24.75	1.65
24	0	15	15	24.75	1.65
25	0	15	15	24.75	1.65
26	0	15	15	22.4	1.4933
27	0	15	15	22.4	1.4933
28	0	15	15	22.4	1.4933
29	0	15	15	22.4	1.4933
30	0	15	15	22.4	1.4933
31	0	15	15	22.4	1.4933
32	0	15	15	22.4	1.4933
33	0	15	15	22.4	1.4933
34	0	15	15	22.4	1.4933
35	0	12	12	22.4	1.8667
36	0	9	9	11.54	1.2822
37	0	6.87	6.87	8.87	1.2911

Source: Approved Mine Plan and Mine Closure Plan

Boreholes present in this proposed external dump area towards south indicate that coal is much deep-seated (more than 300 m). So, in the near future, this area is not considered suitable for economic opencast mining, with present available technology. The area for external dump was unblocked by CMPDI as the coal in this area cannot be extracted through OC Method so soon as per Office Memorandum of Ministry of Coal dated 14.03.2018 (F NO 110/08/2017/NA, O/o Nominated authority, Ministry of Coal, Sub-Summary of 2nd meeting of monitoring committee held on 27.02.2018).

Area was regionally explored coal is much deep-seated at depth of more than 300m this area cannot be extracted through OC Method so soon as per earlier report submitted by CMPDI to MoC subsequent Office Memorandum of Ministry of Coal dated 14.03.2018 (F NO 110/08/2017/NA, O/o Nominated authority, Ministry Of coal, sub-Summary of 2nd meeting of monitoring committee held on 27.02.2018).

2.10.10 EQUIPMENT/MACHINERIES

The fleet of equipment/machineries required to achieve a production capacity of 15 MTPA are given in the following table 2.13.

Table 2:13 Type of Equipment/ HEMM proposed

S.No.	Type of Equipment	Capacity	Unit	Population
1.	Hydraulic Shovel	12	Cubic Meter	7
2.	Rear Dumper	100	tonnes	75
3.	Rotary Drill (250mm)	250	mm	7

4.	Rotary Drill (150-170 mm)	170	mm	4
5.	Ripper Dozer (850HP)	850	Horsepower (HP)	3
6.	Dozer (410 HP)	410	Horsepower (HP)	10
7.	Surface Miner (3000-3800 mm)	3800.000	mm	5
8.	Front End Loader	13	Cubic Meter	3
9.	Rear Dumper(Coal body)	100	tonnes	28
10.	Wheel Dozer (400HP)	400	Horsepower (HP)	2
11.	Telescopic Hyd.Crane	70	tonnes	1
12.	Telescopic Hyd.Crane	30	tonnes	1
13.	Service Crane	8	tonnes	4
14.	Diesel Hyd. Backhoe (0.9-1.2Cum)	1.2	Cubic Meter	2
15.	Front End Loader	3.5	Cubic Meter	4
16.	Tipping Truck	8	tonnes	16
17.	Grader (280 HP)	280	Horsepower (HP)	2
18.	Water Sprinkler (28 KL)	28	KL	6
19.	Vibratory Compactor	0	no	2
20.	Fire Tender	0	no	1
21.	Tyre Handler	0	no	2
22.	Fuel Bowser	0	no	2
23.	Maintenance Van	0	no	2

Source: Approved Mine Plan and Mine Closure Plan

2.10.11 Waste Generation and Management

As discussed above no internal backfilling is possible up to year 10 as mine just reaches the ultimate quarry floor with a nominal area at bottom. As the quarry expands towards the south and north-west, some backfilling can be started but a major part of the waste removed will have to be taken to external dump. From Year14 onwards, possiblyno external dumping is required as total area south of F5-F5 will be available for backfilling.

The RL of top of the external dump at final shape will be 290 m. There will be mainly three tiers of OB dump, each with a maximum height of 30 m, with a maximum total height of 90 m. Another tier of 10m will be constructed on the top inthe southern part of external dump. Active dump slope of Individual tiers will be maintained at 37 degrees, resulting in an overall slope of around 26 degrees with horizontal.

By the end of mine life, 140.25 Mcum OB will remain in external dump, while the remaining 573.3 Mcum OB will be accommodated in internal dump.

Table 2:14 Waste Management (Figures in Million m^3) (Tentative)

Year/Stage (Life of the mine plus post closure period)	OB Removal			External Dump		Internal Backfilling		Embankment	
	Cumulative			Cumulative		Cumulative		Cumulative	
	Topsoil	OB	Total	Topsoil	OB	Topsoil	OB	Topsoil	OB
Up to Base year	0	0	0	0	0	0	0	0	0
Y-1	0	0	0	0	0	0	0	0	0
Y-3	0.054	3.496	3.55	0	3.55	0	0	0.018	0
Y-5	0.228	19.082	19.31	0	19.31	0	0	0.021	0
Y-10	0.612	103	103.61	0	103.61	0	0	0.021	0
Y-15	0.947	222.54	223.49	0.496	140.4	0	82.59	0.021	0
Y-20	1.156	346.08	347.24	0.647	140.25	0	206.34	0.021	0
Y-25	1.365	469.63	470.99	0.647	140.25	0.51	329.58	0.021	0
Y-30	1.573	581.42	582.99	0.647	140.25	0.835	441.26	0.021	0
Y-35	1.782	693.21	694.99	0.647	140.25	1.115	552.98	0.021	0
Y-37	1.866	713.53	715.4	0.647	140.25	1.199	573.3	0.021	0
Post Closure									
Y-40	1.866	713.53	715.4	0.647	140.25	1.199	573.3	0.021	0

Source: Approved Mine Plan and Mine Closure Plan

Solid Waste (Non-hazardous Waste) from Mine:

S. No.	Description	Occupancy	Norms (kg/capita/day)	Waste Generated (kg/day)
1.	**Domestic Solid Waste**			
	Staff	1752	0.123*	215.49

	Visitors	175	0.15	26.25
2.	Horticultural Waste (213.74 acre)		@ 0.2 kg/acre/day	42.74
3.	STP Sludge		Wastewater x 0 .35 x B.O.D difference/1000	13.65
Total Solid Waste Generation = 298.13 kg/day or 108.82 TPA				

*CPCB Annual Report 2021-22

Plastic waste:

Name of the waste	Source	Qty (TPA)	Mode of disposal	Mode of transport
Plastic Waste	Administrative office & service building (packaging, water bottles, food wrappers, PPE, Industrial packaging etc.)	10	Sale to Authorized recycler	Road

E-waste:

Name of the waste	Source	Qty (TPA)	Mode of disposal	Mode of transport
E-waste	Administrative office & service building (computers, phones, printers, control systems, sensors, PLCs, etc)	1.0	Sale to Authorized recycler	Road

Batteries waste:

Name of the waste	Source	Qty (TPA)	Mode of disposal	Mode of transport
Batteries	Equipment, vehicles, personal use, etc	15	Authorized Vendors	Road

Hazardous waste:

Name of the waste	Source	Qty (TPA)	Mode of disposal	Mode of transport
Used Oil / Spent Oil	Mine Machinery	150	Through CPCB /SPCB registered recyclers	Road

2.10.12 TOPSOIL GENERATION AND MANAGEMENT

Table 2:15 Topsoil Management (Figures in Million m^3) – (Tentative)

Year/Stage (Life of the mine plus post closure period)	Topsoil Removal Plan	Topsoil Used				
		Spreading Over Embankment	Spreading Over Backfill area	Spreading Over External OB Dump area	Used in Green Belt area	Total Utilized
Up to Base year	0	0	0	0	0	0
Y-1	0	0	0	0	0	0
Y-3	0.054	0.018	0	0	0	0.018
Y-5	0.228	0.021	0	0	0	0.021
Y-10	0.612	0.021	0	0	0	0.021
Y-15	0.947	0.021	0	0.496	0	0.517
Y-20	1.156	0.021	0	0.647	0	0.667
Y-25	1.365	0.021	0.51	0.647	0	1.178
Y-30	1.573	0.021	0.835	0.647	0	1.502
Y-35	1.782	0.021	1.115	0.647	0	1.782
Y-37	1.866	0.021	1.199	0.647	0	1.866
Post Closure						
Y-40	1.866	0.021	1.199	0.647	0	1.866

Source: Approved Mine Plan and Mine Closure Plan

2.10.13 COAL HANDLING ARRANGEMENT

Mine capacity -15.00 MTPA (2400 tph/3600 tph)
Capacity for Road sale - 3.00 MTPA (900 tph, Peak)
Transportation through rail: 12 MTPA
Nature of coal - Blast free /Drilling blasting (approx. 7 percent)
Effective Working hours – 5/6 hours per shift
Effective Working days- 330 days

2.10.13.1 COAL RECEIPT ARRANGEMENT

For receiving and crushing oversized coal, 12 Numbers (10 Working ,2Standby) of (0-500) tph skid-mounted Impactors with,100 m^3 structural hopper, vibratory grizzly and feeder-belts in a modular unit with skid mounted is provided in two rows at Truck receiving and crushing complex. Both blast-free coal and coal produced through conventional drilling-blasting is brought by trippers/dump trucks from the quarry, shall be unloaded into these structural hoppers. Vibratory grizzly which will directly feed (-)100 mm size coal on to Feeder belts. Coal size more than (-) 100 mm will be crushed in this integral unit and delivers on to the feeder-belts. These feeder-belts will deliver entire coal of (-)100mm sized into conveyor C1A/C1B.This integral unit shall ensure delivery of coal of assured size i.e. (-)100mm for customer.

2.10.13.2 COAL CONVEYING AND STORAGE ARRANGEMENT

Coal will be conveyed up to transfer-house, TH2 at Ground Bunker through series of belt conveyors C1 A/B and C2A/B. At this transfer-house, coal will be discharged on C3A/C3B and Link belt Conveyors feeding to Mechanized Truck Loading Station for road-sale, located parallel to this transfer-house. Coal will be diverted at the rate of 450 tph from this stream with the help of Motorised flow dividers. Two numbers of Truck Loading Station with 150 Tonne surge hopper and 450 tph evacuation rate will be utilised for Road-sale of coal for local customers. Coal will be discharged uniformly in to double-slit ground bunker with the help of tripper-conveyor C3A/C3B. Coal will be sprayed in this 25,000 tph Overhead Ground bunkers for the storage and further dispatch for rail loading. Conveyors up to ground bunkers will be of 1600 mm, 2400 TPH. Link belt conveyors will be 1000 mm, 450 tph. After Ground Bunkers, belt will be of 1800 mm,3600 tph for loading to Surge bins with Rapid Loading System.

2.10.13.3 EVACUATION ARRANGEMENT

Coal will be reclaimed on to 1800 mm, 3600 tph belt conveyor C4A/C4B with the help of four numbers (2 Working,2 Standby) of 3600 tph Plough feeders. From Ground bunker coal will be conveyed, elevated and finally discharges either into 1000 Tonne RCC surge bin-1 or Bridge conveyor C5A/C5B for further discharge into another surge bin-2. Simultaneously, from both the Surge bins, Coal will be

loaded in to wagons with the help of 100 Tonne pre-weigh bin with around 6000 tph loading rate to wagons.

Loading points are located on loop line of proposed Western wing of the Common Rail Corridor.

2.10.13.4 AUXILIARY ARRANGEMENTS

This mine will have dust suppression systems (HPFDS, PWDSS, DS/DE), Fire detection and fighting arrangements, Mine cleaning system and drinking water facilities. Entire plant will be integrated for operation with PLC. Required illumination will be provided all along the mine and strategic points. Two numbers of CHP substation will be utilised for supply of power to all LT and HT loads will be tapped directly from Overhead line.

2.10.13.5 IN-PIT CONVEYING

Mine will reach to its bottom most seam (seam ID) in year 8 or year 9. Access trench and haul road can be permanent only after reaching bottommost seam. In-pit conveying system with reclaim feeder/ truck receiving hopper or any suitable arrangement can be installed after 10th year only. However, installation of such system may reduce internal dumping space.

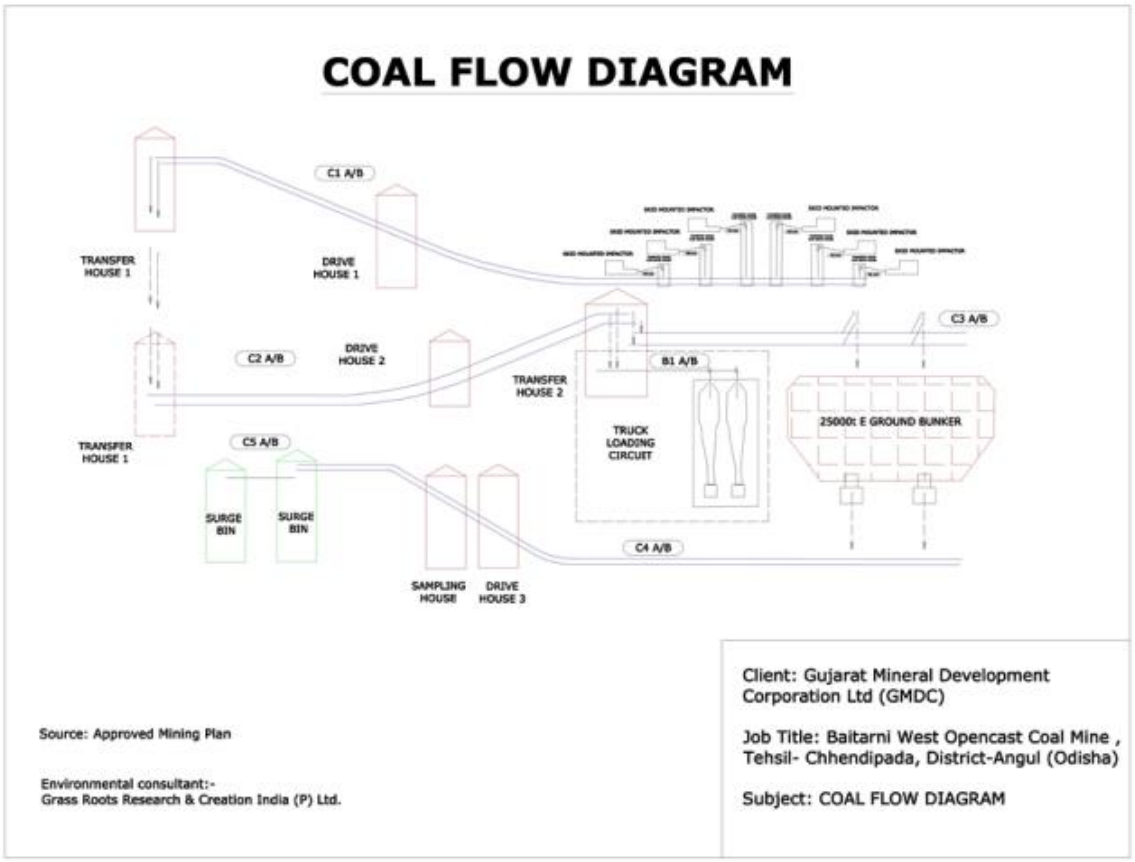


Figure 2-17 Coal Flow Diagram

2.10.14 SITE SERVICES

Service Centre

There will be Service Centers near mines formaintenance, servicing and repair of all mining machinery. Skilled manpower will be available at each Service Centre for repair & maintenance of heavy and light mobile machines at their respective sites.

Canteen

Canteens will be provided separately for each pit and main administrative building. Also, Mobile Canteen Vans will carry and supply refreshments to the employees at their work sites.

Ambulance & Fire Station

Adequate medical facilities including the Occupational Health Centre will be available in the hospital maintained by PP.In case of need, the patient may be referred to super specialty hospitals in citiesall over India with whom GMDC has tie-up. First Aid Rooms will be maintained at each important work site.

One dedicated ambulance and fire brigade will be stationed in the designated place constructed for the purpose to meet the emergency situation in the mines.

Rest shelter, First-Aid station and Latrines

One first aid station will be setup near by the service Centre, field office and stores. A male nurse will be posted dedicatedly to render the first aid as and when required. Sufficient no. of toilets/Latrines will be provided and maintained in the various offices viz. Mining Office, Field Office, Service Centre, and Crushing Plant Office.

Office building

The main administrative offices will be provided with P & T telephone, telefax and wireless communication facilities. Site offices will also be maintained at the Service Centers, crushing & screening plant sites.

Mine Vocational Training Centre

Mine Vocational Training Centres will be provided for imparting training on various aspects of safety, skill development and first aid and for imparting professional training to the regular employees, contractual workers, Apprenticeship training and Vocational training to graduates and diploma holders from various universities / institutions in the field of mining, geology, mechanical, electrical etc. Employees will also be deputed for outside training in HEMM from Original Equipment Manufacturers.

Geological Sampling Laboratory/Chemical laboratory

A geological sampling laboratory will be made for sampling of coal during production and dispatch from the project. This will be equipped with Jaw Crusher,

pulverizer, sieve shakers and other sampling equipment for processing and preparation of various types of samples.

Chemical laboratories will carry out chemical analysis of various types of samples of coal during production and dispatch from the project. The lab will have a valid NABL Accreditation certificate.

2.11 OCCUPATIONAL HEALTH ISSUES

GMDC will establish medical set up to check up and monitor all the Occupational diseases endangering the health of the workmen. The periodical medical examination of all the workmen will be carried out as per statute with the purpose of detecting and keeping records of diseases with specific importance to Coal Worker's Pneumoconiosis.

2.12 DIVERSION/ RECHANNELING OF NALLAH/CANAL/RIVER/ROAD

Supai Nala is a seasonal stream, and no water is available December onwards. A detailed hydrological survey of the Supai Nala, regarding its catchment area, flow volume and length of the stretch to be diverted with water augmentation plan along with development of artificial pond has been provided in Chapter 4. Detailed study report is enclosed as **Annexure IV.**

2.13 REHABILITATION & RESETTLEMENT

Total 7 villages namely Chhendipada Jangal (Part), Handigora (Part), Machhakuta (Part), Machhakuta Jangal (Part), Porapara (Part), Porapara Jangal (Full), Tentuloi-Korasahi (Part) are going to get affected by proposed project. Total Project Affected Families (PAF) is 2664 Nos and Project Displaced Families (PDF) is 1629 Nos (738 Original +891 Extended).

These displaced families will be resettled and rehabilitated socially, culturally and economically as per the latest R&R policy (2006 & subsequent amendments) of the Govt. of Odisha.

As on date site selection for the development of the rehabilitation colony has been completed and is awaiting RPDAC approval. Payment of R&R is going on. The detailed status of R&R is attached as **Annexure VII.**

2.14 DESCRIPTION OF MITIGATION MEASURES INCORPORATED INTO THE PROJECT TO MEET ENVIRONMENTAL STANDARDS, ENVIRONMENTAL OPERATING CONDITIONS, OR OTHER EIA REQUIREMENTS.

Other EIA Requirements

Opencast mining operation in general is associated with a number of hazards/risks.

Some of the various anticipated sources of danger are enumerated as under:

- Slope failure.

- Dangers due to handling and use of explosives and accidents due to fly-rocks following a faulty heavy blast.
- Hazards associated with use of electricity
- Accidents due to unruly operation of HEMM
- Dust hazards
- Fire hazards due to spontaneous heating of coal in stockpiles and exposed benches.
- Danger of water-gas explosion due to rainfall over heated coal seam.
- Fire hazards in stores & workshops where inflammable & highly inflammable materials are stored or used.
- Danger of inundation from surface and/or ground water

Adequate provisions will be made for safe working of the mine in form of design of operational systems, provision of HEMM etc.

Adequate skilled & trained manpower will be provided for compliance of safety provisions. Regular training/refresher courses, "on job" training shall be conducted & mock rehearsals shall be made to make the manpower conversant with various rules, regulations, methods of prevention & combat with hazards.

Slope Stability

Coal/OB benches

The exposed ends of the coal seams and OB shall be left with a safe slope to avoid slope failure and collapse of benches.

Similarly, at the end of mining operation, safe terminal pit slope will be provided to avoid pit failure. Detailed site-specific tests for slope stability will be carried out and site-specific parameters determined. Continuous slope monitoring will be done for mine slope & dump slopes with proper devices during actual mining operations.

OB dump

Due to high depth of the quarry, 140.25 million m^3 of OB has to be dumped externally. Due to non-availability of non-forest land in the vicinity, it is proposed to tentatively locate the external dump on the south of quarry area. External dump is suitably planned by matching the overburden removal and void available in different stages. Internal dumps will be formed in 30m tiers with an overall slope of about 26 degrees.

For better stability of internal dumps, it is suggested to rip the mine floor in strips before backfilling. It is suggested to level the dumps and grade them outward properly to obviate water accumulation.

The following precautions will be taken to reduce the risk of dump failure:

1. OB benches will be made of <30m ht in each tier.
2. The angle of repose of each OB bench will be around 37 degrees.

3. Soil will be scrapped separately, so that it is not mixed in OB rock.
4. The slope of ground is mild and will have no adverse effect.
5. The soil from the foundation ground will be scrapped before starting of OB dumping.
6. Garland drain will be made around OB dump area to avoid water flow during monsoon below the OB dump.
7. Ground water table is generally 3-5m below ground level hence there may be no adverse impact.
8. Levelling, grading and drainage arrangement for top of OB dumps will be done.
9. Technical & biological reclamation will be done.

Safe use of Explosives

Site mixed slurry (SMS) has been proposed to be used for good fragmentation. For transportation of explosives, explosive van of approved type is also envisaged. For proper blasting and minimizing the adverse side effects due to blasting viz. noise, ground vibration back-breaks, air blast and fly rocks etc., the optimal blast design parameters are suggested to be used, after field trials.

Adherence to relevant statutory safety provisions as stipulated by DGMS, Chief Controller of Explosives and others competent authorities shall be made.

Use of electricity

To prevent shock hazards, in use of electricity, proper earthing system has been envisaged. It has been proposed to use restricted earthed neutral system of power supply and adoption of fail-safe electronic relays to minimize shock hazards.

Use of HEMM

Adequate number of trained/skilled operators and maintenance crew will be provided with due consideration of leave/sick provisions.

Properly designed haul roads will be provided away from the general and traffic congestion. The traffic rules as enforced by the DGMS shall be strictly followed by the operators of mobile equipment like rear dumpers, water sprinklers, tippers and other light motor vehicles. All mobile equipment shall be provided with audio-visual alarms.

Safety devices like fire alarm and control, operated by sensors will be inbuilt in the equipment/HEMM. Flashers will be fitted in relevant HEMM. The haul roads will be sufficiently wide to prevent accidents.

Provision for proper illumination of quarry faces, haul roads and other working places will be made as per the statutory guidelines.

Dust pollution control measures

Systematic and regular air quality monitoring is necessary to examine objectively the status of compliance with the statutory standards and for making a real assessment of ambient air quality.

The following measures will be done to contain the pollution arising out of dust emission, within the limits:

- All the drills will be provided with well-designed dust extraction/suppression system;
- Blasting operations will be designed in such a way so that these produce minimum dust;
- Effective use of water sprinklers and dust suppression units during loading, transportation and handling of ROM/processed coal and overburden;
- Dust extraction/suppression system will be installed in coal handling plant;
- Provision of greenbelt around quarry, industrial and residential areas and avenue plantation along the haul roads on surface;
- Black-topping of permanent service roads besides proper maintenance shall be carried out. Wetting of the surface by deploying water tankers/sprinklers to reduce dust generation from haul roads.

Fire due to spontaneous heating in coal benches & ground stocks

The following measures will be taken to avoid spontaneous heating-

- Coal bench slopes and seam outcrops will be overlain with a thin layer of inert rock so as to form an impervious layer.
- Treatment of exposed coal seams & outdoor coal stocks with anti-pyrogenic substances.
- Wider exposure of coal benches for long time shall be avoided.

Fire in Project Stores & Workshops

Sufficient provision shall be made for the prevention & control of fire in the project store, both E&M & HEMM workshops & sub-stations by way of installing fire extinguishers of right type & size. Timely inspection & refilling of fire extinguishers will be done.

Systematic provision of both stores & workshops shall be made so that inflammable & highly inflammable materials do not come in contact with any spark or flame. Adequate number of cautions in the form of hoardings will be displayed near such places.

Inundation

Due care shall be taken to prevent water ingress during mining operations from the higher ground through local streams.

- Possible overflow of water during rainy season, from Kumbhira nallah to mine workings will be guarded against by formation of embankment alongside the water courses.

- Height of embankment shall be kept three meter above HFL. The slope facing nallah will be stone pitched. During implementation, design of the embankment will be properly done considering actual site conditions.
- Careful assessment will be made against the danger from surface water before on the onset of rainy seasons. The necessary precautions will be clearly laid down and implemented. A garland drain will be provided to drain away the surface rainwater from coming into the mine.
- Inspections for any accumulation of rainwater, obstruction in normal drainage and weakening in embankment will be done.

Energy Efficiency Measures

GMDC understands its **responsibility towards the environment** and believes the industry and nature should go hand in hand. In line with the sustainability vision, GMDC is committed to encourage Renewable Energy. **GMDC has installed 200.9 MW Wind power & 5 MW Solar Power** across various locations of Gujarat.

GMDC existing project Offices are installed with 20 KW rooftop solar panels. GMDC has already introduced 19 Electric Vehicles (LMV) at its Corporate Office & iCEM.

Action plan for using renewable sources of energy during mine operation:

S. No.	Action Item	Description	Year	Capacity/ nos.	Rs. In Lakh	Proposed area	Expected Outcome
1	Solar Energy Integration (20% of total energy consumption)	Installation of rooftop solar plant - Exploring optimal placement and sizing of solar installations within the mine area.	1 - 2	20 KW	8	Office building	Reduced reliance on conventional power sources, lower operational costs.
			3 - 5	40 KW	16	Office building, Canteen	
			6 - 10	300 KW	120	VTC, Parking, Workshop, stores etc.	
			10 - 15	1000 KW	400	CHP, railway infrastructure etc.	
			15 - 20	400 KW	160	Conveyor belt infrastructure etc.	
2	Village Solar Installation	Install solar energy systems in nearby villages through Coordination with local village authorities and		100 nos.	20	Nearby community infrastructures, village common places etc. 10 nos./year on requirement basis	Community engagement, positive social impact, access to clean energy for local communities.

		community members					
3	Green Transportation - CNG/Electric Vehicles	Explore all possibilities for transporting coal using CNG/Electric Vehicles Conducting feasibility study for vehicle types, charging/fuelling infrastructure, and operational logistics.					Reduced carbon emissions from transportation, alignment with sustainability goals.
4	Electric Vehicles for Mine & Official Transport	Explore the use of electric vehicles (LMV) for the Baitarni West project in the mines and for official transport.		min. 10% of total vehicles		subject to infrastructure availability	Reduced emissions and fuel costs for internal transportation.
5	Utilization of Solar Pumps & Energy Efficient Machines	Utilize solar pumps and other energy-efficient machines wherever possible within the mine operations.	5	200 KW	80	for dewatering	Optimization of energy consumption, reduced operational expenditure.
Total					**804**		

Measures for Prevention of Illegal Mining and Pilferage of Material

The following measures will be adopted for prevention of illegal mining and pilferage of material.

- Security will be deployed with Check post at entry and exit points, continuous patrolling round the clock, Drone facility to prevent illegal mining and pilferage of minerals.
- The entire loading plant will be secured by fencing and security deployed.
- Coal shall be transported along with valid transit pass only. Accordingly, to prevent illegal mining Online Transport Permit system, Online Portal for dispatch through road, rail and conveyor will be initiated.
- Additional Forest check posts will be established to prevent illegal transportation of coal by road.
- GMDC may think over starting implementation of ERP/SAP for generating online

e-invoice.

2.15 USE OF NATURAL RESOURCES

- This is an opencast mine. No timber is used for the operation of the mine.
- The minimum water requirement has been assessed as the water demand for this project and necessary approval for drawing the water from the competent authority will be obtained.
- The minimum Land has been considered for the mining activities for this project.
- Optimum Electrical Energy requirement has been assessed in the project report.
- Fuel consumption has been assessed for the Coal & OB as per norms.
- Solar energy shall be used to the tune of 20% of the total power consumption.

2.16 SAFETY MANAGEMENT AND CONSERVATION

Adequate provisions have been made for safe working of the mine in form of design of operational systems, provision of safety measures for safe use of explosives, electricity and HEMM etc. Sufficient financial provisions have been made under different heads for procurement of necessary safety equipment.

Adequate skilled & trained manpower has also been provided, for compliance of safety provisions. Regular training/refresher courses, "on job" training shall be conducted & mock rehearsals shall be made to make the manpower conversant with various rules, regulations, methods of prevention & combat with hazards.

Sufficient provision has been made in the approved Project Report for the prevention & control of fire in the project store, both E&M & HEMM workshops & sub- stations by way of installing fire extinguishers of right type & size. Timely inspection & refilling of fire extinguishers will be done.

The exposed ends of the coal seams and OB shall be left with a safe slope to avoid slope failure and collapse of benches. Detailed site-specific tests for slope stability shall be carried out and site-specific parameters determined.

Site mixed slurry (SMS) has been proposed to be used for good fragmentation and obviate storage of bulk quantum of explosives.

For proper blasting and minimizing the adverse side effects due to blasting viz. noise, ground vibration, back-breaks, air blast and fly rocks etc., the optimal blast design parameters are suggested to be used, after field trials.

2.17 ECONOMIC PARAMETERS

Total Capital investment - Rs. 3855 Crs.
Estimated capital investment on EIA/EMP - Rs. 10072.9Lakhs
Estimated cost of Mine Closure - Rs. 321.957 Crs

However, funds for CSR will be allocated at approx. Rs. 1 Cr/Annum (2% of the net profit over the preceding three financial years of the organisation) which shall be used in sectors like healthcare, skill training, development of rural infrastructure etc.

2.18 ASSESSMENT OF NEW & UNTESTED TECHNOLOGY FOR THE RISK OF TECHNOLOGICAL FAILURE.

The Project envisages use of conventionally tested Shovel/Dumper Combination Technology for OB along with use of Surface Miner for extraction of Coal. The Shovel /Dumper Combination for undertaking overburden Removal is being used in Indian Coal Mining Industry since long and is the only preferable technology for opencast mining considering the geo-mining conditions of Indian coalfields.

Further use of Surface Miner for mining of coal is also proven. As the conventional and tested technologies have been envisaged in the present Opencast Coal Mining Project, assessment of risk of technological failure is not required.

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Base line study was carried out for the period 1st Oct.'2023 to 31st Dec.'2023 within the study area of 10 km radius.

Ambient air quality, Ambient noise, Ground water quality, Surface water quality were monitored at 10,8,8,8 and 8 number of locations in the study area.

Predominant wind direction was WNW followed by NW and W.

PM2.5, PM10, SO2, NO2 and CO values were found to be in the range of 20.1 µg /m3 to 45.8 µg /m3, 48.9–83.9 µg/m3, 5 to 10.3 µg/m3, 9.5 to 23.3 µg/m3 and 180 to 680 µg/m3 respectively & within the standard limits.

Day and Nighttime ambient noise levels at all the locations were in the range of 43.1 to 52.9 dB and 38.5 to 43.6 dB respectively.

Ground water and surface water quality were found to be within the standard limits except hardness at few GW locations.

Diversity index was found to be 0.94 in the core zone forest and 0.93 to 1.16 in Buffer zone.

Average Quality of life in the Study area was found to be 0.533.

LULC study shows 62.08 % of the Study area is agricultural land followed by forest land (18.64%).

CHAPTER-3 : DESCRIPTION OF THE ENVIRONMENT

GENERAL

The baseline environmental quality is assessed through field studies within the probable impact zone for various components of the environment viz. Air, Noise, Water, Soil, Ecology and Biodiversity and Socio-economy. The baseline environmental study has been conducted for the study region within 10 km radius. An exhaustive attempt has been made in the current chapter to disclose all possible baseline status of environmental quality in the study area, which further serves as the basis for identification, prediction and evaluation of impacts.

PHYSIOGRAPHIC SETUP

Physiography

Physio graphically the district Angul is divided into three regions:

(i) Northern Mountainous Region
(ii) Central Undulating Plain
(iii) Southern & South-western Mountainous Region.

The variation in land elevations above MSL is shown in **Fig. 3.1**.

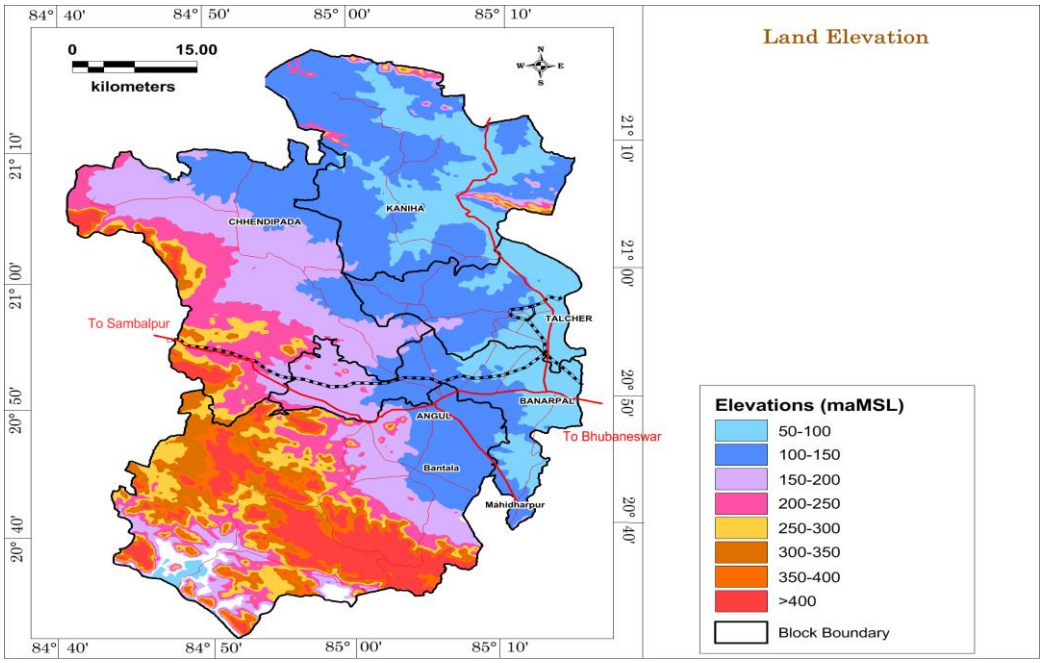


Figure 3-1 Variation in land elevations above MSL of district Angul

3.1 TOPOGRAPHY

The surface topography is generally flat, most of the area is being used for cultivation purposes by the villagers. Around the south and eastern parts, the surface topography is gently undulating with a general northerly slope i.e., the flow direction of Supai and Kumbhira nala.

The highest elevation above Mean Sea Level is about 203 meters in the southwestern corner and the surface elevation gradually decreases to 150 meters in the northeastern and northern part of the block.

Morrum and mud roads are very common in and around villages within the block.

Contour map of the buffer and core zone with 20 meters and 5 meters interval are shown below.

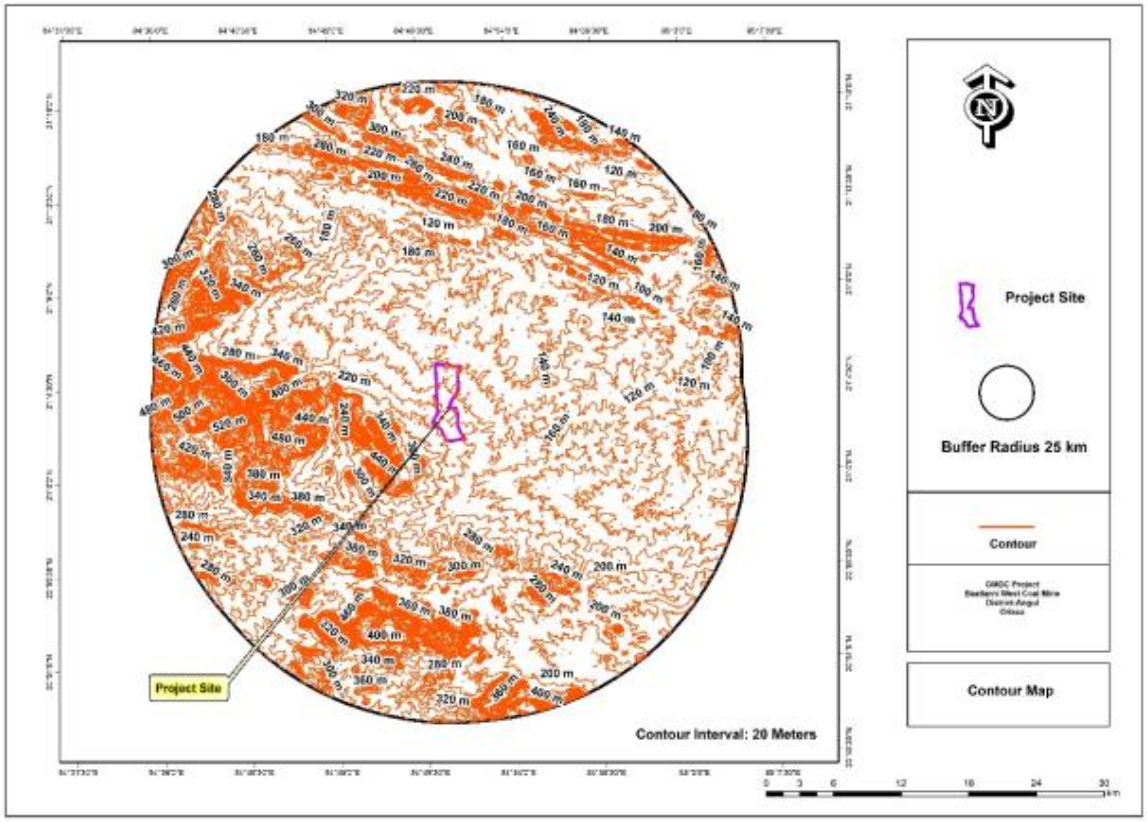


Figure 3-2 Contour Map of the Study area

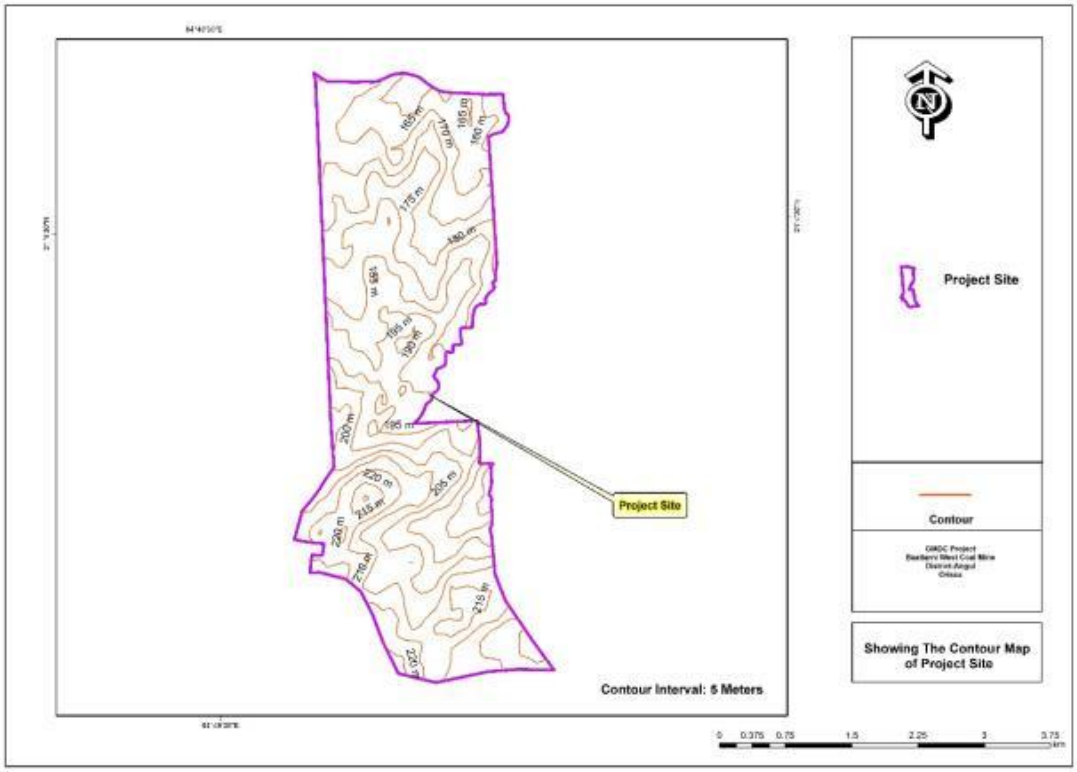


Figure 3-3 Contour Map of the Project site

3.2 DRAINAGE

Brahmani river, flowing in approximately north-south direction along the eastern boundary of the Talcher coalfield, provides the main drainage of the region. Kumbhira nala, which forms the south-eastern boundary of the block, is flowing from south to north. It controls the drainage and topographical pattern around southeastern part of Baitarani West block. A small nala, Supai, is flowing across the block from south to north near northeastern boundary of the block. The Kumbhira and Supai nala drain the area and both are discharging in Tikra nadi, which is running along the northern boundary of the coalfield. Both Supai and Kumbhira nalas are seasonal in nature and feed to Tikra nadi during most part of the year except dry summer seasons. The Tikra Nadi merges in its downstream into Brahmani River at northeastern corner of the coalfield. Small ponds and dug wells are common in this block which are utilised for irrigation and drinking purpose.

The direction of the rivers and Nala located in the 10 km radius of mine lease area are already provided in **Table No. 2-6.**

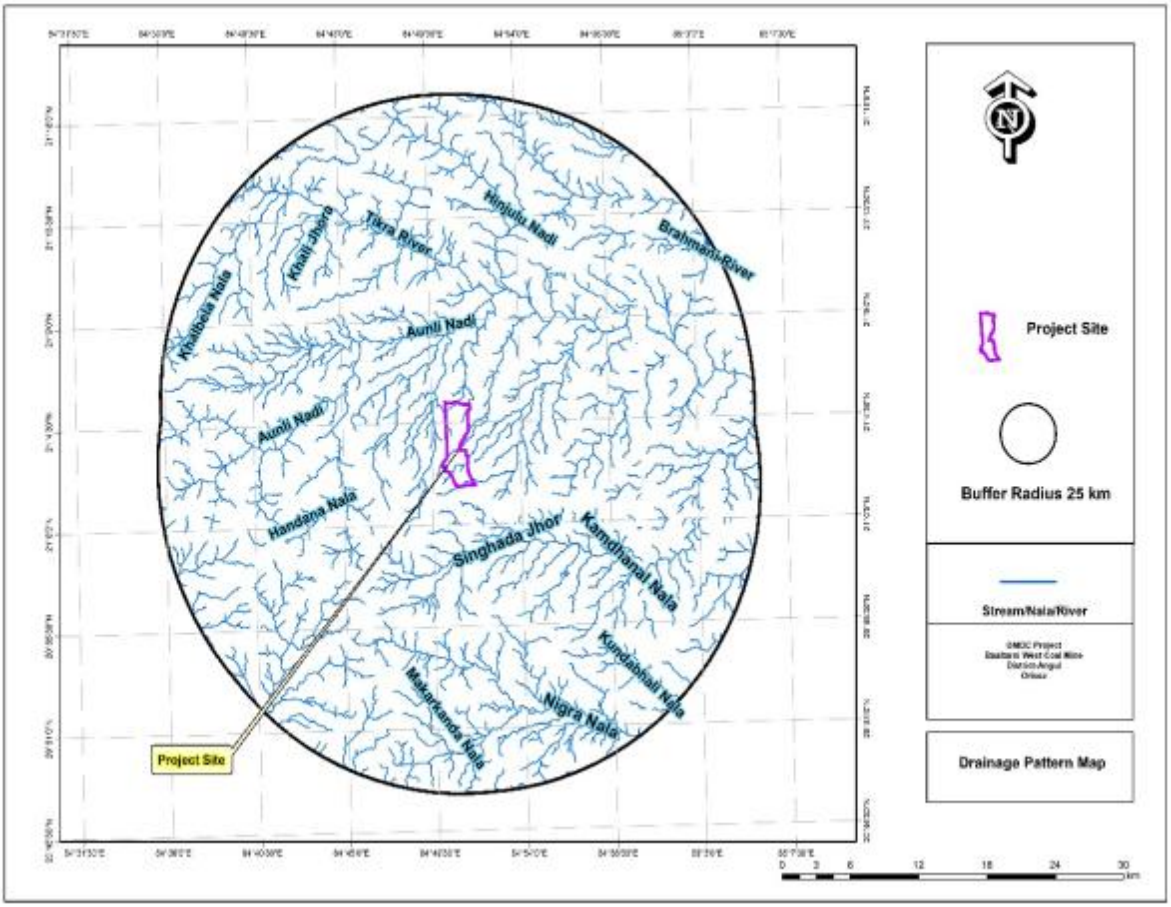


Figure 3-4 Drainage Pattern of the study area

Within the quarry, the faces shall be so laid that water from the working areas shall flow into the sump by gravity. From the sump the water will be pumped out to the surface and will flow into the surface drainage system.

3.3 GEOLOGY & GEOMORPHOLOGY

Regional geology belongs to the part of Talcher Coalfield. The geological setup of the area and other details are discussed in Chapter-2.

3.4 LOCAL GEOLOGY

Out of total 1196 ha, 672.147 ha is fully explored & details are available only for this area is already described in Chapter-2.

The structure of Baitarani West block is highly disturbed in about 2 sq.km area towards the north central part and extreme southern part of the block. A total of 17 faults have been delineated in Baitarni West block.

3.5 GEOMORPHOLOGY

The analysis of geomorphological data and thematic map reveals that the hilly terrains in northern and south-southwestern portions were separated by central undulating plains which comprises of predominantly pediments and shallow buried pediments. The geomorphology of the study area is shown in Figure-3.5.

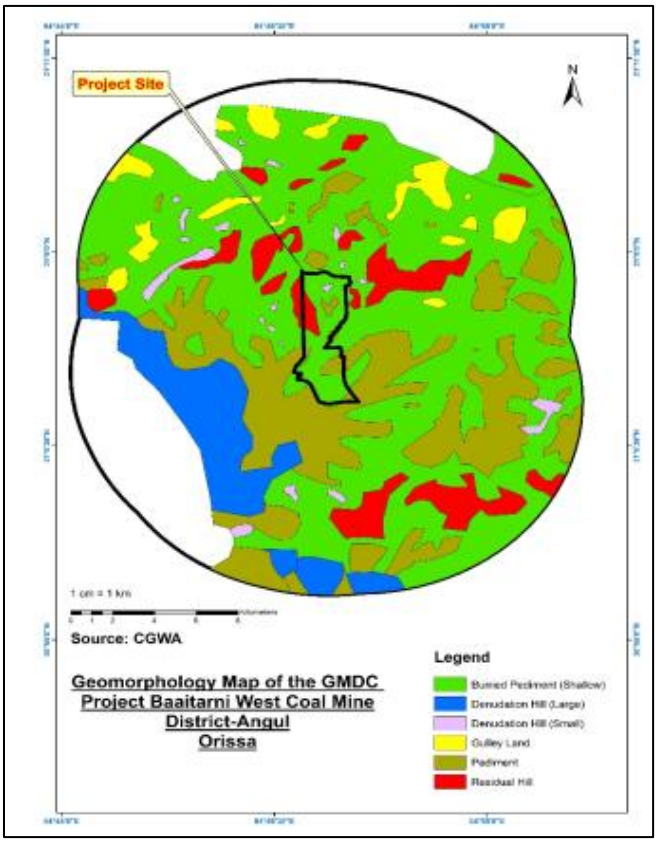


Figure 3-5 Geomorphology of the study area

3.6 SOIL CHARACTERSTICS

It has been observed that a major part of the study area is occupied by LF-Ferric Luvisols followed by NE-Eutric Nitosoil. Afisol includes red sandy soil and red loamy soil. It is porous and friable, tight textured, usually devoid of lime kankars and is

also free of carbonates. It is usually suitable for cultivation of paddy and a large variety of other crops. The thematic map on the soil distribution in the study area is shown in Fig. 3.18.

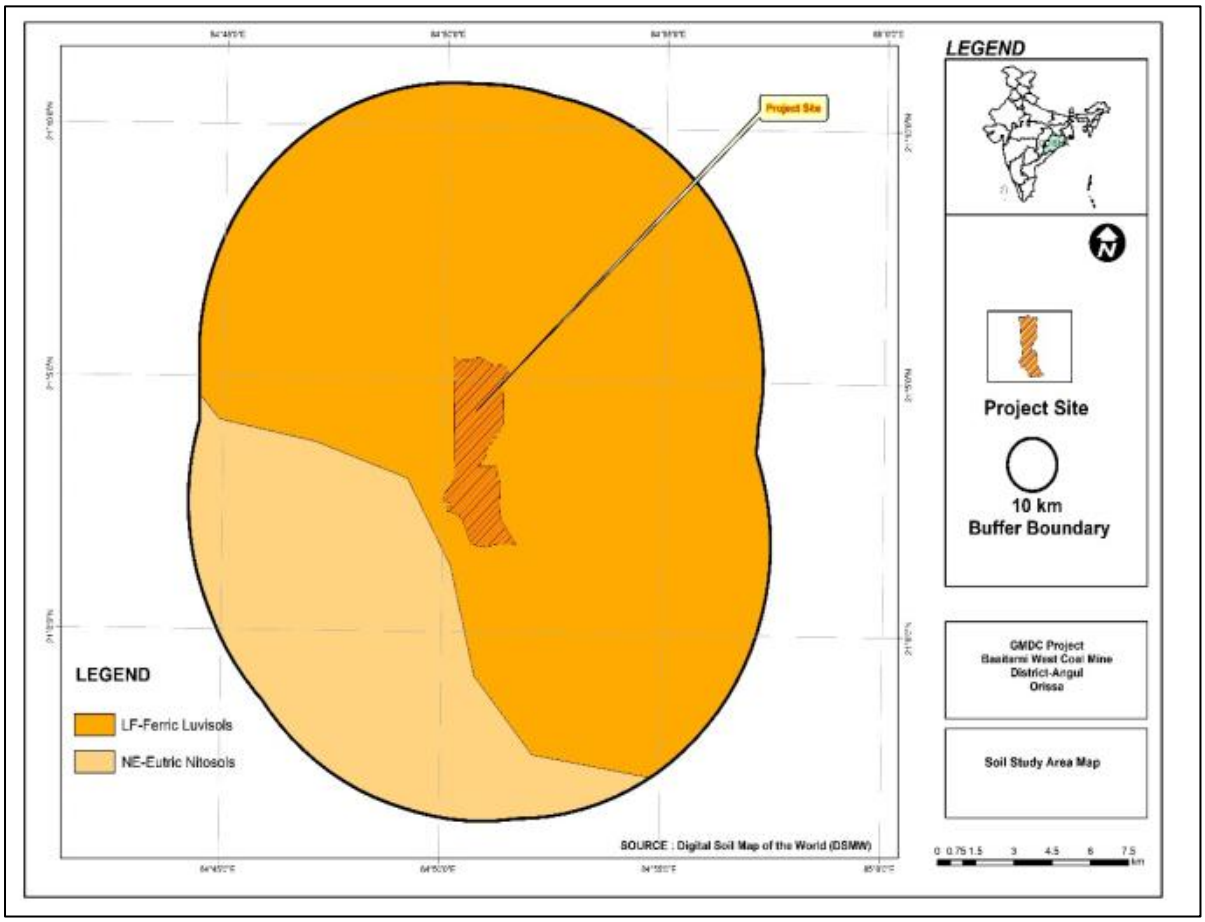


Figure 3-6 Geomorphology of the study area

3.7 SEISMIC STATUS

Angul falls in moderate risk zone.

The project is not falling under seismically active zone or land slide prone area. Chances of flood cannot be ruled out. Besides safety measures, the mine will be well equipped to deal with eventuality as enumerated in CMR,2017 and related DGMS circulars.

The seismic zoning of Odisha falls between zones I to II i.e., low damage risk zone and moderate damage risk zones.

The parts of districts coming under moderate risk zones are: Sundargarh, Jharsuguda, Bargarh, Sambalpur, Deogarh, Angul, Dhenkanal, Jajpur, Cuttack.

3.8 BASELINE STUDY AREA AND PERIOD

The baseline environmental study was carried out for the period 1st Oct.'2023 to 31st Dec.'2023, as well as secondary data also collected from authenticated sources.

Baseline data has been collected by M/s GRC India Training & Analytical Laboratory, Noida {NABL Accredited Lab, Certificate No. TC –7501 & MOEFCC No. S.O. 388 E} in accordance with the Guidelines for EIA issued by the Ministry of Environment, Forests and Climate Change, Govt. of India and CPCB, New Delhi. The impact identification always commences with the collection of baseline data such as Ambient Air Quality, Micrometeorology, Ground and Surface Water Quality, Noise levels, Soil Quality, Land use pattern, Biological Environment and Socio-economic aspects, Solid and Hazardous waste, Risk Assessment, Geology and Hydrology within the study area of 10 km radius.

Table 3:1 Frequency of Environmental Monitoring

Attributes	Sampling		
	Locations	**Parameters**	**Frequency**
A. Air Environment			
Micro-Meteorological Data	Project Site (Entry)	Temperature, Relative Humidity, Precipitation Wind direction, Wind Speed	Hourly data for the period 1st Oct.' 2023 to 31st Dec.' 2023
Ambient Air Quality	10 numbers of locations in the study area of 10 km radius	PM_{10}, $PM_{2.5}$, SO_2, NO_2, CO, Arsenic, Lead, Mercury, Chromium and Silica	24-hour basis, twice a week during study period.
B. Ambient Noise	8 numbers of locations in the study area of 10 km radius	Noise Levels in dB(A)	Once in Study Period
C. Soil Quality	Samples from 8 numbers of locations within 10 km radius area from the project site	Physical, Chemical Characteristics, Soil Texture	Once in Study Period
D. Water			
Ground Water	Samples from 8 numbers of locations within 10 km radius from the project site	Physical, Chemical, Microbiological and Heavy Metal	Once in Study Period
Surface Water	Samples from 8 numbers of locations within 10 km radius from the project site	Physical, Chemical, Microbiological and Heavy Metal	Once in Study Period

3.9 ESTABLISHMENT OF BASELINE STUDY FOR VALUED ENVIRONMENTAL COMPONENTS, AS IDENTIFIED IN THE SCOPE

3.9.1 MICROMETEOROLOGY

The study of micro meteorological data helps to understand the variations in the ambient air quality status in that region. The prevailing micrometeorological condition at project site plays a crucial role in transport and dispersion of air pollutants. The persistence of the predominant wind direction and wind speed at the project site decides the direction and extent of the air pollution impact zone.

The principal variables which affect micrometeorology are horizontal transport and dispersion, convective transport and vertical mixing and topography of the area towards local influences. Micrometeorological data were collected by using the wind monitor as per CPCB guideline which was installed near the project site. All the micrometeorological data were collected on an hourly basis.

3.9.2 SOURCE OF METEOROLOGICAL DATA

Micrometeorological data were collected by using the weather station as per CPCB guidelines which was installed at the project site. All the micrometeorological data were collected on an hourly basis for the period 1st Oct.' 2023 to 31st Dec.' 2023.

3.9.3 TEMPERATURE

The area experiences the sub-tropical to tropical temperate monsoon climate and characterised by a hot summer and general dryness throughout the year except during the southwest monsoon season, i.e., June to September. December is the coldest month with mean daily maximum temperature at 26.9^{0}C and the mean daily minimum temperature at 13.4^{0}C. Both day and night temperature increase rapidly from March and by May the mean daily maximum temperature reaches to 42^{0}C, while the mean minimum temperature is 26.8^{0}C.

3.9.4 HUMIDITY

As per the IMD observation in the district, the air is generally dry except during the monsoon period. In the summer months, the relative humidity is low in the range of 30%. During monsoon, the humidity goes up to 82% or more. Wind velocity in general is low to moderate with some increase in summer and monsoon season. Winds are mostly blown from southwest and northeast direction during monsoon period. In the cold season winds are mainly from west or north. In the summer months, the wind flows from variable directions. The average wind speed is 2.92 m/s. The mean monthly potential evapotranspiration value ranges from 40 mm in December to 326 mm in May.

3.9.5 RAINFALL

The South-west monsoon is the principal source of rainfall in the area. The normal rainfall of the district is 1401.9mm. The rainfall pattern is erratic, and drought is a common feature of the district. The long-term analysis of rainfall data recorded at block headquarters for the period 1995- 2014 (20 years) for Chhendipada was 1126.3mm of avg annual rainfall. The coefficient of variation in rainfall is minimum 22%. Normal rainfall has been received in 55% to 95 % of the years.

3.9.6 WIND SPEED AND WIND DIRECTION

Hourly wind speed data was collected for the period 1st Oct.'2023 to 31st Dec.'2023 with the help of a wind monitor. The rate of dispersion, diffusion and transportation of pollutants in the atmosphere mainly depend on wind speed and its direction. Wind direction and velocity data have been collected during the study

period. During the study period, wind blow dominant was in the direction of NNW-SSE. During the study period the average wind speed was 2.92 m/s.

3.9.7 WINDROSE

Wind rose diagram is a graphical representation of the magnitude and direction of wind speed considering all the directions. From the knowledge of wind rose one can easily predict the direction and extent of spreading of the gaseous and particulate matter from the source. Wind rose diagram has been prepared by using hourly wind velocity and pre-dominant wind direction data and is presented in **Figure 3.7.**

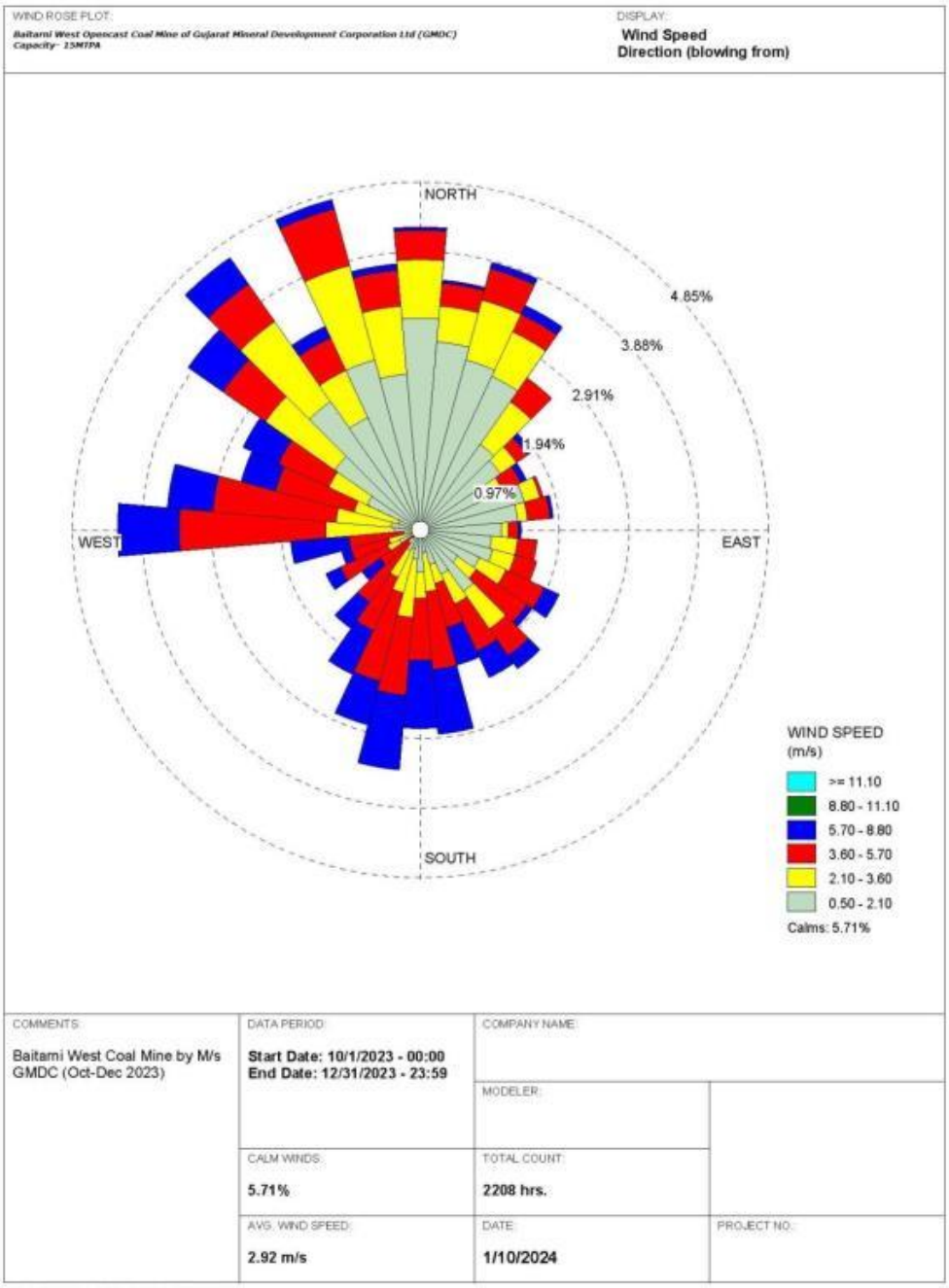


Figure 3-7 Wind Rose Diagram

3.9.8 INTERPRETATION OF MICROMETEOROLOGICAL DATA

During the study period, wind blow dominant was in the direction of NNW-SSE.

3.9.9 AIR ENVIRONMENT



Air is the Earth's atmosphere having the gases in which living organisms live and breathe but air is deteriorating day by day due to the anthropogenic and natural sources. In present scenario Industrialization, Energy production and the burning of fossil fuels has polluted the air environment. Ambient air quality monitoring was carried out for the assessment of the existing status of background air quality in the study area. This will be useful for assessing the conformity of the ambient air quality to the standards even after commencement of the proposed project.

3.9.10 SELECTION OF SAMPLING LOCATIONS

The following points were considered during the selection of Ambient Air Quality Monitoring locations.

- Topography/terrain of the study area
- Regional synoptic scale climatologically norm's
- Densely populated areas within the study area,
- Location of surrounding Industries,
- Representation of regional background
- Facility for Ambient Air Monitoring,
- Representation of valid cross – sectional distribution in downwind direction
- Avoidance of proximity of roads, construction activity or any other perturbing activity which may be temporary in nature, which may lead to some erroneous conclusions
- Availability of manpower, electricity, approach, sturdy structure and protection of samplers and
- Dominant Wind Direction

The monitoring locations have been selected keeping in mind the seasonal wind direction pattern of the study area, accessibility, topography, safety and presence of inhabited localities. The location of stations was selected using the Ambient Air Quality Monitoring Network Design by R.E.Munn such that the stations have potential of getting affected the most. As such, a good number of potential monitoring stations all around the project were identified and the probability of getting affected by the project was calculated. For this, the Coverage Factor method as suggested by R.E.Munn was used to represent probability. Stations falling in the downwind direction with high frequency have a high probability to get polluted. Also, the stations near the mine/activity have also a high probability of getting polluted. The Coverage Factor covers both these factors. Total 10 stations were selected as AAQ monitoring locations. Stations having high coverage

factors were selected along with some having low coverage factor to be used as Control station.



The model is as follows:
Where,
Cij= Coverage Factor of ith Station in j^{th} direction.
In = Intensity of nth Source.
Di = Downwind direction of ith Station.

To establish the baseline status around the project site of the study region, monitoring was conducted for 10 numbers of locations. At the time of location selection previous micrometeorological data (October – December 2022) was referred and consequently the locations where adjusted as per the present windrose (based on a certain period – a month or so) general wind pattern in the study region was considered for the selection of minimum one location in the downwind direction. The previous Windrose for selection of sampling location is enclosed as **Annexure XXII**. However, ambient air monitoring locations were selected in all the directions looking towards the possibility of change in wind pattern during the study period. Ambient Air Quality monitoring locations are presented in **Figure 3-8** and **Table 3-2**.

Table 3:2 Ambient Air Sampling Stations

Code	Location Name	Dist. (km)	Dir. w.r.t project Site	Latitude	Longitude	Selection Criteria
AAQ 1	Project Site	-	-	21° 2' 23.136" N	84° 50' 47.906" E	--
AAQ 2	Near Jamunali Club, Handigoda	0.27	WSW	21° 2' 13.899" N	84° 50' 6.569" E	crosswind
AAQ 3	Village. -Chhendipada	0.6	E	21° 5' 11.326" N	84° 51' 48.863" E	crosswind
AAQ 4	Near Behera sahi new club house	0.65	SE	21° 1' 28.841" N	84° 51' 55.148" E	downwind
AAQ 5	Near Indian Oil, Chhendipada	2	NW	21° 6' 9.272" N	84° 49' 16.913" E	upwind
AAQ 6	Charibati Filling Station, Mamuriasahi	4.5	NE	21° 7' 31.992" N	84° 52' 19.410" E	crosswind
AAQ 7	Odia Sambada - Media house, Chhendipada	6.8	NW	21° 7' 15.360" N	84° 46' 46.047" E	upwind
AAQ 8	Soumya Ranjan Pradhan, Kukurpeta Jungle, Odisha	4.4	E	21° 2'15.34"N	84°54'7.23"E	crosswind
AAQ 9	Akash Bastralaya, Barapada	7.7	SW	20° 58' 55.497" N	84° 54' 46.523" E	crosswind
AAQ 10	Play Field, Madhupur	8	NE	20° 59' 27.026" N	84° 46' 53.322" E	crosswind

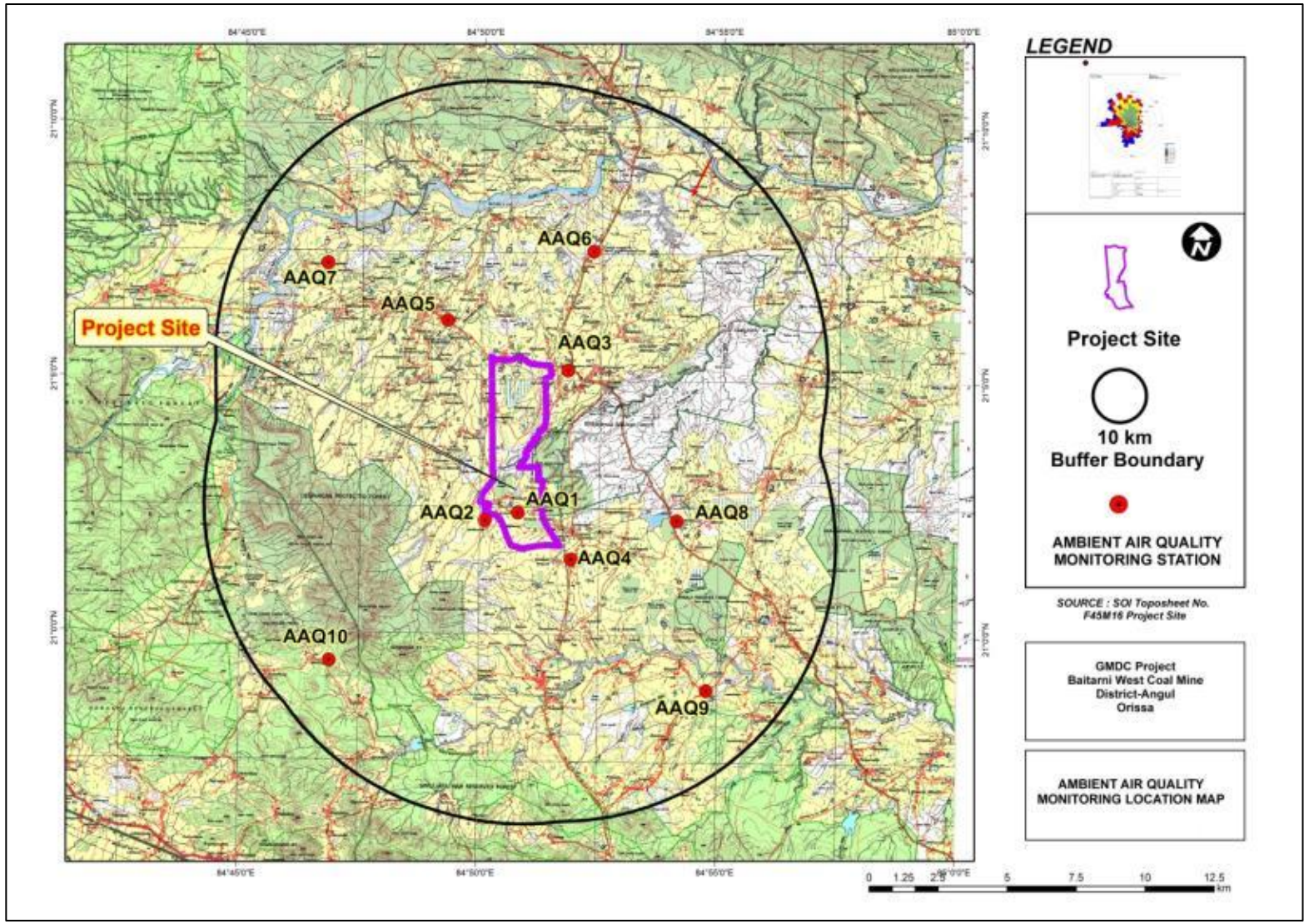


Figure 3-8 Map Showing the Study Region with Locations for Ambient Air Sampling Stations

3.9.11 FREQUENCY AND PARAMETERS FOR SAMPLING

The sampling team consisting of FAE and Technical Assistant were involved to carry out Ambient Air Quality Monitoring at 10 numbers of locations within 10 km radius from the project site. Sampling and analysis was carried out as per CPCB, IS 5182 and EPA and instrument operation manual for the parameters PM2.5, PM10, SO2, NO2, CO, As, Pb, Hg, Cr and Silica. After the completion of sampling, samples were brought to the laboratory in an Ice box and filter box for analysis. Frequency of sampling was twice a week during study period.

3.9.12 METHODOLOGY FOR SAMPLING AND ANALYSIS

Samples were collected by using the PM_{10} and $PM_{2.5}$ micron dust samplers at suitable height from obstruction free area as per the availability of the facility. Sampling and Analysis was carried out as per CPCB Guidelines, instrument operational manual and National Environmental Engineering Research Institute. Details of reference method is presented in **Table 3.3.**

Table 3:3 Methodology for Sampling and Analysis

S. No	Pollutant	Test Method
1.	$PM_{2.5}$	IS 5182 (Part 24): 2019
2.	PM_{10}	IS 5182 (Part-23): 2006, Reaffirmed 2017
3.	SO_2	IS 5182 (Part-2): 2001, Reaffirmed 2017
4.	NOx	IS 5182 (Part-6): 2006, Reaffirmed 2017
5	CO	IS 5182 (Part-10): 1999, Reaffirmed 2014
6	As	GRC-LAB/STP-AIR/04: 2018 & CPCB NAAQ Method, Volume-1, May2011
7	Pb	IS 5182 (Part-22): 2004, Reaffirmed 2014
8	Hg	GRC-LAB/STP-Air/05; NIOSH 6009 Method, 2001
9	Silica	GRC-LAB/STP-Air/06; ASTM D7614-20;2020
10	Cr	GRC-LAB/STP-Air/07;2018

3.9.13 AIR QUALITY OF STUDY AREA

There are no other mines in operation within 10 km radius of the proposed Baitarni West (BW) opencast coal mine.

Only few small industries viz. Behera Bio Briquette, Seven Hill Minerals, Saralia LPG Plant, M/s Pradhan Fly Ash Bricks, Maa Bayani Rice Mill, Dilli Rice Mill, Shriya Mudhi Mill, Tutu Mill are in operation within 10 km radius area of the BW mine.

Ambient air quality monitoring was done at 10 locations within 10 km radius which reflects the baseline status and contribution of all the industries/activities including above mentioned small industries.

Ambient air quality analysis was conducted in the study area including the project site twice in a week for three months (Post Monsoon Season). The analysis results are appended below in Table 3.4.

Table 3:4 Summary of Ambient Air Quality in the Study Area (Date of Sampling & Analysis – 01.10.2023 to 31.12.2023)

Parameters	PM_{10} (µg/m³)	$PM_{2.5}$ (µg/m³)	SO_2 (µg/m³)	NO_2 (µg/m³)	CO (µg/m³)
AAQM Norms	**100**	**60**	**80**	**80**	**4000**
Project Site (AAQ – 1)					
Min	65.3	29.8	5.1	11.6	340
Max	83.9	42.2	7.3	20.26	680
Mean	75.3	36.0	6.0	14.8	534.2
98 Percentile	83.7	42.1	7.1	19.5	666.2
Near Jamunali Club, Handigoda (AAQ- 2)					
Min	60.5	25.2	5.3	9.5	210
Max	76.2	39.8	8.2	23.3	520
Mean	68.2	32.2	6.5	13.9	389.2
98 Percentile	75.3	38.8	8.0	22.2	515.4
Village-Chhendipada (AAQ - 3)					
Min	65.6	32.2	5.2	16.6	310
Max	79.9	43.6	7.9	22.6	580
Mean	73.7	38.1	6.3	20.3	453.8
98 Percentile	78.8	43.4	7.9	22.5	575.4
Near Behera sahi new club house (AAQ - 4)					
Min	55.4	28.1	5.1	10.7	290
Max	78.2	40.8	7.6	20.1	530
Mean	66.7	33.8	6.1	13.6	377.5
98 Percentile	76.2	40.6	7.6	19.0	525.4
Near Indian Oil, Chhendipada (AAQ - 5)					
Min	64.6	31.6	6.6	12.3	380
Max	83.5	43.8	8.9	19.2	620
Mean	76.3	37.4	7.4	15.5	486.7
98 Percentile	82.9	42.9	8.5	18.6	610.8
Charibati Filling Station, Mamuriasahi (AAQ - 6)					
Min	63.5	29.8	5.9	11.5	310
Max	79.8	42.1	10.3	19.2	560

Parameters	PM_{10} (µg/m³)	$PM_{2.5}$ (µg/m³)	SO_2 (µg/m³)	NO_2 (µg/m³)	CO (µg/m³)
AAQM Norms	**100**	**60**	**80**	**80**	**4000**
Mean	73.1	37.2	7.2	14.7	411.3
98 Percentile	79.6	41.9	10.1	19.1	537.0
Odia Sambada - Media house, Chhendipada (AAQ -7)					
Min	56.2	23.6	5.4	10.7	250
Max	77.6	40.9	8.2	18.4	490
Mean	68.8	32.5	6.7	13.6	360.4
98 Percentile	76.4	40.3	8.2	18.0	480.8
Soumya Ranjan Pradhan, Kukurpeta Jungle,Odisha (AAQ - 8)					
Min	56.7	27.3	5	9.7	280
Max	81.2	45.8	8.1	16.0	530
Mean	70.6	35.0	6.6	13.5	414.2
98 Percentile	81.2	45.1	8.0	16.0	525.4
Akash Bastralaya, Barapada (AAQ - 9)					
Min	56.6	25.6	5.2	9.5	240
Max	78.7	40.5	8.3	18.6	430
Mean	69.4	32.3	6.5	13.3	339.6
98 Percentile	78.3	40.2	8.1	18.0	430.0
Play Field, Madhupur (AAQ - 10)					
Min	48.9	20.1	5.1	9.7	180
Max	67.8	34.7	7.8	20.5	390
Mean	58.8	27.1	6.1	13.0	289.6
98 Percentile	66.9	33.7	7.6	20.1	390.0

Summary of Ambient Air Quality

- During the study $PM_{2.5}$ was observed in the range of 20.1-45.8µg/m³. Maximum concentration of $PM_{2.5}$ was found at AAQ-8 i.e. Soumya Ranjan Pradhan, Kukurpeta Jungle, Odisha.
- PM_{10} was observed in the range of 48.9-83.9 µg/m³. Maximum concentration of PM_{10} was found at Project site.
- SO_2 concentration was observed in the range of 5 to 10.3µg/m³, which is well within the standard limit.
- NO_2 concentration in was observed in the range of 9.5-23.3µg/m³, which is well within the standard limit.

- CO concentration was observed in the range of 180 to 680µg/m^3, which is well within the standard limit.
- No heavy metals such as As, Pb, Cr, Hg were observed in the samples
- Silica concentration is less than 0.01 mg/m^3 was observed in samples

Graphical representations of pollutants are given below.

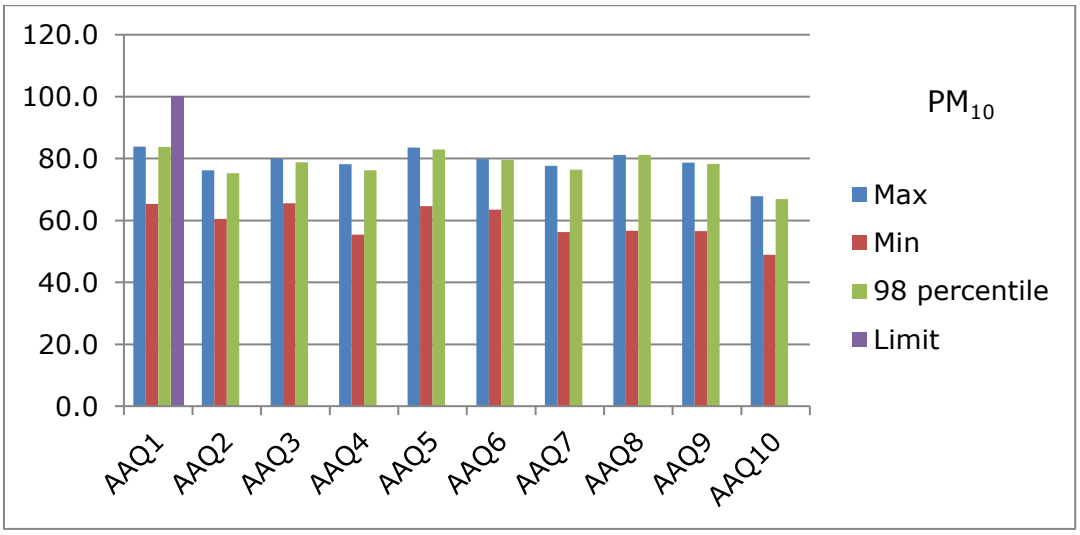


Figure 3-9 PM_{10} Concentration in µg/m^3

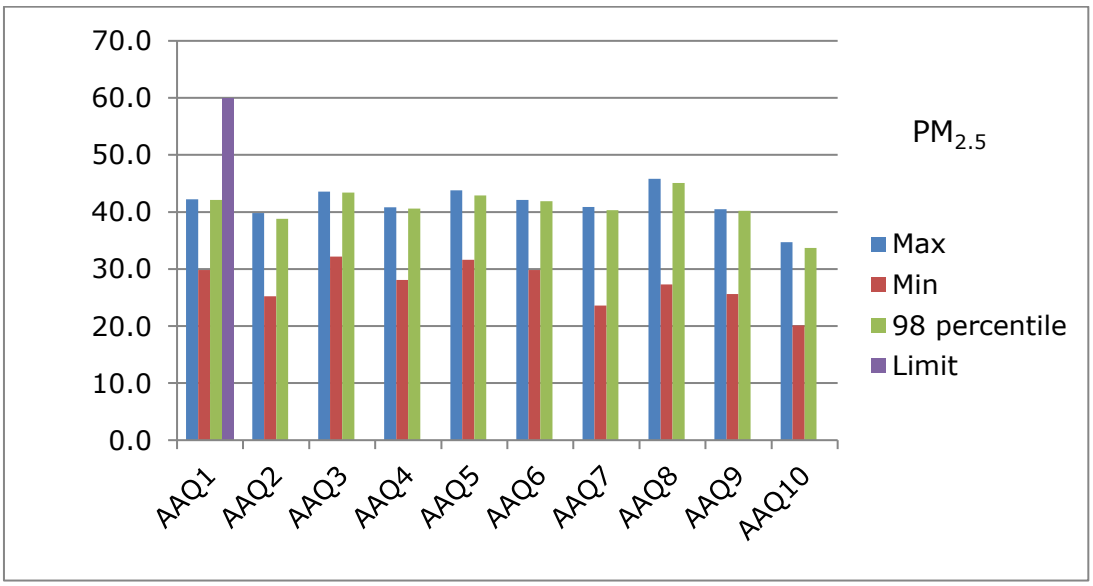


Figure 3-10 $PM_{2.5}$ Concentration in µg/m^3

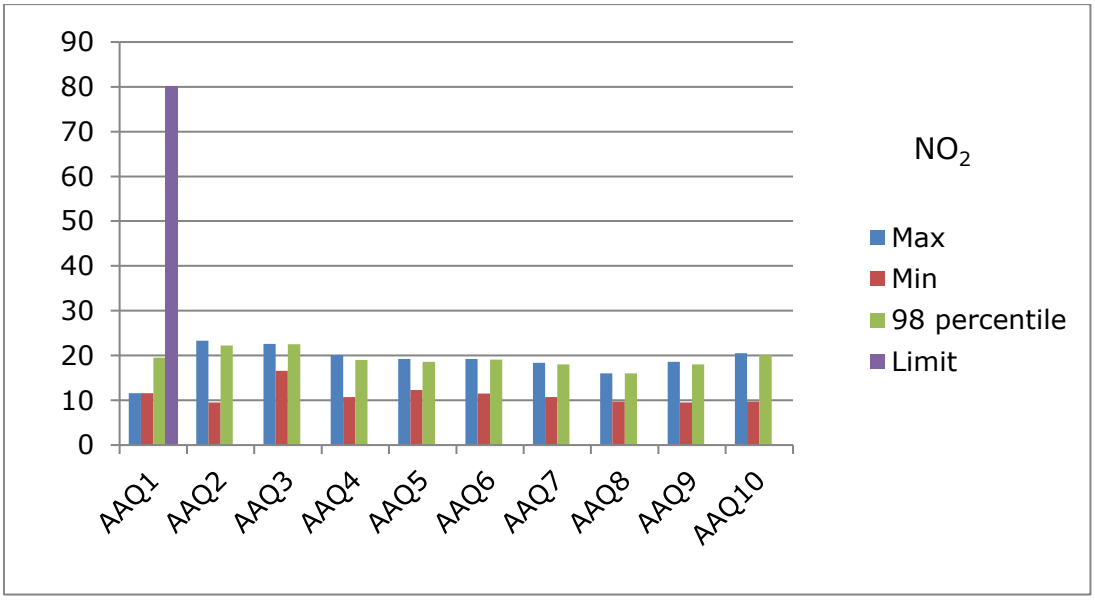


Figure 3-11 NO_2 Concentration in µg/m³

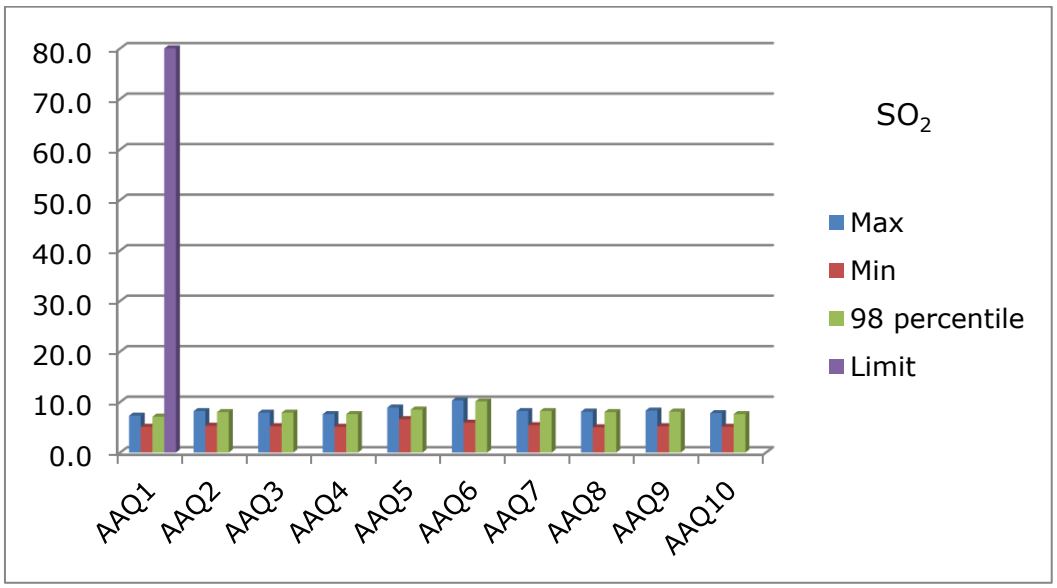


Figure 3-12 SO_2 Concentration in µg/m³

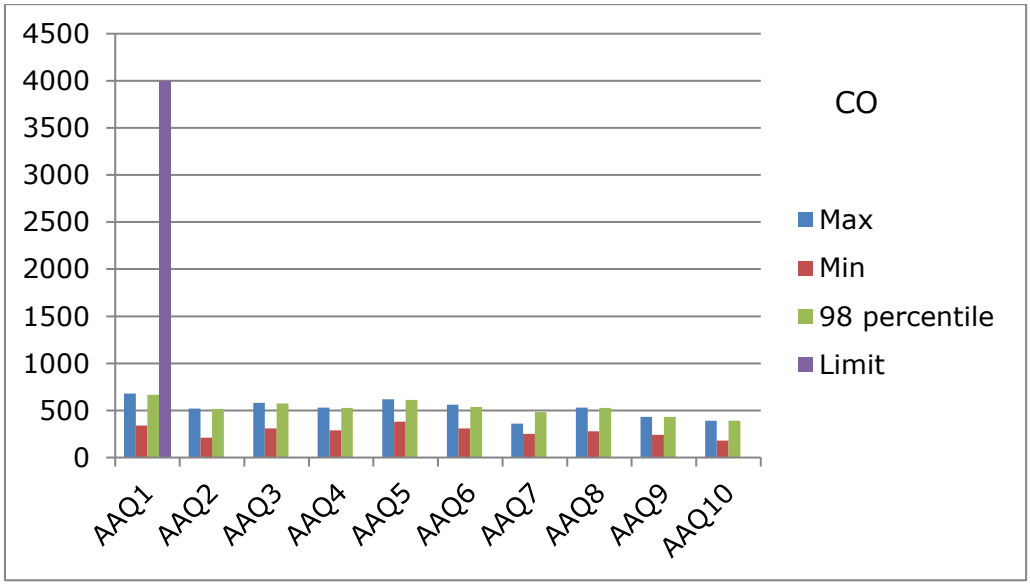


Figure 3-13 CO Concentration in µg/m³

3.9.14 INTERPRETATION OF AMBIENT AIR QUALITY DATA

On the basis of the above results, all the ambient air quality parameters have been found within the standard prescribed limits as per NAAQS standard, 2009. The ambient air quality of monitored locations including the Project site, itself is not an air pollutant sources; however, the traffic induced by the operation of the project or local activities may cause additional air quality impact on sensitive receptors. The emission sources during the initial phase of the project would be the emissions from various sources. PM_{10}, $PM_{2.5}$, SO_2 and NO_2 are the key criteria pollutants for assessment of the air quality impact on this project. Carbon monoxide is one of the primary pollutants emitted by road transport. However, monitoring results of all monitoring stations show that background CO concentration is constantly below the respective criteria. This interpretation relates to the results recorded for monitored locations and monitoring study period.

The results of baseline ambient air quality analysis data of AQ1, AQ4 & AQ5 which are at a distance of 33 km, 33.06 km and 41.97 km respectively from Hakimapada, Angul Monitoring station of CPCB for the period (01.01.24 to 31.01.24) are compared for PM10, PM2.5, SO2, NO2 & CO.

Parameters	Hakimapada, Angul (01.05.24 to 30.06.24)		Project Site (AQ – 1) (01.10.23 to 31.12.23) 33 km from Hakimapada		Near Behera sahi new club house (AQ - 4) (01.10.23 to 31.12.23) 33.06 km from Hakimapada		Near Indian Oil, Chhendipada (AQ - 5) (01.10.23 to 31.12.23) 41.97 km from Hakimapada		AAQM Norms
	Min	Max	Min	Max	Min	Max	Min	Max	
PM_{10} (µg/m³)	16.89	276.17	65.3	83.9	55.4	78.2	64.6	83.5	**100**
$PM_{2.5}$ (µg/m³)	7.31	107.39	29.8	42.2	28.1	40.8	31.6	43.8	**60**
SO_2 (µg/m³)	5.58	57.69	5.1	7.3	5.1	7.6	6.6	8.9	**80**
NO_2 (µg/m³)	2.02	24.51	11.6	20.26	10.7	20.1	12.3	19.2	**80**
CO (µg/m³)	510	1940	340	680	290	530	380	620	**4000**

- From the table it can be clearly observed that the results at Angul (Hakimapada 01.05.24 to 30.06.24) by CPCB is found to be quite high from the baseline monitoring results at AQ1, AQ4 & AQ5. This may be due to various activities at Angul & without proper pollution control measures as Hakimapada being a village at a distance of 33 km with less activity and greenery. The level of pollutants to be found within the permissible limit. However, this will be further compared with the results of summer season May-June 2024 shall be compared & incorporated in the final EIA/EMP.

- The values of SO2, NO2 & CO are well within the standards at all locations.

3.10 NOISE ENVIRONMENT

The proposed project is Coal Mine and hence movement of products, fuels, tools & tackles required for its operation. The movement of personnel from their residence to industry would also result in a moderate increase in the traffic, which would not result in any drastic change in either the existing traffic pattern or noise levels of the area. Background noise levels were measured using integrated sound level meter in 08 locations.

Table 3:5 Noise Monitoring Locations Within the Study Area

Code	Location Name	Distance (km)	Direction w.r.t Project Site	Latitude	Longitude
NQ1	Project Site	-	-	21°2'23.657"N	84°50'48.016" E
NQ2	Maa Bayani Wshg, Fps, Podapada	0.17	N	21°5'28.670"N	84°50'29.147" E
NQ3	Near Panchayat House, Balikudiajungle	0.6	SE	21°1'25.824"N	84°51'46.276" E
NQ4	Near Sarpanch House Chhendipadajungle	0.7	E	21°4'43.081"N	84°51'46.010" E
NQ5	Tentuloi Panchayat Office	0.8	E	21°2'3.684"N	84°51'54.645" E
NQ6	Near Water Purification Company,Sapoinali	1.25	NW	21°5'45.325"N	84°49'33.219" E
NQ7	Bhutia Farm, Chhendipadajungle	1.4	NE	21°6'0.747"N	84°51'35.747" E
NQ8	Near Basantpur Panchyat House	2.4	W	21°3'13.084"N	84°48'50.774" E

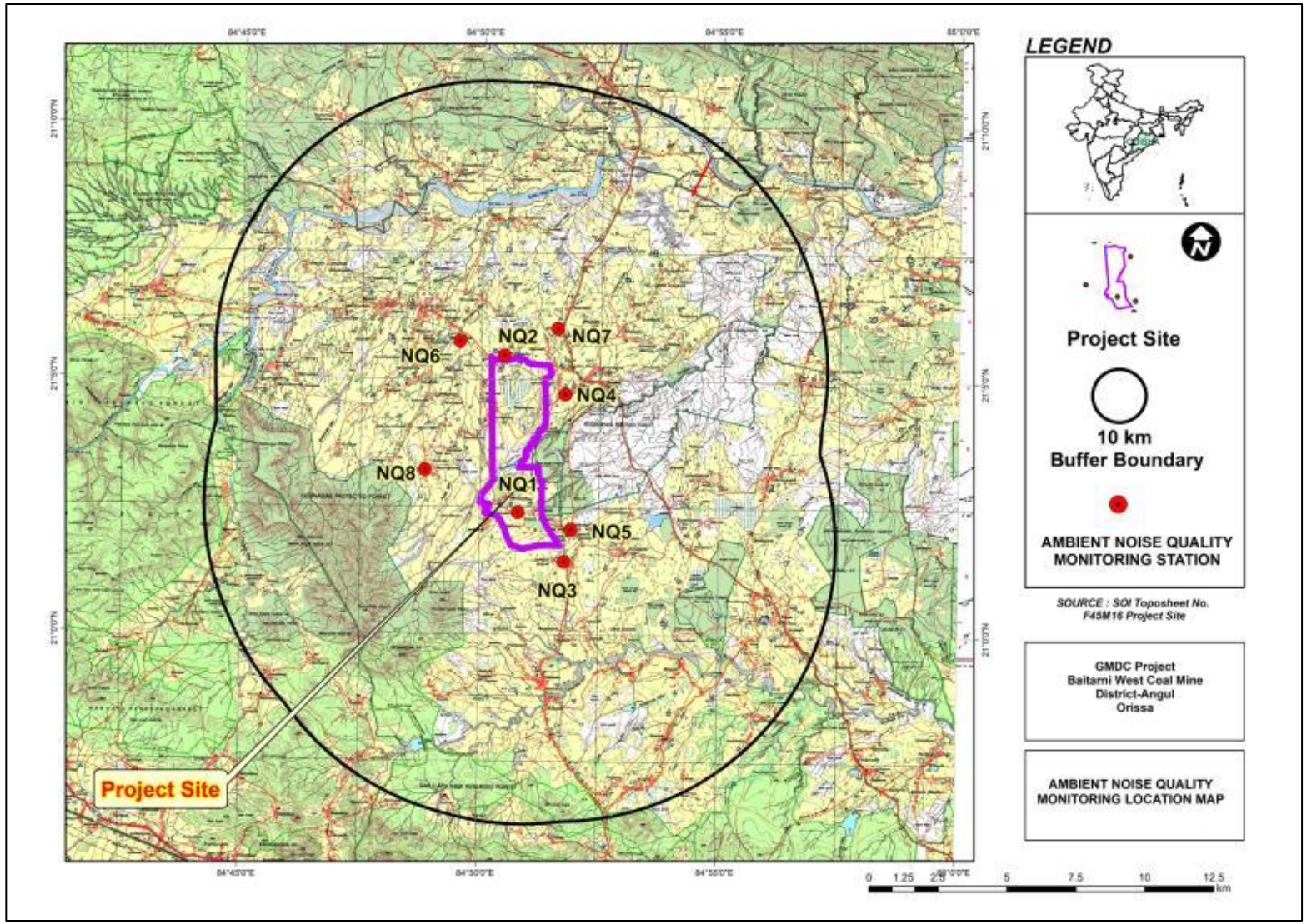


Figure 3-14 Noise Monitoring Locations Map

Table 3:6 Noise Levels in Study Area (Date of Monitoring – 03.10.2023)

Project Site	Zone	Standard Limits (Noise Pollution Regulation & Control, Rules 2000); Leq DB (A)		Observed Value Leq DB(A)	
		Day*	Night **	Day *	Night **
NQ-1-Project Site	Industrial Area	75	70	43.1	38.5
NQ-2-Maa Bayani Wshg, Fps, Podapada	Residential Area	55	45	50.9	40.8
NQ-3-Near Panchayat House, Balikudiajungle	Residential Area	55	45	44.8	39.7
NQ-4-Near Sarpanch House Chhendipadajungle	Residential Area	55	45	43.6	38.8
NQ-5-Tentuloi Panchayat Office	Residential Area	55	45	52.9	43.6
NQ-6-Near Water purification company, Sapoinali	Residential Area	55	45	50.7	39.4
NQ-7- Bhutia Farm, Chhendipadajungle	Residential Area	55	45	43.3	38.7
NQ-8-Near Basantpur Panchyat House	Residential Area	55	45	52.7	42.3
*Day time	(6.00AM TO 10.00PM)				
**Night time	(10.00PM TO 6.00AM)				

Noise standards have been designated for different types of land use i.e. residential, commercial, industrial areas and silence zones, as per 'The Noise Pollution (Regulation and Control) Rules, 2000, Notified by Ministry of Environment and Forests, New Delhi, February 14, 2000. Different standards have been stipulated for day (6 AM to 10 PM) and night (10 PM to 6 AM).

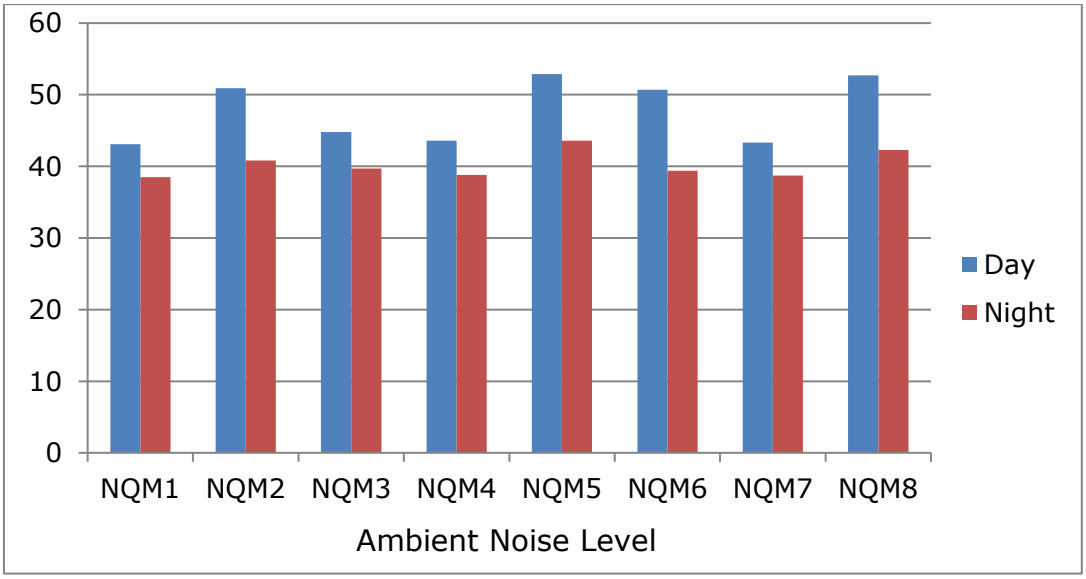


Figure 3-15 Graphical Representation of Noise Level in the Study Area

3.10.1 OBSERVATIONS

The value of noise observed in the residential areas is primarily owing to vehicular traffic. The baseline noise levels have been monitored at different locations as indicated in the table and graphical representation as given above. The noise levels in the area varies from 43.1 to 52.9 dBA during daytime and 38.5 to 43.6 dBA during the night day.

From the results it can be seen that the Day equivalents and the Night equivalents were within the Ambient Noise standards of residential areas standards.

3.10.2 INTERPRETATION OF NOISE DATA

Noise standards have been designated for different types of Noise emitting sources i.e. residential, commercial, industrial areas and silence zones, as per 'The Noise Pollution (Regulation and Control) Rules, 2000, Notified by Ministry of Environment and Forests, New Delhi. Based on noise monitoring data collected from the monitoring locations including project site, it is interpreted that ambient noise quality of studied locations is well within the permissible limits.

3.11 WATER ENVIRONMENT

3.11.1 RECONNAISSANCE SURVEY

The impact has been assessed on randomly selected surface and ground water sources falling within the impact zone. In order to assess the existing water quality, the water samples were collected from eight different locations within the study area and analyzed as per the procedure specified in standard methods for examination of water and wastewater published by American Public Health Association and Bureau of Indian Standards (APHA/BIS).

Selected water quality parameters for water resources of the study area have been used for describing the water environment and assessing the impacts on it. To

assess the water quality impacts, water resources in the impact area have been grouped into 2 classes.

- Ground water resources in the deeper strata of the ground
- Surface water resources

3.11.2 SAMPLING AND ANALYSIS

The water resource in the study area may be classified into two major categories. viz. Surface and Ground water sources. Water samples were collected at 16 demarked locations, out of which 8 samples from ground water sources and 8 samples from surface sources. All the samples were analyzed for parameters as per the procedures specified in Indian Standard (IS)/ APHA such as hardness, alkalinity, dissolve solids, conductivity, inorganic substance, organic substance, heavy metals, microbes etc. Parameters like pH, Temperature and Dissolved Oxygen were analyzed at the time of collection. Results of Ground water samples were compared with Drinking water specification (IS 10500: 2012, Reaffirmed Year: 2018). Surface water samples were also compared with "Water Quality Criteria" published by CPCB for its suitability to use as drinking, outdoor bathing, propagation of wildlife and irrigation purpose.

3.11.3 METHOD OF WATER SAMPLING

The water samples were collected as per the method prescribed in IS-10500:2012, RA 2018. The following procedures were used while water sampling: Before collection of water, the bottles/cans were washed 2-3 times with the same water. For surface water, bottles were lowered to a minimum depth of 30 cm below water surface by using depth sampler. The samples were collected in a sterile sampling container for microbiological analysis. At each point, different sets of water samples were collected so as to cover all the parameters. Dissolved Oxygen is fixed and analyzed in the field itself. Appropriate preservatives were added, depending upon the elements to be analyzed and marked. All the water samples were collected and stored in the ice box and immediately transported to the laboratory.

3.11.4 GROUND WATER AND SURFACE WATER RESOURCES

8 ground water and 8 surface water samples were collected from the study area to assess the water quality of the area during the study period. The ground water samples were drawn from the hand pumps and open wells being used by the villagers for their domestic needs and surface water from the water bodies within 10 km area of project site. Water sampling locations were selected for assessment of the existing status of water environment within the study zone. The details of the locations are given in **Table 3.7.**

Table 3:7 Water Sampling Locations (Ground & Surface Water)

Code	Location Name	Distance	Direction	Latitude	Longitude
GW1	Project Site	-	-	21°2'23.657"N	84°50'48.016"E

GW2	Maa Bayani Wshg, Fps, Podapada	0.17	N	21°4'43.081"N	84°51'46.010"E
GW3	Near Panchayat House,Balikudiajungle	0.6	SE	21°1'25.824"N	84°51'46.276"E
GW4	Near Sarpanch House, Chhendipadajungle	0.7	E	21°5'46.136"N	84°50'31.217"E
GW5	Tentuloi Panchayat Office	0.8	E	21°2'3.684"N	84°51'54.645"E
GW6	Near Water Purification Company,Sapoinali	1.25	NW	21°5'45.325"N	84°49'33.219"E
GW7	Bhutia Farm, Chhendipadajungle	1.4	NE	21°6'0.747"N	84°51'35.747"E
GW8	Near Basantpur Sarpanch House	2.4	W	21°3'13.084"N	84°48'50.774"E

Code	**Location Name**	**Distance**	**Direction**	**Latitude**	**Longitude**
SW1	Kumbhira Nala U/S	0.2	E	21°4'19.842"N	84°51'27.661"E
SW2	Sapui Nala U/S	0.67	N	21°5'42.570"N	84°51'6.390"E
SW3	Gauduni Nala U/S	2.5	E	21°4'25.598"N	84°52'48.660"E
SW4	Kukarpeta Reservoir	3.4	E	21°2'22.680"N	84°53'32.679"E
SW5	Kumbhira Nala D/S	4.7	NE	21°7'18.654"N	84°53'8.690"E
SW6	Aunli Nadi U/S	5.1	N	21°8'11.022"N	84°49'39.247"E
SW7	Aunli Nadi D/S	6.3	N	21°8'37.896"N	84°52'7.371"E
SW8	Gauduni Nala D/S	7	NE	21°6'12.866"N	84°55'25.799"E

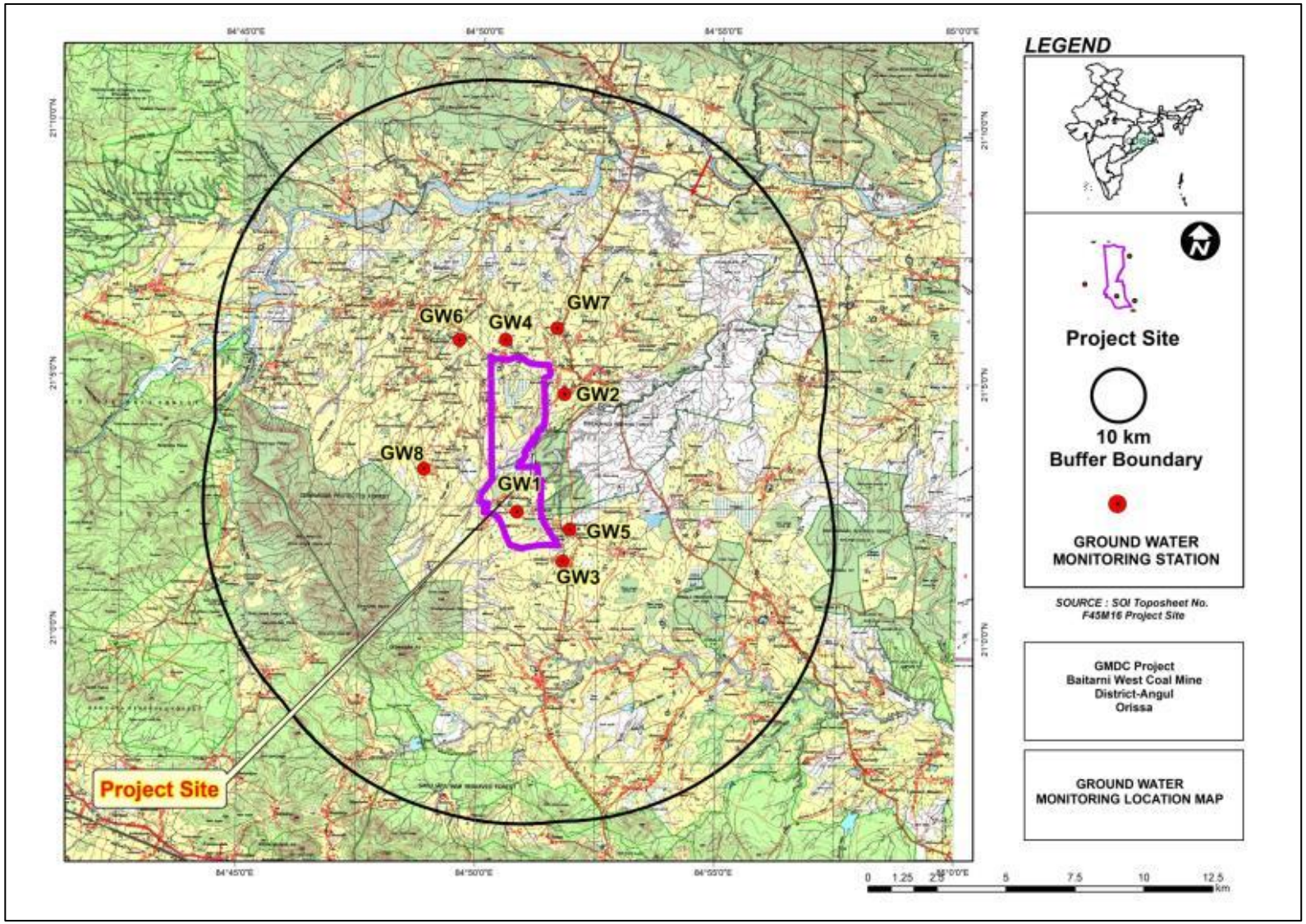


Figure 3-16 Ground Water Quality Sampling Locations

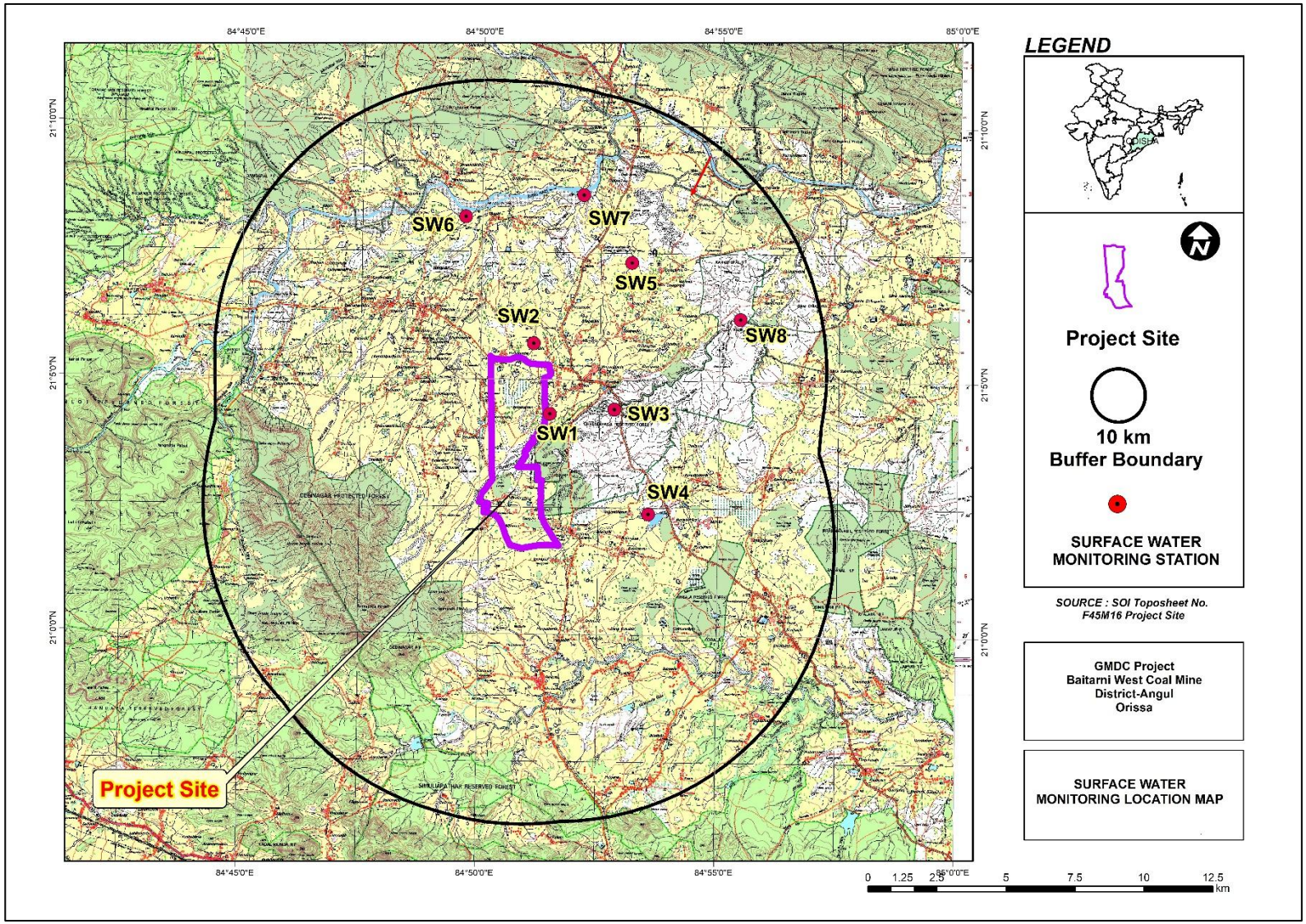


Figure 3-17 Surface Water Quality Sampling Locations

Table 3:8 Ground Water Analysis Result (Date of Sampling – 03.10.2023, Date of Analysis – 06.10.2023 to 20.10.2023)

S. No	Parameter	Unit	Standard Limit (IS-10500:2012, RA 2018)		GW1	GW2	GW3	GW4	GW5	GW6	GW7	GW8
			Desirable Limit	Permissible Limit	Tubewell Near Project site	Maa Bayani Wshg,FPS, Podapada	Near Panchayat House Balikudia Jungle	Near surpanch House Chhendipada Jungle	Tentuloi Panchayat Office	Near Sapoinali Village Water Purification Company	Bhutia Farm Chhendipada Jungle	Near Basantpur Sarpanch House
1	Colour	Hazen	5	15	<5	<5	<5	<5	<5	<5	<5	<5
2	Odour	-	Agreeable	Agreeable	Agreeable	Agreeable	Agreeable	Agreeable	Agreeable	Agreeable	Agreeable	Agreeable
3	Turbidity	NTU	1	5	<1	<1	<1	<1	<1	<1	<1	<1
4	pH	-	6.5-8.5	No Relaxation	7.44	7.41	7.14	7.21	7.43	7.04	7.48	7.09
5	Total Dissolved Solids (TDS)	mg/l	500	2000	730	760	520	495	340	500	180	375
6	Total Hardness	mg/l	200	600	448	493	388	307	246	310	110	263

	(as CaCO3)											
7	Total Alkalinity (as CaCO3)	mg/l	200	600	377	387	296	285	177	297	83	196
8	Chlorides (as Cl)	mg/l	250	1000	113	115	62	65	48	58	30	56
9	Fluoride (as F)	mg/l	1	1.5	0.6	0.7	0.5	0.5	0.7	0.6	0.5	0.7
10	Calcium (as Ca2+)	mg/l	75	200	108	118	93	74	59	74	26	63
11	Magnesium (as Mg2+)	mg/l	30	100	43	47	37	29	24	30	11	25
12	Sulphate (as SO4)	mg/l	200	400	67	78	51	40	36	42	23	34
13	Nitrate (as NO3)	mg/l	45	No Relaxation	7	6	5	8	7	6	5	7
14	Iron (as Fe)	mg/l	1	No Relaxation	0.43	0.44	0.36	0.32	0.33	0.41	0.22	0.33
15	Aluminum (as Al)	mg/l	0.03	0.2	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
16	Copper (as Cu)	mg/l	0.05	1.5	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01

17	Manganese (as Mn)	mg/l	0.1	0.3	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
18	Boron (as B)	mg/l	0.5	1	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
19	Zinc (as Zn)	mg/l	5	15	0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
20	Selenium (as Se)	mg/l	0.01	No Relaxation	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
21	Arsenic (as As)	mg/l	0.01	0.05	<0.01	0.01	0.02	0.01	0.02	0.01	0.01	0.01
22	Cadmium (as Cd)	mg/l	0.003	No Relaxation	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
23	Total Chromium (as Cr3+)	mg/l	0.05	No Relaxation	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
24	Cyanide (as CN)	mg/l	0.05	No Relaxation	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
25	Lead (as Pb)	mg/l	0.01	No Relaxation	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
26	Mercury (as Hg)	mg/l	0.001	No Relaxation	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001

27	Nickel (as Ni)	mg/l	0.02	No Relaxation	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
28	Phenolic Compounds (as C6H5OH)	mg/l	0.001	0.002	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
29	Anionic Detergent (as MBAS)	mg/l	0.2	1	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
30	Silica (as SiO_2)	mg/l	,---	,---	5.8	6.1	4.6	3.5	4.7	4.3	3.9	4.8
31	Phosphate (as PO_4)	mg/l	,---	,---	0.05	0.06	0.03	0.05	0.03	0.02	0.01	0.09
32	Specific Conductivity	µS/cm	,---	,---	1090	1140	780	740	510	750	275	560
Bacteriological Parameter (Date of Sampling – 03.10.2023, Date of Analysis – 06.10.2023 to 09.10.2023)												
1	Total Coliform	MPN/ 100ml	Shall not be detectable in any 100ml Sample		Not Detected (<2)	Not Detected (<2)	Not Detected (<2)	Not Detected (<2)	Not Detected (<2)	Not Detected (<2)	Not Detected (<2)	Not Detected (<2)
2	E. coli	MPN /100 ml	Shall not be detectable in any 100ml Sample		Absent (<2)	Absent (<2)	Absent (<2)	Absent (<2)	Absent (<2)	Absent (<2)	Absent (<2)	Absent (<2)

Table 3:9 Surface water Analysis (Date of Sampling – 03.10.2023, Date of Analysis – 06.10.2023 to 20.10.2023)

S. No.	Parameter	Unit	S.W.1 Kumbhira Nala Upsteam	S.W.2 Sapui Nala	S.W.3 Gauduni Nala Upsteam	S.W.4 Kukarpeta Reservoir	S.W.5 Kumbhira Nala Downsteam	S.W.6 Aunli River Upsteam	S.W.7 Aunli River Downsteam	S.W.8 Gauduni nala Downsteam
1.	Color	Hazen	<5	<5	<5	<5	<5	<5	<5	<5
2.	Turbidity	NTU	14	24	21	10	18	10	12	36
3.	Temperature	^{0}C	22.5	23.2	23.8	22.4	22.3	22.1	22.3	24.9
4.	pH Value	-	7.21	7.17	7.31	7.76	7.64	7.27	7.46	7.09
5.	Dissolved Oxygen	mg/l	4.6	5.5	4.5	5.2	4.3	6.3	6.1	3.2
6.	Biochemical Oxygen Demand (BOD) 3 Days at 27^{0}C	mg/l	3.5	2.9	3.4	3.0	3.7	2.5	2.8	8.2
7.	Chemical Oxygen Demand	mg/l	12	10	13	10.8	14	9.6	10	28
8.	Total Dissolved Solids	mg/l	323	225	320	276	350	152	160	460
9.	Total Hardness (as $CaCO_3$)	mg/l	180	132	180	150	190	63	65	220
10.	Total Alkalinity (as $CaCO_3$)	mg/l	168	110	160	138	174	109	110	230
11.	Chlorides (as Cl)	mg/l	48	40	48	42	53	11	12	76
12.	Fluoride (as F)	mg/l	0.5	0.3	0.4	0.4	0.6	0.2	0.3	0.6
13.	Calcium (as Ca^{2+})	mg/l	43	32	43	36	46	15	16	53
14.	Magnesium (as Mg^{2+})	mg/l	17	13	17	14	18	6	6	21
15.	Sodium (as Na^+)	mg/l	42	26	39	36	46	36	37	75
16.	Potassium (as K^+)	mg/l	4.6	3.5	4.2	3.9	5.2	2.3	2.8	8.4
17.	Sulphate (as SO_4)	mg/l	32	20	34	30	38	8	10	42

18.	Total Kjeldahl Nitrogen (as N)	mg/l	2.1	0.8	1.3	0.9	2.4	1.2	1.3	2.6
19.	Free Ammonia (as NH_3)	mg/l	1.2	1.2	0.6	0.5	1.6	0.2	0.2	1.4
20.	Nitrate (as NO_3)	mg/l	1.7	0.9	1.6	0.7	1.9	0.6	0.7	3.2
21.	Silica (as SiO_2)	mg/l	12.6	8.1	5.4	3.1	14.6	8.6	9.1	18.2
22.	Phosphate (as PO_4)	mg/l	0.9	0.3	0.4	0.4	1.1	0.5	0.6	1.3
23.	Iron (as Fe)	mg/l	0.34	0.31	0.2	0.2	0.53	0.26	0.28	0.6
24.	Aluminum (as Al)	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
25.	Boron (as B)	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
26.	Copper (as Cu)	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
27.	Manganese (as Mn)	mg/l	0.05	<0.01	<0.01	0.03	<0.01	<0.01	<0.01	<0.01
28.	Zinc (as Zn)	mg/l	0.3	<0.01	<0.01	0.25	<0.01	<0.01	<0.01	0.2
29.	Selenium (as Se)	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
30.	Arsenic (as As)	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
31.	Cadmium (as Cd)	mg/l	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
32.	Total Chromium (as Cr^{3+})	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
33.	Cyanide (as CN)	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
34.	Lead (as Pb)	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
35.	Mercury (as Hg)	mg/l	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
36.	Nickel (as Ni)	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
37.	Total Suspended Solids	mg/l	30.6	26	26	26	38.7	32.5	38.2	56
38.	Sodium Adsorption Ratio	-	1	0.7	0.9	0.9	1.1	1.3	1.3	1.6

39.	Phenolic Compounds (as C_6H_5OH)	mg/l	0.002	<0.001	0.003	<0.001	0.003	<0.001	<0.001	0.02
40.	Anionic Detergent (as MBAS)	mg/l	0.01	<0.001	<0.01	0.01	0.02	<0.01	<0.01	0.01
41.	Specific Conductivity	µS/cm	485	340	477	415	520	230	240	690
Bacteriological Parameter (Date of Sampling – 03.10.2023, Date of Analysis – 06.10.2023 to 09.10.2023)										
42.	Total Coliform	MPN/100 ml	1460	350	230	380	1600	940	1120	1600
43.	Faecal Coliform	MPN/100 ml	520	120	29	160	840	320	360	920

3.11.5 SUMMARY OF GROUND WATER QUALITY

The test results were compared with the Drinking Water Specification: **IS-10500:2012, RA 2018** and summarized as under.

- pH was observed in the range of 7.04-7.48 which meets with desirable norms.
- Total dissolved solid were recorded in the range of 180 to 760 mg/l with minimum at GW7 i.e. Bhutia Farm Chhendi and maximum at GW2 i.e. Maa Bayani Wshg,FPS, Podapada.
- Total hardness was in the range of 1110-493 mg/l with minimum at GW7 i.e. Bhutia Farm Chhendi and maximum at GW2 i.e. Maa Bayani Wshg,FPS, Podapada.
- Total Alkalinity was found in the range of 83-387 mg/l with minimum at GW7 i.e. Bhutia Farm Chhendi and maximum at GW2 i.e. Maa Bayani Wshg,FPS, Podapada.
- Iron was found in the range of 0.22-0.44 mg/l with minimum at GW7 i.e. Bhutia Farm Chhendi and maximum at GW2 i.e. Maa Bayani Wshg,FPS, Podapada.
- As microbiological parameters MPN analysis was also carried out and it was found Nil.

3.11.6 INTERPRETATION OF GROUND WATER QUALITY DATA

On the basis of test results, it is conclude that results of all sampling locations met with the desirable limits of Drinking water standard. pH, hardness, alkalinity, chlorides, sulfates and total dissolved solids are below the desirable limits of standard. The Ionic balance computation, considering the relationship between total cations (Ca^{++}, Mg^{++}, Na^{+} and K^{+}) and total anions (HCO_3^{-}, SO_4^{--}, Cl^{-} and NO_2^{--}) for each set of complete analysis of water sample, is observed to be within the acceptability ($\pm$2%) limit. Ground water samples are also compared with the "Water Quality Criteria" published by CPCB, 1979. All the ground water samples are falling under the Category "A" (Drinking water source without conventional treatment but after disinfection).

Comparison of Ground Water Quality Data with the CGWB station at Chhendipada:

The results of baseline ground water analysis data of GW1 and GW2 which are at a distance of 1.78 km and 1.47 km respectively from National Hydrograph Stations located in Chhendipada of CGWB are compared with the Groundwater results (in April 2022) for 13 basic parameters, including pH, Electrical Conductivity, Total Hardness (TH), Alkalinity, Calcium (Ca), Magnesium (Mg), Sodium (Na), Potassium (K), Carbonate (CO3), Bicarbonate (HCO3), Chloride (Cl), Sulphate (SO4), and Nitrate (NO3) to assess according to the guidelines set by the Bureau of Indian Standards (IS 10500:2012).

Ground Water Analysis Result | Values in mg/L

Location	pH	EC	CO3	HCO3	Cl	F	SO4	NO3	Total Hardness	Ca	Mg	Na	K

Chendiapada (CGWB Station)	7.86	780	0	354	60	0.48	5	2	290	66	30	35	17.40
Tubewell Near Project Site (GW1)	7.44	-	-	-	113	0.6	67	7	448	108	43	-	-
Maa Bayani Wshg,FPS, Podapada (GW2)	7.41	-	-	-	115	0.7	78	6	493	118	47	-	-
Desirable Limit	**6.5-8.5**	**-**	**-**	**-**	**250**	**1**	**200**	**45**	**200**	**75**	**30**	**-**	**-**
Permissible Limit	**No Relaxation**	**-**	**-**	**-**	**1000**	**1.5**	**400**	**No Relaxation**	**600**	**200**	**100**	**-**	**-**

- As it is clear from the above table that all the values of all parameters are within the desirable limit except Total Hardness.
- Ca & Mg at GW1 & GW2 though exceeding from the desirable limit but within the permissible limit. However, the result of CGWB station are within the limit.
- Other parameters are comparable with the CGWB results.

3.11.7 SUMMARY OF SURFACE WATER QUALITY

The following description is based on the analysis of the samples:

- During the analysis the pH of the samples was found in the range of 7.09-7.76.
- TDS analysis was also carried out for surface water sample, and it was found in the range of 152-460 mg/L.
- DO measured during analysis was found in the range of 3.2-6.3 mg/L.
- COD measured during analysis was found in the range of 9.6-28 mg/L.
- BOD measured during analysis was found in the range of 2.5-8.2 mg/L.
- MPN test was also carried out for this surface water sample and it was found positive. It indicates towards the fecal contamination in surface water body.

3.11.8 INTERPRETATION OF SURFACE WATER QUALITY DATA

Based on the above test results, it is interpreted that surface water quality does not meet the "Water Quality Standards" as per CPCB, 1972 and falling under the Category "C" (Drinking water source after conventional treatment and disinfection). It is suggest that surface water can be used as alternative source for drinking purposes.

3.12 SOIL ENVIRONMENT

It has been observed that a major part of the area is occupied by Alfisols which includes red sandy soil, red loamy soil and mixed red and black soils. It is porous and friable, tight textured, usually devoid of lime kankars and is also free of carbonates. It is usually suitable for cultivation of paddy and a large variety of other crops. In order to assess the quality of soil in the study area, eight numbers of soil samples were collected during the field survey.

3.12.1 SOIL QUALITY

The present study on soil quality establishes the baseline characteristics in the study area surrounding the project site. The study has been addressed with the following objectives.

- To determine the base line characteristics
- To determine the soil characteristics of proposed project site.
- To determine the impact of industrialization/urbanization on soil characteristics
- To determine the impacts on soils from agricultural productivity point of view.

3.12.2 SOIL SAMPLING LOCATIONS

Details of the soil sampling locations are given in below: -

Table 3:10 Soil Sampling Location

Code	Location Name	Distance (Km)	Direction w.r.t Project Site	Latitude	Longitude
SQ 1	Project Site	-	-	21°2'23.164"N	84°50'47.883"E
SQ 2	Central Office of Sapoisahi	0.04	N	21°5'18.814"N	84°51'17.312"E
SQ 3	Near Chintu Shop, Machhakuta	0.2	NW	21°5'32.886"N	84°50'8.847"E
SQ 4	Jamunali - Club, Handigoda	0.54	SW	21°2'6.068"N	84°50'9.223"E
SQ 5	Tentuloi Panchayat Office, Tentuloi	0.6	E	21°1'59.105"N	84°51'52.095"E
SQ 6	Near Liqour Shop, Chhendipada	0.65	E	21°4'39.216"N	84°51'41.987"E
SQ 7	Near Panchyat House, Chaunriapal	2	W	21°3'56.447"N	84°49'5.994"E

SQ 8	Surya Study Center, Basantapur	2.2	W	21°3'9.838"N	84°48'59.352"E

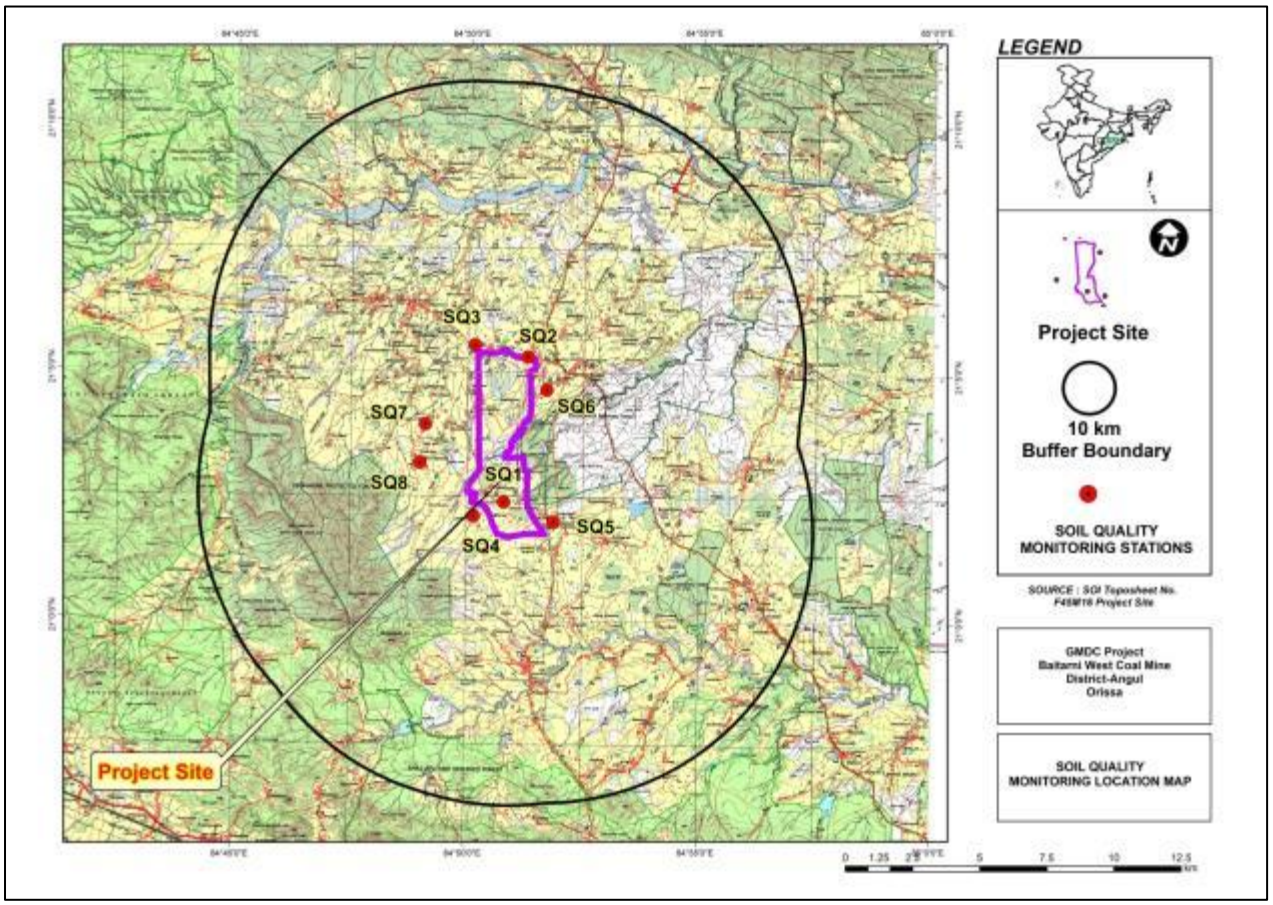


Figure 3-18 Map showing the Study Region with Soil Sampling Locations

Table 3:11 Soil Quality Result (Date of Sampling – 03.10.2023, Date of Analysis – 06.10.2023 to 20.10.2023)

			SQ1	SQ2	SQ3	SQ4	SQ5	SQ6	SQ7	SQ8
S. No.	**Parameter**	**Unit**	**Project Site**	**Central Office of Sapoisahi**	**Near Chintu Shop, Machhakuta**	**Jamunali - Club, Handigoda**	**Tentuloi Panchayat Office, Tentuloi**	**Near Liqour Shop, Chhendipada**	**Near Panchyat House, Chaunriapal**	**Surya Study Center, Basantapur**
1	Texture	-	Sandy Clay Loam	Sandy Loam	Sandy Loam	Sandy Clay Loam	Sandy Loam	Sandy Clay Loam	Sandy Clay Loam	Sandy Loam

2	Particle Size Distribution									
	Sand	%	47.8	63.5	65.7	45.6	65.7	49.6	53.7	63.8
	Silt	%	13.9	18.7	17.5	14.8	16.8	12.8	16.5	17.3
	Clay	%	38.3	17.8	16.8	39.6	17.5	37.6	29.8	18.9
3	pH (1:2 Suspension)	-	6.98	7.16	6.93	7.17	6.96	7.17	7.09	7.16
4	Electrical Conductivity (1:2 Suspension)	µS/cm	269	263	257	228	287	218	268	277
5	Moisture Content	%	13.3	10.1	10.9	12.7	13.9	11.7	11.8	10.7
6	Cation Exchange Capacity	meq/100gm	13.9	13.5	13.6	13.8	13.4	13.9	14.1	13.5
7	Available Potassium (as K)	mg/kg	40	47	49	39	49	43	38	47
8	Exchangeable Sodium (as Na)	mg/kg	69	78	83	63	72	69	77	67
9	Exchangeable Calcium (as Ca)	mg/kg	2093	1993	1983	2047	1977	2054	2108	1987
10	Exchangeable Magnesium	mg/kg	368	374	379	387	375	387	376	383

	(as Mg)									
11	Sodium Absorption Ratio	-	0.36	0.42	0.45	0.33	0.39	0.37	0.40	0.36
12	Organic Matter	%	0.94	0.65	0.63	0.95	0.57	0.89	0.83	0.69
13	Total Nitrogen (as N)	mg/kg	43	39	38	42	44	46	48	41
14	Nitrate (as NO_3)	mg/kg	5.2	4.8	4.3	5.3	4.6	5.5	5.1	4.7
15	Total Phosphorus (as PO_4)	mg/kg	6.3	5.9	5.5	6.1	5.3	6.1	6.2	5.3
16	Iron (as Fe)	mg/kg	4.6	3.8	3.6	4.3	2.9	5.6	4.7	3.4
17	Zinc (as Zn)	mg/kg	1.9	1.3	1.1	1.8	1.3	1.7	2.1	1.1
18	Copper (as Cu)	mg/kg	2.1	0.9	0.8	2.3	1.1	2.3	1.9	0.9
19	Boron (as B)	mg/kg	1.7	1.2	1.1	1.6	1.3	1.7	2.1	1.1
20	Manganese (as Mn)	mg/kg	19.6	13.9	15.6	23.7	9.6	22.8	19.1	13.9
21	Water Holding Capacity	%	32.8	25.8	25.1	32.4	25.9	32.7	32.8	26.3
22	Permeability	cm/hr	1.9	2.3	2.4	1.9	2.4	2.1	2.1	2.3
23	Porosity	%	28.7	39.8	40.6	28.4	39.8	28.4	28.9	40.8
24	Bulk Density	gm/cc	1.38	1.33	1.32	1.36	1.33	1.38	1.36	1.31

3.12.3 OBSERVATIONS

The soil sampling was carried out at 8 locations. Data computations for 8 sampling locations were carried out and summarized as under: based on particle sizes distribution and texture, the soils are mostly sandy loam and sandy clay loam category. The pH value of the soil suspension varied from 7.58 to 7.94. In terms of soil pH the characteristic of the soil is moderately alkaline pH in nature. The Electrical conductivity varied from 218 to 287µMohs/cm. The sodium absorption ratio of soil varied from 0.33 to 0.45. The Cation exchange capacity varied from 13.4 to 14.1 meq/100 gm. The loss on ignition in terms of organic matter varied from 0.57 to 95%, it indicates that soils are medium in organic carbon status. The major nutrient such as Nitrogen, Phosphorus and Potassium level were varied from 38kg/ha to 48 kg/ha., 5.3 to 6.3 kg/ha and 38 to 49 kg/ha respectively. The micronutrients such as copper, zinc, boron and iron are minimum and sufficient for plantation.

Table 3:12 Soil Characteristics

Parameter	Units	Concentration	Level
pH Value	--	<4.5	Very low
		4.5–6.0	Low (Acidic)
		6.0–7.0	Moderate (Normal)
		7.0–8.3	Moderate (Weakly alkaline)
		>8.3	Very High (Alkaline)
Nitrogen(N)	(kg/ha)	<100	Low
		>100-150	Good
		>150-300	Better
Phosphorus(P_2O_5)	(kg/ha)	00-20	Low
		>20-50	Medium
		>50-80	High
Potassium(K_2O)	(kg/ha)	00-150	Low
		151-300	Medium
		>300	High
Organic Carbon	(%)	<0.4	Low
		0.4-0.5	Medium
		0.51-0.8	Average
		2	Preferred

3.12.4 INTERPRETATION OF SOIL ANALYSIS DATA

Based on soil analysis data it is concluded that soils are less alkaline in nature at all sampling locations. The organic carbon status is moderate. The soils are also low in nutrients such as nitrogen, phosphorus and potassium. However, for successful plantation and green belt development the sufficient quantity of organic manure and the quantity of recommended doses of fertilizers should be added.

3.13 LAND USE PATTERN OF STUDY AREA

Land Use/ Land Cover mapping (using Remote Sensing & GIS)

3.13.1 INTRODUCTION

The land use/land cover information relates to the status, spatial distribution & area extent of different landcover/landuse categories. Land cover and land use are terms that are often used interchangeably; however, they have different meanings. Land cover results from a complex mixture of natural and anthropogenic influences and is the composition and characteristics of land surface elements (Cihlar, 2000). In contrast, land use is characterized by economic uses of land and people's relationships with the environment (Avery and Berlin, 1992). For example, a land cover of forest, when considered as a land use, could be a park. To classify land use with satellite imagery often requires the use of supplementary information such as fieldwork. The land cover/ land use classification system is based on the methodology given in 'Manual of Nationwide land use/land cover mapping using Digital Techniques'.

Remote sensing is the science and art of obtaining information about an object, area or phenomena through the analysis of data acquired by a device that is not in contact with the object, area, or phenomena under investigation. The data analysis process involves examining the data using various image processing techniques by a digital computer. Its application in the field of environmental management is of great prominence. The inherently digital nature of remotely sensed data, supporting quantitative & statistical analysis of spectral measurement, led to rapid advancement in the field of digital techniques. With view to facilitate utilization of this modern technology in the management of the resources, a chain of digital image processing steps have been carried for the land use/land cover mapping. The entire investigations have been carried out using ERDAS Imagine digital image processing software and Geographic Information System (ARC GIS).

Satellite Data:

- Landsat 8 Path-140/Row-46 of 30m resolution.

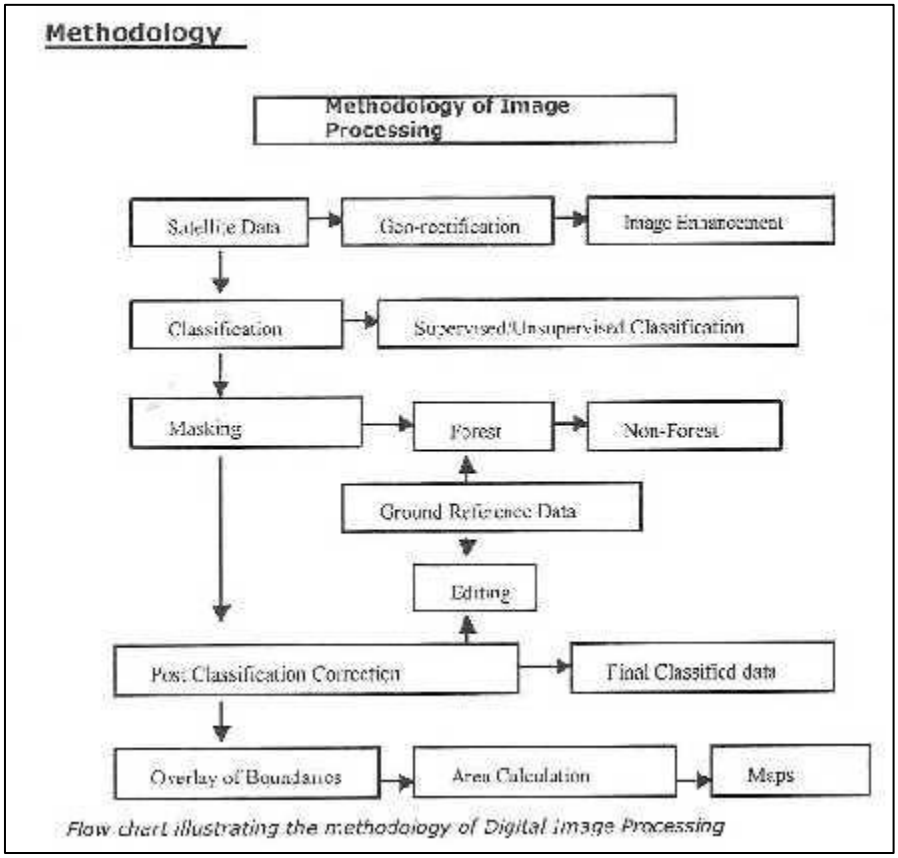
Collateral Data:

- Survey of India Topographical Map
- Ground Truth Information
- Other Collateral Information

Methodology:

Research on remote sensing has been directed for several decades towards image processing & development of methods for digital map generation especially on land use/land cover. The primary aims were to produce thematic maps that could be quickly updated. However, maps obtained from digital automatic classification fails to fully satisfy the purpose for which it is generated. Therefore, digital

classification procedure has been used for generation of maps on land use/land cover from satellite data. The technique is based on stratified approach.



Flow chart illustrating the methodology of Digital Image Processing

The overall methodology for land use/land cover map generation is explained in the flowchart.

ArcGIS Software was used for digital processing of spatial data. Digital image processing techniques were applied for the mapping of the land use/land cover classes of the provided area from the satellite data. The methodology applied comes under following steps:

Image Extraction:

Satellite imageries of study area was obtained from USGS and a subset for the Area of Interest was created through ArcGIS software.

Geo-Rectification:

Geometric correction includes correction for geometric distortions due to sensor, earth geometry variations and conversion of the data to real world coordinates (e.g. Latitude and Longitude) on the Earth's surface. The satellite imagery was geometrically rectified with reference to the geo-referenced topo sheets and vector data.

Image Enhancement:

Image enhancement is one of the important image processing functions primarily done to improve the appearance of the imagery to assist in visual interpretation and analysis. Various options of image enhancement techniques were tried out to get the best image for visual interpretation. Histogram equalized stretch enhancement techniques was applied to the imagery of the study area for better interpretation of different features in the satellite imagery.

Landsat 8 Path-140/Row-46 of 30m resolution has been used for digital classification of land use categories. The subset area of 10 kms radius has been stratified by generating forest mask from topographical map. In non-forest area, the un-supervised classification has been applied. In this particular type of classification spectral classes are grouped first, based solely on the numerical information in the data, and are then matched by the analyst to information classes. Unsupervised classifiers do not utilize training sets as the basis for classification. Rather it involves algorithms called clustering algorithms, that examine the unknown pixels in an image and aggregate them into a number of classes based on the natural groupings or clusters present in the image values. The analyst specifies the desired number of classes. Thus, unlike supervised classification, it does not start with a pre-determined set of classes, however it is neither done completely without human intervention. Forest area has been classified by using NDVI density slicing method. The cultural features like roads, villages and forest boundaries have been drawn from the existing maps.

The land use classified through un-supervised classification of the area and the cultural features of roads, rail and village locations have been overlaid. The land use/ land cover map of the area has been extracted using 10 km. radius mask and area statistics have been generated. The 10 km radius land use /land cover map is shown in Figure 3.21**.**

3.13.2 GENERAL LAND USE/ LAND COVER (10 KM RADIUS AREA)

The land use/ land cover map has been generated on 1:50,000 scale using digital classification of Landsat utilized wherever needed. Based on the methodology developed for the present land use/ land cover, categories have been grouped under following:

Table 3:13 Major land use/land cover categories of the study area

Sl. No.	Category	Area (in ha)	% of the Study area
1	Agricultural land	31439.58	62.08
2	Forest land	9438.82	18.64
3	Open Scrub	5648.49	11.15
4	Settlement	2022.23	3.99

5	Plantation	1288.96	2.55
6	Water Bodies	686.28	1.36
7	Sand	117.59	0.23
	TOTAL	**50641.95**	**100**

Agricultural land:

Based on satellite data and ground truth, the total agricultural land, are classified by using image classification techniques. Existing agricultural area were depicted by utilizing multispectral satellite data. The total agricultural area which is a major part of the study area is about 31439.58 hectares or 62.08 percent of the total study area.

Forest land:

This represents the area under forest which is around 9438.82 hectares or 18.64 percent of the study area. This includes notified forest area and other natural vegetation.

Open Scrub:

Scrub lands with or without scrub can be classified under culturable wastelands and Barren rocky area could be classified into non-culturable lands. Scrub lands are quite similar with degraded forest in image characteristics. These lands are further examined on the ground and cross checked with the forest boundaries to classify into scrub lands. Scrub lands are 5648.49 hectares or 11.15 % of the study area.

Built-up area:

Built-up land includes the urban or rural settlements, roads and railway line etc. No typical spectral signature of settlements could be observed so as to classify under built-up land use. The village locations and their area extent have been extracted from the Landsat data and also from the existing topographical maps. The area occupied by built-up class shown in the classified image is therefore based on the visual interpretation of high-resolution satellite data and topographical maps. The major built-up area is about 2022.23 hectares or 3.99%.

Plantation:

The area covered under plantation is considered under this category. An area of 1288.96 hectares or 2.55 % of the study area is covered under this category.

Water Bodies:

Based on satellite data and ground truth, the total area covered by the river or water bodies is 686.28 hectares or 1.36 % of the total study area.

Sand:

Mostly the riverbed without water is taken under this category. It occupies 117.59 hectares or 0.23 % of the study area.

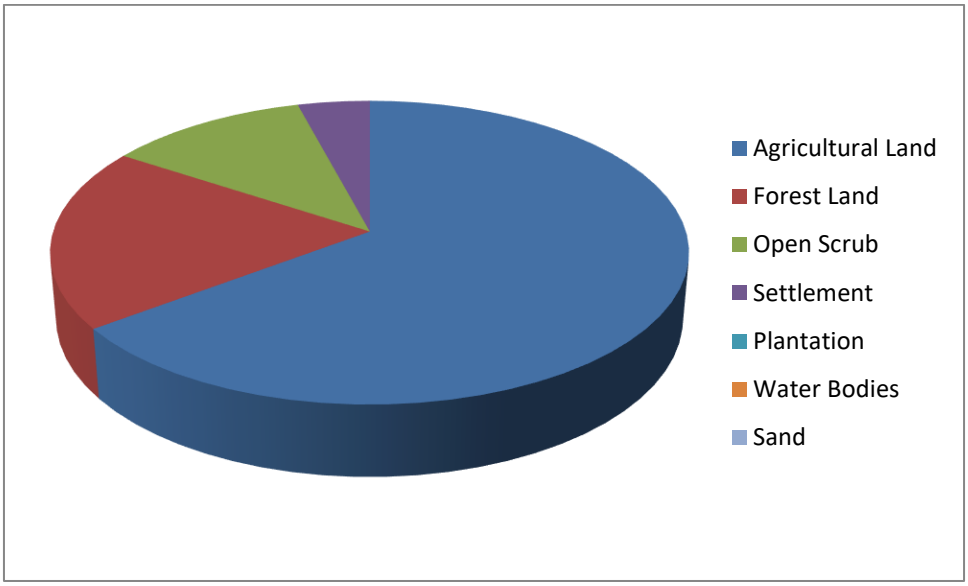


Figure 3-19 Major land use/land cover categories of the study area

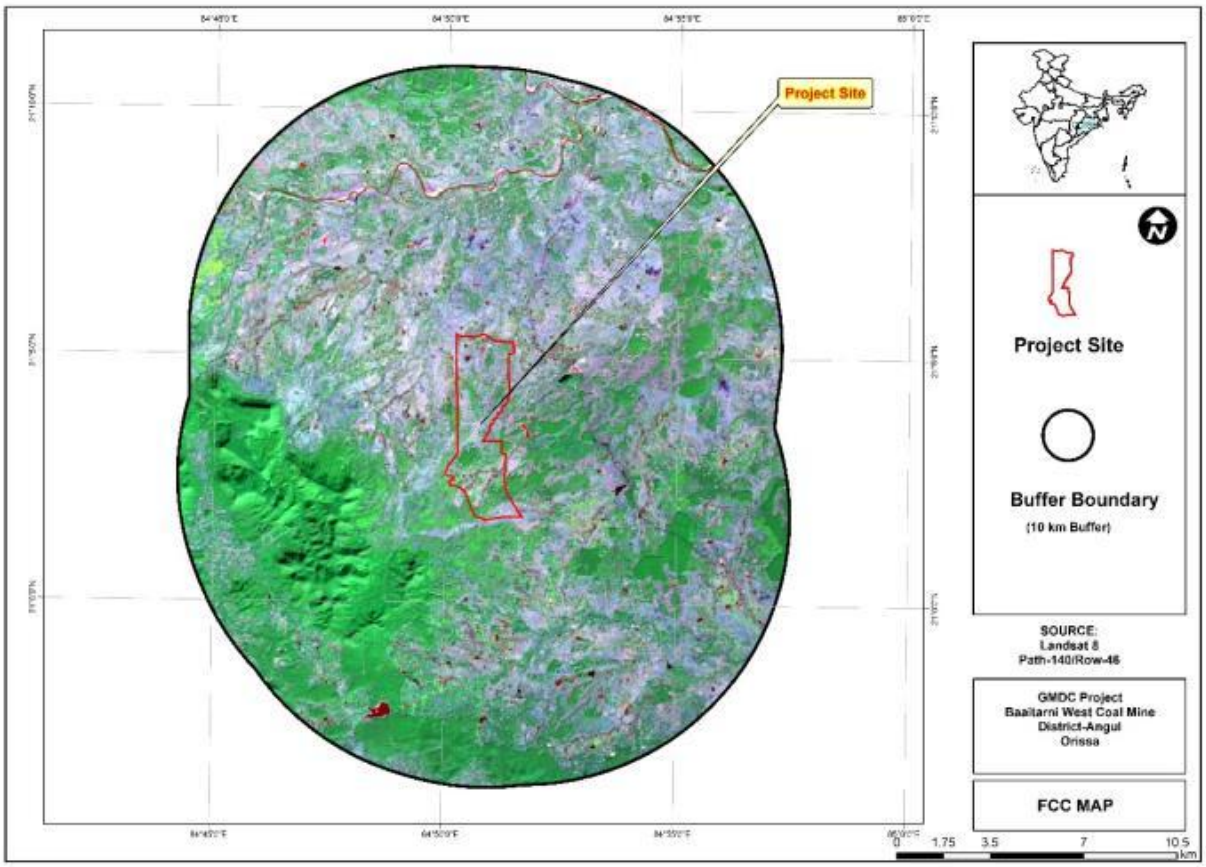


Figure 3-20 FCC Map of the study area

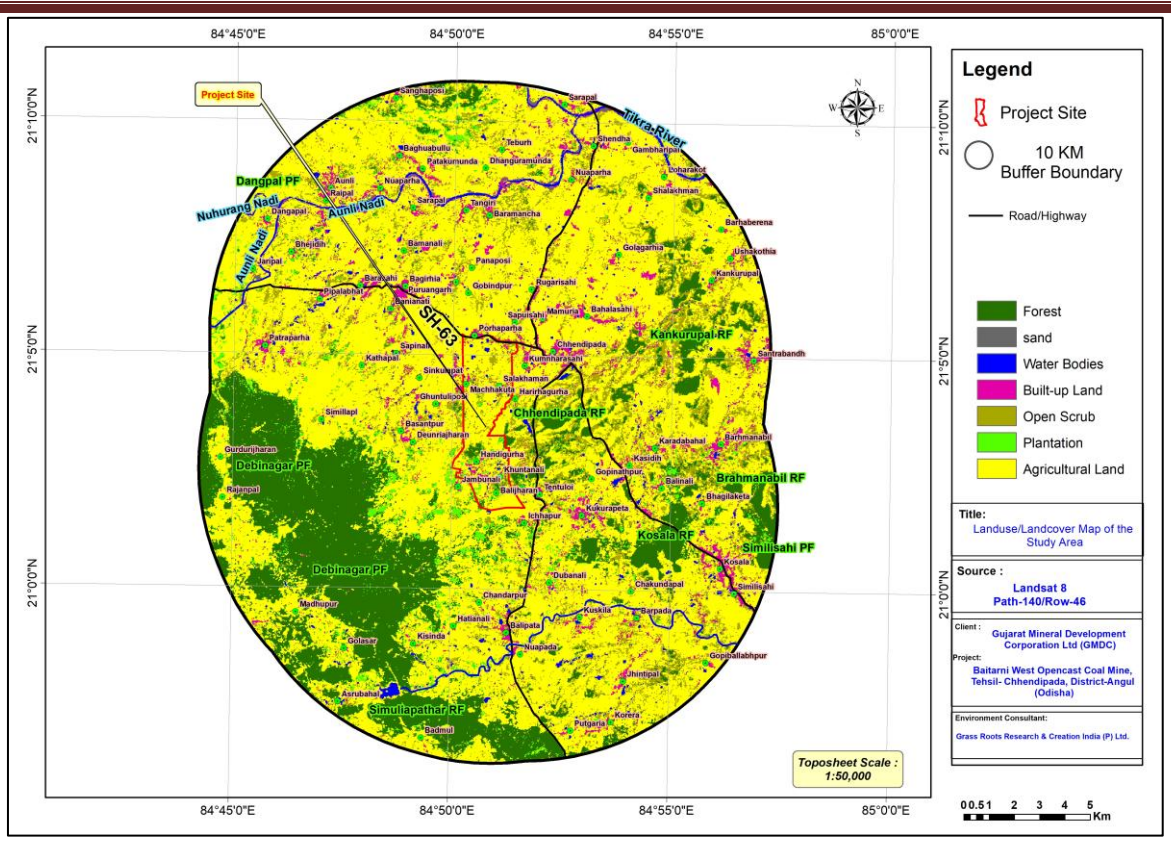


Figure 3-21 LULC Map of the study Area

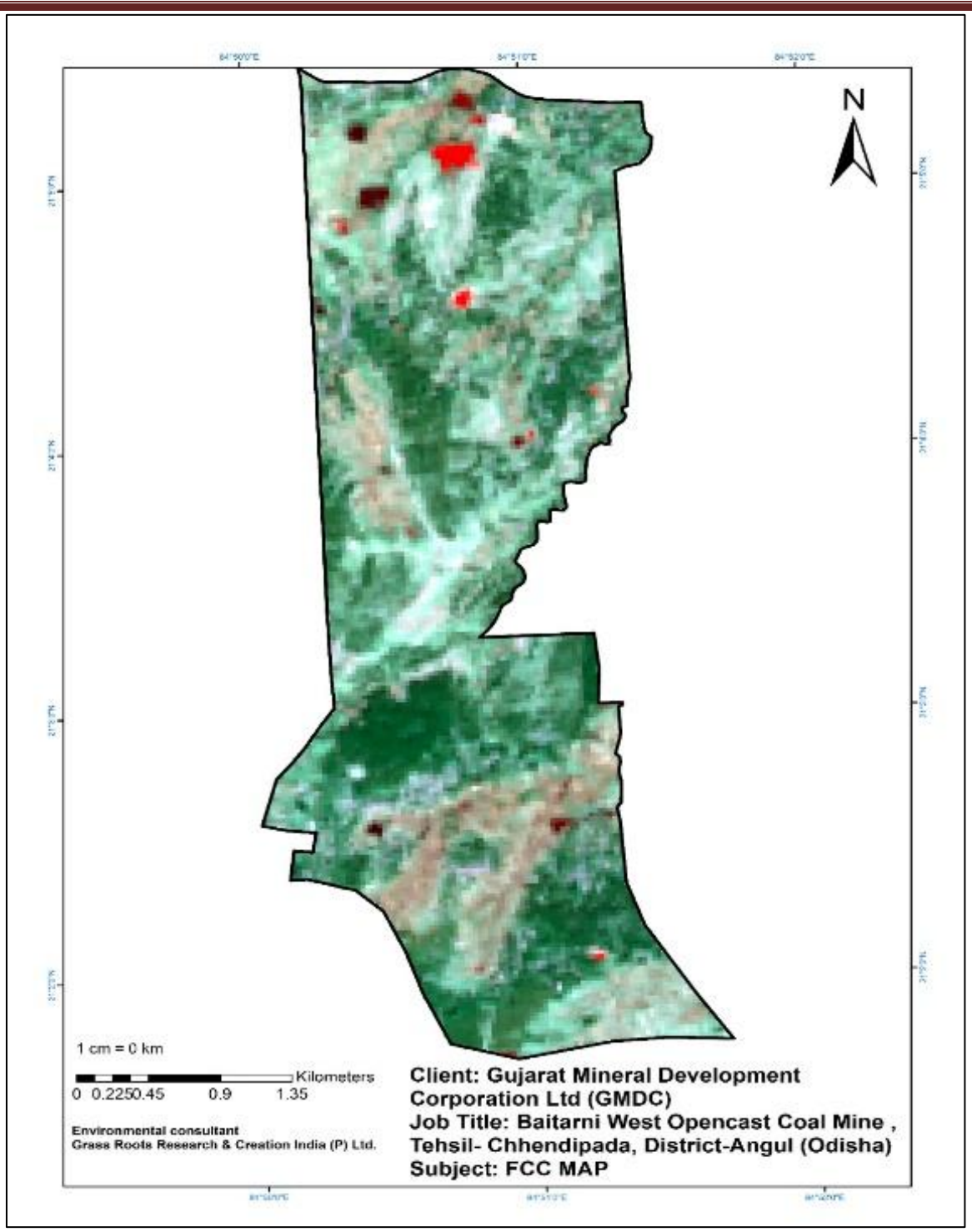


Figure 3-22 FCC Map of the Site

Table 3:14 LULC of the Lease area

LU/LC Class	Area (ha.)
Road	31.4086
Barren	44.3739
Community	5.3926
Grazing	118.8397
Reserve Fores	4.469
Revenue Forest	59.9033
Home Stead	32.7729
Agricultural	784.8499
Water Bodies	35.1422
Others	78.8479
TOTAL	**1196**

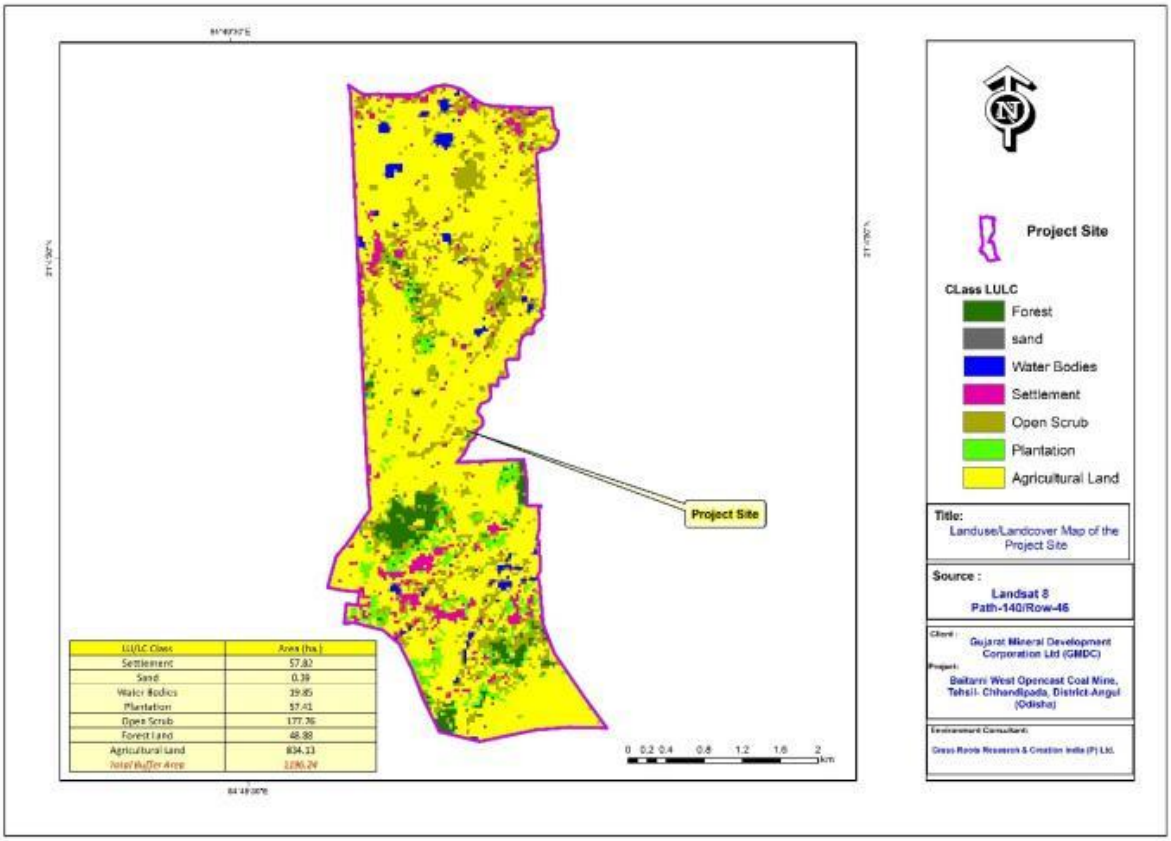


Figure 3-23 LULC Map of the Project Site

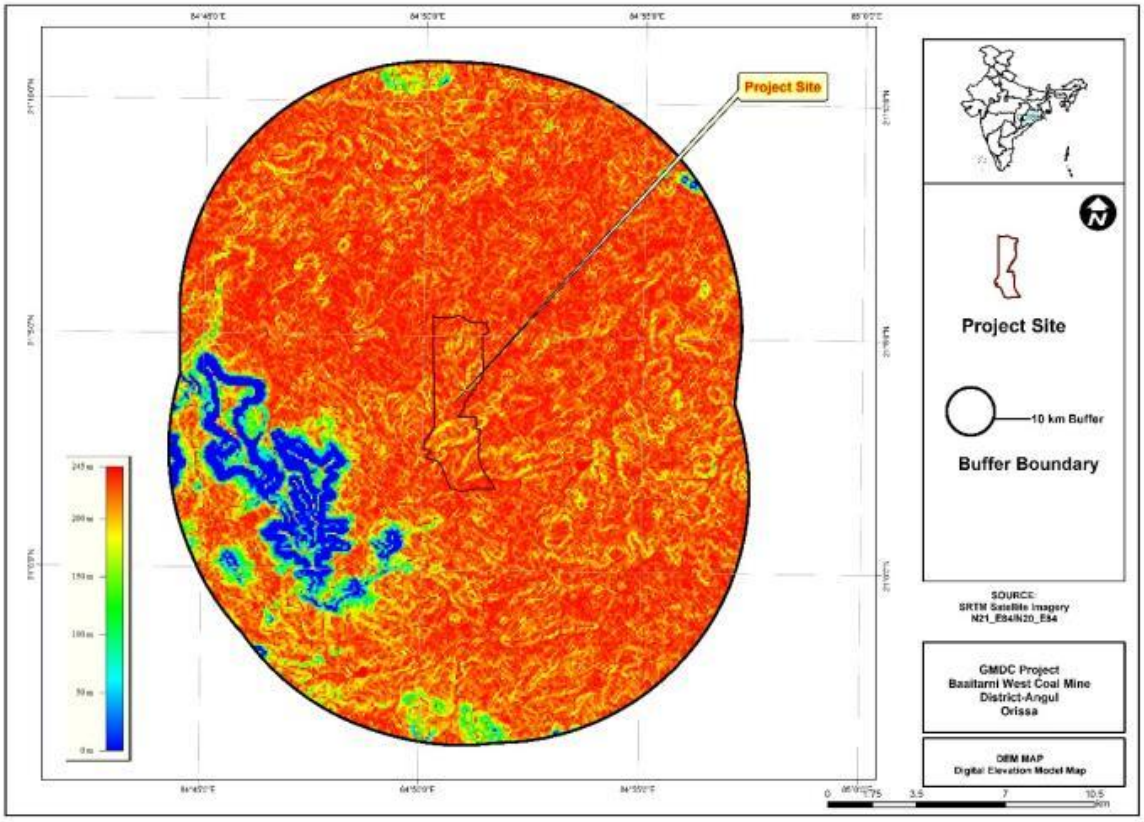


Figure 3-24 DEM Map of the study area

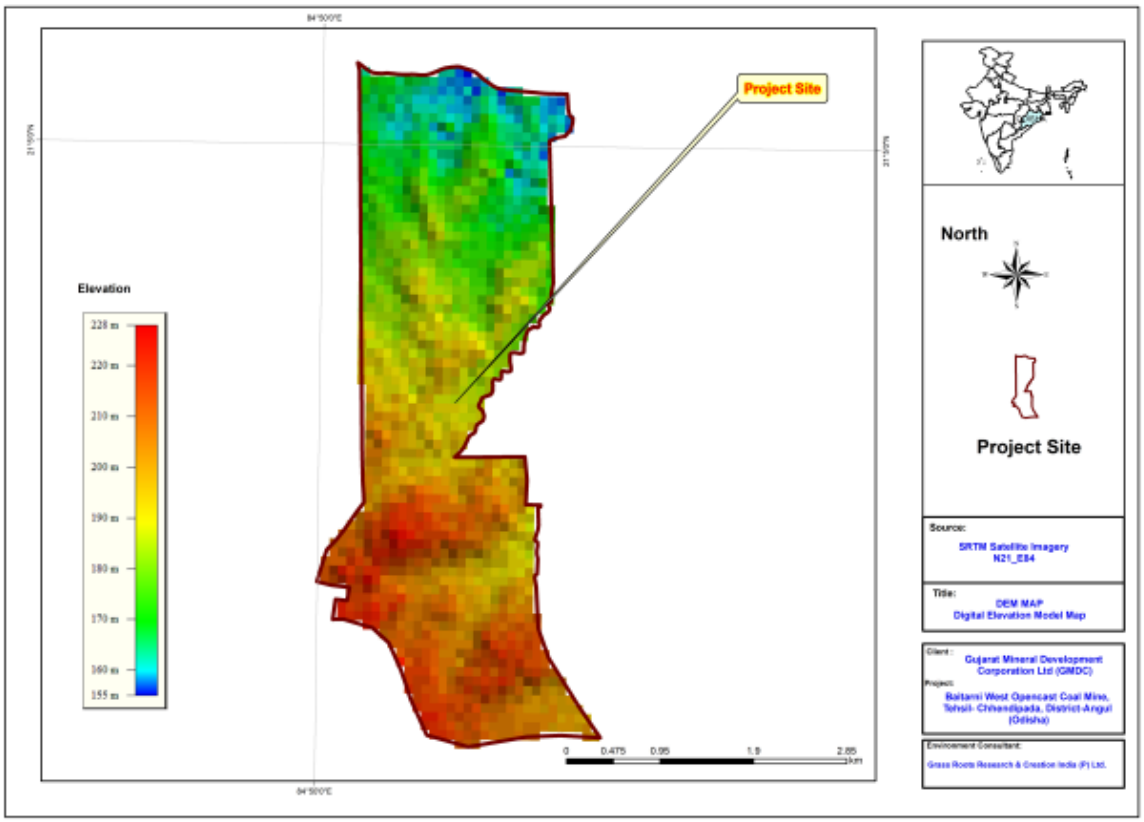


Figure 3-25 DEM Map of the lease area

3.14 SOCIO-ECONOMICS

3.14.1 DESCRIPTION OF THE STUDY AREA (10 km RADIUS AREA)

The study area, also known as Impact Area, has been defined as sum total of core and buffer area with a radius of 10 kilometers from the periphery of the project site. The study is based on secondary data mainly collected from Census 2011 and primary data collected through field study and Field Group Discussions.

The primary survey was conducted through a Questionnaire specially designed to capture firsthand, on the spot, information.

The study area comprises of district Angul in state Odisha.

3.14.2 BASE LINE DATA

Baseline data refers to basic information collected before a project is put under implementation. It is used later to provide a comparison for assessing the impact of the project. The impact may be positive or negative. It may be direct or indirect. Also, there may be no impact and the status will remain the same, unaltered.

Baseline data allows the project proponent to compare the behavior of the project before and after its formal implementation. It helps him/her to conclude whether the interventions are working positively or not.

A baseline study is conducted before the project starts. Any attempt to collect base line data while undertaking actual impact assessment study, is faced with recall error. Hence, the baseline data should be for the period prior to the present study. In the present case the base line data has been collected from secondary and primary sources. It consists of demographic particulars of the study area and various amenities available to the local people.

Socio-Economic Scenario- Rural area (based on Census 2011 data)

The study area comprises of 66 census villages with a total population of 196493. This includes rural sub-district Chhendipada having population 103804. 17 villages fall in core zone (within 2 km from center of site),16 villages fall between 2 to 5 km from site and remaining 33 villages are beyond 5 km.

Demographic features of study area: Rural area

The villages falling within the study area are listed below along with the demographic features as per the Census 2011.

Table 3:15 Summary of Demographic features of Rural area under study

SL.NO.	VILLAGE	No_HH	TOT_P	TOT_M	TOT_F	P_06	M_06	F_06	P_SC	M_SC	F_SC	P_ST	M_ST	F_ST	P_LIT	M_LIT	F_LIT
	VILLAGES IN 0 TO 2 KM																
1	Balijharan	55	227	125	102	37	23	14	1	1	0	219	118	101	150	93	57
2	Chhendipada	1296	5510	2858	2652	515	279	236	1255	628	627	166	82	84	3880	2272	1608
3	Chhendipada -sub dist	24523	103804	53195	50609	11617	6195	5422	21599	10840	10759	9175	4599	4576	69194	40055	29139
4	Chhendipadajungle	403	1806	937	869	194	106	88	104	51	53	47	26	21	1197	696	501
5	Deuria jharan	287	1244	612	632	195	94	101	513	254	259	297	140	157	728	413	315
6	Ghuntulipasi	233	976	487	489	111	60	51	381	188	193	26	9	17	692	390	302
7	Gobindpur	352	1460	711	749	157	84	73	449	212	237	252	129	123	901	496	405
8	Handigoda	363	1389	703	686	140	78	62	249	128	121	303	149	154	906	521	385
9	Ichhapur	354	1316	680	636	153	80	73	233	127	106	80	38	42	1009	565	444
10	Jamubahali	243	1212	626	586	153	87	66	218	108	110	138	68	70	833	481	352
11	Machhakuta	524	2184	1093	1091	220	108	112	382	180	202	186	95	91	1503	835	668
12	Machhakutajungle	38	177	83	94	14	6	8	49	22	27	27	10	17	126	61	65
13	Podapada	430	1841	922	919	204	96	108	636	315	321	171	84	87	1346	752	594
14	Podapadajungle	12	51	25	26	5	3	2	0	0	0	0	0	0	42	22	20
15	Salakhaman	132	735	360	375	110	59	51	14	6	8	647	314	333	424	235	189
16	Tentoloi	304	1384	640	744	181	85	96	948	481	467	220	54	166	1067	517	550
17	Tentuloikodasahi	338	1424	739	685	180	101	79	166	86	80	244	126	118	903	542	361
	SUB TOTAL	29887	126740	64796	61944	14186	7544	6642	27197	13627	13570	12198	6041	6157	84901	48946	35955
	VILLAGES IN 2 TO 5 KM																
18	Bahalsahi	356	1510	786	724	142	86	56	147	82	65	152	85	67	1074	623	451
19	Balipata	1165	4680	2369	2311	530	273	257	1167	582	585	299	144	155	3180	1823	1357
20	Bamanali	53	204	106	98	12	8	4	0	0	0	0	0	0	153	89	64
21	Baramancha	360	1414	705	709	192	91	101	332	169	163	58	32	26	838	479	359
22	Barasahi	278	1181	617	564	130	78	52	375	202	173	4	2	2	802	463	339
23	Basantapur	194	742	382	360	76	35	41	20	11	9	23	13	10	555	328	227
24	Chandrapur	161	681	327	354	80	44	36	125	59	66	252	114	138	463	262	201
25	Dubanali	211	820	427	393	98	61	37	88	45	43	0	0	0	553	330	223
26	Gopinathpur	93	412	201	211	81	41	40	101	54	47	237	113	124	205	113	92
27	Hatianali	194	846	441	405	99	43	56	47	22	25	535	271	264	477	293	184
28	Kasidiha	113	526	276	250	66	33	33	14	7	7	255	135	120	314	194	120
29	Kukurpeta	538	2213	1135	1078	270	149	121	372	185	187	69	36	33	1379	809	570
30	Kuskila	522	2305	1170	1135	270	152	118	616	299	317	46	22	24	1490	846	644
31	Panapasi	35	144	80	64	16	10	6	0	0	0	144	80	64	90	63	27
32	Sapoinali	752	2919	1508	1411	300	159	141	172	80	92	206	101	105	2009	1172	837
33	Tangiri	325	1348	674	674	172	90	82	365	167	198	0	0	0	878	487	391
	SUB TOTAL	5350	21945	11204	10741	2534	1353	1181	3941	1964	1977	2280	1148	1132	14460	8374	6086
	VILLAGES BEYOND 5 KM																
34	Ashrubahal	139	596	303	293	89	56	33	80	43	37	346	174	172	278	164	114
35	Badaberena	158	810	398	412	100	48	52	101	54	47	661	320	341	512	304	208
36	Balinali	175	725	382	343	73	37	36	245	131	114	0	0	0	510	301	209
37	Barapada	548	2230	1171	1059	249	141	108	399	210	189	177	86	91	1395	850	545
38	Bhagilakata	140	583	299	284	61	33	28	185	93	92	87	40	47	385	218	167
39	Bhejidihi	45	208	103	105	22	11	11	0	0	0	0	0	0	140	80	60
40	Brahmanbil	669	3061	1600	1461	322	181	141	620	322	298	0	0	0	2140	1257	883
41	Chakundapal	50	193	103	90	15	8	7	0	0	0	158	81	77	154	86	68
42	Dangapal	391	1644	858	786	200	114	86	260	138	122	296	155	141	1061	624	437
43	Dhangurumunda	16	70	38	32	6	5	1	11	6	5	0	0	0	50	29	21
44	Golagadia	216	909	459	450	127	67	60	224	114	110	175	87	88	596	331	265
45	Golasar	107	433	203	230	69	35	34	11	7	4	0	0	0	306	152	154
46	Gopabalhabpur	15	55	9	46	4	0	4	0	0	0	0	0	0	12	4	8
47	Gundurijharan	35	155	80	75	19	9	10	42	24	18	0	0	0	108	61	47
48	Jaripal	117	523	264	259	63	33	30	125	67	58	32	15	17	306	174	132
49	Jhintipal	468	1872	949	923	262	143	119	330	164	166	257	130	127	1157	664	493
50	Kankarapal	130	517	262	255	62	34	28	155	74	81	0	0	0	296	164	132
51	Karadabahal	249	1035	528	507	102	54	48	199	102	97	0	0	0	683	391	292
52	Kisinda	145	567	292	275	57	27	30	79	39	40	205	106	99	416	236	180
53	Korada	635	2563	1324	1239	368	187	181	557	278	279	351	191	160	1500	886	614
54	Kosala	1294	5455	2865	2590	590	321	269	865	447	418	235	121	114	3996	2295	1701
55	Madhupur	422	1772	905	867	183	92	91	302	147	155	172	93	79	1282	707	575
56	Nuapada	1010	4409	2262	2147	494	280	214	836	419	417	282	137	145	3117	1752	1365
57	Patakumunda	298	1266	666	600	132	78	54	450	230	220	8	5	3	931	525	406
58	Patrapada	1025	4132	2128	2004	475	249	226	1072	545	527	335	172	163	2774	1609	1165
59	Purunagarh	332	1371	687	684	115	53	62	159	75	84	0	0	0	932	543	389
60	Putagadia	161	818	408	410	119	59	60	55	24	31	432	217	215	540	316	224
61	Raipal	279	1224	621	603	203	107	96	90	48	42	635	306	329	687	397	290
62	Rajanpal	48	177	91	86	23	10	13	0	0	0	73	35	38	102	58	44
63	Santarabandha	621	2469	1252	1217	258	140	118	235	118	117	162	82	80	1660	960	700
64	Sarapal	236	940	493	447	107	55	52	179	92	87	0	0	0	669	397	272
65	Similipal	551	2493	1238	1255	288	138	150	246	118	128	547	279	268	1594	892	702
66	Similisahi	594	2533	1314	1219	301	171	130	608	301	307	418	206	212	1674	976	698
	SUB TOTAL	11319	47808	24555	23253	5558	2976	2582	8720	4430	4290	6044	3038	3006	31963	18403	13560
	TOTAL	46556	196493	100555	95938	22278	11873	10405	39858	20021	19837	20522	10227	10295	131324	75723	55601
	AVERAGE	705	2977	1524	1454	338	180	158	604	303	301	311	155	156	1990	1147	842

Contd.

SL.NO.	VILLAGE	P_ILL	M_ILL	F_ILL	TOT_WORK_P	TOT_WORK_M	TOT_WORK_F	MAINWORK_P	MAINWORK_M	MAINWORK_F	MAIN_CL_P	MAIN_CL_M	MAIN_CL_F	MAIN_AL_P	MAIN_AL_M	MAIN_AL_F
	VILLAGES IN 0 TO 2 KM															
1	Balijharan	77	32	45	137	76	61	59	57	2	52	52	0	4	2	2
2	Chhendipada	1630	586	1044	2189	1465	724	1319	1019	300	519	376	143	140	119	21
3	Chhendipada -sub dist	34610	13140	21470	50918	29676	21242	29066	21031	8035	13382	10285	3097	6865	4312	2553
4	Chhendipadajungle	609	241	368	900	496	404	751	447	304	551	307	244	76	38	38
5	Deuria jharan	516	199	317	636	344	292	470	318	152	71	64	7	209	136	73
6	Ghuntulipasi	284	97	187	308	239	69	267	216	51	76	75	1	75	65	10
7	Gobindpur	559	215	344	633	389	244	282	148	134	30	19	11	208	104	104
8	Handigoda	483	182	301	743	418	325	200	168	32	113	111	2	34	25	9
9	Ichhapur	307	115	192	837	420	417	535	349	186	133	99	34	54	29	25
10	Jamubahali	379	145	234	369	324	45	327	297	30	3	3	0	1	1	0
11	Machhakuta	681	258	423	848	606	242	430	394	36	196	193	3	107	105	2
12	Machhakutajungle	51	22	29	58	39	19	38	30	8	19	16	3	3	3	0
13	Podapada	495	170	325	647	454	193	456	378	78	128	121	7	110	87	23
14	Podapadajungle	9	3	6	19	12	7	0	0	0	0	0	0	0	0	0
15	Salakhaman	311	125	186	351	187	164	238	146	92	9	6	3	223	137	86
16	Tentoloi	317	123	194	472	343	129	403	316	87	42	39	3	128	66	62
17	Tentuloikodasahi	521	197	324	579	420	159	371	289	82	123	121	2	109	66	43
	SUB TOTAL	41839	15850	25989	60644	35908	24736	35212	25603	9609	15447	11887	3560	8346	5295	3051
	VILLAGES IN 2 TO 5 KM															
18	Bahalsahi	436	163	273	887	460	427	676	364	312	366	220	146	250	101	149
19	Balipata	1500	546	954	2615	1330	1285	650	571	79	181	171	10	278	241	37
20	Bamanali	51	17	34	113	59	54	73	47	26	54	34	20	1	1	0
21	Baramancha	576	226	350	674	377	297	400	287	113	152	123	29	113	82	31
22	Barasahi	379	154	225	460	324	136	238	179	59	95	89	6	48	25	23
23	Basantapur	187	54	133	278	202	76	231	195	36	140	134	6	3	3	0
24	Chandrapur	218	65	153	404	199	205	24	21	3	7	7	0	4	3	1
25	Dubanali	267	97	170	473	241	232	67	65	2	46	45	1	8	7	1
26	Gopinathpur	207	88	119	136	94	42	103	79	24	17	15	2	62	45	17
27	Hatianali	369	148	221	601	322	279	42	42	0	0	0	0	40	40	0
28	Kasidiha	212	82	130	303	157	146	134	125	9	84	84	0	28	24	4
29	Kukurpeta	834	326	508	1111	623	488	586	489	97	334	269	65	149	126	23
30	Kuskila	815	324	491	1238	639	599	436	359	77	265	218	47	118	95	23
31	Panapasi	54	17	37	86	46	40	70	42	28	18	11	7	24	14	10
32	Sapoinali	910	336	574	1465	882	583	911	738	173	406	337	69	287	214	73
33	Tangiri	470	187	283	699	354	345	354	277	77	128	117	11	63	32	31
	SUB TOTAL	7485	2830	4655	11543	6309	5234	4995	3880	1115	2293	1874	419	1476	1053	423
	VILLAGES BEYOND 5 KM															
34	Ashrubahal	318	139	179	334	171	163	312	163	149	15	9	6	252	134	118
35	Badaberena	298	94	204	364	192	172	233	141	92	84	69	15	137	63	74
36	Balinali	215	81	134	412	223	189	376	212	164	170	169	1	197	37	160
37	Barapada	835	321	514	1196	679	517	586	407	179	363	251	112	72	60	12
38	Bhagilakata	198	81	117	337	174	163	280	150	130	174	96	78	70	28	42
39	Bhejidihi	68	23	45	105	58	47	5	3	2	1	1	0	0	0	0
40	Brahmanbil	921	343	578	1844	951	893	1240	801	439	773	495	278	331	191	140
41	Chakundapal	39	17	22	97	60	37	29	28	1	19	19	0	3	3	0
42	Dangapal	583	234	349	911	524	387	474	335	139	159	151	8	274	153	121
43	Dhangurumunda	20	9	11	47	22	25	5	5	0	1	1	0	1	1	0
44	Golagadia	313	128	185	451	260	191	232	209	23	150	148	2	59	40	19
45	Golasar	127	51	76	279	133	146	114	55	59	77	38	39	32	14	18
46	Gopabalhabpur	43	5	38	31	8	23	1	1	0	0	0	0	0	0	0
47	Gundurijharan	47	19	28	96	46	50	32	32	0	2	2	0	5	5	0
48	Jaripal	217	90	127	309	164	145	217	126	91	106	74	32	78	33	45
49	Jhintipal	715	285	430	968	526	442	843	516	327	417	310	107	250	92	158
50	Kankarapal	221	98	123	157	153	4	7	5	2	0	0	0	2	1	1
51	Karadabahal	352	137	215	511	271	240	456	253	203	207	117	90	213	104	109
52	Kisinda	151	56	95	178	156	22	27	23	4	0	0	0	1	1	0
53	Korada	1063	438	625	1130	601	529	333	286	47	209	207	2	79	48	31
54	Kosala	1459	570	889	2104	1543	561	1829	1431	398	782	535	247	241	214	27
55	Madhupur	490	198	292	757	523	234	473	379	94	149	127	22	148	109	39
56	Nuapada	1292	510	782	1821	1253	568	1466	1148	318	392	321	71	404	299	105
57	Patakumunda	335	141	194	766	393	373	336	311	25	268	257	11	4	4	0
58	Patrapada	1358	519	839	2378	1252	1126	1384	851	533	763	518	245	67	48	19
59	Purunagarh	439	144	295	599	373	226	402	323	79	281	235	46	87	61	26
60	Putagadia	278	92	186	374	237	137	354	224	130	167	112	55	40	14	26
61	Raipal	537	224	313	716	365	351	311	269	42	96	88	8	90	86	4
62	Rajanpal	75	33	42	113	60	53	43	40	3	0	0	0	3	1	2
63	Santarabandha	809	292	517	1041	659	382	718	516	202	400	287	113	167	110	57
64	Sarapal	271	96	175	395	247	148	258	233	25	194	182	12	26	22	4
65	Similipal	899	346	553	1366	755	611	479	321	158	228	169	59	102	54	48
66	Similisahi	859	338	521	1238	747	491	858	557	301	284	182	102	110	48	62
	SUB TOTAL	15845	6152	9693	23425	13779	9646	14713	10354	4359	6931	5170	1761	3545	2078	1467
	TOTAL	65169	24832	40337	95612	55996	39616	54920	39837	15083	24671	18931	5740	13367	8426	4941
	AVERAGE	987	376	611	1449	848	600	832	604	229	374	287	87	203	128	75

SL.NO.	VILLAGE	MAIN_HH_P	MAIN_HH_M	MAIN_HH_F	MAIN_OT_P	MAIN_OT_M	MAIN_OT_F	MARGWORK_P	MARGWORK_M	MARGWORK_F	MARG_CL_P	MARG_CL_M	MARG_CL_F	MARG_AL_P	MARG_AL_M	MARG_AL_F
	VILLAGES IN 0 TO 2 KM															
1	Balijharan	0	0	0	3	3	0	78	19	59	4	2	2	74	17	57
2	Chhendipada	58	32	26	602	492	110	870	446	424	140	58	82	487	256	231
3	Chhendipada -sub dist	2411	1288	1123	6408	5146	1262	21852	8645	13207	3988	1504	2484	12668	5061	7607
4	Chhendipadajungle	3	2	1	121	100	21	149	49	100	61	21	40	77	25	52
5	Deuria jharan	148	81	67	42	37	5	166	26	140	70	14	56	66	7	59
6	Ghuntulipasi	78	43	35	38	33	5	41	23	18	0	0	0	7	4	3
7	Gobindpur	0	0	0	44	25	19	351	241	110	58	52	6	150	71	79
8	Handigoda	26	15	11	27	17	10	543	250	293	258	120	138	228	107	121
9	Ichhapur	109	59	50	239	162	77	302	71	231	136	20	116	67	21	46
10	Jamubahali	2	2	0	321	291	30	42	27	15	0	0	0	2	0	2
11	Machhakuta	31	23	8	96	73	23	418	212	206	60	55	5	120	84	36
12	Machhakutajungle	10	6	4	6	5	1	20	9	11	0	0	0	4	0	4
13	Podapada	24	13	11	194	157	37	191	76	115	18	14	4	90	25	65
14	Podapadajungle	0	0	0	0	0	0	19	12	7	9	9	0	2	1	1
15	Salakhaman	0	0	0	6	3	3	113	41	72	6	1	5	103	39	64
16	Tentoloi	0	0	0	233	211	22	69	27	42	2	1	1	17	5	12
17	Tentuloikodasahi	17	4	13	122	98	24	208	131	77	37	8	29	118	91	27
	SUB TOTAL	2917	1568	1349	8502	6853	1649	25432	10305	15127	4847	1879	2968	14280	5814	8466
	VILLAGES IN 2 TO 5 KM															
18	Bahalsahi	9	5	4	51	38	13	211	96	115	30	9	21	167	79	88
19	Balipata	19	10	9	172	149	23	1965	759	1206	165	69	96	1433	586	847
20	Bamanali	5	1	4	13	11	2	40	12	28	8	1	7	15	4	11
21	Baramancha	55	22	33	80	60	20	274	90	184	133	40	93	97	30	67
22	Barasahi	12	7	5	83	58	25	222	145	77	59	56	3	2	2	0
23	Basantapur	34	12	22	54	46	8	47	7	40	24	1	23	11	2	9
24	Chandrapur	0	0	0	13	11	2	380	178	202	24	22	2	351	153	198
25	Dubanali	4	4	0	9	9	0	406	176	230	9	5	4	358	148	210
26	Gopinathpur	2	2	0	22	17	5	33	15	18	0	0	0	21	9	12
27	Hatianali	0	0	0	2	2	0	559	280	279	3	2	1	551	274	277
28	Kasidiha	7	7	0	15	10	5	169	32	137	1	1	0	166	29	137
29	Kukurpeta	8	7	1	95	87	8	525	134	391	185	55	130	295	63	232
30	Kuskila	25	22	3	28	24	4	802	280	522	66	29	37	658	223	435
31	Panapasi	0	0	0	28	17	11	16	4	12	1	0	1	3	1	2
32	Sapoinali	69	57	12	149	130	19	554	144	410	137	5	132	385	133	252
33	Tangiri	111	89	22	52	39	13	345	77	268	14	4	10	299	62	237
	SUB TOTAL	360	245	115	866	708	158	6548	2429	4119	859	299	560	4812	1798	3014
	VILLAGES BEYOND 5 KM															
34	Ashrubahal	19	3	16	26	17	9	22	8	14	0	0	0	17	4	13
35	Badaberena	3	3	0	9	6	3	131	51	80	4	2	2	103	40	63
36	Balinali	0	0	0	9	6	3	36	11	25	2	0	2	33	10	23
37	Barapada	27	17	10	124	79	45	610	272	338	228	90	138	338	163	175
38	Bhagilakata	2	2	0	34	24	10	57	24	33	20	3	17	33	18	15
39	Bhejidihi	0	0	0	4	2	2	100	55	45	3	3	0	93	50	43
40	Brahmanbil	19	17	2	117	98	19	604	150	454	179	41	138	382	89	293
41	Chakundapal	1	0	1	6	6	0	68	32	36	9	8	1	54	22	32
42	Dangapal	20	16	4	21	15	6	437	189	248	1	0	1	295	69	226
43	Dhangurumunda	2	2	0	1	1	0	42	17	25	0	0	0	41	16	25
44	Golagadia	5	5	0	18	16	2	219	51	168	16	7	9	192	39	153
45	Golasar	1	0	1	4	3	1	165	78	87	142	70	72	20	7	13
46	Gopabalhabpur	1	1	0	0	0	0	30	7	23	2	1	1	5	1	4
47	Gundurijharan	0	0	0	25	25	0	64	14	50	3	3	0	2	2	0
48	Jaripal	13	6	7	20	13	7	92	38	54	12	2	10	53	25	28
49	Jhintipal	43	8	35	133	106	27	125	10	115	5	2	3	116	6	110
50	Kankarapal	0	0	0	5	4	1	150	148	2	0	0	0	150	148	2
51	Karadabahal	0	0	0	36	32	4	55	18	37	0	0	0	47	15	32
52	Kisinda	1	1	0	25	21	4	151	133	18	13	12	1	133	117	16
53	Korada	3	3	0	42	28	14	797	315	482	34	19	15	673	253	420
54	Kosala	83	53	30	723	629	94	275	112	163	128	36	92	73	31	42
55	Madhupur	11	8	3	165	135	30	284	144	140	64	31	33	79	22	57
56	Nuapada	77	31	46	593	497	96	355	105	250	108	24	84	148	29	119
57	Patakumunda	25	23	2	39	27	12	430	82	348	21	8	13	8	0	8
58	Patrapada	184	57	127	370	228	142	994	401	593	189	50	139	175	90	85
59	Purunagarh	2	2	0	32	25	7	197	50	147	7	6	1	165	37	128
60	Putagadia	24	4	20	123	94	29	20	13	7	1	1	0	6	1	5
61	Raipal	65	45	20	60	50	10	405	96	309	63	6	57	258	66	192
62	Rajanpal	0	0	0	40	39	1	70	20	50	0	0	0	1	1	0
63	Santarabandha	14	7	7	137	112	25	323	143	180	174	68	106	102	47	55
64	Sarapal	2	2	0	36	27	9	137	14	123	18	0	18	93	6	87
65	Similipal	44	26	18	105	72	33	887	434	453	103	70	33	541	270	271
66	Similisahi	221	116	105	243	211	32	380	190	190	183	115	68	147	35	112
	SUB TOTAL	912	458	454	3325	2648	677	8712	3425	5287	1732	678	1054	4576	1729	2847
	TOTAL	4189	2271	1918	12693	10209	2484	40692	16159	24533	7438	2856	4582	23668	9341	14327
	AVERAGE	63	34	29	192	155	38	617	245	372	113	43	69	359	142	217

SL.NO.	VILLAGE	MARG_HH_P	MARG_HH_M	MARG_HH_F	MARG_OT_P	MARG_OT_M	MARG_OT_F	NON_WORK_P	NON_WORK_M	NON_WORK_F
	VILLAGES IN 0 TO 2 KM									
1	Balijharan	0	0	0	0	0	0	90	49	41
2	Chhendipada	35	9	26	208	123	85	3321	1393	1928
3	Chhendipada -sub dist	2293	592	1701	2903	1488	1415	52886	23519	29367
4	Chhendipadajungle	3	1	2	8	2	6	906	441	465
5	Deuria jharan	27	2	25	3	3	0	608	268	340
6	Ghuntulipasi	24	16	8	10	3	7	668	248	420
7	Gobindpur	0	0	0	143	118	25	827	322	505
8	Handigoda	19	3	16	38	20	18	646	285	361
9	Ichhapur	67	12	55	32	18	14	479	260	219
10	Jamubahali	2	2	0	38	25	13	843	302	541
11	Machhakuta	65	11	54	173	62	111	1336	487	849
12	Machhakutajungle	9	6	3	7	3	4	119	44	75
13	Podapada	20	5	15	63	32	31	1194	468	726
14	Podapadajungle	0	0	0	8	2	6	32	13	19
15	Salakhaman	0	0	0	4	1	3	384	173	211
16	Tentoloi	0	0	0	50	21	29	912	297	615
17	Tentuloikodasahi	12	1	11	41	31	10	845	319	526
	SUB TOTAL	2576	660	1916	3729	1952	1777	66096	28888	37208
	VILLAGES IN 2 TO 5 KM									
18	Bahalsahi	4	0	4	10	8	2	623	326	297
19	Balipata	118	52	66	249	52	197	2065	1039	1026
20	Bamanali	9	1	8	8	6	2	91	47	44
21	Baramancha	28	13	15	16	7	9	740	328	412
22	Barasahi	90	27	63	71	60	11	721	293	428
23	Basantapur	8	2	6	4	2	2	464	180	284
24	Chandrapur	0	0	0	5	3	2	277	128	149
25	Dubanali	18	7	11	21	16	5	347	186	161
26	Gopinathpur	2	2	0	10	4	6	276	107	169
27	Hatianali	0	0	0	5	4	1	245	119	126
28	Kasidiha	1	1	0	1	1	0	223	119	104
29	Kukurpeta	17	3	14	28	13	15	1102	512	590
30	Kuskila	46	9	37	32	19	13	1067	531	536
31	Panapasi	0	0	0	12	3	9	58	34	24
32	Sapoinali	22	1	21	10	5	5	1454	626	828
33	Tangiri	17	2	15	15	9	6	649	320	329
	SUB TOTAL	380	120	260	497	212	285	10402	4895	5507
	VILLAGES BEYOND 5 KM									
34	Ashrubahal	0	0	0	5	4	1	262	132	130
35	Badaberena	18	4	14	6	5	1	446	206	240
36	Balinali	0	0	0	1	1	0	313	159	154
37	Barapada	21	6	15	23	13	10	1034	492	542
38	Bhagilakata	1	1	0	3	2	1	246	125	121
39	Bhejidihi	0	0	0	4	2	2	103	45	58
40	Brahmanbil	4	4	0	39	16	23	1217	649	568
41	Chakundapal	0	0	0	5	2	3	96	43	53
42	Dangapal	126	108	18	15	12	3	733	334	399
43	Dhangurumunda	0	0	0	1	1	0	23	16	7
44	Golagadia	1	1	0	10	4	6	458	199	259
45	Golasar	0	0	0	3	1	2	154	70	84
46	Gopabalhabpur	4	1	3	19	4	15	24	1	23
47	Gundurijharan	0	0	0	59	9	50	59	34	25
48	Jaripal	16	6	10	11	5	6	214	100	114
49	Jhintipal	2	1	1	2	1	1	904	423	481
50	Kankarapal	0	0	0	0	0	0	360	109	251
51	Karadabahal	0	0	0	8	3	5	524	257	267
52	Kisinda	3	2	1	2	2	0	389	136	253
53	Korada	13	3	10	77	40	37	1433	723	710
54	Kosala	11	4	7	63	41	22	3351	1322	2029
55	Madhupur	6	3	3	135	88	47	1015	382	633
56	Nuapada	24	2	22	75	50	25	2588	1009	1579
57	Patakumunda	321	49	272	80	25	55	500	273	227
58	Patrapada	258	67	191	372	194	178	1754	876	878
59	Purunagarh	0	0	0	25	7	18	772	314	458
60	Putagadia	2	1	1	11	10	1	444	171	273
61	Raipal	37	8	29	47	16	31	508	256	252
62	Rajanpal	1	0	1	68	19	49	64	31	33
63	Santarabandha	13	6	7	34	22	12	1428	593	835
64	Sarapal	12	1	11	14	7	7	545	246	299
65	Similipal	119	36	83	124	58	66	1127	483	644
66	Similisahi	16	16	0	34	24	10	1295	567	728
	SUB TOTAL	1029	330	699	1375	688	687	24383	10776	13607
	TOTAL	3985	1110	2875	5601	2852	2749	100881	44559	56322
	AVERAGE	60	17	44	85	43	42	1529	675	853

ABBREVIATIONS USED

No_HH	Number of Households

TOT_P	Total Population (Persons)
TOT_M	Total Population (Males)
TOT_F	Total Population (Females)
P_06	Population in age group 0-6 years (Persons)
M_06	Population in age group 0-6 years (Males)
F_06	Population in age group 0-6 years (Females)
P_SC	Scheduled Caste Population (Persons)
M_SC	Scheduled Caste Population (Males)
F_SC	Scheduled Caste Population (Females)
P_ST	Scheduled Tribe Population (Persons)
M_ST	Scheduled Tribe Population (Males)
F_ST	Scheduled Tribe Population (Females)
P_LIT	Literates (Persons)
M_LIT	Literates (Males)
F_LIT	Literates (Females)
P_ILL	Illiterates (Persons)
M_ILL	Illiterates (Males)
F_ILL	Illiterates (Females)
TOT_WORK_P	Total Workers (Persons)
TOT_WORK_M	Total Workers (Males)
TOT_WORK_F	Total Workers (Females)
MAINWORK_P	Main Workers (Persons)
MAINWORK_M	Main Workers (Males)
MAINWORK_F	Main Workers (Females)
MAIN_CL_P	Main Workers - Cultivators (Persons)
MAIN_CL_M	Main Workers - Cultivators (Males)
MAIN_CL_F	Main Workers - Cultivators (Females)
MAIN_AL_P	Main Workers - Agricultural Labourers (Persons)
MAIN_AL_M	Main Workers - Agricultural Labourers (Males)
MAIN_AL_F	Main Workers - Agricultural Labourers (Females)
MAIN_HH_P	Main Workers - Workers in Household Industries (Persons)
MAIN_HH_M	Main Workers - Workers in Household Industries (Males)
MAIN_HH_F	Main Workers - Workers in Household Industries (Females)
MAIN_OT_P	Main Workers - Other Workers (Persons)
MAIN_OT_M	Main Workers - Other Workers (Males)
MAIN_OT_F	Main Workers - Other Workers (Females)
MARGWORK_P	Marginal Workers (Persons)
MARGWORK_M	Marginal Workers (Males)
MARGWORK_F	Marginal Workers (Females)
MARG_CL_P	Marginal Workers - Cultivators (Persons)
MARG_CL_M	Marginal Workers - Cultivators (Males)
MARG_CL_F	Marginal Workers - Cultivators (Females)
MARG_AL_P	Marginal Workers - Agricultural Labourers (Persons)

MARG_AL_M	Marginal Workers - Agricultural Labourers (Males)
MARG_AL_F	Marginal Workers - Agricultural Labourers (Females)
MARG_HH_P	Marginal Workers - Workers in Household Industries (Persons)
MARG_HH_M	Marginal Workers - Workers in Household Industries (Males)
MARG_HH_F	Marginal Workers - Workers in Household Industries (Females)
MARG_OT_P	Marginal Workers - Other Workers (Persons)
MARG_OT_M	Marginal Workers - Other Workers (Males)
MARG_OT_F	Marginal Workers - Other Workers (Females)
NON_WORK_P	Non-Workers (Persons)
NON_WORK_M	Non-Workers (Males)
NON_WORK_F	Non-Workers (Females)

The summary of the demographic features is given in **Table No. 3.16** and discussed in the following paragraphs:

i) Population:

The total population of 66 villages is 196493. This includes rural sub-district Chhendipada having population 103804.

The % of Schedule Caste (SC) and Schedule Tribe (ST) to the total population is 20.28 & 10.44 respectively.

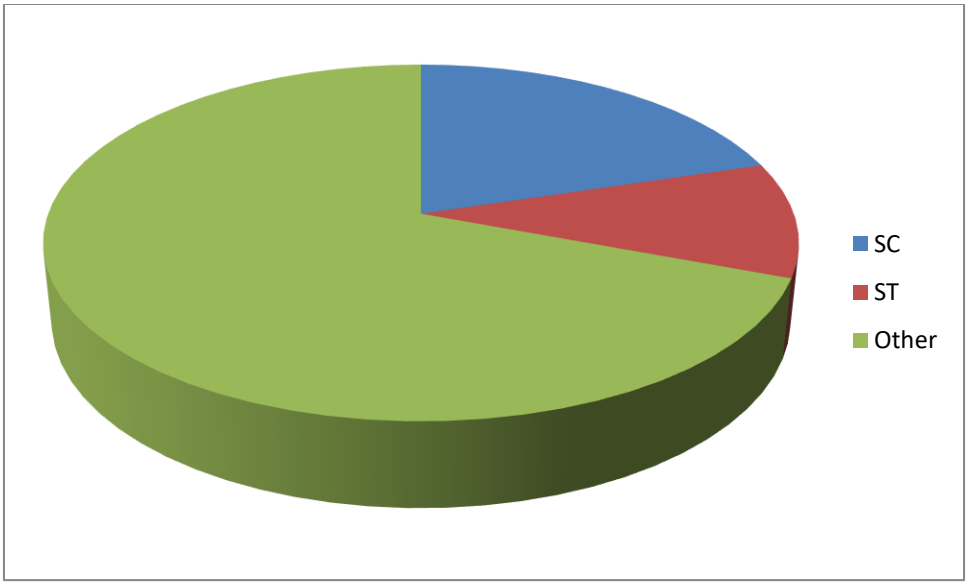


Figure 3-26 Pie Chart depicting the Castes

ii) Village size:

The total households in the 66 villages are 46556. The average household per village is 705. This includes rural sub-district Chhendipada having population 103804 and HH 23726.40.

iii) Household size:

The average household size is 4.22 in rural areas.

iv) Sex Ratio:

The sex ratio of the study area is 954 females per thousand male population in rural area. The percentage of male and female population to the total population is 51.17 and 48.83 respectively.

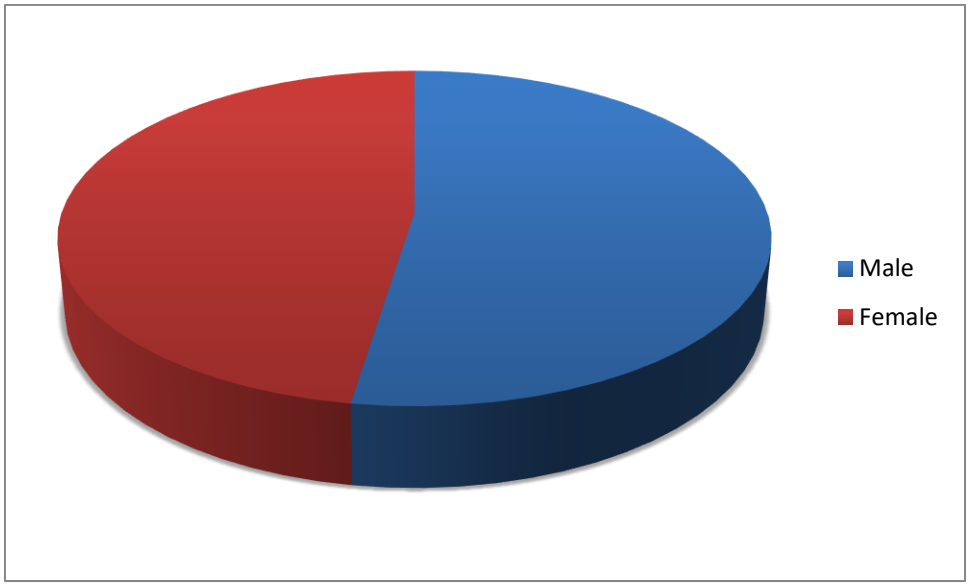


Figure 3-27 Pie Chart depicting the Sex Ratio

v) Literacy: The percentage of literacy in the area is 66.83.

The % male and female literacy to the total population is 38.54 & 28.30 respectively.

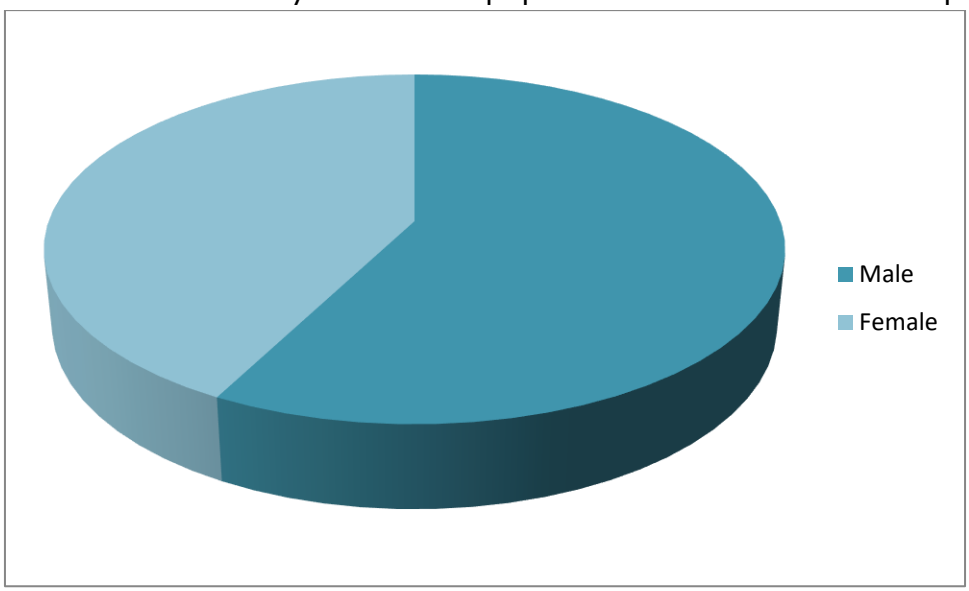


Figure 3-28 Pie Chart Depicting the Literacy Rate

vi) Occupational structure of the study area:

The percentage of total workers to the total population of the rural area is 48.66 in which main workers constitute 27.95% and marginal workers constitute 20.71

% and remaining 51.34 % constitute non workers.

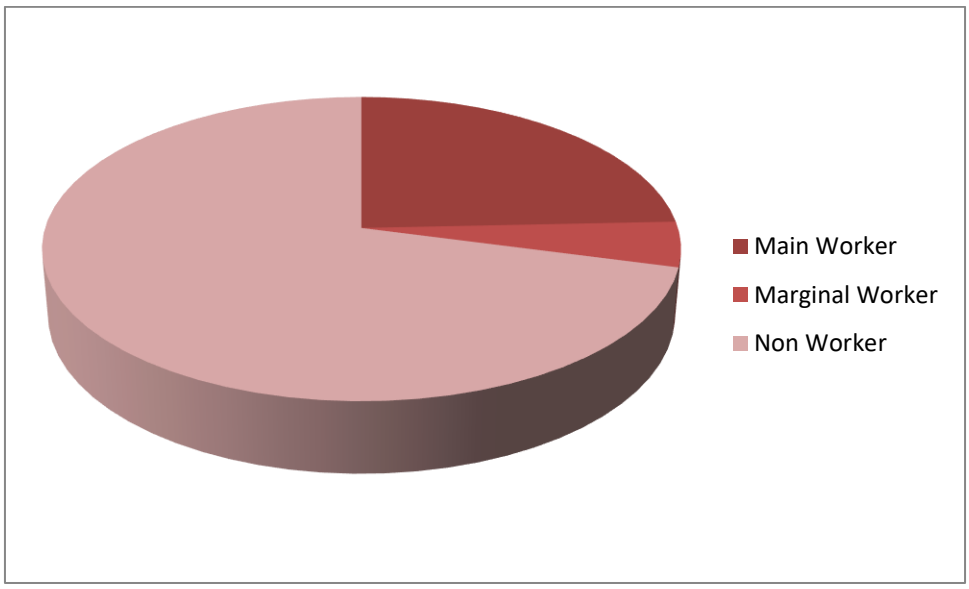


Figure 3-29 Pie Chart Depicting the %of workers

Table 3:16 Demographic features and other Statistics for the Study Area (Rural Area) as per Census 2011

Sl.	Details	No./%
1	Total population	196493
2	No. of Household	46556
3	Average family size	4.22
4	Average no. of household per village	705
5	Average population per village	2977
6	Sex ratio –females per thousand males	954
7	Percent of male population to total population	51.17
8	Percent of female population to total population	48.83
9	Percent of SC population to the total population	20.28
10	Percent of ST population to the total population	10.44
11	Percent of literate population to the total population	66.83
12	Percent of Male literate population to the total population	38.54
13	Percent of Female literate population to the total population	28.30
14	Percent of total main worker to the total population	27.95
15	Percent of total marginal worker to the total population	20.71
16	Percent of non-worker to the total population	51.34

From the table it is clear that Sex Ratio is on a lower side.

Gender gap between male and female literacy is approx. 10 %. Employment is a major concern besides literacy.

Primary Survey Data of the Study Area (villages up to 2 km) (November 2023)

SL. NO.	VILLAGE	No_HH	TOT_P	TOT_M	TOT_F	P_SC	M_SC	F_SC	P_ST	M_ST	F_ST	P_LIT	M_LIT	F_LIT
1	Chhendipada	1400	6000	3200	2800	1400	700	700	190	90	100	4350	2550	1800
2	Chhendipada -subdist	27000	110000	60000	50000	24000	12000	12000	10250	5150	5100	77500	45000	32500
3	Chhendipadajungle	450	2000	1050	950	120	60	60	50	30	20	1350	780	570
4	Handigoda	400	1500	800	700	280	145	135	350	170	180	1000	575	425
5	Machhakuta	600	2500	1250	1250	425	200	225	200	100	100	1700	950	750
6	Machhakutajungle	50	200	100	100	55	25	30	30	10	20	140	70	70
7	Podapada	500	2000	1000	1000	700	350	350	200	100	100	1500	850	650
8	Podapadajungle	15	50	25	25	0	0	0	0	0	0	50	25	25
9	Tentuloikodasahi	350	1600	800	800	200	100	100	275	140	135	1000	600	400

3.14.3 ANGUL DISTRICT AS A WHOLE

General Characteristics of the District: The present Angul District is having 8 Blocks. The present district of Angul with 4.09% of State area and 3.03 % of state population is well known for largest Aluminium company i.e., National Aluminium Co. Ltd, Angul. The significant feature of the district is that there are 5 PSUs besides 4 Large Scale Industries that exist in the district. It is also well known for Talcher Coal Mines and huge deposits of coal in the district. The other feature is that a large portion of the district is under forest which is 42.6 % with varieties of minor and major forest products.

Location & Geographical Area: The district is situated in the central part of Odisha. The district finds its circumferential limits within $20^0$31" and 21^0 41" North Latitude and 84^0 16" and 85^0 23" East Longitude. It is surrounded on the North by Keonjhar and Sundargarh districts, East by Cuttack and Dhenkanal districts, South by Boudh district and West by Sambalpur and Deogarh districts. The total geographical area of the district is 6375 sq.km. The district ranks 11th among the Districts of Odisha area wise.

Topography: Topographically, the district can be divided into three natural tracks. First is a chain of hills running along the northeastern boundary of the district

covering Pallahara. Another chain of hills runs along with south-west boundary covering Athamallik and Angul. The third natural division is a valley of river Brahmani running along with boundary of Talcher through Kaniha touching Pallahara.

Availability of Minerals: Varieties of economic minerals are available in the district which includes Coal, Chromites, Graphite, Manganese, Mica, Keyanite, Granite, Laterite, Sandstone, Quartz etc. However, Coal, Sandstone and Quartz are only commercially exploited. There are 11 Coal Mines covering 10474.336 Hectares, 1 Sandstone mine covering 17.5 Hectares and 1 Quartz mine covering 10.744 Hectares.

Administrative set up: The district has 4 Sub-divisions namely Angul, Athamallik, Pallahada & Talcher; 8 Tehsils; 8 Community Development Blocks; 2 Municipalities; 1 NAC; 23 Police Stations; 8 Fire Stations; 225 Gram Panchayats & 1871 villages. There are 3 towns in the district. The district headquarters is situated at Angul.

District at a glance:

Sl. No.	**Particulars**	**Year**	**Unit**	**Statistics**
1	**Geographical features**			
A	Geographical Data			
	i) Latitude	2019		20^0 31“ to 21^0 41“ (North)
	ii) Longitude	2019		84^0 16“ to 85^0 23“ (East)
	iii) Geographical Area	2019	Sq km	6375
B	Administrative Units	2019		
	i) Subdivisions	,,	No.	4
	ii) Tehsils	,,	No.	8
	iii) Sub-Tehsil	,,	No.	-
	iv) Patwar Circle	,,	No.	-
	v) Panchayat Samitis (CD Block)	,,	No.	8
	vi)Nagar Nigam	,,	No.	0
	vii) Nagar Palika (Municipality & NAC)	,,	No.	3
	viii) Gram Panchayats	,,	No.	225
	ix) Revenue villages	,,	No.	1871
	x) Assembly Area	,,	No.	5
	xi) Police Station (P.S)	,,	No.	23
	xii) Fire Station	,,	No.	8
2.	**Population (Census 2011)**			

A	Sex-wise			
	i) Male Population	2011	No.	655718
	ii) Female Population	,,	,,	618103
	Total Population	,,	,,	1273821
B	Rural Population	,,	,,	1067275
C	Sex Ratio	,,	Females per 1000 males	943
D	Density of Population	,,	Per Sq.km	200
E	Decadal Growth Rate	2001-11	%	11.74
3.	**Agriculture**			
A.	**Land utilization**	2018-19		
	i) Land put to Non-Agricultural Use	,,	000“ Hectare	48
	ii) Barren &Un-cultivable Land	,,	“	7
	iii) Current Fallow	,,	“	24
	iv) Other Fallow	,,	“	17
	v) Culturable Waste	,,	“	19
	vi) Permanent Pasture	,,	“	36
	vii) Misc. trees & Grooves	,,	“	23
	viii) Net area Sown	,,	“	192
B.	**Rainfall**	2018	mm	
	(i) Normal Rainfall	,,	,,	1401.9
	(ii) Actual Avg. Rainfall	,,	,,	1373.4
C.	**Agricultural Production**	2018-19	000“ MT	
	i) Paddy	,,	,,	276.41
	ii) Wheat	,,	,,	0.04
	iii) Maize	,,	,,	15.76
	iv) Ragi	,,	,,	0.06
	v) Moong	,,	,,	16.51
	vi) Biri	,,	,,	17.85
	vii) Kulthi	,,	,,	5.87
	viii) Til	,,	,,	14.82
	ix) Ground Nut	,,	,,	20.22
	x) Mustard	,,	,,	2.73
	xi) Jute	,,	,,	0
	xii) Potato	,,	,,	17.92
	xiii) Sugarcane	,,	,,	1.50
4.	**Forest**	2018-19	Sq.km.	2716.82
5.	**Livestock & Poultry (Livestock Census - 2012)**			
	Cattle			
	i) Cows	2012	No.	436651

	ii) Buffaloes	2012	No.	12287
	Other livestock			
	i) Goat	2012	No.	207271
	ii) Sheep	2012	No.	40503
	iii) Pig	2012	No.	486
	Poultry	2012	No.	556151
6.	**Production of Milk, Egg & Meat**	2018-19		
	i) Milk	,,	000" MT	53.14
	ii) Egg	,,	Lakh. No	274.50
	iii) Meat	,,	„000" MT	7.82
7.	**Fish Production**	2018-19	MT	
	i) Inland Water	,,	,,	21355
	ii) Marine Water	,,	,,	0
	iii) Total	,,	,,	21355
8.	**Railways**			
	i) Length of rail line	2014-15	km	106
	ii) Railway Stations/ PH	2014-15	No.	12
9.	**Roads**			
	(a) National Highway	2018-19	km	286.94
	(b) State Highway	,,	km	179.39
	(c) Major District Road	,,	km	29.58
	(d) Other District Roads	,,	km	1130.47
	(e) Rural road	,,	km	1177.07
	(f) Inter Village Road	,,	km	2093.93
	(g) Intra Village Road	,,	km	2298.06
	(h) Forest Road	,,	km	449.53
10.	**Public Health**			
	(a) Medical College & Dist. Hqrs. Hospital	2018	No.	1
	(b) Sub-Divisional & OtherHospitals	,,	No.	3
	(c) Community HealthCenters	,,	No.	10
	(d) Primary Health Centers	,,	No.	31
	(e) Mobile Health Unit (MHU)	,,	No.	1
	(f) Beds available in Hospitals	,,	No.	392
	(g) Ayurvedic Hospitals & Dispensaries	2018-19	No.	16

	(h) Homeopathic Hospitals & Dispensaries	,,	No.	19
11.	**Education**			
	(a) Primary school	2018-19	No.	992
	(b) Upper Primary school	,,	No.	669
	(c) Secondary school	,,	No.	288
	(d) Colleges (Junior + Degree)	,,	No.	75
	(e) Technical University	,,	No.	0
12.	**Post offices**	As on 01.04.2019	No.	234
13.	**Banking Commercial**			
	(a) Commercial Bank Branches	As on 30.09.2019	No.	176
	(b) Rural Branches	As on 30.09.2019	No.	102
	(c) Urban Branches	As on 30.09.2019	No.	4
	(d) Semi-Urban Branches	As on 30.09.2019	No.	70
14.	**No. of Villages Electrified**	As on 31.03.2019	No.	1654

Source: District at a Glance 2020, Directorate of Economics & Statistics, Odisha & SLBC, Odisha

District at a Glance-Geographical and Population details of the district (Census 2011)

Geographical Area :6375.00 sq.km
Area under Forest :2716.82 sq.km
Total:1,273,821
Male:655,718
Female:618,103
Rural:1,067,275
Urban:206,546
ScheduledCaste:239,552
ScheduledCasteMale:121,459
ScheduledCasteFemale:118,093
ScheduledTribe:179,603
ScheduledTribeMale:89,980
ScheduledTribeFemale:89,623
Population Density: 200 (Per sq.km)

Literacy Households

Total Literate:	869,456

Literate Male:	494,425
Literate Female:	375,031
Total Illiterate:	404,365
Illiterate Male:	161,293
Illiterate Female:	243,072
Total Households:	297,050
Rural Households:	249,733
Urban Households:	47,317

Angul District Religion Data

Description	**Percentage**	**Population 2011**
Hindu	98.81 %	1,258,684
Muslims	0.59 %	7,512
Christian	0.27 %	3,383
Sikh	0.06 %	812
Buddhist	0.02 %	226
Jain	0.00 %	36
Others	0.08 %	971
Not Stated	0.17 %	2,197
TOTAL		1,273,821

ANGUL POPULATION PROJECTIONS ON THE BASIS OF LIKELY POPULATION GROWTH RATE

The last census of Angul was done in 2011 and next census of 2021 has been postponed or cancelled.

Year	**Projected Population**	
2011	1,273,821	12.74 Lakhs
2021	1,400,000	14.01 Lakhs
2022	1,410,000	14.13 Lakhs
2023	1,420,000	14.23 Lakhs
2024	1,430,000	14.32 Lakhs
2025	1,430,000	14.39 Lakhs

Year	Projected Population	
2026	1,440,000	14.46 Lakhs
2027	1,450,000	14.51 Lakhs
2028	1,450,000	14.56 Lakhs
2029	1,450,000	14.60 Lakhs
2030	1,460,000	14.63 Lakhs
2031	1,460,000	14.66 Lakhs

Source -https://www.census2011.co.in/census/district/408-anugul.html

CLIMATE AND RAINFALL

As discussed, the area experiences the sub-tropical to tropical temperate monsoon climate and characterised by a hot summer and general dryness throughout the year except during the southwest monsoon season, i.e., June to September. December is the coldest month.

AGRICULTURE & IRRIGATION

Agriculture occupies a vital place in the economy of Angul district. The total cultivable area of this district is 2,16,403 ha constituting 32.7 percent of the total geographical area of the district. However, the total forest area (legal boundary) is higher than the state average. Within district, the forest area is maximum in the Pallahara block followed by Kaniha and Athamallik blocks.

Land use pattern of different blocks of district Angul during 2007 (Area hectares)

Particulars	Blocks					Total
	Angul	Banarpal	Chhendipada	Kaniha	Talcher	
Forests	9779	954	8955	16540	4635	**40863**
Misc. tree crops & Groves not included in net area sown	537	364	2359	954	376	**4590**
Barren & Uncultivable land	727	90	205	2369	577	**3968**
Land put to non-agricultural use	3432	5005	5307	7151	3486	**24381**
Culturable waste	2755	940	3607	2003	1083	**10388**
Permanent pastures and other grazing land	2547	2451	2470	1271	551	**9290**
Current Fallows	6209	4076	8103	24735	2615	**23476**

Other Fallows	5048	4719	6380	2969	1724	**20840**
Net area sown	17063	14853	22541	18964	6766	**80187**
Total Irrigated Land	2810.32	4160.01	3229.29	767.49	1097.87	**12064.98**

Source: Angul District Plan, TSI-ERA/DISTRICT ADMINISTRATION/ANGUL 2011-12

The agriculture in the district is primarily rain fed because of inadequate irrigation facility. Area irrigated through all sources is only 31% during kharif season and 21% during rabi season as per the available data. The district has 216403 ha of cultivable land. The total irrigation potential is49907 ha.

Most of the cultivated area of the district is covered with double crops like kulthi (kolath),bengalgram (harad), coriander, field pea; and vegetables are taken after harvest of ground nut and early kharif paddy. The **kharif** crops include paddy, maize, ragi, small millets, arhar, biri, mung, ground nut, til, castor, cotton, turmeric, ginger and vegetables like brinjal, tomato, and early cauliflower. On the other hand, **rabi** crops include paddy, wheat, maize, field pea, mung, biri, mustard, sunflower, safflower, niger, potato, onion, garlic, coriander, vegetables, tobacco, sugar cane etc.

The crop rotation practice followed by the farmers in the district are:

- In Upland region: Kulthi and vegetables are taken after harvest of short duration paddy, gram, coriander and groundnut crop.
- In Mid land region: Wheat, onion, garlic, mung, biri, vegetables and groundnut are taken after harvesting of kharif season paddy.
- In Low land region: Paddy and pulses are taken after harvest of rabi season paddy crop.

In assured irrigation farmlands, three crops like paddy-vegetable-pulses, paddy-potato-til and paddy-pulses-groundnut are taken.

Average cropping intensity in the district is around 175 percent. The highest cropping intensity is found in Banarpal block followed by Chhendipada block, whereas,in Talcher block the cropping intensity is almost half as compared to other blocks. This is mainly due to the inadequate irrigation facilities in the block.

Table 3:17 Area Irrigated by different sources (ha)

Block	Area Irrigated by Canal	Area Irrigated by River	Area Irrigated by Dug well	Area Irrigated by Tube well	Area Irrigated by Tank	Area Irrigated by Other Sources	Total Area irrigated	Total Area Unirrigated

Angul	1837.74	50.61	0	0	622.61	299.36	2810.32	27085.02
Banarpal	3777.38	198.78	0	0	113.85	70	4160.01	20387.7
Chhendipada	2522.55	100.3	210	16	110.61	269.83	3229.29	30913.57
Kaniha	438.58	245.07	24.12	0	59.72	0	767.49	15480.58
Talcher	471.99	96.14	0	0	359.62	170.12	1097.87	9203.62
TOTAL	**9048.24**	**690.9**	**234.12**	**16**	**1266.41**	**809.31**	**12064.98**	**103070.49**

Source: Angul District Plan, TSI-ERA/DISTRICT ADMINISTRATION/ANGUL 2011-12

Table 3:18 Source wise irrigation potential in Angul district

Sl. No.	Source	No. of Projects	Ayacut (ha)
1	Major/Medium	02	9138
2	Minor (Flow)	52	15602
3	Minor (Lift)	345	6160
4	Dug well	13730	5636
5	WHS	201	2944
6	Others		10427
	Total	14330	49907

Source: angul.nic.in

There are 2 medium irrigation projects in Angul. The Derjung Medium Irrigation Project is constructed on Ningara and Matalia river in Angul block having total catchment area of 399 sq km with CCA of 7392 ha. The Aunli Medium Irrigation Project is constructed on Aunli river in Chhendipada block**.** Its catchment area is 150 sq km and is having CCA of 1746 ha during Kharif and 300 ha in Rabi.

There are 5 large dams namely, Kansabansa MIP, Kukurpeta MIP, Raijharan MIP, Durgapur MIP and Jhinitipal MIP all of which comes in the Chhendipada block.

Table 3:19 Block wise MIPs in Angul district

Sl. No.	Block	No of Projects	Ayacut Area (ha)
1	Angul	25	2493
2	Banarpal	9	1243
3	Chhendipada	21	6709
4	Kaniha	19	2806

5	Talcher	6	525
	Total	80	13776

Source: Dept. of Water Resources, Minor Irrigation Projects, Odisha 2014

Drainage

Brahmani and Mahanadi are the two major rivers of the district. Both these rivers have numerous perennial and non-perennial tributaries. Most part of the district lies within the Brahmani basin while the Mahanadi basin spreads over Athmalik subdivision and southern part of Angul sub-division.

Economy

The last few years has witnessed a steep rise in the Economy of Angul District, with the setting up of the various public sector undertakings like National Aluminium Company Limited (NALCO), Mahanadi Coal Fields Limited (MCL), National Thermal Power Corporation (NTPC) and Talcher Thermal Power Station (TTPS). Apart from this, many private sectors also established which helped industrialization of the area. Here, there is also the presence of a Talcher heavy water plant.

Talcher Coal area which is regarded as the largest coal filed in Asia also helped to improve revenue and economy of the area.

The District Industries Centre functioning in the district promotes its various industrial activities in micro, small and medium enterprises.

Agriculture is also the major contributor to the District's Economy, providing direct and indirect employment to the people. The rural handicrafts, cottage industries also helped to improve economy of the district.

Medium Enterprises

Sl. No.	Name and address of the unit
1.	M/s. Babamani Roadways & Bore wells, Anand Nagar, Po-Hakimpada, Dist.-Angul.
2.	M/s. Haramohan Company Private Limited, Badabalipasi, Po-Rajdang, Via-Khamar, Angul.
3.	Dinesh Kumar Agarwalla, Vijaya Bhawan, Hulurisingha Chhak, Angul.
4.	M/s. Swain Engineering, Kulad, Po- Kulad, Nalco Nagar, Angul.
5.	M/s. Vaani Conglomerates LLP, 1st Floor, MR Towers, Main Road, Angul
6.	Mangturam Ratanlal, 1st Floor, MR Towers, Main Road, Angul
7.	M/s. Kalinga Guest House & Resorts, 1/A, Industrial Estate, Hakimpada, Angul
8.	M/s Precision Engineering, Deepshikha, Angul
9.	Chandan Publication Pvt. Ltd., Sambad Office, Shed No. - TS3/2,I.E, Angul
10.	Samal Auto India Pvt. Ltd., Angul
11.	Rukmani Infra Projects Pvt. Ltd., Maratira, Po-Tubey, Angul

12.	M/s. Sima garments, Angul
13.	M/s. R K Associates, At/Po- Derang, Via- Kaniha, Angul
14.	M/s. Badrinarayan Transport, At-Baghuabol, Po-Hatatota, Talcher, Angul
15.	M/s. B.N. Transprort & Co., At-Baghuabol, Po-Hatatota, Talcher, Angul
16.	M/s. Siyaram Enterprises, At/Po- Kulad, Ps- Nalco Nagar, Angul

Source: UAM web portal as on 31.03.2020.

Large Scale Industries / Public Sector undertakings:

Sl. No	**Name and address of the unit**	**Item of production**
1.	National Aluminum Company Ltd. (NALCO), NALCO Nagar, Angul	Aluminum
2	National Thermal Power Corporation (NTPC), Kanhia, Angul	Electricity, Thermal
3	Talcher Thermal Power Station, Talcher	Electricity, Thermal
4	Heavy Water Plant, Talcher	Heavy Water
5	Mahanadi Coal Field Ltd. (MCL), Talcher, Angul	Coal
6.	M/s. Shree Metaliks Ltd., Amlapada, Angul	Sponge Iron
7.	M/s. Bindal Sponge Ltd., Sunakhani, Talcher	Sponge Iron, MS Ingot &Bar
8.	M/s. Ganesh Sponge Pvt. Ltd., Krushnachandrapur, Parjang, Angul	Sponge Iron
9.	M/s. CPP Unit of Jindal Steel and Power Ltd., Kerjar, Angul	Power

Source: Directorate of Industries, Cuttack and DIC, Angul.

Major Exportable Items:

The following items have potentiality for exports.

1. Hydrates Rolled Products
2. Aluminum Ingot
3. Sponge Iron
4. Steel Billets, MS Ingot/Bar

Art & Culture

Angul is situated in the centre of the State and also regarded as the heart of Odisha. It gives highest revenue to the state exchequer. It is also one of the most advanced districts of State. The largest coal mine MCL (Mahanadi Coalfields Limited), National Aluminum Company (NALCO), National Thermal Power Corporation (NTPC), Heavy Water Plant etc. are situated in the district. Apart from this industrial scenario, Angul is also famous for Satkosia gorge, beautiful natural dense forest, hills, rivers (Mahanadi, Brhamani) etc. The land is fertile due to flowing of river Mahanadi & Bramhani and its tributaries. Rengali Reservoir on the river Brahmani also helped irrigation facility not only in Angul but also other districts like Dhenkanal, Cuttack, Jajpur and Keonjhar. Due to rapid industrialization, the cosmopolitan culture is visible in many parts of the district

but the traditional art, culture, literature, lifestyle has also made Angul District proud. Its cultural heritage is also unique in the State.

Religion

As mentioned in the population details majority of the inhabitants of the district are Hindus. The Hinduism here has a touch of the worship method of the Adivasi's. The example of shapeless Gramdevati smeared with vermilion in different villages is a pointer towards this.

A specific feature of the religious observance of people in Angul is the worship of miniature tanks, Pillars. Some of the inhabitants also follow Nath & Saiva as their deities. Another important religious practice in the district is Mahima Dharma. Apart from that people belonging to other religions are also staying in the district.

Language

The language of the people is Odia. It has many variants with slight alterations. Some of the tribal use their own language.

Food

The people of the district predominantly use rice as their staple diet. The daily meals consist of boiled rice, dal and vegetables. Chuda (flattened rice), Mudhi (pop rice), Pakhal (watered rice) also are popular food in the district. The tribals consume millet, pulses, beans, sweet potatoes etc.

Dress

Now a days mostly people are using European garments like trousers and shirts. These garments have found their ways even to the remotest village. Elder males of middle class in many cases wear Dhoti and Shirt or Kurta. Poor and even middle-class people males wear only Dhoti And use Gamuchha (napkins) on their upper part of the body.

Women irrespective of caste and social condition wear Saree. Most of them wear Blouse with Saree. The mode of wearing saree varies slightly with the class. The tribals wear small cloths around their waist. The Mahima Sanyasis wear different types of clothing according to their sect, the most prominent being the Saffron colored cloth.

The ornaments worn by women in the district are as follows.

1. Head – Gojikathi, Chauri mundi, Banka Deuria, Kia patra, Saru Kanthi, Champa kadhi, Suki Hara, Adhuli Hara, Tanka Hara etc.
2. Ear- Phasia, Baliguna, Jaulinoli.
3. Nose- Dandi, Notha, Phuli, Naka Chana.
4. Hands- Gazara, Bataphala, Bida, Tikha, Pane, Taeta.
5. Leg- Goda Mudi, Bala, Pauza, Jhuntia, Mala.

Fairs & Festivals

Angul is not only a land of magnificent hill ranges, old monuments, virgin and long riverbeds, downy flakes melodious jungle streams, healing hot springs, luxuriant forests, picturesque landscape, industrial houses but also a land of exciting and exuberant fasts and festivals. All the seasons of the year are marked with a variety of colourful celebrations. The fasts of folk-life and modernity of an industrial society. The traditions and customs, rites and rituals, dances and delicacies of the district are just precipitous, in these occasions people are seen in best of their attire, costumes and pageantry.

The fasts and festivals of the district are mostly of two types i.e., Sarbajanina or community based and Gharoi or family based. The Gharoi or family festivals are those, which are mostly observed at the level of families as per the traditions of their Kula, Gotra and Varna. A group of families consisting of same kinship or lineage also observe them in common. The Gurubara Manabasa, Sudasa Brata, Sabitri Brata, etc. are some such fasts.

The important festival of the district is Durga puja, Gaja Laxmi Puja, Basantika Durga Puja, Dipavali, Shiva ratri & Holi etc. The other festivals include Raja, Nuakhai, Gamha Purnima, Pus Punei are celebrated with traditional fervor and gaiety.

The Sarbajanina festivals are celebrated with pump and ceremony by the people in general at the community level. Some major fasts and festivals of the district are described below.

Hingula Yatra

Hingula Yatra is celebrated with pump and ceremony in Talcher sub-division of the district at the temple of Hingula Thakurani near Gopalprasad. It is regarded as the most significant festival of the district, which draws people from different corners of the state. This is celebrated on Bishnu Damanaka Chaturdasi tithi as per oriya Pajika.

Akshaya Trutiya

This festival is considered as the most ceremonial and auspicious day for sowing of paddy for the current agricultural season. Oblation is offered to Goddess Laxmi. It is also the beginning day of the construction work of Ratha of Lord Jagannath for the ensuing Car Festival.

Chandan Yatra

This starts from Akhaya trutiya and ends on Chandan Purnima. A smaller replica of Lord Jagannath, Balabhadra and Subhadra are taken to the sacred tank for boating. This is aimed for providing comfort and coolness to the Lords from summer. This is celebrated in the Jagannath temples of the district located at Angul, Talamul, Bagedia, Papsara, Khamar and Boinda.

Ratha Yatra

World famous Car festival of Lord Jagannath at Puri is celebrated as one of the most significant festivals of the state. Accordingly, Ratha Yatra is also celebrated with full ceremonial approach at Talcher, Angul, Pallahara and Athamallik with all traditional rites and rituals.

Ganesh Puja

The festival is celebrated on the 4th Day of the bright fortnight of lunar month of Bhadrab. Besides different schools, colleges, business houses Ganesh Puja is celebrated with full gaiety and traditional rituals at Talcher. The festival continues to be celebrated for a period of around 11 days, which attracts people from the whole district. Different prominent cultural troops and musical parties from the neighbouring states participate on this occasion. Talcher Ganesh Puja occupies a prominent position in the state for its colourful and ceremonial observance.

Nuakhai

This is the festival of offering the first rice harvested from the agricultural fields to the God/Goddess before human consumption. It is observed on Bhadraba Sukla Panchami Tithi. Specially in Athamallik subdivision, this festival is celebrated with care and courage.

Viswakarma puja

This is the puja of Lord viswakarma, the main architect of the universe who had fabricated the universe as per the direction of Brahma, the lord of creation. Mostly this puja is observed in Factories, industrial houses, workshops and other related fabrication houses, Engineering establishments. The important celebration of Biswakarma Puja of the district includes TTPS (Thermal), NALCO, F.C.I., Rengali and M.C.L. areas.

Durga Puja

Being the most colourful festival of the state, Durga Puja is celebrated with rites and rituals throughout the district. The Mahanadi Coal Field areas observe Durga Puja in a big way with more than 20 numbers of Puja Mandapas. Besides at Angul on the day of Dusahara the famous 'Ravanpodi' Utsav is also observed to mark the victory of truth over the evil power.

Laxmi Puja / Kumarotsav

Gaja Laxmi puja starts from the day of Kumar Purnima and continues till the next 10 days at Angul. Beautiful images of Goddess Laxmi are prepared in clay and worshiped on brightly decorated puja pendals. This is the most popular and colourful function of the district observed with much pump and ceremony. Besides Angul town, Laxmi puja is also celebrated at Banarpal, Koshala, Athmallik and Pallahara.

Kumarostsav is the day of great pleasure, ecstasy and enjoyment for maiden girls falls on the full moon day of Aswina Purnima. The Moon God is worshiped by the maiden girls on this occasion for a bright future married life.

Kartika Purnima

This is observed on the last day of the lunar month of Kartika. This is specially regarded as the most sacred purnima of the Hindus. Rasa Ustav is also observed to pay honour to Lord Krushna and Radha. To mark the great tradition and cultural heritage of Odisha's trading community, Boita Bandan is celebrated symbolically throughout the district by floating decorative toy boats in the nearby rivers and ponds.

Sripanchami (Saraswati Puja)

This is observed on the fifth day of the bright fortnight of the lunar month of Magha. Devi Saraswati is worshiped by the students as the Goddess of wisdom and knowledge. Besides, clubs, cultural houses and other institutions also celebrate the puja in the district.

Binkei Yatra

This festival is held on the 10th day after the Dolapurnima in the lunar month of Chaitra. This observation is made in the honour of Mother Goddess Binkei who is the saviour of virtuous and killer of vicious people. A great fair is also arranged for three days with a large no. of shops of variety items. People from far off villages gather for offering prayer to the Goddess and marketing for their household items

Music & Dance

Dances are many dimensional, such as, martial, ritualistic, ecstatic and funeral etc. Such dances are always accompanied by music which makes them enjoyable. It is generally said that dance and music are complementary to one another. In the tribal communities there are many types of dances. They are pigeon dance, boar dance, elephant dance, snake dance, bear dance, etc. Now-a-days, dance has degenerated to a great extent. The Changu, a big tambourine, the Ghagudu, a kind of drum and flute are the musical instruments found in their community. In festive occasions as well as in marriage ceremonies, they drink and dance merrily even at times throughout the night. There are many folk songs which are sung at the time of dance.

Danda Nritya

Danda Nritya is the premier and most ancient folk-dance of the district of Angul. This is practiced in all the four sub-divisions of the district. It is a composite synthesis of dance, drama, song and music blended with mythical tales and striking realities of worldly life. This folk-dance is a kind of penance and offering to the almighty for eliciting the blessings of salvation.

This dance commences in between last week of March and 1st of April and ends with Mahabisuva Sankranti i.e., mid of April in each year. The devotees who

undertake this dance are called Bhuktas. The leader of the group is called the Patabhukta. They lead a hard and sacred life with much austerity during these days and remain away from home. They undertake bodily penance and practice Yogatantra to get the divine grace. The ritual trances of the Nritya proceed as follows. The Dandautha is done before 14 to 21 days of Bisuva Sankranti with a series of rituals and worship at the mid-night. The next day morning witnesses the Danda-Yatra in which the symbolic representative of God (Shiva) and Goddesses (Paravati) are taken in procession throughout the village with the royal decorations,
music, dance and sankriatns. At the time of processions, the decorative items like Chhatri, Chhatra, Alata, Chamara and musical instruments like Veri, Turi, Ghanta, Sankha, Dhola are used. The deities that are taken in procession are known as Siva-Parvati and Parava (Mahakali). The Danda Players decorates themselves with sacred thread and yellow clothes and perform the different phases which includes Dhuli danda, Pani danda, Ratri-Ujagar danda as per its traditional rituals.

In this way Danda Nritya becomes a fine synchronization of community enjoyment with devotion, dedication, and consecration. In this form of folk-dance religion, lyric, music, entertainment and striking realities of life are blended together in a unique manner so as to make it a composite whole of Folk-drama. It is a type of mortification undertaken voluntarily by which absolution, confession and satisfaction is conveyed. The whole dance is a kind of penance offered in the honour of Sakti-upasana. Through Sakti-upasana is found in almost all parts of Odisha, Danda Nirtya with its lyric, art, theme and philosophy has been identified as unique over the centuries.

Ravan Chhaya

The tradition of shadow puppet theatre in India is very old and thrives only in the States of Odisha, Andhra Pradesh, Karnataka, Tamilnadu, Kerala and Maharashtra. Each State has its peculiar design, style, colour-scheme and manipulation technique.
The shadow-puppet theatre of Odisha is known as Ravan Chhaya. It is titled thus because, the Chhaya or shadow is considered inauspicious and is therefore related to the character of Ravan who is equated with evil. Among all the styles of India it is the simplest with no colour, and almost no dance or fight sequences. With simple movements, the full unjointed figures depict versions of the Bichitra Ramayana. Accompanied by Khanjani (a tambourine variety of instrument) and cymbals, the musicians sing standing in front of the screen. Though most of the puppet figures are made out of deer skin, those of demoniac characters are made from stag skin. Each puppet is fixed with a thin rod of bamboo for manipulation from behind the white curtain. The light is provided by a large earthen lamp fed with castor oil.

The tradition performers of Ravan Chhaya were from the community of Bhats who received the patronage of the local Rajahs of Pallahara in shape of land grants.

When all other Bhats discontinued their profession, some Gurus have made it alive now. The repertory group now performs with two sets of puppets, the first which are designed and cut according to the old traditional style and the second with the puppets designed and cut on the model of traditional pata painting of Puri. So, 'Ravan Chhaya' is the unique shadow puppet of Angul & Odisha.

Primary Survey:

The villages surrounding the ML area were selected for field survey **(Chhendipada Jangal, Chhendipada RF, Handigora, Machhakuta, Machhakuta Jangal, Porapara, Porapara Jangal, Tentuloi Korasahi** etc.). Apart from the above the villages visited are Mamuriasahi, Chhendipada, Behera Sahi, Madhopur & Nuapada, Pinpal Bahal, Patrapada, Godamandal etc.

Data was collected based on specific designed Questionnaire & focused group discussions. There has been increase in the population & other related factors since 2011 (male population, female population, SC/ST population, literacy, working population, category of workers etc.). While comparing the household survey results with census data 2011 the observations are found to be validated.

Basic Observations during field survey and discussions with villagers

1. The villages in the core zone and buffer zone are having primary amenities and infrastructural facilities but have potential for further improvement. Mostly the villages are electrified, houses are with RCC roof.



2. One can see few houses have roofs with Asbestos and Tin sheets. There are schools, parks, ponds, banking facility and good roads. Some schools are adopted by VEDANTA group. It appears that people are well to do. Cooking medium is wood and gas. There are health and wellness centre. Drainage appears to be a main problem in the villages. Supply of water is through OHT or Handpumps. Ponds are very common and appears to be a tradition which are very useful for ground water recharge.



3. Hinduism is the major religion.
4. Sex Ratio is on a lower side in rural area.
5. Gender gap between male and female literacy is approx. 10 % in rural area.
6. Employment is a major concern besides literacy.



7. The area is running a parallel economy in illegal coal mining. Various groups of scrupulous individuals are digging it illegally.

8. Around 160 villages have been declared ODF(Open Defecation Free), in Angul within four months of the launch of the Swachh Bharat Mission on October 2, 2023. Each household has been provided a toilet. Angul is one of the best performing districts in Odisha for community-led total sanitation (CLTS). Pictures and coordinates of 7,594 toilets in the district have been uploaded on the portal of the ministry of drinking water and sanitation. The information serves as proof of construction of the toilets. Based on this data, the Centre releases Rs 9,000 (the state contributes another Rs 3,000) as cost for each toilet as per the interaction with the school.



9. There is termite problem in almost all the villages.

10. Tikara river is quite large. Anuli river merges with Tikara River.

11. Supai nala appears to be seasonal

12. Dense cultivation of banana is common and can be seen almost in all places. Cashew plantations are seen at some places which are promoted by forest department. Agriculture appears to be the main occupation of the area.



Cow rearing is common in almost all villages



13. Mostly internal village roads are pucca bitumen roads. OSD (Odisha/other district road) was found to be 4 m in width (besides kutcha left over area is approx. 13 m on both sides, total 17 m). SH 63 is approx. 7m in width (besides kutcha left over area with a further width of approx. 14 m on both sides, making total 21 m).

14. Govt. of Odisha, Forest Environment & Climate Change Dept. planted 25000 seedlings in 50 ha @ 500 plants per ha at Chhendipada RF (Patch III) Division Angul (GPS N 21^0 03’ 26” E 84^0 51’54”) in 2022-23. Main species planted are Amla, Jamun, Mango, Bamboo, Aata, Bahada, Kaitha, Bel etc.



15. Avenue plantation of 2016-17 in 5 km range (Godamandal to Patrapada) seen on the way.

16. Kukurpetta MIP(RES) is in Block Chhendipada (GPS N 20^0 02' 10" E 84^0 33'50"), with catchment area 14.89 sq km with max discharge capacity 147.16 m^3/s. No. of canals are 3 ,irrigating 587 ha in Kharif and 121 ha in Rabi.

17. People appeared to be lazy but in festive mood during the survey period.

18. The language is Odia. It has many variants with slight alterations. Some of the tribal use their own language which in not understandable.



19. The people predominantly use rice as their staple diet.

20. Elder males of middle-class wear Dhoti and Shirt or Kurta. Poor and even middle-class people males wear only Dhoti and use Gamuchha (napkins) on their upper part of the body. Women irrespective of caste and social condition wear Saree. Most of them wear Blouse with Saree.



21. Being the most colourful festival, Durga Puja is celebrated with rites and rituals throughout the district. The Mahanadi Coal Field areas observe Durga Puja in a big way with more than 20 numbers of Puja Mandapas as informed by locals during interaction. We observed 4 to 5 small and big Puja Mandapas in Chhendipada Tehsil itself.

22. The inferences is based on the observation during the study period only. As such the inferences cannot be used for any other purpose and is not liable for any legal action.

Quality Of Life

The quality of life (QOL) refers to degree to which a person is healthy, comfortable, and able to participate in various events of his life and enjoy the same to the maximum. Enjoyment has two components viz experience of satisfaction and possession or achievement in the field of education, political power, property, service carrer, business etc.

Project

Baitarni West Opencast Coal Mine project with production capacity of 15 MTPA in an area of 1196 ha located at Villages Chhendipada Jangal, Chhendipada RF, Handigora, Machhakuta, Machhakuta Jangal, Porapara, Porapara Jangal, Tentuloi Korasahi, Tehsil Chhendipada, District Angul (Odisha).



The Baitarni West Coal Block is located in the North central part of Talcher coalfield and falls within Angul district. The details of location are discussed in Chapter-3.

The baseline data was collected for villages within 10 km radius area from the boundary of the project area. Total 66 villages were considered for study (primary and secondary data). Almost 50 % villages were considered for primary data for Quality-of-Life Index.

Table 3:20 List of Villages Falling Within 10 Km from the Boundary Of The Lease

SL.NO.	VILLAGES UP TO 2 KM	SL.NO.	VILLAGES 2 KM TO 5 KM	SL.NO.	VILLAGES 5 KM TO 10 KM
1	Balijharan	1	Bahalsahi	1	Ashrubahal
2	Chhendipada	2	Balipata	2	Badaberena
3	Chhendipada -sub dist	3	Bamanali	3	Balinali
4	Chhendipadajungle	4	Baramancha	4	Barapada
5	Deuria jharan	5	Barasahi	5	Bhagilakata
6	Ghuntulipasi	6	Basantapur	6	Bhejidihi
7	Gobindpur	7	Chandrapur	7	Brahmanbil
8	Handigoda	8	Dubanali	8	Chakundapal
9	Ichhapur	9	Gopinathpur	9	Dangapal
10	Jamubahali	10	Hatianali	10	Dhangurumunda
11	Machhakuta	11	Kasidiha	11	Golagadia
12	Machhakutajungle	12	Kukurpeta	12	Golasar
13	Podapada	13	Kuskila	13	Gopabalhabpur
14	Podapadajungle	14	Panapasi	14	Gundurijharan
15	Salakhaman	15	Sapoinali	15	Jaripal
16	Tentoloi	16	Tangiri	16	Jhintipal
17	Tentuloikodasahi			17	Kankarapal
				18	Karadabahal
				19	Kisinda
				20	Korada
				21	Kosala
				22	Madhupur
				23	Nuapada
				24	Patakumunda
				25	Patrapada
				26	Purunagarh

				27	Putagadia
				28	Raipal
				29	Rajanpal
				30	Santarabandha
				31	Sarapal
				32	Similipal
				33	Similisahi

Total villages = 66.

Quality Of Life Index (QLI):



Quality of life of the people is one of the indices to measure the standard of living of the people in an economy. It is influenced by national and per capita income of the people. Many other factors like consumption, output, health, environment, education, freedom, security, non-violence peaceful atmosphere etc. also influence human welfare directly or indirectly. In other words, none of these factors alone determine the welfare of the people. At the same time, we should note that Income is not itself the single representative factor of welfare.

It is as such necessary that all the determinants of welfare should be combined together to measure the quality-of-life index. But it is not possible that every variable should be accounted for because of conceptual, statistical difficulties. It is, therefore, said that an index comprising of certain selected social factors be made to determine the quality-of-life index.

Quality of life study analyses mainly 10 dimensions which can be measured statistically to represent the various aspects of quality of life. The ten of these dimensions can be considered to reflect people's well-being, which are education, health, income, land, housing condition, fuel used for cooking, sanitation (toilet), clean drinking water, assets, and livestock.

The parameters are given score as given in the following table:

Parameter	**Range= Rs/annum**	**Score**
Income Parameters – (Income in Rs.)	<35000	0.1
	35000 – 50000	0.2
	50000 - 75000	0.3
	75000 – 100000	0.4
	100000 - 200000	0.5

	200000 – 300000		0.6
	300000 – 500000		0.7
	500000 – 800000		0.8
	800000 – 1000000		0.9
	>1000000		1.0
Land	Landless		0
	Marginal		0.25
	Small		0.5
	Medium		0.75
	Big		1
Housing	Floor	Cement	1
		Stone	0.5
		Mud	0.25
	Roof	Cement	1
		Tiles	0.5
		Thatch	0.25
	House Type	Kutcha	0.25
		Semi Pucca	0.5
		Pucca	1
	Room per member	<0.5	0.25
		0.5 – 1	0.5
		=1	0.75
		>1	1
Fuel	LPG		1
	Wood/ Charcoal		0
Toilet	Available		1
	Not Available		0
Water	Available		1
	Not Available		0
Assets availability	Mobility Assets	If any Available	1
		no asset	0
	Asset for Info	If any Available	1
		no asset	0
	Other assets	If any Available	1
		no asset	0
Livestock	Available		1
	Not Available		0
Education	If 0		0
	0-4		0.1
	4-5		0.2
	5-6		0.3
	6-7		0.4

	7-10	0.6
	10-12	0.8
	12-15	0.9
	>15	1.0
Health	= 0	1
	between 0 - 0.25	0.75
	between 0.25 - 0.51	0.5
	between 0.5 – 1	0.25
	If > 1	0

Quality of life (QOL) is one of the components to enumerate the status of socio-economic environment of a region. QOL has been determined for the study area in the core and buffer zones (in and around 10km radius) of the project. Village information from individuals and panchayat level was gathered. A number of socio-economic indicators like food, clothing, sanitation, security, environmental pollution etc. have been considered for the study besides indicators mentioned above.

The QOL has been determined under two categories i.e. subjective evaluation of local inhabitants and objective evaluation by an expert group incorporating the existing conditions with reference to availability and utility of available basic amenities. In the present study assessment of quality of life has been carried out from the survey and observation made during the vist for subjective dimension and by evaluation of existing condition (secondary data) by an expert group.

The population in the study area has been surveyed on the basis of proportionate stratified random sampling. The questionnaire is based on satisfaction level of each individual for the various indicators used, following a scale as:
0.1- Highly Unsatisfactory (HU)
0.3- Not Satisfactory (NS)
0.5- Neutral (NE)
0.8- Satisfactory (SA)
1.0- Highly Satisfactory (HS)

Once objective measures are obtained for each factor, they are transformed to a normal scale varying from 0 to 1 (value function curve) in which "0" corresponds to the lowest or least satisfactory measures and "1" corresponds to the highest. The weights have been assigned to each factor by ranked pair wise technique by an expert group based on the secondary data and general observations.

For each objective measure, a corresponding subjective measure has been developed and obtained for each individual of the sample population, by investigation the rate of satisfaction with the objective measures for each factor. A 0 to 1 scale (value function curve) has been used where 0 corresponds to the lowest level of attitudinal satisfaction. Again, weights have been assigned to each factor from a total of one using ranked pair wise technique. After collecting

the information, the following steps have been followed to calculate the quality of life:

- Ranking of QOL factors i.e. income, food etc.
- Assigned values for subjective attitude of satisfaction levels, i.e. HS, SA, NE, NS, HU have been derived.
- Assigned weight to all the factors by pair wise ranking technique.
- Normalization of weights by taking the total weight as unity.
- Frequency distribution of all the respondents to get QOL (s)
- Adding the values and dividing by the number of respondents to get QOL(s).

QOL is then estimated using the following formulae (USEPA, 1974)

Objective Quality of life

$$QOLo = \sum_{i=1}^{n} QL_1 \times W_1$$

Where,

QOL(o)	=	Objective Quality of life Index.
n	=	Number of QOL factors.
L	=	1, 2, ... n.
QL_1	=	Satisfaction level (assigned by the expert group) for its objective indicator out of one (L).
W_1	=	Normalised weight for the factor.

Subjective Quality of life:

$$QOLs = 1/P \sum_{i}^{m} \sum_{J}^{P} Qlij \times Wi$$

Where,

QOLs	=	Subjective QOL Index.	
P	=	No. of respondents	J= 1, 2,. ..... P
m	=	No. of factors	I= 1, 2,. .....m
QLij	=	Subjective quality index (Vfc) for ith factor assigned by ith respondents.	
Wi	=	Weight of ith factor.	

Quality of life:

$$\frac{QOL(s) + QOL\ 9(o)}{2}$$

QUALITY OF LIFE FACTORS LIST

* Health
* Nutrition
* Housing
* Education
* Employment and working conditions
* Clothing
* Energy
* Transportation
* Communication
* Water supply and sanitation
* Environment and pollution
* Recreation
* Social security
* Human rights

The amenities" like animal, irrigation, drinking water, education, agricultural practices, land, soil type, agricultural production per hectare, industry, self-employment, mining, transportation facilities, per capita income, source of income , environmental conservation practices adopted, rainwater harvesting practices adopted, death and birth rate, availability of fresh food etc." have helped in indexing the overall village facilities and at last the quality of life of the people living in and around 5 & 10 km of the mining area was calculated.

Primary survey was conducted in 50% of total villages and randomly 0.5% of the houses were selected in the villages for survey. On the basis of amenities, resources present in the villages and individual survey the assessment of quality of life (QOL) was done.

Results of QOL Analysis within study area
Socioeconomic Profile within 0-10 km

Sl. No.	Village Name	QOI	Remarks
	0 to 2 km from project area		
1	Balijharan	0.535	Above neutral
2	Chhendipada	0.531	Above neutral
3	Chhendipada -sub dist	**0.650**	Above neutral
4	Chhendipada jungle	0.509	Above neutral
5	Deuria jharan	0.502	Above neutral
6	Ghuntulipasi	0.496	Below neutral
7	Gobindpur	0.507	Above neutral

8	Handigoda	0.551	Above neutral
9	Ichhapur	0.585	Above neutral
10	Jamubahali	0.410	Below neutral
11	Machhakuta	0.409	Below neutral
12	Machhakutajungle	**0.402**	Below neutral
13	Podapada	0.426	Below neutral
14	Podapadajungle	0.437	Below neutral
15	Salakhaman	0.451	Below neutral
16	Tentoloi	0.486	Below neutral
17	Tentuloikodasahi	0.410	Below neutral
	Average (0 to 2 km from project area)	**0.488**	Below neutral
	2 to 5 km from project area		
1	Bahalsahi	0.568	Above neutral
2	Balipata	0.572	Above neutral
3	Bamanali	0.535	Above neutral
4	Baramancha	0.563	Above neutral
5	Barasahi	0.493	Below neutral
6	Basantapur	**0.418**	Below neutral
7	Chandrapur	0.505	Above neutral
8	Dubanali	0.562	Above neutral
9	Gopinathpur	0.537	Above neutral
10	Hatianali	0.615	Above neutral
11	Kasidiha	0.530	Above neutral
12	Kukurpeta	0.521	Above neutral
13	Kuskila	0.532	Above neutral
14	Panapasi	**0.647**	Above neutral
15	Sapoinali	0.533	Above neutral
16	Tangiri	0.606	Above neutral
	Average (2 to 5 km from project area)	**0.546**	Above neutral
1	Ashrubahal	0.578	Above neutral
2	Badaberena	0.492	Below neutral
3	Balinali	0.635	Above neutral
4	Barapada	0.593	Above neutral
5	Bhagilakata	0.583	Above neutral
6	Bhejidihi	0.598	Above neutral
7	Brahmanbil	0.595	Above neutral
8	Chakundapal	0.582	Above neutral
9	Dangapal	0.535	Above neutral

10	Dhangurumunda	0.565	Above neutral
11	Golagadia	0.560	Above neutral
12	Golasar	0.561	Above neutral
13	Gopabalhabpur	0.562	Above neutral
14	Gundurijharan	**0.640**	Above neutral
15	Jaripal	0.533	Above neutral
16	Jhintipal	0.616	Above neutral
17	Kankarapal	0.566	Above neutral
18	Karadabahal	0.497	Below neutral
19	Kisinda	**0.469**	Below neutral
20	Korada	0.560	Above neutral
21	Kosala	0.495	Below neutral
22	Madhupur	0.477	Below neutral
23	Nuapada	0.482	Below neutral
24	Patakumunda	0.573	Above neutral
25	Patrapada	0.616	Above neutral
26	Purunagarh	0.486	Below neutral
27	Putagadia	0.490	Below neutral
28	Raipal	0.569	Above neutral
29	Rajanpal	0.540	Above neutral
30	Santarabandha	0.495	Below neutral
31	Sarapal	0.477	Below neutral
32	Similipal	0.562	Above neutral
33	Similisahi	0.560	Above neutral
	Average (5 to 10 km from project area)	**0.550**	Above neutral
	OVERALL AVERAGE	**0.533**	Above neutral

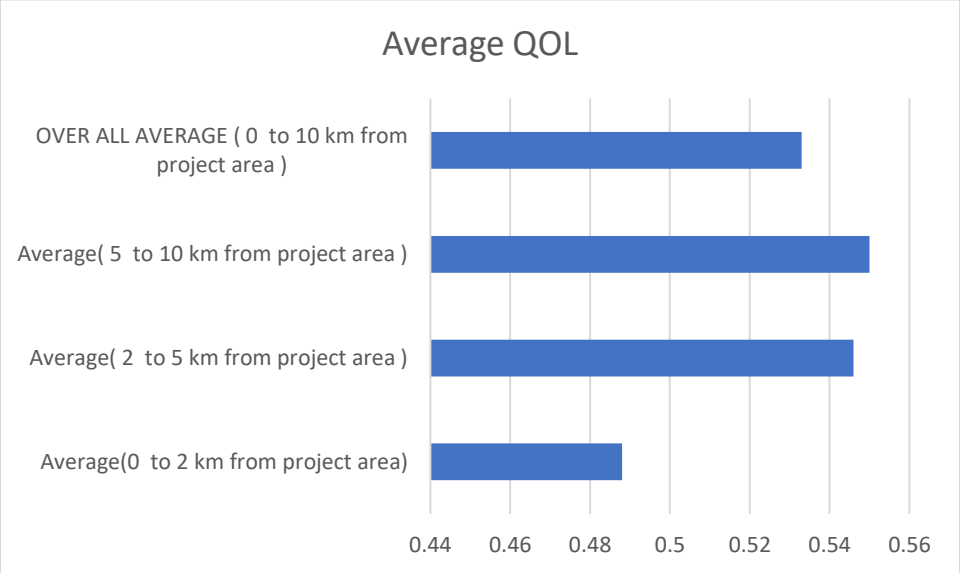
The average Quality of life within the entire study area is 0.533 which is above neutral as different components have different weightage in different areas.

Average Quality of life in the study area

Sl. No.	Area Description	No. of villages in the area	Average QOL	Highest QOL	Lowest QOI
1	Average (0 to 2 km from project area)	17	0.488	0.650	0.402
2	Average (2 to 5 km from project area)	16	0.546	0.647	0.418
3	Average (5 to 10 km from project area)	33	0.550	0.640	0.469

	OVER ALL AVERAGE (0 to 10 km from project area)	66	0.533	0.650	0.402

From the above it is clear that as we go beyond the project area, QOL improves from 0.488 to 0.550. It is also indicated in the figure given below:



Conclusion: The above QOL and QLI are based on the limited survey and supported by secondary data, published papers and other relevant documents of the area. The detailed study is being carried out as a R&R study which will reflect more about the QOL and QLI of the people of the area.

3.5.1 PROBABLE IMPACT OF MINING ON SOCIETY



There is always positive impact of mining if it is legal as development of the area is proportionate to sustained scientific mining.

•Employment opportunities, both direct and indirect are likely to be generated

•Communication and transport facilities are expected to improve.

•Education facilities are expected to improve

•Enhanced Community health care programs will provide better health and medical facilities

•Cultural & sports activities and improvement of social welfare is expected

The socio-economic status of the local people improves substantially due to

employment generation. The land rates in the area improves due to mining.

Educational institutions also come up and lead to improvement in the educational status of the people in the area.
The mining results in growth of the surrounding areas by increasing ancillary development and supporting infrastructure. Special emphases on financial and social benefits are available to the local people. Developments of social amenities are there in the form of medical facilities, education for the underprivileged and creation of self- help groups. State gets revenues in terms of taxes. Business opportunities for the local community is available like transport work etc.

Several skilled, semiskilled and unskilled workers are required for peripheral activities like transport, logistics, engineering, services, commercial services etc.

3.5.2 HEALTH HAZARDS

Inhalation of coal dust often causes inflammation and Fibrosis. The miners and people who live in nearby villages and towns are most affected by inhalation of dust. Cardiopulmonary disease, hypertension, COPD, and kidney disease are found in higher number than normal rates in people who live near the coal block. Hence, both miners and the mining authority must take strictest action to keep them away from inhalation of coal dust when they are engaged in extraction of coal. This is a negative impact of the project.

3.5.3 EMPLOYMENT OPPORTUNITIES

Coal mining is highly labour-intensive. It requires large number of people to operationalize and actualize it.

The proposed mining will provide direct and indirect employment opportunities to job seekers in the region. For mining by opencast method as many as 1752 **persons** will be recruited. Apart from the above there will be employment opportunities for the jobseekers to serve in the rest house and dispensary close to the mining area. Also there will be employment opportunities for those who belong to unorganized sector. This is a positive impact of the project.

3.5.4 INCREASE IN GOVERNMENT REVENUE



The mining results in growth of the surrounding areas by increasing ancillary development and supporting infrastructure. Special emphasis on financial and social benefits are available to the local people. Developments of social amenities are there in the form of medical facilities, education for the underprivileged and creation of self- help groups. State gets revenues in terms of taxes. Business opportunities for the local community is available like transport work etc.

3.16.8 CSR/CER action plan proposed

1. More emphasis on Education specially job-oriented skill development of the youth.
2. Girl Education. BETI BACHAO AND BETI PARAHO considering the gender inequality.
3. Employment generation activities by forming SELF HELP GROUPS. SHG's for agriculture may be considered.
4. Contribution towards Maintenance of Roads /Schools/ Temples/Primary hospitals including value addition in terms of public facilities may be from miners.

CSR activities will be undertaken in Chhendipada Jangal, Handigora, Machhakuta, Machhakuta Jangal, Porapara, Porapara Jangal, Tentuloi-Korasahi.

3.5.5 CONCLUSION

Coal is one of the main sources of energy in India. Accordingly, affordable and sustainable supply of coal is inextricably linked to the goal of ensuring energy security in India. The project will certainly help our country towards energy security.

3.15 ECOLOGY AND BIODIVERSITY

Introduction



Development is must for the nation; however, development needs a price. Price is paid by one kind of people and natural biota and other people rea fruits of prosperity. The disparity between the sufferer and enjoyers increases. The main objective ecological survey is aimed to assess the existing flora and fauna component within study area. Biodiversity is important for ecosystem amenities and hence for all human well-being. At the same time, however, these losses in biodiversity and associated changes in ecosystem services have caused other people to experience declining well-being, with some social groups being pushed into poverty. Across the series of biodiversity measures, current rates of loss exceed those of the historical past by several orders of magnitude and show no indication of deceleration. Biodiversity is deteriorating quickly because of change in land use, climate change, offensive species, overexploitation, and pollution. These result from demographic, economic, socio-political, cultural, technological, and other indirect drivers. While these drivers vary in their importance among ecosystems and regions, current trends indicate an ongoing loss of biodiversity. So, biodiversity assessment is necessary (Millennium Ecosystem Assessment, 2005).

The lease area does not form a part of any National Park or Wildlife Sanctuary or Critical Wildlife habitat. No protected area is situated in the lease area or within the Zone of Influence. The lease area is surrounded by a series of existing mines.



The Forest around the lease & impact zone area are disturbed due to various Mining activities and presence of Wildlife is very sparse. As per the information collected the following flora & fauna are found in the core and Buffer area of the project. No endemic species are found either in the project area & in the ZOI area. The area majorly drained by Aunli and Tikera River.



The ecology & biological study of the area of interest was undertaken with the following objectives:

- To assess the nature and distribution of vegetation in and around the existing project site.
- Analyze the distribution of animal life spectra.
- To quantify the ethano-botanical importance of the plant species.
- Possibility of presence of breeding grounds.



The methodology adopted for above is:

- Compilation of secondary data and literature.
- Generation of primary data by undertaking systematic ecological studies in the area.
- Discussion with local people to elicit information about local flora and fauna.



Study Area: 50641.55 ha.
Core Zone: Project site with 1 km boundary.
Buffer Zone: 10 km radii w.r.t. project site excluding core zone.
Study Period: Oct to Dec 2023
Data Collection: As per QMS of GRC.

Angul District and Study area



Angul is like a bridge between western and coastal halves of Odisha where the prehistoric and protohistoric relics are found in village Bhimakand, Kankili, Kulei, Samal, Sanakerjang, Kaliakata, Paranga, Kerjang, Tikarapara and Pallahara.The geographical location of Angul made the Bhanjas of Angulaka'pattana, the Sulkis of Kodalaka Mandala, the Nandodbhavas of Airavatta Mandala, the Tungas of Yamagartta Mandala rule over it. But all through the rules of different dynasties, Angul has retained her cultural identity which is much more prominent than its political establishments. After independence Angul became one of the important district in the State of Odisha. Rich cultural heritage, cosmopolitan approach, different temples, natural beauty, dense forest, availability of raw materials in its abode make Angul a major district in 21st century.

Athamallik

The tradition regarding the foundation of Athamallik is that the area was divided the territory in 8 divisions, each under a Chieftain called as Mallik as a measure to suppress the unruly tribes. Hence, the Kingdom changed its name from Hondapa to Athamallik which means the land of 8 chieftains. In course of time his successor shifted the capital from Hondapa to Thakurgarh,Nuagaon and finally to Purunagarh (Kaintaragarh) . The palace records show that 42 Kings of this dynasty have ruled the estate. The Chief of Athamallik was officially known as Zamindar and was being addressed as Samanta. The chief of Athamallik was officially recognized as a Raja and he was allowed to have hereditary title of Raja. Devi maheswari is the deity of royal family of Athamallik.

The King appointed some deputy administrators for several area who were Khyatriyas and Athamallik was named after them.

1. Sanjamura – Pandav Mallik
2. Bamur – Rajendra mallik
3. Mindol – Damodar mallik
4. Jamudoli – Balaji Biswal
5. Kaitragarh – Angada Garhtiya

Other three names are still unknown.

The hot spring at Deulajhari is one of the attractive place in this area. Goddess Maheswari is the present deity of Athamallik. Goddess Maheswari is worship with "Maheswari Tantra" and it is obe of the important tantric pitha of the area. God Dolagobinda is also worshipped near this temple. Apart from this, many God and Goddess like Rani Durga, Bandurga, Mahakali, Khambeswari, Narayani, Mahalaxmi, Mahakala and Bhairabi are also worshiped near this area.

Talcher



Talcher is situated on the bank of the river Brahmani and regarded as one of the major Gadjat area of our State. In the dense forest area during the earlier time, Boita Sabara were living peacefully. In the 14th century, Panchdiha Chasa who were mainly farmer community cane to this fertile area and started living here. The King of Talcher Padmanabha Birabara Harichandan who renamed the state as Talcher after the name of the family GoddessTaleswari. Padmanabha Harichandan is remembered as a pious ruler and is said to have constructed many temples in his state. There is a mythology behind the naming of Talcher. King Padmanava once going to the forest and reached in the area where present palace is situated. Near the Taleswari temple, King was decided to establish his capital. But the Khandual Zamindar opposed him. King was disheartened. But in the night whle sleeping, Goddess Hingula told him if the King started worshiping Goddeess Hingula and Goddess Taleswari, then he can win the fight against Khandual Zamindar. But the condition was that the area to be named as Talcher. King then worshipped both Goddess Hingula and Taleswari and won in the fight and named the area as Talcher. He established his capital there. Talcher is now one of the important industrial towns of the district.

Pallahara



It is belived that Pallahara state was established by Satosh Pala before 18th century. Santosh while returning from Puri after pilgrimage, he was nominated to lead the Sabara, Kandha, Malhara, and Jhara people of Pallahara. At that time Pallahara or Palalahada was simply known as Lahada. Santosh stayed there and Pala Sabara had hided him under Pala gada”” while he tried to take control of the area from local tribes. Finally, Santosah Pala and his supporters won the fight and establish his kingdom there. Santosh Pala changed his name as “Ganeswar pala”when he became King of Palalahada. After this as per the tradition whoever became king of Pallahada was known as Ganeswar Pala or Muni Pala. The royla family of this area was known as the Pala family. Though Pala family estabslihed Palalahada, still SAbar community were also held equally responsible for establishment of Palalahada Kingdom.

The area falls under East & Southeastern Coastal Plain **(agro-climatic zone XI)** as per IASRI http://www.iasri.res.in/agridata/12data/chapter1/db2012tb1_2.pdf. The block map is available at https://gisodisha.nic.in/Block/CUTTACK.pdf.

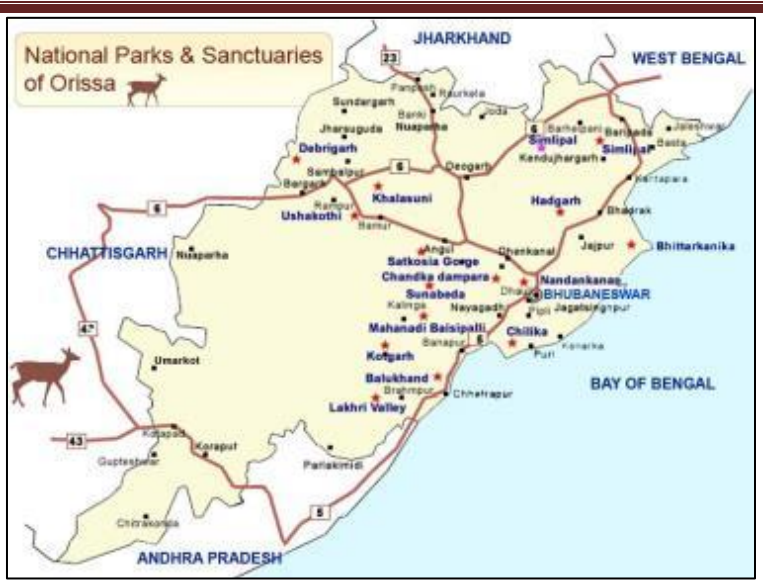


Figure 3-30 Ecological Map of Orissa

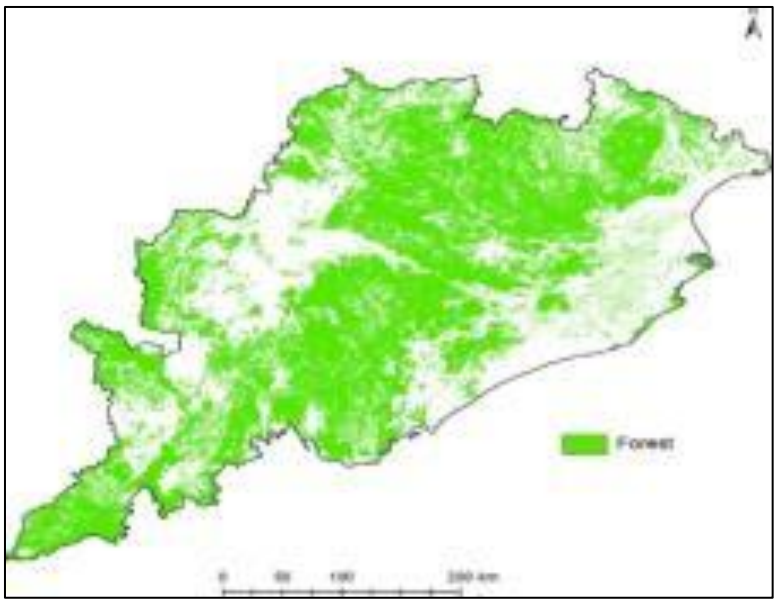


Figure 3-31 Forest Map of Odisha

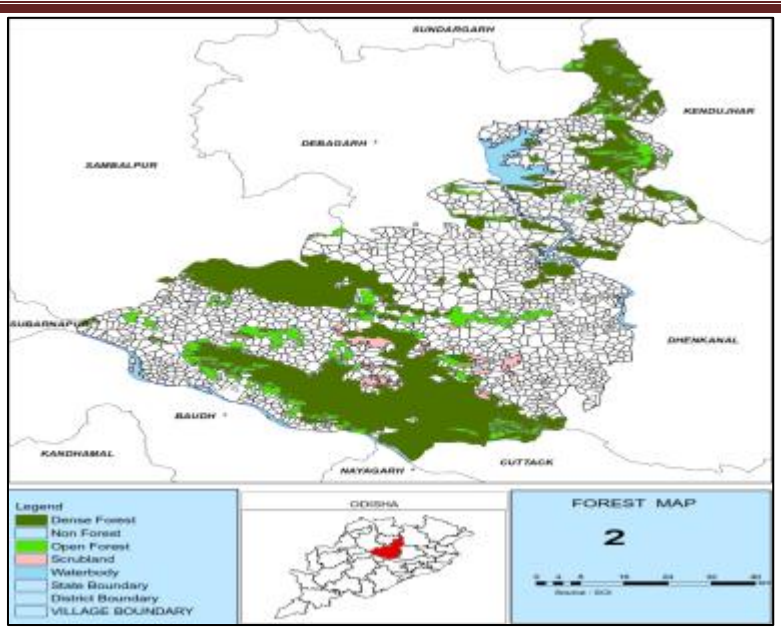


Figure 3-32 Forest map of Angul



Figure 3-33 District Map

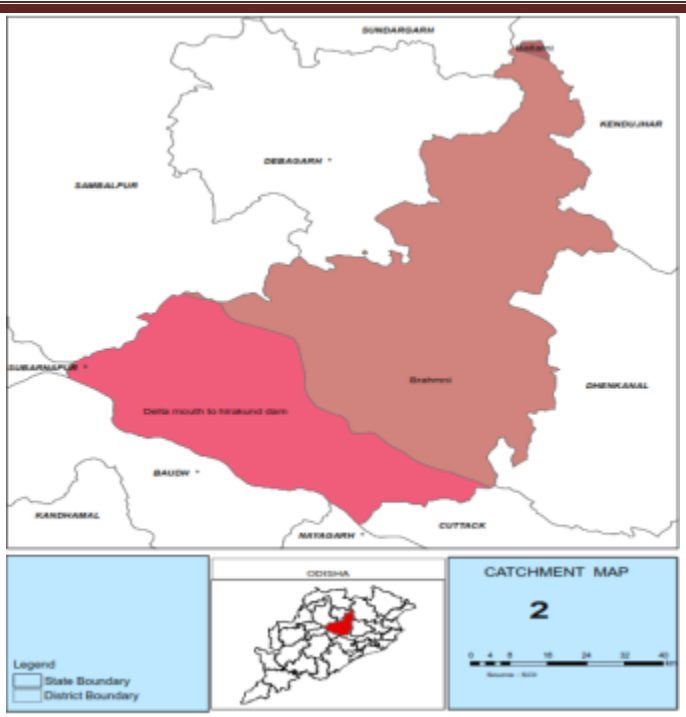


Figure 3-34 Catchment Map of Angul

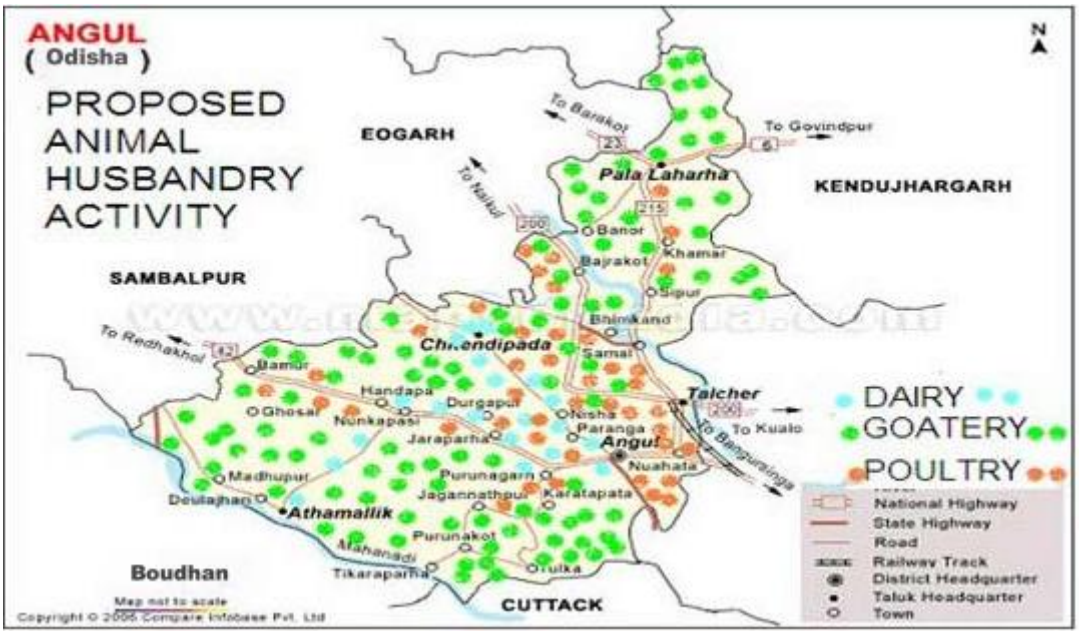


Figure 3-35 Animal Husbandry in Angul

Terrestrial Floral and Faunal Components

Floral Diversity in Study Area

A plant community may be viewed as a vegetational unit characterised by an essentially constant composition with a uniform appearance. The forest of the study area as per revised classification of Indian Forest types belongs to Moist Peninsular Low-Level Sal and Northern Dry Deciduous Forest dominated by Dry Bamboo Brakes. Within the study area, there are plenty of trees because of good soil, partly, flat land and sufficient rainfall. The forest is humid and always full of

green grass. Because of the cultivation done by local farmers, thinning of forest has been observed. The main species existing in the forest are Sal, Mahua, Amla, Bel, Chandan, Bamboo etc. The Gauduni nala is passing though ML area which needs to be diverted which needs complete loss of existing biodiversity of nala (aquatic and terrestrial). The average biodiversity cost of this nala is approx. 6.5 Cr per ha.

To understand the attribute of communities, a phyto-sociological study of the vegetation was conducted in the forest area at three places namely, core zone forest, Debinagar PF and Kosala RF. The sites were selected on the basis of vegetation component and their importance from project viewpoint. At each selected site, random vegetation sampling was done with the help of 10 m x 10 m quadrate size. Presence and absence of species, number of individual species and basal area of each plant species in each species were recorded. Frequencies and densities were calculated following Curtis and Mcintosh (1951) method. Species diversity index was estimated using Shannon-Weiner index (1963).

Shannon-Weiner diversity index is calculated by following formula:

$$\overline{H} = -\Sigma \, Pi \, Log \, Pi$$

Pi = ni/N, where ni is number or biomass or IVI of individual species and N is the total number or biomass or IVI of all individuals.

In the present study IVI (Importance Value Index) has been taken into consideration for evaluation of diversity index. IVI gives total performance of individual species at particular site. IVI is the total relative frequency (RF), relative density (RD) and relative dominance (RDO). Since IVI includes many of the attributes at a time, therefore consideration of IVI was preferred on number for evaluation of Shannon-Weiner diversity index. This diversity index includes both richness of species at any particular site. Higher the value of index, more is the stability of that community. Thus Shannon-Weiner diversity index is an indication of goodness of community.

Phyto-sociological study of tree layer vegetation was conducted at various sites. The parameters such as frequency, density, abundance, relative frequency, relative density, relative dominance and IVI were evaluated for each sampling sites. Shannon-Weiner diversity index was also calculated for each site. Tree flora vegetation analysis results are shown in the tables below.

Table 3:21 Characteristics of Tree Layer Vegetation at Core Zone Forest

Name of Species	F	D	Ab	RF	RD	RDo	IVI
Shorea robusta	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Ficus religoisa	20.0	0.3	1.5	9.52	10.71	58.99	79.22

Holarrhena antidysenterica	30.0	0.4	1.33	14.29	14.29	0.27	28.84
Saraca asoca	30.0	0.4	1.33	14.29	14.29	0.27	28.84
Schleichera aleosa	30.0	0.1	1.0	4.76	3.57	10.04	18.37
Santalum album	30.0	0.4	1.0	14.29	14.29	0.57	29.14
Spondias pinnata	20.0	0.3	1.5	9.52	10.71	2.7	22.94
Madhuca indica	10.0	0.1	1.0	4.76	3.57	0.9	9.23
Aegle marmelos	20.0	0.3	1.5	9.52	10.71	6.36	26.6
Buchanania Ianzan	20.0	0.3	1.5	9.52	10.71	0.31	20.54
Smilax perfoliate	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Morinda pubescens	10.0	0.1	1.0	4.76	3.57	0.1	8.44
Acacia catechu	10.0	0.1	1.0	4.76	3.57	0.1	8.44
Zizyphus mauritiana	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Diversity index = 0.94

Source: Calculated data on primary survey

Table 3:22 Characteristics of Tree Layer Vegetation at Kosala RF

Name of Species	F	D	Ab	RF	RD	RDo	IVI
Acasia Fistula	30.0	0.4	1.33	6.82	8.0	9.4	24.22
Sorea robusta	30.0	0.4	1.33	6.82	8.0	9.4	24.22
Ptescarpus marsupium	20.0	0.3	1.5	4.55	6.0	0.78	11.33
Zizyphus mauitiana	40.0	0.4	1.0	9.09	8.0	1.42	18.51
Diospyors melanoxylon	40.0	0.4	1.0	9.09	8.0	22.75	39.84
Holarrhena antidysentric	20.0	0.2	1.0	4.55	4.0	4.7	13.25
Sizygium cumini	40.0	0.4	1.0	9.09	8.0	5.69	22.78
Terminalia tomentosa	30.0	0.3	1.0	6.82	6.0	7.05	19.87
Azadirachta indica	20.0	0.3	1.5	4.55	6.0	7.05	17.6
Gycosmis pentaphylla	30.0	0.4	1.33	6.82	8.0	2.9	17.72
Semwcaspus anacasdium	40.0	0.4	1.0	9.09	8.0	5.69	22.78
Boswedia searata	30.0	0.4	1.33	6.82	6.0	0.78	15.08
Terminalia chebula	30.0	0.3	1.0	6.82	6.0	0.78	13.6
Emblica officinalis	20.0	0.2	1.0	4.55	4.0	7.02	15.57
Schleiche araleosa	20.0	0.2	1.0	4.55	4.0	18.80	27.35

Diversity Index = 1.16

Source: Calculated data on primary survey.

Table 3:23 Characteristics of Tree Layer Vegetation at Debinagar PF

Name of Species	F	D	Ab	RF	RD	RDo	IVI
Holarrhena antidysentric	40.0	0.5	1.25	9.52	9.26	13.01	31.8
Schleichera aleosa	40.0	0.5	1.25	9.52	9.26	4.02	22.8
Zizyphus mauritiana	50.0	0.7	1.4	11.90	12.96	2.76	27.62
Terminalia tomentosa	50	0.7	1.4	11.90	12.96	2.02	26.89

Horea robusta	40.0	0.6	1.5	9.52	11.11	23.33	43.97
Phoenic ocaulis	40.0	0.5	1.25	9.52	9.26	2.57	21.35
Emblica officinalis	50.0	0.6	1.2	11.90	11.11	23.33	46.35
Ptercarpus marsupium	40.0	0.6	1.5	9.52	11.11	1.74	22.37
Dalbergia latifola	70.0	0.7	1.0	16.67	12.96	27.22	56.85

Diversity Index = 0.93

Source: Calculated data on primary survey

Th results of phyto-sociological studies show that "Mahua" and "Sal" are the main dominating species in the study area. The total number of tree species recorded at various hotspots/quadrats varies from 16 to 24.

Table 3:24 General List of Flora in 10 km Study Area

Sl.No.	Local Name	Species	Family
Tree Species			
1	Acacia/sunajhari	*Acacia auriculiformis*	Mimosaceae
2	Achu	*Morinda tinctoria*	Rubiaceae
3	Amba	*Mangifera indica*	Anacardiaceae
4	Amia	*Emblica officinalls*	Euphorbiaceae
5	Ankula	*Alangium lamarckii*	Alangiaceae
6	Asan	*Teminalia tomentosa*	Combretaceae
7	Aswatha	*Ficus religiosa*	Moraceae
8	Bahada	*Teminalia belerica*	Combretaceae
9	Bandhan	*Ougeinia oojenensis*	Papilonaceae
10	Bara	*Ficus bengalensis*	Moraceae
11	Barabakalia	*Dalbergia paniculata*	Papilonaceae
12	Bel	*Aegle marmelos*	Rutaceae
13	Bhalia	*Semicarpus anacardium*	Anacardiaceae
14	Bheru	*Chloroxylon swietenia*	Meliaceae
15	Bija/Paiasal	*Pterocarpus marsupium*	Fabaceae
16	Bodaka	*Hymenodictyon*	Rubiaceae
17	Chakunda(Sana)	*Cassia siamea*	Ceasalpiniaceae
18	Chara	*Buchnania lanzan*	Anacardiaceae
19	Chhatian	*Alstonia scholaris*	Apocynaceae
20	Dalasinga	*Canthium dicoccum*	Rubiaceae
21	Dhaman	*Grewia tillaefolia*	Tiliaceae
22	Dhaura	*Anogeissus latifolia*	Combretaceae
23	Dimiri(Bai)	*Ficus glomerata*	Moraceae
24	Eucalyptus	*Eucalyptus spp*	Myrtaceae
25	Gohira	*Acacia leucophloea*	Mimosaceae
26	Harida	*Terminalia chebula*	Combretaceae
27	Jamun	*Syzygium cuminii*	Myrtaceae
28	Jhari (Kasaki)	*Ficus tinctoria*	Moraceae
29	Kadamba	*Anthocephalus cadamba*	Rubiaceae

Sl.No.	Local Name	Species	Family
Tree Species			
30	Kakuchia	*Diospyrus sylvatica*	Ebenaceae
31	Kamalagundi	*Mallotus philippensis*	Euphorbiaceae
32	Kanchan/Barada	*Bauhinia purpurea*	Ceasalpiniaceae
33	Kapacia	*Kydia calycina*	Malvaceae
34	Karanja	*Pongamia*	Papilionaceae
35	Karada	*Cleistanthus collinus*	Euphorbiaceae
36	Khakada	*Casearia elliptica*	Flacourtiaceae
37	Kashi	*Bridelia retusa*	Euphorbiaceae
38	Kendu	*Diospyrus melanoxylon*	Ebenaceae
39	Khair	*Acacia catechu*	Mimosaceae
40	Kurum/Haldu	*Adina cordifolia*	Rubaceae
41	Kumbhi	*Careya arborea*	Myrtaceae
42	Karam/Kurum	*Adina cordifoilia*	Rutaceae
43	Kusum	*Schleichera oleosa*	Sapindaceae
44	Mahula	*Madhuca indica*	Sapotaceae
45	Mohi	*Lannea coromandelica*	Anacardiaceae
46	(Raj)Moi	*Bursera serrata*	Burseraceae
47	Mundi	*Mitragyan parviflora*	Rubiaceae
48	Neem	*Azadirachta indica*	Meliaceae
49	Padal	*Stereospemum*	Bignoniaceae
50	Palas	*Butea monosperma*	Papilonaceae
51	Rai	*Dillenia pentagyna*	Dilleniaceae
52	Sal	*Shorea robusta*	Dipterocarpaceae
53	Salai	*Boswellia serrata*	Burseraceae
54	Semul	*Bombax ceiba*	Bombacaceae
55	Sidha	*Lagerstroemia parviflora*	Lythraceae
56	Siris	*Albizzia lebbek*	Mimosaceae
57	Siris (dhala)	*Albizzia procera*	Mimosaceae
58	Sissoo	*Dalbergia sissoo*	Papilonaceae
59	Sissoo(kala)	*Dalbergia latifolia*	Papilonaceae
60	Som	*Soymida febrifuga*	Miliaceae
61	Sunari	*Cassia fistula*	Ceasalpiniaceae
62	Teak/Saguan	*Tectona grandis*	Verbenaceae
63	Tentuli	*Tamarindus indica*	Ceasalpiniaceae
Shrubs/Herbs			
1	Apamarga	*Achynthes aspera*	Amaranthacea
2	Baghaairi	*Litsea glutinosa/Sabira*	Lauraceae
3	Baincha	*Flacourtia indica*	Flacourtiaceae
4	Bajramuli/Bisiripi	*Sida spp.*	Malvaceae
5	Bana-Kolthi	*Atylosia scarabacoides*	Papilionaceae
6	Bana-mirchi	*Croton sparsiflorus*	Euphorbiaceae
7	Bana-nila/Kulthi	*Tephrosia purpurea*	Fabaceae
8	Begunia	*Vitex negundo*	Verbinaceae

Sl.No.	Local Name	Species	Family
Tree Species			
9	Bhuinnimba	*Andrographis*	Acanthaceae
10	Bichhamalia	*Evolvulus nummularius*	Convolvulaceae
11	Chitakutei	*Euphorbia hirta*	Euphorbiaceae
12	Damkurudu	*Gardenia latifolia*	Rubiaceae
13	Dhatki	*Woodfordia fruticosa*	Lythraceae
14	Gandhaguria	*Macaranga peltata*	Euphorbiaceae
15	Kanteikoli	*Ziziphus numularia*	Rhamnaceae
16	Khajuri	*Phoenix spp.*	Palmae
17	Kurei/Keruan	*Holarrhena*	Apocynaceae
18	Koelekha	*Hygrophila spp.*	Acanthaceae
19	Kulthia	*Atylosia scarabaeoides*	Fabaceae
20	Kurudu	*Gardenia spp.*	Rubiaceae
21	Nagaairi	*Lantana camara*	Verbinaceae
22	Pokasunga	*Ageratum conyzoides*	Asteraceae
23	Telkuria	*Ixora arborea*	Rubiaceae
Grasses			
1	Khadika grass	*Aristida setacea*	Gramineae
2	Mutha grass	*Cyperus rotundus*	Cyperaceae
3	Ghasa	*Panicum spp.*	Gramineae (Poaceae)
4	Guguchia	*Chrysopogon aciculatus*	Gramineae (Poaceae)
5	Duba	*Cynodom dactylon*	Gramineae (Poaceae)
6	Sinkola	*Heteropogon contortus*	Gramineae (Poaceae)
7	Kasatandi/Payal	*Saccharum spontaneum*	Gramineae (Poaceae)
8	Sinanath ghas	*Pennisetum Pedicillatum*	Gramineae (Poaceae)
9	Kahr ghas	*Imperata cylindrical*	Gramineae (Poaceae)

Source: Primary survey validated with Secondary Data

Faunal Biodiversity of the Study Area

For the documentation of the faunal biodiversity of the study area with respect to birds, reptiles, amphibians and butterfly species, a baseline survey had been conducted.

Birds: The common birds observed in the study area are listed in the table below.

Table 3:25 Avifauna in study area

S.No.	Common Name	Species (Scientific Name)	Family	Schedule as IWPA, 1972/2022
1	Small green Bee eater	*Merops orientalis*	Meropidae	Schedule II
2	Blue tailed Bea eater	*Merops philippinus*	Meropidae	Schedule II
3	Jungle fowl	*Gallus gallus*	Phasianidae	Schedule II

4	House Fowl	*Gallus domesticus*	Phasianidae	Not Listed
5	Bramhiny Starling	*Sturnus pagodarum*	Sturnidae	Schedule II
6	Indian Myna	*Acridotheres tristis tristis*	Sturnidae	Schedule II
7	Blue Jay (Indian roller)	*Coracias benghatensis*	Coraciidae	Schedule IV
8	Eastern Skylark	*Alauda arvensis*	Alaudidae	Schedule II
9	White browed Bulbul	*Pycnonotus luteolus*	Pycnonotidae	Schedule II
10	Bramhiny Duck	*Tadorna ferruginea*	Anatidae	Not Listed
11	Blossom headed parakeet	*Psittacula roseata*	Psittacidae	Schedule II
12	Rose ringed Parakeet	*Psittacula krameri borealis*	Psittacidae	Schedule II
13	Paddy field pipit	*Anthus rufulus*	Motacillidae	Schedule II
14	Brown rock pipit	*Anthus similes*	Motacillidae	Schedule II
15	Spotted Dove	*Streptopelia chinensis*	Columbidae	Schedule II
16	Oriental magpie Robin	*Copsychus saularis*	Muscicapidae	Schedule II
17	Jungle crow	*Corvus macrorhynchos culminates sykes*	Corvidae	Schedule II
18	House sparrow	*Passer domesticus*	Passeridae	Schedule II
19	Eurasian eagal-owl	*Bubo bubo*	Strigidae	Schedule II
20	BIue rock Pigeon	*Columba livia*	Clumbidae	Schedule II
21	Red vented bulbul	*Pycnonotus cafer*	Pycnonotidae	Schedule II
22	Jungle Myna	*Acridotheres fuscus*	Sturnidae	Schedule IV
23	Common Quail	*Coturnix coturnix*	Phasianidae	Schedule II
24	Black headed oride	*Oriolus xanthornus*	Oriolidae	Schedule II
25	Golden Oriole	*Oriolus oriolus*	Oriolidae	Schedule II
26	Black drongo (King crow)	*Dicrurus macrocercus*	Dicruridae	Schedule II
27	Indian Pond heron	*Ardeola grayii*	Ardeidae	Schedule II
29	House crow	*Corvus splendens*	Corvidae	Schedule II
30	Little Egret	*Egretta garzetta*	Ardeidae	Schedule II
31	Asian Koel	*Eudynamys scolopacea*	Cuculidae	Schedule II
32	Screech owl (Barn owl)	*Tyto alba*	Strigidae	Schedule II
33	Shall Blue Kingfisher	*Alcedo atthis bengalensis*	Alcedinidae	Schedule II
34	White necked stork	*Ciconia episcopus*	Ciconiidae	Schedule II
35	Pariah kite (Black kite)	*Milvus migrans*	Accipitridae	Schedule II

36	White throated Munia	*Lonchura malabarica*	Ploceidae	Schedule II
37	White rumped Munia	*Lonchura striata*	Ploceidae	Schedule II
38	Indian Myna	*Acridotheres tristis*	Sturnidae	Schedule II
39	Common Sandpiper	*Actitis hypoleucos*	Charadriidae	Schedule II
40	Pied crested cuckoo	*Clamator jacobinus*	Cuculidae	Schedule II
41	Indian Plaintive Cuckoo	*Cacomantis passerinus*	Cuculidae	Schedule II
42	Crested Hawk Eagle	*Nisaetus cirrhatus*	Accipitridae	Schedule I
43	Pea-fowl	*Pavo cristatus*	Phasianidae	Schedule I
44	Shikra	*Accipter badius*	Accipitridae	Schedule I

Source: Primary survey validated with Secondary Data

Butterflies from the study area: Butterflies from three families observed during the present study are documented in the table below.

Table 3:26 Butterflies in the Study Area

Scientific Name and Family	Common Name	Relative Abundance
Family Papilionidae		
Papilio polytes	Common Mormon	Common
Family Pieridae		
Eurema hecabe	Common Grass yellow	Very Common
Catopsilia crocale	Common Emigrant	Common
Family: Nymphalidae		
Danaus chrysippus	Plain Tiger	Common
Phalantha phalantha	Common Leopard	Fairy Common
Mycalesis perseus	Common bush brown	Uncommon
Ariadne merione	Common Caster	Uncommon
Precis lemonias	Lemon pansy	Common
Junonia orithya Linnaeus	Blue pansy	Fairy Common

Source: Primary survey validated with Secondary Data

Herpetofauna: In amphibian group, the toads were sighted during the study period. The reptile, Common Garden Lizard and Fan-Throated Lizard, Common rat Snake and were observed in the region is given in the table below.

Table 3:27 Reptiles and Amphibian in Study Area

S.No.	Family	Common Name	Scientific name	Schedule as IWPA, 1972/2022
1.	Agamidae	Common Garden Lizard	*Calotes versicolor* (Cuvier, 1817)	Not listed
2.		Fan-Throated Lizard	*Sitana ponticeriana* (Cuvier, 1817)	Not listed
3.	Bufonidae	Toad	*Bufo bufo* (Gray 1825)	Not listed
4.	Chamaeleonidae	Indian chameleon	*Chameleon calcaratus*	Schedule I
5.	Colubridae	Common Rat Snake	*Ptyas mucosus*	Schedule I
6.	Elapidae	Common Indian Krait*	*Bungarus caeruleus*	Schedule I
7.		Indian Cobra*	*Naja naja*	Schedule I
8.	Viperidae	Russell's Viper	*Vipera russelli*	Schedule I
9.	Sciencidae	Bramhiny	*Mabuya carinata*	Schedule IV
10.	Gekkonidae	Spotted Indian House Gecko	*Hemidactylus frenatus (Gray)*	Schedule II
11.		Tree Gecko	*Hemidactylus typus*	Schedule II
12.	Varanidae	Bengal monitor	*Varanus bengalensis*	Schedule I
13.		Yellow monitor lizard	*Varanus flavescens*	Schedule II
14.	Colubridae	Common wolf snake	*Lycodon aulicus (Linn.)*	Schedule IV
15.		Buffstriped Keelback	*Amphiesma stolatum (Linn.)*	Schedule IV
16.	Elapidae	Banded Krait	*Bungarus fasciatus*	Schedule II
17.	Pythonidae	Indian Python	*Python morulus*	Schedule I
18.	Elapidae	Monocellata Cobra	*Naja naja kuothia*	Schedule I
19.	Elapidae	King Cobra	*Ophiophagus Hannah*	Schedule I
20.	Colubridae	Checkered Keelback	*Xenochrophis piscator*	Schedule I
21.	Testudinidae	Land Tortoise	*Testudo eleonguta*	Schedule I
22.	Emydidae	Water Tortoise	*Lissemys punctata*	Schedule I

*Not sighted but included as per the secondary information from the villagers.
Source: Primary survey validated with Secondary Data

Mammals

The wild mammals observed other than the domesticated ones are given in the table below.

Table 3:28 Mammals in Study Area

S.No.	Family	Common Name	*Scientific name*	Status as per IWPA 1972/2022
1.	Canidae	Small Fox	*Vulpes bengalensis*	Schedule II
2.	Herpestidae	Common Mongoose	*Herpestes edwardsii (É. Geoffroy Saint-Hilaire, 1818)*	Schedule II
3.	Cercopithecidae	Monkey	*Macaca mulatta*	Schedule II
4.		Common langur	*Semnopithecus entellus (*Dufresne, 1797)	Schedule II
5.	Teropodidae	Short nosed Fruit Bat	*Cynopterus sphinx*	Schedule II
6.	Muridae	Common Indian Rat	*Rattus rattus (Linn.)*	Schedule V
7.		Field Rat	*Rattus norvegicus (Linneaus 1758)*	Schedule V
8.	Sciuridae	5 striped Palm Squirrel	*Funambulus pennanti* (Wroughton, 1905)	Schedule IV
9.	Elephantidae	Indian Elephant	*Elephas maximus indicus*	Schedule I
10.	Bovidae	Bison	*Bos gaurus*	Schedule I
11.	Canidae	Jackal	*Canis aureus*	Schedule I
12.	Cervidae	Sambhar	*Cervus unicolor*	Schedule I
13.	Felidae	Wild Cat	*Felis chaus*	Schedule I
14.	Herpestidae	Mongoose	*Herpestres*	Schedule I
15.	Hyaenidae	Striped Hyena	*Hyaena hyaena*	Schedule I
16.	Hystricidae	Indian Porcupine	*Hystrix indica*	Schedule I
17.	Mustelidae	Common Otter	*Lutra lutra*	Schedule I
18.	manidae	Indian Pangolin	*Manis Crassicaudata*	Schedule I
19.	Ursidae	Sloth Bear	*Melursus ursinus*	Schedule I
20.	Felidae	Kalarapartia Bagh	*Panthera pardus Fusca*	Schedule I
21.	Canidae	Common Fox	*Vulpes bengalensis*	Schedule I
22.	Viverridae	Indian Civet	*Viverricula indica*	Schedule I

Source: Primary Survey validated with Secondary Data.

Elephant movement was observed in the study area.

Domestic Animals: The domestic animals *viz*. dog, cow, buffalo, goat, sheep and chicken observed in the study area.

Insect: There is no significant faunal assemblage here except for some insects like Honeybees (*Apis sp*.) and Gum leaf Grasshopper (*Goniaea australasiae)*. Some invertebrates are seen such as earthworms, snails, prawns, crabs, houseflies, butterflies, mosquitoes, wasps, mantis, ants, termites etc.

Aquatic Fauna

The study area mainly drained by Birupa River. During the field survey both inland pond fishes and riverian fishes were recorded with interaction with villagers and fishermen. A detailed checklist of freshwater fishes is given in the table below.

Table 3:29 Fishes within Study area

S.N	Common	Scientific Name	Family
1	Bhakur	*Catla catla (Hamilton)*	Cyprinidae
2	Mrigal	*Cirrhinus mrigala (Hamilton)*	Cyprinidae
3	American rui	*Cyprinus carpio (Linnaeus)*	Cyprinidae
4	Rohu	*Labeo rohita (Hamilton)*	Cyprinidae
5	Bata	*Labeo bata (Hamilton)*	Cyprinidae
6	Kalbans	*Labeo kalbasu (Hamilton)*	Cyprinidae
7	Chanda	*Osteobrama cotio cotio*	Cyprinidae
8	Punthi	*Puntius saphori (Hamilton)*	Cyprinidae
9	Magur	*Clarius batracus (Linnaeus)*	Clariidae
10	Singi	*Heteropneustes fossilis*	*Heteropneusida*
11	Shol	*Channa striatus (Bloch)*	*Channidae*
12	Chang	*Channa orientalis (Schneider)*	*Channidae*
13	Lata	*Channa punctatus (Bloch)*	*Channidae*
14	Koi	*Anabas testudineus (Bloch)*	*Anabantidae*
15	Todi	*Mastacembelus armatus*	*Mastacembelida*
16	Balia	Wallago atu	*Siluridae*

Source: Secondary Data

PLANKTON

The plankton (phytoplankton and zoo plankton) was observed from sample collected and listed below:

Table 3:30 Planktonic Abundance in Aquatic Environment

S. No	Phytoplankton	S.No.	Zooplankton
1	*Ulothrix sp*	1	*Pseudodiaptomus sp*
2	*Pithophora sp*	2	*Keratella sp*
3	*Desmids*	3	*Asplancha sp*
4	*Cosmarium sp*	4	*Brachonus sp*
5	*Pediastrum sp*	5	*Daphnia sp*
6	*Volvox sp*	6	*Cyclops sp*
7	*Navicula sp*	7	*Cypris sp*
8	*Pluerosigma sp*	8	*Diaptomus sp*
9	*Achnanthes sp*	9	*Pseudodiaptomus sp*
10	*Pinnularia sp*	-	
11	*Anabeana sp*	-	
12	*Scenedesmus biguga*	-	
13	*Ankistrodesmus falcatus*	-	
14	*Melosira sp*	-	
15	*Chlorococcum sp*	-	
16	*Chlorella sp*	-	
17	*Nitzschia sp*	-	
18	*Microcystis sp*	-	

Source: Primary Survey

RARE AND ENDANGERED FAUNA OF STUDY AREA

The IUCN Red List is the world's most comprehensive inventory of the global conservation status of plant and animal species. It uses a set of criteria to evaluate the extinction risk of thousands of species and subspecies. These criteria are relevant to all species and all regions of the world. With its strong scientific base, the IUCN Red List is recognized as the most authoritative guide to the status of biological diversity. IUCN, (2008) has evaluated 1976 animal species from India, among them 313 have in recognized as threatened species. Some of the sighted fauna was given protection in Schedule -I by the Indian Wildlife (Protection) Act, 1972 amended in Dec. 2022 by including them in different schedules.

Interpretation

The study area has 6 reserve forests and 2 protected forest & some other forest patches which accompany the rich floral and faunal diversity along with wildlife protected under schedule -I of wildlife protection act 1972 amended 2022. Forest diversion and tree felling will be involved along with diversion of nala in ML area. Biodiversity loss should be estimated before nala diversion and biodiversity conservation plan is recommended. There was paddy crops in agricultural fields observed during site visit on ML area, people practicing agriculture in ML area, ponds for rainwater harvesting and fishing, domestic animals for agriculture practice, wild movement in ML as well as in study area; hence wildlife conservation plan is recommended to implement during entire mining period.

POLLUTION CONTROL MEASURES

Drilling & Blasting shall be done in 7% of the total excavation only. Rest shall be done by Surface Miner.

PP will have its own private railway siding within the ML area and 80% of the total production (15 MTPA) shall be transported through common railway corridor.

Dedicated coal evacuation road for balance 20% through road to meet the local demand.

Out of 1196 ha, 623.710 ha shall be excavated & and subsequently backfilled

Out of 1196 ha of ML area, 1029.583 ha shall be covered under plantation, 38.488 ha under water body, 31.85 ha for public use, 64.3723 ha shall be returned to Forest Department and 31.707 ha will remain as undisturbed at Post Mine Closure.

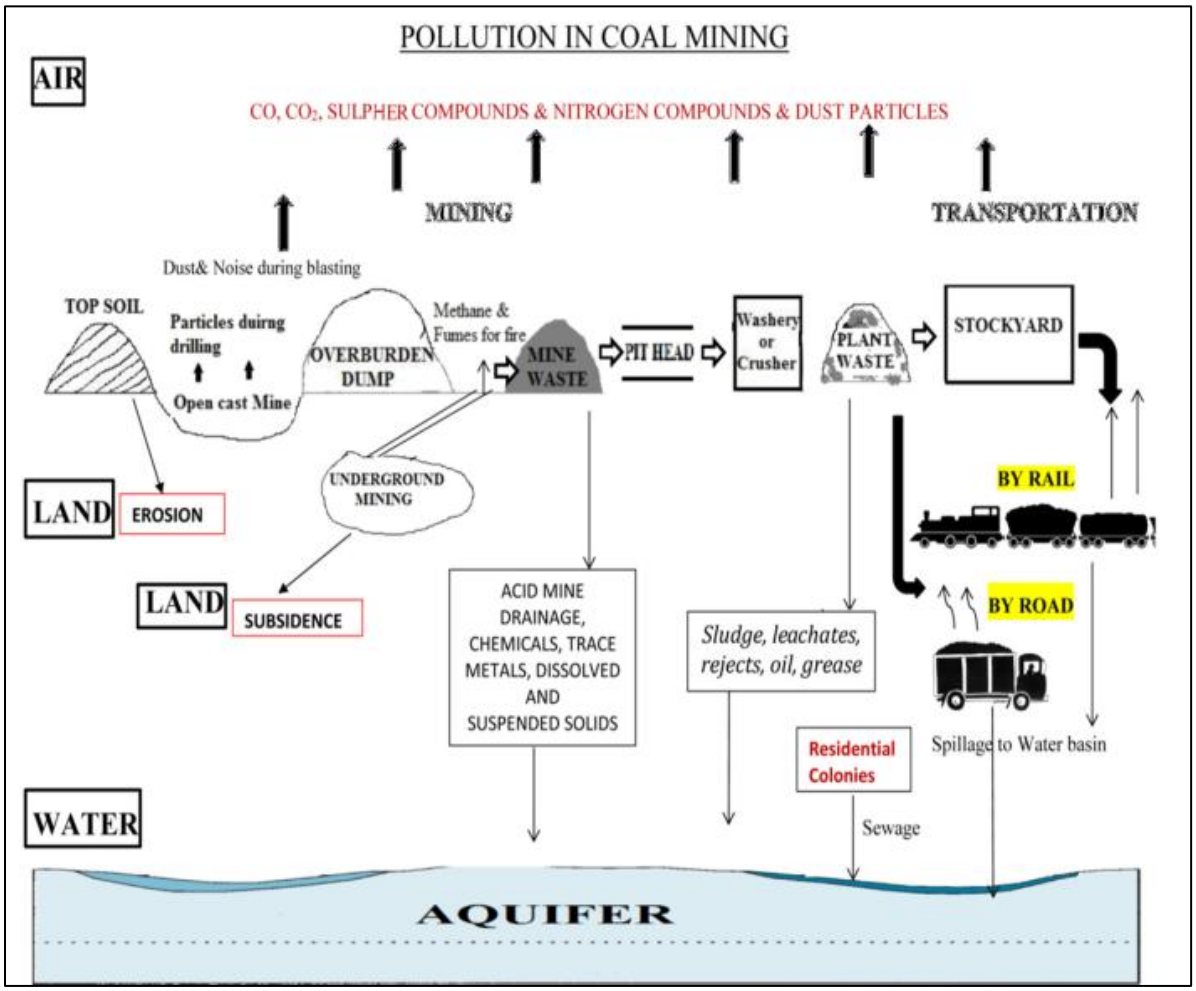
Fixed and mobile water sprinkling system, Mobile water cannon, Wheel washing system, Tarpaulin covered trucks, black toping/concrete road etc. shall be adopted for reducing the air pollution. All relevant engineering measures like check dam, toe walls, garland drains, settling sump, construction of embankment along the Kumbhira stream within the ML area to prevent possible water pollution.

Mine seepage water shall be utilized for industrial use.

CHAPTER-4 : ANTICIPATED ENVIRONMENTAL IMPACTS AND ITS MITIGATION MEASURES

4.1 INTRODUCTION

Environmental impact in the study area is any alteration of environmental conditions or creation of new set of environmental conditions, adverse or beneficial, caused or induced by the impact of project. Prediction involving identification and assessment of potential impacts of the project on surrounding environment is a significant component of EIA studies. The proposed project proposal is for mining of coal with production capacity of 15 MTPA. Coal is a fossil fuel and is composed of mainly carbon. It is extracted predominantly through open cast mining (OCM). OCM disfigures the area and tends to pollute the atmosphere within the locality. The main activities involved in coal mining are drilling of bore holes, blasting and loosening of coal seams, extraction of coal reserve and transportation of coal from mines to the stock yard or to railway siding. Extraction of coal, therefore, involves serious environmental and social concerns, including air, noise and water pollution, land degradation and far-reaching consequences on local bio- diversity. In India most of the coal reserves are located in river basins which are rich in forest cover and are habitats of precious wildlife and indigenous tribal communities. The life cycle of a coal mine involves phases viz. prospecting, exploration, development, exploitation, and finally closure. The impact of coal mining on the environment is maximum during the exploitation phase. In the process of coal mining, transportation and other allied operations, different types of mining pollutants are generated. Mining must, therefore, be carried out in a wise, socially responsible, and environmentally sustainable manner.



All components of the environment are considered and wherever possible impacts are evaluated in quantitative/qualitative terms. Several scientific methods are available to quantify and predict the impact of a project on environmental attributes such as water, air, noise, land, ecological and socio-economic. Such predictions are superimposed over baseline environmental status to derive post project scenario of the environmental conditions.

The resultant (post-project) quality of environmental parameters is reviewed with respect to the permissible limits. Based on the impacts thus predicted, preventive and mitigation measures were formulated and incorporated in the environmental management plan to minimize adverse impacts on environmental quality during and after the project execution.

4.2 IDENTIFICATION & CHARACTERIZATION OF IMPACTS

The wastes and pollutants generated due to various activities of the project may cause impacts on different environmental attributes. The major project activities and the anticipated environmental impacts due to the proposed project are discussed below, under the following categories:

- Impacts due to project location
- Impacts due to project design
- Impacts during construction
- Impacts during operation.

4.2.1 IMPACTS DUE TO PROJECT LOCATION

Mining is a site-specific project. The proposed mining shall be carried out within the mining lease area of 1196 ha.

4.2.2 IMPACTS DUE TO PROJECT DESIGN

Following facilities have been proposed which will have impact on the environment:

- Open cast mine with 15 MTPA production capacity over ML area 1196 ha.

4.2.3 IMPACT DURING CONSTRUCTION

Following civil construction is proposed

4.2.4 RESIDENTIAL COLONY

No township is proposed in this proposal. However, in future if it is required PP will take clearance from respective state SEIAA.

4.2.5 SERVICE BUILDING

Office buildings, sub-station, statutory buildings such as first aid center, rest shelter, canteen would be provided in the southern part of the lease area.

4.2.6 ROADS

- Haul Road: Haul roads suitable for plying 100 tonne rear dumpers with sidedrains and dozer path would be provided. Wherever possible Black topping of permanent haul roads shall be done as provided in mine closure plan.
- Approach road: Approach Road to project site and magazine would be provided.

4.2.7 EMBANKMENT

Embankment of 3 km length & height of 3m above HFL will be provided along the Kumbhira Nala for its protection. Toe-walls, retaining walls with weepholes at strategic stretches will be constructed to arrest the sediments and prevent silting of the water courses by the run-off during rains. Green belt will be developed along the embankment.

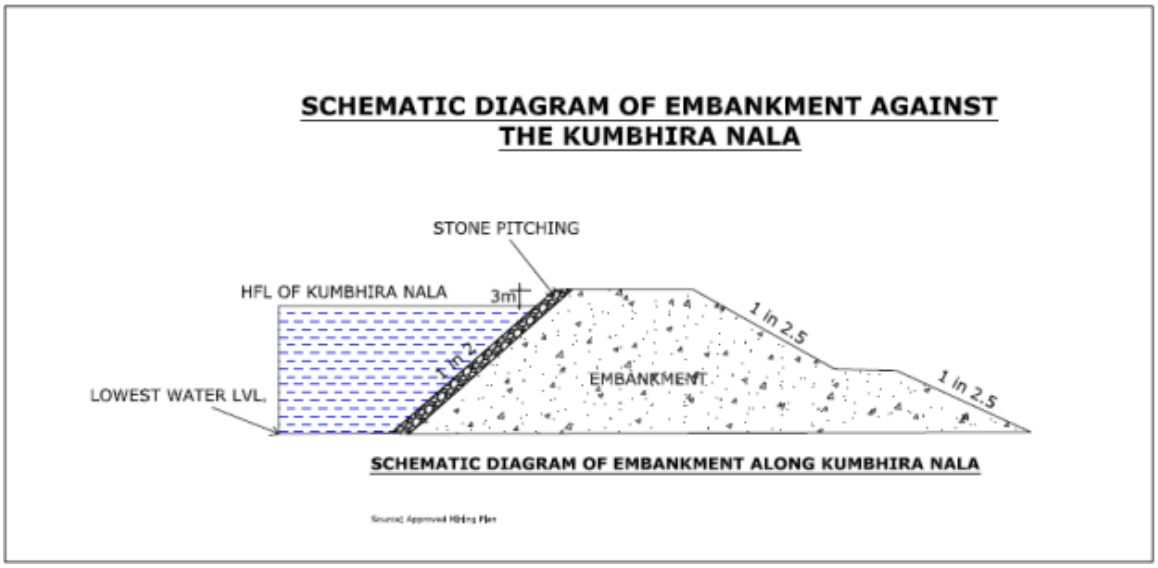


Figure 4-1 Embankment Cross Section

4.2.8 RAILWAY SIDING

It also has been proposed to construct a private railway siding for coal transportation.

The evacuation of coal from proposed rail Terminal at Baitarani West and the despatch to destination stations are as below:

Table 4:1 Source of Origin/Destination of traffic

Commodity	Origin	Destination	Quantity
Outward Coal	Baitarni West Rail Loading Terminal served by Tentuloi station	Different destinations in the state of Gujarat and other destinations over India	15 Million TPA

The above quantity of coal will move to different stations in the state of Gujarat as well as to destinations in state of Odisha, West Bengal, Jharkhand and Andhra Pradesh.

Production of coal from Baitarni West mine has been targeted in FY- 2026 and will increase gradually which will touch to 15 MTPA in FY-2030. Year wise traffic evacuation from Baitarni West mine is given below:

Table 4:2 Year wise production (Proposed)

Production Year	***Financial Year***	***Production (in MTPA)***	***Mode (in MTPA)***	
			Road	***Rail***
Year-1	FY-2026	0.9	0.9	-
Year-2	FY-2027	3.0	3.0	-

Year-3	FY-2028	6.0	3.0	3.0
Year-4	FY-2029	10.5	3.0	7.5
Year-5 to 32	FY 2030 to FY 2057	15.0	3.0	12.0
Year-33	FY2058	12.0	3.0	9.0
Year-34	FY2059	9.0	3.0	6.0
Year-35	FY2060	6.89	3.0	3.89

It is anticipated that Inner Rail Corridor and Outer Rail Corridor are likely to be operational in FY-2027-28 & 2029-30 respectively. The rail terminal of Baitarni West of GMDC will be commissioned in line with the railway corridors of IR.

Till commissioning of Railway corridors in the area, the produced coal will be transported through road , road cum rail and rail mode as below:

(i) In the initial stage of production:
The coal will be dispatched by road to the short lead destinations if any or will be dispatched by Road from Mine head to Jarapada/ Handapa Goods sheds i.e up to nearby rail siding and further by rail to the destinations of the Customers. For this purpose, a meeting has been held on 28-Oct-2024 by ADM (Revenue) for strengthening of Chhendipada-Jharpada Road (24.985 K.M) including 3 by passes and all the coal block allottees in the region have been instructed to pay INR. 35 Cr. GMDC has accepted the proposal and in the process of payment disbursal. Road transportation will be done through trucks covered with tarpaulin.

(ii) From 2nd year till achieving the production level of 3 MTPA:
The ROM Coal will be transported from Mine head to Coal Handing Plant (CHP) by belt conveyors. Afterward movement of Coal from CHP to Public goods siding located at Handapa station and Jarpada station and other nearby stations by Road Public goods siding to different destinations by Rail mode. Road transportation will be done through trucks covered with tarpaulin.

(iii) After commissioning of railway corridors & Rail Terminal of GMDC:
Bare minimum quantity up to 3 MTPA will be dispatched by road to nearby customers and other balance coal from 15 MTPA will be evacuated by rail mode only from GMDC rail terminal.

The CHP and truck loading Silo system and dedicated conveyor arrangement to railway siding with Silo will be constructed at the surfaces. RLS mechanism is proposed for loading to Railway wagons. From mine to CHP and CHP to Rail siding dedicated belt conveyor arrangement has been planned with mechanical loading through Rapid Loading System (RLS). Two numbers of RLS along with dedicated belt Conveyor arrangements have been planned.

Schedule and status of completion of Railway siding as on August 2025:

Activity No.	Activity	Schedule	Status
Status of Feasibility Study Report of Baitarni West Rail terminal			
1	FSR submitted to Railway	18.11.2024	Completed
2	Joint site visit by railway	29.11.2024	Completed
3	In-principal Approval by Railway	18.01.2025	Completed
Status of Detailed Project Report of Baitarni West Rail Terminal			
4	Work order given to agency for preparation and submission of DPR to Railway.	06.11.2024	Completed
5	On getting In Principle Approval for FSR from IR, DPR shall be submitted	17.06.2025	Completed
6	Approval of DPR	30.10.2025	Under progress
7	Rail Terminal Commissioning	FY-2027-28 for the commissioning of IR Inner Rail Corridor and 2029-30 for IR Outer Rail Corridor	Pending

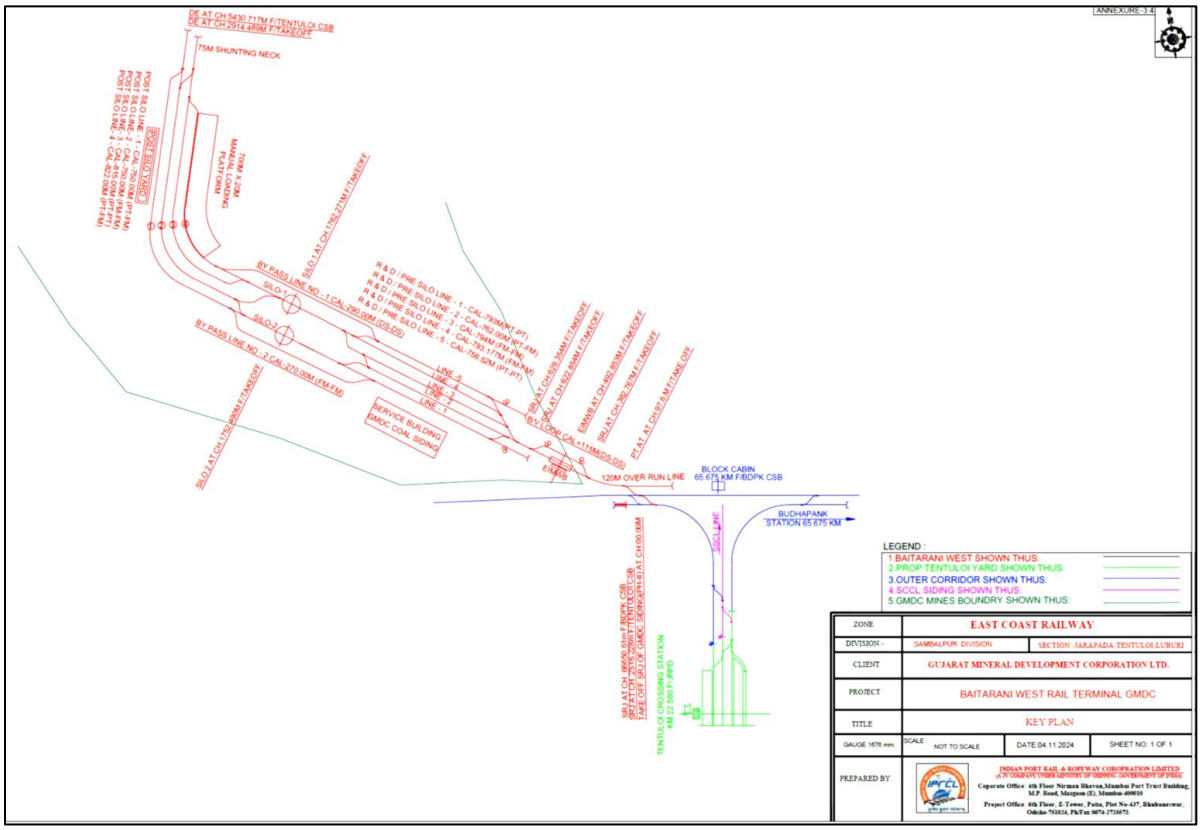


Figure 4-2 Key Plan of Baitarni West Rail Terminal

4.2.9 IN-PIT CONVEYOR BELT

Table 4:3 Time-bound action plan for implementation of in-pit conveyor belt

Year	Production Year	Financial Year	Coal Production (Mt)	Cumm. Area of Pit Excavation (Ha.)	Pit Depth (M)	Phasing For Commissioning & Installation of In-Pit Belt Conveyor	Remark for In-Pit Belt Conveyor
1	-	2024	0		-	-	
2	-	2025	0		-	-	
3	1	2026	0.9	18.02	40	-	Area is not sufficient to construct & operate the in-pit conveyor
4	2	2027	3	45.01	40	-	
5	3	2028	6	76.09	50	-	
6	4	2029	10.5	116.25	95	-	

7	5	2030	15	159.93	95	-	
8	6	2031	15	159.93	100	-	
9	7	2032	15	180.44	110	-	
10	8	2033	15	203.84	120	-	
11	9	2034	15	229.52	160	PHASE-01	Conveyor commissioning starts only after FY 2034 when depth reaches 160 m
12	10	2035	15	254.93	180		
13	11	2036	15	285.42	185		
14	12	2037	15	285.42	190	PHASE-02	Installation will be completed up to 200m depth, once sufficient area becomes available after the mine attains its ultimate depth. Full installation (up to 200 m depth) targeted by FY 2037–2039
15	13	2038	15	315.88	195		
16	14	2039	15	350.51	200		
17	15	2040	15	383.36	210		

The deposit of coal is available at 40 m depth and thereafter at 50m, 95m, 105m, 130m and up to 230m. The Production Plan is 0.9 MTPA in 1st year followed by 3.0 MTPA in 2nd year, 6.0 MTPA in 3rd year, and this will gradually increase to 15 MTPA from 5th year and onwards.

Keeping the gradual increase in mine depth, pit design, extraction of multiple seams, use of machinery for coal extraction like Surface Miner and 10-year production forecast, the coal will be transported to surface from mine face with the combination of the Surface Miner and rear discharge dumpers initially with the all possible measures to be taken to protect the environment like sprinkling of water, use of fog cannon, etc.

Subsequently, looking to the advancement of the technology, alternative option like extraction and transportation of coal with the combination of the Surface Miner and in-pit conveyor for multiple coal seams to be explored and subject to the viability of the logistics, the system to be adopted for transportation of coal from in-pit to the Mine Head.

The CHP and truck loading system, railway siding with Silo and RLS mechanism will be constructed at the surfaces.

Further as the mine depth will increase gradually, the transportation mechanism of coal from mine faces to surface will be done through the suitable mechanism like in pit conveyor belt in an environment friendly manner.

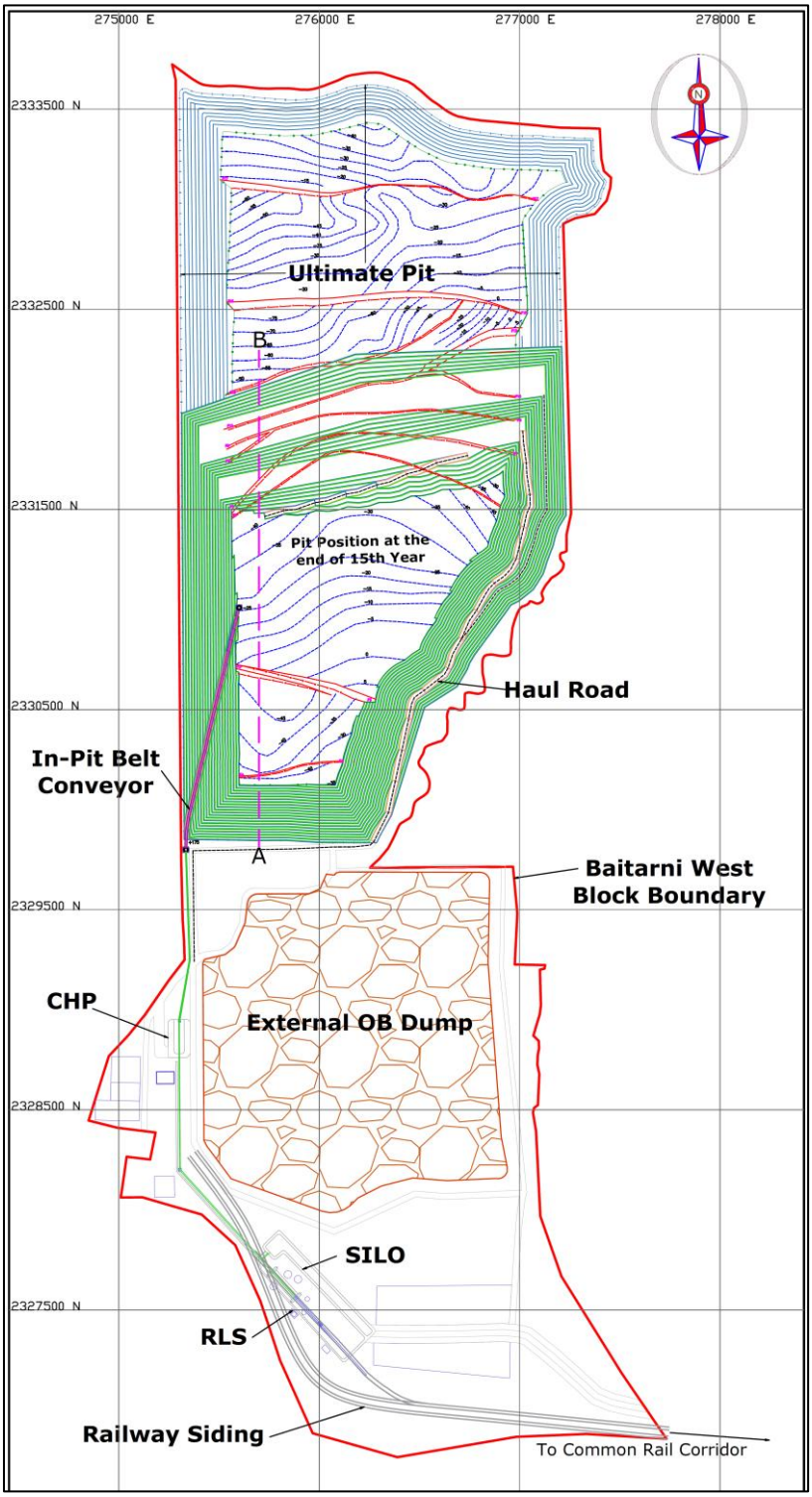


Figure 4-3 Location of In-Pit Belt Conveyor

4.2.10 IMPACT DURING OPERATION PHASE

As discussed above, the major activities at Baitarni West Opencast Coal Mine in the operational phase involve drilling, blasting, excavation, dumping, backfilling, loading, transportation, etc. These activities may affect the environment in varying degrees through natural resources depletion viz. water consumption, release of pollutants (particulates and gaseous emissions), run-off from waste dump areas etc. Air, water, and noise environment may be affected due to these activities. Associated activities, e.g., transportation of materials, operations of HEMM and canteen etc., may also affect air, water, and noise environment. Greenbelt development will have a positive impact not only on flora and fauna but also on the air quality, noise, and soil. Positive impacts on socio-economic environment are expected due to employment, infrastructure development and due to developmental activities by the project management. The major project activities and the affected environmental parameters are given below:

4.2.10.1 PROJECT ACTIVITIES

1. Excavation
2. Drilling & Blasting (only 7% of the total excavation) and Disposal of Over burden
3. Transportation of materials (Coal and OB) and personnel.
4. Water Usage
5. Waste dump management.
6. Waste water management
7. Maintenance of HEMM and Plant

4.2.10.2 POLLUTION SOURCES

1. Air pollution due to Excavation, loading, unloading of Coal and drilling, blasting of OB.
2. Noise & Vibration during blasting
3. Haul road emissions (fugitive dust) during transportation within Mine
4. Gaseous emissions during transportation of coal to the end use plant
5. Water pollution from Waste dump
6. Water pollution from HEMM workshop

4.2.10.3 AFFECTED ENVIRONMENTAL PARAMETERS

1. Air quality
2. Water quality and resource
3. Noise quality and Vibration
4. Soil quality
5. Biological environment
6. Socio-economic conditions

4.3 ANTICIPATED ENVIRONMENTAL IMPACTS & MITIGATION MEASURES DURING OPERATIONAL PHASE

Baitarni West mine is an opencast mining, which will involve excavation of coal, drilling, blasting, Loading/unloading/Dozing, transportation of Coal to the end use plant commercial purpose. The activities related to the operational phase will have varying impacts on the environment and are considered for impact assessment.

Coal Transportation will be through Belt conveyor up to dispatch point/RLS Rail transport thereafter, through the proposed Common Rail Corridor Project in Talcher Coalfield. A small quantity of coal will be dispatched through road for sale to local customers.

Mine capacity: 15.00 MTPA
Transportation through road: 3 MTPA
Transportation through rail: 12 MTPA
Nature of coal: Blast-free/ Drilling blasting (approx. 7%)
Effective Working hours:5/6 hours per shift
Effective Working days:330 days

4.3.1 IMPACT ON LAND USE

Mining involves forest land, Government non-forest land and tenancy or private landfor its developmental as well as operational activities. During exploitation, land degradation due to change in land use takes place. An external dump shall be created to accommodate OB removed to extract coal and is accumulated till internal dumping or backfilling commences. Backfilling can be started from 11th /14th year

The process of land management includes topsoil management, technical reclamation of external OB dump, internal dump / backfilled area, management of void left after completion of extraction, plantation *i.e.,* biological reclamation of technically reclaimed dumps and monitoring progress of reclamation by satellite surveillance.

4.3.1.1 LAND-USE ALTERATION

As indicated earlier the mine lease area of coal mine is 1196 ha, comprising 64.3723 ha of forest land, 222.975 ha of Govt. non-forest land and 908.6527 ha of Tenancy Private Non- Forest land. The details are already discussed in Chapter 2.

4.3.1.2 PROPOSED MINING LAND-USE

The details of the proposed mining land-use are discussed in Table 4.4.

Table 4:4 Proposed Mining Land-use (in ha)

Type	Land use (Proposed)
Excavation Area	623.71

Embankment	16.85
Green Belt	86.5
Undisturbed	31.707
Water Reservoir	7.89
External Dump	215.84
Road And Infrastructure Area	186.783
Other water bodies including Garland Drains, etc	8.38
Safety Zone	16
Topsoil Dump	2.34
Total	**1196**

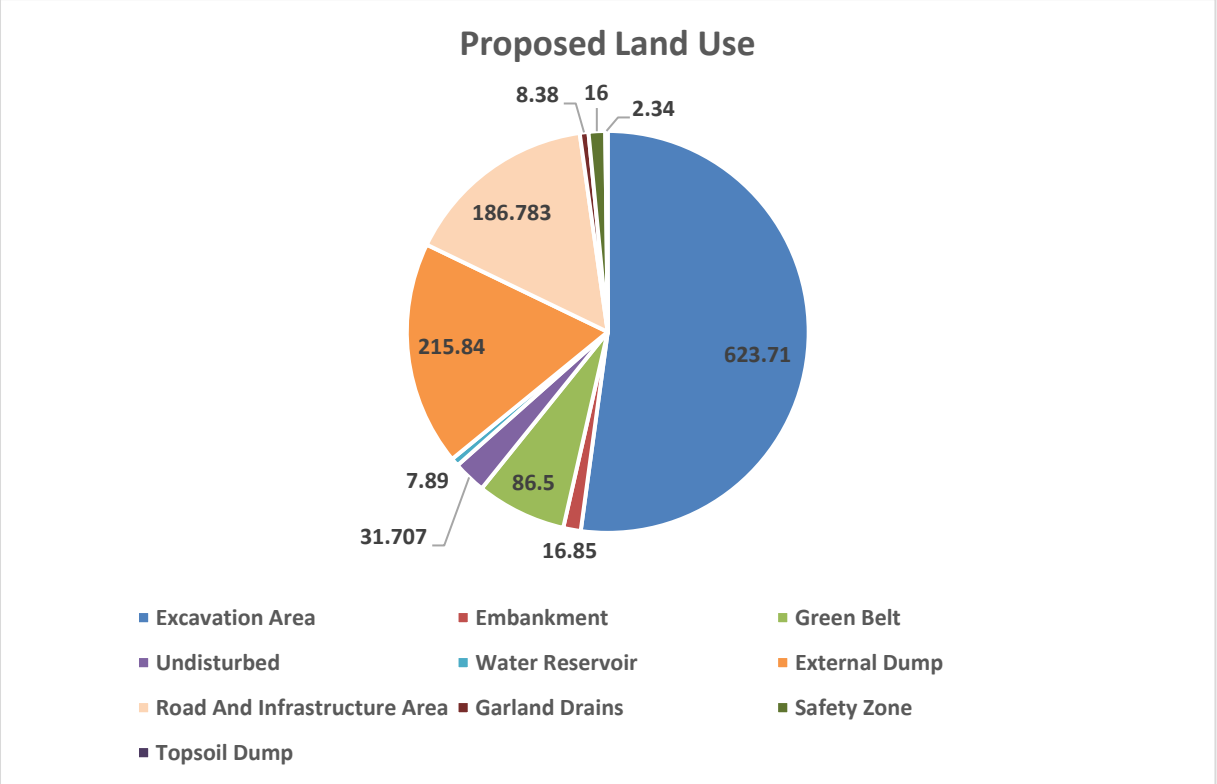


Figure 4-4 Proposed Land Use

4.3.1.3 TOPSOIL MANAGEMENT

Topsoil is the upper and outermost layer of soil, usually the top 5 cm to 20 cm. It has the highest concentration of organic matter and microorganisms wherein most of the earth's biological soil activity occurs. It takes approximately 1000 years for one inch of topsoil deposit to be formed. The topsoil also contains a range of nutrients and trace elements essential for plant growth and may also contain native seeds that are concentrated in the top 50 mm of the soil profile. In order to re-establish native species, the thin layer of topsoil shall be removed during

excavation of coal and stacked separately and preserved carefully for beneficial soil organisms and future vegetation. However, the duration of stockpiling needs to be minimized since excessive time of storage could cause structural degradation, death of seeds and micro- organisms, especially when soil moisture content is high.

Considering the time taken to generate and the importance of topsoil, MoEF&CC stipulated that topsoil be stacked at earmarked specific sites with adequate measures to preserve and be used either concurrently for backfilling or as top layer for reclamation of mined out areas. A record of topsoil indicating the area of stacking along with the date shall be maintained.

It is estimated that a total of 1.866 million m^3 of Topsoil will be generated during life of mine. It is proposed to systematically utilize the generated soil for plantation on backfilled areas and also on external dump areas in phases (Internal dump 1.199 million m^3, external dump 0.647 million m^3, embankment 0.021 million m^3).

The dust generated during blasting operations, loading and unloading operations and vehicular movements normally form heavier particles that would readily settle on very small areas within the mining area. This will have no adverse impact on the surrounding areas. During the rainy season the rainwater run off may spoil the agricultural land of the surrounding area.

Another important aspect is soil erosion from slopes, if not managed properly. Soil erosion may also get accelerated on areas where the overburden will be dumped. As there is neither a toxic effluent nor toxic solid waste from the mines, quality of soil is not expected to be adversely affected.

Impact on soil will be localized i.e. around the mine site. Likelihood of any adverse impact from soil erosion and disturbance in quality is remote. However, the impacts that will occur are reversible as the impacts will be felt in the operational stages of mine operation i.e. till the vegetative cover is developed.

4.3.2 IMPACT ON AIR QUALITY

In coal mines, air pollution starts with removal of massive overburden (OB) which discharges fine particles. It is followed by drilling/ extraction of coal and movement of extracted coal to the coal stock yard and finally transportation of coal to the consumer. These operations generate Respirable Particulate Matter (PM_{10} & $PM_{2.5}$) in the surroundings which are the main sources of air pollution. Emissions of some fugitive gases like carbon dioxide, carbon monoxide, Sulphur dioxide and oxides of nitrogen are also caused by automotive, generators and blasting operations.

The Increase in excavation from the mine may further increase the work-zone dust levels, in absence of any dust control measures.

Various means used by the coal companies to reduce air pollution are installation of monitoring stations to assess the level of pollutants in the air, installation of CHP, piped conveyor belt system, use of SILO for rapid and dust free loading of coal into wagons, installation of Merry-Go-Round, dust suppression by water sprinklers, mist blowers, mechanical brooming, plantation of trees along the route of transportation *etc*.

4.3.2.1 EMISSION FROM AREA SOURCES (FUGITIVE EMISSIONS)

Fugitive emissions are expected from the following sources: Excavation, Transportation and handling of ROM coal and overburden.

4.3.2.2 AIR POLLUTION IMPACT PREDICTION THROUGH DISPERSION MODELLING

The emissions will be mainly Particulate Matter (PM)

Details of Mathematical Modelling

The Gaussian Dispersion Modeling (GDM) is used for prediction of dispersion of air emission and the computation of Ground Level Concentration (GLC) up to a specified distance from source. The fundamental model is given below:

$$c(x,y,z) = \frac{Q}{2\pi\sigma_y\sigma_z u}\exp\left(\frac{-y^2}{2\sigma_y^2}\right)\left(\exp\left(\frac{-(z-h)^2}{2\sigma_z^2}\right)+\exp\left(\frac{-(z+h)^2}{2\sigma_z^2}\right)\right)$$

Where c is a concentration at a given position, Q is the source term, x is the downwind, y is the crosswind and z is the vertical direction and u is the wind speed at the h height of the release. The σy, σz deviations describe the crosswind and vertical mixing of the pollutant. The above equation describes a mixing process that results in a Gaussian concentration distribution both in crosswind and in vertical direction, centered at the line downwind from the source. Gravitational settling and chemical or radioactive decays are neglected.

The model computes the pollutant concentration dispersed in microgram per cubic meter for any point source with the location coordinated x, y and z.

Approach & Methodology

The "Gaussian Dispersion Model" version formed the basic framework of the computer model used for the computations of concentration of pollutants at ground level. Prediction of Impacts on air environment has been carried out by a mathematical model. AERMOD Cloud V.18 dispersion model which is computerized air quality model specifically designed for computing concentration and analysis of the dispersion of emissions.

The model assumptions

- Steady State Condition-Ideal gas, continuous uniform emission rate, homogenous horizontal wind field, representative wind velocity, no directional wind shear in the vertical, infinite plume, no plume history and normal distribution of eddy turbulences.
- Pollutants Characteristics-The pollutant emitted are stable gases or aerosol which remains suspended in the air and particulates in the turbulent movement of the atmosphere and none of the material is removed as the plume advances and diffuses down wind and there is complete reflection at the ground.
- Gaussian distribution - The pollutant material within the plume takes on a Gaussian distribution in both the horizontal cross wind and vertical directions described by empirical dispersion parameters σy and σz.
- It has been assumed that the pollutant does not undergo any physicochemical transformations.
- Gravitational settling of pollutants has not been considered.
- Reflection factor from any surface has not been considered.
- Control measures applied during calculation.

4.3.2.3 EMISSION ESTIMATE FROM MINING AREA

The emissions from the mining operations can be estimated either by sampling or direct measurement or using emission factors. Emission factors are being widely used for estimation of emissions from the Baitarni opencast coal Mining Operations. An emission factor is a representative value that attempts to relate the quantity of pollutant released in the atmosphere with an activity associated with the release of that pollutant. These factors are usually expressed as the weight of pollutant divided by a unit weight, volume, distance, or duration of the activity emitting the pollutant. The parameters that are most likely to influence the particle emissions are measure of source activity, properties of the material being disturbed and climatic parameters.

United States Environmental Protection Agency (USEPA) has developed emission factors for coal mining operation and was published through AP- 42 (1998), based on these emission factors, National Pollution Inventory (NPI), Australian Government also developed emission factors for Suspended Particulate Matter (TSP) and Particulate Matter (PM_{10}) and published the Emission Estimation Technique Manual (EET Manual) for mining and the version 3.1 is updated in January 2012.

4.3.2.4 EMISSION SOURCES AND STRENGTH

The emissions are estimated using the USEPA, AP- 42 Emission factors and NPI emission factors for different mining operations such as drilling, blasting, dozing, loading, dumping, transportation, loading on to the trains, scarping, grading and wind erosion of dumps for both coal and OB operations separately.

As per the National Ambient Air Quality standards notified vide GSR 826 (E), dated November 2009, the monitoring parameters for Particulate Matters for Ambient Air Quality were prescribed as PM_{10} and $PM_{2.5}$.

Hence, the emission estimation for PM_{10} is carried with equation of USEPA and NPI. Emission Factor Equations used for calculation PM_{10} from Excavation, Loading, Transportation, Dumping, Wind Erosion, etc. for both coal and OB (Drilling, blasting only for OB) are discussed below:

BLASTING OPERATION IN OB

As per the USEPA and NPI, the emission factor for coal is also suitable for OB. As such,

$EFTSP = (344*A^{0.8})/(M^{1.9}*D^{1.8})$ kg/blast

Where, EFTSP = Emission factor for TSP in kg/blast

A = Area of blasting in m^2 (OB)

M =Moisture content in OB, %

D = Depth of the hole, m

For PM_{10} fraction, USEPA (1998) estimated that, it will be 52% of TSP (Ref: NPI EET Manual, 2012). Hence, $EFPM_{10}$ = EFTSP x 0.52

DRILLING IN OB

Emissions from the drilling are relatively minor component of the overall emission from the opencast mining. The variables like, depth of hole, diameter of the hole, moisture content of the material being drilled, type of the drilling, wind current at the drilling site will influence the rate of emission. USEPA (1998) has derived emission factor for TSPM 0.59 kg /hole taking into consideration of all the parameters. USEPA (1998) does not provide any emission factor for PM_{10} component. However, the NPI EET Manual (2012) states that the mean fraction of PM_{10}/TSP was 0.52. Hence the best estimation of the emission factor for drilling for PM_{10} is = 0.59*0.52= 0.31 kg/Hole.

EF PM_{10} = 0.31 kg/Hole

Where, EF PM_{10} = Emission Factor for PM_{10}.

EXCAVATION IN COAL

The emission during the excavation i.e., loading of coal on to the trucks is mainly depending on the moisture content. As per the USEPA (1998) and NPI EET Manual, 2012 the emission factor is:

$EFPM_{10} = k*0.0596/(M)^{0.9}$

Where, EFPM10 = emission factor for PM_{10} in kg/tonne

k = 0.75 for PM_{10} concentration

M = Moisture content in (%)

EXCAVATION IN OB

The emission during the excavation i.e., loading of OB on to the dumpers / trucks is mainly depending on the moisture content of OB and mean wind speed.

$EFPM10 = k*0.0016*(U/2.2)^{1.3}/(M/2)^{1.4}$

Where, $EFPM_{10}$ = emission factor for PM_{10} in kg/tonne

k = 0.35 for PM_{10}

U = Mean wind speed in m/s

M = Moisture content in %

BULLDOZING IN COAL

The emission during the bulldozing operation in the coal is mainly depending on the moisture content and silt content of the material. As per the USEPA (1998) and NPI EET Manual, 2012 the emission factor is:

$EFPM_{10} = 6.33(s)^{1.5}/(M)^{1.4}$

Where, $EFPM_{10}$ = emission factor for PM_{10} in kg/h

s = Silt content in (%),

M = Moisture content in (%)

BULLDOZING IN OB

As per the USEPA (1998) and NPI EET Manual, 2012 the emission factor is:

$EFPM_{10} = 0.34(s)^{1.5}/(M)^{1.4}$

$EFPM_{10}$ = emission factor for PM_{10} in kg/h

s = Silt content in (%),

M = Moisture content in (%)

TRANSPORTATION OF COAL (WITHIN LEASE)

When a vehicle travels an unpaved road, the force of the wheels on the road surface causes pulverization of the surface material, particles are lifted and dropped from the rolling wheels and the road surface is exposed to strong air currents in turbulent shear, with turbulent wake behind the vehicle continue to act on the road surface after the vehicle has passed. Various types of road surfaces and vehicle characteristics are likely to have an impact on the particulate emission from unpaved roads. The particulate emission is mainly dependent on the weight of the vehicle and silt content on the road surface.

As per the USEPA (1998) and NPI EET Manual 2012 the emission factor for PM_{10} is:

$EFPM_{10} = k*(s/12)^{0.8*(}W/3)^{0.4}/(M/0.2)^{0.3}$

Where, $EFPM_{10}$=emission factor in kg per vehicle per km (kg/VKT)

k = 0.733 for PM_{10}

s = Silt content of road surface %

W = Vehicle gross mass in tonnes

M = Moisture content, %

TRANSPORTATION OF OB

As per the USEPA (1998) and NPI EET Manual 2012 the emission factor for PM_{10} for OB is also same as coal:

$EFPM_{10} = k*(s/12)^{0.8}*(W/3)^{0.4}/(M/0.2)^{0.3}$

Where, s = silt content of road surface

W = Vehicle gross mass in tonnes

M = Moisture content %

WIND EROSION OF COAL DUMP (STOCK YARD)

Dumping of coal in the coal stock yard also generate dust. Wind provides the mechanical energy for particle entrainment. A certain minimum wind velocity, often called threshold wind velocity is responsible for air borne of the dust from the dumps. For long term emission estimations, average soil moisture content or the number of rainy days, silt content of the dumping material, wind speed is influencing parameters. As per USEPA (1998) and NPI EET Manual 2012, emission factor for PM_{10} for wind erosion of dump is as given below.

EFTSP = 1.9*(s/1.5)*365*((365-p)/235)*(f/15)

WIND EROSION OF OB DUMPS

As per the USEPA (1998) and NPI EET Manual 2012 the emission factor for PM_{10} for wind erosion of dump is:

EFTSP = 1.9*(s/1.5)*365*((365-p)/235)*(f/15)

Taking,

$EFPM_{10}$ = 50% of EFTSP

EF = emission factor for TSPM kg/ha/Year

s = silt content in exposed area of the coal heap, %

p = number of days when rain fall > 0.25 mm

f = % of time that wind speed is > 5.40 m/sec at the mean height of the dump

Table 4:5 Emission of Particulate Matter (PM_{10}) for Peak Production capacities from different mining activities

PM_{10} uncontrolled					**PM_{10} controlled**		
	Activity	**kg/Day /m^2**	**g/sec/ m^2**	**Total (g/sec /$m^{2)}$**	**kg/Day /m^2**	**g/sec/ m^2**	**Total (g/sec/m^2)**
Pit Emissions	Drilling in OB Bench	350	4.050	17.494	35	0.405	1.749
	OB Loading	25.20	0.291		2.52	0.0291	
	Coal Loading	68.18	0.789		6.81	0.0789	
	Blasting OB	967.18	11.194		96.7	1.119	
	Dozing OB	55.04	0.637		5.50	0.0637	
	Dozing Coal	46.07	0.533		4.60	0.0533	
Pit & Haul Road Emissions	Coal transportation on unpaved road	2834.22	32.80	186.15	28.34	3.28	18.61
	OB transportation on unpaved road	13250	153.35		1325	15.33	
Poly Area Emissions only	Coal Unloading	55.91	0.647	2.097	5.591	0.0647	0.2097
	Wind erosion from OB dump, Coal Mine Pits & Coal Stockyards	6.75	0.07		0.67	0.007	
	OB unloading	90	1.04		9	0.104	
	Topsoil Removal	29.40	0.340		2.94	0.034	

Table 4:6 Emission of Particulate Matter ($PM_{2.5}$) for Peak Production capacities from different mining activities

$PM_{2.5}$ uncontrolled					**$PM_{2.5}$ controlled**		
	Activity	**kg/Day /m^2**	**g/sec/ m^2**	**Total (g/sec/m^2)**	**kg/Day /m^2**	**g/sec/ m^2**	**Total (g/sec/m^2)**
Pit Emissions	Drilling in OB Bench	66.5	0.769	3.135	6.65	0.0769	0.313
	OB Loading	2.52	0.029		0.252	0.0029	
	Coal Loading	9.54	0.11		0.954	0.011	
	Blasting OB	154.74	1.79		15.474	0.179	
	Dozing OB	20.36	0.235		2.036	0.0235	
	Dozing Coal	17.50	0.202		1.75	0.0202	
					0	0	
Pit & Haul Road Emissions	Coal transportation on unpaved road	39.67	0.459	2.605	3.967	0.0459	0.260
	OB transportation on unpaved road	185.5	2.146		18.55	0.2146	
					0	0	
Poly Area Emissions only	Coal Unloading	6.15	0.0711	0.2292	0.615	0.00711	0.022
	Wind erosion from OB dump, Coal Mine Pits & Coal Stockyards	0.59	0.0068		0.059	0.00068	
	OB unloading	9.9	0.114		0.99	0.0114	
	Topsoil Removal	3.23	0.0373		0.323	0.00373	

Table 4:7 Calculated Emission rate for various Source group for SO_2 and NO_2

Source Group Name	Emission rate for SO_2	Emission rate for NO_2	Unit
Active Mining Pit including all allied activities like internal OBD/Stock, OB/ Coal, Transportations	**0.00011731833**	**1.8144481**	**g/sec/m²**

4.3.2.5 MODEL OUTPUT

The AERMOD cloud model has been setup and run for study period (3 months) to find out the cumulative maximum ground level concentration of PM_{10} and $PM_{2.5}$ to be generated from the different activities in the mining areas.

Maximum ground level concentration of PM_{10} and $PM_{2.5}$ generated from the different activities in the mining areas is described below:

The predicted incremental concentrations of emissions are presented in **Table 4.8 to 4.11.**

The predicted ground level concentration isopleths for GLCs are shown in the figures.

Table 4:8 Incremental Ground Level Concentrations of PM_{10}

PM_{10}					
Location Code	**Location**	**Baseline Value (µg/m³)**	**Proposed**		**National Ambient Air Quality Standards (NAAQS)**
			Predicted Incremental Conc. (µg/m³)	**Total concentration (µg/m³)**	
AAQ 1	Project Site	83.9	4.307	88.207	100
AAQ 2	Near Jamunali Club, Handigoda	76.2	3.136	79.336	
AAQ 3	Village-Chhendipada	79.9	2.178	82.078	
AAQ 4	Near Behera Sahi New Club House	78.2	1.156	79.356	

AAQ 5	Near Indian Oil , Chhendipada	83.5	1.217	84.717	
AAQ 6	Charibati Filling Station, Mamuriasahi	79.8	1.168	80.968	
AAQ 7	Odia Sambada - Media House, Chhendipada	77.6	1.137	78.737	
AAQ 8	Sounyaranjan Pradhan, Kukurpetajungle, Odishal	81.2	1.188	82.388	
AAQ 9	Akash Bastralaya, Barapada	78.7	1.149	79.849	
AAQ 10	Play Field, Madhupur	67.8	0.135	67.935	

Table 4:9 Incremental Ground Level Concentrations of $PM_{2.5}$

$PM_{2.5}$					
Location Code	**Location**	**Baseline Value ($\mu g/m^3$)**	**Proposed**		**National Ambient Air Quality Standards (NAAQS)**
			Predicted Incremental Conc. ($\mu g/m^3$)	**Total concentration ($\mu g/m^3$)**	
AAQ 1	Project Site	42.2	1.435667	43.63567	60
AAQ 2	Near Jamunali Club, Handigoda	39.8	1.045333	40.84533	
AAQ 3	Village - Chhendipada	43.6	0.726	44.326	
AAQ 4	Near Behera Sahi New Club House	40.8	0.385333	41.18533	
AAQ 5	Near Indian Oil, Chhendipada	43.8	0.405667	44.20567	
AAQ 6	Charibati Filling Station, Mamuriasahi	42.1	0.389333	42.48933	

AAQ 7	Odia Sambada - Media House, Chhendipada	40.9	0.379	41.279	
AAQ 8	Sounyaranjan Pradhan, Kukurpetajungle, Odishal	45.8	0.396	46.196	
AAQ 9	Akash Bastralaya, Barapada	40.5	0.383	40.883	
AAQ 10	Play Field, Madhupur	34.7	0.045	34.745	

Table 4:10 Incremental Ground Level Concentrations of SO_2

SO_2					
Location Code	**Location**	**Baseline Value (µg/m³)**	**Proposed**		**National Ambient Air Quality Standards (NAAQS)**
			Predicted Incremental Conc. (µg/m³)	**Total concentration (µg/m³)**	
AAQ 1	Project Site	7.3	0.44	7.74	80
AAQ 2	Near Jamunali Club, Handigoda	8.2	0.79	8.99	
AAQ 3	Village-Chhendipada	7.9	0.72	8.62	
AAQ 4	Near Behera Sahi New Club House	7.6	0.53	8.13	
AAQ 5	Near Indian Oil, Chhendipada	8.9	0.164	9.064	
AAQ 6	Charibati Filling Station, Mamuriasahi	10.3	0.248	10.548	
AAQ 7	Odia Sambada - Media House, Chhendipada	8.2	0.91	9.11	
AAQ 8	Sounyaranjan Pradhan, Kukurpetajungle, Odishal	8.1	0.78	8.88	

AAQ 9	Akash Bastralaya, Barapada	8.3	0.132	8.432	
AAQ 10	Play Field, Madhupur	7.8	0.63	8.43	

Table 4:11 Incremental Ground Level Concentrations of NO_2

NO_2					
Location Code	**Location**	**Baseline Value ($\mu g/m^3$)**	**Proposed**		**National Ambient Air Quality Standards (NAAQS)**
			Predicted Incremental Conc. ($\mu g/m^3$)	**Total concentration ($\mu g/m^3$)**	
AAQ 1	Project Site	20.26	0.13	20.39	80
AAQ 2	Near Jamunali Club, Handigoda	23.3	0.41	23.71	
AAQ 3	Village-Chhendipada	22.6	0.27	22.87	
AAQ 4	Near Behera Sahi New Club House	20.1	0.15	20.25	
AAQ 5	Near Indian Oil, Chhendipada	19.2	0.13	19.33	
AAQ 6	Charibati Filling Station, Mamuriasahi	19.2	0.12	19.32	
AAQ 7	Odia Sambada - Media House, Chhendipada	18.4	0.0892	18.4892	
AAQ 8	Sounyaranjan Pradhan, Kukurpetajungle, Odishal	16.0	0.0740	16.074	
AAQ 9	Akash Bastralaya, Barapada	18.6	0.106	18.706	
AAQ 10	Play Field, Madhupur	20.5	0.22	20.72	

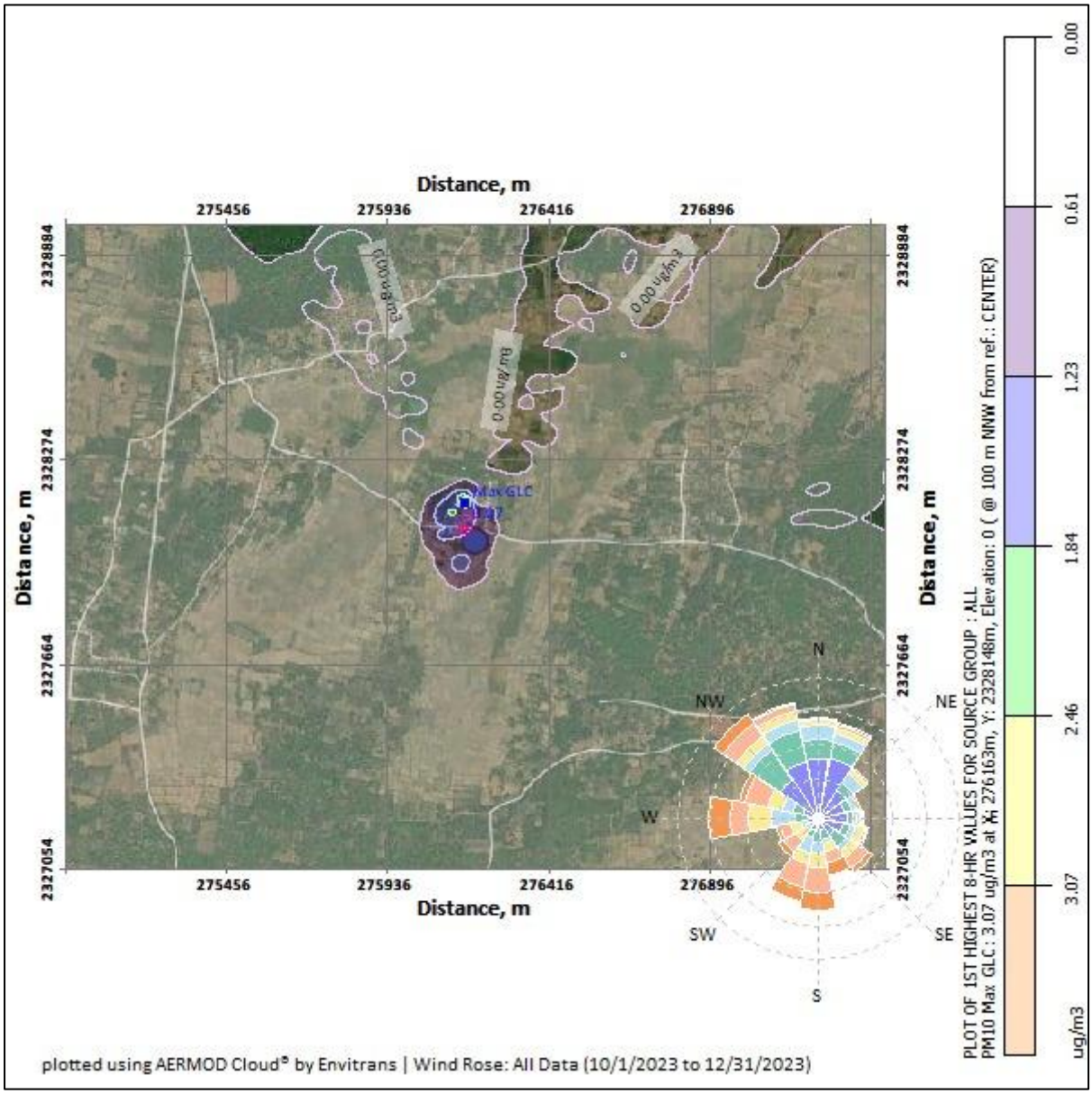


Figure 4-5 Isopleths for 24hr Incremental Conc. (µg/m^3) for PM_{10} (Controlled Condition)

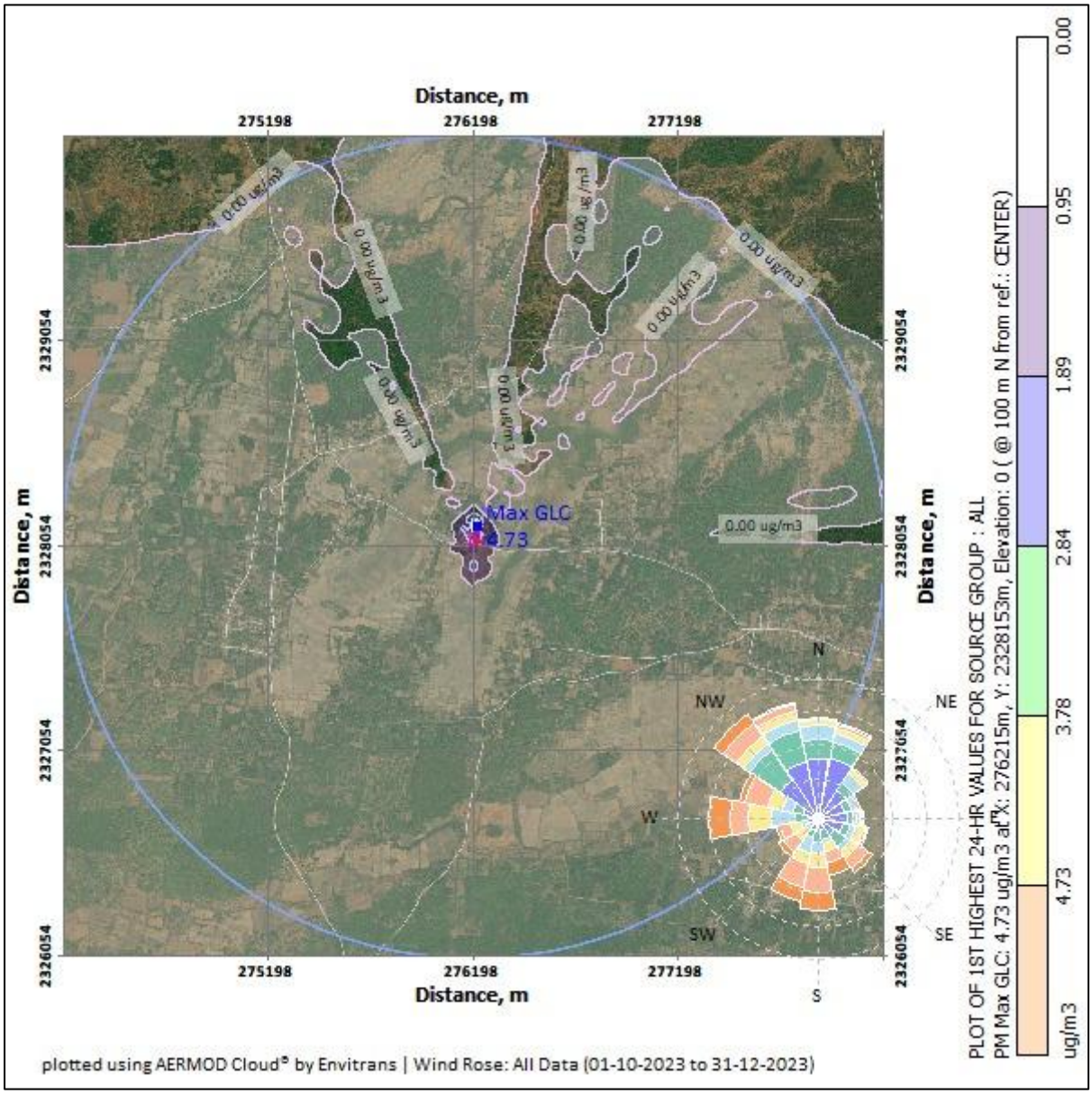


Figure 4-6 Isopleths for 24hr Incremental Conc. (µg/m^3) for $PM_{2.5}$ (Controlled Condition)

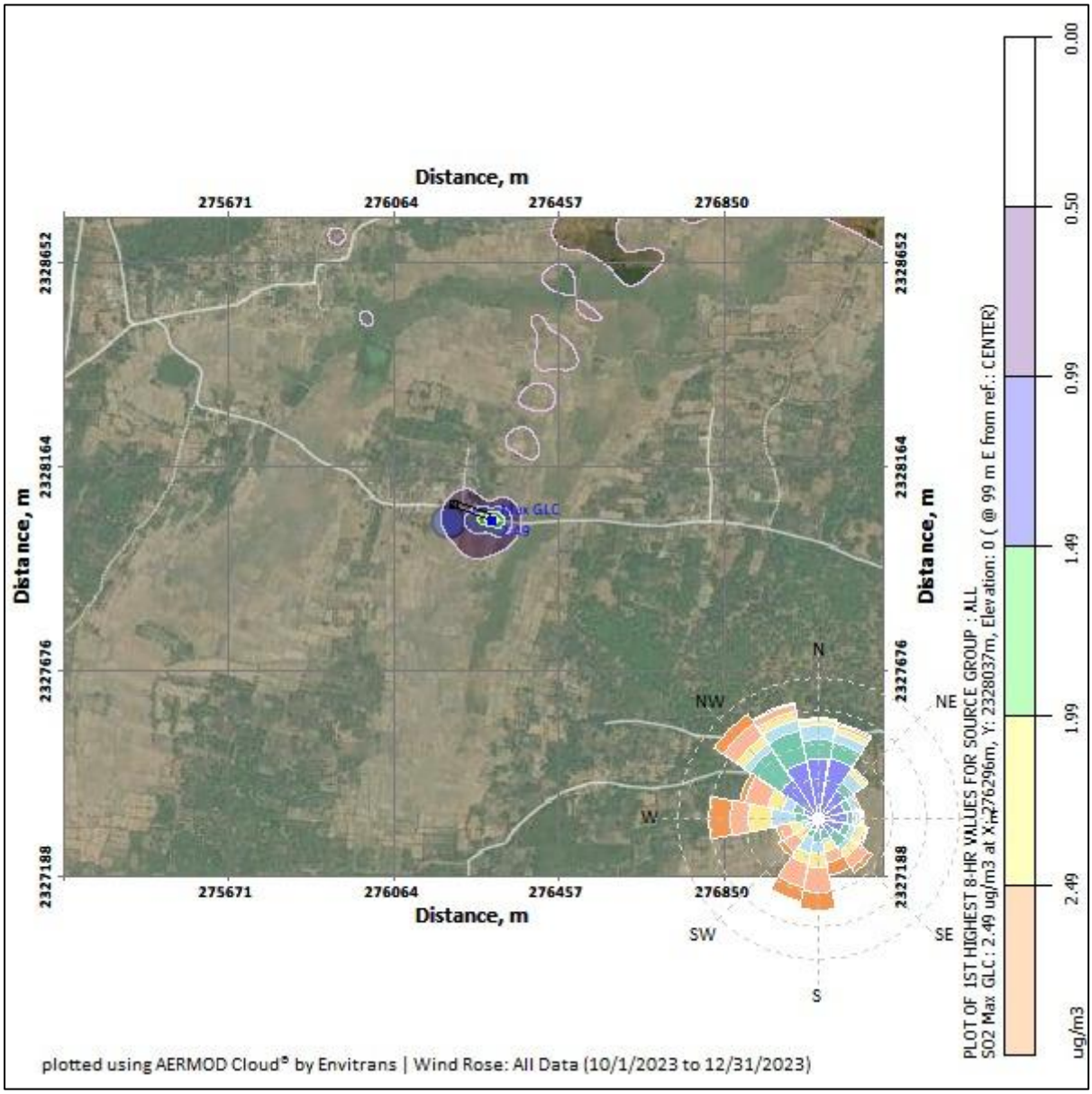


Figure 4-7 Isopleths for 24hr Incremental Conc. (µg/m^3) for SO_2 (Controlled Condition)

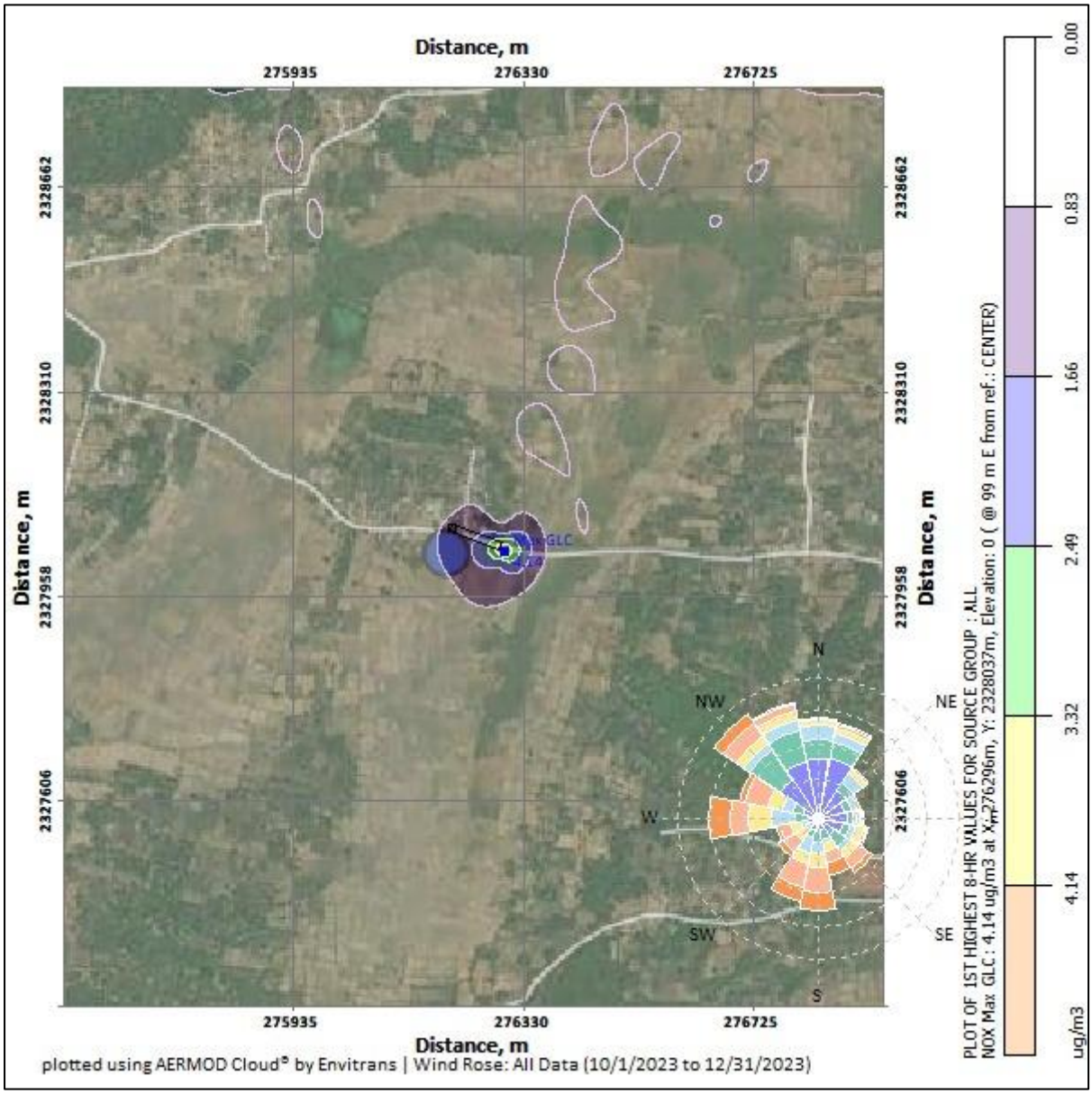


Figure 4-8 Isopleths for 24hr Incremental Conc. (µg/m^3) for NO_2 (Controlled Condition)

4.3.2.6 AIR POLLUTION MITIGATION MEASURES

Dust is the main air pollutant generated from coal mining. The nature of mining operations is such that complete elimination of dust from the mining process would not be possible. However, all preventive measures i.e., control at source like provision of dust collectors, wet drilling & wet cutting, dust suppression during the course of various mining operations, shall be resorted to minimize fugitive dust emission. The details of source wise pollution prevention measures planned to implement are proposed are as follows:

DRILLING

About 93 % of coal will be produced through surface-miner and balance 7% of coal will be produced through drilling-blasting.

Two types of drills have been proposed. 250 mm drills will be deployed for benches in thicker parting and top overburden, 160 mm drills will be deployed for thinner parting horizons.

BLASTING

Blasting is relatively minor contributor to total dust emissions. However, blasting dust in the form of concentrated cloud, that is highly visible and may have potential impact on neighbors at downwind of the blast. The blasting of near-surface weathered materials that contain a high proportion of fines, creates high dust emissions.

Site mixed slurry (SMS) has been proposed to be used for good fragmentation. For proper blasting and minimizing the adverse side effects due to blasting viz. noise, ground vibration back-breaks, air blast and fly rocks etc., the optimal blast design parameters are suggested to be used, after field trials.

Adherence to relevant statutory safety provisions as stipulated by DGMS, Chief Controller of Explosives and others competent authorities shall be made.

It is proposed to do blasting only once in a week for each excavator and only once in a day between shifts.

LOADING OF O.B. AND COAL TO DUMPERS

Since this operation takes place in a bench protected by highwall and since excavator bucket delivers material close to the bucket of the dumper, there is comparatively less generation of dust and chance of spreading to nearby habitation.

HAUL ROADS/TRANSPORTATION OF COAL

Loading of Coal and then transport to its next destination i.e. Coal Handling Plant through designated haul routes. Fugitive dust emissions are caused by the contact of the tyres with the unsealed road surface that affect the total distance travelled. Water sprinklers will be deployed to sprinkle water on haul roads, both for coal and OB to suppress dust from this source. Suitable arrangement for water availability for this purpose will be done.

In addition to the above measures, plantation along both sides of haul roads on the surface of the mine and the green belts formed around mining excavation will largely trap the airborne dust which may escape after sprinkling operation.

In addition to these, the following measures will be adopted to minimize the fugitive dust emissions and gaseous emissions from the haul roads:

- Compaction & gradation and drainage on both sides of haul roads
- Proper maintenance of transport vehicles
- Avoiding overloading and enforcing speed limit on dumpers

4.3.2.7 MEASURES TO MITIGATE CO LEVELS

The concentration of CO in the ambient air is negligible and is far below the prescribed limit of CPCB and is not anticipated to exceed it in future. Since the coal mine operation is open cast hence CO emission is not an issue as in Under Ground coal mine. However, all heavy and light vehicles shall be checked for pollutants concentration in their exhausts regularly and well maintained. Strict vigil will be kept in and around the operational area for any fire which shall be immediately controlled.

4.3.2.8 AIR POLLUTION CONTROL

- Wet drilling method shall be used.
- Timely maintenance of drilling machines shall be done.
- Surface miner shall be used to avoid blasting. However Controlled blasting with advanced initiation system of NONEL method shall be practiced as per DGMS guidelines if at all it is required
- Scientific mining methodology with Shovel and Dumper shall be used for mining.
- Sharp teeth shovels and other soil excavation equipment shall be used.
- Periodical maintenance of HEMM, other vehicles for controlling vehicular emissions shall be done.
- Adequate dust extraction system will be provided during extraction of Ore.
- Surface miners shall be equipped with dust suppression system and operating in effective manner for dust suppression
- Portable water tankers shall also be deployed for water sprinkling in required places for dust Suppression (Loading and Unloading places, Haulage roads)
- Fixed water sprinkling system shall be installed around the coal stock yard area and permanent haulage road, transportation road for dust suppression
- Wind barriers shall be provided around the coal stock yard
- Mobile Fog cannon dust suppression systems shall be engaged for suppression of dust in the loading and unloading areas
- Coal shall be transported through covered trucks.
- Vehicles with PUC certificate shall be deployed in the mine.
- All internal coal transportation roads, temporary mine haulage roads and other material transport roads of the mine shall be maintained properly to avoid ruts and pots hole.
- Maintenance of haul road will be done by regular grading through grader & dozer.
- Mobile water sprinkling tankers will be deployed for dust suppression in the non-

metaled
haulage roads, mining bench, OB dump area and working pit area

- Transportation of coal from mine to outside will be done through Rail network to the great extent (around 80 %). In the initial period, transportation will be done through road network and by covered trucks.
- Trees will be planted all along the coal transportation road and haulage road.
- Mechanical sweepers will be deployed for cleaning of roads.
- Over burden and Soil dump areas to be maintained as per the approved design. Grass will be planted on the completed external dumps to reduce fugitive dusts.
- Adequate firefighting system will be provided in the coal stack yard.
- Water will be stored in identified locations for handling fire emergencies.
- Green belt/plantation shall be developed around the mine, along haul road, quarry boundary, OB stock yard, Topsoil stock yard, coal stockpile area and office building etc.
- Tree translocation will be done inside the mining lease (In the safety zone of the mining lease and in the available vacant areas of the mining lease) to the extent possible.

4.4 IMPACT DUE TO TRANSPORT

4.4.1 TRANSPORTATION OF COAL

Coal after excavation in the mine is transferred to the pit head stock, which is then transferred to the customers either by road (truck) or rail. Transportation by road generates a lot of air pollutants for which EMP emphasized the need to minimize road transportation. For mitigating air pollution, generation of dust is to be controlled at the source with necessary measures *viz.,* CHP, piped conveyor belt system, SILO including Rapid Loading System, Merry-Go-Round (MGR, a system which is a is a closed-circuit dedicated rail transportation system between the production and consumption points. Dust suppression by water sprinklers, mist blowers, mechanical brooming *etc.* Further, dust generation is to be minimized along coal / waste transportation roads and green belt isto be created around the source of dust. On account of various associated activities, there will be increased vehicular traffic on connecting roads. Generation of gaseous emissions is therefore, of primary concern. The best possible measures as discussed above shall be implemented for control of dust.

Impact on Traffic Density/ Enhanced Traffic on Road Adequacy:

With present level of traffic and the increase in existing traffic due to the project during operational phase has been estimated by comparison with the recommendations stipulated by Indian Road Congress (IRC).

4.4.2 TRAFFIC STUDY

The sources likely to be affecting the pollutant concentrations in project area would be emission from vehicles used for transportation. Consideration has been given to minimum transportation by road. As discussed earlier, out of 15 MTPA of Coal 12 MTPA shall be transported through rail and PP will have its own private siding within lease. Balance 3 MTPA of coal shall be transported by road. 45 tonner(net) Trucks shall be used for transportation of coal to meet the local demand. The proponent shall make dedicated coal transportation road (which will not pass through any village having habitation) to connect the MDR connecting to SH-63. It is estimated that a total of 203 Nos. of trucks/day from one side and 406 trucks/day will be arriving/leaving the project site.

The Angul-Chhendipada-Baghadia-Kanloi metalled road runs along the northern boundary of the block.

The existing Major District Road is from Jarapada (in Angul-Sambalpur Highway) up to Tentuloi.

The proposed project will attract commercial vehicles resulting in traffic load which will also increase. Traffic load plays a major role in polluting the air.

Two locations one at SH 63(T1) and one at MDR(T2), **(ODR Tentuloi Chhendipada Road,** before connecting with SH-63) were considered for the study as shown in the photographs.

Traffic data collected continuously for 24 hours by visual observation and counting of vehicles under three categories, viz., heavy motor vehicles, light motor vehicles and two/three wheelers. Two people were deployed simultaneously at each station during each shift- one person on each of the two directions for counting the traffic. At the end of each hour, fresh counting and recording was undertaken. Total numbers of vehicles per hour under the three categories were determined.

Table 4:12 Traffic Scenario

S. No	Type of Vehicles	Existing Vehicular Traffic (ODR Tentuloi Chhendipada Road)		
		Actual Count Per Day	As % of Total Vehicular Count	As PCU
1	Passenger Car	336	25.67	336
2	Cycle, Motorcycle or Scooter	757	57.83	379
3	Truck and Buses	216	16.50	648
Total (Daily Basis)		**1309**	**100**	**1363**

Table 4:13 Traffic Scenario

S. No	Type of Vehicles	Existing Vehicular Traffic (SH-63 Angul to Deogarh)		
		Actual Count Per Day	As % of Total Vehicular Count	As PCU
1	Passenger Car	1148	36.16	1148
2	Cycle, Motorcycle or Scooter	1061	33.41	531
3	Truck and Buses	966	30.43	2898
Total (Daily Basis)		**3175**	**100**	**4577**

Table 4:14 Level of Service Criteria

V/C	LOS	Performance
0.0 - 0.2	A	Excellent
0.2 - 0.4	B	Very Good
0.4 - 0.6	C	Good
0.6 - 0.8	D	Fair/Average
0.8 - 1.0	E	Poor
1.0 & above	F	Very Poor

Table 4:15 Recommendations on Traffic Capacity

S. No	Category of Road	V (PCU/h)	C	Existing V/C Ratio	LOS
1.	**ODR Tentuloi Chhendipada Road**	57	1500	0.03	A
2	**SH-63 Angul to Deogarh**	191	1500	0.12	A

Source: IRC: 106:1990

Calculation of number of trucks

Production capacity = 30, 00,000 TPA

Tons Per day = 9091 TPD

No of trucks = 9091 /45 = 202.02 say 203 trucks both way 406

Part A

It is estimated that 203 Nos. of trucks per day will be required. Traffic load on external road is calculated as under,

Total trucks = 203 trucks/Day

Frequency of trucks (Nos. of trucks x cycle x up and down) = 203 x 1 x 2 = 406 Nos/Day

PCU/Day (Nos. of truck x 3) = 406 x 3 = 1218 PCU/Day

PCU per hour = 51

Part B

No of people to be employed =1752 Nos.
Cars of the employees = 100 Nos.
Two wheelers of the employees = 800 Nos.
Frequency of cars (Nos. of cars x cycle x up and down) = 100 x 1 x 2 = 200 No's/Day
PCU/day (Nos. of car x 1) = 200 x 1 = 200 PUC/Day
PCU per hour = 8.3 say 8

Part C

Frequency of two wheelers (Nos. of two wheelers x cycle x up and down) = 800 x 1 x 2 = 1600 No's/Day
PCU/Day (Nos. of two-wheeler x 0.5) = 1600 x 0.5 = 800 PUC/Day
PCU per hour = 33.33 say 34

Grand Total PCU/h (A+B+C) = 51+8+34 = 93 PCU/h

Table 4:16 Modified Traffic Scenario

S. No		Modified V (PCU/hr)	C	Modified V/C Ratio	Modified V/C Grade
1.	**ODR Tentuloi Chhendipada Road**	57+93= 150	1500	0.1	A
2.	**SH-63 Angul to Deogarh**	191+93=284	1500	0.18	A

The road is capable of absorbing additional truck movement after considering the contribution due to transportation at Baitarni West Coal Block.

Details of Transportation Route

Out of 15 MTPA of Coal, 12 MTPA shall be transported through rail and PP will have its own private siding within lease. Balance 3 MTPA of coal shall be transported by road.

The proponent will make dedicated coal transportation road (which will not pass through any village having habitation) to connect the MDR connecting to SH-63. The proposed dedicated road transportation route is approximately 2.1 km long, as shown in the route map below.

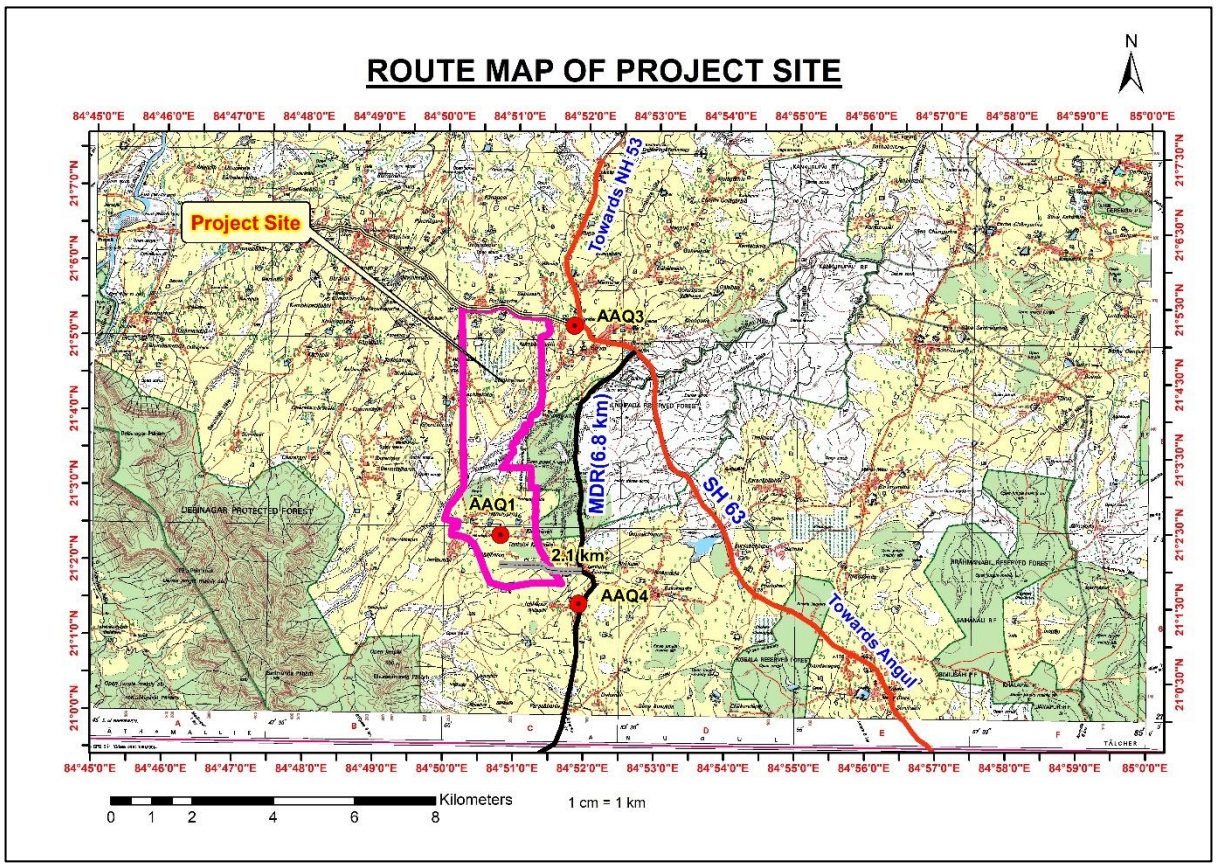


Figure 4-9 Proposed dedicated road transportation route

Tentuloi station will be the serving station for Baitarni West Terminal of GMDC to load 12 MTPA outward coal. The train will be dispatched on main signals from Tentuloi station towards Putagadia of MCRL and then towards Jarapada (JRPD) station or towards Angul as the case may be.

The evacuation of coal from proposed rail Terminal at Baitarani West and the despatch to destination stations are as below:

Table 4:17 Source of Origin/Destination of traffic

Commodity	Origin	Destination	Quantity
Outward Coal	Baitarni West Rail Loading Terminal served by Tentuloi station.	Different destinations in the state of Gujrat and other destinations over India	12 MTPA

Transportation route of rail is shown in below transportation route map.

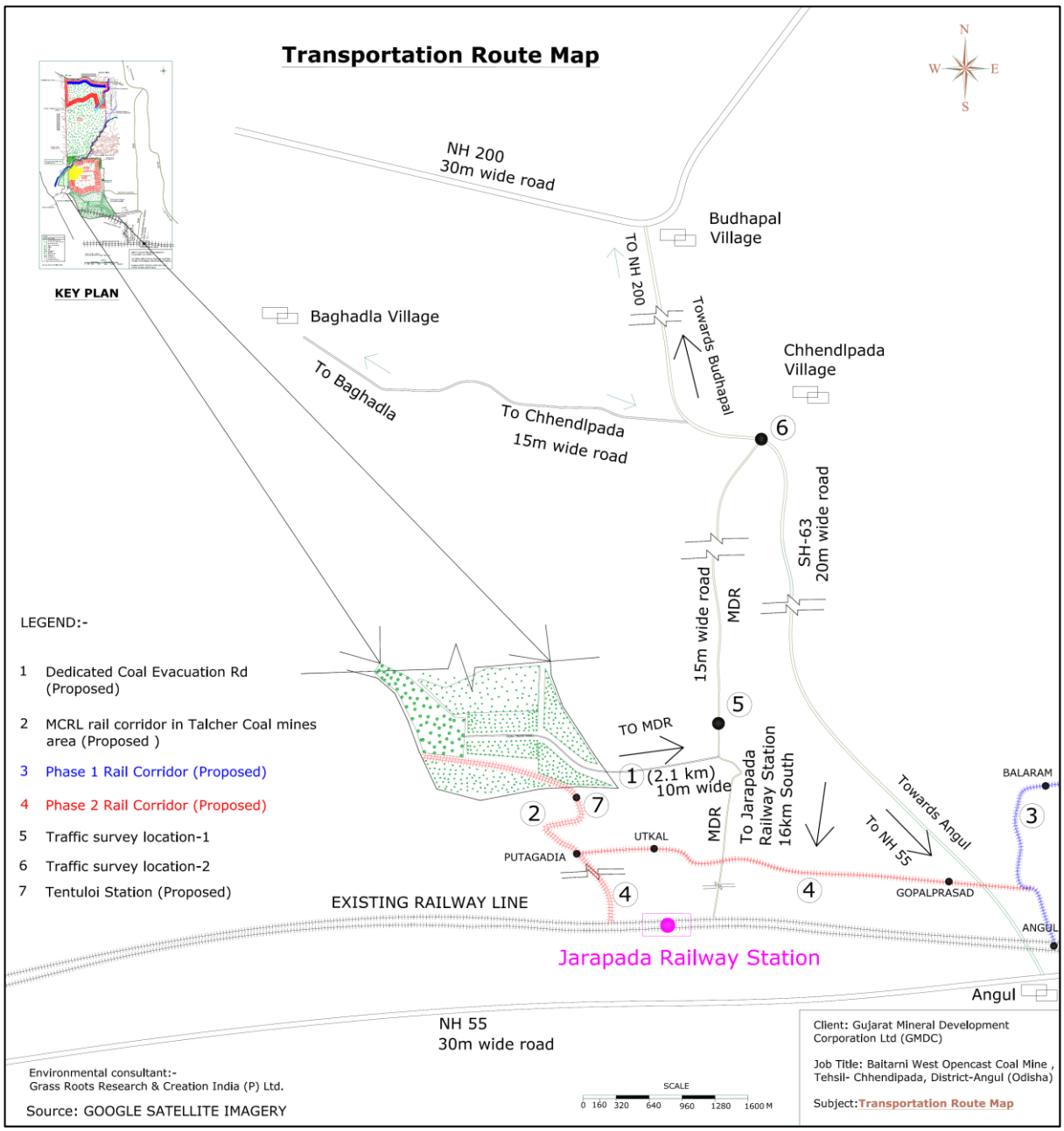


Figure 4-10 Transportation Route Map

Geotagged photographs of road transportation route are shown below:





ODR Tentuloi Chhendipada Road





SH-63 Angul to Deogarh

Figure 4-11 Geotagged photographs of transportation route

4.5 AIR MODELLING

Air Quality Prediction along the coal evacuation transport route.

4.5.1 IMPACT OF OFF-SITE TRAFFIC ON AIR QUALITY

Various related activities increase vehicular traffic on the connecting road. Therefore, the generation of emissions is a primary concern.

4.5.2 THE RATE OF VEHICULAR EMISSION WITHIN A GIVEN STRETCH OF THE ROAD

The extent of these effects depends on:

- Meteorological conditions of that area
- Emission rate for Coal Transportation Road (CTR)

4.5.3 MODEL INPUT DATA

The modelling has been carried out by assuming the truck capacity around 45 tonnes and 330 working days. The prediction of incremental concentrations due to internal traffic assumed are estimated based on site specific meteorological conditions and line source.

The emission rates as inputs to the line source model are calculated based on "Bharat Stage-IV Standards". The inputs used for modelling are given in Table-15.

Formula used:

Emission Rate (g/sec/m2) = (Total Emissions in grams)/ (Area of the Roadway Segment in square meters)/ (Time Period in seconds)

Table 4:18 Emission Rate for modeling Daily Emissions from Transportation Emission factors (gm/km/hr)

S.no	Emission Type	Emission Factor	Emission Rate
1	CO	1.5	0.04752
2	NO_2	3.5	0.105
3	PM_{10}	0.012	0.0003793
4	$PM_{2.5}$	0.008	0.000253

Source: CPCB Bharat Stage IV emission guidelines

4.5.4 MODEL PREDICTIONS

The predicted PM10, PM2.5, CO, and NO_2 concentrations from vehicular traffic are presented in Table 4.19 and isopleths showing max. GLCs are given in Figure-4.12 to Figure-4.15.

Table 4:19 Predicted Incremental Ground Level Concentrations (GLC) due to Additional Traffic load

S.No	Emission Type	Baseline Value ($\mu g/m^3$)	Predicted Incremental Conc. ($\mu g/m^3$)	Total Concentration ($\mu g/m^3$)	NAAQS ($\mu g/m^3$)
1	CO	430	7.10	437.10	4000
2	NO_2	16.4	10.995	27.395	80
3	PM_{10}	66.4	0.36	66.76	100
4	$PM_{2.5}$	32.95	0.24	33.19	60

Note: The maximum addition (incremental) by this mine is given in Table-4.16. The air quality will remain well within the national ambient air quality standards.

4.5.5 RESULTS

The observation from predictions reveals that the maximum GLCs for PM10, PM2.5, NOx and CO are 0.36 µg/m^3, 0.24 µg/m^3, 10.995 µg/m^3 and 7.10 µg/m^3 respectively and all values are very low when compared with NAAQS. It is also concluded that emission due to plying of trucks for 15 MTPA mine production will not cause any significant impact on nearby area.

However, PP will adopt mitigation measures such as water sprinkling and, development of additional green cover along the both side of the road, loaded trucks will be covered with tarpaulin and only PUC certified vehicle will be permitted for transportation.

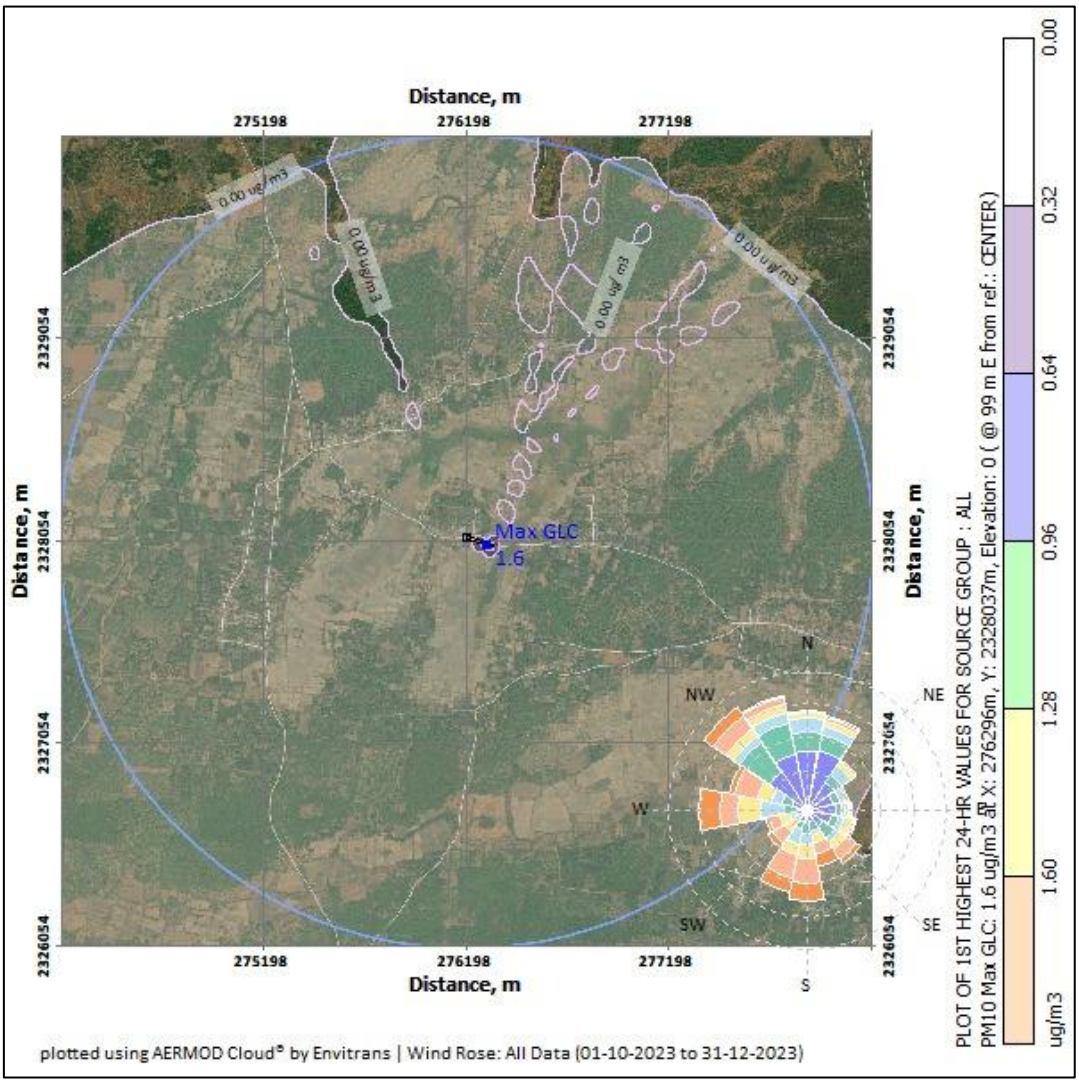


Figure 4-12 Spatial distribution of predicted GLCs of PM_{10} due to transportation

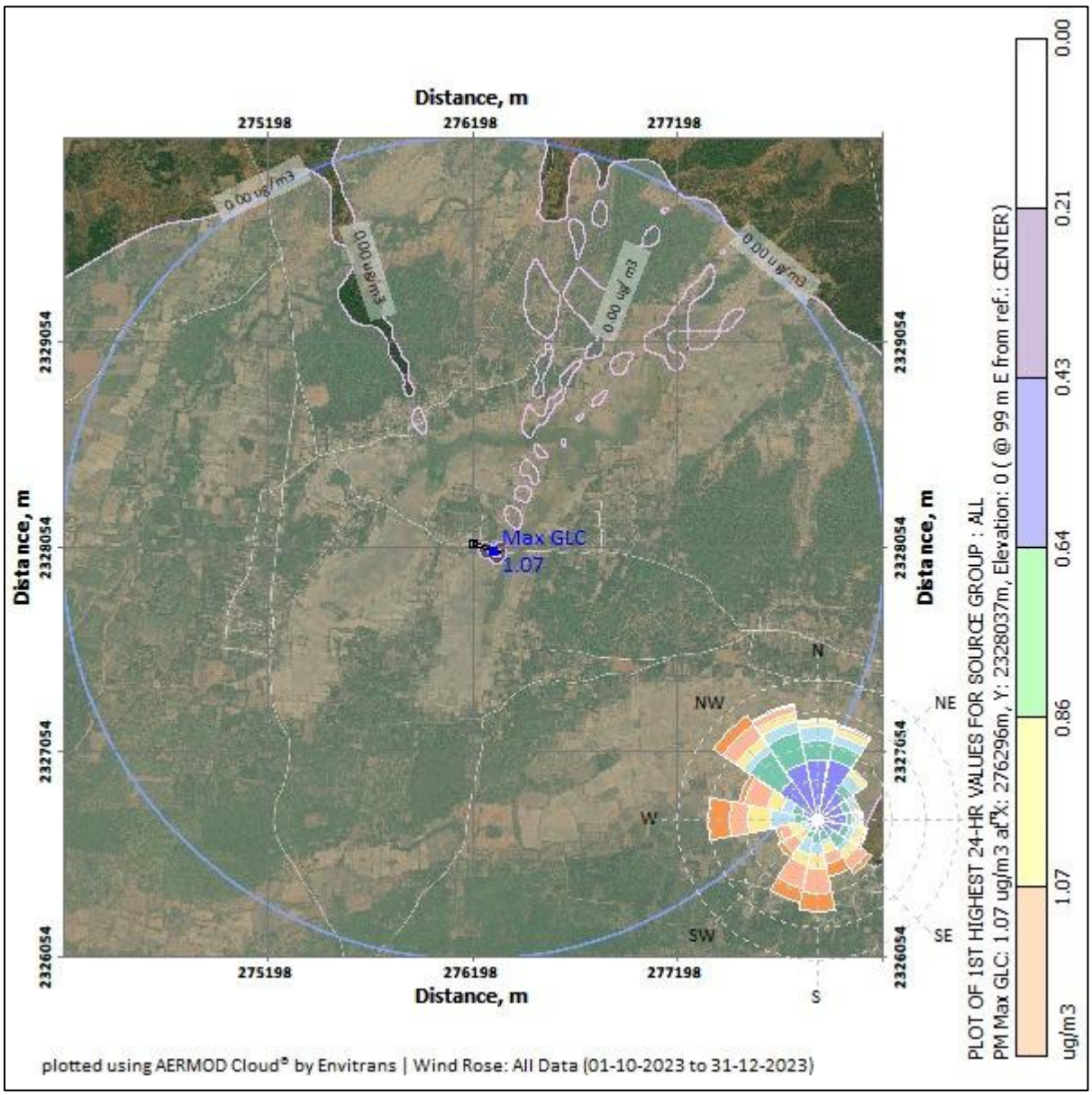


Figure 4-13 Spatial distribution of predicted GLCs of $PM_{2.5}$ due to transportation

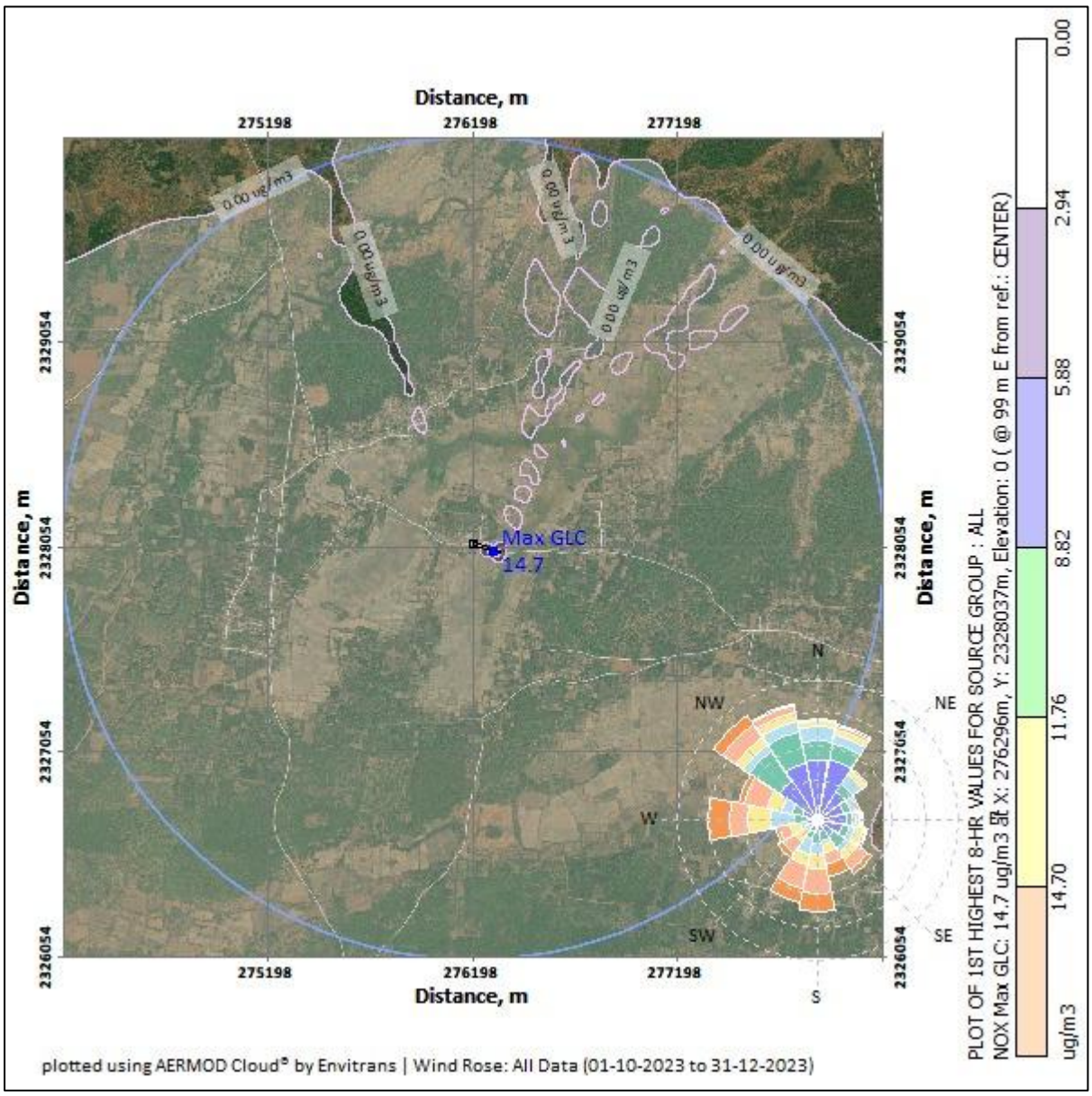


Figure 4-14 Spatial distribution of predicted GLCs of NO_2 due to transportation

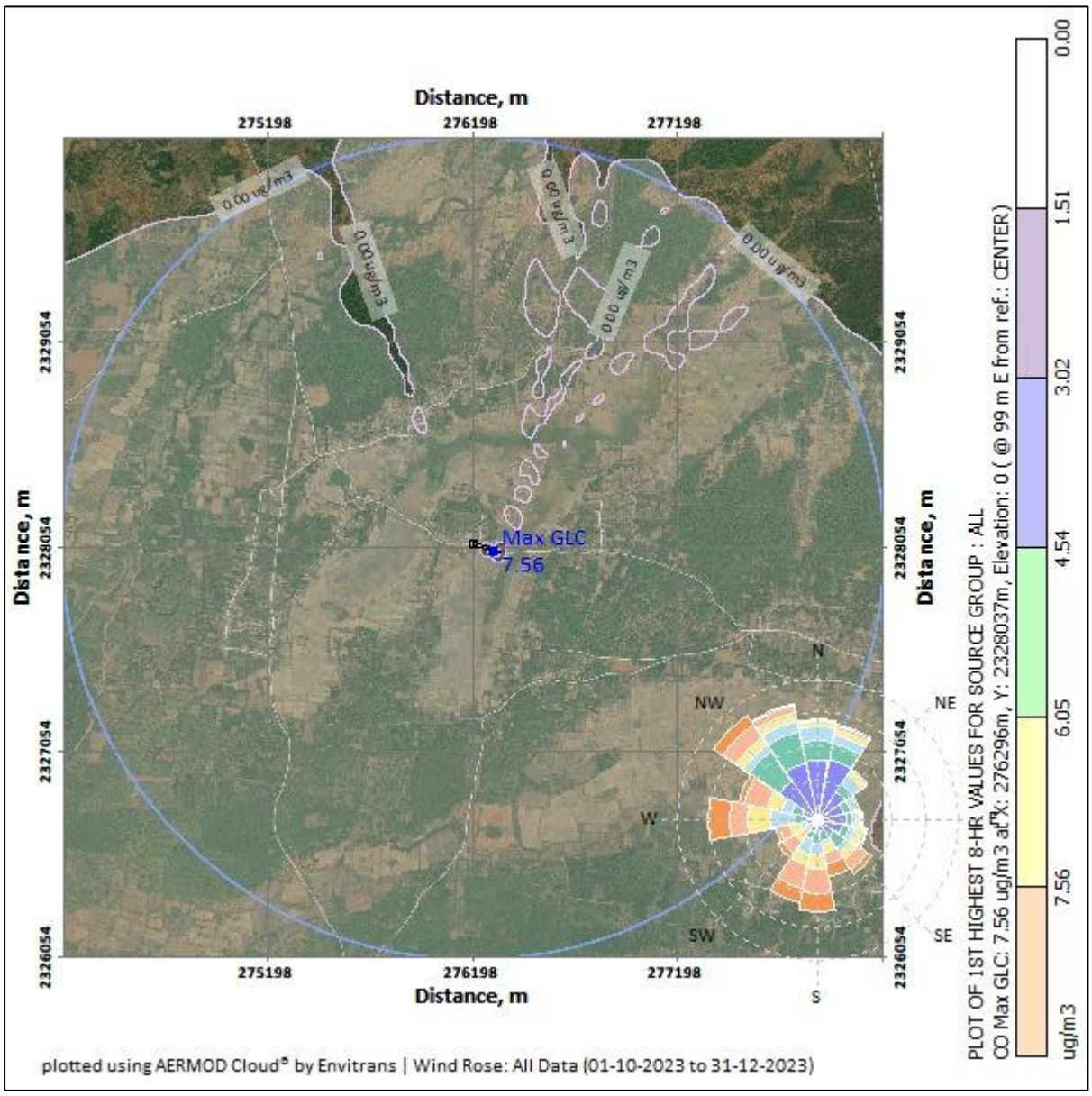


Figure 4-15 Spatial distribution of predicted GLCs of CO due to transportation

Mitigation Measures

- All the trucks used for the transport of raw materials, products and wastes will be completely covered with tarpaulin and ensured no spillage during transportation.
- Total no. of Trucks required to transport of coal will be 203 trucks/day.
- Common Internal haulage roads will be made pucca.
- All the raw material yards will be equipped with water sprinkling systems to avoid fugitive emission during the material handling.
- The movement of trucks will be minimized in peak hours.
- All Truck drivers will be trained for the traffic rules and traffic movement.
- Dedicated coal transportation road shall be made Bituminous to connect the Mine

with Highway for coal evacuation.

- Instant shower system to be installed at the exit point of quarry for wetting the coal.
- Wheel washing system shall be installed at the exit point of mine.
- Vehicles with PUC certificate shall be deployed in the mine.
- Overloading of transport equipment shall be prevented in order to stop spillage.
- Trees will be planted all along the coal transportation road.
- Mechanical sweepers will be deployed for cleaning of roads.

4.5.5.1 IMPACT ON AIR ENVIRONMENT DUE TO TRAFFIC MOVEMENT

The impact on traffic has been studied by using IRC: 64:1990.

The extent of impact on ambient air quality due to increase in traffic, at any given time, will depend upon the rate of vehicular emissions on a given stretch of road and the prevailing wind speed, wind direction and stability class. The impact will have strong temporal dependence as both these factors vary with time. Most of the workers will come by cycle daily to attend work.

The impact of vehicular exhaust due to road traffic on account of transportation by road shall be limited within 25 m distance from either side of the road and thereafter it shall widely disperse and become insignificant beyond 25 m.

4.6 IMPACTS OF WATER POLLUTION AND ITS MITIGATION MEASURES

Mining is likely to have significant effects on ground water as well as surface water. Mining operations can both contaminate and cause severe physical dislocation of aquifers.

The major source of water pollution in coal mines is the suspended solids in the drainage system of mine water and storm water. In some coal mines, the mine water is acidic due to the presence of sulphur / pyrites / pyritic compounds. In this project no washery is proposed. In Heavy Earth Moving Machineries (HEMM) and light vehicles workshops, the workshop floors mix oil and oily matter into water along with dirt that will being washed.

To mitigate water pollution caused by mining activities, measures such as installation of water treatment plants (Effluent Treatment Plant or ETP) for eliminating pollutants from mine water as well as storm water and discharge from the workshops, installation of Sewage Treatment Plant (STP) for domestic wastewater will be adopted.

Total water requirement for the project is as follows:

Table 4:20 Total Water Requirement

S. No	Unit Process	Quantity of Water Used (MLD)
1	CHP	0.76
2	Dust suppression (haul road, stuck yard, fixed and mobile water sprinkler, water cannons etc.	1.74
3	Green belt	0.32
4	Domestic, Drinking, Toilet etc.	0.54
5	Misc.	0.26
	TOTAL	**3.62 MLD**
Note: Initial water requirement is 0.26 MLD and shall be met from the ground water and afterwards seepage water will be used after suitable treatment.		

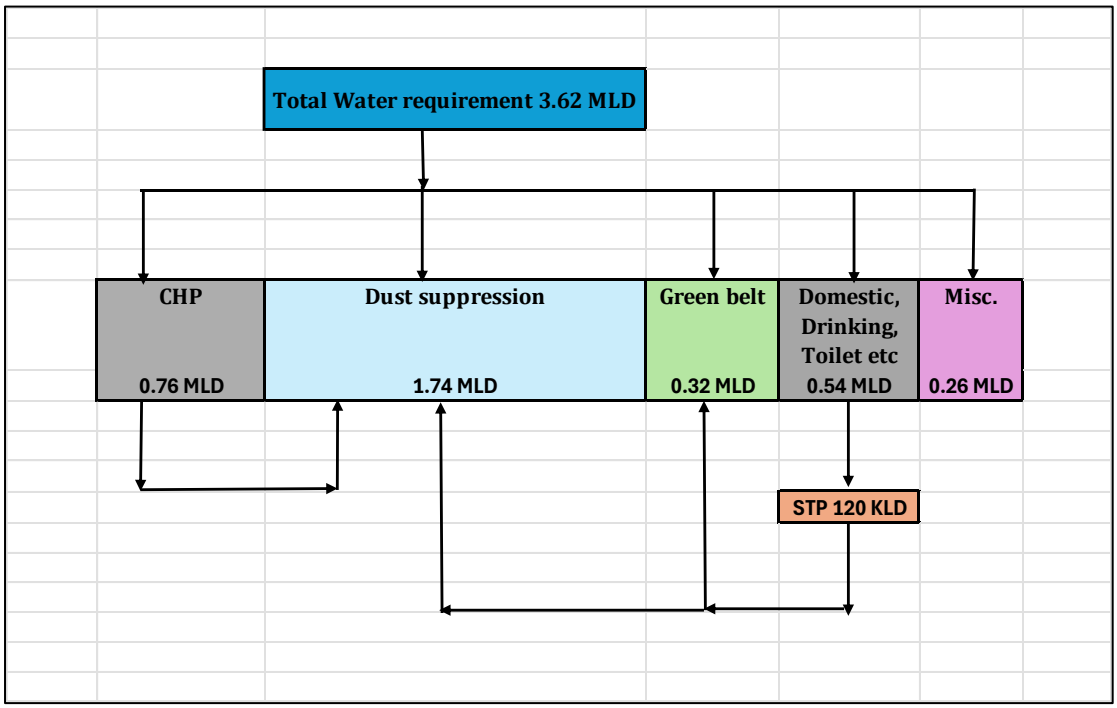


Figure 4-16 Water Balance for total water requirement

Initial water requirement:

Table 4:21 Initial water requirement

S. No	Purpose	Demand for Year 1 & 2 (KLD)

1	Domestic/ Drinking water at Project site	100
2	Green belt development	10
3	Miscellaneous	150
	Grand Total	**260**

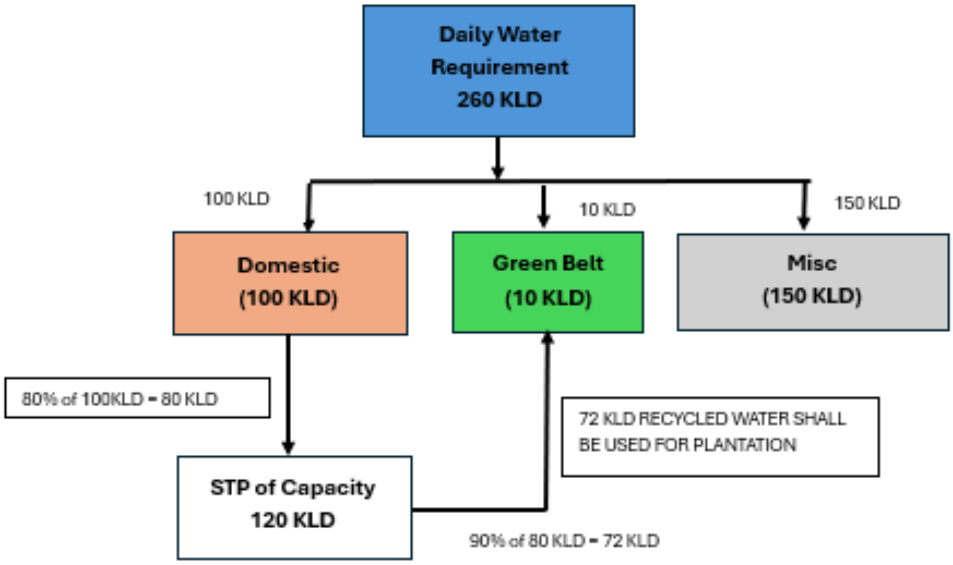


Figure 4-17 Water Balance for Initial water requirement

Impact of Mining on Water Usage

Water is the most important resource. The maximum water requirement for the mine is 3.62 MLD.

Initially 0.26 MLD shall be required for domestic and other purposes. After that PP will use Mine seepage water. Permission of ground water has been obtained vide NOC no. NOC/MIN/OD/2025/534/N dated 29.07.25 & copy of same is attached as **Annexure VIII**. However, Project Proponent will have all the above approval in place before the mine starts production. The mine inflow water will be recycled to meet industrial water demand.

Use of water at various mining operations, run-off water after precipitation, coal stockyard, over mined out areas and pumping discharge of percolated ground water in mine excavation, generate wastes of various quality and quantity, which needs to be addressed to prevent pollution of surface water and contamination of groundwater system.

As per the directives of Odisha State Pollution Control Board (OSPCB), zero discharge of untreated water is to be ensured.

4.6.1 TREATMENT OF ACIDIC MINE WATER

PP will keep track for acidic mine water by collecting and analyzing water samples on regular basis. If found acidic, mine water will be treated in the neutralization tank. However, the PP will give all emphasis for utilization of the OB materials in filling up the mine pits at the earliest in order to prevent the pyritic material from getting exposed to oxygen and moisture thereby preventing the formation of Acid Mine Drainage. The dump shall be covered with grass and plantation to have minimum exposer to water as well as stability.

Effluent management system

Industrial wastewater will be suitably treated in a plant consisting of pre- and post-settling chambers/tanks and oil and grease trap (OGT). The industrial wastewater from workshop shall be treated in the effluent treatment plant. The industrial effluent will be diverted to the effluent treatment plant. Then, treated water will be used for dust suppression in haul road besides washing of dumpers and/or dozers in workshop. The sludge collected from the tank(s) will be utilized as landfill in the decoaled area. The oily sludge from oil and grease trap will be disposed off in the impervious layer lined pit/or to authorized vendors. Oil and grease recovered manually from the trap will be stored in drums safely for disposal through auction to the government authorized facility. There will be no discharge. The surface runoff water shall be pass through garland drain and shall be collected in a sump/settling pond before leaving the ML area.

4.6.2 IMPACT OF MINING ON HYDROLOGY

The Baitarni West Coal Opencast Mine extends over an area of 1196 ha. The proposed mining quarry is situated in the northern part of the lease area, covering 623.71 ha. The mine pit (quarry) area is highlighted and shown in Figure 4.18. Out of the nine (9) watersheds identified, six (6)- W1 to W6 are affected by the mine excavation pit (quarry) and the remaining three (3)- W7 to W9 are not having any physical impact.

Based on the surface hydrological features of the region, it is observed that the major rivers, streams and nalas in the area are either first-order streams or streams that are in the process of forming downstream. From the analysis of terrain and river morphology, it is observed that only Watershed 1 (W1)-Supai Nala and Watershed 5 (W5)-Kumbhira Nala have defined streams, while all other watersheds do not have any defined streams and exhibit only overland flow along the flow accumulation paths.

Supai Nala originates on the eastern side at the center of the quarry and flows towards the northeast, emerging out of the quarry area at the right boundary corner. Inside the lease area, Supai Nala primarily exhibits overland flow with rills in major portions. It forms a defined stream towards the boundary of the lease area as it exits. Kumbhira Nala enters the mine lease area from the left boundary and exits from the right boundary, effectively dividing the mine lease area into two parts: the northern part and the southern part. The entire quarry pit is situated in the northern part, thus ensuring that Kumbhira Nala remains undisturbed. Only a small portion of the catchment contributing flow to Kumbhira Nala near its watershed boundary will be intercepted by the mining activities. The remaining area will continue to contribute flow to the Nala in the same manner as it does under natural conditions, thereby preserving the hydrological balance and minimizing any potential impacts on the stream.

It is observed that the quarry primarily affects the initial catchment area rather than the main body of the stream. It does not intersect with any defined stream at its middle reach. Instead, it intercepts a portion of the catchment area that contributes flow to the stream at its upper reach. This portion of the catchment area primarily consists of first-order streams, which lack distinct paths and flow as overland runoff during precipitation events.

Watershed management plan is attached to **Annexure IV.**

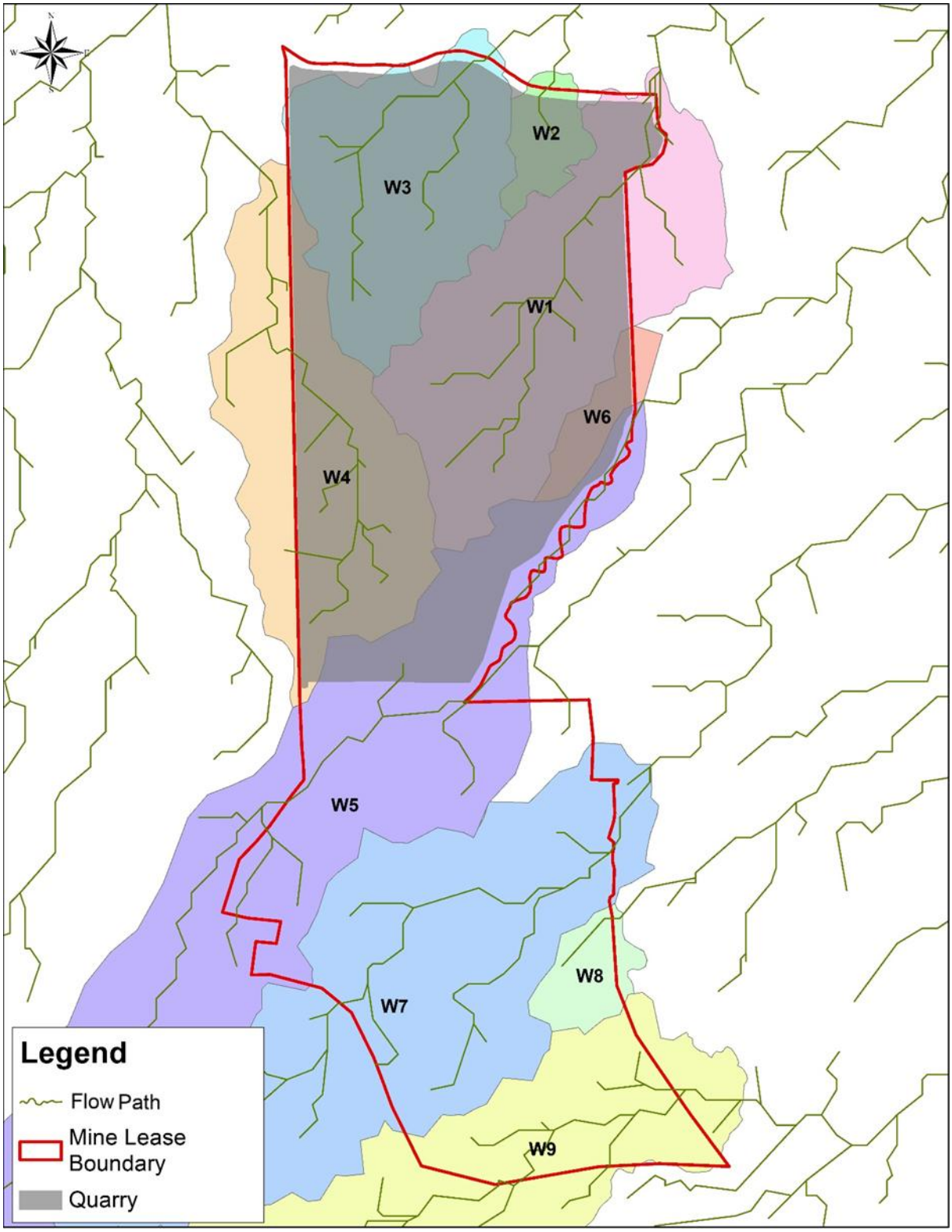


Figure 4-18 Watershed Affected by the Mine Pit (Quarry)

4.6.3 IMPACT OF MINING ON SURFACE WATER

There are two seasonal nalla, Supai and Kumbhira flowing within the lease area.

To arrest the sediments and prevent silting of the water courses by the run-off during rains, toe-walls, retaining walls with weepholes at strategic stretches will be constructed.

Garland drain will convey the water to decantation ponds after settling.

Impact on Existing Users by Mining & Mining Operations

The diversion of natural resources and mining operation can have some impact on existing users. The major activity in the region is agriculture, which relies on surface water during the monsoon season. Post-mining, the surface water within the quarry pit will no longer be naturally available in the river system. To minimize the impact on the quantitative availability of water, a diversion plan has been proposed to intercept the water within the premises and discharge it into the Nala. From a qualitative perspective, water samples will be monitored to ensure they remain within the acceptable limits specified by standard codes. Additionally, due to the open-cast mining operations, there is a high likelihood of increased turbidity. Therefore, sediment traps have been proposed to manage and reduce sediment levels, thereby mitigating potential adverse effects on water quality.

Impact of Diversion on Existing Users

The Supai Nalla originates within the mine lease area, the availability of surface water is only during the occurrence of rainfall. For other period the nalla remains dry. Hence, it is an intermittent stream and is only being used during monsoon season for occasional irrigation of the agricultural land in its periphery.

It is to be noted that the catchment intercepted by the proposed nallah diversion is the origin of the Supai Nallah and hence there are no upstream users. The users who were dependent on the nallah are the people residing in the affected catchment area of Supai Nala, which is proposed to be diverted through proposed swale/quarry pit. The total affected catchment area of Supai Nala at the outfall point of Supai Nala is 2.72 sqkm. Out of which the users residing in an area 2.34 sqkm with be affected by the diversion plan. It is estimated a total population of 501 nos (Chhhendipad Jungle- 373 nos, Porapara-128 nos) will be affected. The rest of the people within ML area and outside of the lease area will be not affected due to the proposed diversion. The water intercepted from this area will flow through the diversion swale and will be available for use for the downstream users.

After diversion, the water intercepted from the area outside of the quarry pit will still be available in the same manner for use for the locality and downstream users.

Impact of Diversion on Hydrology

There will be no considerable loss or gain in the Supai Nala as water will not to diverted to any other stream. It will only be collected through proposed swale from the area outside of the quarry pit and pumped from the area inside of the quarry pit to eventually discharge into Supai Nala at downstream.

Mitigation Measures

To mitigate the potential impacts of the quarry on the catchment area and hydrological features, the following measures are proposed:

i. Surface Water Diversion Channels

The watersheds affected by the quarry primarily consist of initial stream reaches with overland flow. To regulate the flow from areas outside the quarry pit that pass through the mine pit to downstream areas, shallow swale surface water diversion channels will be constructed around the excavation area of mine. These channels are designed to effectively manage and control surface runoff by redirecting water away from active mining or quarrying zones, thereby preserving natural drainage patterns. These channels will be strategically placed along the perimeter at key locations where runoff enters the excavation area. They will follow natural topography and will be positioned to intercept and divert water to the outlet points, thereby minimizing the loss of watersheds that contribute flow downstream to streams.

ii. Retention Ponds (Sump)

To mitigate the impact of surface runoff and manage water flow within the quarry site, retention ponds (or sump) will be constructed within the excavation area. These ponds serve as temporary storage areas for rainwater and runoff, helping to regulate peak flows and reduce sedimentation downstream. By capturing and holding water, retention ponds facilitate sediment settling and improve water quality before discharge. The eastern part of the quarry, being lower, will serve as a sump area when a large section of the quarry is opened.

iii. Silt Fences and Sediment Traps

To minimize sediment runoff and protect water quality during quarry operations, silt fences and sediment traps will be installed strategically across the site. Silt fences are permeable barriers made of geotextile fabric or similar materials, placed along slopes and perimeter areas to intercept sediment-laden runoff. They slow down water flow, allowing sediment to settle while enabling filtered water to pass through. Sediment traps, on the other hand, are excavated depressions or basins designed to capture and retain sediment before runoff enters natural watercourses. Both measures work synergistically to prevent erosion and reduce

sediment transport, thereby preserving downstream water quality and ecological integrity.

iv. Progressive Reclamation Rehabilitation

To ensure responsible land stewardship and minimize long-term environmental impacts, a progressive rehabilitation plan will be integral to the quarry's operations. This plan involves ongoing restoration efforts throughout the mining lifecycle, starting from the early stages of extraction. Areas of the quarry that are no longer in use will undergo systematic reclamation to restore natural landscapes to as much as possible. As approved in the mine closure plan of Baitarani West Coal Mine, the land reclamation will start from Year 3 of post operation till Year 40. The site will undergo both technical and biological reclamation efforts. Technical reclamation will involve backfilling and reshaping of excavated areas totaling 623.71 ha, along with the removal of structures such as temporary buildings and concrete foundations. Concurrently, biological reclamation will encompass extensive plantation efforts across 1,029.583 ha, creating water bodies over 38.488 ha and addressing other ecological enhancements over 31.85 ha.

v. Monitoring and Maintenance

To uphold environmental standards and ensure sustainable practices, stringent measures will be implemented to manage water quality within the quarry site. Regular monitoring of water parameters such as turbidity, pH levels, and pollutant concentrations will be conducted to assess potential impacts on nearby watercourses. Maintenance protocols will include routine inspection and upkeep of erosion control structures, sediment traps, and water diversion systems to prevent sedimentation and mitigate runoff.

FOR ABATING WATER POLLUTION

(a) Seepage water from mine

To prevent surface water contamination, following control measures are proposed in Environmental Management Plan:

- Mine water will be pumped to settling pond for settling and then the clean water will be allowed to pass outside the lease area for irrigation or plantation after internal utilization.
- Leak proof containers will be used for storage and transportation of oil/grease.
- To avoid oil/grease spillage in the store, the container containing oil/grease will be kept in empty open containers of higher volume than these containers.
- The area over which oil/grease is handled will be kept effectively impervious.
- The sewage waste will be treated in a STP.
- All the seepage water will be analysed regularly before discharging outside the lease area.

- The surface and ground water in and around the mine, loading plant and infrastructure will be tested as per the monitoring schedule and appropriate control measures will be adopted, if required.
- All stacking and loading areas will be provided with proper garland drains equipped with baffles to prevent wash offs from reaching the downstream natural channels.

(b) Storm water

Control measures to be adopted are briefly discussed below:

Embankment shall be constructed along the Kumbhira stream (around 3 km length) within the ML area.

- Check dams/Settling ponds will be provided to prevent solids from wash off and screen, if any, from the mine related activities.
- Peripheral bunds will be erected on the outer edge of the abandoned benches before reclamation so that the soil is not carried away by storm water.
- A water gradient of about 1 in 100 will be kept at every bench towards inside of the bench to prevent formation of gullies in the bench slopes causing serious erosion.
- Construction of garland drains around freshly excavated and dumped areas so that flow of water with loose material is prevented.
- The mine water will be passed through specially constructed catch pits to arrest any loose material being carried away with water.
- Any areas with loose debris within the leasehold will be covered with plantation.
- Garland drains will be constructed surrounding the waste dumps and connected to the surface water reservoir to avoid the run-off mixing directly to natural water channels before settling.
- Run-off water from mine pit garland drain will be directed to the settling pond.

4.6.4 DIVERSION PLAN OF SUPAI NALA

This plan details the strategies to divert surface water flows to minimize hydrological impacts on the watersheds affected by quarry. Figure 4.18 shows the watersheds that are getting affected by the mine pit. Two strategies have been proposed to manage surface water: one to intercept the flow from areas outside the quarry pit that pass through the mine pit to the downstream, and another to manage surface flow intercepted by the mine pit inside the quarry boundary.

I. Surface Flow Intercepted from Watersheds Outside of the Quarry Pit

To divert the flow of Supai Nala from the areas contributing towards the mine pit, intercepting swale is proposed along the right boundary of the mine pit. And for channelizing water of Kumbhira Nalla, low-height bunds/embankments will be constructed along the nala at its left bank.

Swales: Swales are shallow, vegetated channels designed to manage water runoff by slowing and redirecting flow, facilitating infiltration, and reducing erosion. The area contributing flow to the swales is from Supai Nala with intercepting area of 0.62 sq km. The surface slope is 1 in 65 to 1 in 100. Based on the flow volume assessment, a maximum runoff depth of 339.4 mm was observed for the month of July. If daily basis is considered, a maximum runoff (flow) depth of 151 mm was observed on 5th of September in 2003-04 (Appendix-F) resulting in a rate of discharge of around 1 cumecs. Therefore, the swale configuration has been proposed to convey a maximum of 1 cumecs, considering the most conservative slope.

Table 4:22: Diversion Channel (Swale) Design Specification

S.N	Particulars	Quantity
1	Discharge	1 cumec
2	Longitudinal Slope	1 in 100
3	Swale Shape	Trapezoidal
4	Side Slope	1:1
5	Bottom Width	0.4 m
6	Top Width	1.2 m
7	Height	0.4 m

Bunds/Embankment: Bunds/Embankments are low-height barriers proposed to be constructed at the left bank of Kumbhira Nalla to channelize the flow away from the mine pit, directing surface water flow more naturally along the intended paths. These bunds will help ensure that water flows away from the quarry.

Table 4:23: Bund/Embankment Design Specification

S. N	Particulars	Quantity
1	Longitudinal Slope	Natural Terrain
2	Shape	Trapezoidal
3	Side Slope	1:1
4	Bottom Width	0.9 m
5	Top Width	0.3 m
6	Height	0.3 m

II. Surface Flow Intercepted from Watersheds Inside the Quarry Pit

In the approved mine plan, mining activities are organized to manage surface water effectively:

- The eastern part of the quarry, being lower, will serve as a sump area when a large section of the quarry is opened.

- The lowest bench will have a gradient of 1 in 200 towards the sump to prevent waterlogging.
- Due to changes in strike direction between lower and upper seams and across faults, bench alignments will be adjusted to ensure proper drainage into the sump.
- For areas where natural drainage to the sump is not feasible, face pumps and stage pumps will be used.

Within the quarry, water from working areas will flow into the sump by gravity. From the sump, water will be pumped to the surface drainage system. During rainfall, inflow into the sump will contain clay and silt. Sufficient slurry water pumps with capacities of 70 lps (54 m head, 75 kW power) and 16 lps (35 m head, 30 kW power) have been proposed to handle slurry water effectively.

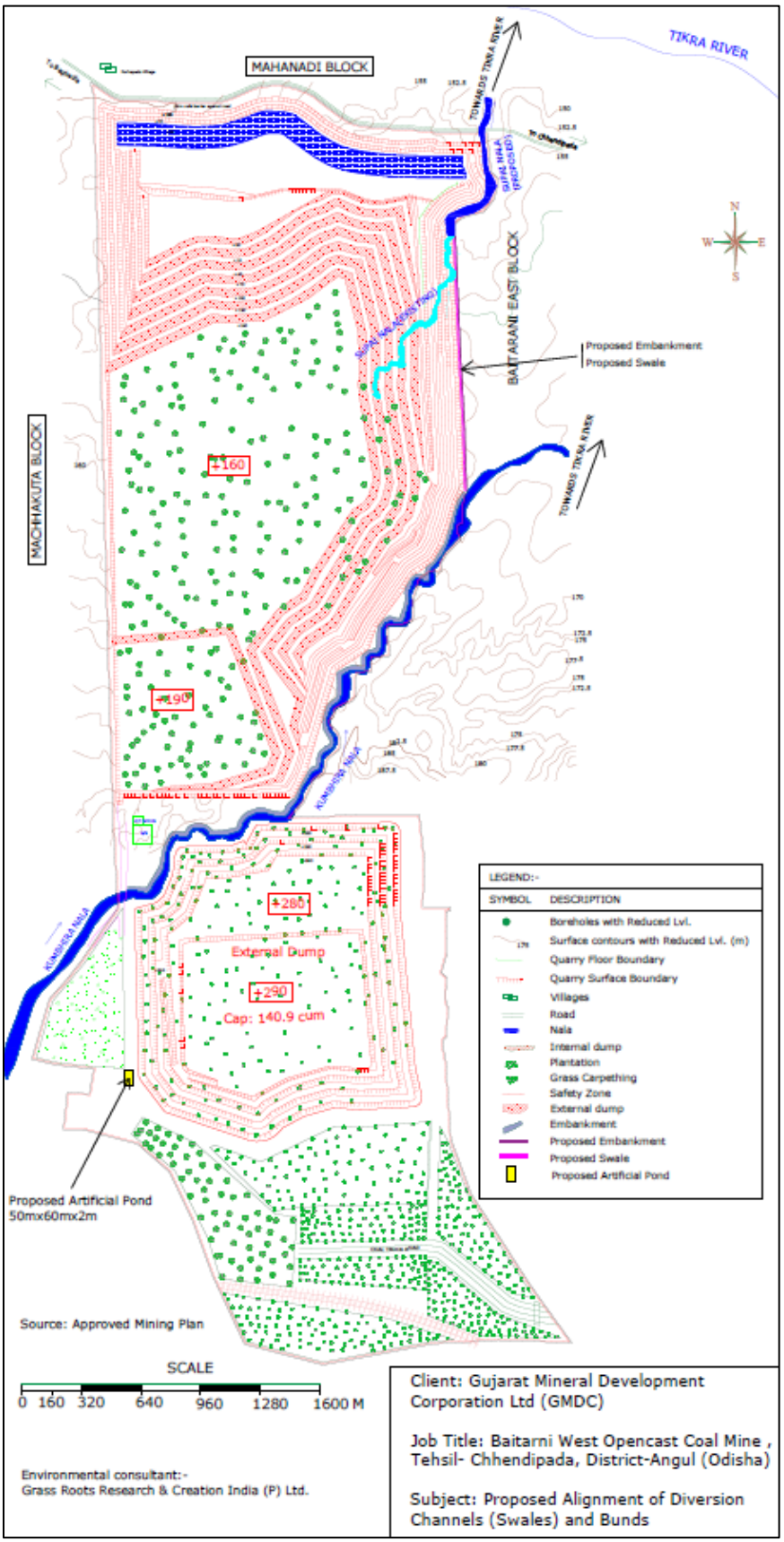


Figure 4-19: Proposed Alignment of Diversion Channels (Swales) and Bunds

Conclusion & Recommendation of Study:

The key findings and recommendations of the report are summarized below.

- The lease area of 1196 ha of Baitarni West Opencast Coal Mine is in the Talcher Coal Field in Chhendipada block of Angul district of Odisha. It focuses its mining operation primarily in the northern part of the lease area covering an area of 623.71 ha.
- Hydrologically the region is in Tikra sub-basin of Brahmani Basin of Odisha. Two dominant streams govern the entire drainage system of the lease area-- Supai Nala and Kumbhira Nala.
- The study carried out in this report identified nine (W1 to W9) critical small watersheds within the lease area based on the flow path direction and terrain condition. Six (W1 to W6) located in the northern part of the lease area are directly impacted by the mining activities, particularly by the excavation pit (quarry), while the remaining three (W7 to W9) remain unaffected.
- The hydrological analysis indicates that the affected watersheds feature first-order streams or channels in the initial stages of development. The quarry intercepts upper catchment areas contributing to Supai Nala and does not intersect with the mainstream at its mid-reach, affecting primarily first-order streams that flow as overland runoff during precipitation events.
- To manage surface water and reduce ecological disruption a detailed diversion plan focusing primarily on surface water diversion has been proposed. Various strategies have been proposed: to intercept the flow of Supai Nala from areas outside the quarry pit that pass through the mine pit to the downstream, diversion channel (Trapezoidal Swales with 1:100 longitudinal slope, 1:1 side slope, 0.4m bottom width, and 1.2m top width to redirect up to 1 cumec of surface runoff) has been proposed, to keep water of Kumbhira Nalla away from the pit, bund (1:1 side slope, 0.9m bottom width, 0.3m top width, and 0.3m height) has been proposed and to manage surface flow intercepted by the mine pit inside the quarry boundary, retention pond (sumps) and then pumping it outside to natural Nala has been proposed. Other measures recommended are silt fences, sediment traps and implementing a progressive rehabilitation plan.
- Diverting the Supai Nala will alter the natural flow within the lease area. The impact can be reduced to minimal by help of Swales and bunds as proposed in the diversion plan.
- For safety of the nala diversion plan, the diversion channels and bunds should be designed with appropriate slopes, depth and width for stability and smooth facilitation of water flow.
- A water augmentation plan has been suggested to recharge the water resources of the region by creating an artificial pond in the south-west boundary of the lease area outside of the quarry pit.
- The potential impact of the mining on the existing users activity has been studied. Major activities over region have been identified as agriculture who

depends on the monsoon rainfall. Both quantitative and qualitative measures have been suggested in the report on how to keep this impact minimal.

In conclusion, the Baitarni West Opencast Coal Mine project integrates meticulous hydrological assessments with strategic mitigation measures to ensure sustainable water management and environmental protection. The project aims to mitigate environmental impact effectively, comply with regulatory standards, and restore ecological balance in the long term.

4.6.5 ESTIMATION OF MINE SEEPAGE AND ADVANCED DEWATERING PLAN

The probable mine inflow from sedimentary bedded deposits can be estimated directly from Darcy's law (Eq. 1). Wherein, the groundwater inflow from different mine faces and the individual aquifer units is assessed, owing to the assumptions that flow is laminar and one phase flow into the mining excavation. However, it can be applied for multiphase flow conditions, if there exists a multi-layered aquifer system and interconnected due to structural controls in a regional scale. Using Darcy's law, one-dimensional inflow may be expressed as:

$$Q = KA\Delta h/\Delta x \quad \textit{...Eq. 1}$$

where,
Q = inflow rate, expressed as cubic meter per day (m3/day), as indicated;
K = hydraulic conductivity of the intercepted aquifer, expressed as meter per day (m/day);
A = cross-sectional area through which flow occurs, determined as the product of the saturated thickness multiplied by the width of the area, expressed as square meter (m2) (i.e., Face length*aquifer thickness);
Δh = change in hydraulic head, expressed as meter
Δx = distance over which the change in hydraulic head is determined, expressed as meter.

Since the coefficient of transmissivity (T) is a product of saturated thickness of the aquifer and the hydraulic conductivity equation attains the form:

$$Q = T*I*L \quad \textit{...Eq. 2}$$

Where,
Q= Flow rate (m3)
b= Aquifer thickness (m)
T= Transmissivity =K*b (m2/day)
A= Area of the pit (m2) (i.e., Face length*aquifer thickness)
L= Perimeter/ average width of the cross-sectional area, L (m)

Accordingly, Mine seepage has been predicted based on Darcy's equation (eq.2) for the exposure of formations/aquifers of Baitarani West Coal mine block as per predicted mine progress up to 5th year of the mine operation.

Total Mine Seepage:

The total mine inflow predictions for different stages of the mine for Baitarni Coal Field is provided in below table 4.24.

Table 4:24 Total Mine seepage

Year	Non-Monsoon seepage from all benches (cubic M)	Monsoon from all benches (cubic M)	Total Yearly seepage (cubic M)	Seepage Cub m Per day
1st year	0	0	0	-
2nd year	0	0	0	-
3rd year	1,92,602	1,39,061	3,31,663	909
4th year	3,59,286	2,45,549	6,04,834	1,657
5th year	4,51,673	3,45,439	7,97,112	2,184
Total	**10,03,561**	**7,30,048**	**17,33,610**	

Estimation of Rainwater Quantum to be Dewatered

Estimation of the rainfall falling on the mouth of the pits has been estimated based on dimensions and monsoon rainfall. The expected evaporation from the mine pit has been reduced from the total runoff and same is provided in the table 4.25 below.

Table 4:25 Total Mine Water Due to Monsoon Rainfall

Mine Pit Rainfall				
Year	**Pit Area (Sq m)**	**Monsoon Rainfall (m)**	**Rainfall Runoff (Cub m)**	**Actual Monsoon Runoff (Cub m /Year) @ ET Loss 55%**
1st year	-	1.1208	-	-
2nd year	-	1.1208	-	-
3rd year	1,80,000	1.1208	2,01,744	90,785
4th year	4,50,000	1.1208	5,04,360	2,26,962
5th year	7,60,000	1.1208	8,51,808	3,83,314

Hence total mine seepage including the rainfall water is given in Table 4.23 below:

Table 4:26 Total Mine seepage including Mine Pit Rainfall

Mine Pit Rainfall				
Year	**Total Yearly seepage (cubic M)**	**Actual Monsoon Runoff (Cub m) @ ET Loss 60%**	**Total Cub m /Year**	**Total Cub m/Day**
1st year	0	-	0	0
2nd year	0	-	0	0
3rd year	3,31,663	1,21,046	4,52,710	1,240
4th year	6,04,834	3,02,616	9,07,450	2,486
5th year	7,97,112	5,11,085	13,08,197	3,584
Total	**17,33,610**	**4,23,662**	**21,57,272**	

4.6.6 IMPACT OF MINE DEWATERING/ ABSTRACTION AND MITIGATION MEASURES

A) IMPACT ON THE GROUND WATER REGIME

In order to assess the groundwater situation of the area, Resource Estimation jointly carried out by CGWB and State (2023) has been considered. The Chhendipada Blocks' annual extractable recharge of groundwater is 7,386.86 ham, and ground water extraction is 4,057.31 ham. The stage of groundwater extraction stands at 54.93% (Safe). This indicates that the groundwater extraction is within sustainable limits and ground water resources in the block are replenished through natural processes such as rainfall and infiltration, but the demand for industrial and domestic use continues to increase suggestive of better water management strategies.

Based on the drinking water withdrawal through 3 wells (260 LKD) Ground water modelling study has indicated that radius of influence will extend to 1000 m from the mine lease area after year 2.

B) IMPACT ON SURFACE WATER SOURCES

The mining of coal by the open cast method causes changes in topography. The change of ground relief in a mine area influences the local drainage. This may alter the drainage at the micro level. Care shall be taken during mining activity to avoid any serious damage to surface water bodies when mining start after 2 years.

C) IMPACT ON WATER QUALITY

The water quality of the mine lease area is good for all purposes as described in section "mine water management plan". **Since, mining is going to take place in the next 2 years, no water will be available as mine seepage**. Hence, there is no expected quality impact due to withdrawal of seepage water.

4.6.7 TREATMENT & DISPOSAL OF WATER FROM MINE

Water pumped out from the mine will be treated in settling ponds before utilization of the same to cater the water requirement of the mining and its allied activities, such as dust suppression, greenbelt development and Workshop and transferring the balance to nearby farmers for agricultural purpose.

Hence, there will be no pollution or contamination of ground water due to coal mining.

The Domestic wastewater will be treated in a STP.

During quarry operations, strata water seeps into the quarry area, which will be pumped out as mine discharge. A part of the mine discharge water will be utilized for meeting water requirements of the project like dust suppression, washing of HEMM, plantation, etc. The excess water from opencast mining activity, if discharged without appropriate treatment, will cause deterioration of quality of receiving water body mainly due to the presence of suspended solids in mine discharge water. Stormwater will cause soil erosion and siltation of watercourses if suitable measures are not taken.

Some of the proposed control measures to be adopted by the project proponent as mentioned in the EIA/EMP and PR reports for controlling water pollution in the proposed project are as follows:

- A suitable garland drain is proposed to arrest the rainwater into the mine. The sump also behaves as a primary settling tank for the mine water. Thereby, the suspended solids, a water will be sent after de-siltation for industrial and drinking purposes and the rest will be discharged onto the surface into the peripheral village ponds for agricultural use etc.
- Creation of water storage areas in the quarry for settlement of suspended solids before pumping water out of the mine.
- Provision of Effluent Treatment Plant with oil and grease traps in HEMM workshop for removal of TSS and Oil and Grease from vehicle wash effluent and its subsequent recycling. Treated water will be used for dust suppression and for raising plantation within the workshop premises.
- Similarly, water running through dumps is expected to get mixed with loose overburden and associated material. Garland drains, check dams and sedimentation tanks will be constructed around the OB dump and the lease area to reduce soil erosion and arrest suspended solids before discharging the run-off water into the natural water regime as well as to prevent stormwater from entering the lease area.
- Besides, the dump slopes shall be formed with suitable inclination (overall slope 28 degrees) so that the water falling over the dump will gravitate down without causing soil erosion. Regular cleaning of drains will be done for de-silting the same. Reclaimed / Backfilled area will be properly graded

and kept slightly sloping against mine workings so that water does not flow into the mine workings.

- The mine discharge, after passing through sedimentation tanks for removal of suspended solids will be used for dust suppression, greenbelt development, etc. and excess water will be let out into nearby streams.
- To avoid the sudden inrush or flow of surface water towards and into the quarry, suitable precautionary measures like protective bunds above the HFL (Highest Flood Level) will be constructed at the applicable places.

Apart from the above, the following suggested control **measures may be adopted** by the Project Proponent.

Wastewater will be generated from HEMM washing. A washing platform may be provided for washing HEMM and the wastewater may be collected in Effluent Treatment Plant for trapping oil and grease. The treated water may be re-used for dust suppression and cleaning purpose etc. This provision may also be made available for contractor vehicles. Domestic effluent from Mine Office, Canteen etc. may be treated in the Sewage Treatment Plant. Treated effluent may be used for horticulture and sewage sludge may be used as manure.

Drainage of excess mine water

Council of Scientific & Industrial Research in collaboration with Central Institute of Mining and Fuel Research (CSIR-CIMFR), Dhanbad supplied, installed and commissioned (November 2014) a pilot plant with capacity to produce 4000 litres potable water per hour from waste mine water, at BCCL. CIL intimated (January 2016) MCL that with a view to providing treated water to the villages for drinking purpose from the surrounding underground mines, a detailed study be conducted so as to standardise the mechanism for utilization of excess mine water and directed MCL to identify the areas where this technology could be implemented. GMDC will further explore the above to implement in this project.

4.6.8 WATER CONSERVATION MEASURES

The proper water conservation measures will be adopted by the project proponent which includes rainwater harvesting, groundwater recharging, reuse and recycle of water. The total quantity of surface runoff available from the rainwater will be stored in the reservoir created by left out pits. Accumulated Rainwater in the mine pit recharges the ground water naturally. Harvested rainwater in the mine pit will be utilized for various activities like greenbelt area development, dust suppression and other activities as and when required.

4.6.9 ASSESSMENT OF IMPACT AND CONTROL MEASURES ON HYDROLOGY & HYDROGEOLOGY

The adverse impacts are changes in ground water flow patterns, lowering of water table, changes in the hydrodynamic conditions of river/underground recharge

basins, reduction in volumes of subsurface discharge to water bodies / rivers, disruption & diversion of water courses / drainages, contamination of water bodies, affecting the yield of water from bore wells and dug wells, etc.

Knowledge of mining impact on ground water is requisite while deciding and designing to protect surface water bodies. Predicted mining impact on water is described below:

4.6.10 GROUNDWATER RESOURCES POTENTIAL

Rainfall is the major recharge source of groundwater. The area (Angul Distt.) experiences an average annual normal rainfall of 1258.5 mm (2019).

The groundwater recharge potential in the study area was estimated by Rainfall - infiltration method and Water Level Fluctuation method. As reported in GEC Report, 2015, for sandstone, rainfall infiltration factor of 12% and 7% for hard rock were considered for the study area. The specific yield considered in the study area for sandstone is 3% and 2% for hard rock. For calculating the domestic water consumption, as recommended by GEC, 2015, the consumption rate in the study area was considered as 70 lpcd. The groundwater balance computation for this study area of 496.02 sq. km has been done and is furnished below:

Table 4:27 Land Use Detail Baitarani West OCP Buffer Zone

Land use	**Area (km^2)**	**Percentage (%)**
Total Forest land	121.35	24.46
Scrubs	122.747	24.69
Total plantation area	2.03	0.41
Total agricultural land	160.28	32.31
Total waste land	74.22	14.96
Total mining area	0.11	0.02
Total settlement area	6.20	1.25
Total water bodv	9.29	1.87
Total area	496.02	100

4.6.11 STAGE OF GROUNDWATER EXTRACTION

In order to assess the groundwater situation of the area, Resource Estimation jointly carried out by CGWB and State (2023) has been considered. The Chhendipada Blocks' annual extractable recharge of groundwater is 7,386.86 ham, and ground water extraction is 4,057.31 ham. The stage of groundwater extraction stands at 54.93% (Safe). This indicates that the groundwater extraction is within sustainable limits and Ground water resources in the block are replenished through natural processes such as rainfall and infiltration, but the demand for industrial

and domestic use continues to increase suggestive of better water management strategies.

4.6.12 IMPACT ON AQUIFER SYSTEM

In general, the area represents two aquifer system up to the explored depth of 200m. Two types of aquifers are demarcated in the area, the top aquifer behaves as unconfined and the deeper aquifer behaves as semi confined to confined.

Aquifer I: An unconfined aquifer occurring across the entire area, except rocky outcrops, formed by weathered mantle atop crystalline and Gondwana formations and alluvial tracts along major river channels. It extends to a maximum depth of 70 m below ground level (bgl).

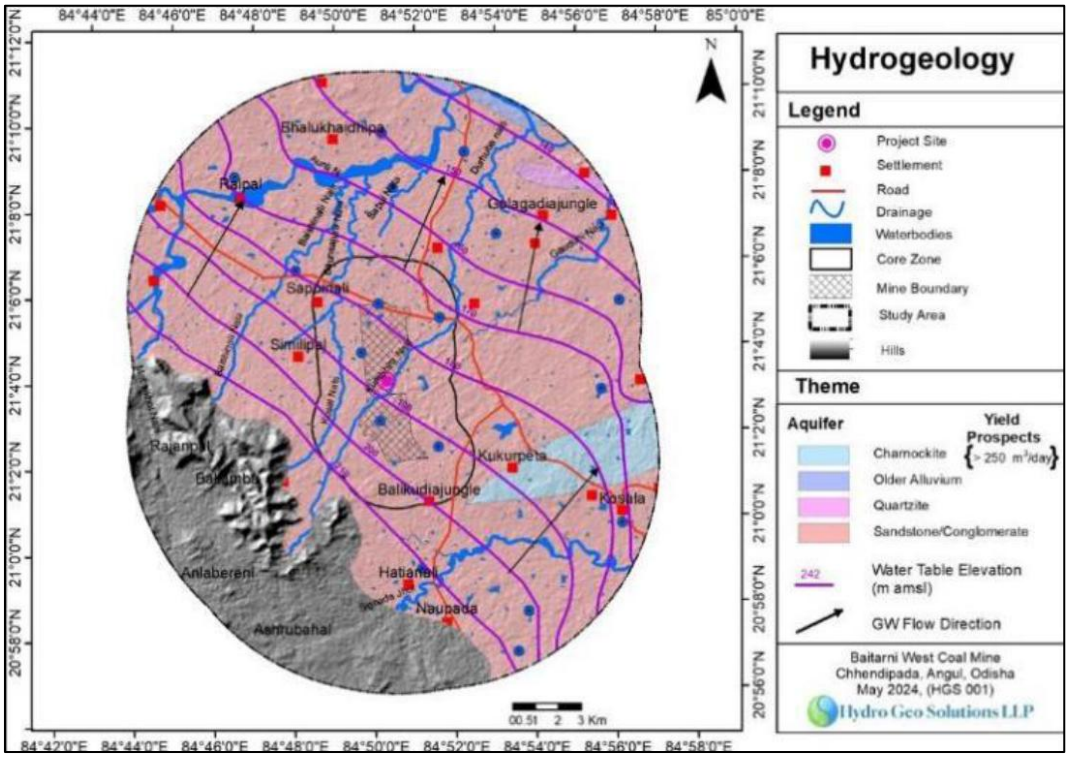


Figure 4-20 Hydrogeological Map of study area

Aquifer II: A semi-confined to confined aquifer found in fracture zones across all rock types. The yield of bore wells varies by rock type, with Granitic rocks offering better yields than Gondwanas, Charnockites, and Khondalites. Most fracture zones are within 0 to 150 m bgl.

The deeper aquifer i.e. Aquifer-II in Eastern Ghat formations viz. Charnockite, Khondalite and Gondwana formations viz. Sandstone and Shale have comparatively poorer yield prospects than the Granitic aquifers.

4.6.13 GROUND WATER FLOW AND AQUIFER INTERACTION WITH SURFACE WATER BODIES

Depth to water level

The depth to ground water level data during pre-monsoon (April) and post-monsoon November) periods of 2023 of Angul District around the study area, as available on INDIAWRIS web portal, have been used to prepare depth to water level and seasonal fluctuation maps.

DEPTH TO WATER LEVEL DURING PRE-MONSOON PERIOD (April) 2023

During April 2023, the depth to water level in and around the study area varied from 1.57 mbgl (Banarpal 2) to 11.06 mbgl (Ugi). A map showing the distribution of water levels during the premonsoon period is presented in Fig. 4.21. As evident from the map, major part of the study area has water level in the range of 5 to 7 m.bgl whereas deeper water levels in the range of 7 to 10 mbgl are observed in the northeastern periphery of the study area.

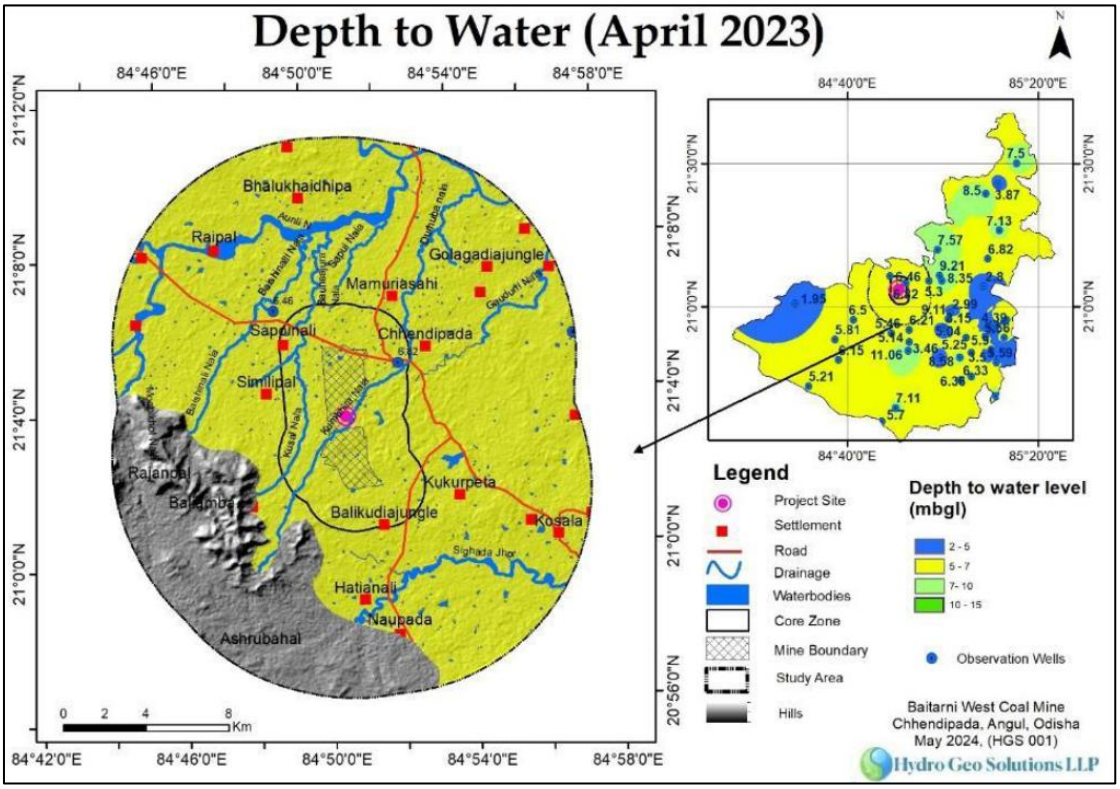


Figure 4-21 Depth To Water Level Map Pre-monsoon April 2023 (Source: CGWB)

DEPTH TO WATER LEVEL DURING POST-MONSOON PERIOD (NOV) 2023

During November 2023, the depth to water level in and around the study area varied from 0.56 mbgl (Banarpal 2) to 7.72 mbgl (Ugi). A map showing the distribution of water levels during the postmonsoon period is presented in Fig. 4.22. As evident from the map, major part of the study area has water level in the range of 2 to 3 m.bgl whereas shallower water levels in the range of 1 to 2 m.bgl are observed in the Northwestern and Northeastern parts of the study area and deeper water levels in the range of 3-4 mbgl are observed at the Northeastern Periphery and southeastern part of the study area.

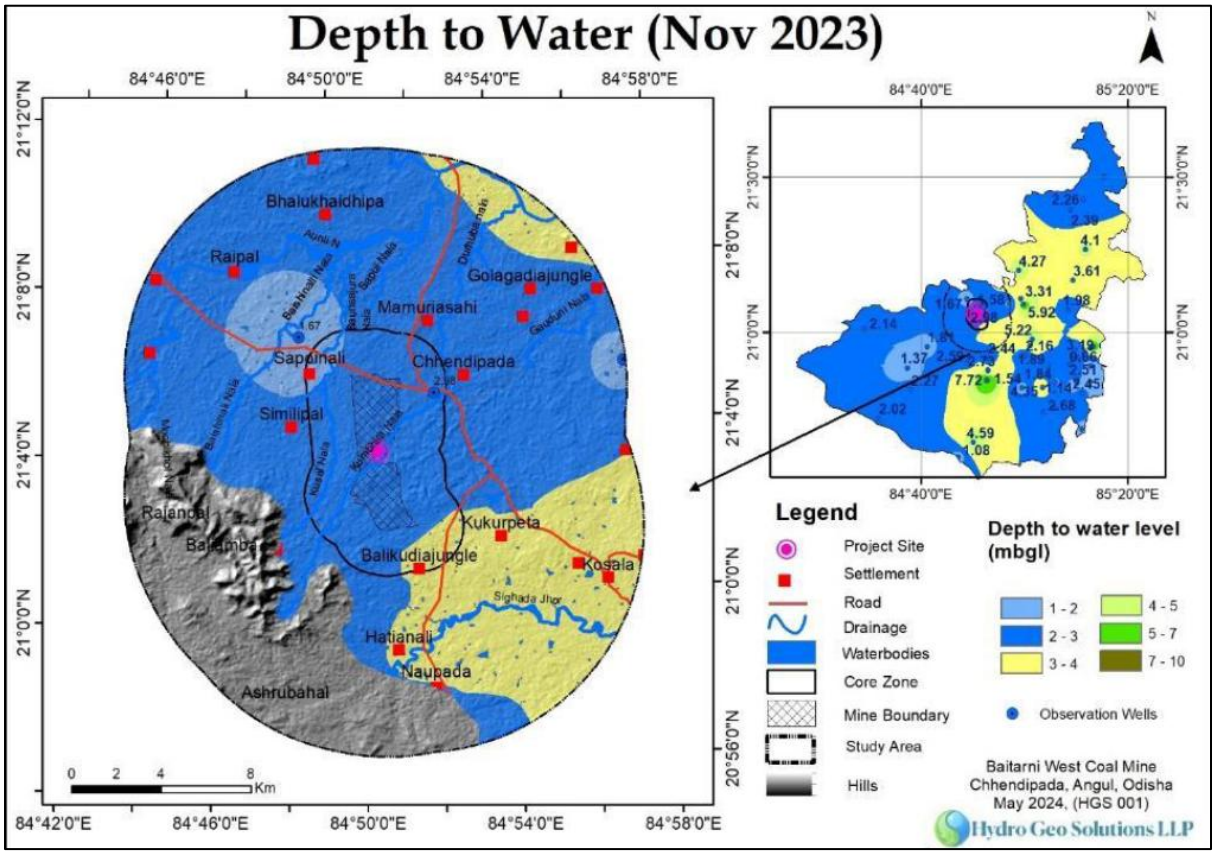


Figure 4-22 Depth To Water Level Map Post-monsoon Nov 2023 (Source: CGWB)

4.6.14 WATER HARVESTING & ARTIFICIAL RECHARGE

ARTIFICIAL RECHARGE

For water augmentation of the nearby region, artificial recharge of groundwater resource by rainwater harvesting will be considered. A retention pond (or sump or artificial pond) will be constructed within the excavation area to intercept the rainfall and surface runoff inside the lease area. A quantum of this water will be used for various mining activities of the project without drawing from ground water source. As well as that it will function as a small balancing reservoir to store rainwater during surplus period and discharge it in deficit period to the river. This will help in indirectly augmenting the water resources of the region. Further, an artificial pond has been proposed in the lease area outside of the quarry pit near

the western boundary. The details of the structure and quantitative recharge by the structure are given in the table below.

Table 4:28 Recharge Computation by Artificial Pond

S. N	Particulars		Details
1	Proposed Location		21° 2'34.02"N, 84°50'24.45"E
2	Artificial Pond Dimension		50 m X 60 m X 2m
3	No. of Structures		1 no
4	Designed Water Spread Area		3000 sqm
5	Average Water Spread Area		1800 sqm
6	No. of Water Available Days (Rainy Days) as per the daily Rainfall Data observed from 2003-04 to 2023-24 for Chhendipada Block RG Station	Monsoon Rainy Days	49 days
		Monsoon Non-Rainy Days	71 days
		Non-Monsoon Rainy Days	11 days
7	Recharge Factor		1.4 mm/day
8	Recharge	Monsoon	302.4 cum
		Non-Monsoon	27.72 cum
		Annual	330.12 cum

The location of the proposed artificial pond is shown in the plant layout map in Figure 4.15. It has been positioned near the western boundary for maximum interception of surface water. Considering the rainy days over region, the water available days for both monsoon and non-monsoon has been considered. And during monsoon non-rainy days has also been considered as recharge will be from the storage of the pond. It has been estimated that the pond will recharge 330.12 cum of water annually into the ground water aquifer.

4.6.15 CONSERVATION MEASURES

The mine water stored in its sumps is utilized to meet the mine's dust suppression, firefighting, washing of HEMM and other industrial water needs.

After the cessation of mining and subsequent backfilling of mine voids, rainfall will help in abundant groundwater recharge, the water levels will recoup and attain normalcy. Thus, the impact of mining on groundwater system may be considered

as a temporary phenomenon. The old mine works also act as water pools and improve the resource availability in the area.
Utilization of treated mine water for irrigation use can also enhance the ground water recharge potential through artificial recharge in the area. Thereby, in post-mining condition, the recharge and source potential in core zone will be much higher than the existing.

Recycling of wastewater at some sources after appropriate treatment to achieve "zero discharge" to the extent possible.

4.6.16 FUTURE STRATEGIES

To assess the impact on local water levels, monitoring of water levels in dugwells and through piezometers will be done at regular frequency.

Monitoring of water quality of surface water and domestic water (dug well/hand pumps) will be continued under routine monitoring.

On analyzing the field data if any impact is noticed, suitable controls measures will be adopted by the project authorities.

Implementation of any other suggestive methods to conserve and augment the groundwater recharge by the regulatory agencies.

Vegetative cover to be enhanced by plantation in the mine areas. This will contain surface runoff and increase the ground water recharge.

4.7 IMPACTS OF NOISE/VIBRATIONS AND MITIGATION MEASURES

4.7.1 PREDICTION OF IMPACT DUE TO THE PROPOSED ACTIVITY

The sound pressure level generated by noise source decreases with increasing distance from the source due to wave divergence. An additional decrease in sound pressure level with distance from the source is expected, due to atmospheric effect or its interaction with objects in the path of transmission. For hemispherical sound wave propagation through homogeneous loss free medium, one can estimate noise levels at various locations, due to different sources using model based on first principle, as per the following equation:

$$Lp2 = Lp1 - 20 \text{ Log } (r2/r1)$$

Where Lp1 and Lp2 are sound pressure levels at points located at distance r1 and r2 from the source Attenuation due to environmental conditions have not been considered. The combined effect of all the sources then can be determined at various locations by logarithmic addition. In first approximation one can assume that for all general population in the villages, every noise source in the mine is a point source. The average equivalent sound power level of such a point source can

be estimated for different distances and directions from hypothetical source by applying following equation:

Lp = Lw - 20 Log r

Where Lw is the sound power level of the source, Lp is sound pressure level at a distance r. A combined noise level Lp (total) of all the sources at a particular place is given by:

Lp (total) = 10 Log (10 Lp1/10) + 10 (Lp2/10) + ---------)

Major noise generating sources were identified from the proposed activity for prediction purposes. The major noise generating sources are Drilling & Blasting, operation of HEMM, vehicular movement & DG sets. The predictions have been made to represent the worst case. The noise levels at various distances were calculated using wave divergence model.

Noise levels due to different sources in the mine is given in table below: -

S. No.	Particulars	Noise levels dB(A)
1.	Drilling & Blasting	112
2.	Operation of HEMM	102
3.	Vehicular Movement	88
4.	DG Sets	82

The resultant Noise levels in addition to noise level monitored due to the project is given in below table:-

Table 4:29 Resultant Noise levels

Station location	Distance from Source (km)	Baseline dB(A)		Predicted Level dB(A)	Resultant Noise (Lp) dB(A)
		Day	Night		
Project Site (NQ1)	Site	43.1	38.5	56.4	56.6
NQ2	0.17	50.9	40.8	42.9	51.5
NQ3	0.6	44.8	39.7	58.4	58.6
NQ4	0.7	43.6	38.8	58.2	58.3
NQ5	0.8	52.9	43.6	74.5	74.5
NQ6	1.25	50.7	39.4	41.1	51.2
NQ7	1.4	43.3	38.7	42.1	45.8
NQ8	2.4	52.7	42.3	42.1	53.1

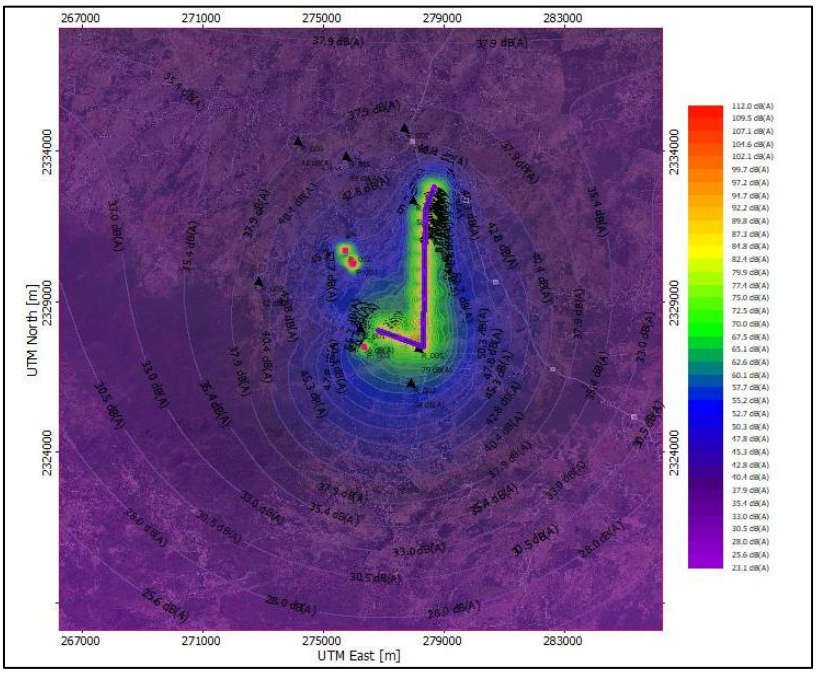


Figure 4-23 Contour of Resultant Noise in dBA

Acoustic enclosures and silencers will be provided to DG sets. All machinery has been manufactured keeping in view of the MOEF&CC/OSHA standards on Noise levels. The Ambient Noise levels will be kept within the standards prescribed by MOEF&CC, GOI vide Notification dated 14-02-2000 under the Noise pollution (regulation & control) Rules, 2000 i.e., ambient noise levels will be kept less than 75 dBA during daytime & less than 70 dBA during nighttime.

4.7.2 MITIGATION MEASURES

Control of Mine Machinery and Units

When attempting to attenuate noise, it should be borne in mind that noise radiates from most machinery as both airborne and as structure-borne sound at the same time. Following's mitigation measures will be adopted:

- Usage of surface miner restricts the usage of drilling and blasting to a great extent.
- Provision of sound-insulated chambers for workers deployed on machines producing higher levels of noise like bulldozers, drills, etc.
- Selection of new low-noise equipment from the manufactures failing which use of additional retrofits if available.
- Providing silencers or enclosures for noise generating machines such as DG sets, compressors etc.
- Attenuation of Structure-Borne Sound: Preventing transmission of vibration from machines to the load-bearing structure can considerably reduce structure-borne sound:

I. Large heavy machines will be mounted on foundations completely separated from buildings or other structures

II. Placing other machines on a stable foundation and where possible using an elastic separation such as rubber blocks or steel springs.

III. Severely vibrating machines may require separate foundations and isolation joints between floor slabs to prevent propagation of structure-borne noise.
Use of Protective equipment- Administrative Control

As part of a hearing protection PP may consider:

- The need for hearing protectors.
- Defining hearing protector areas.
- Selection of hearing protectors
- Issuing of hearing protection to individuals.
- Cleaning, maintenance, and replacement of hearing protectors.
- Training and education of people wearing hearing protectors.
- Provision of job rotation in noisy areas that will reduce personal exposure.
- Planned plant maintenance specifically for noise control.
- Training on noise impact and consequences
- Including Noise aspects in work agreements with contractors.

4.8 IMPACT DUE TO GROUND VIBRATIONS, AIR BLAST & FLY ROCK

4.8.1 GROUND VIBRATION

Vibrations due to blasting may cause damage to the nearby structures if appropriate control measures are not adopted. Fly rock is another possible damage causing outcome of blasting. There are many factors which influence fly rocks. Site mixed slurry (SMS) has been proposed to be used for good fragmentation.

The following control measures have been envisaged to reduce ground vibration within statutory limits:

a) The peak particle velocity (PPV) of ground vibration will be kept below 10mm/s for 8- 25 hz frequency range through optimally controlled blasting techniques, after necessary field trials.
b) Drilling and charging pattern will be formulated, with less explosives charge etc., after field trials.
c) Use of suitable initiating sequence and millisecond detonators.
d) Reduction of amount of explosive charged per day optimally.
e) To contain fly rocks Blasting area will also be muffled, if necessary, to stop fly rocks propagation.
f) Blasting will not be carried out when strong winds are blowing towards habitation areas. Blasting will be done during midday time and never at night.
g) Surrounding villages within 1 km radius of blasting will be regularly inspected for any visual cracks on walls and feedback will be gathered to investigate the reasons for these and for reassessing the charge per delay from time to time
h) Vibration study will also be carried out at appropriate times to firm up most ideal and optimal blasting parameters.
i) Controlled blasting to avoid tension cracks which may endanger the stability of bench slopes in the mine.

Proper care and supervision during blasting by a competent experienced person will be done.

The following measures will be taken to keep the noise at the minimum possible level:

- Proper designing of plant & machinery by providing in-built mechanisms like silencers, mufflers and enclosures for noise generating parts and shock absorbing pads at the foundation of vibrating equipments.
- Silencers provided where necessary.
- Routine maintenance of equipments,
- Enclosures for crusher house, etc.
- Rational deployment of noise generating plant and machinery.
- Greenbelts are created around the quarry, infrastructure sites, service building area and avenue plantation along the haul roads to absorb Noise.
- HEMMs are installed/ kept with soundproof cabins.
- Chute linings in Coal Handling Plant.

4.9 IMPACTS DUE TO SOLID WASTE GENERATION & ITS MANAGEMENT

During the entire life of mine of 37 years (including construction period), around 1.866 million m^3 of topsoil and 713.534 million m^3 of Overburden (OB) will be generated. The entire quantities of waste generation will be utilized for backfilling the excavated areas and external dump. The reclaimed land after backfilling will be biologically reclaimed. The generated topsoil will be utilized for plantation activity on the backfilled area, external dump and embankment. The details of stage wise generation of overburden/topsoil waste are provided in **Chapter2.**

The management of overburden dumps involves the following:

- Stabilization of overburden dumps.
- Construction of retaining boulder walls.
- Construction of garland drains for drainage.
- Provision of good soil mixed with manure and subsequent watering for growth of grass for anchorage on slopes. Plantation mixed with indigenous and fast-growing plants species.
- Degraded areas will be reclaimed and rehabilitated in a phased manner with local plant species.
- Haulage roads will be planted with trees on either side.

4.9.1 IMPACT ON TOPSOIL

Topsoil is proposed to be removed separately and dumped outside the quarry in a manner so as not to lose its fertility. The topsoil would be spread over the reclaimed land, afterward.

Topsoil will be removed and dumped on the area shown on the surface plan. Topsoil will be directly spread over the reclaimed area, external dump and embankment as detailed in Chapter-2.

4.9.2 TOPSOIL GENERATION AND MANAGEMENT

Year wise generation of Topsoil is given in **Chapter 2** of the Report. Topsoil extraction from the lease area may also affect the soil fertility and productivity.

Total topsoil generated during the ensuing period will be used for proposed plantation purposes. Minimum storage time shall be attempted to avoid nutrient loss.

4.10 IMPACTS ON ECOLOGY & BIODIVERSITY AND ITS MITIGATION MEASURES

The ML area does not fall under any Eco-Sensitive Zone, National Park, Sanctuary, Biosphere Reserve, Tiger Reserve and Elephant Corridor. Letter from PCCF is attached as **Annexure XI.**

Details of the species present in the Study Area are given in Chapter 3 of the report.

The project area will be reclaimed by greenbelt development on waste dump & in backfilled area during the planned period and along approach road.

There may be some impact on the biological environment due to the air pollution during transportation of coal the end use plant. The dust deposition on the plants along the approach road may reduce the photosynthesis. This will be taken care by transportation of coal, covered truck with tarpaulin and regular water sprinkling on the approach road. Plantation along the approach road will further reduce the impact. There is no water discharge from the mine, hence impact due to discharge of water in surrounding area is ruled out.

Noise and vibrations due to the blasting operations cannot be avoided. However, the blasting operations is very limited as discussed and will be limited to day hours only and even before 15.00 hrs. so as to minimize disturbance to wildlife. All necessary precautions will be taken to avoid flying off objects outside ML due to blasting operation. A minimum width of 500 m of danger zone will be maintained between the blasting site and mine lease boundary as prescribed by DGMS for such operations.

Ecological Impact Assessment

Given below in Table 4.30.

Table 4:30 Ecological Impact Assessment

Ecological criteria	Identified Impact	Significance of Impact	Magnitude	Duration/Timing/Frequency	Reversibility	Mitigation	Cumulative impact
Zone of influence	Site Clearance	Site clearance require felling of tree cause ecological damage	Moderate	Short time	Irreversible	Open cast mining requires clearing of vegetation after requisite permission and compensatory forestry will be done to compensate the damage. Nala diversion will be done after biodiversity loss estimation and compensatory biodiversity conservation plan.	 Less cumulative impact

	Ecological impact due to fugitive emission	Fugitive dust emission will reduce stomal index of existing trees/herbs/ shrubs etc.	High	Longer time	Reversible	Regular water sprinkling is required to avoid dust emission. Anti-smog gun is highly recommended. The air-borne respirable dust surveys should be conducted monthly as per G.S.R. 978(E) dtd 1st October, 2018. The sample should be collected from the following area: • Within 30m out-bye of the first working face in the intake side of the district. • 30m out bye of the last working face in the return side of mine. • Working faces • Loading/ Unloading/Transfer Points • Bunkers/chutes	 The high impact reduces to low impact.

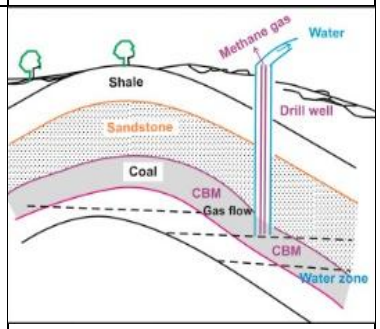
						And other places as per the requirement.	
	Noise & Vibration due to blasting	Wildlife movement will be hindered due to vibration & noise.	Moderate	During operation phase of mine	Reversible	Wet drilling, controlled blasting, use of approved explosive as per DGMS with delayed detonator is highly recommended. The Coal Mines Regulations 2017 should be complied for every activity of u/g mining. Stone dust barriers shall be placed in the full path of the explosion and the shelves shall be so arranged that their collapse in the event of an explosion is not impeded. G.S.R. 984(E) dtd 1st Oct 2018 should be followed for ammonium nitrate	 Moderate impact reduces to zero.

						handling. No smoking and no open flames should be permitted in or near the vans containing blasting agents	
	Poaching/ Hunting	Animal-human conflict will create nuisance and it allows illegal hunting of small animals like hares etc.	Moderate	During Operation phase	Irreversible	No worker/person allowed to hunt/kill the animal in any area. No hunting policy should be included in Integrated corporate policy and adequate training should be provided to deal with reptiles/avifauna and mammals in the area.	 Moderate impact should be reduced to zero by providing the training.
	High Noise due to vehicular traffic	Vehicular movement will cause hinderance of wildlife movement in the area.	Moderate	Longer time	Irreversible	The vehicle with valid PUC should be utilized, proper maintenance of trucks etc should be done frequently, speed should be	

						fixed 20-30 km/hr in forest area, NO U TRUN & NO HORN policy should be followed in forest area/approach road. Adequate training should be provided.	Moderate impact reduces to low.
Accessibility	Approach Road construction	Site clearance and tree felling will be involved in construction of new road, may have impact ecological balance.	High	Short time	Irreversible	Conditions for Haul Road G.S.R. 976(E) dtd 1st October, 2018 should be followed. All roads for trucks, tippers, dumpers or other mobile machinery shall be constructed to suit their load capacity and maintained in good condition. No vehicle other than HEMM shall be used on haul roads except between designated points and when permitted	 High Impact reduce to Low impact

						in writing by the manager.	
	Animal collisions due to transportation	Vehicular traffic may cause animal collision. Leopard movement was observed in the study area.	High	During Transportation	Irreversible	Animal Rescue Team (ART) followed by Quick Response Team (QRT) should be in place. Emergency numbers should be displayed along the approach road. Sign board with proper labeling should be displayed in mammal prone area.	 High Impact reduces to low.
Zone of influence	Noise due to workers during shift change	Huge number of workers are required to work, during shift change; noisy environment may affect the movement of reptiles, avifauna & other fauna.	Low	Short time	Reversible	Silence zone should maintain with low pitch during talk. No loud speaker or high noise celebration allowed within premises.	 No impact

	Transportation of explosive in Bulk	Risk of explosion during transportation, cause the habitat destruction	High	During Transportation of explosive – short time	Irreversible	G.S.R. 982(E) dtd. 1st Oct. 2018 should be followed for transportation of explosive in bulk. The quantity of explosives to be transported in bulk at one time to the site of blasting shall not exceed the actual quantity required for use in one round of shots, and also not before 30 minutes of the commencement of charging of holes. Only a vehicle duly approved by the Competent Authority shall only be used for transport of explosives in bulk. The vehicle shall be properly earthed	 High impact reduces to low impact

						with chain links while loading.	
	Transportation of Coal from pit bunker to cement plants	It leads the noise and dust emission which may cause detrimental effect on existing ecological conditions.	Moderate	During Operation phase	Irreversible	The place has no overhanging sides or prominent undercuts. The roof and sides are adequately supported and kept white-washed. The place is kept clean, is free from loose debris and is adequately fenced. The place is adequately ventilated.	 Low impact
	Emission of Methane gas	Rise in temperature may affect the crop as well as wildlife of the area.	Moderate	During Operation phase	Irreversible	G.S.R. 980(E) dtd 1st October, 2018 should be followed for ventilation. No auxiliary fan shall be installed at a point within or less than 4.5m from the nearer side of the entrance to the	 Methane gas is greenhouse gas which

						place to be ventilated by it. The percentage of CH4 in the return air in any heading ventilated by an auxiliary fan shall not exceed 0.5%. Gas detector should install and handheld gas detector should be provided to supervisors while working in the mine.	is responsible for temperature rise in the area. Moderate impact reduces to low impact by taking proper mitigation measures.
	Unused/fused/waste detonators	Hazardous Waste may affect the soil fertility and ecology.	Moderate	During Operation phase	Irreversible	The waste should be collected properly and handled as per Hazardous wate rule 2016.	 No cumulative impact

4.11 GREENBELT DEVELOPMENT

4.11.1 PLANTATION FOR GREEN COVER

Biological reclamation by way of broadcasting of grass seeds and plantation / afforestation shall be undertaken for stabilization of OB dumps against erosion and to put the land to best use. Tree plantation shall be taken up on external OB dumps and on back filled / internal dump areas including terraced slope, vacant land and avenue plantation as a remedial measure to mitigate air and noise pollution. Developing a heterogeneous mix of forest with local species having combined properties like medicinal, timber yielding and fruit bearing, so as to ensure perennial green cover and high survival rate.

At present a total area of 64.3723 ha is under forests. At the end of the life of the mine the equivalent area will be returned back as forests.

Apart from this the PP will start Miyawaki plantation which will further increase the density of trees per hectare.

Table 4:31 Biologically Reclaimed Area (in ha)

Year/Stage	**Biologically Reclaimed Area**				**Forest land (Return)**	**Un Disturbed/ To be left for Public/com Use**	**Total**
(Life of the mine plus post closure period)	**Plantation**	**Water Body**	**Public/ Company Use**	**Total**			
Up to Base year	0	0	0	0	0	0	0
Y-1	0	0	0	0	0	1071.3	1071.3
Y-3	8.09	0	0	8.09	0	876.2	884.29
Y-5	27.84	0	0	27.84	0	451.58	479.42
Y-10	64.46	0	0	64.46	0	339.54	404
Y-15	159.55	0	0	159.55	0	205.42	364.97
Y-20	192.52	0	0	192.52	0	110.62	303.14
Y-25	305.75	0	0	305.75	0	50.417	356.17
Y-30	328.65	0	0	328.65	0	31.707	360.36
Y-35	436.38	0	0	436.38	0	31.707	468.09
Y-37	574.35	0	0	574.35	0	31.707	606.06
Y-40	1029.583	38.488	31.85	1099.9	64.3723	31.707	1196

The Green Belt shall:

- ✓ Compensate the loss of vegetation due to setting up of the mine

- ✓ Prevent the spread of fugitive dust generated due to mining and allied activities
- ✓ Attenuate noise generated by the mine
- ✓ Reduce soil erosion
- ✓ Increase green covers and improve aesthetics

4.12 MINE CLOSURE PLAN

The mine closure plan is one of the most important requirements in the environmental management of mining projects. The closure operation is a continuous series of activities right from the commencement to decommissioning of the project. Therefore, the progressive mine closure plan is specifically included in the mining plan. The primary aim is to ensure that the following broad objectives along with the abandonment of the mine can be successfully achieved:

- ➢ Creation of a productive and sustainable after-use for the site, acceptable to mine owners, regulatory agencies, and most importantly to the community.
- ➢ Protection of public health and safety of the surrounding habitation.
- ➢ Minimization of environmental damage.
- ➢ Conservation of valuable attributes and aesthetics.
- ➢ Counterbalancing the adverse socioeconomic impacts.

4.13 IMPACT ON LAND USE AND LANDSCAPE

4.13.1 IMPACT ON LAND USE PATTERN

The major direct impacts on existing land use during the pre-mining phase are the removal of vegetation and resettlement of displaced population. There may also be land-use changes with respect to agriculture, fisheries, recreation sites, housing, forestry areas, etc. Land reclamation/restoration of mined-out lands may give rise to enhanced beneficial land use.

There exist major environmental impacts due to landscape disruption particularly visuals (unsightly huge dumps, voids, mine structures, subsidence, mine fires, etc.). During mining and post-mining phases drastic changes in landscape with landforms take place. The major associated impacts are soil-erosion, loss of topsoil, change in complete geology, creation of huge dumps & voids, disposal of wastes, deforestation, etc.

Irrespective of the type of mining used for extracting coal, mining invariably results in enormous land disturbance - e.g. large-scale excavation, removal of topsoil, dumping of solid wastes, cutting of roads, creation of derelict land, etc. Opencast mining has more potential impact on land than underground mining. With improved technology, opencast coal mining is being used extensively because of its cost effectiveness and productivity, though it results in large-scale land disturbance. The alteration in land use pattern due to infrastructure is not to be considered a true change as these facilities can be utilized for some other purposes after the mining operation is over. The alteration in land use pattern due

to activities of quarrying and external dumping of 08 materials may be considered as true change in land use pattern.

4.13.2 Control Measures to Restore Land Use & Landscape

Backfilling will be started from 14th year and there will be simultaneous backfilling up to mine life.

During the process, the geometrical shape of the dumps is altered to make them amenable to effective biological reclamation and also to provide safety and stability.

The face slopes of the dump will be maintained at the natural angle of repose of the material and at an overall slope angle of 26° to 28°. Suitable arrangements will be made for collection of stormwaters.

Appropriate garland drain is to be provided to collect run-off.

Backfilled area is to be reclaimed biologically and technically.

At the initial stage scrapped topsoil will be stored separately and reused as early as possible to layover the backfilled area for biological reclamation.

External OB dump will be re-handled during post-closure.

Arboriculture is to be carried out in vacant areas.

Proper afforestation/plantation is to be carried out for greenbelt development.

The details of post mining land use and reclamation & rehabilitation measures are discussed below.

4.13.3 POST MINING LAND USE

After completion of the mining operations, the area which are inaccessible and are dangerous (excavated, dumps etc.) will be fenced. In the post mining scenario, proper approach to the water body will be made for men and animals and the route will be thrown open for the public.

The Mining lease area allotted is 1196 ha, out of total mining lease area total excavation area proposed is 623.710 ha.

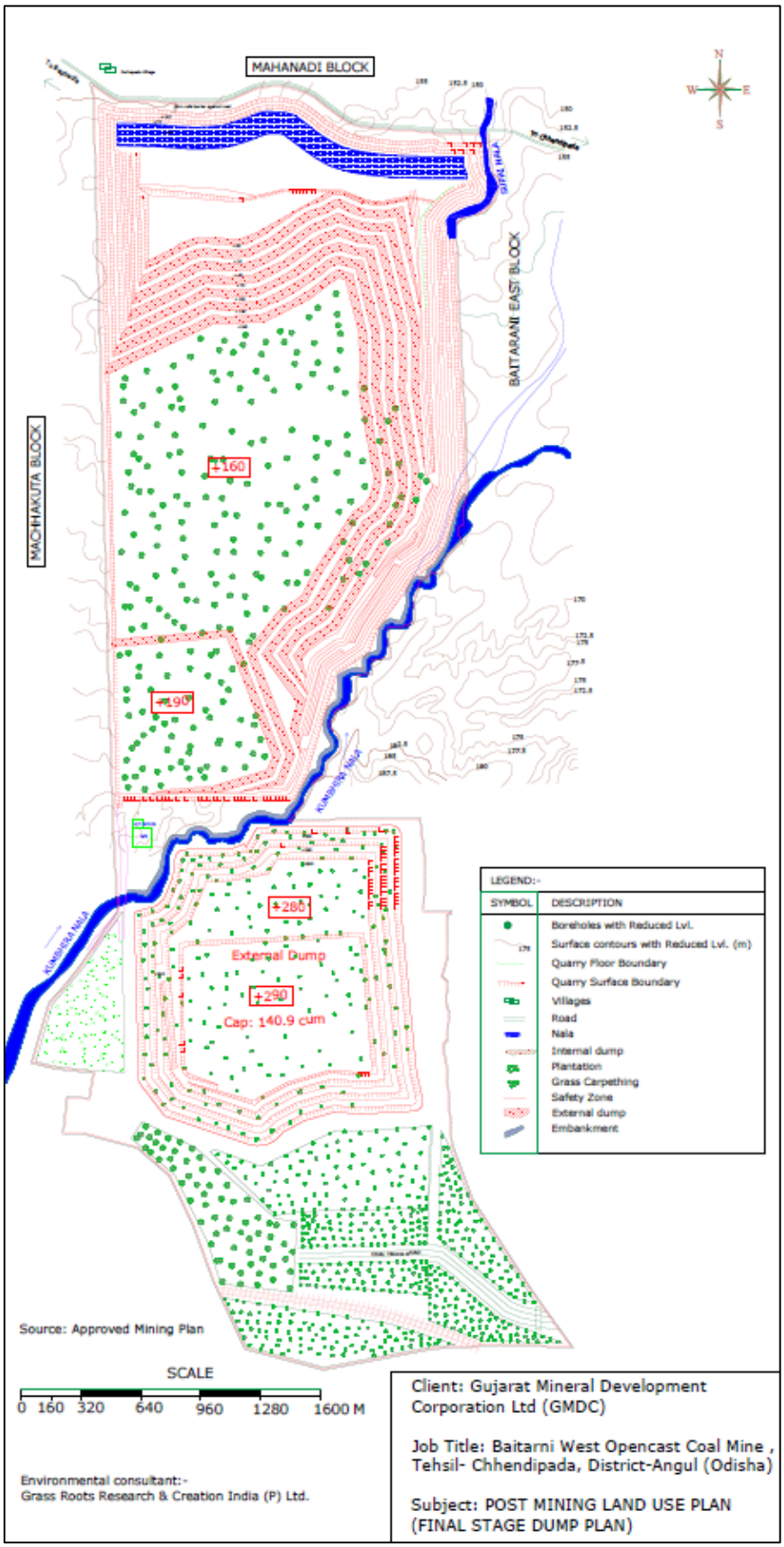


Figure 4-24 Post Mining Land use map & Final stage dump plan

Table 4:32 Post Mining Land-use (in ha)

Type	**Land Use (Post Closure)**					
	Plantation	**Water Body**	**Public/ Company Use**	**Forest Land (Returned)**	**Undisturbed**	**Total**
Backfilled Area	537.12	22.218	0	64.3723	0	623.71
Embankment	10	0	6.85	0	0	16.85
Green Belt	86.5	0	0	0	0	86.5
Undisturbed	0	0	0	0	31.707	31.707
Water Reservoir Near Pit	0	7.89	0	0	0	7.89
External Dump	215.84	0	0	0	0	215.84
Road And Infrastructure Area	161.783	0	25	0	0	186.783
Other water bodies including Garland Drains, etc	0	8.38	0	0	0	8.38
Safety Zone	16	0	0	0	0	16
Topsoil Dump	2.34	0	0	0	0	2.34
Total	**1029.583**	**38.488**	**31.85**	**64.3723**	**31.707**	**1196**

Source: Approved Mine Plan

4.13.4 LAND DEGRADATION AND TECHNICAL RECLAMATION

Table 4:33 Tentative Land Degradation and Technical Reclamation (Cumulative Area ha)

Year/Stage	Land Degraded				Technically Reclaimed Area			
(Life of the mine plus post closure period)	Excav	Dump (External)	Infra/others	Total	Back fill	Dump (External)	Others	Total
Up to Base year	0	0	0	0	0	0	0	0
Y-1	0	0	15	15	0	0	0	0
Y-3	18.02	18.79	87.86	124.67	0	9.4	17.57	26.97
Y-5	76.09	65.65	178.06	319.8	0	39.39	53.42	92.81
Y-10	203.84	215.84	324.75	744.42	0	149.43	65.42	214.85
Y-15	315.88	215.84	324.75	856.46	87.28	215.84	98.13	398.88
Y-20	450	215.84	324.75	990.58	144	215.84	123.83	481.3
Y-25	544.8	215.84	324.75	1085.4	274.2	215.84	123.83	611.5
Y-30	605	215.84	324.75	1145.6	320	215.84	123.83	657.3
Y-35	623.71	215.84	324.75	1164.3	390	215.84	123.83	727.3
Y-37	623.71	215.84	324.75	1164.3	428.5	215.84	123.83	765.8
Post Closure								
Y-40	623.71	215.84	324.75	1164.3	623.71	215.84	324.75	1164.3

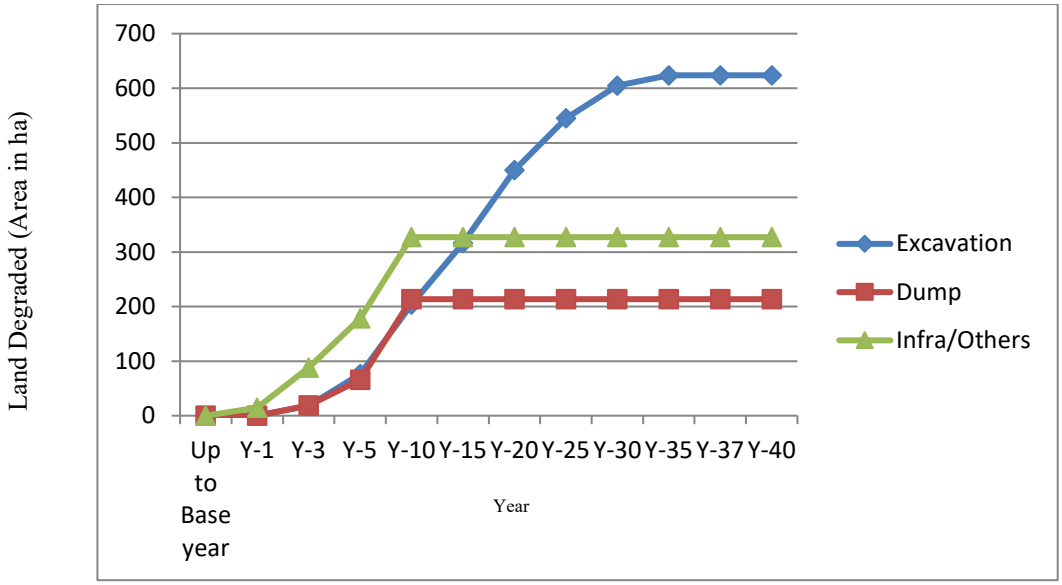


Figure 4-25 Land Degraded vs Year Graph

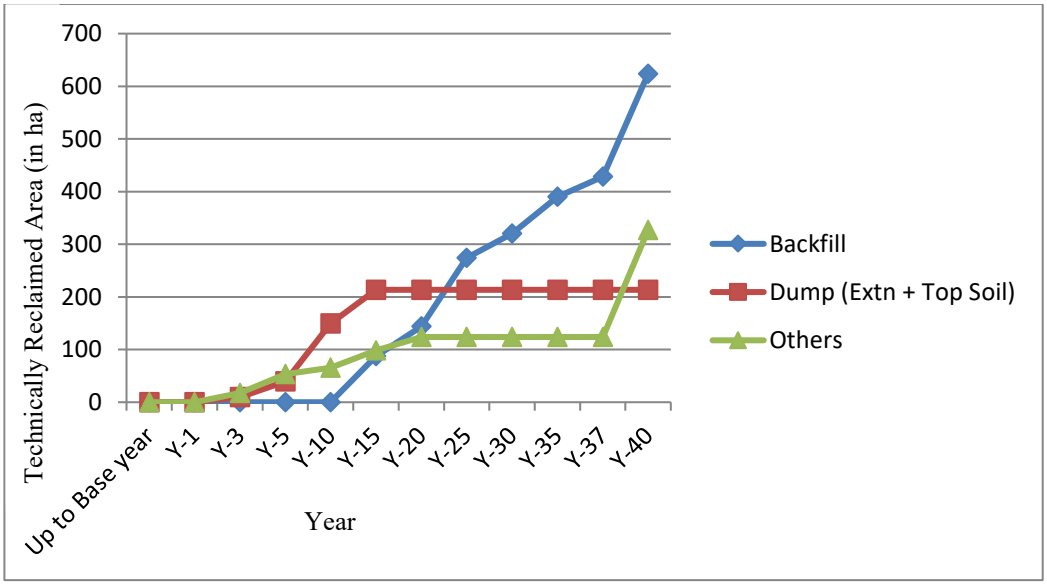


Figure 4-26 Technically Reclaimed Area vs Year Graph

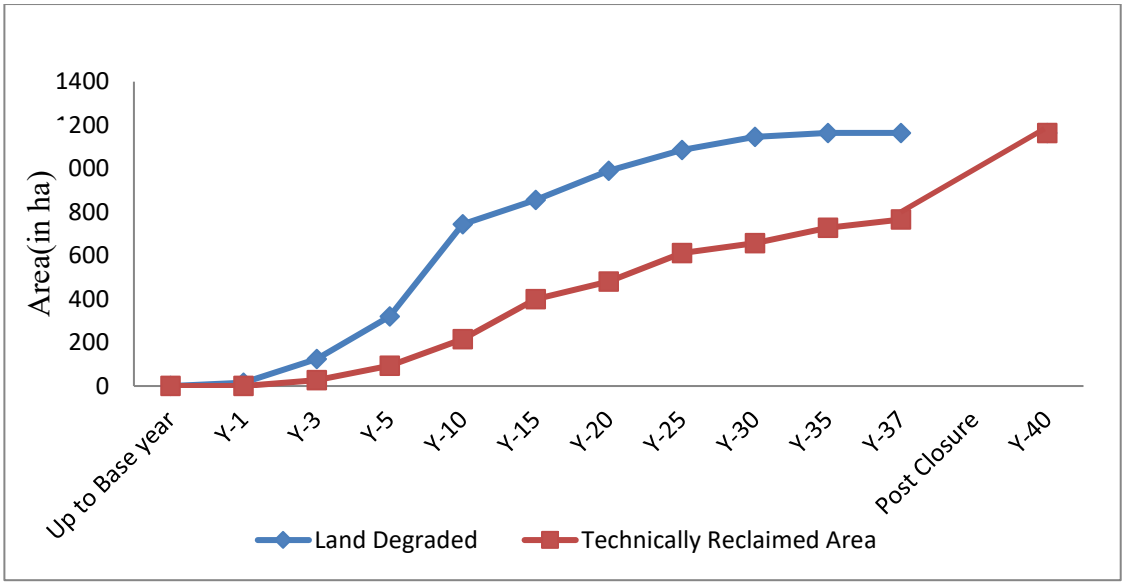


Figure 4-27 Year Wise Total Land Degraded vs Technically Reclaimed Graph

4.13.5 RECLAMATION & RESTORATION OF THE LEASE AREA

The entire infrastructure area proposed for the mine will be dismantled / demolished at the post closure plan. The waste material will be disposed in the low-lying areas. The resultant area, after clearing the waste from underneath the dismantled/ demolished facilities will be technically and biologically reclaimed to render it fit for use.

Restoration of Land used for Infrastructure

1. Buildings for short-term use should preferably be prefabricated structures that can be easily dismantled at the time of closure and reclamation. Such structures should be totally removed from the site and may be disposed of at an approved facility or reused elsewhere. Foundations comprised of concrete should be broken up, buried or removed in accordance with applicable waste management regulations.
2. The coal handling plant including belt conveyors and other structures will be dismantled and land will be levelled for reuse.
3. Fixed equipment and structures such as weigh scales may also have concrete pads or footings and these should be broken up, buried on site or removed in accordance with applicable waste management regulations.
4. All power lines, cables, towers and guy wires should be removed as well as any concrete footings or slabs where appropriate.
5. All roads, culverts and drains should be removed, and original drainage restored as much as possible in accordance with future land use planning.

Disposal of Machinery

1. HEMMs which will have balance life may be deployed in some other project of the company if possible, or otherwise will be disposed off. CHP, workshop at the end

of mine life will be dismantled/ disposed off. Every effort will be made to restore the area to economic utilization value as per the mine closure plan.

2. The transmission line and substation will be used by neighboring mines or local community. The transmission line and transformers will be dismantled and removed from the site if not required for purposeful use by the local community or nearby mines.

Safety & Security

1. Fencing around the abandoned quarry will be done as per details given in D.G.M.S Circulars.
2. After closure of the mine, the lighting arrangements will be kept maintained at all locations which are not required to be demolished or dismantled like sub-stations, transformers, community services, pump- houses, water-treatment/ filtration plants, waterlines, power lines, roads etc. to be utilized for the neighboring projects and at critical places for safety point of view.
3. For slope stability arrangement,the geometrical shape of the dumps is altered to make it amenable to effective biological reclamation and also to provide safety and stability. High wall will be maintained and stabilized as per norms.
4. The guidelines/instructions from DGMS will be followed in case of discontinuance of mine operation, if any.

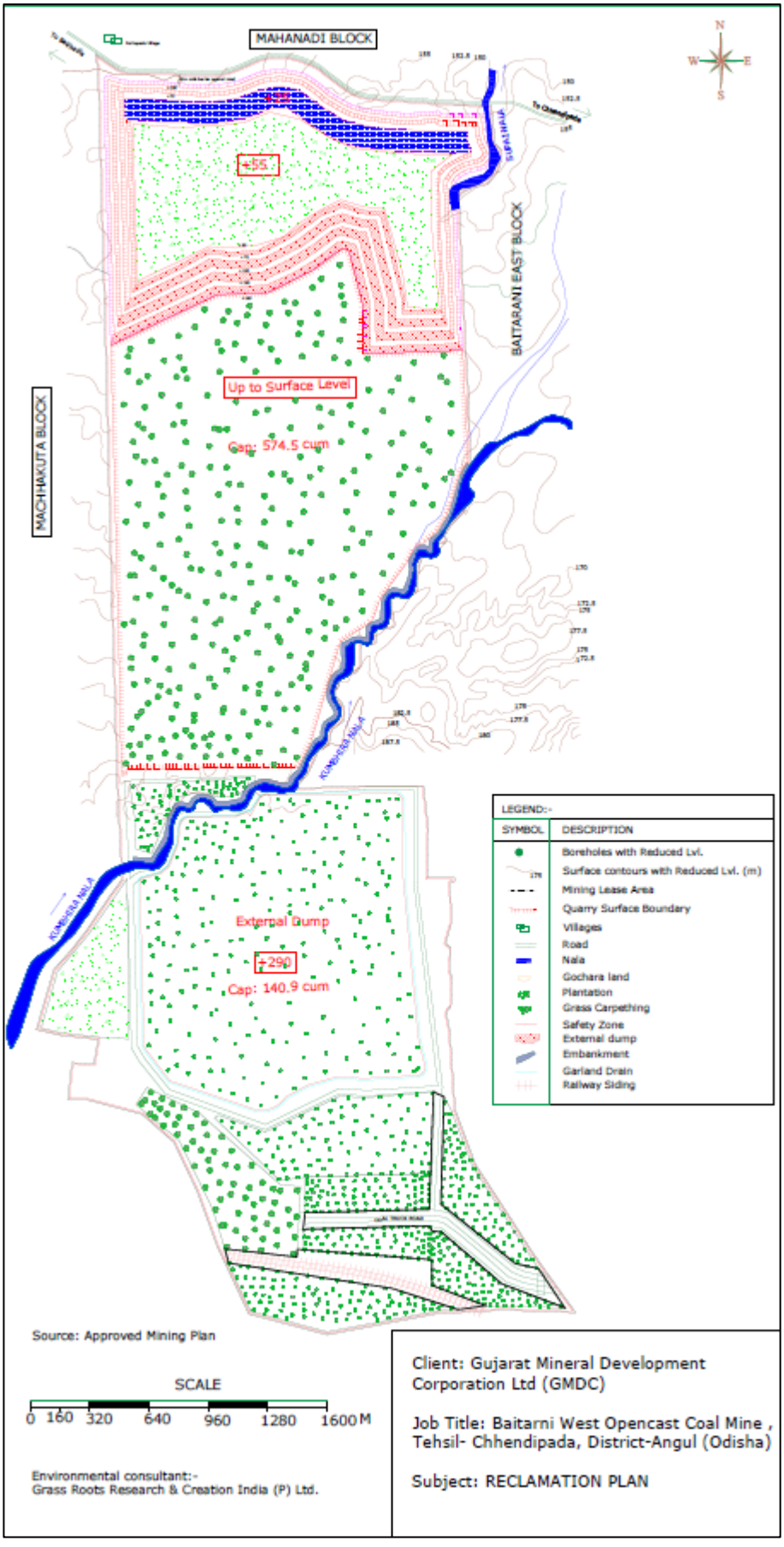


Figure 4-28 RECLAMATION PLAN (FINAL MINE CLOSURE PLAN)

4.14 CARBON FOOTPRINT STUDY

COAL PRODUCTION

Raw coal (coking and non-coking) production in India increased from 341.272 MT in 2002-03 to 777.31 MT in 2021-22 implying an annual growth rate of just over 8 percent. Most of the growth came from an increase in production of thermal coal. Historically this has meant the import of only coking coal as reserves were not adequate. However, as demand for power accelerated, coal was put under open general license (OGL) in 1993 that initiated thermal coal imports. Until the mid-2000s, volume of coking coal imports exceeded volume of thermal coal imports. This changed in 2005-06 when India imported 21.695 MT of thermal coal compared to 16.891 MT of coking coal. It was attributed to consumer (thermal power generators) preference for coal quality. Thermal coal imports accelerated with construction of imported coal-based coastal power plants. Between 2002-03 and 2019-20 (pre-pandemic year), coking coal imports increased from about 12.947 MT to 51.833 MT, whilst thermal coal imports grew from just 10.313 MT to over 196.704 MT in the same period.

Indian coal is characterized having high ash content and low heat value in nature compared to coals of US or Australian origin. More than 90 percent of coal mined in India is produced from opencast mines contributing to inert contamination. Run of mine (ROM) coal extracted from opencast mines in India typically has ash contents in the range of 35 percent to as high as 50%, low and reducing calorific value (2500- 4500 kcal/kg). Presence of high ash content is reported to lead to faster wear and tear of power plant components, difficulty in pulverisation, poor emissivity and flame temperature, low radiative transfer, generation of excessive amounts of fly-ash containing large amounts of un-burnt carbon, etc. Further, transportation of high-ash coal across long distances leads to increased cost because large quantities of non- combustible inert materials are also transported. It also leads to increased freight service demand causing excess pressure on the rail transportation. Transportation of inert materials also lead to additional consumption of energy due to rail and road transportation contributing to emission of carbon-dioxide (CO_2) and other green- house gases (GHG) from the mode of transport (rail and road) (Prasad, 2019).

Coal Mining Emissions

The opencast mine will use Electric driven mining equipment like draglines, rope shovels and RBH drills along with diesel driven mining equipment like rear dumpers, loaders, dozers and other support equipment which will run on diesel. In addition to above, power will be required for running the mine infrastructure like mine water pumping, lighting etc.

The mine end emissions are calculated on three broad categories.

- Carbon dioxide emissions from coal mines
- Methane emissions from coal mines
- CO_2 emissions from electricity

A coal production programme has been developed in line with the coal requirement of power plants. Operation of the Heavy Earth Moving Machinery has been planned according to the mine operations. The Mine Plan has been approved by the Ministry of Coal, Government of India

A) Carbon dioxide emissions from coal mine:

Four factors to estimate the extraction based (CO_2) emissions from coal mining:

- Production
- Coal type conversion factor
- Effective CO_2 emissions factor
- Exclusion factor

The CO_2 emissions for a coal mine can be calculated with the following formula:
Annual CO_2 (in million tonnes) = Yearly production * (1- Yearly exclusion factor) * Coal type conversion factor * Effective CO_2 emissions factor

- Production: 15.0 million tonnes per annum
- Coal type conversion factor: 25.8 TJ/kt*
- Effective CO_2 emissions factor: 94600 CO2/TJ*
- Conversion factor: 10^6 (tonne of CO_2/tonne of coal) *
- Exclusion factor: 0.017

*As per IPCC guidelines, 2006 for National Greenhouse Gas Inventories

The annual CO_2 for the mine will be = 15,000,000 * (1-0.017) * (25.8 *94600)/10^6

= 3,59,87,826.6 tCO_2

B) Methane emissions from Coal mine:

Coal mine methane refers to methane (CH_4) released from the coal and surrounding rock strata due to mining activities. Three factors to estimate the CH_4 emissions from coal mines:

- Production
- Gas content at mining depth
- Emission factor coefficient

The CH_4 emissions for a coal mine can be calculated with the following formula:
Annual CH_4 (in million cubic meters) = Production * Gas content at mining depth * Emission factor coefficient

- Production: 15 million tonnes per annum
- Gas content at mining depth: 2.0 m^3/tonne (Up to mine depth of 245m)
- Emission factor coefficient: 1.5

The annual CH_4 for the mine will be = 15 * 2.0 * 1.5

= 45 million cubic meters per year

= 4,50,00,000 t/year

Methane emissions estimated in million cubic meters can be converted into CO_2equivalent (CO_2e) based on the global warming potential of methane.

The Intergovernmental Panel on Climate Change (IPCC)'s Fifth Assessment Report (2013) provides the global warming potential (GWP) for methane averaged over 20- and 100-year timespans.

- GWP on a 20-year horizon: 84
- GWP on a 100-year horizon: 28

This means releasing 1 kg of methane is equivalent to releasing 84 kg of CO_2 averaged on a 20-year horizon and releasing 1 kg of methane is equivalent to releasing 28 kg of CO_2 on a 100-year horizon.

Therefore, CO_2 equivalent of methane = 28 * 4,50,00,000 t/year
= 126,00,00,000 tCO_2

C) CO_2 emissions from electricity:

To calculate the CO_2 emissions from electricity consumption in India, we have used the formula:

Annual CO_2 emissions = Electricity Consumption (kWh) * Carbon Intensity (kgCO_2/kWh)
= 8,875.2 kwh * 0.82 kgCO_2/kwh
= 7,277.66 kgCO_2
= 7.27 tCO_2e

Hence total annual Emission of GHG during operation phase of the project is summarized in the below given table:

S.No.	Activity	GHG t/year
1.	CO_2 emissions from coal mine	4,50,00,000
2.	CH_4 emissions from coal mine	126,00,00,000
3.	CO_2 emissions from electricity	7.27
Total		**130,50,00,007**

MITIGATION MEASURES:

Following mitigation measures will be adopted to bring down the emission of GHG:

i. 20% of the auxiliary load will be switched or substituted to Solar Power Grid
ii. If there is provision of open access in Gujarat, proponent will procure and purchase power from Indian Energy Exchange, so that renewable power can be procured and utilised through the Green-Term Ahead Market. Further it will reduce dependency on conventional power supply or burning of fossil fuels.

The Green-Term Ahead Market (G-TAM) is a new market segment for trading in renewable energy following the CERC approval. The new market segment features contracts such as Green-Intraday, Green-Day-ahead Contingency (DAC), Green-Daily and Green-Weekly. The matching mechanism is continuous/spot trading for Green-Intraday, Green-DAC and Green-Daily contracts whereas double sided open auction process to be implemented for Green-Weekly. The operations to be carried out in accordance with the Procedure for Scheduling Bilateral Transactions through Power Exchange issued by Power System Operation Corporation Ltd; CERC Power Market

Regulations, 2010; CERC Open Access in inter-State Transmission Regulations, 2008; CERC Indian Electricity Grid Code Regulations, 2010 as amended from time to time and the Byelaws, Rules and Business Rules of the Exchange.

Market Features:
A market mechanism to facilitate accomplishment of national renewable energy capacity addition and effective integration of green energy in the country.

Trading of 15-minute time block in G-Intraday and G-DAC; whereas trading of certain time of blocks in G-Daily and G-Weekly.

All contracts under G-TAM are at national level.
Continuous trading in G-Intraday, G-DAC and G-Daily; whereas double sided open auction bidding process for G-Weekly.

Exchange to manage risk management leveraging bank balance, requisite margin, including any additional margin as specified for the respective trading segment or the type of contracts There will be several products available for trading under G-TAM segment.

CARBON SEQUESTERING

Carbon is found in all living organisms and is the vital building block for life. Carbon is found in various forms, mostly as plant biomass and organic matter in soil. Trees capture CO_2 by fixing carbon during photosynthesis process and accumulating extra carbon as biomass. Plants grow through the natural process of photosynthesis, in which carbon dioxide is captured and stored in cells of plants.

One of the most burning issues in the modern era is the problem of change in climatic conditions and harmful role of greenhouse gases, which plays an important role in the changing temperatures at the international level. Trees act as a sink for CO2 by fixing carbon during photosynthesis and storing excess carbon as biomass.

The possible sinks include plants, soils, carbonate minerals, geological formation and ocean. It is well known that the Greenhouse effect and Global warming can be reduced by planting more trees that sequestrates more carbon.

Methodology

Biomass and CO_2 in standing woody biomass of selective five tree species from wood stock area was calculated in a study by VIT and this is used for the project.

Measurement of tree height and diameter at breast height (DBH):

To estimate biomass of different trees, non-destructive method was used. The biomass of tree was estimated on the basis of DBH and tree height. DBH can be determined by measuring tree Girth at Breast Height (GBH), approximately 1.3 meter above the ground. The GBH of trees having diameter greater than 10 cm were measured directly by measuring tape.

Above Ground Biomass (AGB) Of Tree:

The above ground biomass of tree includes the whole shoot, branches, leaves, flowers and fruits. It is calculated using the following formula:

AGB kg = Volume of tree ($m^{3)}$ X wood density kg/m^3

Volume of tree: $V = \pi r^2 H$

Where V = volume of the cylindrical shaped tree in m^3

r = Radius of the tree in meters

H = Height of the tree in meter

Radius (r) of the tree is calculated from GBH of tree. Wood density is used from Global density database.

The standard average density is 0.6 gm/cm^3.

Assuming radius of tree 0.20 m & height of tree 8 m.

Volume of tree = $3.14 * (0.20)^2 * 8$

= 10.04 m^3

GB kg = Volume of tree (m^3) X wood density kg/m^3

= 10.04 m^3 * 600 kg/m^3

= 6024 kg

Estimation Of Below Ground Biomass (BGB):

The below ground biomass (BGB) includes all biomass of live roots excluding fine roots. The BGB has been calculated by multiplying AGB X 0.26 factors as the root:

BGB is calculated by following formula:

BGB (kg/tree) = AGB (kg/tree) X 0.26

= 6024 * 0.26
=1566.24kg/tree

Estimation of Total Biomass

Total biomass is the sum of the above and below ground biomass.
TB = AGB + BGB (kg/tree).
= 6024 + 1566.24
= 7590.24 kg/tree

Estimation of Carbon

Generally, for any plant species 50% of its biomass is considered carbon. Hence Carbon=Biomass X 50 %
= 7590.24 * 50%
= 3795.12 kg/tree

Determination of the Weight of Carbon dioxide Sequestered in the Tree:

The weight of CO_2 is C + 2 X O = 43.99915

Hence the ratio of CO_2 to C is calculated as: 43.99915/12.001118 = 3.6663. Therefore, in order to determine the weight of carbon dioxide sequestered in the tree, the weight of carbon in the tree is multiplied by 3.6663.

CO_2 sequestered (kg/tree) = weight of carbon * 3.6663
= 3795.12 * 3.6663
= 13,914.04 kg/tree

Considering the average life of a tree as 50 years.

So, CO_2 sequestrated kg/tree per year = 13914.04/50
= 278.28 kg/tree per year

Greenbelt over 1029.583 ha area will be developed. Greenbelt will be developed considering the proposed plantation @ 2500 trees per hectare. Plantation of about 25,74,000 nos. of native species is envisaged over a period of 37 years of Mine Life in backfilled area, safety zone and infrastructure.

Based on the above, the Carbon sequestration in 5 types of trees will be as follows:

CO_2 sequestered kg/tree per year	**No. of trees**	**Project CO_2 sequestered (TPA)**
278.28	25,74,000	71,62,92,720

Summary:

The project will emit 130,50,00,007 t/yr. of GHG in the form of CO_2 eq. per year. Out of this, about 71,62,92,720 t/yr. of CO_2 will be sequestered by plants. Hence a balance of 130,50,00,007 – 71,62,92,720 = 58,87,07,287 t/yr. of GHG will be emitted to the atmosphere.

Total Greenhouse gas emissions from the project is calculated as 130,50,00,007 t/year with Carbon sequestration potential of 71,62,92,720 t/yr. of C02 from Plantation.

Following measures will further reduce the GHG:

For sequestering the GHG, PP will develop an area of 1029.583 ha out of 1196 ha as plantation including greenbelt. Trees will be planted at the rate of 2500 trees/ha.

Apart from the above, the following measures will be taken to further reduce the GHG:

- PP will explore possibilities of plantation in nearby village in consultation with local authorities.
- The possibility of Miyawaki plantation shall be explored to further reduce the GHG.
- Native plant species with carbon sequestration potential shall be planted to create carbon sinks.
- Out of 15 MTPA, 80% will be dispatched through railway.
- Only 3 MTPA will be dispatched through the road to meet the demand of local users.
- All possibilities will be explored for the transportation of 3 MTPA by CNG/Electric Vehicle.
- GMDC has already introduced 19 Electric Vehicles (LMV) at its Corporate Office & iCEM. The possibility usage of EVs shall be explored to the extent possible.
- Around 20% of the total power demand is proposed to be met through solar power.
- Transportation of Coal from pit to railway siding through conveyor.
- Fixed water sprinkler at coal stock yard
- HEMM will be efficient & the latest technology to consume less diesel.
- Solar pumps and energy efficient machines, etc. will be utilized wherever possible.

4.15 ASSESSMENT OF ANNUAL MINE CLOSURE COST

The Progressive Mine Closure Plan approved by the Ministry of Coal provides the tentative cost estimate done for the Final Mine Closure on approximate basis. The exact figures shall depend on the Wholesale Price Index during the time of mine closure.

Table 4:34 Abandonment Cost: Cost of Activities to be taken up for closure of the mine

Head	Activities	Unit	Quantity	Rate Rs/Unit	Amount Rs Cr
Progressive Closure	Water quality management	Ls	36	10160	3.71
	Air quality management	Ls	96	43639	42.47
	Barbed wire fencing around dump	m	14896	858.06	1.28
	Barbed wire fencing around the pit	m	10931	858.06	0.94
	Topsoil Management	Million m^3	1.866	55	10.26
	Technical And Biological Reclamation of Mined out land and OB Dump	ha	755.3	1057500	79.87
	Plantation over virgin area including green belt	ha	274.28	960000	26.33
	Manpower Cost and Supervision	No	7	130416.67	3.38
	Total wall around the dump	m	14896	8090	12.05
	Garland drain	m	10931	145	0.16
	Garland drains around the dump	m	14896	145	0.22
	Any other Activity	No	8	10052	0.82
	Any other Activity - 2	Nos	48	2375	1.16
Dismantaling of infrastructure & Disposal/ rehabilitation of mining Machinery	Dismantaling of workshop	Ls	150000	750	11.25
	Rehabilitation of the dismantled facilities	Ls	0	0	1.13
	Dismantaling of pump and pipes/ other facilities.	Ls	0	0	1
	Rearranging water pipeline to dump top park/Agriculture land	Ls	0	0	0.2
	Dismantaling of power lines.	m	3000	150	0.05
Safety and Security	Barbed wire fencing around dump	m	14896	858.06	0
	Barbed wire fencing around the pit	m	10931	858.06	0

	Concrete wall with Masonry pillars around the pit	m	10931	11060	12.09
	Concrete wall fencing around the water body	m	5000	11060	5.53
	Toe wall around the dump	m	14896	8090	0
	Garland Drain	m	10931	145	0
	Garland Drain around the dump	m	14896	145	0
Technical and Biological Reclamation of mined out land and OB Dump	Topsoil management	Million m^3	1.866	55	10.26
	Terracing, blanketing with soil and vegetation of External OB Dump	ha	0	0	20
	Peripheral road, gates, viewpoint, cemented steps on bank	m	10931	3100	3.41
	Expenditure on development of Agriculture land	Ls	0	0	5
	Landscaping and Plantation	Ls	256	500000	12.59
Post Closure management and supervision	Post mining water quality management	Ls	24	10160	0.07
	Post mining air quality management	Ls	72	43639	0.94
	Manpower Cost and supervision	Ls	10	4840000	14.52
	Manpower Cost and supervision	No	24	2375	0.02
Others	Entrepreneurship development (vocational/skill l development training for sustainable income of affected people)	No	618	0.01	6.18
	Golden Handshake/Retrenchment benefits to 100 employees of OC	No	50	0.48	14.25

	Onetime financial grant to societies/ institutions/ organizations which is dependent upon the project	Ls	0	0	13.46
	Continuation of other services like running of school etc.	No	1	250000	11.1
Total					**325.7**

Financial Assurance: Amount to be deposited in Escrow account as a security against the mine activities to be carried out for the closureof the mine.

Table 4:35 Amount to be deposited in Escrow account

WPI as on Apr-19	121.1
WPI as on base date Apr-21	131.7
Escalation rate of Closure cost	1.088
	OC
Base Cost "Rs. Crs/Ha	0.09
Closure Cost "Rs. Crs/Ha"	0.098
Project Area "Ha"	1196
Amount to be deposited into Escrow Account "Rs. in Crs	117.208
Amount already deposited into Escrow Account "Rs. in Crs	0
Net Amount to be deposited into Escrow Account "Rs. in Crs	117.208
Rate of compounding of Annual Closure Cost	5.00%
Balance Life of the project "in Yrs	37
Annual Closure Cost "Rs. in Crs"	3.168
Amount to be deposited into Escrow Account after compounding @ of 5% "Rs. in Crs"	321.957

The details of the Year Wise Annual Closure Cost (Rs Cr.) to be deposited into the Escrow Account in accordance with the Guidelines of Ministry of Coal and Approved Mine Closure Plan are provided in below mentioned table:

Table 4:36 Amount to be deposited in Escrow account (Year wise breakup)

Year	OC- (Rs Crore)
1	3.168

2	3.326
3	3.493
4	3.667
5	3.851
6	4.043
7	4.245
8	4.458
9	4.681
10	4.915
11	5.16
12	5.418
13	5.689
14	5.974
15	6.272
16	6.586
17	6.915
18	7.261
19	7.624
20	8.005
21	8.406
22	8.826
23	9.267
24	9.731
25	10.217
26	10.728
27	11.264
28	11.828
29	12.419
30	13.04
31	13.692
32	14.377
33	15.095
34	15.85
35	16.643
36	17.475
37	18.348
Total	321.957

Mining will be carried out in phased manner initiating afforestation/reclamation work in the mined-out area i.e. continuation of mining activities from one phase to other indicating the sequence of operations depending on the geo-mining conditions of the mine. Up to 80% of the total deposited amount including interest

accrued in the Escrow account may be released after every five years in line with the periodic examination of the Closure Plan. The amount released will be equal to expenditure incurred on the Progressive mine closure.

The balance amount at the end of the Final Mine Closure will be released on compliance of all provisions of Closure Plan duly signed by the Company to the effect that said closure of mine has complied all statutory rules, regulations, orders made by the Central or State Government, statutory organizations, court etc. and duly certified by the Coal Controller.

4.16 PUBLIC HEALTH IMPLICATIONS OF THE PROJECT

The company will take all preventive measures for the villagers in the local area for better health. The company will provide the following for betterment of health.

1	Weekly visit of Doctor with Para medical staff and medicines.
2	Providing Ambulance for patients for nearest hospital and as and when required.
3	Health awareness programme in village schools.
4	Supply of blankets to the villagers.
5	Safe drinking water supply to villagers through Boring, OHT, pipeline & water tanker.
6	Water treatment done by Bleaching powder & Alum which will be added to water before 12 hrs of supply.
7	Anti Malaria Treatment like Fogging, anti-larva spray & indoor residual spray done weekly, fortnightly & monthly.
8	Awareness programmes among school children for use of pipeline supplied water for drinking purpose instead of taking pond or nala/river water.
9	Water vat in the villages will be cleaned on regular interval.

4.17 SPECIFIC PROGRAMME FOR UPLIFTMENT OF SC/ST POPULATION OF THE STUDY AREA

As per Census 2011, the percentage of ST population to the total population is 10.04%. Similarly, the SC population to the total population is 20.28%. The peripheral/ CSR activities proposed will also focus on upliftment of SC/ST population.

The company will focus on the following activities

1) Education & communication
2) Health care
3) Livelihood
4) Sports & culture
5) Infrastructure & development

Peripheral Development work proposed is given in the following table: (Rs lacs)

Proposed budget allocation for five years for Baitarni West Block		
Sr. No.	**CSR Intervention Sector**	**Total Amount (Rs. in Lakhs/Annum)**
1	Healthcare and Sanitation	100
2	Quality and Inclusive Education/Skill Training	
3	Development of Rural Infrastructure	
4	Environment Sustainability and Water Conservation	
5	Development of Sports Ecosystem	
6	Conservation of Art and Culture	

4.18 PUBLIC CONSULTATION

Public Hearing was conducted on 24-09-2024 at 10.00 A.M. at Govt. Land in front of Jagannath Temple, Chhendipada Tahsil of Angul district in accordance with the Ministry of Environment, Forest & Climate Change, Govt. of India, EIA Notification No. S0-1533(E) dt.14.09.2006. Action Plan of Public Hearing discussed in Chapter 7.

4.19 MERCURY CONTENT

Mercury is one of the natural and harmful components of coal.

- Mercury levels shall be analyzed by PP regularly in all coal seam samples by its own laboratory and by third party at certain intervals.
- Mercury levels will be analyzed in blood samples of its employees to keep records periodically.

4.20 HAZARDOUS SUBSTANCE MANAGEMENT

Hazardous wastes in coal mines include used / spent oil and wastes / residue containing oil arising out of the process of industrial operation using mineral / synthetic oil as lubricant in hydraulic systems or other applications, chemical

sludge from wastewater treatment and oil and grease skimming residue resulting from the process of purification of air, water and wastewater.

Hazardous Wastes (Management, Handling and Trans boundary Movement) Rules, 2008, which was in vogue till March 2016, and Hazardous and other Wastes (Management and Trans boundary Movement) Rules, 2016 (Rules) which was applicable thereafter defined hazardous waste as any waste which could endanger health or environment. Their handling, generation, collection, storage, packaging and transportation required authorization from the SPCB under the Rules. The Rules also stipulated that these wastes could be stored only up to ninety days.

4.21 CORPORATE SOCIAL RESPONSIBILITY

Mining of coal has an adverse impact on the ecosystem and biodiversity in and around the areas where it is in operation. Mitigation measures have been suggested to reduce the impact and the same has been discussed in Table 4.34. Therefore, projects need to be designed on the principle of sustainable development with due consideration for environment, conservation, safety, quality and aspirations of the community around it. Expenditure on CSR is required to be incurred for activities relating to protection and safeguard of environment and for maintaining ecological balance.

Approx. Rs 1 Cr/Annum (2% of the net profit over the preceding three financial years of the organisation) is to be earmarked for activities under CSR. The amount is to be spent for community development under CSR activities.

4.22 WILDLIFE CONSERVATION PLAN

Angul District has a presence of wildlife. River Mahanadi forms the southern boundary of the district. Sizable population of wildlife is noticed on both sides of Mahanadi River across Satkoshia Gorge. Satkoshia Wildlife sanctuary is a very rich source of wildlife and is situated 58 kms from the site. The rich fauna including presence of Tigers and Elephants make it unique.

At the same time Angul District is also an Industrial Belt. A number of Industries have come up in District. Therefore, the presence of Wildlife along with Industries and Mining activities creates a conflicting situation. Hence effort should be made to minimizing the conflict to the extent it relates with the Mining activities.

The forests in this area are scattered and interspersed with human habitation. These forests are therefore subjected to heavy pressure for collection of fuel wood, small timber, building materials and for cattle grazing.

Although some of the forests are protected through VSS, the biotic pressure is not completely eliminated. In such areas the ground cover is completely removed as

fuel, which makes the soil devoid of humus content and the water retention capacity of the soil also reduces.

The NTFP collection is limited to Mohua Flower, Mohua Seeds and Kendu Leaves. In this area 14% of the population depends on collection of this product.

Kenduleaves collection is made through Forest Department in time tasted systematic manner. Whereas Mohua collection is done in an old fashion where the ground under the tree is cleaned by burning and the flowers are collected. Sometimes the fire is not properly extinguished which results in spreading the same to the nearby forest areas. No systematic or organized procedure has been developed for collection of mohua flowers. The spreading of fire to the forest area leads to destruction of Wildlife habitat and also kills several wild animals. Such forest fire also bums all the organic matter of the soil and makes the soil refractory type ultimately leading to heavy soil erosion and further destruction of forest.

Due to the operation of various mining activities the Wildlife present within the lease area is threatened. Besides the Wild-life habitat is reduced to the extent of forest area involved. The Wildlife Management Plan therefore aims at providing safe passage to the existing wild animals in the lease area to the nearby forest areas. lnspite of the project activities some small animals will continue to live in the project area. Certain measures are therefore required to be taken within the project area for the survival of these animals.

The Environmental Management Plan prescribes certain control measures for conservation of Flora & Fauna. These are:

- Enhancement of Forest area due to biological reclamation, arboriculture/afforestation, green belt development / avenue plantation.
- Waterbody creation in the mine pits will subsequently serve as a Game Pond.
- The principle of zero discharge will be followed.
- The mining area as well as the Impact Zone will be treated for habitat improvement so that more fodder and water will be available for the wild animals.

The proposed mitigation measures are therefore:

a) Interventions to be implemented by the project authorities inside the project area.

Noise control

► Lighting

The core area of the mining activity will be sufficiently illuminated. Such illumination distracts the animals from the core mining area, which would otherwise be dangerous to their lives.

The use of vehicular Headlights which disturb the animals is restricted and only dippers are allowed within the mining area.

These activities have been included in the mining process and hence no special funds are required.

4.23 SITE SPECIFIC WILDLIFE MANAGEMENT PLAN

The steps as discussed in the previous sections regarding waste management, air, water, soil, noise pollution control measures will also act as a part of WMP.

4.24 SITE SPECIFIC MANAGEMENT PLAN

There will be large scale activities within the mining area, and it will be sufficiently illuminated with bright light so that the wild animals will avoid the area.

It is also proposed to install Solar Electric fencing to prevent animals from entering the project area. The project proponent will take up this activity at his own cost.

In spite of the precautions taken in case of accidental fall of any wild animal in the mining pit, the workers will be educated to inform the local Forest authorities and act as per their advice.

Relevant provisions of the Environment Management Plan for the project as already described will help in the conservation plan.

- Control of Air Pollution
- Control of Water Pollution
- Control of Noise Pollution
- Land Management and Reclamation Plan
- Control measures to reduce impact on Biodiversity
- Traffic Movement Control Plan

Since the area is a part of Talcher coal field few mines are in operation and more mines will also start operation in future. Hence a regional WLCP shall be prepared at DFO level. PP will contribute proportionately as per the DFO letter for WLCP implementation.

4.25 IMPACT ON SOCIO-ECONOMIC PROFILE

The major adverse impact will be displacement and rehabilitation/ resettlement of affected people including change in culture, heritage & related features. The crime and illicit activities also prop-up due to sudden economic development of the area.

Socio Economic Impact of the Project and Safety Measures

The proposed project will be helpful in generation of employment opportunities which will help in raising the standard of life of the nearby residents. As regards to assessment of change in living standard, it has been largely linked with gross income of an area due to various economic activities.

The project proponent proposes to ensure their participation in social activities of the surrounding community through awareness and welfare programs and CSR

activities. The activities include upliftment of social, economic and health related aspects along with efforts to address the basic needs of local communities.

S. No	Component	Positive impact	Negative impact
A.	**Socio-economic condition**		
1.	**Economy/ Employment**	The development and operation of the mine will create direct employment of about 1752. Development of ancillary and accessory industries is foreseen. Transport and other service sectors would also develop manifold. Considering employment generation of this magnitude, the proposed mining project will have positive impact on the local economy.	People will migrate to this segment both from within and outside the district as well as the state due to the creation of new employment opportunities and non-availability of skilled manpower.
2.	**Education**	Improvement in the existing infrastructure of education around the project area.	Nil
3.	**Cultural Aspects**	There are no such places related to ecological sensitivity and scientific or geological importance, which can be adversely impacted by the proposed project.	Nil
4.	**Aesthetics**	Being open cast mine, there would be change in the Landscape. The landscape will be improved due to various measures, like tree plantations proposed to be undertaken by the project proponent. With the proposed development of green belt around the mining project, the aesthetics of the study area around the project site would improve.	Nil
B. Health, Water and Sanitation			
1.	**Human health**	Land, water and air environment are not affected as the project site is an open cast mine. The people in the vicinity are not affected by any major diseases	Nil

		caused by environmental pollution. There are no qualitative changes in the plants and animal products used as food. Hence, proposed mining activity will not have any adverse impact on the human health.	
2.	**Sanitation**	Adequate hygiene and sanitation will be maintained in mining area.	Nil
C.	**Impacts on agriculture**	Due to addition in the number of employees and some induced population growth, the demand for agricultural and medicinal herbs products will increase.	Loss of agriculture land
D.	**Animal Husbandry and Pastureland development**	Improvement in farm animals, their health facilities, vaccination and productivity of animal products is expected. Besides, better economic condition will facilitate improvement in quality of animal breed, upkeep of pastureland and quantity of fodder.	Nil

The project is likely to create positive impacts due to the creation of employment opportunities both direct and indirect. There are certain economic opportunities arising out of the need for goods and services in the project area. The economic opportunities include preference to nearby village people in employment in the project through the following avenues, subject to their suitability/eligibility:

i) Employment with contracting agencies
ii) Allotment of shops/kiosk,
iii) Award of petty contracts and small civil works such as removal and shifting of loose earth, construction of boundary wall, drains etc.
iv) Vehicle hiring, PCOs, Kiosk
v) Gardening and maintenance of greenery around the mines and township areas
vi) Construction works and removal of debris near mines and township
vii) Transportation of men, material and other goods and equipment
viii) Lifting & shifting of other material such as sorting of coal, stores material etc.
ix) Road maintenance, water management, grassland development etc.

Healthcare Facilities

GMDC will set up dispensaries/ hospitals for its own employees near the project site.

Every worker is to be periodically checked up once every five years keeping proper record of their health profile including X ray and laboratory tests by establishing and extending pathological facilities. The importance of such periodical medical examination is to detect and prevent occupational diseases like Pneumoconiosis and Tuberculosis. The mining activities expose workers to some injuries and health hazards. Incidence of occupational disease and injuries and health hazards will be recorded in those health centers hospitals.

Economic Impact

- Loss of agricultural land
- General improvement of economy of the area
- Increase in revenue of the state exchequer

Impact On Village Habitation

Total 7 villages namely Chhendipada Jangal (Part), Handigora (Part), Machhakuta (Part), Machhakuta Jangal (Part), Porapara (Part), Porapara Jangal (Full), Tentuloi-Korasahi (Part) are going to get affected by proposed project. Total Project Affected Families (PAF) is 2664 Nos and Project Displaced Families (PDF) is 1629 Nos.

These displaced families will be resettled and rehabilitated socially, culturally and economically as per latest R&R policy (2006 &subsequent amendments) of Govt. of Odisha)

4.26 IMPACT ON BIODIVERSITY

Impact assessment due to Open Cast Coal Mining

The Survey was carried out by the Technical Team of GRC India and discussed in detail in Chapter 3. The team visited and observed the 10 km range of Baitarni West Coal mine (15.0 MTPA). In the study, all types of flora from natural as well as artificial plantation and fauna have been surveyed. The impact on flora & fauna is not that serious. The buffer zone of the project is an Indian Elephant (Elephas maximus) movement area (migratory path). Indian Elephant is included in the Schedule I of the Wildlife (Protection) Act, 1972. There will not be much effect on the fauna of the area due to the mining activities. A comprehensive Regional Wildlife conservation plan (WLCP) is under preparation at DFO level. PP will deposit the proportionate amount as per the direction of Forest Department. Further Site-Specific Conservation Plan has been prepared & approved by PCCF & Chief Wildlife Warden, Odisha. Copy of SSWLCP is attached as **Annexure XVI.** A budgetary provision of 37.26 Cr (Including 5% of the total financial outlay of the SSWCP to be deposited in Wildlife Society of Odisha) is earmarked for 10 years.

The impact assessment of activities under Baitarni West Coal mine (15.0 MTPA), various environmental attributes was carried out.

Suggestive Measures:

The balance in the regional population will be maintained in the natural course, owing to existing undisturbed forest areas in the vicinity of the project. The increase in green cover due to implementation of various measures like biological reclamation of the backfilled area, arboriculture/ afforestation adopted by the mine establishment will be a bonus, though expected in the distant future.

The mine will be a "zero-discharge" one. No adverse impact on the downstream aquatic life of surface watercourses is expected.

The project is not likely to have an impact on the faunal species diversity within the terrestrial and aquatic habitats.

Impact on Floral Species:

The major adverse impacts due to pre-mining and mining phases are loss of habitat, biodiversity, flora & fauna, fisheries & other aquatic life, migration of wildlife, and overall disruption of the ecology of the area. Due to mining activity, the land use will undergo progressive changes involving quarrying, transportation, etc. The physical and/or biological reclaimed land would transform over time which would have a positive impact on the environment.

As per Simpson's biodiversity index, biodiversity in Core zone ranged between 0.00 - 0.76, which is lower than buffer zone forest 0.80 - 0.86. It shows that there are more species in the quadrat sampling area. The value calculated through this index reflects the good biodiversity in the study area as a whole. Therefore, due to mining activities, there would be direct as well as indirect impact on ecology and biodiversity. The Project Proponent therefore needs to provide adequate measures for minimizing the impact on biodiversity and also carry out extensive plantation and afforestation in an open area for creating and augmenting natural resources.

Terrestrial Flora:

Following anticipated impact are analysed for observed terrestrial flora

- Loss of forest resources, economically important plants
- Impact on natural flora and its regeneration
- Impact on flora due to air emissions, noise and vibration, illumination, vehicular movement, wastewater discharges, changes in land use pattern etc.

In the study area, the villagers cultivate Cashew trees and Sun flowers as a commercial crop. Mahua (Madhucalongifolia var. /atifo/ia (Roxb.) A. Chev.) tree is distributed uniformly where villagers collect Mahua flowers, dry it and sell the produce. The margins of agriculture lands also contain several trees like tamarind,

mango, Jack, Jamun, Polash etc. These will be uprooted because of mining activity. Arjun is nearly threatened (NT) species and and Asna is an endemic species of the area. Plants belonging to the Sal based deciduous forest are present. The buffer zone vegetation contains large patches of wild forest and agriculture land. Forest department has planted Bamboo, Acacia catechu, Acacia auriculiformis and Teak etc in the surrounding area.

Aquatic Flora:

The enhancement of forest area occurs due to measures like biological reclamation of the backfilled area, arboriculture/afforestation, compensatory afforestation, creation of greenbelt and avenue plantation.

Water body created by the final voids is beneficial to flora as the area is prone to water scarcity.

This project will have no detrimental impact on the diversity of floral species within terrestrial and aquatic habitats.

Impact on Fauna:

The buffer zone of the project is an Indian Elephant (Elephas maximus) movement area (migratory path). Indian Elephant is included in the Schedule I of the Wildlife (Protection) Act, 1972. There will not be much effect on the fauna of the area due to the mining activities. The project is not likely to have an impact on the faunal species diversity within the terrestrial and aquatic habitats.

Green Belt development

Creation of a greenbelt with local species will enhance the project area. The greenbelt shall be developed around the quarry, infrastructure sites, service building area besides avenue plantation on both sides of the roads.

The purpose of a green belt around the mining site is to capture the fugitive emissions, attenuate the noise generated and improve aesthetics. Green vegetation cover is beneficial in many ways in terms of conservation of biodiversity, retention of soil moisture, recharge of ground water and maintaining pleasant microclimate of the region. In addition, vegetation cover can also absorb pollutants from the environment and helps in effective pollution control.

Visual/Aesthetic Impacts & Control Measures

The following visual impacts are anticipated:

- Due to excavation of coal and OB as well as transportation, the area may have spots of eyesore, unless proper mitigatory measures are adopted.
- Water logging may occur due to changes in the drainage pattern of the area.
- Loss in aesthetic value of the surrounding environment.
- Physical and biological reclamation of mined-out land to the extent possible as discussed above to ameliorate the environment.

- Regular supervision to detect the spots of eyesore and to remove the same immediately.

Ecosystem Services and Carrying Capacity implication due to proposed mine:

A) Ecosystem Services Report:

1. Provisioning Services

Provisioning services refer to tangible products provided by ecosystems that directly benefit human communities.

- Food Production
    - Baseline: The mine lease (ML) area supports agricultural activities, including paddy cultivation and banana harvesting. Freshwater fish (e.g., Rohu, Catla, Mrigal) are sourced from local ponds and rivers (Aunli and Tikera). Domestic animals such as cows, buffaloes, goats, sheep, and chickens are raised for food and livelihoods.
    - Significance : High for local communities, particularly in rural Angul, where agriculture and fishing supplement diets and income.
    - Impact: Mining will clear 1196 Ha, eliminating paddy and banana cultivation within the ML area. Pond fishing may decline due to water pollution or diversion of the Supai Nala. Mitigation includes surface water diversion channels at the right boundary of the mine, retention ponds (or sump) construction within the excavation area, installation of silt fences and sediment traps strategically across the site, etc.
- Raw Materials
    - Baseline: Forests provide timber (e.g., Sal, Teak), fuelwood (e.g., Mahua, Acacia), and non-timber forest products (NTFPs) like fruits (Amla, Bel), leaves, and resins. Bamboo is abundant for construction and crafts.
    - Significance: Moderate; these materials support local construction, fuel needs, and small-scale economies.
    - Impact: Tree felling will reduce timber and NTFP availability. Compensatory afforestation may restore some supply post-mining, but short-term losses are anticipated.
- Freshwater
    - Baseline: The ML area is drained by the Supai Nala, with ponds used for rainwater harvesting and fishing. The broader study area relies on the Aunli and Tikera rivers.
    - Significance: Critical for drinking, irrigation, and aquatic ecosystems, especially in a monsoon-dependent region (Angul: ~1400 mm rainfall).
    - Impact: Nala diversion and pumping of water may strain local water availability. Pollution from runoff could degrade water quality, though mitigation (e.g., sedimentation ponds) is planned. Nalas are seasonal in nature. One Nala (Kumbhira) shall not be diverted).
- Medicinal Resources
    - Baseline: Ethno-botanical plants like Neem (Azadirachta indica), Amla (Emblica officinalis), and Harida (Terminalia chebula) are used in traditional medicine.
    - Significance: Moderate; supports local healthcare practices.

- Impact: Vegetation clearance will reduce access to medicinal plants within the ML area, though buffer zone forests may still provide some resources.
- Mitigation: Afforestation program with native species.

2. Regulating Services

Regulating services maintain environmental conditions and processes essential for ecosystem stability.

- Air Quality Regulation
  - Baseline: Forests (Moist Peninsular Low-Level Sal and Northern Dry Deciduous) absorb CO_2 and filter dust. Tree density in Kosala RF and Debinagar PF supports this function (Shannon-Weiner indices: 1.16 and 0.93, respectively).
  - Significance: High, given Angul's existing industrial pollution (e.g., from nearby mines).
  - Impact: Clearing 1196 Ha and fugitive dust from mining (blasting, hauling) will reduce air filtration capacity and increase PM10/PM2.5 levels. Mitigation (water sprinkling, anti-smog guns) may lower impacts, but cumulative effects with nearby mines could persist.
- Climate Regulation
  - Baseline: Forests sequester carbon (Sal forests: ~50–100 tC/ha), moderating local temperatures. Wetlands and nalas contribute to microclimate stability.
  - Significance: Moderate; critical in a warming climate (Angul summers reach ~40°C).
  - Impact: Deforestation and methane emissions from coal mining (CH_4 < 0.5% in return air, per mitigation) will increase greenhouse gas contributions. Reforestation post-mining could restore some capacity after 37 years.
- Water Regulation and Purification
  - Baseline: Forests and nalas regulate runoff and recharge groundwater. Ponds and rivers filter sediments and pollutants.
  - Significance: High; prevents flooding and sustains aquifers in a monsoon-heavy region.
  - Impact: Nala diversion and overburden dumps (e.g., 60 Mtpa waste at 4:1 stripping ratio) may disrupt hydrology, increasing erosion and sedimentation. Mitigation includes diversion channels and reclamation, but temporary degradation is likely.
- Soil Stability and Erosion Control
  - Baseline: Vegetation (trees, shrubs, grasses like Cynodon dactylon) stabilizes soil on flat and partly hilly terrain.
  - Significance: Moderate; prevents landslides and maintains agricultural productivity.
  - Impact: Excavation and overburden removal (e.g., 40 million m^3/year) will destabilize soil, increasing erosion until reclamation restores vegetation (70–80% targeted). Cumulative impacts with adjacent mines may amplify this.
- Pollination and Pest Control
  - Baseline: Insects (e.g., honey bees, butterflies like Common Mormon) pollinate crops and forests. Birds (e.g., Black Drango) and bats (e.g., Short-nosed Fruit Bat) control pests.
  - Significance: Moderate; supports agriculture and forest regeneration.

- Impact: Habitat loss and noise/vibration from blasting may displace pollinators and predators, reducing crop yields temporarily. Mitigation (wildlife conservation plans) aims to minimize this.

3. Cultural Services

Cultural services provide non-material benefits like recreation and spiritual value.

- Recreational and Aesthetic Value
  - Baseline: Forests, rivers, and ponds offer scenic beauty and fishing opportunities. The hot spring at Deulajhari (Athamallik) attracts visitors.
  - Significance: Moderate; supports local tourism and relaxation.
  - Impact: Mining will degrade aesthetics within 1196 Ha, though buffer zone features (e.g., Deulajhari) remain. Post-reclamation parks could restore some value.
- Spiritual and Cultural Significance
  - Baseline: Sacred trees (e.g., Ficus religiosa) and sites linked to Goddesses Maheswari, Hingula, and Taleswari hold cultural importance.
  - Significance: High for local identity and rituals (e.g., Maheswari Tantra in Athamallik).
  - Impact: Tree felling may disrupt sacred groves and compensatory planting aim to preserve cultural elements.
- Educational and Scientific Value
  - Baseline: Diverse flora (e.g., 62 tree species) and fauna (e.g., Indian Elephant, Schedule I species) offer research opportunities.
  - Significance: Moderate; supports ecological studies.
  - Impact: Biodiversity loss reduces research potential, though conservation plans may maintain some study value in the buffer zone.

4. Supporting Services

Supporting services underpin other ecosystem functions.

- Habitat Provision
  - Baseline: Forests (six reserved, two protected) and nalas support wildlife, including Schedule I species (e.g., Indian Elephant, Indian Cobra). Bird diversity (41 species) and aquatic fauna (15 fish species) thrive here.
  - Significance: High; sustains regional biodiversity.
  - Impact: Habitat loss from 1196 Ha clearance and nala diversion will displace species (e.g., elephant movement noted). Mitigation (wildlife corridors, no-hunting policies) aims to reduce fragmentation, but cumulative impacts with nearby mines are a concern.
- Nutrient Cycling
  - Baseline: Forests and plankton (e.g., Ulothrix, Daphnia) recycle nutrients, supporting soil fertility and aquatic ecosystems.
  - Significance: Moderate; sustains agriculture and fisheries.
  - Impact: Soil disturbance and water pollution will disrupt cycling, though reclamation and waste management (per Hazardous Waste Rules 2016) aim to restore it post-mining.
- Primary Production

- Baseline: Vegetation (Sal, Mahua) and phytoplankton (e.g., Chlorella) drive energy flow through photosynthesis.
- Significance: High; foundational to food webs.
- Impact: Vegetation loss reduces production within the ML area, though buffer zone forests maintain some capacity.

B) Carrying Capacity:

Factors affecting the carrying capacity due to the proposed Baitarani West Opencast Coal Mine can be ecological, socioeconomic, non-adherence to post environmental clearance conditions, infrastructure and resource etc.

- Ecological: 64.3723 ha of forest land shall be diverted due to the proposed mine. Approximately 11000 trees including forest and non-forest are existing at present. A major number of trees will be going to be felled.
- Water resources: Ground water pumping leads to ground water depletion due to continuous mine operation.
- Air and Noise Pollution: Increase emission due to mining may further deteriorate the baseline status. Noise due to blasting and machinery may affect the noise level in nearby villages.
- Land Degradation & Soil Erosion: Loss of topsoil and land degradation may remain as it is for a long time of 40 years.
- Social Impacts: Displacement & Livelihood Loss, Inadequate R&R (Rehabilitation & Resettlement) policies could worsen poverty
- Health Risks: Increased respiratory diseases (e.g., silicosis, asthma) due to coal dust. Water contamination leading to skin diseases and gastrointestinal issues
- Economic Impacts: Both positive and negative impacts will be there. Employment generation, infrastructure development, and revenue for the state. Long-term loss of agricultural productivity, tourism potential, and increased healthcare costs.
- Health care, Education support will have positive impacts.

Mitigation Measures:

Afforestation & Biodiversity Conservation and Compensatory plantations. Water Management- Sustainable extraction, rainwater harvesting, land reclamation and rehabilitation, dust suppression, electric vehicles in mining, and stricter emission norms, fair R&R, skill development, and health monitoring programs as detailed in EIA/EMP will reduce or keep the impact under control.

The Site-Specific Conservation Plan has been prepared & approved by PCCF & Chief Wildlife Warden, Odisha. Copy of the conservation plan along with budgetary provision for 10 years & timeline is attached as **Annexure XVI.** Total budget earmarked for SSWLCP is 37.26 Cr for 10 years. Summary of the intervention are mentioned in the below table:

S. No.	Interventions	Total Cost of intervention	Remarks
A. Cost Implementation by Forest Department			
1.	Habitat Improvement	113846800	For enrichment of habitat of elephants,

			Creation of water bodies and water harvesting structures and Construction of permanent check dams
2.	Anti-Depredation Activities including Rescue and Release	3250000	Rescue, release, and postmortem of wild animals, covering vehicle hire charges, manpower engagement, feeding costs, medical expenses, and other related costs.
3.	Protection and Surveillance	40000000	Early warning system, Anti Depredation equipment's, Construction of Elephant proof Trench (EPT) and other expenses
4.	Integrated Forest Fire Management	10500000	Forest fire awareness and IEC activities, Capacity building training, Purchase of fire Blowers
5.	Research, Training and Development	4800000	Manager for the Monitoring and Supervision of the IEC activities.
6.	Publicity, Awareness and Entry Point Activities	95750000	Publicity, Awareness, Entry-point and IEC activities, Purchase of hand-held rechargeable torches, Cattle immunisation Camps, Solar Lights or electric street lights, etc
7.	Infrastructure	22000000	T-Fencing of Open Wells, Construction of composite unit, etc
8.	Review, Monitoring and Supervision	5600000	Evaluation of SSWLMP plan, Purchase of

			Drones, Monitoring, Supervision, Learning, Documentation and other contingencies etc
Sub Total		**295746800**	
9.	Add 20% for cost escalation	59149360	
Total (A)		**354896160**	
B. Cost Implementation by User Agency			
10.	5% of the total project cost will be deposited by the User Agency into the account of the Wildlife Society of Odisha	17744808	
Total (B)		**17744808**	
Grand Total (A+B)		**372640968**	

4.27 Environmental Impact Matrix

The environment impact matrix incorporating all the environmental components and project activities (with in-built control measures is presented in Table 4.34. The impact score is calculated by multiplying the sum of all proposed activities against each environmental component by its corresponding Parameter importance value (PIV).

4.27.1 IMPACT MATRIX WITH QUANTITATIVE RATING

The total impact score is – 860, which falls under category below 1000. It indicates that the impacts are not very significant if the benefits are accounted for. But negative impacts are there. Hence mitigation measures are required side by side to carry out the project activities eco-friendly. Major affected environmental attributes are air quality, noise level, land use, trade & commerce, economic aspect and public utility which are due to the project activities. Thus, due to the matrix, we identify the environmental attributes likely to be affected by the project activities. The impacts may be beneficial or adverse. These will be analyzed in detail during assessment of the impacts.

Table 4:37 Impact on Present State Environment due to Mining with Mitigation Measures

S.N.	Environment	Parameter importance value	Site preparation	Quarrying including drilling and blasting	Loading & unloading	Transportation	Disposal of over burden / rejects	Civil works	Social measures	Water sprinkling	Improved mines management	Reclamation soil through conservation &	Impact score
1.	Air quality	100	-0.8	-1.0	-0.8	-1.0	-1.2	-0.2	--	0.8	0.8	1.2	-310
2.	Surface water resource	75	-0.2	-0.6	--	-0.4	-0.4	-0.4	--	0.4	--	0.4	-90
3.	Ground water resource	50	--	-2.0	--	--	-0.8	-0.4	--	--	0.2	0.4	-130
4.	Water quality	75	--	-0.6	--	--	-0.6	-0.8	--	--	0.2	--	-135
5.	Noise level	75	-0.2	-1.2	-0.8	-1.6	--	--	--	--	0.4	--	-255
6.	Land use & soil characteristics	50	-0.4	-0.8	--	-0.4	-0.6	-0.6	--	--	0.2	0.4	-100
7.	Health	50	--	-0.8	-0.4	--	--	0.4	0.6	0.4	0.2	0.4	20
8.	Public utility/Literacy/Social awareness	75	--	--	--	0.6	--	0.2	0.2	0.2	--	0.4	120
9.	Human settlement	75	-1.4	--	--	-0.4	--	--	0.4	--	0.2	0.4	-60
10.	Aesthetic beauty	50	-0.8	-0.8	--	--	-0.4	--	0.8	--	0.6	0.8	10
11.	Employment	75	0.2	0.6	0.4	0.4	--	0.6	0.2	--	--	0.2	210
12.	Wildlife	50	-0.8	-0.6	-0.2	-0.4	--	--	--	--	0.2	0.4	-70
13.	Forest /Vegetation	50	-1.4	-0.4	--	-0.6	-0.4	-0.4	--	--	0.4	0.6	-110
14.	Geological resource	50	--	-1.6	--	--	--	--	--	--	--	--	-80
15.	Economic aspect/ trade/commerce	100	--	0.2	--	0.6	--	0.4	--	--	0.2	0.4	180
Total												**-860**	

Summary of Ecosystem Services and Carrying Capacity Implications

- Pre-Mining Value: The 1196 Ha ML area provides significant provisioning (food, water), regulating (air, water), and supporting (habitat) services, with moderate cultural value. The broader 50,641.55 Ha study area amplifies these benefits, despite disturbance from existing mines.
- Mining Impacts: Over 37 years, mining will temporarily eliminate most services within 1196 Ha due to land clearance, nala diversion, and pollution (dust, noise, methane). Cumulative effects with adjacent Talcher Coalfield mines may strain regional carrying capacity (e.g., water: 50 million m^3/year demand vs. supply; air quality: increased PM levels).
- Mitigation Potential: Compensatory afforestation, water management, and wildlife plans aim to restore ~70–80% of services post-mining. Production (15 Mtpa) aligns with geological feasibility (e.g., 468.27 Mt over 37 years), but ecological limits (e.g., biodiversity loss, water stress) require strict adherence to mitigation measures.

Recommendations

1. Biodiversity Offset: Establish a conservation area in the buffer zone to offset the Rs 6.16 Cr biodiversity loss from nala diversion
2. Water Management: Monitor groundwater recharge and river flows to ensure mining does not exceed the water resource carrying capacity.
3. Reclamation Monitoring: Track revegetation success (e.g., Sal, Mahua) to restore regulating and supporting services by year 37.
4. Community Support: Compensate for lost food production and NTFPs with alternative livelihoods during the life of the mine (LOM).

Timebound action plan to comply activities under carrying capacity study, cumulative impact assessment study, ecosystem services study and biodiversity study

S. No.	Activity	Capital Cost (in Lakhs)	Year
1.	Surface water diversion of Supai Nala	Included in EMP	10th – 37th Year of Mining Operation
2.	Retention ponds (or sump) construction within the excavation area	125	1st – 37th Year of Mining Operation
3.	Installation of silt fences and sediment traps strategically across the site	17.62	1st – 37th Year of Mining Operation
4.	Water sprinkling/Anti-smog guns/Dust Suppression	150	1st – 37th Year of Mining Operation
5.	Phased plantation in 1029 ha	2474	3rd - 40th Year of Mining Operation
6.	Excavation and overburden removal	5	1st – 37th Year of Mining Operation
7.	Wildlife conservation plan	3726.40	1st –10th Year of Mining Operation
8.	Post-reclamation parks	51.26	38th – 40th Year

9.	Monitor groundwater recharge and river	5	1st – 37th Year of Mining Operation
10.	Rainwater harvesting	75	1st – 37th Year of Mining Operation
11.	Best R&R package	72793.20	3rd – 5th Year of Mining Operation
12.	Skill development	20	1st – 37th Year of Mining Operation
13.	Health monitoring programs	20	1st – 37th Year of Mining Operation

Conclusion

The proposed mining project in the 1196 Ha ML area will significantly impact ecosystem services, particularly provisioning, regulating, and supporting services. It will also affect the carrying capacity to the certain extent as discussed above. While mitigation measures are outlined, strict compliance and continuous monitoring are essential to minimize long-term ecological and socio-economic consequences. Prioritize biodiversity offsets, water resource sustainability, and community compensation to ensure ecological resilience and equitable development.

Baitarani West OCP is a coal mine so site specific.

Mine methodology is Opencast Fully mechanized.

Surface miner shall be used for mining of coal.

Shovel / Dumper Combination for Overburden Removal which will avoid drilling and blasting to a great extent.

Only 7% of total excavation shall be done by drilling and blasting.

CHAPTER-5 : ANALYSIS OF ALTERNATIVES TECHNOLOGY & SITE

5.1 INTRODUCTION

The EIA Notification 2006 stipulates the requirement of analysis of alternatives for the Project being appraised in terms of technology and site. The Alternative Analysis in Environmental Impact Assessment is basically an exercise to assess whether the objectives of the proposed Project can be achieved through another cost-effective way having lower environmental & social costs. In the present case, no alternative of site is done as mine is **site specific**.

Coal reserves of desired quality and adequate quantity are already proven in the coal block under consideration.

Considering the above **no other alternative site was considered** for the Coal Mining Project.

5.2 PROCESS/TECHNOLOGY ALTERNATIVE

The basic two methodologies available for coal mining are Underground Mining and Opencast Mining. Apart from many other factors, the choice between these technologies mainly depends on the geo-mining conditions of the coal deposit. The details of the comparative assessment between these two technologies are provided below. This is an opencast mine.

5.2.1 MINING METHODOLOGY

5.2.1.1OPEN CAST MINING METHOD

For the Baitarni opencast mine, highly mechanized conventional mining methodology with Shovel / Dumper Combination for Overburden Removal and Surface Miner for coal mining is envisaged.

The Opencast Mining methodology involves complete removal of topsoil and overburden rock to expose the underlying coal seams to facilitate coal mining. The advantages of Opencast Mining over Underground Mining are high extraction percentage (about 40%) which aids conservation

of mineral, safe operations and flexibility in terms of quality control. Due to availability of sophisticated Heavy Earth Moving Machineries (HEMM), the method is also relatively less labour intensive with high Output per Manshift (OMS).

5.2.1.2 EQUIPMENT SELECTION

The following options are available for consideration of suitable equipment for the Project:

1. **Deployment of Dragline (not selected):**

The possibility of deployment of Dragline for the Project was examined. Mining by Dragline Technology is suitable under following circumstances:

- Dragline is suitable for flat deposits, to permit back dumping of OB in de-coaled area. The OB is usually dumped on seam floor very near the coal bench, leaving space sufficient only for water drainage and also to reduce mixing of OB with coal. If the coal seam gradient is not flat, the dumped OB will slide towards the coal area preventing coal extraction besides being dangerous.
- Dragline is suitable for deposits with Strike Length more than 1.5 Kms to 2.0 Kms to avoid frequent shifting of Dragline from one end to other. Frequent shifting of Dragline reduces machine efficiency as well as it is a costly affair.
- Dragline is suitable for the deposit which is free from geological disturbances. A Dragline System works with a rigid operational geometry and frequent changes in the geometry may be difficult to implement without heavy loss of efficiency.
- Dragline is more suitable for thick single seam working rather than multi-seam working.
- Dragline is suitable for the deposit large enough to ensure the mine life of about 25 years or more so that heavy capital investment on the System could be recovered.

Hence, the Dragline system was not selected.

2. **Deployment of Bucket Wheel Excavator (not selected):**

The possibility of deployment of Bucket Wheel Excavator for the Project was examined. Mining by Bucket Wheel Excavator Technology is suitable under following circumstances:

- Bucket Wheel Excavator is suitable for high volume operations in comparatively soft rock strata.
- Bucket Wheel Excavators are suitable for thick single seam workings.
- Bucket Wheel Excavators are suitable for the deposit large enough to ensure the mine life of more than 25 years so that heavy capital investment on the System could be recovered.

The suitability of Bucket Wheel Excavator Technology on the basis of above parameters was evaluated. Due to occurrence of multiple coal horizons and high capital investment, deployment of Bucket Wheel Excavator is ruled out.

3. **Deployment of Surface Miner(selected):**

Surface Miner is suitable for flat and thin seams. The limiting gradient is 1 in 10 or flatter. Also, Surface Miner requires a large coal exposure which is possible only with sufficient flat deposits. Use of Surface Miner (within built dust extraction and water sprinkling) is proposed for mining of coal because of following reasons:

- Environment friendly operations.
- One surface miner replaces equipment for drilling, blasting, crushing, loading and auxiliary works like Secondary Blasting.
- Coal Mining without drilling & blasting and hence lesser vibration and fly rock issues.
- Seams are relatively flatter, and both have strike length and Dip rise length
- No coal crushing is required; this minimizes re-handling & dust generation.
- Enhanced quality of ROM (Run Off Mine) product by highly selective mining.
- Stable, clean surfaces and benches
- Reduced noise and dust generation.
- Improved exploitation of the deposit.
- Improved safety.

4. Deployment of Shovel Dumper Combination for Overburden(selected):

Shovel / Dumper Technology is the most flexible technology for opencast mining. This technology can be used in conditions of varying thickness of seams and partings. The flexibility of the operations enables geological disturbances to be negotiated without much loss of efficiency.

Shovel/Dumper System offer convenient mining operations to deal with sudden occurrences of unworkable or poor-quality patches. It also offers flexibility for easy transition to any other technology or equipment configuration. The technology is well known and easier to get skilled manpower. Presently the Shovel / Dumper combination is mainly used with drilling and blasting for OB removal.

For Baitarni Mining Project, the combination of Shovel / Dumper Technology is much more convenient due to the presence of multiple seams. Further, this System can deal with sudden occurrence of unworkable and poor-quality zones and offers flexibility for easy upgradation.

5.3 ASSESSMENT OF NEW AND UNTESTED TECHNOLOGY

The Project envisages use of conventionally tested Shovel/Dumper Combination Technology along with use of Surface Miner.

The Shovel / Dumper Combination for undertaking Overburden Removal is being used in Indian Coal Mining Industry since long and is the only preferable technology for opencast mining considering the geo-mining conditions of Indian coalfields.

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CHAPTER-6 : ENVIRONMENTAL MONITORING PROGRAM

An effective Environment Management Program offers an opportunity for maintaining and reviewing the sustainable development of the environment.

Monitoring schedule of environmental parameters in details are mentioned in this chapter.

A separate and dedicated Environment Management Cell (EMC) comprising of qualified personnel directly reporting to the head of the Project will be there for speedy implementation and monitoring of the EMP.

For better environmental monitoring and management an independent laboratory with the facility of pollution monitoring and analysis equipment shall be set up at the mine site.

The environmental management activities of the project will be carried out under the overall supervision of the GM, followed by executives to assist G.M. for smooth functioning of the project.

Environmental audits shall be undertaken for systematic documentation, periodic and objective evaluation of how well environmental organization, management systems and equipment are performing with the aim of helping to safeguard the environment.

6.1 INTRODUCTION

Environment Management Cell

A mine starts its operation after obtaining various clearances and permissions from different statutory bodies. Once the mining activities are made operational, various pollution control and mitigation measures and other activities are required to be undertaken asper the EMP, EC, FC and CTE and CTO. To ensure that the compliance mechanism is fully and effectively operational, it is necessary that the mitigation measures required for control of pollution are appropriately monitored at different levels. Environment Management Cell plays an important role in this regard.

MoEF&CC, while granting EC to projects, stipulated a separate Environment Management Cell (EMC) comprising of qualified personnel directly reporting to the head of the Project.

Environment Monitoring Program is an important aspect of the monitoring during mine development and operation stage. This is the technical aspects of monitoring the effectiveness of mitigation measures (including measurement methodologies, frequency, location, data analysis, reporting schedules, emergency procedures, detailed budget & procurement schedules). It essentially involves regular monitoring of the prescribed environmental parameters of the Project Study Area.

A properly planned and implemented Environment Monitoring Program:

- Provides up-to-date status of the environment of the Project Area.

- Provides an early indication of the failures, if any, of the implemented environmental control measures or practices for the Project.
- Monitor the performance of the Project and the effectiveness of the implemented mitigation measures.
- Determines the compliance of the Project with regulatory requirements, government policies and national norms.
- Allows for planning of remedial measures in case of occurrence of any unexpected problem or unaccepted environmental issue.

An effective Environment Management Program offers an opportunity for maintaining and reviewing the sustainable development of the environment.
It is the part of the overall Environment Management System and includes an organization structure, planning activities, responsibilities, practices, procedures, process and resources for developing, implementing, achieving, reviewing and maintaining the environmental policy.

6.2 SCOPE OF THE ENVIRONMENT MONITORING PROGRAMME

The Environment Monitoring Program suggested for Baitarni West Coal Block Project shall be carried out throughout the Mine Life and also for minimum three to five years post-mine closure.

1) Collection of environment quality data for Key Environmental Parameters in the Project Area at regular intervals in accordance with prescribed methodologies.
2) Analysis & interpretation of the collected data.
3) Comparison of the extent & severity of the environmental actual impacts of the Project against the predicted impacts.
4) Assessment of the effectiveness of the implemented EMP Measures.
5) Assessment of the requirement of the additional EMP Measures, if any
6) Assessment of the overall Environment Protection Measures of the Project vis-à-vis applicable regulatory requirements and conditions of Environment Clearance granted by MoEF&CC.
7) Reporting on the compliance of the Environmental Clearance conditions along with suggested Environmental Monitoring Program to all the concerned Authorities including the Regional Office of MoEF&CC at prescribed regular intervals.

6.3 KEY PARAMETERS TO BE MONITORED

Considering the nature and size of the Project, the following Key Environmental Parameters have been identified for regular monitoring in the Project Study Area of 10 km under the Environment Monitoring Program:

1. Air Quality & Meteorology
2. Surface Water Quality
3. Ground water Quality
4. Noise Level
5. Soil Quality

6.4 ENVIRONMENTAL MONITORING PROGRAMME FOR KEY PARAMETERS

The details of the suggested Environment Monitoring Program are provided in **Table 6.1.**

Table 6:1 Environment Monitoring Program

S. No.	Particulars	Frequency of Monitoring	Duration of sampling	Parameters required to be monitored
1. Water & Wastewater quality				
A.	Ground Water quality in the area.	Once in 3 months	Grab sampling	As per IS: 10500-2012 RA 2018
B.	Surface Water Quality	Once in 3 months	Composite sampling	As per EPA Rules, 1996 (pH, TSS, BOD, COD, Free Chlorine, ammonia, oil & grease, Coliform count)
2. Air Quality				
	Continuous Ambient air Monitoring	Continuous	Continuous	Particulate Matter (PM_{10}, $PM_{2.5}$) SO_2, NO_2 and CO
	Manual AAQ Monitoring	Monthly	24 hrs.	Particulate Matter (PM_{10}, $PM_{2.5}$) SO_2, NO_2 and CO
	Monitoring of Heavy Metals	Once in six months	---	Hg, Ni, Cd and Cr.
	Fugitive emissions (Process area and RMH Yard)	Monthly	24hrs continuously	Particulate Matter
	Stack emissions monitoring	Monthly	-	Particulate Matter (PM_{10}, $PM_{2.5}$)
3. Noise level monitoring				
A	Near the blasting / drilling site	Fortnightly	24 hours	Noise levels dBA (Leq)
B	Near loading and unloading points	Fortnightly	24 hours	
C	Along the haul roads for Transportation	Fortnightly	24 hours	
4.Ambient Noise Levels				
A.	Ambient Noise Levels	Fortnightly	24 hr continuous	Noise levels in dB(A)

			With one hour interval	
5. Soil Quality				
A	Waste Dump Area Nearby Agricultural Land	Annually	One grab sample	Color, textural class, grain, size distribution, pH, electrical conductivity, bulk density, porosity, Infiltration rate, moisture retention capacity, organic matter, Na, N, K, PO_4, SO_4, etc.

Apart from the Key Environment Parameters, monitoring of certain Project Parameters which play major role in restoration of the Project Area is also important. For Baitarni Coal Mining Project regular monitoring of following Project Parameters has been suggested:

1. Plantation
2. Land Reclamation &Backfilling
3. Occupational Health

Apart from the above any stipulation in the EC letter/NOC/Consent/CGWA/MOC etc.by the statutory agency shall also to be done.

6.4.1MONITORING OF PLANTATION

The Project involves extensive plantation programme as one of the measures to mitigate the adverse environmental impacts to the possible extent. Therefore, regular monitoring of the plantation programme shall be very important.

Total plantation will be on 1029.583 ha post closure of the mine besides 64.3723 ha of forest land will be returned back.

Greenbelt will be developed considering the proposed plantation @ 2500 trees per hectare.

Regular monitoring of plantation programme in terms of number of trees to be planted per annum, plant growth, and maintenance of plantation and survival rate shall be done on a half yearly basis.

The details of the Plantation Programme shall be included in the Half Yearly Returns to be submitted to the Regional Office of MoEF&CC.

6.4.2 MONITORING OF OCCUPATIONAL HEALTH

Occupational health and safety monitoring programs shall verify the effectiveness of prevention and control strategies. The selected indicators shall be representative of the most significant occupational, health, and safety hazards, and the implementation of prevention and control strategies. The performance and achievements of the OHSMS (Occupational health & Safety management system) responsible for all management of all environments, Health & Safety aspects shall be reassessed on an annual basis. Further periodical medical examinations of the workers engaged in mining & allied activities shall be undertaken once in every year for occupational diseases and hearing impairment. All the records pertaining to the Occupational Health of the workers shall be maintained as required under the provisions of the Mines Rules, 1955 and the relevant DGMS Circulars.

6.5 MONITORING SCHEDULE OF ENVIRONMENTAL PARAMETERS

The mining in the Project Area will start after obtaining all the Statutory Clearances/Permissions. The anticipated life of the opencast mine is 37 years. The details of the envisaged schedule for monitoring the environmental parameters are given below:

Air Quality, Water Quality and Noise Level Monitoring shall be done regularly as per the prescribed standards in consultation with the State Pollution Control Board by taking measurements near the mine and residential areas preferably close to mine.

For air quality monitoring, continuous monitoring shall be done for two days per week and analytical checks made for $PM_{2.5}$, PM_{10}, SO_2, NO_2 and CO. Water quality and soil quality shall be monitored seasonally. Noise levels shall be monitored periodically.

For effective management of the environment, it is envisaged to have an organizational set-up under the administrative supervision of the Mines Management where responsibilities can be delegated to technical personnel like Environmental Engineer/Mining Engineer, Geologist/Chemist and Horticulturist with regard to specific aspects of Environment Management Plan with reporting to Top management for quick implementation of the EMP and short comings ear marked in the monitoring.

6.6 ENVIRONMENTAL CELL

For effective implementation of the recommended Environmental Monitoring Programme, it is necessary to establish and develop adequate facilities for sampling and analysis on permanent basis together with adequate instrumentation facilities and development of competent trained manpower.

A separate Environmental Cell shall be there to ensure effective monitoring of the Key Environmental Aspects of the Project on a day-to-day basis. The organizational structure of the Environmental Cell is provided in **Figure 6.1.**

The Environment **Officer** should be given due responsibility for the implementation of the reclamation plan with the supervision of Manager. The Environmental Management Cell consists of Assistant Manager (Environment) as the In-Charge of the cell with one member as plantation supervisorand one Lab Chemist (In-Charge of Environment Lab) for cooperation of various functions. They are assisted by ten personnel (outsourced) as workers/ plantation helpers for monitoring and analysis of environmental parameters, safety and training, occupational health, surveillance program and plantation.

Environmental Management Cell

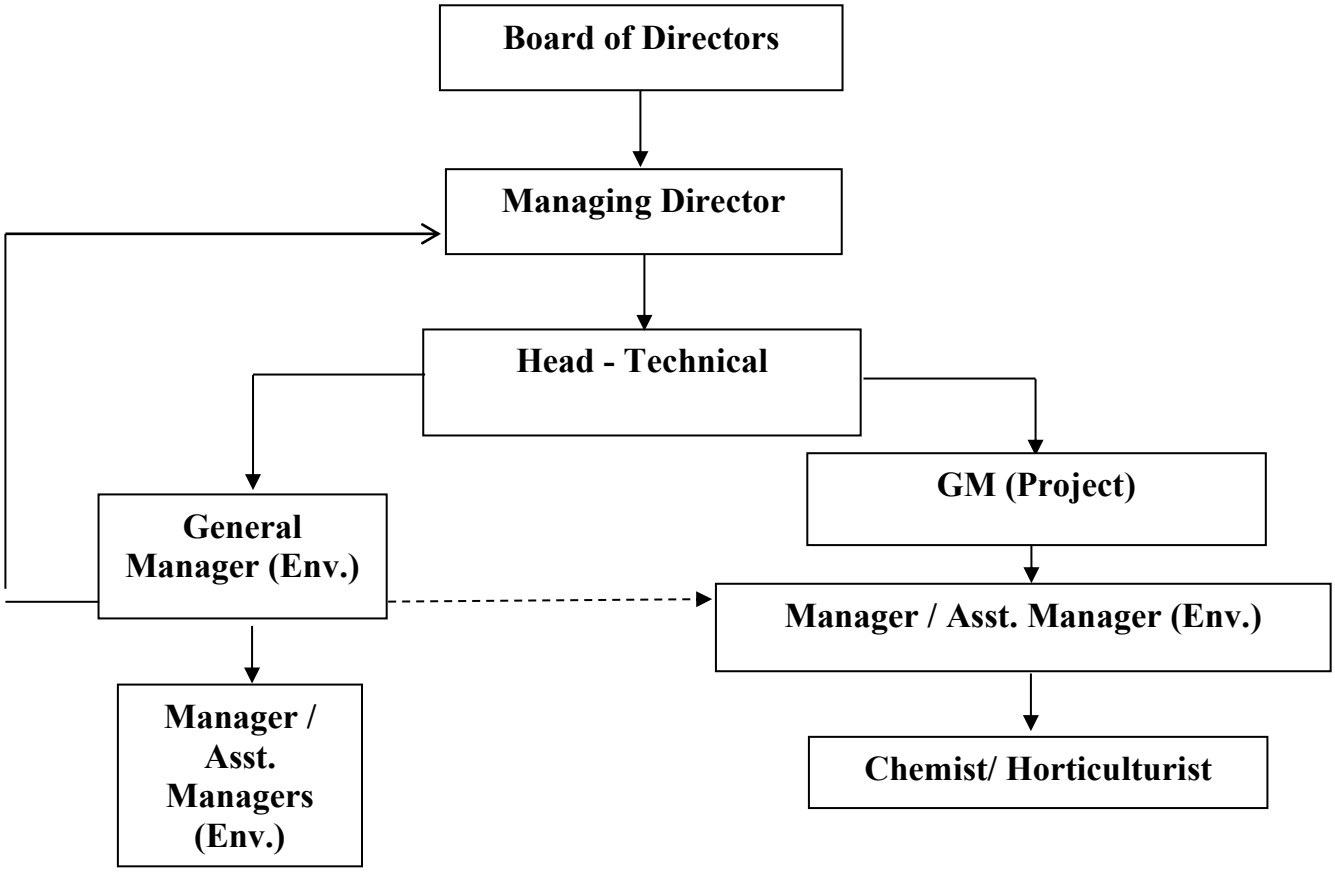


Figure 6-1 Organogram of Environmental Cell

6.7 BUDGETARY PROVISIONS FOR MONITORING

On regular basis, Environment Management Cell will inspect the necessity and availability of the materials, technologies, services and maintenance works. The Cell will make appropriate budget for the purpose. Regular record review for any change in financial requirement of environment management will be done and

appropriate budgetary provisions will be made. Along with other budgets, the Budget for environmental management will be prepared and revised regularly as per requirement. The budget will include provisions for: -

- Environmental Monitoring Program;
- Laboratory works for Environmental management activities;
- Emergency Purchase of necessary material, equipment, tools, services;
- Greenbelt development;
- Social and Environmental Welfare and Awareness programs / training (CER);
- Annual Environmental statement.

Adequate expenditure towards capital and recurring expenditure required for Environment Protection including Environment Monitoring has been planned by PP. The details of the same have been provided in chapter 10.

CORPORATE LEVEL

GMDC, the owner of this project, has established the Environment Department at its HQs, Ahmedabad. This Department provides necessary support that are required for environmental management at project level.

PROJECT LEVEL

The environmental management activities of the project will be carried out under the overall supervision of the GM, following executives will assist G.M. for smooth functioning of the project:

(a) DGM (Mines)/Mine Manager - He is responsible for overall control of coal production, safety, environment, land reclamation, plantation etc of the mine and the persons working therein. He provides all the financial and technical support for carrying out the above activities.

(b) DGM (Mines)/Production I/C Responsible for overall production, safety, supervision including environment management, monitoring of safety, air pollution control, noise pollution control, plantation, backfilling, land reclamation activities.

(c) DGM (Mines)/Safety I/C Responsible for all the mechanical maintenance of HEMM like dumpers, shovels, water sprinklers, dozers, graders, fire tenders etc.

(d) Manager (Electrical) - Responsible for uninterrupted power supply in the mine, general lighting in the working face, Haul Roads, OB dump, CHP, workshop. He is responsible for the working of all pumps in the monsoon season, maintenance of all light vehicle etc.

(e) Manager (Civil) - Responsible for construction and maintenance of effluent treatment plants, water supply, black topping & metalling of roads & haul roads,

cleaning and overall housekeeping of the CHPs, railway siding, construction and maintenance of check dams, garland drains, etc.

(f) Manager (HR) - Responsible for payment of compensation to land oustees, employment, rehabilitation and resettlement.

(g)Manager/Assistant Manager (Environment) -Responsible for safety of the mine and workers, control of air pollution, water pollution, noise pollution, ground vibration, plantation etc.

The organizational structures for mine as well as environmental management are given in Fig.-6.1.

The following items will be considered under the monitoring schedule of the EMC:

Slope Failure

Regular examination must be carried out to look for slope failure on open cast mine faces, ore andoverburden benches etc. Any abnormal condition, if observed, must be brought immediately to thenotice of the concerned department.

Land Erosion

Regular observations to check land erosion will be made around the mining and dumpingareas.

Drainage

The effectiveness of drainage system depends upon the proper cleaning of drains and sumps. Any blockage due to silting or accumulation of loose materials will be checked on a regular basis. Stone pitching, brick mounds etc. on drain shall also be monitored.

Re-vegetation and Green belt development

Schedule, planned for green belt development will be checked after every year and any alteration required will be implemented. Post plantation status will be regularly checked in every season. Phase-wise development in the areas of plantation including rate of growth, survival rate etc. will be recorded systematically.

Environmental Quality Monitoring

Facility for monitoring surface runoff water, mine discharge water and ground infiltration will be provided. Ground water treated and untreated mine water and surface water bodies and wells must be monitored at regular intervals. Air quality should be monitored according to CPCBguidelines,and dust prevention control actions are to be taken according to prevailing condition.

Socio-Economic Development

The Environmental Department must be in regular touch with surrounding villages to monitor the implementation of various developmental schemes made by the mine authorities. The following measures may be suggested for creating cordial working atmosphere at mining sites:

- Preparation of an environmental awareness and education package with reference to mining activities should be undertaken for disseminating the correct information to the local population. This would help in reducing misapprehension in the minds of the local people as well as the NGOs who influence the local people.
- Confidence in NGOs regarding the functioning of the Project in the area needs be generated.
- Settlement if any, disputes regarding displacement of population need to be done.
- Nearby villages for social welfare work in the vicinity should be adopted by the company. This would generate ample goodwill for the company.
- Efforts for implementation of infrastructure in the labor colony should be made by way of perusing different schemes available with Rural Development Department of State Govt.
- Important records to be maintained by Environmental Management Cell may be summarized as given in **Table 6.2.**

A team consist of Sociologist, HR etc. shall be constituted to have regular interaction with the villagers within the ML and nearby villages to develop a friendly relationship. Project will extend help in their emergency situation, cultural and economic activities so that they should not feel PP as an outsider. This will be a win win situation for both.

Table 6:2 Important Records to be maintained by EMC

Sl. No.	Particulars
1.	Field monitoring results for air, water, noise, soil, meteorology.
2.	Inspection record of slope failure, land erosion, drainage, socio-economic development
3.	Format to record/monitor plantation measures.
4.	Nursery records
5.	Environmental related standards and norms
6.	Records pertaining to statutory consents, approvals
7.	Code of actions for pollution code in defined areas
8.	Periodic medical examinations records
9.	Records pertaining to laboratory equipment maintenance and calibration
10.	Environmental complain register

11.	Records of water and electricity consumption
12	Periodic progress records
13.	Environmental audit records
14.	Records of annual budgetary requirement and allocation for pollution control

6.8 LABORATORY FACILITIES AND EQUIPMENT

For better environmental monitoring and management an independent laboratory with the facility of pollution monitoring and analysis equipment shall be set up at the mine site. The laboratory shall have the provision for fume hood and cold room. A separate air-conditioned dust-free room shall be provided for the instruments. It is suggested to setup well equipped laboratory with the following facilities at the site for effective internal monitoring and analysis work. List of Equipment Required for Environmental Laboratory is given in **Table 6.3.**

Table 6:3 List of Equipment Required for Environmental Laboratory

Sl. No.	**Particular**	**No.**
1.	Particulate Sampler (PM_{10}& $PM_{2.5}$) & Gaseous sampler	3 Sets
2.	Hygrometer	2
3.	Electronic Chemical Balance	1
4.	Water testing kit/ Soil testing Kit	1
5.	Glass ware/chemicals/Filter Paper	--
6.	Distillation unit	1
7.	Noise meter	1
8	pH Bench top meter	1
9	BOD Incubator	1
10	Hot Plate	1
11	Oven	1
12	Spectrophotometer	1
13	COD Digester	1
14	Magnetic Stirrer	1
15	Muffle furnace	1

6.9 ENVIRONMENTAL AUDITING

Preserving the ecological environment surrounding the activity is most crucial. This can be achieved by auditing and identifying the operating conditions that prevail in the mines. Environmental audits are defined as a basic management tool comprising of systematic documented, periodic and objective evaluation of how well environmental organization, management systems and equipment are performing with the aim of helping to safeguard the environment by (i) facilitating management control of environmental practices, (ii) assessing compliance with

company policies, which include meeting regulatory requirements. The main objectives of environmental audit can be summarized as:

- To ensure that cost effective systems of environmental protection are sufficient to meet current constraints.
- To ensure that the standard of environmental protection management is sufficient to promote good relations with local, national communities.
- To minimize actual or potential liabilities.
- To provide reassurance and make recommendations to management on the above.

Basic steps involved in the environmental audit are presented in Table 6.4. It is recommended that the Project should adopt an effective environmental audit program to keep vigil on the effectiveness of implemented Environmental Management Plan.

Table 6:4 Steps Involved in Environmental Audit

Pre- auditactivities	**Activities at the site**	**Post audit activities**
Obtain Preliminary Information Through Questionnaire Survey *Review and Identify Main Areas of Consideration* Prepare AuditTeam ↓ Organize Resources Visit Program *Allocate SpecificTask to Team Members*	Interview with Cross Section of Staffs Record verification Field Examination **Material and Balance** Process Input Process output Inclusion of data on Process Flow Sheets Material Balance and Water Balance study **Waste Flow** Waste Flow identification Listing the details Pre-treatment Final Treatment Details of disposal **Monitoring & Analysis** Monitoring Schedule Design for Solid Waste&Wastewater Sampling, Treatment Facility suitability Analysis of Identified ParametersType &Frequency of Sampling Sample Analysis **Concluding Session** Preparation of the draft report Draft Report Presentation Discussion with the Management	Assess Performance of the Waste TreatmentFacilities Detect Snags in Waste Generation, Handling & Discharge Waste Segregation and Classify Waste Minimizing Actions Assess Capabilities ofStaff Best Practices Formulation for the WasteManagement Final Report Preparation Including Remarks Time Schedule forAction Follow ups

CHAPTER-7 : ADDITIONAL STUDIES

7.1 PUBLIC HEARING

Notice for public hearing was advertised in newspaper of The New Indian Express, Pragativadi on 23.08.2024 by Odisha SPCB for general information regarding public hearing for Environmental Clearance of Baitarni West Opencast Coal Mine with production capacity of 15 MTPA in an area of 1196 ha Located at Village Chhendipada Jangal, Chhendipada RF, Handigora, Machhakuta, Machhakuta Jangal, Porapara, Porapara Jangal, Tentuloi-Korasahi under Chhendipada Tehsil of Angul district, Odisha by M/s Gujarat Mineral Development Corporation Ltd (GMDC).

Mr. Pratap Pritimaya, O.A.S (SAG), Additional District Magistrate, Angul presided over the meeting along with Er. Ramesh Kumar Ekka, Regional Officer, State Pollution Control Board, Angul, Odisha.

The proceedings of the Public Hearing have been forwarded to The Secretary, MoEF&CC vide letter 15811/IND-II-PH-1209 Dated 05.10.2024 by State Pollution Control Board, Odisha. Copy of the proceedings is attached as **Annexure I.**

Public Hearing is an essential tool to involve various stake holders for their views of the project.

Public Hearing was conducted by Odisha State Pollution Control Board on 24.09.2024. The issues raised in the PH is addressed with timeline and fund earmarked on top priority.

It is a OC coal mine so risk and disaster will be there during operation in the form of

- **Accidents due to operation of HEMM**
- **Failure of Bench/Pit Slope**
- **Inundation of Working during monsoon**
- **Accidents due to improper connections/handling of explosives**
- **Fly Rocks due to blasting**
- **Accidents due to Slope Failure etc.**
- **However, mine will follow the rules and regulation as per the DGMS guideline.**

A continuous and integrated process of planning, organizing, coordinating and implementing measures which are necessary, or expedient shall be followed.

Occupational health and safety shall be addressed as per the norm.

On-site and Off-site emergency and disaster management plan shall be prepared for any emergency situation.



ରାଜ୍ୟ ପ୍ରଦୂଷଣ ନିୟନ୍ତ୍ରଣ ବୋର୍ଡ, ଓଡ଼ିଶା

(ଜଙ୍ଗଲ, ପରିବେଶ ଓ ଜଳବାୟୁ ପରିବର୍ତ୍ତନ ବିଭାଗ, ଓଡ଼ିଶା ସରକାର)

ପରିବେଶ ଭବନ, ଏ/୧୨୮, ନୀଳକଣ୍ଠ ନଗର, ୟୁନିଟ-୮, ଭୁବନେଶ୍ୱର-୭୫୧୦୨୨

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ବିଜ୍ଞାପନ

CAD-909: ଏତଦ୍ଦ୍ୱାରା ସମସ୍ତଙ୍କ ଅବଗତି ନିମନ୍ତେ ଜଣାଇ ଦିଆଯାଉଅଛି ଯେ, **ମେସର୍ସ ଗୁଜୁରାଟ ମିନେରାଲ ଡେଭେଲପମେଣ୍ଟ କର୍ପୋରେସନ ଲିମିଟେଡ୍, ଓଡ଼ିଶା ର ଅନୁଗୁଳ ଜିଲ୍ଲା, ଛେଣ୍ଡିପଦା ତହସିଲ ଅନ୍ତର୍ଗତ ଛେଣ୍ଡିପଦା ଜଙ୍ଗଲ, ହାଂଡିପର, ମାଛକୁଟା, ମାଛକୁଟା ହଙ୍ଗଲ, ପରାପାର, ପରାପାର ଜଙ୍ଗଲ, ତେନ୍ତୁଲୋଇ-କୋରାସାହି ଗ୍ରାମରେ ୧୨୯୬ ହେକ୍ଟର ପରିମିତ ସଂଚଳରେ ସ୍ଥିତ ବୈତରଣୀ ଥ୍ରେଷ୍ଟ ଓପେନକାଷ୍ଟ କୋଇଲା ଖଣି ରୁ ବାର୍ଷିକ ୧୫ ନିୟୁତ ଟନ କୋଇଲା ଉତ୍ପାଦନ ପାଇଁ** ପରିବେଶୀୟ ମଞ୍ଜୁରୀ ନିମନ୍ତେ ଆବଶ୍ୟକ ପଡୁଥିବା ସର୍ବସାଧାରଣ ଶୁଣାଣୀ ପାଇଁ ଭାରତ ସରକାରଙ୍କ ପରିବେଶ, ଜଙ୍ଗଲ ଏବଂ **ଜଳବାୟୁ ପରିବର୍ତ୍ତନ ମନ୍ତ୍ରଣାଳୟ** (MoEF&CC) ଦ୍ୱାରା ପରିବେଶୀୟ ମଞ୍ଜୁରୀ ନିମନ୍ତେ ଆବଶ୍ୟକ ପଡୁଥିବା ସର୍ବସାଧାରଣ ଶୁଣାଣୀ ନିମନ୍ତେ ଓଡ଼ିଶା ପ୍ରଦୂଷଣ ନିୟନ୍ତ୍ରଣ ବୋର୍ଡକୁ ଆବେଦନ କରିଛନ୍ତି।

ଭାରତ ସରକାର ପରିବେଶ, ଜଙ୍ଗଲ ଏବଂ ଜଳବାୟୁପରିବର୍ତ୍ତନ ମନ୍ତ୍ରଣାଳୟର ପରିବେଶୀୟ ମଞ୍ଜୁରୀ ନିମନ୍ତେ ଅଧିସୂଚନା ସଂଖ୍ୟା-ଏସ.ଓ ୧୫୩୩(ଇ), ତା.୧୪.୦୯.୨୦୦୬ ରିଖ ଅନୁଯାୟୀ ରାଜ୍ୟ ପ୍ରଦୂଷଣ ନିୟନ୍ତ୍ରଣ ବୋର୍ଡ, ଓଡିଶା, ଭୁବନେଶ୍ୱରକୁ ସର୍ବସାଧାରଣ ଶୁଣାଣି କରିବାର କ୍ଷମତା ପ୍ରଦାନ କରିଛନ୍ତି। ଏଥି ନିମନ୍ତେ ବୋର୍ଡ, ସର୍ବସାଧାରଣଙ୍କ ସମେତ ପ୍ରକୃତ ଅଧିବାସୀ, ପରିବେଶୀୟ ସଂସ୍ଥା ଏବଂ ପ୍ରସ୍ତାବିତ ପ୍ରକଳ୍ପ ସ୍ଥାପନ ହେଉଥିବା ସ୍ଥାନରେ ଜନବର୍ଗ, ପ୍ରକଳ୍ପ ଯୋଗୁଁ ଉଚ୍ଛେଦ ହେଉଥିବା ବାସିନ୍ଦା, ପ୍ରକଳ୍ପ ଯୋଗୁଁ ପ୍ରଭାବିତ ହେଉଥିବା ବାସିନ୍ଦାମାନଙ୍କଠାରୁ ପ୍ରକଳ୍ପ ସ୍ଥାପନ ଯୋଗୁଁ ପରିବେଶୀୟ ପ୍ରଭାବ ସମ୍ପର୍କୀୟ ପ୍ରସ୍ତାବ, ମତାମତ, ମନ୍ତବ୍ୟ ଓ ଆପତ୍ତି ପତ୍ର ଆହ୍ୱାନ କରୁଛନ୍ତି। ଏଥି ନିମନ୍ତେ ବ୍ୟକ୍ତିର ଅର୍ଥ କେବଳ ନିମ୍ନମତେ ବୁଝାଯାଉଛି:

(କ) ପରିବେଶୀୟ ମଞ୍ଜୁରୀ ଦ୍ୱାରା ପ୍ରଭାବିତ ହେଉଥିବା ବା ହେବାକୁ ଥିବା ଯେ କୌଣସି ବ୍ୟକ୍ତି।

(ଖ) ପ୍ରକଳ୍ପ ସମ୍ବନ୍ଧିତ ପରିବେଶୀୟ ମଞ୍ଜୁରୀ ପାଇଁ ଆଗତ ଦରଖାସ୍ତ ପରିପ୍ରେକ୍ଷୀରେ ତହିଁ ଉପରେ କର୍ତ୍ତୃତ୍ୱ ଜାହିର କରିପାରୁଥିବା କୌଣସି ବ୍ୟକ୍ତି ବିଶେଷ।

(ଗ) ପରିବେଶ କ୍ଷେତ୍ରରେ କାର୍ଯ୍ୟ କରୁଥିବା କିମ୍ବା / ଏବଂ ପ୍ରସ୍ତାବିତ ପ୍ରକଳ୍ପ ଦ୍ୱାରା ପ୍ରଭାବିତ ହେଉଥିବା (ପଞ୍ଜିକୃତ ବା ଅପଞ୍ଜିକୃତ) ଯେ କୌଣସି ବ୍ୟକ୍ତି ସମୂହଙ୍କ ସଂଘ।

(ଘ) ପ୍ରସ୍ତାବିତ ପ୍ରକଳ୍ପର ସ୍ଥାନୀୟ ସୀମାଭୁକ୍ତ ନିକଟବର୍ତ୍ତୀ ଅଞ୍ଚଳରେ ଯେ କୌଣସି ସ୍ଥାନୀୟ କର୍ତ୍ତୃପକ୍ଷ।

ଉପରୋକ୍ତ ବ୍ୟକ୍ତିମାନେ ସେମାନଙ୍କର ପ୍ରସ୍ତାବ, ମତାମତ, ମନ୍ତବ୍ୟ ଏବଂ ଆପତ୍ତି ଇତ୍ୟାଦିକୁ ସଦସ୍ୟ ସଚିବ, ରାଜ୍ୟ ପ୍ରଦୂଷଣ ନିୟନ୍ତ୍ରଣ ବୋର୍ଡ, ଓଡ଼ିଶାଙ୍କୁ ଲିଖିତ ଆକାରରେ ଏହି ବିଜ୍ଞାପନ ପ୍ରକାଶନର ୩୦ ଦିନ ମଧ୍ୟରେ ରେଜିଷ୍ଟ୍ରି ଡାକ ଯୋଗେ ଜଣାଇ ପାରିବେ। ଏହାଛଡା ଉକ୍ତ ପ୍ରସ୍ତାବିତ ପ୍ରକଳ୍ପ ଉପରେ ଲିଖିତ ବା ମୌଖିକ ଭାବେ ମତାମତ ଦେବା ପାଇଁ ଇଚ୍ଛୁକ ବ୍ୟକ୍ତିମାନେ **ତା.୨୪.୦୯.୨୦୨୪ ରିଖ ବିବା ୧୦.୦୦ ଘଟିକା ସମୟରେ ଅନୁଗୁଳ ଜିଲ୍ଲା, ଛେଣ୍ଡିପଦା ତହସିଲ ଅନ୍ତର୍ଗତ ଜଗନ୍ନାଥ ମନ୍ଦିର ସମ୍ମୁଖ ସରକାରୀ ଜମି ରେ** ଧାର୍ଯ୍ୟ ସର୍ବସାଧାରଣ ଶୁଣାଣୀ ସମୟରେ ମଧ୍ୟ ଉପସ୍ଥାପନ କରିପାରିବେ।

ପ୍ରସ୍ତାବିତ ପ୍ରକଳ୍ପ ବିଷୟରେ ବିଶଦ ଭାବରେ ଜାଣିବା ପାଇଁ ପରିବେଶୀୟ ପ୍ରଭାବ ମୂଲ୍ୟାୟନ ବିବରଣୀ (Environmental Impact Assessment) ଏବଂ ଏହାର ନିର୍ବାହୀ ସାରାଂଶ (Executive Summary) ଓଡିଆ ଏବଂ ଇଂରାଜୀ ଭାଷାରେ ନିମ୍ନଲିଖିତ ସ୍ଥାନମାନଙ୍କରେ ମିଳିପାରିବ।

୧. ଜିଲ୍ଲାପାଳଙ୍କ କାର୍ଯ୍ୟାଳୟ, **ଅନୁଗୁଳ**।
୨. ଜିଲ୍ଲା ଶିଳ୍ପ କେନ୍ଦ୍ର, **ଅନୁଗୁଳ**।
୩. ଜିଲ୍ଲା ପରିଷଦ ମୁଖ୍ୟ ନିର୍ବାହୀ ଅଧିକାରୀ, **ଅନୁଗୁଳ**।
୪. ମୁଖ୍ୟ କାର୍ଯ୍ୟାଳୟ, ରାଜ୍ୟ ପ୍ରଦୂଷଣ ନିୟନ୍ତ୍ରଣ ବୋର୍ଡ, ଓଡିଶା, ପରିବେଶ ଭବନ, ଏ/୧୨୮, ନୀଳକଣ୍ଠ ନଗର, ଭୁବନେଶ୍ୱର-୭୫୧୦୨୨।
୫. ଆଞ୍ଚଳିକ କାର୍ଯ୍ୟାଳୟ, ରାଜ୍ୟ ପ୍ରଦୂଷଣ ନିୟନ୍ତ୍ରଣ ବୋର୍ଡ, ଓଡିଶା, **ଅନୁଗୁଳ**।
୬. ଜଙ୍ଗଲ, ପରିବେଶ ଓ ଜଳବାୟୁ ପରିବର୍ତ୍ତନ ବିଭାଗ, ଓଡ଼ିଶା ସରକାର, ଭୁବନେଶ୍ୱର।

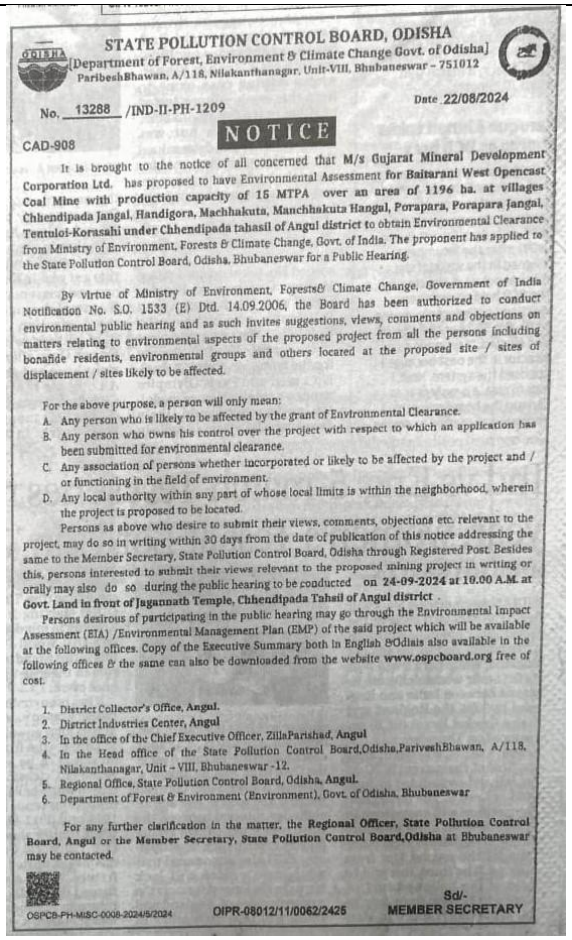
ଏତଦ୍ ବ୍ୟତୀତ ପରିବେଶୀୟ ପ୍ରଭାବ ମୂଲ୍ୟାୟନ ବିବରଣୀର ନିର୍ବାହୀ ସାରାଂଶ (**Executive Summary**) ଓଡ଼ିଆ ଏବଂ ଇଂରାଜୀ ଭାଷାରେ ରାଜ୍ୟ ପ୍ରଦୂଷଣ ନିୟନ୍ତ୍ରଣ ବୋର୍ଡ, ଓଡ଼ିଶାର ୱେବସାଇଟ **www.ospcboard.org** ରୁ ମଧ୍ୟ ମିଳିପାରିବ।

ଏ ବିଷୟରେ ସବିଶେଷ ବିବରଣୀ ପାଇଁ ରାଜ୍ୟ ପ୍ରଦୂଷଣ ନିୟନ୍ତ୍ରଣ ବୋର୍ଡ, ଓଡ଼ିଶାର **ଆଞ୍ଚଳିକ କାର୍ଯ୍ୟାଳୟ, ଅନୁଗୁଳ** ସ୍ଥିତ ଆଞ୍ଚଳିକ ଅଧିକାରୀ କିମ୍ବା ଭୁବନେଶ୍ୱରସ୍ଥିତ ମୁଖ୍ୟ କାର୍ଯ୍ୟାଳୟରେ ସଦସ୍ୟ ସଚିବଙ୍କ ସହ ଯୋଗାଯୋଗ କରାଯାଇପାରେ।

ସ୍ୱା/- ସଦସ୍ୟ ସଚିବ

OIPR-08012/11/0063/2425

Pragativadi Dated 23.08.24



STATE POLLUTION CONTROL BOARD, ODISHA

[Department of Forest, Environment & Climate Change Govt. of Odisha]

ParibeshBhawan, A/118, Nilakanthanagar, Unit-VIII, Bhubaneswar - 751012

No. 13288 /IND-II-PH-1209 Date 22/08/2024

CAD-908

NOTICE

It is brought to the notice of all concerned that **M/s Gujarat Mineral Development Corporation Ltd.** has proposed to have Environmental Assessment **for Baitarani West Opencast Coal Mine with production capacity of 15 MTPA over an area of 1196 ha. at villages Chhendipada Jangal, Handigora, Machhakuta, Manchhakuta Hangal, Porapara, Porapara Jangal, Tentuloi-Korasahi under Chhendipada tahasil of Angul district** to obtain Environmental Clearance from Ministry of Environment, Forests & Climate Change, Govt. of India. The proponent has applied to the State Pollution Control Board, Odisha, Bhubaneswar for a Public Hearing.

By virtue of Ministry of Environment, Forests& Climate Change, Government of India Notification No. S.O. 1533 (E) Dtd. 14.09.2006, the Board has been authorized to conduct environmental public hearing and as such invites suggestions, views, comments and objections on matters relating to environmental aspects of the proposed project from all the persons including bonafide residents, environmental groups and others located at the proposed site / sites of displacement / sites likely to be affected.

For the above purpose, a person will only mean:

A. Any person who is likely to be affected by the grant of Environmental Clearance.
B. Any person who owns his control over the project with respect to which an application has been submitted for environmental clearance.
C. Any association of persons whether incorporated or likely to be affected by the project and / or functioning in the field of environment.
D. Any local authority within any part of whose local limits is within the neighborhood, wherein the project is proposed to be located.

Persons as above who desire to submit their views, comments, objections etc. relevant to the project, may do so in writing within 30 days from the date of publication of this notice addressing the same to the Member Secretary, State Pollution Control Board, Odisha through Registered Post. Besides this, persons interested to submit their views relevant to the proposed mining project in writing or orally may also do so during the public hearing to be conducted **on 24-09-2024 at 10.00 A.M. at Govt. Land in front of Jagannath Temple, Chhendipada Tahsil of Angul district** .

Persons desirous of participating in the public hearing may go through the Environmental Impact Assessment (EIA) /Environmental Management Plan (EMP) of the said project which will be available at the following offices. Copy of the Executive Summary both in English &Odiais also available in the following offices & the same can also be downloaded from the website **www.ospcboard.org** free of cost.

1. District Collector's Office, **Angul.**
2. District Industries Center, **Angul**
3. In the office of the Chief Executive Officer, ZillaParishad, **Angul**
4. In the Head office of the State Pollution Control Board,Odisha,ParibeshBhawan, A/118, Nilakanthanagar, Unit - VIII, Bhubaneswar -12.
5. Regional Office, State Pollution Control Board, Odisha, **Angul.**
6. Department of Forest & Environment (Environment), Govt. of Odisha, Bhubaneswar

For any further clarification in the matter, the **Regional Officer, State Pollution Control Board, Angul** or the **Member Secretary, State Pollution Control Board,Odisha** at Bhubaneswar may be contacted.

Sd/-
MEMBER SECRETARY

OSPCB-PH-MISC-0008-2024/9/2024 OIPR-08012/11/0062/2425

The New Indian Express Dated 23.08.24

Figure 7-1 Notice for Public Hearing published on 23.08.24 in Pragativadi & in The New Indian Express



Figure 7-2 Photographs of Public Hearing

Table 7:1 Issues Raised in the Public Hearing and Commitments by M/s Gujarat Mineral Development Corporation Ltd (GMDC)

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
1.	Sri Agasti Behera, Hon'ble MLA, Chhendipada	He greeted all officials and public present in the meeting. He informed that proposed mining project has been previously allocated to Orissa Mining Corporation, Odisha, however the OMC surrendered the mine. Further, this mine has been allocated to M/s. GMDC by Govt. of India through auction. He also stated that during operation of mines, water crisis will arise in the nearby affected area, however the project proponent has presented the usable steps to be taken by the company. He also informed that the proposed mining project will create an employment opportunity to the local youths and females of the affected areas. He further informed that the proposed project will offer the highest packages in India for land	• Employment for Local Youth & Females • Compensation to land losers	The project is likely to generate 1752 job opportunities. In addition to this, the project will create more indirect job opportunities. Apart from this, employment-oriented skill development training program under ''Project Samarthya'' is also being organized to create more job opportunities for the youth and middle-aged group. An attractive R&R package has been communicated to all the affected families, which is much better than the package specified in Odisha R&R policy 2006 & the previous RPDAC. The package includes cash compensation in lieu of homestead land and structures.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		losers. Finally, he urged the local people to support and welcomed the proposed project	• Concern for drinking water	We assure that we will provide safe drinking water facility in the local villages under CSR activity.
2.	Smt. Susama Raul, Ex-Vice Chairman, Chhendipada	She conditionally supported the proposed mining project and urged before the project proponent to give emphasis on abatement of air pollution, water pollution of water bodies, river and nallahs situated around 50 Km radius of the proposed project. She also demanded to undertake massive plantation drive in affected forest areas, protection of Wildlife and to provide compensation to land losers. Also, she demanded before the project proponent to provide technical education and training to local youths in order to get engagement in the proposed company.	• Pollution Control	We will adopt environmental protection measures in proposed mining project for control of air pollution (PM, SOx, NOx), water pollution and soil pollution. We will also carry out environmental quality monitoring to monitor the pollution level.
			• Plantation Drives	Massive plantation drives in the project and surrounding areas for better local environment will be done.
			• Protection of Wildlife	We will take adequate steps for forest conservation and the protection of wildlife.
			• Compensation to land losers	An attractive R&R package has been communicated to all the affected

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
				families, which is much better than the package specified in Odisha R&R policy 2006 & the previous RPDAC. The package includes cash compensation in lieu of homestead land and structures.
			• Employment for Local Youth & Females	The project is likely to generate 1752 job opportunities. In addition to this, the project will create more indirect job opportunities. Apart from this, employment-oriented skill development training program under ''Project Samarthya'' is also being organized to create more job opportunities for the youth and middle-aged group.
3.	Sri Chandan Kar, Podapada	He expressed his dissatisfaction over the proposed project and informed that the environment of the Angul district is not in good condition. He informed that local environment of the proposed project area will become similar to the local	• Compensation to land losers	An attractive R&R package has been communicated to all the affected families, which is much better than the package specified in Odisha R&R policy 2006 & the previous RPDAC. The package includes cash

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		environment of Raijharan mine area. Finally, he suggested that the project proponent should discuss with the land losers in order to provide one package for one project policy for all land losers.		compensation in lieu of homestead land and structures.
4.	Sri Sudipta Pradhan, Sarpanch, Chhendipada G.P	He stated that the mining activity will impact local environment. He demanded before the project proponent that special attention should be given in order to provide one project- one package policy for the land losers and 80% employment to the local youths and villagers. He also suggested that the mining activity shall not pollute natural environment such as river and nallah etc. He highlighted the presently ongoing CSR activities in local area by M/s. GMDC and expressed his trust over the proposed proponent and informed that all demands of the local villagers will be fulfilled in the near	• Compensation to land losers • Employment for Locals	An attractive R&R package has been communicated to all the affected families, which is much better than the package specified in Odisha R&R policy 2006 & the previous RPDAC. The package includes cash compensation in lieu of homestead land and structures. The project is likely to generathe1752 job opportunities. In addition to this, the project will create more indirect job opportunities. Apart from this, employment-oriented skill development training program under ''Project Samarthya'' is also being organized to create more job

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		future through resettlement and rehabilitation policy.		opportunities for the youth and middle-aged group.
			• Pollution Control	We will adopt environmental protection measures in proposed mining project for control of water pollution. To protect the Kumbhira Nala, an embankment will be constructed all along the nallah within the mine lease (ML) area and plantation will be done along the embankment.
5.	Sri Ashok Kumar Pradhan. Chhendipada	He demanded before the proposed project proponent to provide drinking water facility, focus on wildlife protection, focus on improvement of air quality, create an employment opportunity to local youths as per educational qualification. He also urged before the project proponent to protect the Kumbhira nallah from getting water pollution by this proposed mining project and the company should	• Concern for drinking water	We will provide safe drinking water facility in the local villages and help in improvement of village roads
			• Wildlife protection	We will take adequate steps for forest conservation and the protection of wildlife.
			• Pollution control	We will adopt environmental protection measures in proposed mining project for control of air pollution (PM, SOx, NOx.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		construct separate road for mining activity, so that local people shall not get affected by the mine	• Employment for Locals	The project is likely to generate the 1752 job opportunities. In addition to this, the project will create more indirect job opportunities. Apart from this, employment-oriented skill development training program under ''Project Samarthya'' is also being organized to create more job opportunities for the youth and middle-aged group.
			• Protection of water bodies	Embankment of 3 km length & height of 3m above HFL will be provided along the Kumbhira Nala for its protection. Toe-walls, retaining walls with weepholes at strategic stretches will be constructed to arrest the sediments and prevent silting of the water courses by the run-off during rains. Green belt shall be developed along the embankment.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
			• Road traffic and safety	Limited quantity of coal will be transported by road to meet the local demand. Bulk of the coal will be evacuated via rail. Further the movement of trucks will be minimized in peak traffic hours. Wheel washing system shall be installed at the exit point of mine. Overloading of transport equipment shall be prevented in order to stop Spillage. Trees will be planted all along the coal transportation road within mine lease area.
6.	Sri Jasobant Prasad Behera, Chhendipada	He stated that the mining project will surely impact the local environment, however special attention must be given, so that the natural environment and local people may not be get affected by this proposed mining project. He supported the project and expressed his faith on project proponent that all demands of local villagers and land losers will be fulfilled.	• Pollution control	We will adopt environmental protection measures in the proposed mining project for control of pollution. Massive plantation drives will be provided in the project and surrounding areas for better local environment.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
7.	Sri Madhusudan Dehury, Chhendipada	He supported the proposed mining project. He urged before the project proponent to fulfil demands of the villager such as actual compensation to loser and to provide employment to local youths. He also urged before the project proponent to take steps to rejuvenating the nallah and river in order to get relief from water crisis which may takes place due to proposed mining project	• Compensation to land losers	An attractive R&R package has been communicated to all the affected families, which is much better than the package specified in Odisha R&R policy 2006 & the previous RPDAC. The package includes cash compensation in lieu of homestead land and structures.
			• Employment for Locals	The project is likely to generate 1752 job opportunities. In addition to this, the project will create more indirect job opportunities. Apart from this, employment oriented skill development training program under ''Project Samarthya'' is also being organized to create more job opportunities for the youth and middle aged group.
			• Protection of water bodies	Embankment of 3 km length & height of 3m above HFL will be provided along the Kumbhira Nala for its protection. Toe-walls, retaining walls

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
				with weepholes at strategic stretches will be constructed to arrest the sediments and prevent silting of the water courses by the run-off during rains. Green belt shall be developed along the embankment.
8.	Sri Choudhury Pradhan, Chhendipada	He is hopeful that the proposed project will fulfil all the demands of local people, such as creation of employment for the local youths and development of local area. The company should provide exact cost of land to every land loser, steps should be taken to abate the pollution and for the protection of environment. Further, he stated that after completion of mining activity, the mine void should be filled up with soil and plantation work must be increase and forest should be generated in that area.	• Employment for Locals • Development of Local Area • Land Compensation	The project is likely to generate 1752 job opportunities. In addition to this, the project will create more indirect job opportunities. Apart from this, employment-oriented skill development training program under ''Project Samarthya'' is also being organized to create more job opportunities for the youth and middle aged group. Will be done under Corporate Social Responsibility CSR) funds for the benefits of the local people. An attractive R&R package has been communicated to all the affected families, which is much better than

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
				the package specified in Odisha R&R policy 2006 & the previous RPDAC. The package includes cash compensation in lieu of homestead land and structures.
			• Pollution control	We will adopt environmental protection measures in proposed mining project for control of air pollution (PM, SOx, NOx), water pollution and soil pollution.
			• Plantation	We will take up massive plantation drives in the project and surrounding areas for better local environment
9.	Sri Durga Ch. Mohapatra, Chhendipada	He informed that mining activity will create pollution in the local area, so steps shall be taken for the prevention of pollution. He demanded to provide an employment to the local people and steps must be taken for rehabilitation and resettlement of the local land losers	• Pollution Control	Regular maintenance of vehicles and machinery; Water sprinkling on haul roads within ML area; Plantation in and around the proposed mine and Green Belt Development will be done.
			• Employment for Locals	The project is likely to generate 1752 job opportunities. In addition to this,

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
				the project will create more indirect job opportunities. Apart from this, employment oriented skill development training program under ''Project Samarthya'' is also being organized to create more job opportunities for the youth and middle aged group.
			• Compensation to land losers the land losers	An attractive R&R package has been communicated to all the affected families, which is much better than the package specified in Odisha R&R policy 2006 & the previous RPDAC. The package includes cash compensation in lieu of homestead land and structures.
10.	Sri Dilip Kumar Behera, Chhendipada Jungle	He supported the proposed mining project and demanded before the company authority to provide safe drinking water facility, construction of roads for local villagers, look after the protection of forest and to carry out plantation activities	• Safe Drinking Water Facility & Construction of roads	Will be provided under Corporate Social Responsibility (CSR) funds for the benefit of the local people.
			• Protection of forest	We will take adequate steps for forest conservation.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
			• Plantation activities	We will take up massive plantation drives in the project and surrounding areas for better local environment
11.	Sri Deepak Kumar Pradhan, Chhendipada	He welcomed the project in order to create an employment opportunity and increase financial status of the local villagers. However, he demanded before the district administration to give emphasis on protection of environment and wildlife, conservation of forest and rehabilitation and resettlement of land losers.	• Employment for Locals	The project is likely to generate 1752 job opportunities. In addition to this, the project will create more indirect job opportunities. Apart from this, employment oriented skill development training program under ''Project Samarthya'' is also being organized to create more job opportunities for the youth and middle aged group.
			• Protection of environment and wildlife, conservation of forest	We will take adequate steps for protection of environment, forest conservation and the protection of wildlife.
			• Rehabilitation and	An attractive R&R package has been communicated to all the affected families, which is much better than

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
			resettlement of land losers	the package specified in Odisha R&R policy 2006 & the previous RPDAC. Beside this, a suitable R&R colony with high quality infrastructure is being planned.
12.	Sri Gopabandhu garnayak, Podapada	He informed that the mining area covering one side of Podapada area where mango trees, Mahua trees and cashew plants are there, and the local people are depending on these for their livelihood. He stated that, the water is flowing from north parts to south parts of Podapada village, if the mining activities will be carried out at the northern side of the Podapada villages then mining water will be flow towards the south direction and water logging will be takes place and local people will get affected.	• Livelihood • Direction of Flow	Employment opportunity will be provided to the affected people for their livelihood. To regulate the flow from areas outside the quarry pit that pass through the mine pit to downstream areas, shallow swale surface water diversion channels will be constructed around the excavation area of the mine. These channels are designed to effectively manage and control surface runoff by redirecting water away from active mining or quarrying zones, thereby preserving natural drainage patterns.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
13.	Sri Sanjaya Behera, Chhendipada	He alleged against the proposed project for not undertaking survey work directly and demanded that benefits should be given to all the land losers by the project proponent.	• Benefits to land losers	Land compensation payment as per RFCTLARR Act. The land rate has been communicated to landowners. An attractive R&R package has also been communicated to all the affected families, which is much better than the package specified in Odisha R&R policy 2006 & the previous RPDAC.
14.	Smt. Rojalin Dash, Korasahi	She supported the proposed mining project and stated that we need clean water and clean environment, and employment for local youths. Further, she informed that as per environment concern the Project proponent must carry out more plantation activities to reduce environmental pollution and steps shall be taken to protect the environment.	• Clean water and clean environment • Employment for Locals	We will provide safe drinking water facility in the local villages and will take all measures for environment protection. The project is likely to generate 1752 job opportunities. In addition to this, the project will create more indirect job opportunities. Apart from this, employment-oriented skill development training program under ''Project Samarthya'' is also being organized to create more job opportunities for the youth and middle aged group.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
			• Plantation activities • Protect the environment	We will take up massive plantation drives in the project and surrounding areas for better local environment. We will adopt environmental protection measures in proposed mining project for control of air pollution (PM, SOx, NOx), water pollution and soil pollution.
15.	Sri Udaya Nath Behera, Chhendipada	He opposed the mining project and stated that it will impact local environment, people, forest, wildlife, water bodies, and we will lose our own land and this coal mining will bring destruction to the local environment. Further stated that we have fundamental rights for forest, land and water and hence forest, wildlife and environment must be protected. Environment protection is our first priority	• Protection of Environment, forest and wildlife.	We will take adequate steps for forest conservation and protection of Environment & Wildlife. We have proposed Capital Expenditure of INR 10072.9 Lakhs and Recurring Expenditure of INR 315.7 Lakhs for implementation of the suggested Environment Management Plan such as Continuous Air Monitoring, SW & GW Monitoring, Solar lighting arrangement, Rainwater Harvesting, Soil Preservation, Green Belt

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
				Development, Ecological Conservation Measures, etc.
16.	Sri Jitendra Kumar Sahoo, Chhendipada	He opposed the mining project, and further alleged that the company and also district administration has not done any discussion with local people, so we are not ready to give our mother land to the company.	• R&R Issue	R&R process is already in discussion with the village people & it will be done as per R&R policy.
17.	Sri Bidyadhar Sahoo, Chhendipada	He informed that the proposed mining project will impact the local environment, forest and wildlife, rivers and natural nallahs nearer to proposed mining area	• Impact on local environment, forest and wildlife, rivers and natural nallahs	We will take adequate steps for forest conservation and will adopt preventive measures for the protection of local environment, wildlife, rivers & natural nallahs.
18.	Sri Sangram Majhi, Chhendipada Jungle	He stated that previously many industries have affected the local environment of the Angul district. Further stated the priority must be given for protection of environment, forest and wildlife. He opposed the proposed mining project and requested not to allow the industries for mining of coal and informed that	• Protection of Environment, forest and wildlife.	We will take adequate steps for forest conservation and for the protection of environment & wildlife. We have proposed Capital Expenditure of INR 10072.9 Lakhs and Recurring Expenditure of INR 315.7 Lakhs for implementation of the suggested Environment Management Plan such as

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		we don't want to give our own land for mining activities.		Continuous Air Monitoring, SW & GW Monitoring, Solar lighting arrangement, Rainwater Harvesting, Soil Preservation, Green Belt Development, Ecological Conservation Measures, etc.
19.	Sri Prabodh Swain, Angul	He opposed the project and stated that first priority should be focus on protection of environment, then for industry and mining. He also stated that environment must be protected.	• Protection of environment	Refer Point No. 18
20.	Sri Bijayananda Chaulia, Angul	He alleged that the summary of the mining project which has been uploaded in the Odisha State Pollution Control Board website in one language (Odia) only not in both the languages, whereas executive summary should be uploaded in both Odia and English, as per sl. 18, mine lease is for 37 years, they have not mentioned about including construction work or excluding construction work, number of family affected and number of family	• Summary of the mining project not uploaded in both the languages • Mine lease for 37 years is for including construction work or excluding	Odisha State Pollution Control Board uploads summary only in Odia Language for all projects in Odisha. Mine lease for 37 years is for including construction work.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		displaced was not mentioned in the executive summary and he also alleged that no details about number & types of machineries to be used for the mining and transportation purpose.	construction work • Number of family affected, and number of family displaced. • Types of machineries to be used for the mining and transportation purpose	Total Project Affected Families (PAF) is approx. 2664 Nos and Project Displaced Families (PDF) is approx. 1629 Nos. Types of machineries to be used for Opencast Mining are Surface Miner, pay loader & tipper and Shovel-Dumper combination for OB removal.
21.	Smt. Sasmita Behera, Chhendipada	She stated the proposed mining project will impact local environment, pollution will increase, affect socio-economic value, and cultural diversity etc. She also stated that previously many industries and mines has been established over the Angul area, however they are not providing employment to the -local	• Impact local environment, pollution will increase, affect socio-economic value, and cultural diversity	We will adopt preventive measures for the protection of local environment. We have proposed Capital Expenditure of INR 10072.9 Lakhs and Recurring Expenditure of INR 315.7 Lakhs for implementation of the suggested Environment Management Plan such as Continuous Air Monitoring, SW & GW

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		youths, like that this proposed mining project will not provide an employment opportunity to the local youths of the affected areas.	• Employment for Locals	Monitoring, Solar lighting arrangement, Rainwater Harvesting, Soil Preservation, Green Belt Development, Ecological Conservation Measures, etc. The project is likely to generate 1752 job opportunities. In addition to this, the project will create more indirect job opportunities. Apart from this, employment-oriented skill development training program under ''Project Samarthya'' is also being organized to create more job opportunities for the youth and middle-aged group.

Table 7:2 Action plan to address issues raised in public hearing as per MoEF&CC O.M. dated 30/09/2020

S. No.	Issue Raised during PH	Physical activity and action plan	Budget	Year wise Breakup		
				1st Year	2nd Year	3rd Year
1.	Development of Local Area	Formulate village development program under consultation with local panchayat and district administration for need-based community development activities which would be in addition to the development plans being undertaken by state and central government.	The various activities for development of local area shall be undertaken by GMDC under its CSR with provision of adequate budget. GMDC had already undertaken various activities (summarized under) for improvement of Rural Infrastructure, improvement of Community health, conservation of Arts & Culture etc in the region. **Development of Local Area covered under CSR Budget.** ➢ Help in the improvement of village roads ➢ Construction/Renovation of Sabhagruha ➢ Construction/Renovation of Community Hall ➢ Construction/Renovation of Rahaswadi ➢ Construction of Mandap ➢ Community Infrastructure ➢ Mobile Medical Unit (MMU) services ➢ Establishment of Physiotherapy Centre			

S. No.	Issue Raised during PH	Physical activity and action plan	Budget	Year wise Breakup		
				1st Year	2nd Year	3rd Year
2.	Protection of water bodies	Embankment will be constructed all along the Kumbhira nallah within the mine lease (ML) area. Plantation will be done along the embankment of Kumbhira nallah.	**1000 Lakhs** ➢ Construction of Embankment: 1000 Lakhs ➢ Plantation along the embankment of Kumbhira nallah: **Budget covered under EMP**	**335 Lakhs**	**335 Lakhs**	**330 Lakhs**
3.	Pollution Control	Truck Mounted Tanker with Mist Spray Sprinkling Arrangement for Haul Roads/Mine/Safety zone etc. Mobile Water Mist Spray Sprinkler for transport route. For control of pollution, massive planation drive and water sprinklers in affected villages will be done.	➢ Truck Mounted Tanker with Mist Spray Sprinkling Arrangement for Haul Roads/Mine/Safety zone etc.: **Budget covered under EMP** ➢ Mobile Water Mist Spray Sprinkler for transport route: **Budget covered under EMP** ➢ Plantation of trees/Greenbelt development: **Budget covered under EMP**			
4.	Skill Development	Willing and employable youths will be identified in consultation with gram	➢ Skill development of the local area shall be undertaken by GMDC under its CSR. A project name "Samarthya" has already			

S. No.	Issue Raised during PH	Physical activity and action plan	Budget	Year wise Breakup		
				1st Year	2nd Year	3rd Year
	/Employment to the local people	panchayat of nearby villages. Technical education & training will be provided to Local Youths Project Samarthya for Skill Development	been introduced by GMDC with employment-oriented skill development programs for the youths. The training programs like Assistant Electrician Training and General Duty Assistant Training are under progress in the region with **Budgetary provisions under CSR**			
5.	Compensation to land losers	R&R Package will be provided to the displaced families after the approval from RPDAC (Rehabilitation & Periphery Development Advisory Committee)	**Budget covered under R&R Plan**			
6.	Protection of wildlife	Wildlife protection & conservation shall be done. A comprehensive Regional Wildlife conservation plan (WLCP) is under preparation at DFO level. PP will deposit the proportionate amount as per the direction of Forest Department. Further Site-	**Budget covered under WLCP**			

S. No.	Issue Raised during PH	Physical activity and action plan	Budget	Year wise Breakup		
				1st Year	2nd Year	3rd Year
		Specific Conservation Plan has been prepared & approved by PCCF & Chief Wildlife Warden, Odisha. A budgetary provision of 37.26 Cr (Including 5% of the total financial outlay of the SSWCP to be deposited in Wildlife Society of Odisha) is earmarked for 10 years.				
7.	Concern for drinking water	Drinking water arrangement shall be made in the villages as per the requirement	**Budget covered under CSR**			
Total			**Rs. 1000 Lakhs**	**Rs 335 Lakhs**	**Rs 335 Lakhs**	**Rs 330 Lakhs**

Action Plan for Written Representations

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
22.	Balipata villagers, Manglu sahoo, Sarat kumar Garnaik, Barun kumar sahu, Chandrakanta Garnayak, Pradeep swain, Lochan Pradhan, Basanta Barik, Golekha Pradhan	I am writing this letter to inform that as per letter no. 13285 dtd. 22.08.2024 by SPCB Odisha, due to large population in our village periphery area which will be damaged for the mining therefore we are opposing this public hearing. The displaced people of affected villages will be rehabilitated in clean and good environment locations whereas we the people of periphery villagers will be deprived from these; therefore, GMDC should declare first that what steps have been taken to provide clean air, water, forest & good environment for the affected areas. We have no objection to Baitarani Open cast mining if prior steps are taken for its permanent solutions to these issues. If our periphery affected villages have to face Toxic climate, severe water shortages,	• Pollution Control	Air quality, water quality, and forest environment will be regularly monitored, and reports will be submitted in compliance with the norms of the CPCB and MoEF&CC. All efforts will be made to ensure that these parameters remain within the stipulated standards. We have proposed Capital Expenditure of INR 10072.9 Lakhs and Recurring Expenditure of INR 315.7 Lakhs for implementation of the Environment Management Plan. Hydrological studies have been conducted, and the stage of groundwater extraction stands at 54.93% (Safe). Pumping operations will only be carried out after obtaining the necessary permission from the Central Ground Water

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		difficulties in farming in future due to mining, in such case we will not hesitate to make strong oppose in future. Therefore, we are requesting you to kindly consider our grievances for better solutions before commencement of mining.		Authority (CGWA). Conservation measures like rainwater harvesting, use of rainwater/seepage water shall be used for internal consumption etc.
23.	Mr. Bijayananda Chaulia, Hakimapada, Angul	Coal Mining is a high-risk industry specially, in the case of opencast mining Where, a mineral itself is part of the surface soil or rock or fairly close to the surface. Even though, mining is regarded as a crucial economic activity worldwide, it has a significant negative impact on environment that inevitably leads to serious degradation on ecological and aesthetic values of the landscape. Coal mining causes extensive	• Degradation to natural ecosystems	Massive plantation drives in the project (about 25,74,000 nos. of native species is envisaged over a period of 37 years of Mine Life in backfilled area, safety zone and infrastructure) and surrounding areas for better local environment

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		degradation to natural ecosystems such as forests and can scar the landscape irreparably. Damage to humans, animals and plants, too occurs due to habitat destruction and environmental contamination.		will be done. Rs 2474 Lakhs (Capital Cost) earmarked for the same. A comprehensive Regional Wildlife conservation plan (WLCP) is under preparation at DFO level. PP will deposit the proportionate amount as per the direction of Forest Department. Further Site-Specific Conservation Plan has been prepared & approved by PCCF & Chief Wildlife Warden, Odisha. A budgetary provision of 37.26 Cr (Including 5% of the total financial outlay of the SSWCP to be deposited in Wildlife Society of Odisha) is earmarked for 10 years.
		Impacts of mining activities on environment and climate change Land, in various forms, is a finite natural resource. The necessity of land is ever increasing due to rapid population growth. The irony of the	• Adverse impact of opencast mining on land	Backfilled areas will be reclaimed biologically and technically. At the initial stage scrapped topsoil will be stored separately and reused as early as possible to layover the

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		situation is nothing but the adverse impact of opencast mining on land is soil contamination or Soil destruction. In the course of removing the desired mineral material, original soil become lost or buried by wastes. Opencast mining operations involve the removal of huge quantities of overburden, dumping, and backfilling of the excavated area. These overburden deposits of waste material containing low-grade ores are represented as mine spoil dumps. The gradual increase in such landscapes may endanger the agro-forest productivity as well as the ecological balance of the area in particular. Further, the generated waste rocks and tailings in huge quantities will become sources of pollution.		backfilled area for biological reclamation. Proper afforestation/plantation is to be carried out for greenbelt development. An area of 1029.583 ha (25,74,000 nos. of native species) will be designated for plantation, as detailed in the EIA/EMP. Rs 2474 Lakhs earmarked for the same.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		Moreover, it has been recently declared by the United States Environmental Protection Agency (US EPA) that the imperfect management of wastes produced in the course of mining and reclamation works is detrimental to environmental and human health. The effect of wastes due to mining and processing activities on ecosystems can be observed in groundwater, surface water, and soil and the following points may put human health in danger (Passariello et al. 2002): A. inhalation of aerosols containing high levels of metals. B. percutaneous absorption following skin contact. C. Use of contaminated water. D. Consumption of food from contaminated areas.	• Protection of Environment	We will take adequate steps for protection of Environment. We have proposed Capital Expenditure of INR 10072.9 Lakhs and Recurring Expenditure of INR 315.7 Lakhs for implementation of the Environment Management Plan such as Continuous Air Monitoring, SW & GW Monitoring, Solar lighting arrangement, Rainwater Harvesting, Soil Preservation, Green Belt Development, Ecological Conservation Measures, etc. All necessary precautionary measures, including the installation of Garland drains, the use of geotextiles, and other relevant measures outlined in the Environmental Impact Assessment (EIA) and Environmental Management Plan (EMP), will be strictly implemented to prevent water contamination. Post EC monitoring will be done regularly as

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		E. The plying of heavy vehicles on the roads would cause substantial dust nuisance, generate serious noises and above all the area would be more prone to accidents and mishaps. In this backdrop, the above mining activities by GMDC if permitted, would be detrimental to our lives (the seven villagers), flora and fauna and the environment of our area. In view of the above, we therefore, strongly condemn the open caste mining by GMDC and as such vehemently oppose the public hearing. We further request your good self to kindly save the environment and thereby allow us (the simple gullible villagers) to remain in the midst of our natural habitat and the flora and fauna.		per the prescribed standards in consultation with the State Pollution Control Board. All operations will be carried out in strict adherence to the guidelines set by the Directorate General of Mines Safety (DGMS) and other competent authorities. Limited quantity of coal will be transported by road to meet the local demand. Bulk of the coal will be evacuated via rail. Further the movement of trucks will be minimized in peak traffic hours. Wheel washing system shall be installed at the exit point of mine. Overloading of transport equipment shall be prevented in order to stop Spillage. Trees will be planted all along the coal transportation road within mine lease area. A comprehensive Regional Wildlife conservation plan (WLCP) is under preparation at DFO level. PP will deposit the proportionate amount as

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		Such kind act of yours would be highly solicited.		per the direction of Forest Department. Further Site-Specific Conservation Plan has been prepared & approved by PCCF & Chief Wildlife Warden, Odisha. A budgetary provision of 37.26 Cr (Including 5% of the total financial outlay of the SSWCP to be deposited in Wildlife Society of Odisha) is earmarked for 10 years.
24.	Ms. Poorva Chaulia, Hakimapada, Angul	The proposed coal mines conducted public hearing at Chhendipada on 24th Sep 2024 without proper procedure. 1. Pallisabha was not organised before 2. Executive Summary uploaded in OSPCB site is only in Oriya, no English copy is available in site.	• Pallisabha was not organised before • Summary of the mining project not uploaded in	The public hearing was conducted in accordance with the EIA Notification 2006 and its subsequent amendments and as per the norms set by the Odisha State Pollution Control Board. Odisha State Pollution Control Board was uploaded the summary.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
			English language	
		3. No. of family to be displaced is not mentioned in Executive Summary	• Number of family affected, and number of family displaced.	Total Project Affected Families (PAF) is approx. 2664 Nos and Project Displaced Families (PDF) is approx. 1629 Nos mentioned in Chapter 7 of the EIA report.
		4. No. & make of machine to evacuate coal is not mentioned clearly	• No. & make of machine to evacuate coal is not mentioned clearly	List of machineries will be used for Opencast Mining-like Surface Miner, pay loader & tipper and Shovel-Dumper combination is mentioned in Chapter 2 of EIA.
		5. The genuine land losers were not allowed to participate in the Public hearing	• Land losers were not allowed to participate in	The public hearing was conducted in accordance with the EIA Notification 2006 and its subsequent amendments through the Regional Office of OSPCB Angul.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		Hence Public hearing shall be stood cancelled the GMDC should not be allowed to operate.	the Public hearing	
25.	Mr. Arjun, Chhendipada	The Public hearing organised today at Chhendipada of Baitarni West Coal mine of GMDC was a complete false & unethical 1. The Executive Summary uploaded in the OSPCB website has no proper information about the no. of family likely to displaced	• Number of family affected, and number of family displaced.	Total Project Affected Families (PAF) is approx. 2664 Nos and Project Displaced Families (PDF) is approx. 1629 Nos.
		2. There is no details about the no. & types of machinery to be used for the mining & transportation proposed.	• No. & make of machine to evacuate coal is not mentioned clearly	List of machineries will be used for Opencast Mining like Surface Miner, pay loader & tipper and Shovel-Dumper combination is mentioned in Chapter 2 of EIA/EMP.
		3. The Executive Summary should be uploaded in both English & Oriya.	• Summary of the mining	Odisha State Pollution Control Board was uploaded the summary.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
			project not uploaded in both the languages	
		4. The common land losers were not allowed to put their view in front of the committee. 5. Selected people by certain group were allowed to put their view. So the Public Hearing shall be cancelled & shall be organised in the affected villages.		The Regional Officer, OSPCB Angul has conducted the Public hearing as per the norms/requirements at suitable place, easily approachable to everyone.
26.	Mr. Kumar Sahoo, Chhendipada	We the people residing in nearly area of the proposed coal mining sites bring the following few line for your kind attention that the proposal coal mines is not at all environmental friendly and violating all Human Rights Law. 1. That, a lot of transporting vehicle will be run on the road which will be life threatening	• Road traffic and safety	Limited quantity of coal (20%) will be transported by road to meet the local demand. Bulk of the coal will be

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		for the life of school, college going children as well as adults.		evacuated via rail. Further the movement of trucks will be through a dedicated coal evacuation road will not pass through any village. Wheel washing system shall be installed at the exit point of mine. The trucks shall be covered by Tarpaulin to avoid any spillage. Overloading of transport vehicles shall be prevented in order to stop Spillage. Trees will be planted all along the coal transportation road within mine lease area. The traffic rules as enforced by the DGMS will be strictly followed. A traffic study was conducted for the ODR Tentuloi-Chhendipada Road and SH-63 (Angul to Deogarh). The analysis revealed that the Level of Service (LOS) for the roads was found to be **B**, indicating a very good condition with minimal congestion and smooth traffic flow. Furthermore, the study also considered the potential impact of

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
				increased traffic from the Baitarni West Coal Block. However, it was determined that this contribution would not lead to any significant change in the existing Level of Service, meaning that the road conditions and traffic flow are expected to remain unchanged despite the additional transportation demands.
		2. That, there is a large patch of agricultural land on both side of the connecting road. The agricultural pad will be damaged due to the movement of transporting vehicle.		Adequate safety norms shall be adapted by the Mine.
		3. That, a lot of dust emission will caused due to flying of trucks on the approach road. 4. That, such activity will lead to the incident of theft,	• Dust emission	Dedicated coal transportation road shall be made Bituminous/Tar to connect the Mine with nearest main road for coal evacuation. Out of total 15 Million TPA, only 3 Million TPA will be dispatched by road. Coal will be

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		misbehaving with ladies and public nuisance will take place in the area. 5. That, it will lead to lowering of water level. 6. That, coal excerption and transportation has an adverse impact in biodiversity water turbidity, water table levels, land scape and on climate through carbon dioxide emission from transportation. Keeping all the above complained in mind, we request the authority not to allow the operation of coal mining as it will damage a lot to the life and liberty of the people living near the proposed site (Violating all	• Lowering of water level	transported using covered trucks to prevent dust and minimize environmental impact. Trees will be planted all along the coal transportation road. Hydrological studies have been conducted, and the stage of groundwater extraction stands at 54.93% (Safe). Pumping operations will only be carried out after obtaining the necessary permission from the Central Ground Water Authority (CGWA). Conservation measures like rainwater harvesting, use of rainwater/seepage water shall be used for internal consumption etc. A comprehensive Regional Wildlife conservation plan (WLCP) is under preparation at DFO level. PP will deposit the proportionate amount as per the direction of Forest Department. Further Site-Specific

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		environmental and Human Right Law)		Conservation Plan has been prepared & approved by PCCF & Chief Wildlife Warden, Odisha. A budgetary provision of 37.26 Cr (Including 5% of the total financial outlay of the SSWCP to be deposited in Wildlife Society of Odisha) is earmarked for 10 years.
27.	Mr. Ashok Kumar Sahoo, Chhendipada, Angul	Coal mines have impending dangers and risk for which a disaster management plan shall be prepared with an aim of taking precautionary steps to avert disasters and also take such action after the disaster, which is still not published or applied by the Govt. authority. The following natural and industrial problems will be encountered during the mining operation. 1. Inundation due to excessive rains.	• Disaster Management Plan • Inundation due to excessive rains	Adherence to relevant statutory safety provisions as stipulated by DGMS shall be made. A comprehensive disaster management plan (DMP) is prepared for mitigating risks, ensuring worker safety, and minimizing the environmental impact of accidents & same is discussed in Section 10.7 of Chapter 7 of EIA. Adequate pumping arrangements shall be made. Rainwater will be stored in mine pit & settling pond,

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		2. Accident by heavy machinery.	• Accident by heavy machinery • Slope failures	etc. which will be used for internal consumption. Following mitigative measures are suggested to reduce the impact by machinery: a) Formulation of safe operating procedure and implementing the same b) Creating safe working conditions, providing all safety measures, devices etc. c) Training the operator and others about the equipment, the safe operating procedures, vocational, safety training etc. d) Proper preventive maintenance check etc. Guarding of moving/rotating parts, preventing entry near moving parts, maintaining all guards in proper place.

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		3. Slope failures at the mines faces.		Regular inspections will be conducted to monitor potential slope failures on open cast mine faces, as well as on ore and overburden benches. In the event that any abnormal conditions are identified during these inspections, they must be promptly reported to the relevant department for immediate attention as well as corrective & preventive action.
		4. Accident of school going children and other people.	• Accident of school going children and other people	Speed limits, dedicated coal transportation route will be followed to avoid accidents.
		5. Member of family residing near (Ariel distance within 900 metre) the proposed site will suffer from various diseases due to sound and air pollution.	• Pollution control	We will adopt environmental protection measures in the proposed mining project for control of sound & air pollution. Massive plantation drives in the project (about 25,74,000 nos. of native species is envisaged over a period of 37 years of Mine Life in backfilled area, safety

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
				zone and infrastructure) and surrounding areas for better local environment will be done. Rs 2474 Lakhs (Capital Cost) earmarked for the same. Pollution levels will be strictly controlled and maintained in accordance with the standards and norms set by the Pollution Control Board.
28.	Mr. Nilubehbra, Chhendipada Angul	Due to coal mining activity in the area the dust emission will be caused which will affect the adjacent agriculture land and degradation of environment shall take place in the surrounding area. We are depending on agriculture. Our life and liberty will be in danger due to pollution. Due to this reason the said coal mining should not be allowed to operates.	• Pollution control	We will adopt environmental protection measures in the proposed mining project for control of pollution. Massive plantation drives in the project (about 25,74,000 nos. of native species is envisaged over a period of 37 years of Mine Life in backfilled area, safety zone and infrastructure) and surrounding areas for better local environment will be done. Truck Mounted Tanker with Mist Spray Sprinkling Arrangement, Mobile Water Mist Spray Sprinkler,

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
				fog cannon, etc. will be installed around the coal stockyard area, permanent haulage roads, and transportation routes to control dust and ensure effective dust suppression. We have proposed Capital Expenditure of INR 10072.9 Lakhs and Recurring Expenditure of INR 315.7 Lakhs for implementation of the suggested Environment Management Plan such as Continuous Air Monitoring, SW & GW Monitoring, Solar lighting arrangement, Rainwater Harvesting, Soil Preservation, Green Belt Development, Ecological Conservation Measures, etc.
29.	Biswamitra sahoo (Secretary), Gobrdhana Nayak (President),	We all the residents of the region through the Anchalika surakhya mancha Chendipada are informing that on the day of the said public hearing we came in protest of about 5000 women and men and stood		A public hearing was conducted in accordance with the EIA Notification 2006 and its subsequent amendments. The Additional District Magistrate chaired the public hearing

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
	Subasini Pradhan, Benudhar Gharnayak, Jhunu Behera, Dalimbha Majhi, Sudhkar Majhi, Saroj Majhi, Pratap Gadanaik, Sambhu Gadanaik, Shyam Sundar Pradhan, Upendra Gardnaik, Deba Garanaik, Sanjay Kumar Sahu, Souvagini Garnaik, Ichabati Nayak,	outside the platform. As per advertisement dtd 24/09/2024 for public hearing was held in connection with the environmental clearance of the proposed project at Chendipada was one sided. The affected people were not given a chance to enter and speak In the said meeting, the public hearing was quickly closed after the speeches of a few people including our M.L.A, District Councilor, Sarpanch and Samiti members. This is a gross insult to democracy in a democratic country. We regret to inform you that none of the villages were informed about obtaining the company license (GMDC) nor was any notification issued under section 7 of the Act and subsequently the company and the administration carried out arbitrary genocide without any discussion with the villagers to acquire the land. Some of the employees of the company are tempting simple villagers to sign Aadhaar cards, documents and signature in plain papers.		to ensure equal participation of all the stakeholders present. Kumbhira Nala and Supai Nala (Gurkha Nala) pass through the ML. Supai Nala will be diverted, while Kumbhira Nala will remain undisturbed. All necessary precautionary measures as mentioned in the EIA will be implemented to ensure the conservation of water resources. Sec 7 refers to Land acquisition, Land Rights have been obtained by Gujarat Mineral Development Corporation under Sec 11(1) of CBA & henceforth Sec 19 of CBA has been published. As per the forest record there are 29 numbers Schedule I Species namely Indian Elephant, Bengal Monitor, Indian Chameleon, Common Rat

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
	July Pradhan, Jamuna Maji, Jatak Maji, Ranjita Maji, Basanti Maji, Rita Maji, Sukanti Maji, Nibedita Maji, Duhita Maji, Basanti Maji, Joysna Nayak, Arati Maji, Mally Nayak, Abanti Maji, Dalimba Maji, Santi Dehury	Chendipada village and surrounding villages are a densely populated area and 90 percent of the total population will be affected. Apart from this, Angul district has been declared as one of the hottest regions in the country due to the number of factories and mines that have been built in the entire region. There is a passage for animals such as Elephants, bears, Monkeys and there are two water projects in this Chendipada jungle, the Kumbhira Canal Project and the Gurkha Canal Project, for which a lot of agricultural land will be lost due to the irrigation. Therefore, we respectfully request you to consider this matter carefully and provide us with proper justice. we will be grateful for you.		Snake, Common Indian Krait, Indian Cobra, Russell's Viper, etc. listed in the study area. A comprehensive Regional Wildlife conservation plan (WLCP) is under preparation at DFO level. PP will deposit the proportionate amount as per the direction of Forest Department. Further Site-Specific Conservation Plan has been prepared & approved by PCCF & Chief Wildlife Warden, Odisha. A budgetary provision of 37.26 Cr (Including 5% of the total financial outlay of the SSWCP to be deposited in Wildlife Society of Odisha) is earmarked for 10 years.
30.	Biswamitra sahoo (Secretary), Gobrdhana	We, the 70 members of the community of Chendipada Tehsil Chendipada Jungle Village, Police Station No. 75, District Angul, declare that, as per the advertisement, a public hearing was		A public hearing was conducted in accordance with the EIA Notification 2006 and its subsequent amendments. The Additional District Magistrate chaired the public hearing

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
	Nayak (President), Anchalika Surakhya Mancha, Chhendipada	held in Chendipada on 24/09/2024 in the context of the environmental clearance of the proposed project was one sided. The affected villagers were not allowed to enter the stage and not given the opportunity for speeches. In the said meeting, the public hearing was quickly closed with response of a few people including our M.L.A, District Councilor, Sarpanch and Samiti members. Apart from this, Angul district has been declared as one of the hottest regions in the country due to the number of factories and mines that have been built in the entire region. There is a passage of wild animals like elephants, bears, monkeys and there are two water projects in this Chendipada forest area, namely the Kumbhira nala project and the Gurkha canal project, for which a lot of agricultural land will be lost due to the irrigation. So we all public through Anchalika Surakhya Mancha, Chhendipada opposing this public hearing & we		to ensure equal participation of all the stakeholders present. There are no ecologically sensitive areas within the Mine Lease (ML) area. Kumbhira Nala and Supai Nala (Gurkha Nala) pass through the ML. Supai Nala will be diverted, while Kumbhira Nala will remain undisturbed. All necessary precautionary measures as mentioned in the EIA will be implemented to ensure the conservation of water resources. As per the forest record there are 29 numbers Schedule I Species namely Indian Elephant, Bengal Monitor, Indian Chameleon, Common Rat Snake, Common Indian Krait, Indian Cobra, Russell's Viper, etc. listed in the study area. A comprehensive Regional Wildlife conservation plan (WLCP) is under preparation at DFO

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
		will grateful for you if you could consider this & hold the public hearing. And we should be given full protection in a democratic nation.		level. PP will deposit the proportionate amount as per the direction of Forest Department. Further Site-Specific Conservation Plan has been prepared & approved by PCCF & Chief Wildlife Warden, Odisha. A budgetary provision of 37.26 Cr (Including 5% of the total financial outlay of the SSWCP to be deposited in Wildlife Society of Odisha) is earmarked for 10 years.
31.	Sri/Srimati Udayanath Dehuri father Chaitan Dehuri, Village Chendipada, Post Chendipada, Thana Chendipada, Tehsil	The proposed open cast coal project is a risky project Our crops, vegetables and fruits live in a peaceful environment with humans, and are a transit route and habitat for animals and birds. The above-mentioned are very fertile and is used for agricultural purposes for many crops including paddy, wheat, maize, pulses and a variety of vegetables. I demand the withdrawal of proposal issued for land acquisition and environmental clearance or the establishment of the coal project without taking my		Mining operations will commence only after obtaining the necessary statutory clearances, including Environmental Clearance (EC), Forest Clearance (FC), Consent to Establish (CTE), Consent to Operate (CTO), and other applicable approvals. All mining activities will be carried out in accordance with the Mine Plan approved by the MOC. Land Rights have been obtained by Gujarat Mineral Development

S. No.	Name and address of the Participants	Question/Issues Raised	Category of Issue Raised	Commitment by Project Proponent
	Chendipada, District Angul	opinion. I cannot transfer this land for coal mining at any price. Therefore I informed you to take appropriate action		Corporation under Sec 11(1) of CBA & henceforth Sec 19 of CBA has been published. An attractive R&R package has been communicated to all the affected families, which is much better than the package specified in Odisha R&R policy 2006 & the previous RPDAC. The package includes cash compensation in lieu of homestead land and structures. Rs. 727.94 crores have been earmarked for R&R.

Table 7:3 Summary of Amount Expenditure

Sr. No.	Activities	Total Proposed Budget for PH commitments (in Lakhs)	Amount already spent under CSR (in Lakhs)
1.	Development of Local Area (Rural infrastructure, Community Health, Art & Culture, Education & Sports etc.)	-	410.22
2.	Protection of water bodies	1000	-
3.	Pollution Control	-	-
4.	Skill Development /Employment to the local people	-	67.39
5.	Compensation to land losers	-	-
6.	Protection of wildlife	-	-
7.	Concern for drinking water	-	4.3
	Total	**1000**	**481.91**

Budget/ Scheme for the Public Hearing Commitments

As per Ministry's OM dated 30.09.2020, the Ministry has decided to deliberate on the commitments made by the project proponent to address the issues raised in the public hearing held on 24.09.2024. Hence, the activities proposed by M/s. Gujarat Mineral Development Corporation Ltd (GMDC) to address the issues raised in the public hearing vis-à-vis budget for the same.

Total budget for implementation of the activities is INR 1000 Lakhs. The overall estimated project cost of the proposed mine is Rs. 3855.00 Crores.

7.2 RISK ASSESSMENT AND DISASTER MANAGEMENT PLAN

The Disaster Management Act, 2005 defines the term "disaster" as under:

"Disaster" means a catastrophe, mishap, calamity or grave occurrence in any area, arising from natural or manmade causes, or by accident or negligence which results in substantial loss of life or human suffering or damage to, or destruction of, property, or damage to, or degradation of environment, and is of such a nature or magnitude as to be beyond the coping capacity of the community of the affected area".

Further, the Disaster Management Act, 2005 defines the term "disaster management" as under "Disaster management "means a continuous and

integrated process of planning, organizing, coordinating and implementing measures which are necessary or expedient for:

i. Prevention of danger or threat of any disaster;
ii. Mitigation or reduction of risk of any disaster or its severity or consequences;
iii. capacity-building;
iv. preparedness to deal with any disaster;
v. prompt response to any threatening disaster situation or disaster;
vi. assessing the severity or magnitude of effects of any disaster;
vii. evacuation, rescue and relief;

The objective of the Risk Assessment and Disaster Management Plan (RADMP) is to describe the Project Proponent's emergency preparedness capability, resources mobilization capability and capability to respond to various types of exigencies that could take place in an event of happening of any disaster at the Mine Site.

The Disaster Management Plan for Baitarni Coal Mine described in this chapter covers the hazard identification due to various activities, the risk involved due to the hazards and the necessary mitigative measures for the same.

As far as this mining project is concerned, the risk involved is comparatively less mainly because the method of operation is opencast mining. A detailed study has been conducted in existing mine. Identification and assessment of risk and recommendation of measures to prevent damage to life property against such risks has been made. The major risks involved are:

- Operation of main mining & other equipment
- Accidents due to fire, electrical hazards
- Flooding of mine/mine inundation
- Accidents due to other activities

7.3 OPERATION OF MINING AND OTHER EQUIPMENT

The accident due to the operation of mining and other equipment may be due to various reasons as per **Table 7.4** given below**.**

Table 7:4 accident due to the operation of mining and other equipment

S. No.	Hazardous Activity	Associated Hazards
1	Removal of OB & Mining of Coal using HEMM	Accidents due to operation of HEMM
		Failure of Bench/Pit Slope
		Inundation of Working during monsoon
		Accidents due to improper connections/handling of explosives

2	Drilling & Blasting	Accidents due to improper storage of explosives
		Fly Rocks due to blasting
3	Loading/Unloading &Transportation of Coal & Overburden	Accidents due to operation of HEMM
4	Mining Activity & Overburden Dumping	Accidents due to Slope Failure

7.3.1 RISK ASSESSMENT OPERATION OF REMOVAL OF OB & MINING OF COAL

The operations for removal of OB and mining of coal shall involve use of HEMM like Hydraulic Excavators, Rear Dumpers, Trucks, Surface Miners, Dozers etc. along with a number of ancillary equipment. During continuous operations of the HEMM mechanical failures and/or human errors may result in disastrous accidents.

Further, the Mine shall be intersecting the water table and local aquifer and there will be in flow of ground water into the working during the operations. Also, during monsoon season the rare chances of heavy surface run off entering into the workings. Such a situation may result in inundation of the mine workings unless proper precautions are taken.

7.3.2 RISK ASSESSMENT FOR DRILLING & BLASTING

The Project envisages some drilling & blasting also for removal of overburden. About 93 % of coal will be produced through surface-miner and balance 7% of coal will be produced through drilling-blasting.

Operation of diesel/electrically operated Heavy Duty Drills is a skilled work and if not done properly may result in accident. Further, most of the accidents from blasting occur due to fly rocks mainly due to overcharging of the shot holes. Overcharging also creates ground vibrations which may lead to displacement of adjoining areas. Dust and noise are also problems commonly encountered during drilling and blasting operations. Further, wrong connections, improper handling and storage of explosives may also result in an accident.

Site mixed slurry (SMS) has been proposed to be used for good fragmentation. For transportation of explosives, explosive van of approved type is also envisaged. For proper blasting and minimizing the adverse side effects due to blasting viz. noise, ground vibration back-breaks, air blast and fly rocks etc., the optimal blast design parameters are suggested to be used, after field trials.

Adherence to relevant statutory safety provisions as stipulated by DGMS, Chief Controller of Explosives and others competent authorities shall be made.

7.3.3 RISK ASSESSMENT FOR THE OPERATIONS OF LOADING /UNLOADING & TRANSPORTATION OF COAL

The Project involves a Targeted capacity of 15 MTPA with associated OB Removal. The operation of loading, transportation and unloading of such huge quantities at the designated place shall involve use of HEMM. The operation of HEMM is skilled work and if not done properly may result in an accident. Synchronization of the HEMM Fleet to handle such volume of material needs special attention. Most of the accidents, during operation of the HEMM, occur due to oversight or negligence of the operator while operating or reversing the machine during operations.

7.3.4 RISK ASSESSMENT FOR MINING ACTIVITY & OVERBURDEN DUMPING OPERATION

Mining activity and Overburden dumping involves operation of various machineries and equipment including HEMM mishandling of which may result in accidents, Slope failure etc.

7.3.4.1 ACCIDENT DUE TO SLOPE FAILURES

The stability of slope mainly depends on

a) Design of the slope
b) Petrographic composition
c) Jointing and bedding places of the rock
d) Extent of which it is saturated with water

The risk associated with failure of slopes may be mitigated by adoption of preventing measures as detailed below:

- Due to the high depth of the quarry, 140.25 Mcum of OB has to be dumped externally. Due to non-availability of non-forest land in the vicinity, it is proposed to tentatively locate the external dump on the south of quarry area.
- External dump is suitably planned by matching the overburden removal and void available in different stages. Internal dumps will be formed in 30m tiers with an overall slope of about 26 degrees.

For better stability of internal dumps, it is suggested to rip the mine floor in strips before backfilling. It is suggested to level the dumps and grade them outward properly to obviate water accumulation.

The following precautions will be taken to reduce the risk of dump failure:

1. OB benches will be made of <30m ht in each tier.
2. Active dump slope of Individual tiers will be maintained at 37 degrees, resulting an overall slope of around 26 degrees with horizontal.
3. Soil should be scrapped separately, so that it is not mixed in OB rock.
4. The slope of the ground is mild and will have no adverse effect.
5. The soil from the foundation ground should be scrapped before starting OB dumping.

6. Garland drains to be made around OB dump area to avoid water flow during monsoon below the OB dump.
7. Ground water table is generally 3-5m below ground level hence may have no adverse impact.
8. Leveling, grading and drainage arrangement for top of OB dumps will be done.
9. Technical & biological reclamation will be done.

The exposed ends of the coal seams and OB shall be left with a safe slope to avoid slope failure and collapse of benches.

Similarly, at the end of mining operation, safe terminal pit slope will be provided to avoid pit failure. Detailed site-specific tests for slope stability should be carried out and site-specific parameters determined. Continuous slope monitoring should be done for mine slope & dump slopes with proper devices during actual mining operations.

7.3.4.2 ACCIDENT DUE TO OPERATION OF MACHINERIES

Accident due to operation of mining machineries may be due to various reasons such as:

- Improper operation of equipment
- Improper Condition of the equipment
- Inadequate knowledge, training and lack of safety regulation
- Unauthorized and mal operation
- Danger of being caught in the moving parts

The consequent effect due to these accidents may result in equipment damage, production loss, injury and loss of life. The following mitigative measures are suggested to mitigate the above impact:

a. Formulation of safe operating procedure and implementing the same
b. Creating safe working conditions, providing all safety measures, devices etc.
c. Training the operator and others about the equipment, the safe operating procedures, vocational, safety training etc.
d. Proper preventive maintenance check etc.
e. Guarding of moving/rotating parts, preventing entry near moving parts, maintaining all guards in proper place.

7.3.4.3 ACCIDENTS DUE TO FIRE

Fires in Coal Blocks may occur due to various reasons such as

1. Spontaneous combustion of coal
2. Electrical short-circuiting
3. Accidental or open fires

1. Spontaneous Combustion of Coal

Coal has an extraordinary character by which it absorbs atmospheric oxygen and gets heated if exposed for longer periods greater than the incubation period. Fires

due to spontaneous heating can occur in an opencast Coal Block if exposed for a long time. Coal benches have prominent cleats or over breaks from blasting or faults nearby permitting establishment of an air flow circuit of broken coal remains in heap on top of coal benches for longer periods.

There may be instants of fire due to spontaneous combustion in coal dumps and in rare cases in OB dumps. In coal dumps, exposed coal is stored for longer periods, which may lead to spontaneous heating thereby promoting fires.

2. Fires due to electrical short circuiting

Electrical short circuiting may occur in any equipment /industry using electricity. This occurs mainly due to improper wiring, overheating of cables etc. These fires range from simple sparking to major fires involving the whole machinery.

3. Accidental or Open Fires

Accidental or open fires are chemical reactions where a fuel reacts with atmospheric oxygen. The fuel may be in the form of any combustible material like coal, diesel, lubricating oil, cotton waste etc.

The presence of fuel and oxygen will lead to fire only when the reactants are heated to the ignition temperature. These factors viz. fuel, oxygen and heat constitute the combustible triangle and all the three must be there to start a fire or for burning to continue. The main danger associated with fires in mine apart from human loss comes from the use of HEMM as well as large electrical equipment and appliance in the mines by way of damages to tyres, hydraulics hoses, power cables etc. In certain cases, costly machinery like dozers, loaders, dumpers, shovels etc. may be totally damaged by accidental fires.

Precautions to be taken in dealing with fires

As already discussed, to start a fire or for burning to continue, fuel, oxygen and heat are required. Preventive as well as fire-fighting measures are based on the principle of removing one or more of the factors in the combustion triangle.

The precautionary measures may include:

1. Providing Pipe ranges from high head pumps installed in the sump to enable supply of water at high pressure near fire prone areas.
2. Phasing the production to dispatches thus keeping a minimum coal stock.
3. Proper maintenance of electrical cables and other wiring and periodic inspection of the same.
4. Constant sprinkling of water over coal dumps to quench fires if any and to cool down the hot zones.
5. Preventing to the maximum extent, contamination of coal with waste material and in Unavoidable circumstances, through mixing of coal from seams less than 0.5 m thickness with overburden material whereby the concentration of coal is reduced.

7.3.4.4 FLOODING OF MINE/MINE INUNDATION

Water inflow in opencast pits may be due to

- Direct rainfall in the mine pit
- Seepage from the strata, backfilled area
- Flooding of nearby natural drainage system and overflowing into the mine area.

The major problem posed due to flooding are

a) Drowning of men and machinery
b) Collapsing of mine benches
c) Damages to roads and ramps
d) Contamination of exposed coal

This ultimately affects productivity of the mine. The following preventive control measures will be carried out to avoid/prevent flooding of mines.

- Providing adequate de-watering facility comprising a battery of large capacity pumps to pump direct rainfall and seepage water to keep the working areas and haul roads dry as far as possible.
- Maintaining a surface plan with contours showing particularly low-lying areas, lakes water courses, bore holes, well etc.
- Leave sufficient barrier along natural drainage
- Construction of drain all around the active mine area to prevent any surface run off water from entering the pit.
- Creating excess pumping capacity to take care of additional pumping necessities from the mine sump during monsoon.

7.3.4.5 ACCIDENTS DUE TO OTHER ACTIVITIES

Besides accident may occur due to other activities like blasting, rolling of boulders, noise hazards, Dust hazards etc.

The effects due to these hazards can be injury, damage to equipment, occupational health hazards, loss of life etc.

By adopting proper blasting techniques, training, adopting proper noise, dust control measures, these effects can be controlled. To prevent people falling in the pits and quarries near settlements and roads should be fenced off.

7.4 DISASTER MANAGEMENT PLAN

A Disaster Management Plan is aimed to ensure safety of life, protection of environment, protection of installation, restoration of production and salvage operations in the order of priorities.

For effective implementation of the Disaster Management Plan, it shall be widely circulated and training of personnel along with rehearsals/mock drills shall be organized.

The Disaster Management Plan aims to:

1. Organize the rescue operation and provide medical treatment to injured
2. Minimize damage to property and the environment
3. Contain and ultimately bring the accident under control
4. Provide authoritative information to the relatives and news media
5. Secure the safe rehabilitation of affected area
6. Preserve relevant records and equipment for the subsequent inquiry into the cause and circumstances of the emergency
7. Optimize operational efficiency for rehabilitation and render medical help and restore normalcy at the earliest.

The Disaster Management Plan needs to outline various actions to be undertaken in pre-disaster and post-disaster situations. The same has been detailed below: -

7.4.1 PRE –DISASTER ACTIONS

This stage occurs when prior information is available about a situation that may lead to a disaster in near future: Organizing Employee / Public Awareness Programmes is very important. The workers / employees as well as general public need to be informed about potential hazards likely to occur in the mining area. Emphasis may be given to the following aspects:

- Pamphlets and booklets containing details Do's & Don'ts in the event of crisis /emergency situations and hazards associated with coal mining be prepared and be made available.
- Permanent notice boards may be fixed at all the suitable places in the area displaying information, maps, escape routes, precautions to be taken and emergency communication details of supervisor.
- Help from local youth organizations, voluntary organizations, educational institutions may be sought to conduct educational sessions to make people aware about the safety measures and rescue operations in the event of a disaster.

7.4.1.1 EMERGENCY ORGANIZATION

An Emergency Organization is to handle the situation in case of happening of any accident/disaster. A Senior Executive, who has control over the activities of the mine, would be heading the Emergency Organization. He shall be designated as **Site Controller**. A Mine Manager shall be designated as the **Incident Controller** (IC). The Incident Controller would constitute a Team responsible for controlling the incident. **Shift In-charge** would be the **Reporting Officer**, who would bring the incident to the notice of the Incident Controller and Site Controller.

In addition, **Emergency Coordinators** would be appointed who would be responsible for activities like firefighting, rescue, rehabilitation, transport and providing essential support services. For this purposes, security in-charge, personnel department, essential services personnel shall be engaged.

7.4.1.2 EMERGENCY COMMUNICATION

Whoever notices an emergency situation such as fire /accident etc. would inform his immediate superior and **Emergency Control Center (ECC).** The person on duty in the ECC would appraise the situation to the Site Controller. Site Controller would verify the situation from the Incident Controller of that area or the shift In-charge and take a decision about an impending on - site emergency. This would be communicated to all Incident Controllers, Emergency Coordinators. Simultaneously, the emergency warning system would be activated on the instructions of the Site Controller.

7.4.1.3 RESPONSIBILITIES OF KEY PERSONNEL DURING EMERGENCY

A. Site Controller

On receiving information about emergency, he would rush to emergency control center and take charge of ECC and assesses the magnitude of the situation on the advice of incident controller and decide:

i. Whether the affected area needs to be evacuated;
ii. Whether personnel who are at assembly points need to be evacuated;
iii. Declares Emergency and orders for operation of emergency siren;
iv. Organizes announcement by public address system about location of emergency;
v. Maintains a continuous review of likely development and assesses the situation in consultation with Incident Controller and other Key Personnel as to whether shutting the mine operation required and if evacuation of personnel is required;
vi. Directs personnel for rescue, rehabilitation, transport, fire brigade, medical and other designated mutual support systems locally available, for meeting emergencies;
vii. Controls evacuation of affected areas, if the situation is likely to go out of control or effects are likely to go beyond the mine boundary, informs the District Emergency Authority, Police, Hospital and seeks their intervention and help;
viii. Informs the statutory authorities;
ix. Gives a public statement if necessary;
x. Keeps record of chronological events and prepares an investigation report and preserves evidence; and

B. Incident Controller

i. Assembles the incident control team;
ii. Directs operations within the affected areas with the priorities for safety to personnel, minimum damage to property, environment and minimum loss of materials;
iii. Directs shutting down operations and areas likely to be adversely affected by the emergency;
iv. Seeks help of all key personnel;
v. Provides advice and information to the Fire and Security Officer and the Local Fire Services as and when they arrive;
vi. Ensures that all non-essential workers/staff of the affected areas evacuated to the appropriate assembly points, and the areas are searched for causalities;

vii. Preserve the evidences for any likely inquiry later into the cause and circumstances which caused do escalate to the emergency;
viii. Co-ordinates with emergency services at the site;
ix. Provides tools and safety equipment to the team members;
x. Keeps in touch with the team and advises them regarding the method of control to be used; and
xi. Keeps the Site Controller of Emergency informed of the progress of operation.

C. Emergency Coordinator –Rescue, Fire Fighting

i. On learning about emergency, rushes to ECC;
ii. Helps the incident Controller in containment of the emergency;
iii. Ensure fire pumps in operating conditions and instructs pump house operator to be ready for any emergency with stand by arrangement;
iv. Guides the firefighting crew i.e. firemen, trained mine personnel and security staff;
v. Organizes shifting of fire-fighting facilities to the emergency site, if required;
vi. Takes guidance from the Incident Controller for firefighting as well as assesses the requirements of outside help;
vii. Controls the traffic at the incident area;
viii. Directs the security staff to the incident site to handle emergency operations under his guidance and supervision;
ix. Evacuates the people in the mine or in the nearby areas as advised by Site Controller;
x. Searches for casualties and make necessary arrangements;
xi. Assembles evacuation team;
xii. Make arrangements for safety equipment for the members of the team;
xiii. Decides which paths the evacuated workers should follow; and
xiv. Maintains law and order in the area, and if necessary, seeks the help of police.

D. Emergency Coordinator-Medical, Mutual-Aid, Rehabilitation, Transport and Communication

i. In the event of failure of electric supply and internal telephones, sets up communication point and establishes contact with the Emergency Control Center (ECC).
ii. Organizes medical treatment to the injured and if necessary, shifts them to nearby hospitals;
iii. Mobilizes extra medical help from outside, if necessary;
iv. Keeps a list of qualified first aid providers of the factory and seek their assistance;
v. Maintains first aid and medical emergency requirements;
vi. Makes sure that all safety equipment's are made available to the emergency team;
vii. Assists Site Controller with necessary data in coordinating emergency activities;
viii. Assists Site Controller in updating emergency plan, organizing mock drills, verification of inventory of emergency facilities and furnishing report to Site Controller;
ix. Ensures transportation facility;
x. Controls rehabilitation of affected areas on discontinuation of emergency.

7.4.1.4 EMERGENCY FACILITIES

Emergency Control Plan

This will be situated in an area away from the places of fire and shall be provided with the following facilities:

a) Master plan of the mines
b) Portable gas detectors
c) First aid boxes
d) Gas masks
e) Telephone line with STD facility
f) Loud hailers
g) Emergency lighting system
h) Stretchers
i) Transport Facility

Assembly Points

A number of assembly points depending upon the mine location would be identified wherein employees who are not directly connected with the disaster management would be assembled for safety and rescue. Emergency breathing apparatus, minimum facilities like water etc. would be organized.

Communication System

Different types of alarms to differentiate types of emergencies will be provided. Alarms will be followed by an announcement over the public address system. In case of failure of the alarm system, communication will be by telephone operator who will make announcement in industrial complex through public address system which shall be installed. Walkie- talkie systems for communication are very useful during emergency.

Firefighting facilities

Fire-fighting equipment suitable for emergency should be maintained in each operation areas of the mine as per statutory requirements. It shall consist of:

a) Hydrant system for all fire prone areas of mine.
b) Automatic High velocity water spray systems for stockpile area.
c) Fire extinguishers to be placed near office buildings and other places which are prone to fire related hazards.
d) Fire detection systems at office buildings shall be installed.

Other Facilities

Location of Windsock: On the top of the administrative block, windsocks would be installed to indicate direction of wind for emergency escape.

Emergency medical facilities: Stretchers, gas masks and general first aid materials for dealing with chemical burns, fire burns etc. is to be maintained in the medical center as well as in the emergency control room. Medical practitioners

help would be sought. Government hospitals shall be approached for emergency help.

Ambulance: An ambulance with the availability of driver in all the shifts is to be ensured to transport injured or affected persons. A number of persons trained in providing first aid shall be deployed in each shift.

Mock Drills: Emergency preparedness is an important aspect of planning in Industrial Disaster Management. Personnel would be trained suitably and prepared mentally and physically in emergency response through carefully planned, simulated procedures. Similarly, the key personnel and essential personnel would be trained in the operations.

Important Information: Important information such as names and addresses of key personnel, essential employees, medical personnel, transporters address, address of those connected with Off Site Emergency such as Police, Local Authorities, Fire Services, District Emergency Authority shall be prepared and maintained.

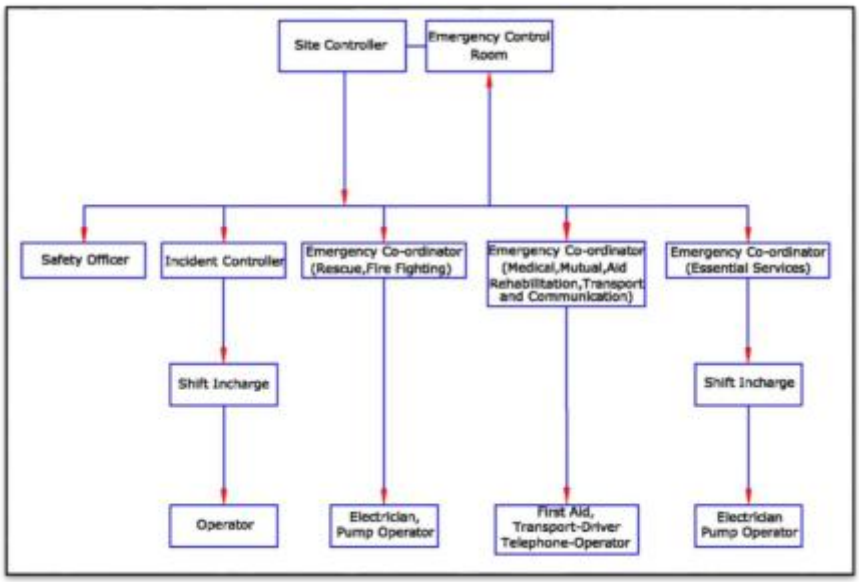


Figure 7-3 THE ON-SITE EMERGENCY ORGANIZATION CHART FOR VARIOUS EMERGENCIES

7.4.1.5 OFF-SITE EMERGENCY PREPAREDNESS PLAN

The task of preparing the off-site emergency plan lies with the District Collector. The main aspects which should be included in the off-site emergency plan are:

a) **Organization:** Details of command structure, warning systems, implementation procedures, emergency control centers, names and appointments of incident controller, site main controller, their deputies and other key personnel.
b) **Communication:** Identification of personnel involved, communication center, call signs, network, lists of telephone numbers.

c) **Specialized Knowledge:** Details of specialist bodies, firms and people upon whom it may be necessary to call e.g., those with specialized knowledge of fire control.
d) **Voluntary Organizations:** Details of organizers, telephone numbers, resources etc.
e) **Chemical Information:** Details of the hazardous substances stored or procedure on each site and a summary of the risk associated with them.
f) **Meteorological Information:** Arrangements for obtaining details of weather conditions prevailing at the time and weather forecasts.
g) **Public Arrangements and Information:** Transport, evacuation centers, emergency treatment of injured, first aid, ambulances, temporary mortuaries. Arrangements for communicating with the media and information to relatives, etc.
h) **Assessment of Emergency:** Arrangements for :(a) collecting information on the causes of the emergency; (b) reviewing the efficiency and effectiveness of all aspects of the emergency plan.

7.5 POST DISASTER ANALYSIS AND EVALUATION

When the emergency is over, the team will carry out a detailed analysis of the causes of the accident, evaluate the influence of various factors and take necessary measures for the future. At the same time, the adequacy of the Disaster Preparedness Plan will be evaluated and shortcoming if any will be rectified for subsequent improvement of the plan.

7.6 OCCUPATIONAL HEALTH AND SAFETY

Occupational Health and Safety aspects need more attention during the operational phase of the mine. The Coal Mining operations is invariably associated with the Occupational Health Issues.

7.6.1 IDENTIFICATION OF WORK-RELATED OCCUPATIONAL HEALTH HAZARDS

The problem of occupational health during the operation phase is primarily due to dust and noise which could affect breath and hearing of the working personnel. However, other incidental and ancillary activities associated with mining may also pose occupational health hazards to the personnel. The details of the activities and associated principal environmental and occupational risks are provided in **Table 7.5.**

Table 7:5 Details of the activities and associated principal environmental and occupational risks

S. No.	Hazardous Activities	Expected Hazards	Severity of Injury
1	Removal of Overburden	Exposure to high noise level	Hearing Problems
		Exposure to dust	Diseases related to dust

	and Mining of Coal		And free silica
		Exposure to accidents due to HEMM	Physical Injury of severe Nature
2	Drilling & Blasting	Exposure to high noise level	Hearing Problems
		Exposure to dust	Diseases related to dust And free silica
		Exposure to ground vibrations	Psychological Disorder
		Exposure to Fly Rocks	Physical Injury of severe Nature
		Exposure to accidents to explosives	Physical Injury of severe Nature
3	Loading/Unloading and Transportation of Coal & Overburden	Exposure to high noise level	Hearing Problems
		Exposure to dust	Diseases related to dust and free silica
		Exposure to accidents due to HEMM	Physical Injury of severe nature
4	Storage of Explosives and Petroleum Products	Exposure to accidents due to improper handling of explosives/ petroleum products	Physical Injury of severe nature
		Exposure to toxic fumes	Asphyxiation
		Exposure to accidents due to fire	Physical Injury of severe nature
5	Storage of Coal	Exposure to dust	Diseases related to dust and free silica
		Exposure to accidents due to Spontaneous Heating	Physical Injury of severe Nature and/or burns
6	Use of Electricity	Exposure to electrical shocks	Physical Injury of severe nature

7.6.2 MEASURES FOR MITIGATION OF OCCUPATIONAL HEALTH HAZARDS-PROJECT PERSONNEL

The issues associated with Occupational Health during the operational phase can be addressed by providing Personal Protection Equipment (PPEs) to the workers. Use of the following equipment is proposed as per requirement of the activity:

- Industrial safety helmet and crash helmets;
- Zero power goggles with cut type filters on both sides and blue color glasses;
- Welder's equipment for eye and face protection;
- Cylindrical type earplug;
- Earmuffs;
- Canister gas mask;
- Fluorescent Jacket;
- Safety belt/lineman's safety belt;
- Acid/alkali proof rubberized hand gloves;
- Electrically tested electrical resistance hand gloves; and
- Industrial safety shoes with steel toe.

7.6.3 MEASURES FOR MITIGATION OF HEALTH HAZARDS-GENERAL PUBLIC

The general public residing in the adjoining areas of the mine would also be subjected to issues related to Health Hazards. Considering the mitigative measures to be implemented during the operation of the mine, it is expected that the impact of mining on the population residing in the Study Area of 10 km shall be minimal. However, to monitor and also keep check on the issues associated with Occupational Health of the general public, the following measures shall be implemented.

i. Health checkups of residents near the Mine shall be carried out periodically. In case any person or a group of persons is found to be suffering from any ailment, directly related to the Coal Mining Operations, proper medical treatment shall be arranged.
ii. Surface Water Management shall be adopted to ensure that run-off from the mining area does not adversely affect natural water streams or other water bodies.
iii. All water bodies, e.g., wells and surface water sources in the vicinity of the mine, shall be periodically tested for any pollution related to mining operations and remedial action taken, if warranted.
iv. Operators of all transport vehicles shall be instructed not to honk unnecessarily while passing through villages or nearby schools.

7.7 RESETTLEMENT AND REHABILITATION

A separate study has been done for R & R.

Total 7 villages namely Chhendipada Jangal- (Part), Handigora- (Part), Machhakuta - (Part), Machhakuta Jangal- (Part), Porapara- (Part), Porapara Jangal - (Full), Tentuloi-Korasahi- (Part) are going to get affected by proposed project. Total Project Affected Families (PAF) is 2664 Nos and Project Displaced Families (PDF) is 1629 Nos (738 Original + 891 Extended).

Village wise PAFs/PDFs details are shown in the table below.

Name of the Village	No of PAFs	No of PDFs
Chhendipada Jungle	1166	429
Handigoda	215	551
Machhakuta	269	337
Machhakuta Jungle	87	0
Podapada	444	145
Podapada Jungle	197	43
TentuloiKodasahi	286	124
Total	**2664**	**1629**

Source: Field Survey

Year-wise Displacement

Sl. No.	Mining Year	No of Families going to be displaced
1	0-1 Yr	76
2	1- 3 Yrs	75
3	3- 5 Yrs	387
4	5- 8 Yrs	374
5	Above 8yrs	717
Total		**1629**

These displaced families will be resettled and rehabilitated socially, culturally and economically as per the latest R&R policy (2006 & subsequent amendments) of Govt.of Odisha. As on date site selection for the development of the rehabilitation colony has been completed and is awaiting RPDAC approval. Payment of R&R is going on. Detailed R&R plan is enclosed as **Annexure VII.**

7.8 SOCIAL IMPACT ASSESSMENT

According to the caste distribution of affected families, majority of the PAFs/PDFs belong to Other Backward Caste (OBC) followed by Scheduled Caste (SC), Scheduled Tribes (ST), and General Caste.

Table 7:6 Caste –wise distribution of PAFs/ PDFs (As per Socio Economic Survey)

Ethnic Group	No of PAFs	No of PDFs (Original)
SC	577	89
ST	203	55
OBC	1718	561

General	166	33
All	**2664**	**738**

Socio-Economic Status of Project in detail is described in R&R plan.

Statutory Obligations

The main legal obligations which the mine owner is bound to comply with, respect to mine closure, as per the special conditions imposed by the various authorities.

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CHAPTER-8 : PROJECT BENEFITS

8.1 INTRODUCTION

All the Projects have their own advantages and disadvantages. Normally, every Industrial Project has the potential to result primarily in economic benefits to the local community and also to the Government.

Upcoming of the project will result primarily in economic benefits to the local community and to the Government.

727.94 Cr is proposed for R & R implementation.

Estimated CSR cost is Rs. Approx. 1 Cr/Annum (2% of the net profit over the preceding three financial years of the organisation) which shall be used in sectors like healthcare, skill training, development of rural infrastructure etc.

GMDC has already contributed 481.91 Lakhs as on date 31.08.25 on the development of area.

There will be improvement in both Physical and Social Infrastructure.

- **Increase in Employment Potential, Contribution to the Exchequer (both State and Central Govt.).**
- **Post-mining Enhancement of Green Cover.**

Improvement of Electrical Power Generation and consequently rise in electric power consumption there by improvement in overall economic growth of the country

The project brings overall improvement in the locality, neighborhood and the State by bringing industry, roads, employment and hence improving living standard and economic growth.

The project will enhance the following physical infrastructure facilities in the adjoining areas. The project will give direct employment to people. Local people will be preferred for providing job opportunity. The mine management will recruit skilled, semi-skilled & unskilled workers. The overall effect will improve the buying power of employees up to some extent and thus a higher standard of living viz. better education, improved health and sanitation facilities, housing and acquisition of consumer durables.

Revenue Share, Royalty and other statutory dues will be paid to the government against the mineral so it will increase the exchequer of the government leading to improvement in economic condition of the area including the transport structure, medical infrastructure, educational infrastructure and improvement of other civic amenities and will get a gradual boost in future.

GMDC shall follow The Companies (Corporate Social Responsibility Policy) Amendment Rules, 2021 (GazetteNotification dated 22.01.2021).

8.2 SOCIAL BENEFITS

8.2.1 EMPLOYMENT

Total 1752 persons will be employed during operational phase, in the skilled, semi-skilled and unskilled category. Preference will be given to local population for employment in the semi-skilled and unskilled category.

Manpower requirement assessment has been carried out on the basis of prevailing norms of operation in Indian coal mine i.e. 3 shift operation.

Proposed manpower includes the following:

1) Operational manpower for coal winning
2) Maintenance manpower for equipment's
3) Project office, supervision & common manpower
4) Services and infrastructure facilities manpower
5) Environment & reclamation manpower

There will be a total manpower of 1752 shall be required for proposed production capacity of 15 MTPA out of this around 100 manpower will be from GMDC and around 1652 will be through MDO contractor.

Indirect employment around 5000 will be generated through handling of material, transport etc., which will improve social & economic standard of the people in the locality.

8.3 ENVIRONMENTAL BENEFITS

After mining of mineral from the area, the whole area will be reclaimed and will be planted with trees and some part of the void will be converted to water reservoir. Plantation and water conservation are the needs of this hour. It has been proposed that out of 1196 ha area, 1029.583 ha will be under plantation, 38.488 ha will be under water bodies. 64.3723 ha will be returned to forest dept. Remaining area is undisturbed area and area for public use.

The afforestation program is of paramount importance for Eco-friendly development. The proposed plantation program benefits the environment through supplementing oxygen to the atmosphere and combat air pollution effectively. It not only improves the aesthetic beauty and landscape resulting in harmonizing and amalgamating the physical structures of the mines with surrounding environment, but also acts as pollution sink. It also checks soil erosion, makes the ecosystem more complex and functionally stable and make the climate more conducive.

All measures will be adopted at each project development phase for environment management and have been described in previous chapters. Therefore, it is safe to say that the project is not likely to cause any significant impact on the ecology of the area, as adequate preventive measures will be adopted to contain the various pollutants within permissible limits.

Greenbelt & plantation would be developed in 1029.583 ha area preferring the local plant species by the end of life of mine.

8.4 IMPROVEMENT IN PHYSICAL INFRASTRUCTURE & COMMUNITY DEVELOPMENT

The proposed mine will improve the physical infrastructure of the adjoining areas. This would include the following:

- Improvement of road network for better communication.
- Strengthening of existing community facilities through the CSR
- Programmes being carried out by GMDC.
- Pumping of mine water may augment the water availability after due treatment.
- Gainful post-closure land utilization of mine lease area.
- Skill development & capacity building like vocational training, income generation programmes and enterpreneurship development program.
- Literacy programme, adult education, formation of Village Working Group (VWG), mahila mandal, etc.

Pre-Mining Land Use

SI. No	Type of Land	Within ML Area (ha)
a)	Agricultural	784.8499
b)	Forest	64.3723
c)	Barren	44.3739
d)	Grazing	118.8397
e)	Water bodies	35.1422
f)	Roads	31.4086
g)	Homestead	32.7729
h)	Community/ Other use	84.2405
Total		**1196.00**

8.5 IMPROVEMENT IN SOCIAL INFRASTRUCTURE

There would be some obvious changes in various environmental parameters due to mining activity, increased economic activities and creation of new employment opportunities, infrastructural development, and better educational and healthcare facilities. The following are the specific impacts.

Socio-Economic

Overall, there will be positive impact in socio-economic areas due to increased economic activities, creation of new employment opportunities, infrastructural development and better educational and healthcare facilities. The impact in the Core Zone and Buffer Zone is due to the following:

- **Population Dynamics**

Due to direct and indirect employment potential, there is scope of migration of people into project area and in the peripheral regions, from nearby areas. Mining activities, acceleration of the economic activities and urbanization along with

creation of new employment opportunities and business may change the population dynamics of the area.

- **Standard Of Living**

The people will come in contact with migrated people. This may encourage higher aspirations among the people of the area. Accelerated economic activities and urbanization may increase quality of life and standard of living.

- **Health Care Facilities**

GMDC will undertake awareness programme and community activities, like village health camps, medical aids, family welfare camps, eye camp, immunization camp, etc.

8.6 EMPLOYMENT POTENTIAL

There is a possibility of creation of direct and indirect employment opportunities due to working of this mine. Overall, this will have positive impact on socio-economic profile of the area.

GMDC will undertake skill development & capacity building programmes like vocational training, income generation and entrepreneurship development.

8.7 OTHER TANGIBLE BENEFITS

The State & Central Government will be benefitted directly from the proposed mine through increased revenue from royalties, excise duty etc.

8.8 CORPORATE ENVIRONMENT RESPONSIBILITY (CER)

Sustainable development has many important facets/components like social, economic, environmental, etc. All these components are closely interrelated and mutually re-enforcing. Therefore, the general structure of EIA document, under Appendix-Ill to the notification, prescribes inter-alia public consultation, social impact assessment and R&R action plan besides environment management plan (EMP).

Government of India through Ministry of Environment, Forest and Climate Change has issued an office memo vide F.No.22-65/2017-IA.III Dated: 30/09/2020 where it has issued the guidelines that the cost of Corporate Environmental Responsibility will be a part of EMP.

As per MoEF&CC's O.M dated 30.09.2020 (F.No. 22-65/2017-IA.III) it was decided that henceforth the Expert Appraisal Committee or State Level Expert Appraisal Committee shall deliberate on the commitments made by the project proponent to address the concerns raised during the public consultation and prescribe specific condition(s) in physical terms while recommending the proposal, for grant of prior environment clearance instead of allocation of funds under Corporate Environment Responsibility.

8.9 CORPORATE SOCIAL RESPONSIBILITY (CSR)

GMDC is committed to fulfill its obligation under Corporate Social Responsibility (CSR). Other than undertaking development works itself in the vicinity of GMDC Mines, huge amount of funds has also been placed with various district administrations for development activities such as Community Infrastructure, Mobile Medical Unit (MMU), Notebook Distribution, Drinking Water, etc. in the concerned districts. GMDC has recently formulated and adopted CSR Policy wherein upto 2% of the net profit of the Organization is set aside for CSR activities.

Estimated CSR cost is approx. Rs. 1 Cr/Annum (2% of the net profit over the preceding three financial years of the organisation) which shall be used in sectors like healthcare, skill training, development of rural infrastructure etc.

GMDC has laid well defined CSR Policy with a vision to improve the quality of life of the communities in the working area of GMDC.

The focus areas are:

- Water conservation and environment
- Micro Irrigation and Agriculture
- Health and Sanitation
- Education
- Employment Enhancement
- Rural Infrastructure

GMDC Gramya Vikas Trust (GMDC-GVT), *an initiative of GMDC Ltd. is presently working on various CSR projects towards improving the quality of lives of various communities in the operational area.*

GMDC has a dedicated CSR Budget wherein min. of 2% of the net profit over the preceding three years of the Organization is set aside for CSR activities. An estimated CSR cost is of approximately Rs. 1 Cr/Annum is likely to be used in sectors like improvement of Community Infrastructure, Safe Drinking Water, Skill Building, Sports, Art & Culture, Rural Infrastructure, Health, education and Focus on Renewable Energy etc.

Table 8:1 Proposed CSR Budget

Sr. No.	CSR Activities	Estimated Capital Cost (Rs. In Lakhs)	Recurring Cost (Rs. In Lakhs)
1.	Community/Rural Infrastructure	30	3
2.	Safe Drinking Water	5	1
3.	Skill Development	20	2

4.	Sports & Education	5	1
5.	Art & Culture	5	0.5
6.	Health	20	2
7.	Focus on Renewable Energy	5	0.5
	Total	**90**	**10**

Table 8:2 CSR Fund Utilized in Various Head

Sr. No.	**Activities**	**Amount Spent (Rs. Lakhs)**	**Name of the Villages**
1.	Art & Culture	40.67	Kuntanali, Podapada, Chhendipada, Machakutta, Handigoda, Kumbhar sahi, Rugudi Sahi, Harijan Sahi, Dolomandap Sahi.
2.	Community Infrastructure Development	4.62	Chendipada, Kuntanali, Machakutta, Podapada Jungle, Sapohi sahi, Maji sahi, Podapada, Nakki sahi, Kumbhar sahi, Patharmunda Sahi, Telisahi, Rugudu Sahi, Harijan Sahi, Handiguda, Jamunali, Chendipada Block H.Q, Angul District H.Q
3.	Education	16.19	Machhakutta, Chhendipada
4.	Healthcare	149.00	Dehury Sahi, Machakutta, Rugudi, Shai, podapada Jungle, Naki sahi, Nala Sahi, Handiguda, Jamunali, Kumbhar Sahi, Harijan Sahi, Machakuta Jungle, Gochayat Sahi, Chhendipada Jungle, Bantaburi Sahi, Sapoi Sahi.
5.	Others (Renovation of Police Statton, Installation of Solar Lights)	32.29	Chhendipada
6.	Rural Culture	1.10	Chhendipada
7.	Rural Infrastructure Development	136.17	Chhendipada, Podapada Jungle, Machhakuta, Podapada, Handigoda. Tentuloi Kodasahi
8.	Sports	9.83	Handigoda, Chhendipada
9.	Hearse Van	12.37	Donated to Chhendipada Gram Panchayat
10.	Skill Development Training under Project Samarthya	67.39	Chhendipada & Machhakuta
11.	Notebook Distribution	12.28	Govt School of Tentuloi Kodasahi, Handiguda, Kumbhar Sahi, Kuntanali, Machakuta, Podapada, Chendipada, Sapoi Sahi, Jamunali and Private School of Bantaburi Sahi, Sapoi Sahi, Forest office road, Bhalupala sahi.
	Total	**481.91**	



Figure 8-1 Photographs of CSR Activities Done

8.10 CONCLUSION

The management likely to be recruit highly skilled and semi-skilled workers from the nearby villages based on availability. The project proponent will also train the local people for suitable jobs. The project activity and the management will support the local Panchayat and provide other forms of assistance for the development of public amenities in this region. The company management will contribute to the local schools and dispensaries for the welfare of the villagers. A suitable combination of trees that can grow fast and have good leaf cover will be

adopted to develop the greenbelt. Overall, the proposed project will be beneficial in the development of the nearby area and enhancing the living standards of locals and with implementation of proper mitigation measures there won't be any significant impact on the ecology and environment due to adoption of scientific ways of coal mining and pollution control measures.

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CHAPTER-9 : ENVIRONMENTAL COST BENEFIT ANALYSIS

Total cost of the project is Rs. 3855.00 Cr. Cost of production is Rs.337.11 per tonne.

Environmental Cost-Benefit Analysis (CBA) is a systematic approach to evaluating the economic, social, and environmental impacts of a project or policy by comparing its total benefits against its total costs, including hidden environmental externalities.

CB Ratio = benefits/costs = 30.75

9.1 INTRODUCTION

The Cost-Benefit Analysis (CBA) is a Decision-Making Tool. It is widely used for evaluating a Project for its social and environmental impacts in terms of costs versus financial benefits it offers. Only when the benefits exceed the costs, the Project is expected to have Net Public Benefit and therefore can be recommended for implementation.

9.2 ENVIRONMENTAL COST BENEFIT ANALYSIS

Total cost of the project is Rs. 3855.00 Cr. Cost of production is Rs.337.11 per tonne. CSR cost is approx. Rs.1/Annum. R&R cost is Rs. 727.94 crores. NPV cost is 6.16 Cr & Compensatory Afforestation cost is 5.19 Cr.

9.3 INTRODUCTION

The cost estimate for EIA-EMP of this project is given below along with the break-up:

Table 9:1Cost Estimate of EMP

S. No	Activity	Capital Cost (in Lakhs)	Recurring Cost (in Lakhs)
1.	Truck Mounted Tanker with Mist Spray Sprinkling Arrangement (25m) for Haul Roads/Mine/Safety zone etc.	150	10
2.	Mobile Water Mist Spray Sprinkler for transport route	150	3
3.	Fixed type Mist spray at Coal stockpile/crusher/Transfer Points etc.	85	2
4.	Fixed type Mist Spray System (long range) at proposed Railway Siding	50	1
5.	Wind Barrier Wall & Vertical Greenery System at proposed Railway Siding	5	0.5
6.	Mechanized Sweeping of Roads Electrically Operated	30	1
7.	Dust arrester equipped with drilling machine	350	5.25
8.	Garland Drain	38	1
9.	Boulder Check Plug	5	0.075
10.	Mine Water Sedimentation Pond & Pumps	125	2
11.	Noise Level Indicator (Fixed)	24.5	0.3675
12.	Continuous Air Monitoring Station	150	10
13.	Piezometer for Hydrogeological Monitoring (including drilling)	50	0.75
14.	Vibration Monitoring Equipment	10	0.15
15.	SW&GW Monitoring	5	0.075
16.	Solar lighting arrangement	50	5
17.	Rainwater Harvesting	75	1.125
18.	Soil Preservation	17.5	0.2625
19.	Green Belt Development including Plantation over backfilled, other area within ML and Fencing for green patch.	2474	221.62
20.	Fire tender	37.5	2
21.	Advance Firefighting equipment, Fire extinguisher, smoke detectors (for office and workshop), firefighting Drones, fire Automatic	200	3

	Fire Detection and alarm system etc,		
22.	Personnel Protection Equipment (goggles, gloves, helmets, dust mask, safety boots)	5	2
23.	Topsoil Management	1026	1
24.	Awareness Programme (Display Boards (Digital) etc.	30	1
25.	Ecological Conservation Measures	50	2
26.	Barbed Wire Fencing for protection of void	94	0.375
27.	CCTV camera for monitoring loading and transport, mine blast, fire, dust generation monitoring,	10	0.15
28.	Equipment for Plantation O & M like water tanker, tractor with trolly, other equipment	50	2
29.	Biodiversity Conservation Measures	3726.40	10
30.	Action plan for Water Conservation (Supai Nala Diversion and Conservation, Construction of Kumbhira Nala Embankment, etc.)	1000	27
Total		**10072.9**	**315.7**

Total EMP Cost (Capital Cost) = Rs. 10072.9 Lakhs
Total EMP (Recurring Cost) = Rs. 315.7 Lakhs

Apart from the above there is also EMP cost already earmarked as the post closure mine cost.

S. No	Activity	Capital Cost (in Cr)	Recurring Cost (in Cr)
1.	Safety & Security like barbed wire fencing around dump, garland drain etc.	17.62	1.70
2.	Technical and Biological Reclamation of mined out land and OB dump like Terracing, blanketing with soil and vegetation of External OB Dump, topsoil management, landscaping and plantation etc.	51.26	5.12

3.	Post Closure Management and supervision like post mining water quality management, manpower cost and supervision etc.	15.55	1.55
4.	Others like entrepreneurship development, retrenchment benefits etc.	44.99	4.50
TOTAL		**129.42**	**12.87**

9.4 EXPENDITURE TO BE INCURRED TOWARDS CSR

Estimated CSR cost – Approx. Rs. 1 Cr/Annum.

However, the funds for CSR will be allocated at approx. Rs. 1 Cr/Annum (Minimum 2% of the net profit over the preceding three financial years of the organisation) which shall be used in sectors like healthcare, skill training, development of rural infrastructure etc.

9.5 COMPENSATORY AFFORESTATION AND WILDLIFE MANAGEMENT COST

Out of the proposed mining lease area of 1196.00 ha, forest land is 64.3723 ha. An amount of Rs. 519.13 Lakhs has been provided for Compensatory Afforestation cost. An amount of Rs. 3726.40 lakh has been proposed for wildlife conservation plan which will be met from revenue budget.

9.6 RAINWATER HARVESTING

Due emphasis has been given for rainwater harvesting. A lump sump amount Rs. 25.00 lakhs has been kept for the infrastructural facilities for storage of rainwater for harvesting which will be made from the revenue budget.

9.7 COMMUNITY DEVELOPMENT FOR PERIPHERAL VILLAGES

A lump sum amount towards Community Development for Peripheral Villages is Rs. 2798.00 lakhs.

9.8 BIOLOGICAL RECLAMATION COST

A provision of Rs. 237.62 lakhs have been made for biological reclamation, block and avenue plantation.

9.9 MINE CLOSURE COST

As per the guidelines, the closure cost for opencast mine will be six lakhs per hectare (at WPI of August 2009) of the total project area and this rate will stand modified based on the wholesale price index as notified by the Government of India from time to time.

WPI as on	I	April-19	121.1
WPI as on base date	I	April-21	131.7
Escalation rate of closure cost			1.088

Base Rate of Closure Cost "Rs. Crs./Ha"	0.09
Closure cost "Rs. Crs/Ha"	0.093567
Project Area in Ha	1196.00
Amount to be deposited into Escrow Account "Rs. In Cr.	117.208

Rate of compoundinq of Annual Closure Cost	5.00%
Balance life of the project "In yrs"	37
Annual Closure Cost (To be deposited in 1st Year)	3.168
Amount to be deposited into Escrow Account after compounding *av* of 5% "Rs. In Crs."	**321.957**

Net Amount to be deposited into Escrow Account Rs. 117.208 Cr.

9.10 ENVIRONMENTAL COST BENEFIT ANALYSIS

Environmental Cost-Benefit Analysis (CBA) is a systematic approach to evaluating the **economic, social, and environmental impacts** of a project or policy by comparing its **total benefits** against its **total costs**, including **hidden environmental externalities.**

It helps to determine the **net benefit of a project to the society** when accounting for **pollution, ecosystem damage, and climate effects**.

1. **Key Information/Assumptions**
   - Max Annual Coal Production: 15 million tonnes
   - Life of Mine: 37 years
   - **Lease Period - 40 Years (including 3 years of Mine Closure Period)**
   - Market Price of Coal: Approx. Rs 3000/tonne
   - Cost of Coal production: Rs 337.11/tonne
   - Forest Land: 64.3723 hectares (Shall be returned back in post EC)
   - PAF - 2664, PDF - 1629
   - Carbon Emissions: 2.5 kg CO_2 per tonne of coal burned (common emission factor for many types of coal)
   - Social cost of carbon emission is Rs 2500/t CO_2e
   - Forest carbon stock of 100 tonnes per hectare (tC/ha)
   - Habitat Fragmentation Cost/Loss of animal husbandry (considered in R &R plan)

- CA (Compensatory Afforestation) Cost: Rs. 5.19 Cr
- Biological reclamation will encompass extensive plantation efforts across 1029.583 ha, creating water bodies over 38.488 ha and addressing other ecological enhancements over 31.85 ha**:** Rs 24.74 Cr for plantation in 1029.583 ha and its upkeep and safety.
- Wildlife conservation plan: Site Specific Rs 35.49 Cr + Rs 1.77 Cr (5% of the total financial outlay of the SSWCP to be deposited in Wildlife Society of Odisha) = 37.26 Cr
- EMP Cost (Capital Cost) = **Rs 100.72 Cr** (including RWH cost of Rs 75 Lakhs)
- EMP (Recurring Cost) = **Rs 3.15 Cr**
- Post Mine Closure Cost (Capital Cost) = **Rs 1.2942 Cr**
- Post Mine Closure (Recurring Cost) = **Rs 0.0128 Cr**
- CSR cost- approx. **Rs.1Cr /Annum**
- Addressing Public Hearing issues -**Rs 10 Cr**
- Community development **Rs 27.98 Cr**
- MINE CLOSURE COST (Abandonment Cost)- Amount to be deposited in Escrow account Rs **321.957 Cr.**
- **Mine closure** cost is a give and take cost so not calculated.
- Total cost of the project is Rs. 3855.00 Cr.
- R&R cost is Rs. 727.94 Cr

1. Economic Benefits (Quantified)

A. Direct overall Economic Gains (GMDC & Society) **(Rs Crore)**

Component	**Calculation**	**Avg. Annual Value**	**Total (37 yrs)**
Revenue from Coal Sales	**	3370.14(124695/37 & rounding)	124695
Royalty to Govt. (14%)	3370.14*0.14	471.82	17457.33
District Mineral Foundation (DMF)	@ 10% of Royalty	47.18	1745.66
National Mineral Exploration Trust (NMET)	@ 2% of Royalty	9.44	349.15
Employment (Direct: 1752 jobs)	Avg. salary ₹5L/year × 1752 jobs	87.6	3241.2
Indirect Jobs (3x multiplier) Service sector, transport sector etc.	Avg. income ₹5L/year × 1752 x3 jobs	262.8	9723.6

Total economic benefits	**4248.98**	**157211.94**

These benefits may further extend depending upon further exploration.

B. Other Expenses (Rs crore)

Component	Calculation	Avg. Annual Value	Total (37 yrs)
EMP Cost (Capital) = Rs 100.72 Cr (including RWH cost of Rs 75 Lakhs)		2.72	100.72
EMP (Recurring Cost)		3.15	116.55
Post Closure Total EMP Cost (Capital)		43.14 (per year for 3 years)	129.42 (in 3 years after 37 years)
Post Closure Total EMP Cost (Recurring)		0.0128 (per year for 3 years)	0.0384 (3 years after 37 years)
CSR cost- approx. Rs.1Cr /Annum (Minimum 2% of the net profit over the preceding three financial years)		1	37
Public issues		0.27	10 (it is time bound)
Community development		0.76	27.98
Total expenses		**51.05**	**421.70**

2. Environmental & Social Costs (Quantified)

A. Deforestation & Biodiversity Loss (Rs crore)

Component	Calculation	Avg. Annual Value	Total (37 yrs)
Forest Carbon Stock Loss	64.3723 ha × 100 tC/ha × ₹2,500/tCO2e	1.61	59.57
Biodiversity Damage	₹5 Cr/year	5	185

Compensatory Afforestation (land for CA 67.821 ha)		(-)0.14	(-)5.19
Biological reclamation		(-)2.22	(-)**82**
Wildlife conservation plan		(-)**1.00**	(-)**37.26**
Total Deforestation Cost		**3.25**	**120.12**

B. Air & Water Pollution Health Costs (Rs Crore)

Component	**Calculation**	**Avg. Annual Value**	**Total (37 yrs)**
Health Impact (respiratory diseases)	20,000 cases × ₹50,000/case	100	3700
Water Contamination remediation	₹10 Cr/year for remediation	10	370
Total Pollution Cost		110	4070

(Assumes 20,000 additional cases of asthma, lung disease due to mining-related pollution)

C. Climate Change Impact (Carbon Emissions) (Rs Crore)

Component	**Calculation**	**Avg. Annual Value**	**Total (37 yrs)**
CO2 Emissions (coal burning)	Avg. prod. MT × 2.5 kg CO2/tonne × ₹2,500/tCO_2e/1000##	7.91	292.67
Total Climate Cost		7.	292.67

D. Social Costs (Displacement & Livelihood Loss) (Rs Crore)

Component	**Calculation**	**Total**
R&R Package for PDFs as per GMDC provision	1629 PDF*Rs 4425000	720.8325
Assistance to Artisan PAFs as per RFCTLARR Act 2013	159 Artisan PAFs*Rs 25000	0.3975
Assistance to Ag. Labourer PAFs as per RFCTLARR Act 2013	4 Ag Labourer PAFs*Rs 500 X 200 days	0.04

Assistance to SC (577) and ST (203) PAFs for Agri-allied Activities like land development, water management, backyard Poultry etc	SC/ST 780 *Rs 50000	3.90
Assistance to physically challenged people	Physically challenged 140 *Rs 15000	0.21
Assistance to women headed PAFs for Income Generation Activities	Alternative livelihood 394 *Rs 10000	0.394
Alternative livelihood support through skill development/ training, financial assistance to undertake self-employment activities (both individual and Group)	Alternative livelihood 1079 *Rs 20,000	2.158
TOTAL	**727.94**	

4. Summary of Costs vs. Benefits (Rs Crore)

Category	Total (37 yrs)
Total Benefits	157211.94+421.70 =1,57,633.64
Total Costs	120.12+4070+292.67+727.93=5210.72
Net Benefits	1,52,422.92

Note: This is a simplified model; real-world costs may vary due to policy changes, inflation, and unaccounted externalities

5. BENEFIT COST RATIO

Economic benefits (Rs 1,57,633.64 Cr) outweigh costs (Rs 5210.72 Cr) in monetary terms.

Ratio works out to be 30.25. Thus, the project gives positive Benefit/Cost Ratio. The monetary returns of the Project are positive over the environmental losses.

Further considering plantation over 67.821 ha of CA land and reduction of 20% of total power consumption through solar and other means (Solar power and reduction of diesel consumption etc.) there will be a saving of Rs 2625 Cr**.**

Considering Green energy and reduction in diesel consumption Economic benefits further improve resulting in increased Ratio tilted towards benefits than costs.

Summary of Costs vs. Benefits (Rs crore) - with use of solar power and reduction in diesel consumption

Category	Total (37 yrs)
Total Benefits	157633.64+2625=1,60,258.64
Total Costs	5210.72
Net Benefits	1,55,047.92

CB Ratio = benefits/costs =30.75

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CHAPTER-10 : ENVIRONMENTAL MANAGEMENT PLAN

The Environmental Management Plan includes four major elements viz Planning, Implementation, Checking, Management Review.

The Environmental Management System (EMS) shall be developed.

Total cost of the project is Rs. 3855.00 Cr.

Cost of production is Rs.337.11 per tonne

CSR cost is approx. Rs.1 Cr/Annum

R&R cost is Rs. 727.94-crores

Total EMP Cost (Capital Cost) = Rs. 10072.9 Lakhs
Total EMP (Recurring Cost) = Rs. 315.7 Lakhs

Apart from the above EMP cost already earmarked in the post closure mine cost is Rs 129.42 Cr and recurring cost is Rs.12.87 Cr.

At the end of life of mine 1029.583 ha will be under plantation.

Concerned department of forest will implement the WLCP as per the proposed regional conservation plan.

An EMC is to be set up to look after the effectiveness of mitigation measures.

10.1 INTRODUCTION

The environmental management plan (EMP) is required to minimize adverse environmental impacts by designing an administrative structure for ensuring that mitigative measures are implemented and their effectiveness monitored, after approval of the EIA implementing suggested mitigation measures with timelines and responsibilities during the project life cycle.

EMP consists of a set of institutional measures to be taken up during implementation and operation of a project to eliminate adverse environmental impacts or reduce them to an acceptable level. Chapter IV dealt with environmental impacts of this project and proposed mitigative measures to minimize adverse impacts as well as to improve the existing environment quality. This chapter mainly discusses the administrative aspects of ensuring that mitigative measures are implemented and their effectiveness is monitored after approval of EMP. It comprises of Quality Environment Health and Safety (QEHS) Policy of company, establishment of an Environment Management Cell (EMC) and developing mechanism of reporting of non-compliances, if any to company board, conduct root cause analysis of non-compliances and propose appropriate mitigation measures to ensure avoidance of adverse Penal action arising out of environment and other statutory regulations. Good practices of EMP will ensure to keep all the environmental parameters of the project area (Air, Water, soil, biological diversity, Socio economic) within statutory limits and sustainable manner.

10.2 ELEMENTS OF ENVIRONMENTAL MANAGEMENT PLAN

Environmental Management Plan includes four major elements:

Planning: This includes identification of environmental impacts, legal requirement, commitments and policies, setting environmental objectives and environment, health, safety and social compliance requirements.

Implementation: This comprises of resources available for the project, accountability of employees, contractors and documentation of measures taken.

Checking: Measurement & Evaluation: This includes regular inspection, audits, monitoring corrective actions and record keeping; and

Management Review: Actions are taken to continually improve the environment, health, safety, and social performance of the organization.

In order to develop the **Environmental Management System (EMS),** the company shall identify the impacts, plan and set targets to reduce the negative impacts, implement steps to achieve the targets effectively and efficiently, evaluate the results of implementations, take steps to bridge the gaps between planned and achieved objectives, and continually improve the environmental performance.

The important components of the proposed EMS are as follows:

- Environmental Policy
- Operation and Emergency Procedures

10.2.1 QUALITY ENVIRONMENT HEALTH AND SAFETY POLICY (QEHS)

The policy applies to all GMDC projects in operations and managed sites, including new acquisitions, corporate offices and research facilities and to all new and existing employees and contractor employees, business partners suppliers and consultants who should be required to comply with the policy when they act on behalf of GMDC. This Policy is applicable to the entire operational lifecycle (including exploration and planning, evaluation, operation, and closure) as well as applicable to downstream operations (including distribution, logistics, sale of products & services up to the customer).

GMDC will:

- Comply with applicable national, regional, and local HSE regulations and statutory obligations. In the absence (or lack) of appropriate legislation, industry best practices and applicable international standards will be used; Strict monitoring and compliance of the conditions stipulated in consent for establishment issued by Odisha Pollution Control Board.
- Develop, implement, and improve HSE management systems, with commitments and beliefs, consistent with world-class standards;
- Promote a positive HSE culture through effective communication, participation and consultation with employees and business partners;

- Ensure that all facilities are designed, constructed, operated and closed in compliance with all applicable laws and regulations and in alignment with accepted international practice;
- Raise awareness and train employees, business partners, suppliers and other stakeholders to adopt principles and practices in alignment with our policies;
- Communicate and provide appropriate information to ensure that the policy is understood, implemented, maintained and available to all working under the control of the organization.
- Prevent, minimize and safeguard interests in terms of ill health, safety, resource conservation and environmental protection.
- All the time comply with applicable environment, health and safety legal requirements and fulfill other compliance obligations.
- Develop competencies to continually incorporate best practices.
- Periodically review the policy to ensure that it remains relevant and appropriate.

The detailed QEHS Policy is enclosed as **Annexure XXIII.**

Documentation

The Company shall formulate and implement documented procedures, and maintain records of all operations, policies, objectives, targets, responsibilities and information on environmental performance and its impacts. These will be utilized for verifying and demonstrating environmental performance to the staff, regulatory authorities and the community, and can also be used at a later stage for continual improvement of performance.

10.2.1.1 OBJECTIVES AND TARGETS

The Company shall set feasible, specific, measurable and effective objectives and targets for continual improvement of the environment. Stringent long-term objectives will be formulated to encourage and enhance stable and continuous growth and performance. The targets and objectives shall be continuously and timely revised so as to ensure better results. The objectives and targets shall be based on the impacts identified and observed during implementation and operation of the project.

10.2.1.2 CONSULTATION

Consultation with staff, local communities, associated experts/contractors, relevant organization and managerial bodies shall be conducted at regular interval or as and when required for reviewing the environmental performance, share opinions and provide suggestive measures for the possibility of further improvement of EMS and EMP. In order to develop the commitment of staff, the company shall take steps like making the staff aware of the policies and objectives, conducting awareness training and getting feedback from the staff members. Consultation with the public will also help to improve public perception of the company, one of the benefits of implementing an EMS.

10.2.2 OPERATIONAL AND EMERGENCY PROCEDURES

Operational and emergency procedures mainly aim at ensuring that they are congruent with the organization's environment objectives and targets and suggest required steps in case of any diversion.

10.2.2.1 TRAINING

Training shall focus on developing trained and competent staff. This will include familiarizing the staff with all their responsibilities, duties and work instructions for the tasks associated with them for implementation of EMS. For this, the company shall undertake various measures in order to infuse the staff with the necessary skill and motivation required for effective and efficient implementation of EMP.

iCEM:

iCEM, "International Center of Excellence in Mining Safety and Automation (iCEM), Ahmedabad is an initiative of Government of Gujarat (GoG) being promoted by GoG through GMDC Ltd as Principal Promoter.

The centre is envisioned to focus on Skill Development, Corporate Development, Demonstration of Technology and Best Practices, Education and Training, Research & Development, Technology Incubation & Entrepreneurship Development in Mining sector through continued innovation and excellence to facilitate industrial development in a joint venture with industrial partners.

GMDC, through iCEM is in continually working on capacity building through various training programs. A training module on EMS is showcased. The classroom training will be accompanied with practical on-site trainings.

Training Module on Environmental Management for GMDC

Day 1	Inaugural Session	Environmental Aspects of Mining	Environmental Legislation	
Day 2	Water Pollution & its Control	Hydrogeological studies	Air pollution Monitoring and assessment	Air pollution Control
Day 3	Land Use Planning and Land Reclamation	Land Use Planning and Land Reclamation	Forest and Wildlife and its Conservation	
Day 4	Environmental Impact Assessment		Social impact Assessment and R&R	
Day 5	Mine Closure	ESG	Interactive Session with the Participants	Valedictory Session

Other Mass Awareness Programs being undertaken by GMDC:

1. Celebration of WED
2. Celebration of Vanamahotsava
3. Celebration of Safety Week

10.2.2.2 REVIEWS, AUDITS AND COMPLIANCE MONITORING

This component will mainly target towards smooth and delicate functioning of the activities. Prior and essential reviewing of audit and monitoring compliance shall be done so as to achieve the stated objectives and to modify and steer the policies laid down with time in order to meet the goals. Monitoring is also one of the vital components which ensures whether all the requirements are met or not and everything is carried on in a systematic and planned way.

10.2.2.3 CONTINUAL IMPROVEMENT

This is the most important component as everything is based on the concept of continuous development. An EMS is said to be effective only when it leads to continual growth and improvement to the organization by reviewing and monitoring the targets and objectives set by the company. The policy and procedures laid down by the company shall be constantly monitored and checked and any required changes shall be made and implemented.

10.2.3 ANTICIAPTED POTENTIAL IMPACTS & SUGGESTED MITIGATION MEASURES

A summary of the anticipated potential impacts and the details of the suggested mitigation measures are provided below mentioned **Table 10-1.**

Table 10:1 Summary of Anticipated Potential Impacts & Suggested Mitigation measures

Parameters	Anticipated Impacts	Suggested Mitigation/ Management Measures
Topography	Due to change in topography the Natural drainage pattern shall change.	Creation of large Water Body: The creation of large water body will make up losses by increase in seepage into groundwater. Efforts will be made to restore the land (with adequate plantation around 90 % of total lease area as per Mine Closure Plan.
Drainage	Reduction in surface run off. Increase in ground water run off (base flow).	The increase in ground water runoff (base flow) will be gainfully used in mining activity and balance will contribute to local surface water need.

Air Quality	Emission of Particulate Matter, Dust and Gaseous Pollutants viz. NOx, SO_2 due to mining operations viz. drilling, blasting, loading and transportation and movement of HEMM.	Wet drilling, Controlled blasting – Optimization of charge per hole and charge per round, use of surface miner, with dust extraction, water sprinkling, fog cannon. Wind shield etc Regular maintenance of vehicles and machinery; Water sprinkling on haul roads within ML area; Plantation in and around the proposed mine and Green Belt Development. Afforestation of completely mined out area, with minimum gap between excavation and afforestation.
Water Regime	Mining induced increase in rate of infiltration and ground water recharge. Mined area will function as rainwater harvesting system.	Regular monitoring of water levels and quality of water in existing open well and borewells to study the ground water in the project area. Rainwater harvesting shall be done.
Surface Water Quality	Discharge of mine runoff during rains to surface water channels	Water channels/drains carrying the rainwater from the mine will be provided with baffles and settling pits to arrest the suspended solids;
	Wash off from waste dumps and coal stockpiles during rainy season	Retaining walls will be provided at the toe of dumps and unstable OB benches within the mine to prevent wash off from dumps and sliding of material from benches.
	Oil spillage from maintenance workshops	The workshop effluent will be routed through oil & grease trap and treated to the discharge standards through ETP and reused in the workshop.
	Waste water discharge from office building etc.	The domestic sewage from the canteen and toilets will be treated in a STP.
Ground water Quality	Contamination due to discharge of wastewater generated from workshop and offices.	ETP will be provided and regular monitoring of water level around the mine area will be carried out and corrective measures shall be undertaken.
Noise and Vibrations	Noise/Vibration generation due to drilling, blasting,	Surface miner shall be used to restrict blasting. However, in case it is required, drilling parameters like

	operation of HEMM, workshop and vehicular movement.	burden, depth, diameter and spacing will be properly designed to give proper blast. Controlled blasting with proper spacing, burden and stemming will be carried out. Site mixed slurry (SMS) has been proposed to be used for good fragmentation; Personal Protective Equipment (PPE) like ear muffs/ear plugs will be provided to the operators of HEMM and persons working near HEMM; Provision of sound insulated chambers for the workers deployed on machines (HEMM) producing high levels of noise will be made; Exposure time of workers to the higher noise levels would be minimized; Prime movers/diesel engines will be properly maintained; A buffer barrier of tree belt will be provided along the periphery of the mine to attenuate noise;
Land /Soil Environment	Change in land use pattern due to excavation, overburden dumps, soil extraction etc.	Concurrent backfilling and land reclamation is proposed to be carried out as soon as sufficient de-coaled area will be available. It is estimated that a total of 1.866 million m^3 of topsoil and 713.53 million m^3 of Overburden (OB) will be generated during mining and the same will be stacked in OB dumps and Topsoil Dumps. The waste stored in this OB Dump shall be utilized for backfilling in the de-coaled mine pits. After back-filling, area shall be leveled & planted with local species in consultation with local forest officials. The void area of the excavated pit will ultimately become a water body. This void area will also be suitably sloped, bunded and fenced.

Flora and Fauna	As per the forest record there are 29 numbers Schedule I Species Namely Indian Elephant, Bengal Monitor, Indian Chameleon, Common Rat Snake, Common Indian Krait, Indian Cobra, Russell's Viper, etc. listed in the study area. No wildlife sanctuary or national park exists in the study area.	Though no endemic species reported in the project area, every attempt will be made to protect and conserve the forest ecosystem. A comprehensive Regional Wildlife conservation plan (WLCP) is under preparation at DFO level. PP will deposit the proportionate amount as per the direction of Forest Department. Further Site-Specific Conservation Plan has been prepared & approved by PCCF & Chief Wildlife Warden, Odisha. A budgetary provision of 37.26 Cr (Including 5% of the total financial outlay of the SSWCP to be deposited in Wildlife Society of Odisha) is earmarked for 10 years.
Occupational /Public Health	Health problems including respiratory disease due to air and noise pollution. Health hazards in the workplace.	Dust masks would be provided to the workers to prevent inhalation of RSPM thereby reducing the risk of lung diseases and other respiratory disorders. Earplugs will be provided to all the operators of HEMM. Regular health check-ups including lung function test and audiometry test of workers will be carried out. Earmuffs will be provided to the workers working in high noise area. Training on health and safety will be imparted to all the workers so that they develop habit of using the protective equipment. Regular Medical Camps for villagers, labours, employees & their family members would be organized. A dispensary will be set up for the workers. Various health awareness and family welfare programmes for the benefit of the general public in project region will be organized.
Socio-economic	The project will create positive impacts both direct and indirect on the	As far as possible locally available workforce in skilled and unskilled category will be employed in the

	economy of the region and generate employment.	project. The basic amenities/facilities viz. education, safe drinking water supply, sanitation, health care will be done under CER/CSR activities based on the needs of the region. Regular environmental awareness programs would be organized by the project proponent to bring forth the environmental management measures being undertaken and the beneficial aspects of the project for improving their quality of life.

10.3 GREENBELT DEVELOPMENT

At the end of life of mine 1029.583 ha will be under plantation, 38.488 ha under water body, 31.85 ha under public use, 64.3723 ha under forest (to be returned) and 31.707 ha will remain untouched showing almost the whole lease area shall be returned to more environmentally friendly and aesthetic value.

An area of 1029.583 ha shall be covered under plantation at conceptual stage. Greenbelt will be developed considering the proposed plantation @ 2500 trees per hectare.

Apart from this the PP will explore Miyawaki plantation which will further increase the density of trees per hectare.

The Green Belt shall:

- ✓ Compensate the loss of vegetation due to setting up of the mine
- ✓ Prevent the spread of fugitive dust generated due to mining and allied activities
- ✓ Attenuate noise generated by the mine
- ✓ Reduce soil erosion
- ✓ Increase green covers and improve aesthetics

Native species (Local) shall be planted in consultation with the Forest Department. In addition to the above, green belt shall also be developed in vacant areas, around office buildings, workshop, stores etc. along the roadsides and on the external dump area.

10.4 PLAN FOR IMPLEMENTATION & MONITORING OF THE SUGGESTED MITIGATION MEASURES

A comprehensive programme for Environment Monitoring has already been dealt in Chapter 6 of the EIA/EMP.

For effective implementation of the recommended Environmental Monitoring Programme, it is necessary to establish and develop adequate facilities for sampling and analysis on permanent basis together with adequate instrumentation facilities and development of competent trained manpower.

10.4.1 ENVIRONMENT MANAGEMENT CELL

A separate Environmental Cell to ensure effective monitoring of the Key Environmental Aspects of the Project on a day-to-day basis is proposed. Further, internal quality control and assurance of the monitoring technique as well as analytical procedure together with periodic external audit shall also be planned. The organogram of EMC is given in **Chapter-6.**

The responsibilities of the EMC include the following:

- Monitoring & measurement of compliance obligations;
- Developing the green belt/Plantation;
- Ensuring conservation of resources (water, power & fuel);
- Proper implementation of pollution control measures;
- Implementation of QEHS;
- Conducting Internal Audits;
- Formulation and implementation of action plans for non-conformance;
- Interacting with project proponent and reporting performance of QEHS; and
- Interacting with statutory agencies.

10.4.2 ENVIRONMENT MANAGEMENT PLAN BUDGET

The implementation of the suggested Environment Protection Measures shall involve Capital & Recurring Expenditure. As the implementation of the protection measures shall be a continuous process during the life of the mine, the capital expenditure shall not be required in one go. It would be required in phases. The Recurring Expenditure shall be regularly required on an annual basis.

Capital Cost of EMP is Rs. 10072.9 Lakhs and Recurring Cost of EMP is Rs. 315.7 Lakhs. The details of the estimated expenditure for implementing the pollution control measures are provided in Table 9.1.

Capital Cost: Total cost of the project is Rs. 3855.00 Cr. Cost of production is Rs.337.11 per tonne, CSR cost is approx. Rs.1 Cr/Annum. R&R cost is Rs. 727.94 crores.

The Capital Project Cost for the next five years of the total 3855.00 Cr is given as follows:

Table 10:2 Capital Project Cost proposed

S. No.	Year	Project Cost (in Lakhs)					
		2023-24	2024-25	2025-26	2026-27	2027-28	2028-29

	Particulars						
1	Land	_	48930	45875	45650	45675	45000
2	Building	_	_	14000	14200	10000	10000
3	Plant & Machinery	_	_	200	2000	2000	3000
4	Furniture & Fixtures	75	_	75	200	_	_
5	Office Appliances	4	14	20	22	_	_
6	Vehicles	_	0	260	200	_	_
7	Electrical Equipment's	1	2	265	150	_	_
8	Others	2500	4055	1055	1070	_	_
		2580	53001	61750	63492	57675	58000

10.5 CONCLUSION

It is safe to say that the project is not likely to cause any significant impact on the ecology of the area, as adequate preventive measures have been proposed in EMP with its expenditure details. Green belt development around the area will also be taken up as an effective pollution mitigation measure which will also help in increasing the aesthetic beauty of the area. An EMC is to be set up to look after the effectiveness of mitigation measures.

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CHAPTER-11 : SUMMARY & CONCLUSION

This Chapter provides the concise summary of the Key Project Parameters along with overall justification for implementation of the project and explanation of how adverse effects have been mitigated as required under the generic structure of Appendix III of the EIA Notification 2006. Further, the conclusion on the studies undertaken is also included in the Chapter.

11.1 DETAILED CHRONOLOGY ON ALLOCATION OF COAL BLOCK

- Baitarni West Coal block of Talcher Coalfield with production capacity of 15 MTPA in an area of 1196 ha by M/s Gujarat Mineral Development Corporation Ltd. located at Village Chhendipada Jangal, Chhendipada RF, Handigora, Machhakuta, Machhakuta Jangal, Porapara, Porapara Jangal, Tentuloi-Korasahi Tehsil Chhendipada District Angul (Odisha).
- Baitarni West Coal Mine of GMDC comes under TALCHER Coal Field was earlier allotted to Gujarat Power Corporation Ltd. (GPCL), Kerala State Electricity Board (KSEB) and Orissa Hydro Power Corporation (OHPC) by Ministry of Coal, Govt. of India, vides letter no. 13016/8/2007-CA-I dated 25.07.2007. Later on Supreme Court of India through its judgment dated 25th August, 2014 read with its order dated 24th Sept 2014 cancelled allotment of 204 coal blocks including Baitarni West Coal Mine.
- Accordingly, its allotment to the prior allottee was cancelled in pursuant to the Supreme Court Judgment. MoC vide order no. 13016/2014-CA-III. dt. 16.03.2016 notified Baitarni West Coal Mine for allotment to the State PSUs for sale of coal. Based on the above order of the MoC, GoI, Govt. of Odisha vide letter no. 2831/SM, dtd.05.04.2016 advised OMC Ltd a Gold category state PSU to apply for allocation of Baitarni West Coal Mine.
- Accordingly, OMC Ltd entered into the allotment agreement with the nominated authority on dt. 24.08.2016 after fulfilling all the conditions as required by allotment authority. The nominated authority MoC, GoI issued the allotment order to OMC Ltd. Vide order no. 103/3/2016/N.A dtd.29.09.2016. Again, a Corrigendum No. 01 for the said allotment is issued by MOC vide 103/16/2015/NA dated 14.08.2020.
- As per the earlier approved Mining Plan vide letter dt. 07-10-2009, Form-I and Prefeasibility report for Baitarni West Coal Mine (15.0 MTPA) capacity was prepared and considered in 19th Expert Appraisal Committee (EAC) meeting held on 26th September, 2017.
- EAC had recommended Terms of Reference (TOR) for Baitarni West Coal Mine (15MTPA) vide letter no. J-11015/81/2017-IA.II (M) dated 29th September, 2017 issued by MoEF&CC, New Delhi. However, due to some changes in the Mining Plan, Final EIA/EMP could not be submitted to MoEF&CC.
- Accordingly, as per the changes a revised Mining Plan of Baitarni (W) has been prepared and approved by MoC vide letter no. SWCS/Baitarni West Coal Mine/OR/AN/2021/CCO-MCBA dt. 10-11-2021.

- Further an application for fresh TOR was submitted (by OMC) along with Form 1 and PFR vide Online Proposal No. IA/OR/CMIN/234993/202 dated 21st October, 2021.
- Accordingly Approved Mining plan and mine closure plan (first modification) is transferred to Gujarat Mineral Development Corporation vide vesting order NA-104/4/2023-NA, dated 08/06/2023.

11.2 PROJECT DETAILS

Baitarni West Opencast Coal Mine is an opencast coal mining project with production capacity of 15 MTPA in an area of 1196 ha of M/s GMDC is located at Villages Chhendipada Jangal, Chhendipada RF, Handigora, Machhakuta, Machhakuta Jangal, Porapara, Porapara Jangal, Tentuloi Korasahi, Tehsil Chhendipada, District Angul (Odisha).

The proposed project activity falls under the category of "Mining of Minerals" and categorized as "Category-A" under project activity 1(a) of EIA Notification 2006 & its amendment till date.

Table 11:1 Salient features of the project

A	**General Details**	
1	Name of the Company	**M/s Gujarat Mineral Development Corporation Ltd (GMDC)**
2	Project Location	Village-Chhendipada Jangal, Chhendipada RF, Handigora, Machhakuta, Machhakuta Jangal, Porapara, Porapara Jangal, Tentulo - Korasahi, Tehsil- Chhendipada, District-Angul (Odisha)
3	Coordinates	Co-ordinates: NW: 21°05'26.87"N, 84°50'12.08"E NE: 21°05'17.37"N, 84°51'26.13"E SW: 21°01'44.84"N, 84°50'39.59"E SE: 21°01'44.74"N, 84°51'40.70"E
4	Category of project as per EIA Notification 2006	1 (a), Mining of Minerals "A".
5	Proponent Details	Registered Address: Gujarat Mineral Development Corporation Ltd (GMDC), Khanij Bhawan, 132 ft Ring Road, Vastrapur, Ahmedabad, Gujarat, 380052 Correspondence Address: Gujarat Mineral Development Corporation Ltd (GMDC), Khanij Bhawan, 132 ft Ring Road, Vastrapur, Ahmedabad, Gujarat, 380052 E-mail: gmdcghala@gmail.com
6	Type of Project	1 (a), Mining of Minerals "A"
7	Production capacity	15 MTPA
B	**Mine details**	

24	Forest land	64.3723 ha
25	Non-forestland	222.975 ha
26	Tenancies	908.6527 ha
27	Area:	Block Area: 1196 ha Mining Lease Area: 1196 ha
28	Technology	Opencast mining is proposed by Surface miners (SM) -FEL-Truck for coal and drill blast for OB.
29	Area of the block	1196 ha
30	Gross geological Reserve	662.94 MT
31	Net Geological Reserve	602.1106 MT
32	Blocked Reserve	117.0413 MT
33	Mineable Reserve	485.06 MT
34	Extractable Reserve	468.27MT
35	% of Extraction/ recovery	77.771%
36	Reserve depleted (Till the base date)	0 MT
37	Balance Extractable Reserve	468.27 MT
38	Average grade	G12(3946 CV)
39	Ultimate depth of Mining (m)	Maximum-245 Minimum-160
40	Bench width	24-36 m for overburden and 24-34 m for coal
41	Bench height	10-12 m
42	Mining technology	Opencast Mining Method of Coal winning by Surface Miner, pay loader& tipper and OB removed by Shovel-Dumper combination.
43	Total Life of mine	37 years including construction period
44	Period of mining lease	Land rights have been obtained by Gujarat Mineral Development Corporation under Sec 11(1) of CBA & henceforth Sec 19 of CBA has been published.
45	**R & R Involved**	Total 7 villages namely Chhendipada Jangal (Part), Handigora (Part), Machhakuta (Part), Machhakuta Jangal (Part), Porapara (Part), Porapara Jangal (Full), Tentuloi-Korasahi (Part) are going to get affected by proposed project. Total Project Affected Families (PAF) is 2664 Nos and Project Displaced Families (PDF) is 1629 Nos. These displaced families will be resettled and rehabilitated socially, culturally and economically

<table>
<tr><td></td><td></td><td>as per the latest R&R policy (2006 & subsequent amendments) of the Govt. of Odisha.</td></tr>
<tr><td>46</td><td>No. of affected families</td><td>2664 & 1629 no of PAFs & PDFs respectively due to proposed project.</td></tr>
<tr><td>C</td><td colspan="2">Details of Seams</td></tr>
<tr><td>1</td><td>No. of seams</td><td>18</td></tr>
<tr><td>2</td><td>Seams available as per GR</td><td>Seam-IX, Seam- I A TOP, Seam- I A BOTT, Seam- I B, Seam- I C, Seam- I D, Seam- IG, Seam- IE, Seam-IF, Seam-X, Seam-VI/VII/VII, Seam-IV/V, Seam-III
TOP(TOP), Seam-III TOP (BOTT), Seam-III TOP (T&B Comb.), Seam-III Middle, Seam- III BOTT (TOP), Seam-III BOT (BOT)</td></tr>
<tr><td>3</td><td>Seams not considered for Mining with Reasons</td><td>
<table>
<tr><th>S. No.</th><th>Seams</th><th>Reason</th></tr>
<tr><td>1</td><td>Seam- IG</td><td>Thin, unworkable in major area, occurs around 50m below seam ID</td></tr>
<tr><td>2</td><td>Seam- IE</td><td>Thin, unworkable, not considered in GR</td></tr>
<tr><td>3</td><td>Seam-IF</td><td>Thin, unworkable, not considered in GR</td></tr>
<tr><td>4</td><td>Seam-X</td><td>Incrops, Local development, Deteriorated. Not considered for reserve estimation.</td></tr>
</table>
</td></tr>
<tr><td>4</td><td>Thickness of seam</td><td>0.08 m to 33.94 m</td></tr>
<tr><td>5</td><td>Grade of Coal</td><td>G-12</td></tr>
<tr><td>6</td><td>Stripping Ratio</td><td>1.5278 m^3/tonne</td></tr>
<tr><td>7</td><td>Average Gradient</td><td>2^O to 7°</td></tr>
<tr><td>D</td><td colspan="2">Details of Dumps</td></tr>
<tr><td>1</td><td>No of Dumps</td><td>1 External OB Dump of 215.84 ha with maximum total height of 90 m and quantity of OB 140.25 million m^3 and Internal OB Dump of 623.71 ha with quantity of OB 573.3 million m^3 is envisaged for backfilling.
By the end of mine life, 140.25 Mcum OB will remain in external dump, while the remaining 573.3 Mcum OB will be accommodated in internal dump. The top RL of internal dump will</td></tr>
</table>

		be 190m towards north and central part of the quarry and 220m towards southern area. 1 Topsoil dump of area 2.34 ha with quantity of topsoil 1.866 Mcum. However at the Final Stage, Topsoil shall be utilized for plantation in external dump 0.647 Mcum, internal dump 1.199 Mcum & embankment 0.021 Mcum.
2	Area of Dump	Internal dump = 623.71 ha External Dump = 215.84 ha
3	Height of Dump	There will be mainly three tiers of OB dump, each with a maximum height of 30 m, with a maximum total height of 90 m. RL of top of the external dump at final shape will be 290 m.
E	**Environmental Sensitivity**	
1	Nearest District Headquarter	Angul 43 kms from southeastern boundary in SE Direction
2	Nearest Airport	Biju Pattanaik International Airport, 131.7 Kms in SE Direction
3	Nearest Railway station	Angul 42 kms from southeastern boundary in SE Direction
4	Nearest National/ State Highway	SH63 which is 0.8 Km in E Direction
5	Nearby Water Bodies	Rivers/Nalla: Kumbhira Nala flows from south-east (seasonal). Supai Nala flows (seasonal) in the core zone. Gauduni Nala – 1.2 km, E Kusal Nala – 1 km, W Bainsajaria Nala – 1.5 km, NW Shagarhia Nala – 3.0 km, NW Singhda Jhor – 5.0 km, S Aunli Nadi – 5.5 km, N Mandalia Nala – 9.5 km, W Nuhurang nadi -9 km, NW Tikra river- 9 km, NNE
6	Nala Diversion	Supai stream is a seasonal stream, and no water is available December onwards. A detailed hydrological survey of the Supai Nala, regarding its catchment area, flow volume and length of the stretch to be diverted with water augmentation plan along with development of artificial pond has been provided. The Nala shall be diverted in the 10th year of mine operation.

7	National Park/ Wildlife sanctuary/ Biosphere reserve/ Tiger Reserve/ migratory routes for Birds/Corridor	Following RF/PF are present within 10km study area as per SOI Toposheet. i) Chhendipada RF – Core zone ii) Debinagar PF – 2.5 km, SW iii) Kamkurupal RF – 5.0 km, E iv) Kosala RF – 4.6 km, ES v) Simulipathar RF – 7.8 km, S vi) Brahmanabil RF- 8.8 km, E vii) Dangpal PF- 9.7 km, NW viii) Khalapal RF- 9.3 km, E ix) Bamur Forest- 9.3 km, W x) Katara RF – 14.3 km , SSW Letter from PCCF is attached as **Annexure XI.**
F	**Project Cost**	Total cost of the project is Rs. 3855.00 Cr. Cost of production is Rs.337.11 per tonne, CSR cost is approx. Rs.1 Cr/Annum. R&R cost is Rs. 727.94 crores.
G	**Manpower requirement**	1752 direct employment
H	**Water requirement**	3.62 MLD (potable 1.02 MLD, industrial water 2.6MLD). Initial requirement of potable water and industrial water shall be 0.26 MLD which will be met out from ground water. Afterwards the demand of water will be met from the mine inflow water.
I	**Water source**	Through borewell and mine seepage water.
J	**Power requirement**	For the initial mine operation, 11 KV power supply shall be obtained through a link from the Handigoda rural line, within the mine lease boundary of the coal block. The proposed 33 KV transmission power line is still at the conceptual stage, as the alignment of the 132 KV line originating from the Boinda Substation under OPTCL is yet to be finalized.
K	**EMP Cost**	Capital Cost: Rs 10072.9 lakhs Recurring Cost: Rs 315.7 lakhs
L	**CSR Cost**	Approx. Rs. 1 Cr/Annum (2% of the net profit over the preceding three financial years of the organisation) which shall be used in sectors like Healthcare, skill training, development of rural infrastructure etc.

Table 11:2 DESCRIPTION OF BASELINE ENVIRONMENT

S. No.	Parameters	Baseline status
	Oct 2023 – Dec 2023	
1.	**Ambient Air Quality**	
i.	$PM_{2.5}$	20.1μg/m^3 to 45.8μg/m^3
ii.	PM_{10}	48.9–83.9 μg/m^3
iii.	SO_2	5 to 10.3μg/m^3
iv.	NO_2	9.5 to 23.3μg/m^3
v.	CO	180 to 680μg/m^3
All results have been found within the AAQ Standard Limit		
2.	**Noise Level Monitoring**	
i.	Day Time (06:00 Am to 10:00 PM)	43.1dB to 52.9dB
ii.	Nighttime (06:00 Am to 10:00 PM)	38.5dB to 43.6dB
3.	**Soil Quality and Characteristics**	
i.	pH	7.58 to 7.94
ii.	Electrical Conductivity	210 to 263 μmhos/cm
iii.	Organic Matter	0.63 % to 0.86 %
iv.	Total Nitrogen	37 mg/kg to 47mg/kg
v.	Phosphorus	4.2 mg/kg to 5.9 mg/kg
vi.	Bulk Density	1.31 to 1.33 gm/cc
4.	**Ground Water**	
i.	pH	7.04 to 7.48
ii.	Total Hardness	110 to 493 mg/L
iii.	Total Dissolved Solids	210 to 260 mg/L.
Water quality of study region is within the drinking water standard as per 10500:2012, except total hardness which at few locations which exceed desirable limit but are within permissible limit.		
5.	**Surface water**	
i.	pH	7.19 to 7.86
ii.	DO	5.1 to 6.1 mg/L.
iii.	COD	4.8 to 10.6 mg/L.
iv.	BOD	2.4 to 5.3 mg/L.
From the test result of surface water and comparison study with Inland surface water classification (CPCB Standards), it is interpreted that this water can be used for drinking after being suitably treated.		

Ecology and Biodiversity

The study area has 6 reserve forests and 2 protected forest & some other forest patches which accompany the rich floral and faunal diversity along with wildlife protected under schedule -I of wildlife protection act 1972 amended 2022. Forest diversion and tree felling will be involved along with diversion of nala in ML area. Biodiversity loss should be estimated before nala diversion and biodiversity conservation plan is recommended. There was paddy crops in agricultural fields

observed during site visit on ML area, people practicing agriculture in ML area, ponds for rainwater harvesting and fishing, domestic animals for agriculture practice, wild movement in ML as well as in study area; hence wildlife conservation plan is recommended to implement during entire mining period. All measures will be adopted at each project development phase for environment management and have been described in previous chapters.

Greenbelt would be developed in 1029.583 ha area preferring the local plant species.

Socio-economic condition

Coal is one of the main sources of energy in India. Accordingly, affordable and sustainable supply of coal is inextricably linked to the goal of ensuring energy security in India. The project will certainly help our country towards energy security.

The proposed coal mining project of M/s GMDC is a right step towards Indianisation. Instead of Importing coal from China and other countries at high Cif prices it is appropriate to use coal deposited in our own mines.

In power mix production and supply, the maximum contribution is of thermal power. Its generation depends upon availability of coal and its shortage causes electricity outage not only in residential areas but also in industrial production. It is hoped that the proposed open cast Coal Block in district Angul, Odisha will increase supply of coal to the power units across the country, which in turn will ensure stable power supply to industries and other sectors.

11.3 ANTICIPATED ENVIRONMENTAL IMPACTS AND MITIGATION MEASURES

The proposed project is not anticipated to raise the concentration of the pollutants beyond prescribed limits. However, the measures are suggested to mitigate any harmful impacts of pollutants, like the plantation of trees along roads especially near settlements to help to reduce the impact of dust on the nearby villages; planning, transportation routes of material so, as to reach the nearest paved roads by shortest route; regular water sprinkling on unpaved roads to avoid dust generation during transportation etc. The local people have been provided with either direct employment or indirect employment.

The mining operations may cause surface water pollution due to wash off from dumps and soil erosion. Proper control measures as mentioned in the EMP, which are essential to prevent the flow of suspended matter from the mine and dump should be undertaken.

It is proposed to liquidate the reserves by conventional benching method of opencast with shovel-dumper combination for overburden removal and use of

surface miner for coal removal. Out of the 1196 ha of mining lease area, at the end of life of mine 623.71 ha will be backfilled.

It is estimated that a total of 713.53 million m^3 of OB will be generated during mining and the same will be stacked in OB dumps.

The topsoil is removed before excavating the area and stored for future use. It is suitably protected from erosion and infertility by planting fodder grass and leguminous plants during temporary storage. The Topsoil will be utilized for Spreading over the backfilled and dump area.

Under CSR activities it is proposed to undertake activities as per the felt needs of surrounding villages like improvement of sanitation facilities, roads, drinking water, health, education etc. The quantity of explosive used for blasting is unlikely to create any strong vibration. Impact due to vibration on the surface structures is not anticipated due to the controlled blasting techniques.

The impact on the socio economic of surrounding area will be positive; expansion of existing mine production capacity will directly create employment. Preference will be given to the local resident of the area for employment as per the eligibility. There will be significant employment generation in secondary and tertiary sectors. The impact due to mining on flora and fauna is discussed in chapter 3 and 4. A comprehensive Regional Wildlife conservation plan (WLCP) is under preparation at DFO level. PP will deposit the proportionate amount as per the direction of Forest Department. Further Site-Specific Conservation Plan has been prepared & approved by PCCF & Chief Wildlife Warden, Odisha. A budgetary provision of 37.26 Cr (Including 5% of the total financial outlay of the SSWCP to be deposited in Wildlife Society of Odisha) is earmarked for 10 years.

11.4 ENVIRONMENT MANAGEMENT PLAN

A Comprehensive Environment Management Plan including development of Green Belt over 1029.583 ha area has been suggested. Plantation of about 25,74,000 nos. of native species of is envisaged over a period of 37 years of Mine Life.

Capital Expenditure of INR 10072.9 Lakhs and Recurring Expenditure of INR 315.7 Lakhs is envisaged for implementation of the suggested Environment Management Plan.

11.5 OVERALL JUSTIFICATION FOR IMPLEMENTATION OF THE PROJECT

Baitarni West Coal Mine of M/s GMDC Limited gives downstream integration of supply chain providing leverage on supply of coal from own source which ultimately helps in increased efficiency and flexibility of Steel/Power generation for existing and increased demand in future.

Considering the Demand/Supply Position of coal, extraction of coal from Baitarni West Coal Mine of M/s GMDC Limited is economically beneficial for the national economy.

Further, the Project shall be creating additional employment opportunities, peripheral development of the mining area and also contributing to the State Exchequer.

11.6 EXPLANATION ON HOW ADVERSE EFFECTS ARE MITIGATED

The EIA/EMP Report has established the Base Line Environment of the Project Area and has assessed anticipated impacts of the Project on the overall ecology & environment. Accordingly, general as well as specific mitigation measures for management of the Key Environmental Parameters have been suggested. Further, specific measures towards monitoring and implementation of the Environment Management Plan along with details of the funds required towards implementation of the Pollution Control Measures are also included in the Report. By implementing the suggested Environment Management Plan adverse effects of the Project can be mitigated.

11.7 CONCLUSION

The proposed Baitarni West Coal Mine is bound to have certain negative impacts on the local environment. However, with proper mitigation measures and an effective implementation of the Environment Management Plan as suggested in this EIA/EMP Report and further control measures as recommended by MoEF&CC, CPCB and State Pollution Control Board, the negative impacts may be minimized to a significant extent.

The Project on the other hand shall bring in positive and beneficial impacts in terms of growth in regional economy by transforming the economy from predominantly agricultural to significantly industrial, increase in government earnings and revenues and accelerate the pace of overall development of the region.

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CHAPTER-12 : DISCLOSURE OF CONSULTANT ENGAGED

12.0 GENERAL

M/s GMDC Limited has appointed M/s GRC India Pvt. Ltd., for carrying out this Environmental Impact Assessment Study as per the EIA notification, 2006 as amended till date. M/s GRC India Pvt. Ltd. has approved EIA coordinators and Field Area Experts for undertaking Environmental and related studies in Eighteen (18) approved sectors by NABET, Quality Council of India, New Delhi.

12.1 INTRODUCTION OF ORGANIZATION

GRC India is an ISO 9001:2015, 14001:2015 & ISO 45000:2018 certified pioneer environmental consultancy organisation in India. It has been accredited by National Accreditation Board of Education & Training (NABET), Quality Council of India (QCI), which is the highest accreditation authority in India. Accreditation is the formal recognition by an accreditation authority to carry out a specific service in accordance with the standards and technical regulations as prescribed in the scope of accreditation. The accreditation permits the organisation to carry out the Environment Impact Assessment (EIA) Studies for obtaining an Environmental Clearance for various development projects. This has been granted following a rigorous inspection based on a number of quality parameters applicable to concerned Environmental Functional Areas and the required infrastructure facilities available in the organisation, as prescribed by the MoEF&CC, Govt. of India in its guidelines.

The accreditation certificate number NABET/EIA/24-27/RA 0354 is valid till 16th August 2027. A copy of NABET certificate is attached as **Annexure XXIV** with EIA report.

Current Status of ACO is available on below given link:-
http://eia.nabet.qci.org.in/Accredited_EIA_Consultant.aspx

ADDRESS & CONTACTS

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Registered Office

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CSC-1, Mayur Vihar Phase 3, New Delhi – 110096,
Ph: 91-11-22622031

12.3 EIA CORDINATOR AND TEAM MEMBER

The EIA/EMP report has been prepared under the guidance of following Coordinators & Functional Area Experts:

EIA Coordinator	**Dr. K.L Satapathy**	**Coordinated the EIA Study and instructed FAEs and PAs for proper data generation, assessment and report writing**
Associate to EIA Coordinator	**Ms. Mudita Tomar Singh**	
FAE-AP	Dr. Dhiraj Kr Singh Ms. Monika Gupta (Category B)	Identifying the sources of emissions and mitigation measures. Inventorisation of point source stack emission details Site specific micro-meteorology monitoring. Ambient Air Quality (AAQ)monitoring impact predictions and mitigations
FAE-WP	Mr. Sonu Mr. Saurabh Gola (Cat-B)	Surface water and ground water quality monitoring and assessment. impacts on water Environment and mitigations. Identification, characterization of effluent streams and treatments thereof. Water balance and conservation measures
FAE-AQ	Dr. K.L. Satapathy Ms. Monika Gupta (Category B)	Visited the site and surroundings, assessed the meteorological data, did predictions on air quality and suggested mitigation measures.
FAE-EB	Dr. Ashok Kr Rathoure	Biological environment status in respect of terrestrial fauna and aquatic eco system. Impact on ecological environment.
FAE-SE	Mr. Rajiv Rastogi	Determination of baseline human environment. Impact on socioeconomic environment & mitigation methods.
FAE-HG	Prof. Tapan Majumder	Ground water resource assessment. Impact on ground water potential and mitigation measures for avoiding ground water contamination.
FAE-GEO	Prof. Tapan Majumder	Topography and geological aspects Developing geological maps

EIA Coordinator	**Dr. K.L Satapathy**	**Coordinated the EIA Study and instructed FAEs and PAs for proper data generation, assessment and report writing**
Associate to EIA Coordinator	**Ms. Mudita Tomar Singh**	
FAE-RH	Mr. K. D. Choudhury	Assessed the process details, identified risks / hazards and suggested mitigation measures.
FAE-SHW	Dr. Dhiraj Kr Singh	Non-hazardous solids, wastes generation, recycling and disposal. Storage and management of hazardous solid waste
FAE- SC	Mr. N P S Varde	Monitoring, analysis & characterization of soil. Assessment of impact on soil quality and mitigation measures.
FAE – NV	Dr. K.L. Satapathy	Analysis of ambient noise quality data. Impact due to plant noise and abatement measures.
FAE – LU	Mr. N P S Varde	Analysis of data related to land-use pattern. Land-use map development. Impact on land environment in respect to landform change.

12.4 LABORATORY INVOLVED FOR BASELINE MONITORING AND OTHER ANALYSIS

Baseline Monitoring during 1st Oct 2023 to 31st Dec 2023 and Analysis has been done by M/s GRC India Training & Analytical Laboratory (A Unit of GRC) Accredited by NABL and Recognized by MoEF&CC, New Delhi in accordance with the Guidelines for EIA issued by the Ministry of Environment Forests and Climate Change, Govt. of India.

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